LARGE LOCAL VOLUMES OF THREEFOLD SINGULARITIES

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Abstract. We affirmatively answer Questions 5.3 and 5.4 in Yuchen Liu's paper on optimal bounds for local volumes of threefold singularities. We prove that, among the local volumes of klt threefold singularities, those that are at least 9 are exactly the ten values listed in Question 5.3. We also prove that, among klt threefold singularities that are not finite quasi-étale quotients of local complete intersection singularities, the vertex of the anti-canonical cone over the first Hirzebruch surface has the largest local volume, and that up to analytic isomorphism it is the only such singularity with the largest local volume.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

The normalized volume of a valuation over a klt singularity was introduced in [Li18]. The local volume $\widehat{\mathrm{vol}}(X \ni x)$ of a klt germ $X \ni x$ is the infimum of the normalized volumes of the valuations of $K(X)$ centered at x (see Definition 2.6). We refer to the surveys [LLX20, Zhu25]. By the stable degeneration theorem [Blu18, LX18, LWX21, XZ21, BLQ24, XZ25], $\widehat{\mathrm{vol}}(X \ni x)$ is computed by a valuation that is unique up to scaling, and $X \ni x$ degenerates to a klt cone with a torus action whose local volume equals $\widehat{\mathrm{vol}}(X \ni x)$ (see Theorem 2.12 and Definition 2.8).

Let $X \ni x$ be a klt germ of dimension n. Then $\widehat{\mathrm{vol}}(X \ni x) \leq n^n$, and equality holds if and only if x is a smooth point of X [LX19, Theorem 1.6]. Spotti and Sun proposed the ODP gap conjecture [SS17, Conjecture 5.5] (see also [XZ26a, Problem 33]): if x is a singular point of X, then

$$\widehat{\mathrm{vol}}(X \ni x) \leq 2(n-1)^n,$$

and equality holds if and only if $X \ni x$ is an ordinary double point. The ODP gap conjecture was known in the following cases:

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(1) When $n = 3$, including the equality case [LX19, Theorem 1.3(1)].
(2) When $X \ni x$ is a local complete intersection [Liu22, Theorem 2.12].
(3) When $X \ni x$ is toric [MS24, Theorem 2].
(4) When $X \ni x$ is the vertex of the affine cone $C(V, L) = \operatorname{Spec} \bigoplus_{k\geq 0} H^0(V, kL)$, where V is a smooth K-semistable Fano manifold and L is an ample line bundle on V with $-K_V \sim rL$ for some positive integer r [LM25, Theorem 4.17].

The ODP gap conjecture has recently been settled in all dimensions [OAI26, Theorem 1.1], which we do not use.

Local volumes are discrete away from 0. For a fixed positive integer n and a fixed finite set $\Gamma \subset [0, 1]$, the set of local volumes $\widehat{\operatorname{vol}}(X \ni x, B)$ of klt pairs $(X \ni x, B)$ of dimension n with $B \in \Gamma$ is discrete away from 0 [XZ24, Theorem 1.2], which answers [LLX20, Question 6.12]. Earlier results in lower dimensions are [HLQ23, Zhu24, LMS26], and for a DCC set $\Gamma \subset [0, 1]$ the set of local volumes satisfies the ACC [HLQ24, Theorem 1.2]. In particular, by [XZ24, Theorem 1.2] with $n = 3$ and $B = 0$, the set of local volumes of klt threefold germs in $[9, 27]$ is finite, and Question 1.1 below asks which values it contains.

We now turn to threefolds. Yuchen Liu [Liu25] obtained the following optimal bounds for a klt threefold germ $X \ni x$ (see Theorem 2.17). If $X \ni x$ is Gorenstein canonical and is not a hypersurface germ, then $\widehat{\operatorname{vol}}(X \ni x) \leq 9$, and equality holds if and only if $X \ni x$ is a cyclic quotient singularity of type $\frac{1}{3}(1, 1, 1)$ [Liu25, Theorem 1.1]. Moreover, $\widehat{\operatorname{vol}}(X \ni x) \geq 9$ if and only if $X \ni x$ is a hypersurface germ of type cA_k (see Definition 2.14) for some $k \leq 2$, or a cyclic quotient singularity of type $\frac{1}{2}(1, 1, 1)$, $\frac{1}{3}(1, 1, 0)$, $\frac{1}{3}(1, 1, 1)$, or $\frac{1}{3}(1, 1, 2)$ [Liu25, Theorem 1.2]. Liu also proved that

$$\widehat{\operatorname{vol}}(X \ni x) \leq 9\operatorname{mld}(X \ni x),$$

and that equality holds if and only if $X \ni x$ is a cyclic quotient singularity of type $\frac{1}{r}(1, 1, 1)$ for some positive integer r [Liu25, Theorem 1.4], which answers [LLX20, Question 6.16] in dimension 3. The inequality was extended to all dimensions [Han26, Theorem 1.1]: $\widehat{\operatorname{vol}}(X \ni x, B) \leq n^{n-1}\operatorname{mld}(X \ni x, B)$ for any klt pair $(X \ni x, B)$ of dimension n.

Liu asked the following two questions.

Question 1.1 ([Liu25, Question 5.3])**.** Let $\widehat{\operatorname{Vol}}_3$ be the set of the local volumes $\widehat{\operatorname{vol}}(X \ni x)$ of all klt threefold germs $X \ni x$. Then

$$\widehat{\operatorname{Vol}}_3 \cap [9, 27] \supset \left\{9, \frac{2048}{225}, \frac{250}{27}, \frac{343}{36}, 6\sqrt{3}, \frac{32}{3}, \frac{27}{2}, \frac{125}{9}, 16, 27\right\}.$$

Is $\widehat{\operatorname{Vol}}_3 \cap [9, 27]$ equal to the set on the right-hand side?

The ten values in Question 1.1 come from [Liu25, Example 5.2], which relies on [Li18, LL19, LX18, LST25], [LX20, Example 7.1], and [CS19, Theorem 8.1]. A smooth point has local volume 27. The germs of $\{uv + z^2 + t^{k+1} = 0\} \subset \mathbb{A}^4$ at the origin (the A_k singularities) have local volumes 16, $\frac{125}{9}$, and $\frac{27}{2}$ for $k = 1$, $k = 2$, and $k \geq 3$, respectively (see Proposition 3.1), and the germ of $\{uv + z^2 = 0\} \subset \mathbb{A}^4$ at the origin (the transversal A_1 singularity), that is, the cyclic quotient singularity $\frac{1}{2}(1, 1, 0)$, has local volume $\frac{27}{2}$. The germ of $\{uv + z^3 + t^k = 0\} \subset \mathbb{A}^4$ at the origin has local volume $\frac{4(3+k)^3}{9k^2}$ for $3 \leq k \leq 6$, that is, $\frac{32}{3}$, $\frac{343}{36}$, $\frac{2048}{225}$, and 9. The germs of $\{uv + z^2t + t^{k-1} = 0\}$ at the origin with $k \geq 5$ (the D_k singularities) have local volume $6\sqrt{3}$, and the germ of $\{uv + z^3 + zt^3 = 0\}$ at the origin (the E_7 singularity) has local volume $\frac{250}{27}$. For a finite group $G \subset \mathrm{GL}_3(\mathbb{C})$ acting freely in codimension one, the finite degree formula [XZ21, Theorem 1.3] for the Galois morphism $\mathbb{C}^3 \to \mathbb{C}^3/G$ implies $\widehat{\operatorname{vol}}(\mathbb{C}^3/G \ni 0) = \frac{27}{|G|}$. Thus the cyclic quotient singularity $\frac{1}{2}(1, 1, 1)$ has local volume $\frac{27}{2}$, and the cyclic quotient singularities of types $\frac{1}{3}(1, 1, 0)$, $\frac{1}{3}(1, 1, 1)$, and $\frac{1}{3}(1, 1, 2)$ have local volume 9.

The local volumes of smooth germs, of hypersurface germs of type cA_1, which are the A_k singularities and the transversal A_1 singularity (Proposition 3.2(1)), and of the four cyclic quotient

singularities in [Liu25, Theorem 1.2] are listed above. Thus, by [Liu25, Theorem 1.2], Question 1.1 concerns the local volumes of hypersurface germs of type cA_2, and [Liu25, Example 5.2(3)] writes the formula $\frac{4(3+k)^3}{9k^2}$ only for $3 \leq k \leq 6$. The hypersurface germs of type cA_2 form infinitely many analytic isomorphism classes, and for the germs $X_k \ni 0$ below with $k \geq 7$, the weights making the equation quasi-homogeneous do not give the local volume. For any integer $k \geq 3$, let $f_k := uv + z^3 + t^k$ and let $X_k \ni 0$ be the germ of $\{f_k = 0\} \subset \mathbb{A}^4$ at the origin ($X_k \ni 0$ is an isolated singularity, since the partial derivatives $v, u, 3z^2, kt^{k-1}$ of f_k vanish simultaneously only at the origin, hence normal [Har77, Proposition II.8.23(b)], and a general hyperplane section of X_k through 0 is an A_2 singularity by a direct check on the section $t = au + bv + cz$, so $X_k \ni 0$ is of type cA_2 and terminal [KM98, Corollary 5.38]). The space T^1 of first-order deformations of $X_k \ni 0$ satisfies

$$\dim_{\mathbb{C}} T^1 = \dim_{\mathbb{C}} \mathbb{C}[u,v,z,t]/(f_k, \partial_u f_k, \partial_v f_k, \partial_z f_k, \partial_t f_k) = \dim_{\mathbb{C}} \mathbb{C}[z,t]/(z^2, t^{k-1}) = 2(k-1)$$

[Har10, Proposition 3.10] (the module T^1 is supported at the origin and is compatible with localization [Har10, Exercise 3.5] and, since completion is flat and $(f_k)/(f_k^2)$ is a finitely presented module, with completion [Mat86, Theorems 7.11 and 8.8]). Thus the germs $X_k \ni 0$ with $k \geq 6$ are pairwise not analytically isomorphic (see Definition 2.4). Let w be the monomial valuation on $\mathbb{A}^4$ with weights $(\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ on (u, v, z, t). The weights $(\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ make f_6 quasi-homogeneous of weight 1, and $w(t^k) = \frac{k}{6} \geq 1$ with equality only for $k = 6$, so $w(f_k) = 1$ for any $k \geq 6$. Proposition 2.20 (cf. [Liu25, Proposition 4.1]) with the weights $(\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ gives

$$\widehat{\mathrm{vol}}(X_k \ni 0) \leq \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{3} + \frac{1}{6} - 1\right)^3 \cdot \left(\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{6}\right)^{-1} = 9$$

for any $k \geq 6$. The weights $(3, 3, 2, 1) = 6 \cdot (\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ are used in Section 3. The weights $(\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{k})$ make f_k quasi-homogeneous of weight 1, and the right-hand side of Proposition 2.20 for the weights $(\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{k})$ is $\frac{4(3+k)^3}{9k^2}$. For $3 \leq k \leq 6$ it equals $\widehat{\mathrm{vol}}(X_k \ni 0)$ [Liu25, Example 5.2(3)], while for $k \geq 7$ it is larger than $9 \geq \widehat{\mathrm{vol}}(X_k \ni 0)$, and for $k = 7$ it equals $\frac{4000}{441}$. Therefore, for $k \geq 7$, the quasi-homogeneous weights do not give $\widehat{\mathrm{vol}}(X_k \ni 0)$, and the method of [Liu25, Example 5.2(3)], which writes down the local volume from the weights making the equation quasi-homogeneous, does not extend to the family $\{X_k \ni 0\}_{k \geq 7}$. On the other hand, $\widehat{\mathrm{vol}}(X_k \ni 0) \leq 9$ for any $k \geq 6$, and together with [Liu25, Theorem 1.2], $\widehat{\mathrm{vol}}(X_k \ni 0) = 9$ (see Section 3).

Let

$$C_{\mathbb{F}_1} := \mathrm{Spec} \bigoplus_{m \geq 0} H^0(\mathbb{F}_1, \mathcal{O}_{\mathbb{F}_1}(-mK_{\mathbb{F}_1}))$$

be the anti-canonical cone over the Hirzebruch surface $\mathbb{F}_1$ (see Definition 2.10), with vertex o. The local volume of $C_{\mathbb{F}_1} \ni o$ is computed in [Blu18, Section 8.3], and $C_{\mathbb{F}_1} \ni o$ already appears in [MS06, Section 7]. See also [MSY06, Section 3] for the determination of the Reeb vector by volume minimization on toric Sasakian cones.

Question 1.2 ([Liu25, Question 5.4])**.** Let $C_{\mathbb{F}_1} \ni o$ be the anti-canonical cone over the Hirzebruch surface $\mathbb{F}_1$ with vertex o. Then $C_{\mathbb{F}_1} \ni o$ is Gorenstein canonical and $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = \frac{46+13\sqrt{13}}{12}$. Is $\frac{46+13\sqrt{13}}{12}$ the largest local volume of klt threefold germs that are not LCIQ, that is, that are not finite quasi-étale quotients of local complete intersection singularities?

Throughout the paper, we write

$$q := \frac{46 + 13\sqrt{13}}{12} \approx 7.74.$$

We now explain why the methods of [Liu25] do not reach Question 1.2. In [Liu25, Section 4.4], the bound 9 of [Liu25, Theorem 1.1] is obtained by induction on the number of prime divisors E over X with $\mathrm{center}_X E = \{x\}$ and $a(E, X) = 1$. Each step passes from $X \ni x$, after replacing

it by an étale neighborhood in one case, to a germ $Y' \ni y'$ with $K_{Y'} = g^*K_X$ for a projective birational morphism $g : Y' \to X$, and uses the inequality

$$\widehat{\mathrm{vol}}(X \ni x) < \widehat{\mathrm{vol}}(Y' \ni y') \leq 9.$$

The germs $Y' \ni y'$ occurring there may be LCIQ germs (see Definition 2.13) of local volume 9 or 8. For example, the vertex of the anti-canonical cone over $\mathbb{P}^2$ is the cyclic quotient singularity $\frac{1}{3}(1,1,1)$, of local volume $\frac{27}{3} = 9$ [XZ21, Theorem 1.3]. The vertex of the anti-canonical cone over $\mathbb{P}^1 \times \mathbb{P}^1$ is the quotient of the three-dimensional ordinary double point by ± 1, of local volume $\frac{16}{2} = 8$ [LX19, Theorem 1.3(1)], [XZ21, Theorem 1.3]. The germ $(\mathbb{C}^2/\mu_3) \times \mathbb{C}$ of a transversal A_2 singularity is a quotient of $\mathbb{C}^3$, of local volume $\frac{27}{3} = 9$ [XZ21, Theorem 1.3]. Since $Y' \ni y'$ may be LCIQ even if $X \ni x$ is not, a better bound for germs that are not LCIQ cannot be applied to $Y' \ni y'$, and the inequality only gives $\widehat{\mathrm{vol}}(X \ni x) < 9$.

The minimizer of $\widehat{\mathrm{vol}}$ on $C_{\mathbb{F}_1} \ni o$ is unique up to scaling [XZ21, Theorem 1.1] and is not a multiple of a divisorial valuation [Blu18, Section 8.3]. In particular, $\widehat{\mathrm{vol}}(\mathrm{ord}_E) > q$ for any prime divisor E over $C_{\mathbb{F}_1}$ with center o. Moreover, let $C_{\mathbb{P}^1\times\mathbb{P}^1} \ni o$ be the anti-canonical cone over $\mathbb{P}^1 \times \mathbb{P}^1$ with vertex o. Then $C_{\mathbb{F}_1} \ni o$ and the LCIQ germ $C_{\mathbb{P}^1\times\mathbb{P}^1} \ni o$ have equal Hilbert functions of their graded coordinate rings and equal embedding dimensions, neither of them is a local complete intersection, and the exceptional divisors of the blow-ups of their vertices have equal normalized volumes. Indeed, let S be $\mathbb{P}^1 \times \mathbb{P}^1$ or $\mathbb{F}_1$. Then S is a smooth del Pezzo surface with $(K_S^2) = 8$, and by the Riemann–Roch theorem and Kodaira vanishing,

$$\dim_{\mathbb{C}} H^0(S, \mathcal{O}_S(-mK_S)) = 1 + \frac{m(m+1)}{2}(K_S^2) = (2m+1)^2$$

for any integer $m \geq 0$. In particular, the embedding dimension of $C_S \ni o$ is 9 (for cones see [Kol13, Section 3.1]: the blow-up of the vertex has exceptional divisor $E \cong S$, and the section ring of $-K_S$ is generated in degree one, since S is a smooth projective toric surface [CLS11, Proposition 4.3.3 and Corollary 2.2.13]). The ideal of $C_S \subset \mathbb{A}^9$ contains no linear forms and contains $45 - 25 = 20$ linearly independent quadrics, so it needs at least 20 generators at o. Since $20 > 9 - \dim C_S = 6$, $C_S \ni o$ is not a local complete intersection (otherwise its ideal in the regular local ring of $\mathbb{A}^9$ at o would be generated by a regular sequence of length 6 [Mat86, Theorem 21.2(ii)]). Moreover, the exceptional divisor E of the blow-up of C_S at o is isomorphic to S, $a(E, C_S) = 1$, and

$$\widehat{\mathrm{vol}}(\mathrm{ord}_E) = \mathrm{vol}(\mathrm{ord}_E) = (K_S^2) = 8,$$

which is the value of the upper bound $\widehat{\mathrm{vol}}(X \ni x) \leq \widehat{\mathrm{vol}}(\mathrm{ord}_E) = (K_E^2)$ of [Liu25, (4.2)] for $X \ni x = C_S \ni o$ (see Lemma 2.19(2)). However, $C_{\mathbb{P}^1\times\mathbb{P}^1} \ni o$ is LCIQ of local volume 8, while $C_{\mathbb{F}_1} \ni o$ is not LCIQ and $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q < 8$ (see Construction-Proposition 4.2).

Question 1.2 asks only for the largest value, and the germs attaining it need a separate argument. In [Liu25, Proposition 4.5], the equality case of [Liu25, Theorem 1.1] is obtained as follows: an exceptional divisor E over X with $a(E, X) = 1$ and $E \cong \mathbb{P}^2$ is a Kollár component (see Definition 2.18) computing $\widehat{\mathrm{vol}}(X \ni x)$, $X \ni x$ degenerates isotrivially to the cone over $\mathbb{P}^2$ with polarization $\mathcal{O}_{\mathbb{P}^2}(3)$, and the rigidity of the quotient singularity $\frac{1}{3}(1,1,1)$ implies $X \ni x \cong \frac{1}{3}(1,1,1)$. Neither ingredient is available for $C_{\mathbb{F}_1} \ni o$. No divisorial valuation computes $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o)$ [Blu18, Section 8.3], and $C_{\mathbb{F}_1} \ni o$ is not rigid, since the deformation functor of the completion of $C_{\mathbb{F}_1}$ at o has the miniversal hull $\mathbb{C}[\epsilon]/(\epsilon^2)$ [Alt94, (9.1)(iv)–(9.2)] (see Proposition 4.4(2)).

In this paper, we answer Questions 1.1 and 1.2 completely, and we classify the klt threefold germs that are not LCIQ and have local volume q.

Main results. Our first main theorem answers Question 1.1 affirmatively.

Theorem 1.3. *Let $\widehat{\mathrm{Vol}}_3$ be the set of the local volumes of all klt threefold germs. Then*

$$\widehat{\mathrm{Vol}}_3 \cap [9,27] = \left\{9, \frac{2048}{225}, \frac{250}{27}, \frac{343}{36}, 6\sqrt{3}, \frac{32}{3}, \frac{27}{2}, \frac{125}{9}, 16, 27\right\}.$$

Our second main theorem bounds the local volumes of klt threefold germs that are not LCIQ and determines the germs attaining the bound.

Theorem 1.4. *Let $X \ni x$ be a klt threefold germ that is not LCIQ. Then*

$$\widehat{\mathrm{vol}}(X \ni x) \le \frac{46+13\sqrt{13}}{12},$$

and equality holds if and only if $X \ni x$ is analytically isomorphic (see Definition 2.4) to $C_{\mathbb{F}_1} \ni o$. Moreover, $C_{\mathbb{F}_1} \ni o$ is not LCIQ.

Theorem 1.4 answers Question 1.2 affirmatively. By [Liu25, Theorem 1.2] (see Theorem 2.17(3)), a klt threefold germ of local volume at least 9 is either a hypersurface germ, which is a local complete intersection, or a cyclic quotient germ of one of the four types listed there. Each of the four cyclic groups acts on $\mathbb{A}^3$ freely in codimension one: a non-trivial element of μ_3 acting with weights $(1,1,0)$ fixes only the line $\{x_1 = x_2 = 0\}$, and for the other three types a non-trivial element fixes only the origin. Thus any klt threefold germ of local volume at least 9 is LCIQ, and the content of Theorem 1.4 is that no non-LCIQ klt threefold germ has local volume in $[q, 9)$, except the germs analytically isomorphic to $C_{\mathbb{F}_1} \ni o$.

We now explain where the difficulty of the upper bound in Theorem 1.4 is. Let $X \ni x$ be a non-LCIQ klt threefold germ with $\widehat{\mathrm{vol}}(X \ni x) \ge q$, and let v be a valuation computing $\widehat{\mathrm{vol}}(X \ni x)$. Then $X \ni x$ is Gorenstein, and it degenerates to the Gorenstein cone $X_0 := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$ with vertex x_0 and $\widehat{\mathrm{vol}}(X_0 \ni x_0) = \widehat{\mathrm{vol}}(X \ni x)$. The proof is divided according to the rational rank r of v, that is, the dimension of the $\mathbb{Q}$-vector space spanned by the values $v(f)$ for $f \in \mathcal{O}_{X,x}\backslash\{0\}$, and $r \le 2$. If $r = 2$, then X_0 carries an action of a two-dimensional torus. Apart from the non-LCIQ toric cones, which reach q only at $C_{\mathbb{F}_1} \ni o$, such cones are LCIQ or have local volume at most $\frac{27}{4}$, which is smaller than q by more than 0.98. The equality case uses only the case $r = 2$, since q is irrational while the local volume is rational when a valuation of rational rank one computes it.

If $r = 1$, then a positive multiple of v is a divisorial valuation, and the coordinate ring of X_0 carries a positive $\mathbb{Z}$-grading whose degree-zero part is $\mathbb{C}$ and whose degrees generate $\mathbb{Z}$. Let $V := \widehat{\mathrm{vol}}(X \ni x)$, and let α be the degree of a homogeneous generator of the canonical module of X_0. Then α is a positive integer, equal to the log discrepancy of the valuation defined by the grading. If $\alpha = 1$, then $V \le 7$, and the main work is the case $\alpha \ge 2$. A local count at the points of $X_0\backslash\{x_0\}$ with non-trivial stabilizer in $\mathbb{G}_m$ only shows

$$V \le \frac{9\alpha}{\alpha+1},$$

which excludes $\alpha \le 6$, since $\frac{54}{7} < q < \frac{63}{8}$. A local description of the centers of crepant divisors over x_0 works only for bounded α: it excludes points as centers when $\alpha \le 12$, and two curves as centers when $\alpha \le 39$. The bound for curves is also an input of the global argument below: if two curves are centers of crepant divisors over x_0, then $\alpha \ge 40$ and the order of the stabilizer in $\mathbb{G}_m$ of a general point of one of them is less than $\frac{40\alpha}{39}$ (Proposition 10.13(2)), and the global argument uses the bound on the stabilizer order to exclude the second curve (Proposition 10.17).

Since $\frac{9\alpha}{\alpha+1} \to 9$ as $\alpha \to +\infty$, we then argue globally on the quotient S of $X_0\backslash\{x_0\}$ by $\mathbb{G}_m$, with the boundary $\Delta = \sum_P (1 - \frac{1}{m_P})P$, where m_P is the order of the stabilizer of a general point of X_0 over the prime divisor P, using complements of (S, Δ) and Kawamata–Viehweg vanishing. The global argument works only for $V > \frac{54}{7}$, and $q - \frac{54}{7}$ is only about 0.025. It needs a complement of (S, Δ) of index at most 6, which is where the germs with one or two crepant divisors enter (see part (c) of the sketch of the proof below). For general log del Pezzo pairs with standard

coefficients, the index of a complement cannot be bounded by 6: for four lines $L_1, \dots, L_4$ in general position in $\mathbb{P}^2$, the smallest index of a complement of $(\mathbb{P}^2, \frac{1}{2}L_1 + \frac{2}{3}L_2 + \frac{10}{11}L_3 + \frac{12}{13}L_4)$ is 66 [Pro01, Example 10.3.1].

Sketch of the proof. The proof of Theorem 1.4 has four parts. Let $X \ni x$ be a klt threefold germ which is not LCIQ, with $V := \widehat{\mathrm{vol}}(X \ni x) \geq q$, and let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$ (Definition 2.6).

(1) $X \ni x$ is Gorenstein canonical and degenerates, in a flat family of finite type, to the Gorenstein, K-semistable, non-LCIQ cone $X_0 := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$ with vertex x_0 and $\widehat{\mathrm{vol}}(X_0 \ni x_0) = V$ (Proposition 5.8), and v has rational rank one or two (Proposition 6.4).
(2) If v has rational rank two, then a two-dimensional torus acts on X_0, and $V \leq q$ by Theorem 7.3.
(3) If v has rational rank one, then $\mathbb{C}[X_0]$ carries a positive $\mathbb{Z}$-grading whose degree-zero part is $\mathbb{C}$ and whose degrees generate $\mathbb{Z}$, and the canonical module of X_0 is generated by a homogeneous element of degree α, a positive integer (Lemma 10.2), and $V \leq 7$ if $\alpha = 1$ (Proposition 10.6). If $\alpha \geq 2$, then $V \leq q$ by Theorem 10.3, whose proof uses a complement of (S, Δ) (Construction-Proposition 10.4) of index at most 6 (Proposition 10.14, which depends on Theorems 8.1 and 9.1) and two crepant divisors over x_0 with crepant point centers (Definition 10.11 and Proposition 10.17).
(4) If $V = q$, then v has rational rank two and $X_0 \cong C_{\mathbb{F}_1}$ (Proposition 11.3), and Propositions 4.4(3) and 5.9, applied to the family in (1), show that $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$.

The key new step is the end of the proof of Theorem 10.3. There, Kawamata–Viehweg vanishing on a higher model of S, with a klt boundary obtained by perturbing the complement, implies that some homogeneous element of $\mathbb{C}[X_0]$ vanishes on the $\mathbb{G}_m$-orbit over one crepant point center and not on the orbit over the other, while Proposition 5.6 forces any non-zero homogeneous element of that degree to vanish on neither orbit. Thus the case $\alpha \geq 2$ is proved by contradiction, without an explicit bound below q.

(a) Degeneration to a Gorenstein non-LCIQ cone. The stable degeneration theorem [XZ25, Theorems 1.1 and 1.2] (Theorem 2.12) provides X_0, but Theorems 7.3 and 10.3 need X_0 to be Gorenstein and $X_0 \ni x_0$ to be not LCIQ. The Gorenstein property may fail on the special fiber of a flat family, and an LCIQ special fiber has to be shown to force LCIQ nearby fibers. We degenerate $X \ni x$ through the extended Rees algebra of a divisorial valuation w with $\operatorname{gr}_w \mathcal{O}_{X,x} \cong \operatorname{gr}_v \mathcal{O}_{X,x}$ [XZ25, Corollary 4.10]. The family is a product away from its central fiber, which is a reduced prime divisor, so X_0 is Gorenstein (Lemma 5.4). For the stable degeneration, the Gorenstein property of X_0 also follows from [NO25, Proposition 4.1], see also [XZ26b, Lemma 2.8]. Proposition 5.5 adds that the canonical module of X_0 has a generator homogeneous with respect to the value group.

Since $\frac{125}{18} < q \leq V < 9$ (Proposition 5.8(1)), if $X_0 \ni x_0$ were LCIQ, then so would be $X \ni x$ (Proposition 5.3(3)). If $X_0 \ni x_0$ were a local complete intersection, then the nearby fibers of the family would be local complete intersections. If $X_0 \ni x_0$ were LCIQ and not a local complete intersection, then it would be the quotient of an ordinary double point by one of two involutions (Proposition 5.3(1)), and its double cover on the punctured germ would extend over the family (Proposition 5.3(2)).

(b) Cones of complexity one. Unlike the toric case, which Lemma 6.3(3) reduces to finitely many lattice polygons, we have no finite list of cones with a two-dimensional torus action, whose Cox rings [ABHW18, Construction 2.1] can have arbitrarily many blocks of variables. The Cox ring is a polynomial ring in variables y_{ij}, grouped into blocks indexed by i, possibly with further free variables, modulo trinomial relations among one monomial in each block (Proposition 7.6 and [ABHW18, Construction 2.1]). We first turn the LCIQ property, a condition on all quasi-étale covers, into a condition on the Cox ring: if X_0 is $\mathbb{Q}$-factorial at x_0, or if the exponents are constant

within each block and the Cox ring has no free variables, then a quotient of the spectrum of the Cox ring is a quasi-étale cover of X_0 by a complete intersection (Propositions 7.5 and 7.6).

We then bound V by normalized volumes of Reeb valuations attached to points of the degree polyhedron of the p-divisor of X_0 (Construction-Proposition 7.7 and Lemma 7.8(2)), without determining the minimizer. For a primitive canonical character (Definition-Lemma 7.2), non-$\mathbb{Q}$-factoriality implies a lower bound on the size of the degree polyhedron (Proposition 7.12). Otherwise, Proposition 5.6(2) for a Cox monomial bounds the exponents in a block with two different exponents.

(c) Complements of the quotient pair. When $\alpha \geq 2$, (S, Δ) is only known to be a K-semistable log del Pezzo pair with standard coefficients, and Shokurov's complements of index $N \leq 6$ [Pro01, Corollary 8.4.2], [BL23, Proposition 5.3] exist for non-exceptional pairs (Definition 10.8). Since $V \geq q > \frac{54}{7}$, any crepant divisor over x_0 restricts to a divisor over S of log discrepancy less than $\frac{1}{7}$ (Lemma 10.10(2)), and an exceptional (S, Δ) has at most two such divisors by [Pro01, Corollary 10.1.2] (Lemma 10.10(4)). If $V > q$, then Theorems 8.1 and 9.1 on arbitrary Gorenstein canonical germs imply that at least three crepant divisors exist, so (S, Δ) is not exceptional. Since log discrepancies with respect to the complement Δ^+ belong to $\frac{1}{N}\mathbb{Z}$ with $\frac{1}{N} > \frac{1}{7}$, the restrictions of all crepant divisors become lc places of (S, Δ^+) (Proposition 10.14).

In the proof of Theorem 9.1, where [Liu25, proofs of Propositions 4.7 and 4.9] drop the different of a Kollár component $\mathbb{P}(1,1,3)$ and bound the local volume only by $\frac{25}{3} > q$, we keep as the different half of the curve along which the exceptional divisor E over x is not normal, which is the conductor of the normalization of E (Proposition 9.17 and Lemma 9.6(3)), and the volume of the Kollár component is computed in Proposition 9.17(6) and, for Lemma 9.6(3), by Lemma 9.4(3).

(d) From the cone back to the germ. Identifying X_0 does not identify $X \ni x$: the minimizing cone of the D_k singularity with $k \geq 5$ is the D_∞ singularity [Liu25, Example 5.2(4)], which is not isolated. Although $C_{\mathbb{F}_1}$ is not rigid, its miniversal hull is $\mathbb{C}[\epsilon]/(\epsilon^2)$ (Proposition 4.4(2)), and the image of ϵ in a reduced ring has square zero, hence is zero. Thus any formal deformation of the completion of $C_{\mathbb{F}_1}$ at o over a reduced complete local base is trivial (Proposition 4.4(3)), and so is the completion of the family in (1) at the vertex of X_0. Proposition 5.9 then implies that the family is an analytic product near the section, and since $C_{\mathbb{F}_1} \ni o$ is an isolated singularity and the non-LCIQ germ $X \ni x$ is singular, the section stays at the singular point.

Finally, Theorem 1.3 follows from [Liu25, Theorem 1.2] and the normal forms of germs of types cA_1 and cA_2 (Proposition 3.2). The named normal forms have the local volumes of Proposition 3.1, and the remaining family of cA_2 germs has local volume 9 by the weight $(3,3,2,1)$ in Proposition 2.20 and the lower bound in Theorem 2.17(4).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce LCIQ singularities and recall the known bounds on local volumes of threefold singularities that we use.

In Section 3 we prove Theorem 1.3 by normal forms of hypersurface singularities of types cA_1 and cA_2, and Theorem 1.3 is used again in Section 9, where the fact that no klt threefold germ has local volume in $(\frac{125}{9}, 16)$ handles the case in which the exceptional divisor of the blow-up of the maximal ideal is a prime divisor that is not Cartier.

In Section 4 we compute the local volume of $C_{\mathbb{F}_1} \ni o$, show that $C_{\mathbb{F}_1} \ni o$ has no non-trivial finite quasi-étale cover and is not LCIQ, and prove that the completion of $C_{\mathbb{F}_1}$ at o has no non-trivial formal deformation over a reduced complete local base, although it has a non-zero first-order deformation.

In Section 5 we reduce to Gorenstein germs by index-one covers, show how the LCIQ property behaves under degenerations preserving the local volume, show that the associated graded ring of a valuation computing the local volume of a Gorenstein germ is Gorenstein, prove an inequality between such a valuation and the orders of vanishing of functions along prime divisors through

the point, which is used in Sections 7 and 10, degenerate a germ to the cone given by a valuation computing the local volume, and deduce analytic isomorphisms from formal ones.

In Section 6 we prove the upper bound of Theorem 1.4 for toric singularities, with equality only for $C_{\mathbb{F}_1} \ni o$, and we show that a valuation computing the local volume of a klt threefold germ of local volume at least q has rational rank at most 2.

In Section 7 we study Fano cones of complexity one, that is, threefold Fano cones with a faithful action of a two-dimensional torus, which arise when the valuation computing the local volume has rational rank 2, and prove the upper bound of Theorem 1.4 for them.

In Section 8 we classify the Gorenstein canonical threefold germs with exactly one crepant divisor and local volume greater than 7.

In Section 9 we show that a Gorenstein canonical threefold germ with exactly two crepant divisors has local volume at most $\frac{250}{33}$, and, together with Section 8, that a non-LCIQ Gorenstein canonical threefold germ of local volume greater than q has at least three crepant divisors, which is used in Section 10.

In Section 10 we study quasi-regular Fano cones, which arise when the valuation computing the local volume has rational rank 1, and prove the upper bound of Theorem 1.4 for them.

In Section 11 we prove Theorem 1.4.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and germs.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. A *germ* $X \ni x$ consists of a normal quasi-projective variety X and a closed point $x \in X$, and we shrink X around x whenever convenient. We say that $X \ni x$ is klt (resp. Gorenstein, canonical, smooth) if X is klt (resp. Gorenstein, canonical, smooth) near x, and that $X \ni x$ is a *hypersurface germ* if the embedding dimension of X at x is at most $\dim X + 1$. For a pair $(X \ni x, B)$, we say that $(X \ni x, B)$ is lc (resp. klt) if (X, B) is lc (resp. klt) near x.

Definition 2.3. Let $(X \ni x, B)$ be an lc pair, and let $\mathfrak{b} \subset \mathcal{O}_X$ be a non-zero ideal. The *log canonical threshold* of $\mathfrak{b}$ with respect to $(X \ni x, B)$ is

$$\operatorname{lct}(X \ni x, B; \mathfrak{b}) := \inf_E \frac{a(E, X, B)}{\operatorname{mult}_E \mathfrak{b}},$$

where E runs over the prime divisors over X such that $x \in \operatorname{center}_X E$ and $\operatorname{mult}_E \mathfrak{b} > 0$. For a $\mathbb{Q}$-Cartier Weil divisor $D \geq 0$ on X, we define $\operatorname{lct}(X \ni x, B; D) := \inf_E a(E, X, B)/\operatorname{mult}_E D$, where E runs over the prime divisors over X such that $x \in \operatorname{center}_X E$ and $\operatorname{mult}_E D > 0$. We write $\operatorname{lct}(X \ni x; \mathfrak{b}) := \operatorname{lct}(X \ni x, 0; \mathfrak{b})$ and $\operatorname{lct}(X \ni x; D) := \operatorname{lct}(X \ni x, 0; D)$.

Definition 2.4. Let $X \ni x$ and $X' \ni x'$ be germs. We say that $X \ni x$ and $X' \ni x'$ are *analytically isomorphic* if $\widehat{\mathcal{O}}_{X,x} \cong \widehat{\mathcal{O}}_{X',x'}$ as $\mathbb{C}$-algebras.

Lemma 2.5. *Let $X \ni x$ and $X' \ni x'$ be germs. Then the following hold.*

1. *If $X \ni x$ and $X' \ni x'$ are analytically isomorphic, then $x \in X$ and $x' \in X'$ have a common étale neighborhood.*
2. *$X \ni x$ and $X' \ni x'$ are analytically isomorphic if and only if the complex analytic germs of X at x and of X' at x' are isomorphic.*

Proof. (1) is [Art69, Corollary 2.6]. If $x \in X$ and $x' \in X'$ have a common étale neighborhood, then the complex analytic germs of X at x and of X' at x' are isomorphic, since an étale morphism induces local isomorphisms of the associated complex analytic spaces [Gro71, Exposé XII, Proposition 3.1]. Conversely, if the complex analytic germs of X at x and of X' at x' are isomorphic, then $\widehat{\mathcal{O}}_{X,x} \cong \widehat{\mathcal{O}}_{X',x'}$, since $\widehat{\mathcal{O}}_{X,x}$ is isomorphic to the completion of the local ring of X^{an} at x [Ser56, §6, Proposition 3], and similarly for X'. This proves (2). □

2.2. **Local volumes.** The local volume of a klt singularity was introduced in [Li18].

Definition 2.6. Let $(X \ni x, B)$ be a klt pair of dimension n. We denote by $\operatorname{Val}_{X,x}$ the set of real valuations of the function field $K(X)$ which are centered at x, and by $a(v, X, B) \in (0, +\infty]$ the log discrepancy of $v \in \operatorname{Val}_{X,x}$, which extends the log discrepancy of divisorial valuations, see [JM12, Section 5] for smooth varieties and [Blu18, Sections 2.5 and 7.1] for pairs. For $v \in \operatorname{Val}_{X,x}$ and $\lambda \in \mathbb{R}_{\geq 0}$, let $\mathfrak{a}_\lambda(v) := \{g \in \mathcal{O}_{X,x} \mid v(g) \geq \lambda\}$ and $\mathfrak{a}_{>\lambda}(v) := \{g \in \mathcal{O}_{X,x} \mid v(g) > \lambda\}$, and let

$$\operatorname{vol}(v) := \lim_{m \to +\infty} \frac{\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{a}_m(v)}{m^n/n!}$$

be the *volume* of v, where the limit exists, see [Blu18, Section 2.8]. The *normalized volume* of v is $\widehat{\operatorname{vol}}(v) := a(v, X, B)^n \operatorname{vol}(v)$ if $a(v, X, B) < +\infty$, and $\widehat{\operatorname{vol}}(v) := +\infty$ otherwise. The *local volume* of $(X \ni x, B)$ is

$$\widehat{\operatorname{vol}}(X \ni x, B) := \inf_{v \in \operatorname{Val}_{X,x}} \widehat{\operatorname{vol}}(v),$$

and we write $\widehat{\operatorname{vol}}(X \ni x) := \widehat{\operatorname{vol}}(X \ni x, 0)$. We say that $v \in \operatorname{Val}_{X,x}$ is a *minimizer*, or that v *computes* $\widehat{\operatorname{vol}}(X \ni x, B)$, if $\widehat{\operatorname{vol}}(v) = \widehat{\operatorname{vol}}(X \ni x, B)$.

Proposition 2.7. *The following hold.*

1. *If $X \ni x$ and $X' \ni x'$ are analytically isomorphic klt germs, then $\widehat{\operatorname{vol}}(X \ni x) = \widehat{\operatorname{vol}}(X' \ni x')$ [HLQ23, Proposition 2.24].*
2. *Let $f : (Y \ni y, B_Y) \to (X \ni x, B)$ be a finite Galois morphism between klt pairs with $f^{-1}(x) = \{y\}$ and $K_Y + B_Y = f^*(K_X + B)$. Then $\widehat{\operatorname{vol}}(Y \ni y, B_Y) = \deg(f) \cdot \widehat{\operatorname{vol}}(X \ni x, B)$ [XZ21, Theorem 1.3].*
3. *Let $g : Y \to X$ be a projective birational morphism between klt varieties with $K_Y = g^* K_X$, and let $y \in \operatorname{Exc}(g)$ be a closed point with $x := g(y)$. Then $\widehat{\operatorname{vol}}(X \ni x) < \widehat{\operatorname{vol}}(Y \ni y)$ [LX19, Corollary 2.12].*

2.3. **Fano cones and stable degenerations.**

Definition 2.8. Let X be a normal affine variety with a faithful action of a torus $T = \operatorname{Spec} \mathbb{C}[M]$, and write $\mathbb{C}[X] = \bigoplus_{u \in M} \mathbb{C}[X]_u$ for the weight decomposition. The action is *good* if $\mathbb{C}[X]_0 = \mathbb{C}$ and the cone generated by the weights u with $\mathbb{C}[X]_u \neq 0$ is strictly convex. In this case X has a unique T-fixed point x, called the *vertex*. The *Reeb cone* is the set of $\xi \in \operatorname{Hom}(M, \mathbb{R})$ with

$\langle u, \xi\rangle > 0$ for any $u \neq 0$ with $\mathbb{C}[X]_u \neq 0$, and an element ξ of the Reeb cone is called a *Reeb vector*. The *Reeb valuation* $\mathrm{wt}_\xi \in \mathrm{Val}_{X,x}$ is defined by $\mathrm{wt}_\xi(g) := \min\{\langle u, \xi\rangle \mid g_u \neq 0\}$ for $g = \sum_u g_u$. If X is klt, then the pair (X, ξ) is called a *Fano cone*. A Reeb vector ξ is *quasi-regular* if $\mathbb{R}_{>0}\xi$ contains a non-zero element of $\mathrm{Hom}(M, \mathbb{Z})$. A Fano cone (X, ξ) is *K-semistable* if wt_ξ computes $\widehat{\mathrm{vol}}(X \ni x)$.

By [LX18, Theorems 1.1 and 1.3], K-semistability of (X, ξ) in the sense of Definition 2.8 is equivalent to K-semistability defined by test configurations, see [LWX21, Definitions 2.14, 2.16, 2.22, and 2.23].

Lemma 2.9 ([LX18, Lemma 2.18])**.** *Let (X, ξ) be a Fano cone with respect to a torus $T = \operatorname{Spec}\mathbb{C}[M]$, and assume that the canonical module of $\mathbb{C}[X]$, with the M-grading for which the weight of dg equals the weight of g for any homogeneous $g \in \mathbb{C}[X]$, is generated by a homogeneous element of weight $u \in M$. Then $a(\mathrm{wt}_\xi, X) = \langle u, \xi\rangle$.*

Definition 2.10. Let S be a normal projective surface such that $-K_S$ is an ample Cartier divisor. The *anti-canonical cone* over S is

$$C_S := \operatorname{Spec} \bigoplus_{m\geq 0} H^0(S, \mathcal{O}_S(-mK_S)),$$

and the *vertex* of C_S is the closed point $o \in C_S$ defined by the ideal $\bigoplus_{m>0} H^0(S, \mathcal{O}_S(-mK_S))$.

Definition 2.11. A *log Fano pair* is a projective klt pair (X, B) such that B is a $\mathbb{Q}$-divisor and $-(K_X + B)$ is ample. Let (X, B) be a log Fano pair, and let $L := -(K_X + B)$. For a prime divisor F over X, choose a projective birational morphism $h : X' \to X$ from a normal variety X' such that F is a prime divisor on X', and let

$$\beta_{(X,B)}(F) := a(F, X, B) - \frac{1}{\mathrm{vol}(L)} \int_0^{+\infty} \mathrm{vol}(h^*L - tF)dt.$$

Here $\mathrm{vol}(\cdot)$ denotes the volume of [ADL24, Definition 2.6], and $\beta_{(X,B)}(F)$ does not depend on h by [ADL24, Remark 2.7]. We say that (X, B) is *K-semistable* if $\beta_{(X,B)}(F) \geq 0$ for any prime divisor F over X.

By [ADL24, Theorem 2.9(1)], K-semistability of a log Fano pair in the sense of Definition 2.11 is equivalent to K-semistability defined by test configurations.

Theorem 2.12. *Let $X \ni x$ be a klt germ. Then the following hold.*

1. *There exists a minimizer v of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$. It is quasi-monomial and unique up to scaling.*
2. *The associated graded ring $\mathrm{gr}_v \mathcal{O}_{X,x} := \bigoplus_{\lambda\geq 0} \mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v)$ is a finitely generated $\mathbb{C}$-algebra. Let $M \subset \mathbb{R}$ be the subgroup generated by the degrees λ with $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$. Then $X_0 := \operatorname{Spec}\mathrm{gr}_v \mathcal{O}_{X,x}$ is a normal klt variety with a good action of the torus $\operatorname{Spec}\mathbb{C}[M]$, and the inclusion $M \subset \mathbb{R}$ defines a Reeb vector ξ of X_0.*
3. *Let $x_0 \in X_0$ be the vertex. Then (X_0, ξ) is a K-semistable Fano cone, $a(\mathrm{wt}_\xi, X_0) = a(v, X)$, $\mathrm{vol}(\mathrm{wt}_\xi) = \mathrm{vol}(v)$, and*

$$\widehat{\mathrm{vol}}(X_0 \ni x_0) = \widehat{\mathrm{vol}}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(X \ni x).$$

Theorem 2.12 is the stable degeneration theorem. In Theorem 2.12(1), the minimizer exists by [Blu18, Main Theorem], it is quasi-monomial by [Xu20, Theorem 1.2], see [JM12, Section 3.1] for quasi-monomial valuations, and it is unique up to scaling by [XZ21, Theorem 1.1]. Theorem 2.12(2) and (3) follow from [XZ25, Theorems 1.1 and 1.2] and [LX18, Lemma 2.58 and Theorem 1.3].

2.4. **LCIQ singularities.**

Definition 2.13. A finite surjective morphism $f : Y \to X$ between normal varieties is *quasi-étale* if f is étale in codimension one. Let $X \ni x$ be a klt germ. We say that $X \ni x$ is *LCIQ* if there exist a normal germ $Y \ni y$ which is a local complete intersection, and a finite group G acting on Y with y fixed and freely in codimension one, such that $X \ni x$ is analytically isomorphic to the germ of Y/G at the image of y. In particular, any local complete intersection klt germ, for example a smooth germ, is LCIQ. For a finite subgroup $G \subset \mathrm{GL}_n(\mathbb{C})$ acting on $\mathbb{A}^n$ freely in codimension one, the germ of $\mathbb{A}^n/G$ at the image of the origin, which is klt by [KM98, Proposition 5.20(4)] since $\mathbb{A}^n \to \mathbb{A}^n/G$ is étale in codimension one, is LCIQ. For example, the cyclic quotient singularity of type $\frac{1}{2}(1,1,1)$, where μ_2 acts on $\mathbb{A}^3$ by -1 freely outside the origin, is LCIQ.

Thus the non-LCIQ hypothesis of [Liu25, Question 5.4] excludes the quotients of $\mathbb{A}^n$ in Definition 2.13. Definition 2.13 describes exactly the finite quasi-étale quotients of local complete intersection singularities in [Liu25, Question 5.4], since for a finite group G acting faithfully on a normal variety Y, the quotient morphism $Y \to Y/G$ is étale exactly over the images of the points of Y with trivial stabilizer, hence quasi-étale if and only if G acts freely in codimension one, and for $y \in Y$ the germ of Y/G at the image of y is analytically isomorphic to the germ of Y/G_y at the image of y, where the stabilizer G_y of y fixes y, see [Gro71, Exposé V, Proposition 2.2 and Corollaries 2.3, 2.4].

2.5. **Crepant divisors and Kollár components.**

Definition 2.14. Let $X \ni x$ be a klt germ. A *crepant divisor* over $X \ni x$ is a prime divisor E over X with $\mathrm{center}_X E = \{x\}$ and $a(E, X) = 1$. We denote by $\mathrm{crex}(X \ni x)$ the number of crepant divisors over $X \ni x$, which is finite by [KM98, Proposition 2.36(2)] (applied to $(U, 0)$ for a klt open neighborhood U of x, for which $\mathrm{totaldiscrep}(U) > -1$, and $a(E, X) = 1$ means that the discrepancy of E is $0 < 1 + \mathrm{totaldiscrep}(U)$). A cDV threefold germ $X \ni x$ [KM98, Definition 5.32] is *of type* cA_k if a general hyperplane section of X through x has a Du Val singularity of type A_k at x, where a smooth germ is of type cA_0.

Theorem 2.15 ([Liu25, Theorem 2.19 and Lemma 2.20])**.** *Let $X \ni x$ be a Gorenstein canonical threefold germ.*

(1) $\mathrm{crex}(X \ni x) = 0$ *if and only if $X \ni x$ is a cDV hypersurface germ.*

(2) Assume that $X \ni x$ is not a hypersurface germ. Then $\mathrm{crex}(X \ni x) > 0$*, the blow-up $Y \to X$ of the maximal ideal of x is a crepant projective birational morphism, and Y is normal and Gorenstein canonical. Moreover, the embedding dimension of $X \ni x$ is equal to the multiplicity of $X \ni x$ plus one, and a general hyperplane section of X through x is a Gorenstein elliptic surface singularity.*

Theorem 2.15(1) and the statements of (2) other than the last one are [Liu25, Theorem 2.19], which relies on [KM98, Theorems 5.34 and 5.35], see also [Rei80, Corollary 2.10 and Theorem 2.11]. The statement on the general hyperplane section is [Liu25, Lemma 2.20], which relies on [Rei80, Corollary 2.10] and [KM98, Theorems 5.34 and 5.35]. Here a Gorenstein elliptic surface singularity is an elliptic surface singularity in the sense of [KM98, Definition 4.48], see also [Liu25, Section 2.4], and [KM98, Theorem 4.57] gives the multiplicity and the embedding dimension of such singularities.

Lemma 2.16. *Let Y be a normal threefold with canonical singularities, and let $\Gamma \subset \mathrm{Sing}\, Y$ be an irreducible curve. Then there exists a prime divisor G over Y with* $\mathrm{center}_Y G = \Gamma$ *and* $a(G, Y) = 1$.

Proof. Since Y is canonical, $a(F, Y) \geq 1$ for any exceptional prime divisor F over Y, and only finitely many exceptional prime divisors $F_1, \ldots, F_k$ over Y satisfy $a(F_i, Y) = 1$ [KM98, Proposition 2.36(2)] (applied to the klt pair $(Y, 0)$, for which $\mathrm{totaldiscrep}(Y) = 0$ since Y

is canonical, and $a(F_i, Y) = 1$ means that the discrepancy of F_i is $0 < 1$). Suppose that $\operatorname{center}_Y F_i \neq \Gamma$ for any i. Since $\dim \operatorname{center}_Y F_i \leq 1$, the set $\operatorname{center}_Y F_i \cap \Gamma$ is finite for any i, so $U := Y \setminus \bigcup_{i=1}^{k} \operatorname{center}_Y F_i$ is an open subset with $U \cap \Gamma \neq \emptyset$, and $a(F, U) > 1$ for any exceptional prime divisor F over U, since an exceptional prime divisor over U with log discrepancy one would be one of the F_i. Hence U is terminal, and U is smooth in codimension two [KM98, Corollary 5.18], which contradicts $U \cap \Gamma \subset \operatorname{Sing} U$. □

The following bounds on local volumes of threefold germs are known.

Theorem 2.17. *Let $X \ni x$ be a klt threefold germ.*

1. *$\widehat{\operatorname{vol}}(X \ni x) \leq 27$, and equality holds if and only if $X \ni x$ is smooth. If $X \ni x$ is singular, then $\widehat{\operatorname{vol}}(X \ni x) \leq 16$, and equality holds if and only if $X \ni x$ is an ordinary double point* [LX19, Theorems 1.3 and 1.6].
2. *If $X \ni x$ is Gorenstein canonical and is not a hypersurface germ, then $\widehat{\operatorname{vol}}(X \ni x) \leq 9$, and equality holds if and only if $X \ni x$ is a cyclic quotient singularity of type $\frac{1}{3}(1,1,1)$* [Liu25, Theorem 1.1].
3. *$\widehat{\operatorname{vol}}(X \ni x) \geq 9$ if and only if $X \ni x$ is a hypersurface germ of type cA_k for some $k \leq 2$, or a cyclic quotient singularity of type $\frac{1}{2}(1,1,1)$, $\frac{1}{3}(1,1,0)$, $\frac{1}{3}(1,1,1)$, or $\frac{1}{3}(1,1,2)$* [Liu25, Theorem 1.2].
4. *If $X \ni x$ is a singular hypersurface germ with $\widehat{\operatorname{vol}}(X \ni x) > 8$, then $X \ni x$ is of type cA_1 or cA_2, and in particular $\operatorname{crex}(X \ni x) = 0$. Conversely, $\widehat{\operatorname{vol}}(X \ni x) \geq 9$ for any germ $X \ni x$ of type cA_1 or cA_2* [Liu25, Proposition 4.2].

Definition 2.18. Let $X \ni x$ be a klt germ. A *plt blow-up* of $X \ni x$ is a projective birational morphism $h : X' \to X$ from a normal variety X' which is an isomorphism over $X \setminus \{x\}$, such that $h^{-1}(x)$ is a prime divisor P, $-P$ is $\mathbb{Q}$-Cartier and h-ample, and (X', P) is plt [KM98, Definition 2.34]. We call P a *Kollár component* over $X \ni x$. In this case P is normal [KM98, Proposition 5.51], and we write $(K_{X'} + P)|_P = K_P + \operatorname{Diff}_P(0)$, where $\operatorname{Diff}_P(0)$ is the different [Kol13, Definition 4.2], which is effective by [Kol13, Proposition 4.5(1)].

Lemma 2.19. *The following hold.*

1. *Let $h : X' \to X$ be a plt blow-up of a klt threefold germ $X \ni x$ with Kollár component P. Then $\widehat{\operatorname{vol}}(\operatorname{ord}_P) = a(P, X) \cdot ((K_P + \operatorname{Diff}_P(0))^2)$. In particular, if $K_{X'} = h^* K_X$, then $\widehat{\operatorname{vol}}(\operatorname{ord}_P) = ((K_P + \operatorname{Diff}_P(0))^2)$* [LX20, Lemma 2.11].
2. *Let $X \ni x$ be a Gorenstein canonical threefold germ, and let $f : Y \to X$ be a projective birational morphism from a normal variety Y with $K_Y = f^* K_X$ which is an isomorphism over $X \setminus \{x\}$, such that $E := f^{-1}(x)_{\mathrm{red}}$ is a Cartier divisor and $-E$ is f-ample. Then E is Gorenstein, and $\widehat{\operatorname{vol}}(X \ni x) \leq \widehat{\operatorname{vol}}(\operatorname{ord}_{E'}) \leq (E^3) = (K_E^2)$ for any irreducible component E' of E, where K_E is a canonical divisor of E. If E is irreducible, then $\widehat{\operatorname{vol}}(\operatorname{ord}_E) = (K_E^2)$.*

Proof. Part (1) is [LX20, Lemma 2.11]. We prove (2). Since X is Gorenstein, K_X is Cartier, so possibly shrinking X around x, we may assume that $K_X \sim 0$. Then $K_Y = f^* K_X \sim 0$. By [KM98, Lemma 2.30], $a(F, Y) = a(F, X)$ for any prime divisor F over Y, so Y is canonical. Thus Y has rational singularities, and hence Y is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. Since K_Y is Cartier, Y is Gorenstein. Since $E \geq 0$ is a Cartier divisor on Y, E is Gorenstein and $\omega_E \cong \omega_Y(E)|_E$ [Kol13, (4.1.3)]. As $K_Y \sim 0$, we have $K_E \sim E|_E$, and so

$$(K_E^2) = ((E|_E)^2) = (E^3).$$

Let E' be an irreducible component of E, and let $m \geq 1$ be an integer. Since f is an isomorphism over $X \setminus \{x\}$ and $f_* \mathcal{O}_Y = \mathcal{O}_X$, the ideal $f_* \mathcal{O}_Y(-mE)$ consists of the functions g with $\operatorname{ord}_{E_i}(g) \geq m$ for any irreducible component E_i of E. In particular,

$$f_* \mathcal{O}_Y(-mE)_x \subset \mathfrak{a}_m(\operatorname{ord}_{E'}).$$

Since $-E$ is an f-ample Cartier divisor and $-E|_E$ is ample, by Serre vanishing [Har77, Theorem III.8.8(c)], applied to f and to $E \to \operatorname{Spec}\mathbb{C}$, there exists an integer $k_0 \geq 0$ such that $R^1 f_*\mathcal{O}_Y(-(k+1)E) = 0$ and $H^i(E, \mathcal{O}_E(-kE)) = 0$ for any $i > 0$ and any $k \geq k_0$. The exact sequence

$$0 \to \mathcal{O}_Y(-(k+1)E) \to \mathcal{O}_Y(-kE) \to \mathcal{O}_E(-kE) \to 0$$

implies that

$$\dim_{\mathbb{C}} f_*\mathcal{O}_Y(-kE)/f_*\mathcal{O}_Y(-(k+1)E) \leq h^0(E, \mathcal{O}_E(-kE)) < +\infty$$

for any $k \geq 0$, with equality for $k \geq k_0$, and $h^0(E, \mathcal{O}_E(-kE)) = \chi(E, \mathcal{O}_E(-kE))$ for $k \geq k_0$. Since $E = f^{-1}(x)_{\mathrm{red}}$ is a reduced Cartier divisor on Y contained in a fiber of the projective morphism f, E is a projective scheme of pure dimension 2. Thus by asymptotic Riemann–Roch [Laz04, Theorem 1.1.24 and Remark 1.1.26],

$$\chi(E, \mathcal{O}_E(-kE)) = \frac{(E^3)}{2}k^2 + O(k).$$

Summing over $0 \leq k \leq m-1$, we obtain

$$\dim_{\mathbb{C}} \mathcal{O}_{X,x}/f_*\mathcal{O}_Y(-mE)_x = \frac{(E^3)}{6}m^3 + O(m^2).$$

Therefore

$$\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{a}_m(\mathrm{ord}_{E'}) \leq \frac{(E^3)}{6}m^3 + O(m^2),$$

and so $\mathrm{vol}(\mathrm{ord}_{E'}) \leq (E^3)$. Since $K_Y = f^*K_X$, we have $a(E', X) = 1$. Thus

$$\widehat{\mathrm{vol}}(X \ni x) \leq \widehat{\mathrm{vol}}(\mathrm{ord}_{E'}) = \mathrm{vol}(\mathrm{ord}_{E'}) \leq (E^3) = (K_E^2).$$

Assume that E is irreducible. Let $g \in \mathfrak{a}_m(\mathrm{ord}_E)$. Since E is the unique prime divisor on Y whose image is x, the divisor of f^*g near E is at least mE, and since Y is normal, f^*g is a section of $\mathcal{O}_Y(-mE)$ near E. Thus $f_*\mathcal{O}_Y(-mE)_x = \mathfrak{a}_m(\mathrm{ord}_E)$, and so $\mathrm{vol}(\mathrm{ord}_E) = (E^3)$ and $\widehat{\mathrm{vol}}(\mathrm{ord}_E) = (K_E^2)$ (cf. [Liu25, Inequality (4.2)]). □

Proposition 2.20 ([Liu25, Proposition 4.1])**.** *Let $X \ni x$ be a klt hypersurface germ of dimension $n \geq 2$ with $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[z_1, \ldots, z_{n+1}]]/(F)$. For $w = (w_1, \ldots, w_{n+1}) \in \mathbb{R}^{n+1}_{>0}$, let v_w be the monomial valuation on $\mathbb{C}[[z_1, \ldots, z_{n+1}]]$ with weights w. Then*

$$\widehat{\mathrm{vol}}(X \ni x) \leq \Big(\sum_{i=1}^{n+1} w_i - v_w(F)\Big)^n \cdot \frac{v_w(F)}{\prod_{i=1}^{n+1} w_i}.$$

In Proposition 2.20, F is any formal power series in the completed local ring, and F need not be a polynomial.

3. Local volumes at least nine

In this section, we prove Theorem 1.3. Theorem 1.3 is used once more in Section 9, in the proof of Proposition 9.22.

3.1. **Normal forms.** In this subsection, we recall the local volumes of the threefold germs of type A, D, and E, and we prove a normal form for germs of type cA_1 and cA_2. We use the names A_k, A_∞, D_k, D_∞, E_6, E_7, E_8, and transversal A_k ($k \geq 1$) for threefold hypersurface germs as in [Liu25, Definitions 2.12–2.15], written in the coordinates u, v, z, t. For example, the completed local ring of a transversal A_k germ is isomorphic to $\mathbb{C}[[u, v, z, t]]/(uv + z^{k+1})$.

Proposition 3.1 ([Liu25, Examples 2.7 and 5.2])**.** *Let $X \ni x$ be a klt threefold germ.*

(1) If $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u, v, z, t]]/(uv + z^2 + t^{k+1})$ for an integer $k \geq 1$, then $\widehat{\mathrm{vol}}(X \ni x)$ is 16 for $k = 1$, $\frac{125}{9}$ for $k = 2$, and $\frac{27}{2}$ for $k \geq 3$.

(2) If $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u, v, z, t]]/(uv + z^2)$, then $\widehat{\mathrm{vol}}(X \ni x) = \frac{27}{2}$.

(3) *If* $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+z^3+t^k)$ *for an integer* $3 \le k \le 6$, *then* $\widehat{\mathrm{vol}}(X \ni x) = \frac{4(3+k)^3}{9k^2}$, *that is,* $\frac{32}{3}$, $\frac{343}{36}$, $\frac{2048}{225}$, *and* 9 *for* $k = 3,4,5,6$, *respectively.*

(4) *If* $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+h)$, *where* $h = z^2t + t^{k-1}$ *for an integer* $k \ge 5$ *or* $h = z^2t$, *then* $\widehat{\mathrm{vol}}(X \ni x) = 6\sqrt{3}$.

(5) *If* $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+z^3+zt^3)$, *then* $\widehat{\mathrm{vol}}(X \ni x) = \frac{250}{27}$.

Moreover, if $X \ni x$ *is analytically isomorphic to* $\mathbb{A}^3/G$ *at the origin for a finite subgroup* $G \subset \mathrm{GL}_3(\mathbb{C})$ *acting freely in codimension one, then* $\widehat{\mathrm{vol}}(X \ni x) = \frac{27}{|G|}$.

The germs in (1), (2), and (4) are the A_k, transversal A_1, D_k, and D_∞ germs, the first three germs in (3) are the D_4, E_6, and E_8 germs, and the germ in (5) is the E_7 germ. The values in (1) are due to [Li18, Section 5], [LL19, Example 4.7], and [LX20, Example 7.1(2)], the first three values in (3) to [CS19, Theorem 8.1(I)] and [LST25, Example 2.12], the last value in (3) to [LST25, Theorem 1.4] for the K-semistability and to [LX18, Theorem 1.3] for the value, the values in (4) to [LX18, Appendix, Example 4.1], and the value in (5) to [Li18, Appendix I in arXiv:1511.08164] and [LX20, Example 7.1(3)]. The last value in (3) is also given by Proposition 3.2(2) below, since the series z^3+t^6 belongs to the last series there. The value for quotient germs is [LX20, Example 7.1(1)]. It also follows from Proposition 2.7(2) applied to the quotient morphism $\mathbb{A}^3 \to \mathbb{A}^3/G$.

The local volume of the germs in the last series of Proposition 3.2(2) below, which contains infinitely many analytic types (for instance the germs X_k, $k \ge 6$, of the introduction, which have $\dim T^1 = 2(k-1)$), is determined in Proposition 3.2 itself rather than by Proposition 3.1.

Proposition 3.2. *Let* $X \ni x$ *be a klt threefold germ of type* cA_1 *or* cA_2, *and write* $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+g(z,t))$, *where* $d := \operatorname{ord} g$ *is* 2 *or* 3, *respectively.*

(1) *If* $d = 2$, *then* $\widehat{\mathcal{O}}_{X,x}$ *is isomorphic to* $\mathbb{C}[[u,v,z,t]]/(uv+z^2+t^{k+1})$ *for an integer* $k \ge 1$, *or to* $\mathbb{C}[[u,v,z,t]]/(uv+z^2)$.

(2) *If* $d = 3$, *then* $\widehat{\mathcal{O}}_{X,x}$ *is isomorphic to* $\mathbb{C}[[u,v,z,t]]/(uv+h)$ *for at least one of the following series* h: z^2t+t^{k-1} *for an integer* $k \ge 4$, z^2t, z^3+t^4, z^3+zt^3, z^3+t^5, *and* $z^3+p(t)z+s(t)$ *with* $p \in t^4\mathbb{C}[[t]]$ *and* $s \in t^6\mathbb{C}[[t]]$. *In the last case* $\widehat{\mathrm{vol}}(X \ni x) = 9$.

In particular, if $X \ni x$ *is of type* cA_2 *and* $\widehat{\mathrm{vol}}(X \ni x) > 9$, *then* $\widehat{\mathcal{O}}_{X,x}$ *is isomorphic to* $\mathbb{C}[[u,v,z,t]]/(uv+h)$ *for one of the first five series* h *in (2).*

Proof. The series g exists by the splitting lemma [GLS07, Theorem I.2.47] (the coordinate change constructed for [GLS07, Theorem I.2.47] is formal, so the splitting lemma holds verbatim for formal power series), and $\operatorname{ord} g$ is 2 or 3 because a general hyperplane section of X through x has a Du Val singularity of type A_1 or A_2 at x, see also [Liu25, Section 4.1].

Let $e \in \mathbb{C}[[z,t]]$ be a unit and φ an automorphism of $\mathbb{C}[[z,t]]$. Then φ extends to an automorphism of $\mathbb{C}[[u,v,z,t]]$ fixing u and v, and

$$uv + e \cdot g = e \cdot \big((e^{-1}u) \cdot v + g\big),$$

where $e^{-1}u, v, z, t$ are again coordinates of $\mathbb{C}[[u,v,z,t]]$. Thus we may replace g by $e \cdot \varphi(g)$ for any such e and φ, and we keep the notation g after each replacement. For an integer $m \ge 1$, any unit of $\mathbb{C}[[t]]$ has an m-th root in $\mathbb{C}[[t]]$, and we use such roots without further comment.

We also use the following normal form for semiquasihomogeneous series [AGV85, Section 12.6]. Let w_z, w_t be positive integer weights of z, t, and let $h_0 \in \mathbb{C}[z,t]$ be a quasihomogeneous polynomial of weight D such that $\mathbb{C}[[z,t]]/(\partial h_0/\partial z, \partial h_0/\partial t)$ is spanned by monomials of weight less than D. Then for any $f \in \mathbb{C}[[z,t]]$ whose monomials all have weight greater than D, there exists an automorphism φ of $\mathbb{C}[[z,t]]$ such that $\varphi(h_0+f) = h_0$. A direct calculation shows that the hypothesis holds for the four pairs

$$(z^2t+t^3;1,1), \quad (z^3+t^4;4,3), \quad (z^3+zt^3;3,2), \quad (z^3+t^5;5,3) \tag{3.1}$$

of a polynomial h_0 and weights w_z, w_t: the four quotients are spanned by $1, z, t, t^2$, by $z^i t^j$ with $i \le 1$ and $j \le 2$, by $1, z, zt, t, t^2, t^3, t^4$, and by $z^i t^j$ with $i \le 1$ and $j \le 3$, respectively. The plane curve germs defined by the polynomials in (3.1) are the simple singularities of types D_4, E_6, E_7, and E_8 in the notation of [Arn72].

(1) Suppose that $d = 2$. By the splitting lemma [GLS07, Theorem I.2.47], we may assume that $g = z^2 + c(t)$ for some $c \in t^2\mathbb{C}[[t]]$. If $c = 0$, then $g = z^2$. Otherwise, $c = t^m\lambda(t)$ for an integer $m \ge 2$ and a unit $\lambda \in \mathbb{C}[[t]]$. Taking $t\lambda(t)^{1/m}$ as a new coordinate t, we may assume that $c = t^m$. Thus (1) holds with $k = m - 1 \ge 1$.

(2) Suppose that $d = 3$, and let $g_3 \ne 0$ be the homogeneous part of g of degree 3. Then g_3 is a product of three linear forms. We distinguish three cases.

Case 1: the three linear factors of g_3 are pairwise non-proportional. Since $\mathrm{PGL}_2(\mathbb{C})$ acts transitively on triples of distinct points of $\mathbb{P}^1$, after a linear change of coordinates and multiplication by a non-zero constant, we may assume that

$$g_3 = z^2t + t^3 = t(z + \sqrt{-1}t)(z - \sqrt{-1}t).$$

Then $g - g_3 \in (z, t)^4$, which consists of series whose monomials have weight greater than 3 for the weights $(1, 1)$. By (3.1), we may assume that $g = z^2t + t^3$, which is the first series in (2) with $k = 4$.

Case 2: g_3 has exactly two non-proportional linear factors. After a linear change of coordinates and multiplication by a non-zero constant, we may assume that $g_3 = zt^2$. Let

$$G(s, t) := t^{-3}g(ts, t) \in \mathbb{C}[[s, t]].$$

Since $\operatorname{ord} g = 3$ and $g_3(ts, t) = st^3$, we have $G(s, 0) = s$. By the formal implicit function theorem, there exists $\sigma \in t\mathbb{C}[[t]]$ such that $G(\sigma(t), t) = 0$. Let $\beta(t) := t\sigma(t) \in t^2\mathbb{C}[[t]]$. Then $g(\beta(t), t) = t^3G(\sigma(t), t) = 0$, hence $g = (z - \beta(t)) \cdot g'$ for some $g' \in \mathbb{C}[[z, t]]$. Since $\beta \in t^2\mathbb{C}[[t]]$, the lowest degree homogeneous part of $z - \beta(t)$ is z, so the lowest degree homogeneous part of g' is t^2. We take $t' := z - \beta(t)$ and $z' := t$ as new coordinates. Then $g = t' \cdot g'$, where the lowest degree homogeneous part of g' is z'^2. By the Weierstrass preparation theorem [GLS07, Theorem I.1.6] applied to g' with respect to z', we may assume that

$$g = t' \cdot (z'^2 + c_1(t')z' + c_0(t')),$$

where $c_1, c_0 \in \mathbb{C}[[t']]$. Since g' and $z'^2 + c_1(t')z' + c_0(t')$ differ by a unit factor and the lowest degree homogeneous part of g' is z'^2, the homogeneous part of $z'^2 + c_1(t')z' + c_0(t')$ of degree 2 is z'^2, so we have $\operatorname{ord} c_1 \ge 2$ and $\operatorname{ord} c_0 \ge 3$. Replacing z' with $z' - \frac{1}{2}c_1(t')$, we may assume that $g = t' \cdot (z'^2 + c(t'))$ for some $c \in t'^3\mathbb{C}[[t']]$. If $c = 0$, then $g = t'z'^2$. Otherwise, $c = t'^m\lambda(t')$ for an integer $m \ge 3$ and a unit $\lambda \in \mathbb{C}[[t']]$. Taking $t'\lambda(t')^{\frac{1}{m+1}}$ and $z'\lambda(t')^{-\frac{1}{2(m+1)}}$ as new coordinates t' and z', we may assume that $g = t'z'^2 + t'^{m+1}$. Renaming z', t' as z, t, we obtain the series z^2t if $c = 0$, and the first series in (2) with $k = m + 2 \ge 5$ otherwise.

Case 3: the three linear factors of g_3 are proportional. After a linear change of coordinates and multiplication by a non-zero constant, we may assume that $g_3 = z^3$. By the Weierstrass preparation theorem [GLS07, Theorem I.1.6] with respect to z, we may assume that

$$g = z^3 + c_2(t)z^2 + c_1(t)z + c_0(t),$$

where $c_2, c_1, c_0 \in \mathbb{C}[[t]]$. Since the new series g is monic in z and differs from the previous one by a unit factor, the homogeneous part of g of degree 3 is still z^3, so $\operatorname{ord} c_2 \ge 2$, $\operatorname{ord} c_1 \ge 3$, and $\operatorname{ord} c_0 \ge 4$. Replacing z with $z - \frac{1}{3}c_2(t)$, we may assume that $g = z^3 + p(t)z + s(t)$, where

$$p = c_1 - \tfrac{1}{3}c_2^2 \quad \text{and} \quad s = c_0 - \tfrac{1}{3}c_1c_2 + \tfrac{2}{27}c_2^3.$$

Thus $\operatorname{ord} p \ge 3$ and $\operatorname{ord} s \ge 4$. We consider four subcases, which exhaust all possible values of $\operatorname{ord} p \in \{3, 4, \dots\} \cup \{+\infty\}$ and $\operatorname{ord} s \in \{4, 5, \dots\} \cup \{+\infty\}$.

- Suppose that $\operatorname{ord} s = 4$. Replacing t with a non-zero multiple of t, we may assume that the coefficient of t^4 in s is 1. For the weights $(4, 3)$, we have $g = z^3 + t^4 + f$, where the

monomials of $p(t)z$ have weight at least 13 and the monomials of $f - p(t)z$ have weight at least 15. By (3.1), we may assume that $g = z^3 + t^4$.

- Suppose that $\operatorname{ord} s \geq 5$ and $\operatorname{ord} p = 3$. Replacing t with a non-zero multiple of t, we may assume that the coefficient of zt^3 in g is 1. For the weights $(3, 2)$, we have $g = z^3 + zt^3 + f$, where the monomials of f have weight at least 10. By (3.1), we may assume that $g = z^3 + zt^3$.
- Suppose that $\operatorname{ord} p \geq 4$ and $\operatorname{ord} s = 5$. Replacing t with a non-zero multiple of t, we may assume that the coefficient of t^5 in s is 1. For the weights $(5, 3)$, we have $g = z^3 + t^5 + f$, where the monomials of $p(t)z$ have weight at least 17 and the monomials of $f - p(t)z$ have weight at least 18. By (3.1), we may assume that $g = z^3 + t^5$.
- Suppose that $\operatorname{ord} p \geq 4$ and $\operatorname{ord} s \geq 6$. Then g is the last series in (2).

It remains to compute $\widehat{\operatorname{vol}}(X \ni x)$ in the last case, that is, when $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u, v, z, t]]/(F)$ with $F := uv + z^3 + p(t)z + s(t)$, $p \in t^4\mathbb{C}[[t]]$, and $s \in t^6\mathbb{C}[[t]]$. Let $w := (3, 3, 2, 1)$ be the weights of (u, v, z, t). The monomials uv and z^3 have weight 6, and any monomial of $p(t)z + s(t)$ has weight at least 6, so $v_w(F) = 6$. By Proposition 2.20,

$$\widehat{\operatorname{vol}}(X \ni x) \leq (3 + 3 + 2 + 1 - 6)^3 \cdot \frac{6}{3 \cdot 3 \cdot 2 \cdot 1} = 9.$$

Since $X \ni x$ is of type cA_2, Theorem 2.17(4) implies that $\widehat{\operatorname{vol}}(X \ni x) \geq 9$. Thus $\widehat{\operatorname{vol}}(X \ni x) = 9$, and (2) holds.

Finally, if $X \ni x$ is of type cA_2 and $\widehat{\operatorname{vol}}(X \ni x) > 9$, then $\widehat{\mathcal{O}}_{X,x}$ is not isomorphic to $\mathbb{C}[[u, v, z, t]]/(uv + h)$ for h in the last series of (2), so h is one of the first five series by (2). □

3.2. **Proof of Theorem 1.3.**

Proof of Theorem 1.3. Let

$$\Lambda := \left\{9, \frac{2048}{225}, \frac{250}{27}, \frac{343}{36}, 6\sqrt{3}, \frac{32}{3}, \frac{27}{2}, \frac{125}{9}, 16, 27\right\}.$$

We first show that $\widehat{\operatorname{Vol}}_3 \cap [9, 27] \subset \Lambda$. Let $X \ni x$ be a klt threefold germ such that $\widehat{\operatorname{vol}}(X \ni x) \geq 9$. By Theorem 2.17(3), $X \ni x$ is a hypersurface germ of type cA_k for some $k \leq 2$, or a cyclic quotient singularity of type $\frac{1}{2}(1, 1, 1)$, $\frac{1}{3}(1, 1, 0)$, $\frac{1}{3}(1, 1, 1)$, or $\frac{1}{3}(1, 1, 2)$.

If $X \ni x$ is of type cA_0, then $X \ni x$ is smooth, and $\widehat{\operatorname{vol}}(X \ni x) = 27$ by Theorem 2.17(1).

Suppose that $X \ni x$ is a cyclic quotient singularity of one of the four types, and let $G \subset \mathrm{GL}_3(\mathbb{C})$ be the corresponding cyclic group, so that $|G| = 2$ for $\frac{1}{2}(1, 1, 1)$ and $|G| = 3$ for the other three types. For $\frac{1}{3}(1, 1, 0)$, the non-trivial elements of G fix exactly the line $\{x_1 = x_2 = 0\} \subset \mathbb{A}^3$, which has codimension two. For the other three types, the non-trivial elements of G fix only the origin. Thus G acts freely in codimension one, and Proposition 3.1 implies that $\widehat{\operatorname{vol}}(X \ni x) = \frac{27}{|G|}$, which is $\frac{27}{2}$ or 9.

Suppose that $X \ni x$ is of type cA_1. By Proposition 3.2(1) and Proposition 3.1(1)(2), $\widehat{\operatorname{vol}}(X \ni x) \in \{16, \frac{125}{9}, \frac{27}{2}\}$.

Suppose that $X \ni x$ is of type cA_2. By Proposition 3.2(2), either $\widehat{\operatorname{vol}}(X \ni x) = 9$, or $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u, v, z, t]]/(uv + h)$ for one of the first five series h in Proposition 3.2(2). We consider the five series one by one.

- The series $z^2t + t^3$ (the first series in Proposition 3.2(2) with $k = 4$) and the series $z^3 + t^3$ (Proposition 3.1(3) with $k = 3$) are both products of three distinct linear forms, so they differ by a linear change of coordinates and a non-zero constant factor, which we absorb into u. Thus $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u, v, z, t]]/(uv + z^3 + t^3)$, and $\widehat{\operatorname{vol}}(X \ni x) = \frac{32}{3}$ by Proposition 3.1(3).
- If $h = z^2t + t^{k-1}$ for an integer $k \geq 5$, or $h = z^2t$, then $\widehat{\operatorname{vol}}(X \ni x) = 6\sqrt{3}$ by Proposition 3.1(4).

- If h is z^3+t^4, z^3+zt^3, or z^3+t^5, then $\widehat{\mathrm{vol}}(X \ni x)$ is $\frac{343}{36}$, $\frac{250}{27}$, or $\frac{2048}{225}$, respectively, by Proposition 3.1(3)(5).

Thus $\widehat{\mathrm{vol}}(X \ni x) \in \Lambda$ in all cases, and $\widehat{\mathrm{Vol}}_3 \cap [9,27] \subset \Lambda$.

We now show that $\Lambda \subset \widehat{\mathrm{Vol}}_3 \cap [9,27]$. Since $\Lambda \subset [9,27]$, it suffices to show that $\Lambda \subset \widehat{\mathrm{Vol}}_3$. The smooth germ $\mathbb{A}^3 \ni 0$ has local volume 27 by Theorem 2.17(1). For the other nine values, consider the germs at the origin of the hypersurfaces in $\mathbb{A}^4$ defined by

$$uv+z^2+t^2,\ uv+z^2+t^3,\ uv+z^2,\ uv+z^3+t^3,\ uv+z^2t,$$
$$uv+z^3+t^4,\ uv+z^3+zt^3,\ uv+z^3+t^5,\ uv+z^3+t^6.$$

Each of the nine hypersurfaces is singular at most along the t-axis, which has codimension two, so it is regular in codimension one, and hence it is normal [Har77, Proposition II.8.23(b)]. A general hyperplane section of each of them through the origin has a Du Val singularity of type A_1 or A_2 at the origin, by a direct check on the section $t = au+bv+cz$, see also [Liu25, the second paragraph after Definition 2.14], so each of them is klt near the origin by inversion of adjunction [KM98, Theorem 5.50]. By Proposition 3.1(1)–(5), the local volumes of the nine germs are 16, $\frac{125}{9}$, $\frac{27}{2}$, $\frac{32}{3}$, $6\sqrt{3}$, $\frac{343}{36}$, $\frac{250}{27}$, $\frac{2048}{225}$, and 9, respectively. Thus $\Lambda \subset \widehat{\mathrm{Vol}}_3 \cap [9,27]$, and Theorem 1.3 follows. □

4. The anti-canonical cone over the first Hirzebruch surface

In this section, we prove that the vertex o of $C_{\mathbb{F}_1}$ is a klt germ of local volume q which has no non-trivial quasi-étale cover and is not LCIQ (Construction-Proposition 4.2), and that $C_{\mathbb{F}_1}$ has no non-trivial formal deformation over a reduced base (Proposition 4.4). The first result shows that the upper bound in Theorem 1.4 is attained, and the second is used in the classification of the equality case.

4.1. **The local volume and quasi-étale covers.** In this subsection, we describe $C_{\mathbb{F}_1}$ as an affine toric variety, compute the local volume of $C_{\mathbb{F}_1} \ni o$, and show that $C_{\mathbb{F}_1} \ni o$ is not LCIQ.

A germ which is not a local complete intersection may still be LCIQ, as the following example shows.

Example 4.1. The germ $C_{\mathbb{P}^1\times\mathbb{P}^1} \ni o$, which is klt by [Kol13, Lemma 3.1(3)] since $\mathbb{P}^1\times\mathbb{P}^1$ is smooth and $-K_{\mathbb{P}^1\times\mathbb{P}^1}$ is ample, is not a local complete intersection (see Section 1), but it is LCIQ. Indeed, $-K_{\mathbb{P}^1\times\mathbb{P}^1} = \mathcal{O}(2,2)$, so the coordinate ring of $C_{\mathbb{P}^1\times\mathbb{P}^1}$ is the even-degree part of the section ring of $\mathcal{O}(1,1)$. Thus $C_{\mathbb{P}^1\times\mathbb{P}^1}$ is the quotient of the quadric cone $\{xy=zw\} \subset \mathbb{C}^4$ by μ_2 acting by -1 (the image of $\mathbb{P}^1\times\mathbb{P}^1$ under the embedding by $\mathcal{O}(1,1)$ is the quadric surface $\{xy=zw\} \subset \mathbb{P}^3$, which is projectively normal [Har77, Exercise II.8.4(b)(c)], so the section ring of $\mathcal{O}(1,1)$ is $\mathbb{C}[x,y,z,w]/(xy-zw)$), the quotient map is étale outside the origin, and $\{xy=zw\}$ is a normal hypersurface.

Therefore, to show that $C_{\mathbb{F}_1} \ni o$ is not LCIQ, we exclude all quasi-étale covers of $C_{\mathbb{F}_1} \ni o$, using that the link of o is simply connected and the Zariski–Nagata purity theorem.

Construction-Proposition 4.2. Let $\sigma_{\mathbb{F}_1} \subset \mathbb{R}^3$ be the cone spanned by $(1,0,1)$, $(1,1,1)$, $(0,1,1)$, and $(-1,-1,1)$, and let $\xi_{\mathbb{F}_1} := \left(\frac{4-\sqrt{13}}{3}, \frac{4-\sqrt{13}}{3}, 1\right)$. Then $C_{\mathbb{F}_1}$ is isomorphic to the affine toric variety of $\sigma_{\mathbb{F}_1}$ with respect to the lattice $\mathbb{Z}^3$, under an isomorphism which sends o to the torus-fixed point, and $\xi_{\mathbb{F}_1}$ is a Reeb vector of the torus action. The germ $C_{\mathbb{F}_1} \ni o$ is a normal isolated Gorenstein canonical threefold singularity with

$$\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q = \frac{46+13\sqrt{13}}{12},$$

and the Reeb valuation $\mathrm{wt}_{\xi_{\mathbb{F}_1}}$ computes $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o)$. Moreover, any finite surjective morphism $f : \widetilde{W} \to W$ from a connected normal complex analytic germ $\widetilde{W}$ onto the complex analytic germ

W of $C_{\mathbb{F}_1}$ at o, such that f is étale outside a closed analytic subset of $\widetilde{W}$ of codimension at least two, is an isomorphism, and $C_{\mathbb{F}_1} \ni o$ is not a local complete intersection. In particular, $C_{\mathbb{F}_1} \ni o$ is not LCIQ.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we identify $C_{\mathbb{F}_1}$ with the affine toric variety of $\sigma_{\mathbb{F}_1}$ and show that the anti-canonical ring of $\mathbb{F}_1$ is generated in degree one.

For any integer $m \geq 0$, let $R_m := H^0(\mathbb{F}_1, \mathcal{O}_{\mathbb{F}_1}(-mK_{\mathbb{F}_1}))$, and let $R := \bigoplus_{m\geq 0} R_m$, so that $C_{\mathbb{F}_1} = \operatorname{Spec} R$. The surface $\mathbb{F}_1$ is the blow-up of $\mathbb{P}^2$ at a torus-fixed point, so $\mathbb{F}_1$ is the smooth projective toric surface whose fan in $\mathbb{R}^2$ has the rays spanned by

$$v_1 := (1,0), \quad v_2 := (1,1), \quad v_3 := (0,1), \quad v_4 := (-1,-1).$$

Since $-K_{\mathbb{F}_1}$ is the sum of the four torus-invariant prime divisors [CLS11, Theorem 8.2.3], R_m has the basis $\{\chi^w \mid w \in mP \cap \mathbb{Z}^2\}$ [CLS11, Proposition 4.3.3], where

$$\begin{aligned} P &:= \{w \in \mathbb{R}^2 \mid \langle w, v_i\rangle \geq -1 \text{ for } 1 \leq i \leq 4\} \\ &= \{(w_1, w_2) \in \mathbb{R}^2 \mid w_1 \geq -1,\ w_2 \geq -1,\ -1 \leq w_1 + w_2 \leq 1\}, \end{aligned}$$

and the product of sections is the product of characters. Let $M := \mathbb{Z}^3$, and write $u = (u_1, u_2, u_3)$ for elements of M. For $u_3 \geq 0$, a point $u \in \mathbb{R}^3$ satisfies $\langle u, (v_i, 1)\rangle \geq 0$ for $1 \leq i \leq 4$ if and only if $(u_1, u_2) \in u_3 P$ (for $u_3 = 0$ both sides say $(u_1, u_2) = 0$, since $v_1 + v_3 + v_4 = 0$ and v_1, v_3 form a basis), and no u with $u_3 < 0$ satisfies $\langle u, (v_i, 1)\rangle \geq 0$ for $1 \leq i \leq 4$, since $v_1 + v_3 + v_4 = 0$ and adding the three inequalities for $i = 1, 3, 4$ gives $3u_3 \geq 0$. Thus

$$\sigma_{\mathbb{F}_1}^\vee = \{u \in \mathbb{R}^3 \mid u_3 \geq 0,\ (u_1, u_2) \in u_3 P\},$$

and the $\mathbb{C}$-linear map sending $\chi^{(u_1,u_2)} \in R_m$ to $\chi^{(u_1,u_2,m)}$ is an isomorphism of $\mathbb{C}$-algebras $R \to \mathbb{C}[\sigma_{\mathbb{F}_1}^\vee \cap M]$. Since $0 \cdot P = \{0\}$, the ideal $\bigoplus_{m>0} R_m$ is spanned by the characters χ^u with $0 \neq u \in \sigma_{\mathbb{F}_1}^\vee \cap M$, that is, it is the ideal of the torus-fixed point. Since $\sigma_{\mathbb{F}_1}^\vee \cap M$ is saturated, $C_{\mathbb{F}_1}$ is normal [CLS11, Theorem 1.3.5], and $\dim C_{\mathbb{F}_1} = 3$.

Any lattice polygon is normal [CLS11, Corollary 2.2.13], so for any integer $m \geq 1$, any point of $mP \cap \mathbb{Z}^2$ is a sum of m points of $P \cap \mathbb{Z}^2$. Thus R is generated by R_1. For $h \in \mathbb{Z}$ with $-m \leq h \leq m$, the points $w \in mP \cap \mathbb{Z}^2$ with $w_1 + w_2 = h$ are those with $-m \leq w_1 \leq h + m$. Hence

$$\dim_{\mathbb{C}} R_m = \#(mP \cap \mathbb{Z}^2) = \sum_{h=-m}^{m} (h + 2m + 1) = (2m+1)^2. \tag{4.1}$$

In particular, $\dim_{\mathbb{C}} R_1 = 9$ and $\dim_{\mathbb{C}} R_2 = 25$.

Step 2. In this step, we show that $C_{\mathbb{F}_1}$ is Gorenstein and canonical, and that $\xi_{\mathbb{F}_1}$ is a Reeb vector.

Let $m_0 := (0,0,1) \in M$. Then $\langle m_0, (v_i, 1)\rangle = 1$ for $1 \leq i \leq 4$, so $K_{C_{\mathbb{F}_1}}$ is Cartier [CLS11, Proposition 8.2.12]. Since $C_{\mathbb{F}_1}$ is a normal affine toric variety, $C_{\mathbb{F}_1}$ is Cohen–Macaulay [Hoc72, Theorem 1], hence Gorenstein. Thus $C_{\mathbb{F}_1}$ is canonical [CLS11, Proposition 11.4.11], and in particular $C_{\mathbb{F}_1} \ni o$ is klt.

The weights of the torus action on R are the elements of $\sigma_{\mathbb{F}_1}^\vee \cap M$, the weight zero part of R is $\mathbb{C}\chi^0$, and $\sigma_{\mathbb{F}_1}^\vee$ is strictly convex because $\dim \sigma_{\mathbb{F}_1} = 3$. Thus the torus action is good in the sense of Definition 2.8, with vertex o. Let $t_0 := \frac{4-\sqrt{13}}{3}$, so that $0 < t_0 < 1$ and $\xi_{\mathbb{F}_1} = (t_0, t_0, 1)$. For any $0 \neq u \in \sigma_{\mathbb{F}_1}^\vee \cap M$, we have $u_3 \geq 1$ and $-u_3 \leq u_1 + u_2 \leq u_3$, so

$$\langle u, \xi_{\mathbb{F}_1}\rangle = u_3 + t_0(u_1 + u_2) \geq (1 - t_0)u_3 > 0.$$

Thus $\xi_{\mathbb{F}_1}$ is a Reeb vector, and the toric valuation of $\xi_{\mathbb{F}_1}$ is

$$\operatorname{wt}_{\xi_{\mathbb{F}_1}}\left(\sum_u c_u \chi^u\right) = \min\{\langle u, \xi_{\mathbb{F}_1}\rangle \mid c_u \neq 0\}.$$

Step 3. In this step, we show that $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q$ and that $\mathrm{wt}_{\xi_{\mathbb{F}_1}}$ computes $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o)$.

By [Blu18, Theorem 1.4], $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o)$ is the infimum of $\widehat{\mathrm{vol}}(\mathrm{wt}_\xi)$ over $\xi \in \mathrm{Int}(\sigma_{\mathbb{F}_1})$. By [Blu18, Section 8], $a(\mathrm{wt}_\xi, C_{\mathbb{F}_1}) = \langle m_0, \xi\rangle$, and $\mathrm{vol}(\mathrm{wt}_\xi)$ is 3! times the Euclidean volume of $\{u \in \sigma^\vee_{\mathbb{F}_1} \mid \langle u, \xi\rangle < 1\}$, normalized by M. Since $\widehat{\mathrm{vol}}(\mathrm{wt}_{c\xi}) = \widehat{\mathrm{vol}}(\mathrm{wt}_\xi)$ for any real number $c > 0$ and $\langle m_0, \xi\rangle > 0$, we may assume that $\xi = (a, b, 1)$. Then $a(\mathrm{wt}_\xi, C_{\mathbb{F}_1}) = 1$, and writing $u = (kw_1, kw_2, k)$ with $w \in P$ and $k \geq 0$, whose Jacobian is k^2, we obtain

$$\widehat{\mathrm{vol}}(\mathrm{wt}_{(a,b,1)}) = 6\int_P \int_0^{(1+aw_1+bw_2)^{-1}} k^2 dk dw = 2\int_P \frac{dw}{(1+aw_1+bw_2)^3}.$$

The polygon P is invariant under $(w_1, w_2) \to (w_2, w_1)$, so $\widehat{\mathrm{vol}}(\mathrm{wt}_{(a,b,1)}) = \widehat{\mathrm{vol}}(\mathrm{wt}_{(b,a,1)})$. Since $r \to r^{-3}$ is convex on $(0, +\infty)$, the function $(a, b) \to \widehat{\mathrm{vol}}(\mathrm{wt}_{(a,b,1)})$ is convex on the convex set of (a, b) with $(a, b, 1) \in \mathrm{Int}(\sigma_{\mathbb{F}_1})$. Thus

$$\widehat{\mathrm{vol}}(\mathrm{wt}_{(a,b,1)}) \geq \widehat{\mathrm{vol}}(\mathrm{wt}_{(t,t,1)}) \quad \text{with } t = \frac{a+b}{2},$$

and it suffices to minimize on the diagonal. Since $-1 \leq w_1 + w_2 \leq 1$ on P with both bounds attained, $(t, t, 1) \in \mathrm{Int}(\sigma_{\mathbb{F}_1})$ if and only if $-1 < t < 1$. For $h \in [-1, 1]$, the points $w \in P$ with $w_1 + w_2 = h$ are those with $-1 \leq w_1 \leq h + 1$. Changing coordinates to (w_1, h), whose Jacobian is 1, we obtain

$$\widehat{\mathrm{vol}}(\mathrm{wt}_{(t,t,1)}) = 2\int_{-1}^{1} \frac{h+2}{(1+th)^3} dh = \frac{4(2-t)}{(1-t^2)^2} \quad \text{for } -1 < t < 1. \tag{4.2}$$

Moreover,

$$\frac{d}{dt}\frac{4(2-t)}{(1-t^2)^2} = \frac{4(-3t^2+8t-1)}{(1-t^2)^3}.$$

The roots of $3t^2 - 8t + 1$ are $t_0 = \frac{4-\sqrt{13}}{3} \in (-1, 1)$ and $\frac{4+\sqrt{13}}{3} > 1$, so the derivative is negative on $(-1, t_0)$ and positive on $(t_0, 1)$. Thus the minimum of (4.2) on $(-1, 1)$ is attained exactly at $t = t_0$, and a direct calculation using $3t_0^2 - 8t_0 + 1 = 0$ shows that

$$\widehat{\mathrm{vol}}(\mathrm{wt}_{\xi_{\mathbb{F}_1}}) = \frac{4(2-t_0)}{(1-t_0^2)^2} = \frac{46+13\sqrt{13}}{12} = q.$$

Hence $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q$, and $\mathrm{wt}_{\xi_{\mathbb{F}_1}}$ computes $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o)$. The value q and the minimizer $\mathrm{wt}_{\xi_{\mathbb{F}_1}}$ are also computed in [Blu18, Section 8.3].

Step 4. In this step, we show that o is an isolated singular point of $C_{\mathbb{F}_1}$ and that the link of o is simply connected.

Let B be the negative section of $\mathbb{F}_1$ and A a fiber of the ruling $\mathbb{F}_1 \to \mathbb{P}^1$. Then $(B^2) = -1$, $(A \cdot B) = 1$, $(A^2) = 0$, and $-K_{\mathbb{F}_1} = 2B + 3A$, so

$$(-K_{\mathbb{F}_1} \cdot B) = -2 + 3 = 1.$$

The polygon P has the four vertices $(-1, 0)$, $(-1, 2)$, $(2, -1)$, $(0, -1)$, and its edges have inward normals $v_1, \ldots, v_4$, so the normal fan of P is the fan of $\mathbb{F}_1$ and $-K_{\mathbb{F}_1}$ is the divisor D_P of P. Hence $-K_{\mathbb{F}_1}$ is ample [CLS11, Proposition 6.1.10(a)], and it is very ample since $\mathbb{F}_1$ is smooth [CLS11, Theorem 6.1.15]. Moreover, R is generated by R_1 (**Step 1**). Thus the nine characters $\chi^{(w_1,w_2,1)}$ with $w \in P \cap \mathbb{Z}^2$ define a closed embedding $C_{\mathbb{F}_1} \subset \mathbb{C}^9$ whose image is the affine cone over the image of the anti-canonical embedding $\phi : \mathbb{F}_1 \to \mathbb{P}^8$. Hence $C_{\mathbb{F}_1}\backslash\{o\}$ is the complement of the zero section in the total space of $\mathcal{O}_{\mathbb{P}^8}(-1)|_{\phi(\mathbb{F}_1)} \cong \mathcal{O}_{\mathbb{F}_1}(K_{\mathbb{F}_1})$, and in particular $C_{\mathbb{F}_1}\backslash\{o\}$ is smooth. Since R is graded and generated by R_1, the cotangent space of $C_{\mathbb{F}_1}$ at o is R_1, and $\dim_{\mathbb{C}} R_1 = 9 > 3$. Thus o is an isolated singular point of $C_{\mathbb{F}_1}$.

Let $S^{17} \subset \mathbb{C}^9$ be the unit sphere and $\mathcal{L} := C_{\mathbb{F}_1} \cap S^{17}$. For any $\rho > 0$, the map $(z, r) \to rz$ is a homeomorphism from $\mathcal{L} \times (0, \rho)$ onto

$$U_\rho := \{z \in C_{\mathbb{F}_1} \mid 0 < |z| < \rho\},$$

since $C_{\mathbb{F}_1} \subset \mathbb{C}^9$ is stable under multiplication by positive real numbers. The restriction of the Hopf fibration $S^{17} \to \mathbb{P}^8$ to $\mathcal{L}$ is a circle bundle $\mathcal{L} \to \phi(\mathbb{F}_1)$. Since $(-K_{\mathbb{F}_1} \cdot B) = 1$, $\phi(B)$ is a line $\ell \subset \mathbb{P}^8$. Thus the preimage of ℓ in $\mathcal{L}$ is the unit sphere S^3 of the 2-dimensional linear subspace of $\mathbb{C}^9$ corresponding to ℓ, and $S^3 \to \ell$ is the Hopf bundle [Hat02, Example 4.45]. Since $\pi_1(S^3) = 0$, a fiber of $\mathcal{L} \to \phi(\mathbb{F}_1)$ over a point of $\phi(B)$ is null-homotopic in S^3, hence in $\mathcal{L}$. By the long exact sequence of homotopy groups of a fiber bundle [Hat02, Theorem 4.41 and Proposition 4.48], the sequence

$$\pi_1(S^1) \to \pi_1(\mathcal{L}) \to \pi_1(\mathbb{F}_1)$$

is exact, and the first map is zero. Since $\mathbb{F}_1$ is an S^2-bundle over S^2, the long exact sequence of homotopy groups of this S^2-bundle [Hat02, Theorem 4.41 and Proposition 4.48] implies that $\pi_1(\mathbb{F}_1) = 0$. Thus $\pi_1(\mathcal{L}) = 0$, and $\pi_1(U_\rho) = 0$ for any $\rho > 0$.

Step 5. In this step, we show that $C_{\mathbb{F}_1} \ni o$ is not a local complete intersection.

Let $\mathfrak{m} := (z_1, \dots, z_9) \subset \mathbb{C}[z_1, \dots, z_9]$, and let I be the kernel of the surjection $\mathbb{C}[z_1, \dots, z_9] \to R$ sending $z_1, \dots, z_9$ to the nine characters $\chi^{(w_1, w_2, 1)}$ with $w \in P \cap \mathbb{Z}^2$. Then I is homogeneous, $I_1 = 0$, and by (4.1),

$$\dim_{\mathbb{C}} I_2 = \dim_{\mathbb{C}} \mathbb{C}[z_1, \dots, z_9]_2 - \dim_{\mathbb{C}} R_2 = 45 - 25 = 20.$$

Since I has no non-zero element of degree at most one, $\mathfrak{m}I$ has no non-zero element of degree at most two. Thus the degree two part of the graded vector space $I/\mathfrak{m}I$ is I_2, and $\dim_{\mathbb{C}} I/\mathfrak{m}I \geq 20$. Let $T := \mathbb{C}[z_1, \dots, z_9]_{\mathfrak{m}}$, a regular local ring with $\dim T = 9$, so that $\mathcal{O}_{C_{\mathbb{F}_1}, o} = T/IT$ and $\dim \mathcal{O}_{C_{\mathbb{F}_1}, o} = 3$. Since $IT/\mathfrak{m}IT \cong I/\mathfrak{m}I$, any set of generators of IT has at least 20 elements by Nakayama's lemma. If $\mathcal{O}_{C_{\mathbb{F}_1}, o}$ were a complete intersection, then IT would be generated by a T-regular sequence [Mat86, Theorem 21.2(ii)], of length $\dim T - \dim \mathcal{O}_{C_{\mathbb{F}_1}, o} = 6 < 20$, a contradiction.

Step 6. In this step, we show that any finite surjective morphism $f : \widetilde{W} \to W$ as in the statement is an isomorphism.

Let $\tilde{o} \in \widetilde{W}$ be the point of the germ $\widetilde{W}$. Since f is a finite morphism of germs, possibly shrinking the representatives, we may assume that $W = \{z \in C_{\mathbb{F}_1} \mid |z| < \rho\}$ for some $\rho > 0$, that $\widetilde{W}$ is a connected normal complex space with $f^{-1}(o) = \{\tilde{o}\}$, that $f : \widetilde{W} \to W$ is finite and surjective, and that f is étale outside a closed analytic subset $Z \subset \widetilde{W}$ with $\operatorname{codim}_{\widetilde{W}} Z \geq 2$. Let $U_\rho := W \backslash \{o\}$ as in **Step 4**, and let $\widetilde{W}^\circ := \widetilde{W} \backslash \{\tilde{o}\}$. Since $\widetilde{W}$ is normal, $\widetilde{W}$ is locally irreducible, so $\widetilde{W}^\circ$ is connected.

We show that f is a local biholomorphism at any point $\tilde{w} \in \widetilde{W}^\circ$. Let $p := f(\tilde{w})$. Since U_ρ is smooth, $\mathcal{O}_{U_\rho, p} \cong \mathbb{C}\{x_1, x_2, x_3\}$ is a regular local ring. Since f is finite and $\widetilde{W}$ is normal, $\mathcal{O}_{\widetilde{W}, \tilde{w}}$ is a normal local domain which is finite over $\mathcal{O}_{U_\rho, p}$, and since $\dim \mathcal{O}_{\widetilde{W}, \tilde{w}} = \dim \mathcal{O}_{U_\rho, p} = 3$, the homomorphism $\mathcal{O}_{U_\rho, p} \to \mathcal{O}_{\widetilde{W}, \tilde{w}}$ is injective. The module of relative differentials $\Omega := \Omega_{\mathcal{O}_{\widetilde{W}, \tilde{w}} / \mathcal{O}_{U_\rho, p}}$ is a finite $\mathcal{O}_{\widetilde{W}, \tilde{w}}$-module whose support is contained in the germ of Z at $\tilde{w}$, since Ω is the stalk of the coherent sheaf $\Omega_{\widetilde{W}^\circ / U_\rho}$, which vanishes where f is étale, by Rückert's Nullstellensatz. Thus $\dim \mathcal{O}_{\widetilde{W}, \tilde{w}} / \operatorname{Ann}(\Omega) \leq 1$, so $\operatorname{Ann}(\Omega)$ has height at least two, and $\Omega_{\mathfrak{q}} = 0$ for any prime ideal $\mathfrak{q} \subset \mathcal{O}_{\widetilde{W}, \tilde{w}}$ of height one. For such $\mathfrak{q}$, the prime ideal $\mathfrak{p} := \mathfrak{q} \cap \mathcal{O}_{U_\rho, p}$ has height one, $(\mathcal{O}_{U_\rho, p})_{\mathfrak{p}}$ is a discrete valuation ring, and $(\mathcal{O}_{\widetilde{W}, \tilde{w}})_{\mathfrak{q}}$ is a localization of the torsion-free finite $(\mathcal{O}_{U_\rho, p})_{\mathfrak{p}}$-module $(\mathcal{O}_{\widetilde{W}, \tilde{w}})_{\mathfrak{p}}$, hence flat over $(\mathcal{O}_{U_\rho, p})_{\mathfrak{p}}$. Thus $\operatorname{Spec} \mathcal{O}_{\widetilde{W}, \tilde{w}} \to \operatorname{Spec} \mathcal{O}_{U_\rho, p}$ is étale at any point of codimension one, and by the Zariski–Nagata purity theorem [Gro71, Exposé X, Théorème 3.1], it is étale at the closed point. Since both residue fields are $\mathbb{C}$, the finite étale local homomorphism $\mathcal{O}_{U_\rho, p} \to \mathcal{O}_{\widetilde{W}, \tilde{w}}$ is an isomorphism. Thus f is a local biholomorphism at $\tilde{w}$.

Hence $f|_{\widetilde{W}^\circ} : \widetilde{W}^\circ \to U_\rho$ is a finite local homeomorphism, that is, a finite covering map. Since U_ρ is simply connected (**Step 4**) and $\widetilde{W}^\circ$ is connected, $f|_{\widetilde{W}^\circ}$ is a homeomorphism, hence biholomorphic. Let $j : U_\rho \to W$ be the inclusion. Since the germ of $C_{\mathbb{F}_1}$ at o is normal [Gro71, Exposé XII, Proposition 2.1] and $\operatorname{codim}_W\{o\} = 3$, any holomorphic function on U_ρ extends to W by the Riemann extension theorem on normal complex spaces [GR84, Chapter 7, §4.2], that is, $j_*\mathcal{O}_{U_\rho} = \mathcal{O}_W$. Since $\widetilde{W}^\circ$ is dense in the reduced space $\widetilde{W}$, we obtain

$$\mathcal{O}_W \subset f_*\mathcal{O}_{\widetilde{W}} \subset j_*\mathcal{O}_{U_\rho} = \mathcal{O}_W.$$

Thus f is an isomorphism.

Step 7. In this step, we show that $C_{\mathbb{F}_1} \ni o$ is not LCIQ.

Suppose that $C_{\mathbb{F}_1} \ni o$ is LCIQ. Since $C_{\mathbb{F}_1} \ni o$ is klt (**Step 2**), by Definition 2.13, there exist a normal germ $Y \ni y$ which is a local complete intersection and a finite group G acting on Y with y fixed and freely in codimension one, such that $C_{\mathbb{F}_1} \ni o$ is analytically isomorphic to the germ of Y/G at the image $\bar{y}$ of y. Let $\pi : Y \to Y/G$ be the quotient morphism, and let $Y' \subset Y$ be the closed subset of points with non-trivial stabilizer. Then $\operatorname{codim}_Y Y' \geq 2$, and π is étale on $Y\backslash Y'$ since G acts freely on $Y\backslash Y'$. By Lemma 2.5(2), the complex analytic germ of Y/G at $\bar{y}$ is isomorphic to W. Since $\pi^{-1}(\bar{y}) = \{y\}$, composing the germ of π^{an} at y with an isomorphism from the complex analytic germ of Y/G at $\bar{y}$ onto W, we obtain a finite surjective morphism onto W from the complex analytic germ of Y at y, which is étale outside the germ of $(Y')^{\mathrm{an}}$. The complex analytic germ of Y at y is normal [Gro71, Exposé XII, Proposition 2.1], hence connected. By **Step 6**, the complex analytic germs of Y at y and of $C_{\mathbb{F}_1}$ at o are isomorphic, so $\widehat{\mathcal{O}}_{Y,y} \cong \widehat{\mathcal{O}}_{C_{\mathbb{F}_1},o}$ by Lemma 2.5(2). Since $\mathcal{O}_{Y,y}$ is a complete intersection, so are $\widehat{\mathcal{O}}_{Y,y}$, $\widehat{\mathcal{O}}_{C_{\mathbb{F}_1},o}$, and $\mathcal{O}_{C_{\mathbb{F}_1},o}$ [BH98, Definition 2.3.1 and Theorem 2.3.3], which contradicts **Step 5**. □

The following rational bounds for q are used in the later sections.

Lemma 4.3. *The number* $q = \frac{46+13\sqrt{13}}{12}$ *satisfies*

$$7 < \frac{250}{33} < \frac{54}{7} < \frac{116}{15} < q < 8.$$

Proof. Since $18^2 = 324 < 325 = 13 \cdot 5^2$ and $(13\sqrt{13})^2 = 2197 < 2500 = 50^2$, we have $\frac{18}{5} < \sqrt{13} < \frac{50}{13}$. Hence $q > \frac{1}{12}\left(46 + 13 \cdot \frac{18}{5}\right) = \frac{116}{15}$ and $q < \frac{46+50}{12} = 8$. Moreover, $\frac{116}{15} > \frac{54}{7}$ since $116 \cdot 7 = 812 > 810 = 54 \cdot 15$, $\frac{54}{7} > \frac{250}{33}$ since $54 \cdot 33 = 1782 > 1750 = 250 \cdot 7$, and $\frac{250}{33} > 7$ since $250 > 231 = 7 \cdot 33$. □

4.2. **Deformations.** In this subsection, we prove that $C_{\mathbb{F}_1}$ has no non-trivial formal deformation over a reduced complete local base.

The first cotangent cohomology of $C_{\mathbb{F}_1}$ at o is one-dimensional (Proposition 4.4(1) below), so $C_{\mathbb{F}_1}$ is not rigid, and a degeneration of a germ to $C_{\mathbb{F}_1} \ni o$ does not by itself imply that the germ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$. What we use is that the miniversal hull of the deformation functor is $\mathbb{C}[\epsilon]/(\epsilon^2)$ (Proposition 4.4(2) below).

Proposition 4.4. *Let* $\sigma_{\mathbb{F}_1}$ *and* $\xi_{\mathbb{F}_1}$ *be as in Construction-Proposition 4.2, and identify* $C_{\mathbb{F}_1}$ *with the affine toric variety of* $\sigma_{\mathbb{F}_1}$ *with respect to* $\mathbb{Z}^3$. *Let* $R := \bigoplus_{m\geq 0} H^0(\mathbb{F}_1, \mathcal{O}_{\mathbb{F}_1}(-mK_{\mathbb{F}_1}))$, *so that* $C_{\mathbb{F}_1} = \operatorname{Spec} R$, *let* $\widehat{R}$ *be the completion of* R *at* o, *let* $m_0 := (0,0,1) \in \mathbb{Z}^3$, *and let* T^1_R *be the first cotangent cohomology of* R, *with the natural* $\mathbb{Z}^3$*-grading, normalized so that a homogeneous element of* T^1_R *of degree* d *sends a homogeneous equation of degree* u *to a function of degree* $u+d$. *Then the following hold.*

(1) $\dim_{\mathbb{C}} T^1_R = 1$, *and* T^1_R *is concentrated in degree* $-m_0$, *and* $\langle -m_0, \xi_{\mathbb{F}_1}\rangle = -1$.

(2) The deformation functor of $\widehat{R}$ *has the miniversal hull* $\mathbb{C}[\epsilon]/(\epsilon^2)$, *and the Kodaira–Spencer map of the corresponding family is an isomorphism onto* T^1_R. *In particular, no non-zero first-order deformation of* $\widehat{R}$ *extends over* $\mathbb{C}[s]/(s^3)$.

(3) *Let $(\Lambda, \mathfrak{n})$ be a reduced complete Noetherian local $\mathbb{C}$-algebra with residue field $\mathbb{C}$, and let $\mathcal{A}$ be a flat formal deformation of $\widehat{R}$ over Λ which is topologically of finite presentation, that is, $\mathcal{A} = \Lambda[[z_1, \dots, z_n]]/I$ for an integer $n \geq 0$ and an ideal I, together with an isomorphism $\mathcal{A}/\mathfrak{n}\mathcal{A} \cong \widehat{R}$, such that $\mathcal{A}/\mathfrak{n}^{k+1}\mathcal{A}$ is flat over $\Lambda/\mathfrak{n}^{k+1}$ for any $k \geq 0$ and the natural map $\mathcal{A} \to \varprojlim_k \mathcal{A}/\mathfrak{n}^{k+1}\mathcal{A}$ is an isomorphism. Then $\mathcal{A} \cong \Lambda \widehat{\otimes}_{\mathbb{C}} \widehat{R}$ as Λ-algebras.*

Proof. (1) Let $g : \mathbb{Z}^3 \to \mathbb{Z}^3$ be the linear map $(s_1, s_2, s_3) \to (s_3 - s_1, s_3 - s_2, s_3)$. Then $\det g = 1$, $g^2 = \mathrm{id}_{\mathbb{Z}^3}$, and g sends $(1, 0, 1)$, $(1, 1, 1)$, $(0, 1, 1)$, and $(-1, -1, 1)$ to $(0, 1, 1)$, $(0, 0, 1)$, $(1, 0, 1)$, and $(2, 2, 1)$, respectively. Thus $g(\sigma_{\mathbb{F}_1})$ is the cone over $Q_4 \times \{1\}$, where $Q_4 := \mathrm{Conv}\{(0, 0), (1, 0), (2, 2), (0, 1)\}$ is the polygon of [Alt94, (9.1)(iv)]. The dual map $u \to u \circ g$ sends (u_1, u_2, u_3) to $(-u_1, -u_2, u_1 + u_2 + u_3)$, so it fixes m_0, and m_0 is the degree which takes the value 1 on $Q_4 \times \{1\}$, as in [Alt94, (6.2)]. Hence g induces an isomorphism from $C_{\mathbb{F}_1}$ onto the affine toric variety of [Alt94, (9.1)(iv)] which preserves the $\mathbb{Z}^3$-gradings and m_0.

By Construction-Proposition 4.2, o is an isolated singular point of $C_{\mathbb{F}_1}$. Since the singular locus of $C_{\mathbb{F}_1}$ is closed and stable under the torus, and the closure of any torus orbit contains o, the singular locus is $\{o\}$, that is, $C_{\mathbb{F}_1} \backslash \{o\}$ is smooth. Thus T^1_R is a finitely generated R-module supported at o, and $\dim_{\mathbb{C}} T^1_R < +\infty$. By [Alt94, (6.3)], T^1_R is concentrated in degree $-m_0$. By [Alt94, (7.2), Theorem and Corollary], the family over the base space of [Alt94, (2.3)–(2.4)] is a versal deformation of $C_{\mathbb{F}_1} \ni o$, and by [Alt94, (6.2)], the Kodaira–Spencer map of the family is an isomorphism from the tangent space of the base space onto T^1_R. By [Alt94, (9.1)(iv)–(9.2)], for Q_4 the base space is $\operatorname{Spec} \mathbb{C}[\epsilon]/(\epsilon^2)$. Thus $\dim_{\mathbb{C}} T^1_R = 1$. Since the third coordinate of $\xi_{\mathbb{F}_1}$ is 1, $\langle -m_0, \xi_{\mathbb{F}_1} \rangle = -1$.

(2) The deformation functors of R and of $\widehat{R}$ have hulls by Schlessinger's criterion [Sch68, Theorem 2.11], see also [Har10, Theorem 16.2]. Since $C_{\mathbb{F}_1} \backslash \{o\}$ is smooth, T^1_R and T^2_R are finitely generated R-modules supported at o, so they coincide with the corresponding modules of $\widehat{R}$. Thus the morphism from the deformation functor of R to the deformation functor of $\widehat{R}$ is bijective on tangent spaces and injective on obstruction spaces, hence smooth, that is, strongly surjective in the sense of [Har10, Section 15] (by the standard smoothness criterion: a morphism of deformation functors which is surjective on tangent spaces and injective on obstruction spaces is smooth), and a hull of the deformation functor of R is also a hull of the deformation functor of $\widehat{R}$ [Sch68, Proposition 2.5(ii) and Definition 2.7]. By [Alt94, (7.2), Theorem and Corollary] and [Alt94, (9.1)(iv)–(9.2)], the base space of the versal deformation of $C_{\mathbb{F}_1} \ni o$ is $\operatorname{Spec} \mathbb{C}[\epsilon]/(\epsilon^2)$ (see the proof of (1)). Thus the hull is $H := \mathbb{C}[\epsilon]/(\epsilon^2)$, and by [Alt94, (6.2)], the Kodaira–Spencer map of the corresponding family $\mathcal{V}$ is an isomorphism onto T^1_R.

Suppose that a non-zero first-order deformation of $\widehat{R}$ extends to a deformation over $\mathbb{C}[s]/(s^3)$. By versality, the deformation over $\mathbb{C}[s]/(s^3)$ is induced from $\mathcal{V}$ by a local homomorphism $\varphi : H \to \mathbb{C}[s]/(s^3)$, and we write $\varphi(\epsilon) = c_1 s + c_2 s^2$ with $c_1, c_2 \in \mathbb{C}$. Since the Kodaira–Spencer map is an isomorphism and the first-order deformation is non-zero, $c_1 \neq 0$. Thus $\varphi(\epsilon)^2 = c_1^2 s^2 \neq 0$, which contradicts $\epsilon^2 = 0$ in H.

(3) Let $f_1, \dots, f_r$ generate the kernel I_0 of a surjection $\mathbb{C}[y_1, \dots, y_e] \to R$ of $\mathbb{C}$-algebras which sends each y_i to a homogeneous element of positive degree. Then $\widehat{R} = \mathbb{C}[[y_1, \dots, y_e]]/(f_1, \dots, f_r)$. Since $\mathcal{V}$ is flat over H, we may write

$$\mathcal{V} = H[[y_1, \dots, y_e]]/(f_1 + \epsilon g_1, \dots, f_r + \epsilon g_r)$$

with $g_1, \dots, g_r \in \mathbb{C}[[y_1, \dots, y_e]]$, and then the Kodaira–Spencer image $\kappa \in T^1_R$ of $\partial/\partial\epsilon$ is the class of the homomorphism sending f_l to g_l for $1 \leq l \leq r$. By (1) and (2), $\kappa \neq 0$ and $T^1_R = \mathbb{C}\kappa$.

For any integer $k \geq 0$, let $\Lambda_k := \Lambda/\mathfrak{n}^{k+1}$ and $\mathcal{A}_k := \mathcal{A}/\mathfrak{n}^{k+1}\mathcal{A}$. Then Λ_k is a local Artinian $\mathbb{C}$-algebra with residue field $\mathbb{C}$, and $\mathcal{A}_k = \Lambda_k[[z_1, \dots, z_n]]/I\Lambda_k[[z_1, \dots, z_n]]$ is a deformation of $\widehat{R}$ over Λ_k. For a local homomorphism $\varphi : H \to \Lambda_k$, write $\varphi^*\mathcal{V} := \mathcal{V} \otimes_{H,\varphi} \Lambda_k$. We construct, by induction on k, local homomorphisms $\varphi_k : H \to \Lambda_k$ and isomorphisms $\psi_k : \varphi_k^*\mathcal{V} \to \mathcal{A}_k$ of

deformations of $\widehat{R}$, such that φ_{k+1} and ψ_{k+1} reduce to φ_k and ψ_k modulo $\mathfrak{n}^{k+1}$. For $k=0$, let φ_0 be the residue map and ψ_0 the given isomorphism $\widehat{R} \cong \mathcal{A}/\mathfrak{n}\mathcal{A}$.

Assume that φ_k and ψ_k are constructed. Let $J := \mathfrak{n}^{k+1}/\mathfrak{n}^{k+2}$ be the kernel of $\Lambda_{k+1} \to \Lambda_k$. Then $\mathfrak{n}J = 0$, and $J^2 = 0$ since $2k+2 \geq k+2$. Since $\mathcal{V}$ is versal, the morphism $\Phi_{\mathcal{V}}$ from $\mathrm{Hom}(H, -)$ to the deformation functor of $\widehat{R}$ is smooth, that is, strongly surjective in the sense of [Har10, Section 15] (see also [Har10, Proposition 15.2(a)]). Applying the smoothness of $\Phi_{\mathcal{V}}$ to $\Lambda_{k+1} \to \Lambda_k$, φ_k, and $\mathcal{A}_{k+1}$, there exists a local homomorphism $\varphi' : H \to \Lambda_{k+1}$ which lifts φ_k. The smoothness of $\Phi_{\mathcal{V}}$ also gives an isomorphism $\varphi'^*\mathcal{V} \cong \mathcal{A}_{k+1}$ of deformations, but its reduction modulo $\mathfrak{n}^{k+1}$ need not be ψ_k, since the deformation functor takes values in sets of isomorphism classes. We therefore keep only φ', and below we modify φ' to φ_{k+1} and construct an isomorphism ψ_{k+1} which reduces to ψ_k.

Let $\mathcal{B}_0 := \mathbb{C}[[y_1, \dots, y_e]]$, $\mathfrak{b}_0 := (f_1, \dots, f_r) \subset \mathcal{B}_0$, and $\mathcal{B} := \Lambda_{k+1}[[y_1, \dots, y_e]] = \Lambda_{k+1} \otimes_{\mathbb{C}} \mathcal{B}_0$. Since $\mathfrak{n}J = 0$, we have $J\mathcal{B} = J \otimes_{\mathbb{C}} \mathcal{B}_0$, and $xF = x\bar{F}$ for any $x \in J\mathcal{B}$ and any $F \in \mathcal{B}$ with image $\bar{F} \in \mathcal{B}_0$. Since $\mathcal{B}$ is flat over Λ_{k+1}, for any ideal $\mathfrak{b} \subset \mathcal{B}$ with $\mathcal{B}/\mathfrak{b}$ flat over Λ_{k+1} and any ideal $\mathfrak{a} \subset \Lambda_{k+1}$, the long exact sequence of $\mathrm{Tor}_\bullet^{\Lambda_{k+1}}(-, \Lambda_{k+1}/\mathfrak{a})$ for $0 \to \mathfrak{b} \to \mathcal{B} \to \mathcal{B}/\mathfrak{b} \to 0$ implies $\mathfrak{b} \cap \mathfrak{a}\mathcal{B} = \mathfrak{a}\mathfrak{b}$. Let $F'_l := f_l + \varphi'(\epsilon)g_l$ for $1 \leq l \leq r$ and $\mathfrak{b}' := (F'_1, \dots, F'_r) \subset \mathcal{B}$, so that $\varphi'^*\mathcal{V} = \mathcal{B}/\mathfrak{b}'$ and $\varphi'^*\mathcal{V} \otimes_{\Lambda_{k+1}} \Lambda_k = \varphi_k^*\mathcal{V}$. Let $q_k : \mathcal{B} \to \mathcal{A}_k$ be the composition of the quotient map $\mathcal{B} \to \varphi_k^*\mathcal{V}$ with ψ_k.

We first present $\mathcal{A}_{k+1}$ by lifts of $F'_1, \dots, F'_r$. Since $\mathcal{A}_{k+1}$ is a complete local ring, there exists a homomorphism of Λ_{k+1}-algebras $\theta : \mathcal{B} \to \mathcal{A}_{k+1}$ such that $\theta(y_i)$ lifts $q_k(y_i)$ for $1 \leq i \leq e$, so that the composition of θ with $\mathcal{A}_{k+1} \to \mathcal{A}_k$ is q_k. Since q_k is surjective and $J\mathcal{A}_{k+1}$ is nilpotent, θ is surjective. Let $\mathfrak{b}'' := \ker\theta$. Since $\ker q_k$ is the image of $\mathfrak{b}'$ in $\mathcal{B}/J\mathcal{B} = \Lambda_k[[y_1, \dots, y_e]]$, we have $\mathfrak{b}'' + J\mathcal{B} = \mathfrak{b}' + J\mathcal{B}$. Thus there exist $F''_l \in \mathfrak{b}''$ with $\delta_l := F''_l - F'_l \in J\mathcal{B}$ for $1 \leq l \leq r$. Since $\mathcal{A}_{k+1}$ is flat over Λ_{k+1}, $\mathfrak{b}'' \cap J\mathcal{B} = J\mathfrak{b}''$, so $\mathfrak{b}'' = (F''_1, \dots, F''_r) + J\mathfrak{b}''$, and $\mathfrak{b}'' = (F''_1, \dots, F''_r)$ since $J^2 = 0$. In particular, $J\mathfrak{b}'' = J \otimes_{\mathbb{C}} \mathfrak{b}_0$.

We next show that $f_l \to \delta_l$ defines a homomorphism $\mathfrak{b}_0/\mathfrak{b}_0^2 \to J \otimes_{\mathbb{C}} \widehat{R}$. Let $h_1, \dots, h_r \in \mathcal{B}_0$ with $\sum_l h_l f_l = 0$, and let $\tilde{h}_l \in \mathcal{B}$ be lifts of h_l. Since $\varphi'^*\mathcal{V}$ is flat over Λ_{k+1}, $\sum_l \tilde{h}_l F'_l \in \mathfrak{b}' \cap \mathfrak{n}\mathcal{B} = \mathfrak{n}\mathfrak{b}'$, so after subtracting elements of $\mathfrak{n}\mathcal{B}$ from $\tilde{h}_1, \dots, \tilde{h}_r$, we may assume that $\sum_l \tilde{h}_l F'_l = 0$. Then

$$\sum_l h_l \delta_l = \sum_l \tilde{h}_l \delta_l = \sum_l \tilde{h}_l F''_l \in \mathfrak{b}'' \cap J\mathcal{B} = J \otimes_{\mathbb{C}} \mathfrak{b}_0.$$

Thus $\sum_l h_l f_l \to \sum_l h_l \delta_l$ is a well-defined $\mathcal{B}_0$-linear map $\mathfrak{b}_0 \to J \otimes_{\mathbb{C}} \widehat{R}$, and it vanishes on $\mathfrak{b}_0^2$. By [Har10, Proposition 3.10], T^1_R is the quotient of $\mathrm{Hom}_R(I_0/I_0^2, R)$ by the R-span of the homomorphisms $f_l \to \partial f_l/\partial y_i$ for $1 \leq i \leq e$. Since $\widehat{R}$ is flat over R, I_0/I_0^2 is a finitely presented R-module, and $(I_0/I_0^2) \otimes_R \widehat{R} = \mathfrak{b}_0/\mathfrak{b}_0^2$, the quotient of $\mathrm{Hom}_{\widehat{R}}(\mathfrak{b}_0/\mathfrak{b}_0^2, \widehat{R})$ by the $\widehat{R}$-span of the homomorphisms $f_l \to \partial f_l/\partial y_i$ is $T^1_R \otimes_R \widehat{R}$, which is T^1_R since T^1_R is a finite-dimensional R-module supported at o. Since $\dim_{\mathbb{C}} J < +\infty$, we have

$$\mathrm{Hom}_{\widehat{R}}(\mathfrak{b}_0/\mathfrak{b}_0^2, J \otimes_{\mathbb{C}} \widehat{R}) = \mathrm{Hom}_{\widehat{R}}(\mathfrak{b}_0/\mathfrak{b}_0^2, \widehat{R}) \otimes_{\mathbb{C}} J.$$

Since $T^1_R = \mathbb{C}\kappa$, there exist $j \in J$ and $d_1, \dots, d_e \in J\mathcal{B}$ such that

$$\delta_l - jg_l - \sum_{i=1}^{e} d_i \frac{\partial f_l}{\partial y_i} \in J \otimes_{\mathbb{C}} \mathfrak{b}_0 \quad \text{for } 1 \leq l \leq r. \tag{4.3}$$

Let $\varphi_{k+1} : H \to \Lambda_{k+1}$ be the local homomorphism with $\varphi_{k+1}(\epsilon) = \varphi'(\epsilon) + j$. It is well defined, since $\varphi'(\epsilon) \in \mathfrak{n}\Lambda_{k+1}$ and $\mathfrak{n}J = 0$ imply

$$(\varphi'(\epsilon) + j)^2 = \varphi'(\epsilon)^2 = 0,$$

and φ_{k+1} lifts φ_k. Let $\mathfrak{b}^+ := (F'_1 + jg_1, \dots, F'_r + jg_r)$, so that $\varphi_{k+1}^*\mathcal{V} = \mathcal{B}/\mathfrak{b}^+$. Let τ be the automorphism of the Λ_{k+1}-algebra $\mathcal{B}$ with $\tau(y_i) = y_i - d_i$ for $1 \leq i \leq e$, whose inverse sends y_i

to $y_i + d_i$. Since $J^2 = 0$, the Taylor expansion implies $\tau(F) = F - \sum_i d_i \partial \bar{F}/\partial y_i$ for any $F \in \mathcal{B}$. In particular, τ induces the identity on $\mathcal{B}/J\mathcal{B}$, $\tau(\delta_l) = \delta_l$, and by (4.3),

$$\tau(F_l'') = F_l' + \delta_l - \sum_{i=1}^{e} d_i \frac{\partial f_l}{\partial y_i} \in F_l' + jg_l + J \otimes_{\mathbb{C}} \mathfrak{b}_0 \quad \text{for } 1 \le l \le r.$$

Since $J \otimes_{\mathbb{C}} \mathfrak{b}_0 = J\mathfrak{b}^+$, we obtain $\tau(\mathfrak{b}'') \subset \mathfrak{b}^+$ and $\mathfrak{b}^+ \subset \tau(\mathfrak{b}'') + J\mathfrak{b}^+$, so $\mathfrak{b}^+ = \tau(\mathfrak{b}'')$ since $J^2 = 0$. Thus $\theta \circ \tau^{-1}$ induces an isomorphism $\psi_{k+1} : \varphi_{k+1}^* \mathcal{V} = \mathcal{B}/\mathfrak{b}^+ \to \mathcal{A}_{k+1}$ of deformations. Since τ induces the identity on $\mathcal{B}/J\mathcal{B}$ and the composition of θ with $\mathcal{A}_{k+1} \to \mathcal{A}_k$ is q_k, ψ_{k+1} reduces to ψ_k modulo $\mathfrak{n}^{k+1}$.

Since $\Lambda = \varprojlim_k \Lambda_k$, the homomorphisms φ_k define a local homomorphism $\varphi : H \to \Lambda$, and since $\mathcal{A} \to \varprojlim_k \mathcal{A}_k$ is an isomorphism, the isomorphisms ψ_k define an isomorphism

$$\varprojlim_k \varphi_k^* \mathcal{V} \cong \mathcal{A}.$$

Let $a := \varphi(\epsilon)$. Then $a^2 = 0$, and since Λ is reduced, $a = 0$. The rings Λ_k may contain non-zero elements of square zero, but $\varphi_k(\epsilon)$ is the image of a in Λ_k, so $\varphi_k(\epsilon) = 0$ for any $k \ge 0$. Thus φ_k factors through the residue map $H \to \mathbb{C}$, and $\varphi_k^* \mathcal{V} = \Lambda_k \otimes_{\mathbb{C}} \widehat{R}$. Hence

$$\mathcal{A} \cong \varprojlim_k \Lambda_k \otimes_{\mathbb{C}} \widehat{R} = \Lambda \widehat{\otimes}_{\mathbb{C}} \widehat{R}$$

as Λ-algebras. □

5. Covers, minimizers, and degenerations

In this section, we prove the results used in Section 11 to pass from a germ to its minimizing cone and back: properties of index-one covers, the preservation of LCIQ singularities under volume-preserving degenerations, the associated graded rings of minimizers, the degeneration to the minimizing cone, and a criterion deducing analytic isomorphisms from formal ones. We also prove an inequality for minimizers (Proposition 5.6), which is used in Sections 7 and 10.

5.1. **Index-one covers.**

In this subsection, we use the index-one cover to reduce non-LCIQ germs of large local volume to Gorenstein canonical germs.

Lemma 5.1. *Let $X \ni x$ be a klt threefold germ, let r be the Cartier index of K_X at x, and let $V := \widehat{\mathrm{vol}}(X \ni x)$. Let $\pi : (Y \ni y) \to (X \ni x)$ be the index-one cover of K_X* [KM98, Definition 5.19]. *Then the following hold.*

1. *The morphism π is a connected cyclic quasi-étale cover of degree r with $\pi^{-1}(x) = \{y\}$, the germ $Y \ni y$ is Gorenstein canonical, and $\widehat{\mathrm{vol}}(Y \ni y) = rV$.*
2. *If $rV > 9$, then $Y \ni y$ is a hypersurface germ, and hence $X \ni x$ is LCIQ.*
3. *If $X \ni x$ is not LCIQ and $r \ge 2$, then $V \le \frac{9}{r} \le \frac{9}{2}$.*
4. *If $X \ni x$ is not LCIQ and $V > \frac{9}{2}$, then $X \ni x$ is Gorenstein canonical, $X \ni x$ is not a hypersurface germ, and $V < 9$.*

Proof. (1) Possibly shrinking X around x, we may assume that X is affine and $rK_X \sim 0$. By [KM98, Definitions 2.52 and 5.19],

$$Y = \mathrm{Spec}_X \bigoplus_{i=0}^{r-1} \mathcal{O}_X(-iK_X),$$

where the multiplication is induced by an isomorphism $\mathcal{O}_X(-rK_X) \cong \mathcal{O}_X$, and π is étale over the open subset of X where K_X is Cartier. In particular, $\deg \pi = r$, and π is étale over any codimension one point of X. Since $\pi_* \mathcal{O}_Y$ is reflexive and π is étale over the codimension one points of the normal variety X, Y satisfies (S_2) and (R_1), so Y is normal by Serre's criterion. Let $G := \mu_r$ act on Y such that $g^* s = g^i s$ for any $g \in \mu_r$ and any local section s of $\mathcal{O}_X(-iK_X)$. Then

$\mathcal{O}_X(-iK_X)$ is the χ^i-eigensheaf of $\pi_*\mathcal{O}_Y$, where $\chi : G \to \mathbb{C}^*$ is the inclusion, and $(\pi_*\mathcal{O}_Y)^G = \mathcal{O}_X$, that is, $X = Y/G$.

Let $S := (\pi_*\mathcal{O}_Y)_x$. The maximal ideals of the semilocal ring S correspond to the points of $\pi^{-1}(x)$, and since $S^G = \mathcal{O}_{X,x}$, G acts transitively on $\pi^{-1}(x)$. Suppose that $\pi^{-1}(x)$ has more than one point. Then the stabilizer H of a point $y_1 \in \pi^{-1}(x)$ is a proper subgroup of the cyclic group G, so there exists an integer $0 < j < r$ such that χ^j is trivial on H. The function e on $\pi^{-1}(x) = G \cdot y_1$ defined by $e(g \cdot y_1) = \chi^j(g)$ is well defined and nowhere zero. By the Chinese remainder theorem, there exists $a' \in S$ whose residue at any point of $\pi^{-1}(x)$ is the value of e. Let

$$a := \frac{1}{r} \sum_{g \in G} \chi^j(g)^{-1} g^* a'.$$

Since $e(g \cdot p) = \chi^j(g)e(p)$ for any $g \in G$ and $p \in \pi^{-1}(x)$, the residue of a at any point of $\pi^{-1}(x)$ equals the residue of a' there. Thus a is a unit of S, and $a \in \mathcal{O}_X(-jK_X)_x$, the χ^j-eigenspace of S. For any $c \in \mathcal{O}_X(-jK_X)_x$, the element ca^{-1} is G-invariant, so $c \in \mathcal{O}_{X,x} \cdot a$. Hence $\mathcal{O}_X(-jK_X)$ is free of rank one at x, that is, jK_X is Cartier at x, which contradicts the minimality of r. Therefore $\pi^{-1}(x) = \{y\}$, and the residue field of y is $\mathbb{C}$ since it is finite over $\mathbb{C}$.

Since $\pi_*\mathcal{O}_Y$ is torsion free, any connected component of Y dominates X, and since π is finite, any connected component of Y contains y. Thus Y is connected, hence integral as Y is normal. Since $X = Y/G$ and $\deg \pi = r = |G|$, G acts faithfully on Y and $K(Y)/K(X)$ is a Galois extension with group G. Thus π is a connected cyclic quasi-étale cover of degree r, and $K_Y = \pi^* K_X$.

By [KM98, Proposition 5.20(4)], Y is klt. Hence Y has rational singularities and is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. By [KM98, Definition 5.19], $K_Y = \pi^* K_X$ is Cartier and linearly equivalent to zero near y, so $\mathcal{O}_Y(K_Y) \cong \mathcal{O}_Y$ near y, and $Y \ni y$ is Gorenstein. Since K_Y is Cartier, $a(E, Y)$ is an integer for any prime divisor E over Y, and since $a(E, Y) > 0$, we have $a(E, Y) \geq 1$. Thus $Y \ni y$ is canonical. Finally, π is a finite Galois morphism between klt germs with $\pi^{-1}(x) = \{y\}$ and $K_Y = \pi^* K_X$, so $\widehat{\mathrm{vol}}(Y \ni y) = rV$ by Proposition 2.7(2).

(2) Assume that $rV > 9$. By (1), $Y \ni y$ is Gorenstein canonical with $\widehat{\mathrm{vol}}(Y \ni y) = rV > 9$, so $Y \ni y$ is a hypersurface germ by Theorem 2.17(2). In particular, $Y \ni y$ is a normal local complete intersection germ. Since π is quasi-étale and Galois with group G, G acts on Y freely in codimension one, and G fixes y as $\pi^{-1}(x) = \{y\}$. Since $X = Y/G$, $X \ni x$ is LCIQ by Definition 2.13.

(3) If $X \ni x$ is not LCIQ, then $rV \leq 9$ by (2). Thus $V \leq \frac{9}{r} \leq \frac{9}{2}$ when $r \geq 2$.

(4) Assume that $X \ni x$ is not LCIQ and $V > \frac{9}{2}$. By (3), $r = 1$, so π is the identity, and $X \ni x$ is Gorenstein canonical by (1). Any hypersurface germ is a local complete intersection germ, hence LCIQ (Definition 2.13), so $X \ni x$ is not a hypersurface germ. By Theorem 2.17(2), $V \leq 9$, and if $V = 9$, then $X \ni x$ is a cyclic quotient singularity of type $\frac{1}{3}(1, 1, 1)$, that is, the quotient of $\mathbb{A}^3$ by μ_3 acting by scalars, which acts freely outside the origin. In that case $X \ni x$ is LCIQ, a contradiction. Thus $V < 9$. □

5.2. **Deformations of LCIQ singularities.** In this subsection, we classify the LCIQ germs which are not local complete intersections and have local volume in $(\frac{125}{18}, 9)$, and we study their deformations.

Lemma 5.2. *Let $X \ni x$ be a Gorenstein klt threefold germ. Assume that $\widehat{\mathrm{vol}}(X \ni x) > \frac{125}{9}$. Then $X \ni x$ is smooth or an ordinary double point. If $X \ni x$ is an ordinary double point, then*

$$\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[z_1, z_2, z_3, z_4]]/(z_1^2 + z_2^2 + z_3^2 + z_4^2).$$

Proof. Since K_X is Cartier near x, $a(E, X)$ is a positive integer for any exceptional prime divisor E over X, so $X \ni x$ is canonical. Since $\widehat{\mathrm{vol}}(X \ni x) > \frac{125}{9} > 9$, $X \ni x$ is a hypersurface germ by Theorem 2.17(2). Assume that $X \ni x$ is singular. Then the embedding dimension of X at x is 4,

and by the Cohen structure theorem,

$$\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[z_1, z_2, z_3, z_4]]/(F)$$

for some F of order $m \geq 2$. We bound $\widehat{\mathrm{vol}}(X \ni x)$ by Proposition 2.20.

If $m \geq 3$, then the weights $w = (1,1,1,1)$ satisfy $v_w(F) = m$, so

$$0 < \widehat{\mathrm{vol}}(X \ni x) \leq (4-m)^3 m.$$

Thus $m = 3$ and $\widehat{\mathrm{vol}}(X \ni x) \leq 3$, a contradiction.

Thus $m = 2$. Let F_2 be the quadratic part of F. If F_2 is non-degenerate, then $X \ni x$ is an ordinary double point by the formal Morse lemma. Suppose that F_2 is degenerate. After a linear change of coordinates, $F_2 \in \mathbb{C}[z_1, z_2, z_3]$. Let $w = (3,3,3,2)$. Any monomial of degree at least three has weight at least 6, and any monomial of F_2 has weight 6, so $v_w(F) = 6$. Proposition 2.20 implies that

$$\widehat{\mathrm{vol}}(X \ni x) \leq (3+3+3+2-6)^3 \cdot \frac{6}{3 \cdot 3 \cdot 3 \cdot 2} = \frac{125}{9},$$

a contradiction. □

A degeneration of a klt germ to the minimizing cone preserves the local volume, but it is not clear that the germ is LCIQ when the cone is LCIQ, since a finite quotient presentation of the cone has to be deformed along the family. When the local volume is greater than $\frac{125}{18}$, only two quotients of the ordinary double point have to be deformed. Indeed, by (1) of the following proposition, such a germ with local volume less than 9 which is LCIQ but not a local complete intersection is the quotient of an ordinary double point by an involution. When the local volume is at least 9, no quotient presentation has to be deformed, since by Theorem 2.17(3) any klt threefold germ of local volume at least 9 is a hypersurface germ or a cyclic quotient singularity, hence LCIQ. The bound $\frac{125}{18}$ is half of the bound $\frac{125}{9}$ of Lemma 5.2, which is applied to the double cover. Part (3) of the following proposition is the form used in Proposition 5.8(3).

Proposition 5.3. *Let $Q := \{z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0\} \subset \mathbb{A}^4$, and for $k \in \{3,4\}$ let ι_k be the involution of $\mathbb{A}^4$ which changes the signs of $z_1, \ldots, z_k$ and fixes the other coordinates. Then the following hold.*

(1) Let $X \ni x$ be a klt threefold germ which is not a local complete intersection, and assume that $\frac{125}{18} < \widehat{\mathrm{vol}}(X \ni x) < 9$. If $X \ni x$ is LCIQ, then $\widehat{\mathrm{vol}}(X \ni x) = 8$, and $X \ni x$ is analytically isomorphic to the germ of $Q/\langle \iota_3 \rangle$ or of $Q/\langle \iota_4 \rangle$ at the image of the origin. Both quotient morphisms are étale outside the origin.

(2) Let $(T, 0)$ be a smooth pointed curve, $f : \mathcal{X} \to T$ a flat morphism of finite type, and σ a section of f, such that the germ $\mathcal{X}_0 \ni \sigma(0)$ is analytically isomorphic to the germ of $Q/\langle \iota_3 \rangle$ or of $Q/\langle \iota_4 \rangle$ at the image of the origin. Then, after shrinking T around 0 in the analytic topology, any germ $\mathcal{X}_t \ni \sigma(t)$ with $t \neq 0$ is analytically isomorphic to $\mathcal{X}_0 \ni \sigma(0)$, to a smooth germ, or to the germ of $\mathbb{A}^3/\{\pm 1\}$ at the origin, and the last case occurs only for ι_3. In particular, if $\widehat{\mathrm{vol}}(\mathcal{X}_t \ni \sigma(t)) = 8$, then $\mathcal{X}_t \ni \sigma(t)$ is analytically isomorphic to $\mathcal{X}_0 \ni \sigma(0)$, and $\mathcal{X}_t \ni \sigma(t)$ is LCIQ.

(3) Let $(T, 0)$, $f : \mathcal{X} \to T$, and σ be as in (2), but assume only that $\mathcal{X}_0 \ni \sigma(0)$ and the germs $\mathcal{X}_t \ni \sigma(t)$ for $t \neq 0$ close to 0 are klt threefold germs with $\widehat{\mathrm{vol}}(\mathcal{X}_t \ni \sigma(t)) = \widehat{\mathrm{vol}}(\mathcal{X}_0 \ni \sigma(0)) > \frac{125}{18}$. If $\mathcal{X}_0 \ni \sigma(0)$ is LCIQ, then $\mathcal{X}_t \ni \sigma(t)$ is LCIQ for any $t \neq 0$ close to 0.

Proof. The proof is long, so we divide it into several steps. **Steps 1 and 2** prove (1), **Steps 3–5** prove (2), and **Step 6** proves (3).

Step 1. In this step, we show that, under the assumptions of (1), $X \ni x$ is analytically isomorphic to the quotient of an ordinary double point by an involution, and $\widehat{\mathrm{vol}}(X \ni x) = 8$.

Let $V := \widehat{\mathrm{vol}}(X \ni x)$. By Definition 2.13, there exist a normal germ $Y \ni y$ which is a local complete intersection and a finite group G acting on Y with y fixed and freely in codimension

one, such that $X \ni x$ is analytically isomorphic to the germ of Y/G at $x' := \pi(y)$, where $\pi : Y \to Y/G$ is the quotient morphism. Since G acts freely in codimension one, G acts faithfully, π is quasi-étale and Galois of degree $\deg \pi = |G|$, $K_Y = \pi^* K_{Y/G}$, and $\pi^{-1}(x') = \{y\}$.

Since $Y \ni y$ is a local complete intersection, $Y \ni y$ is Gorenstein [Mat86, Theorem 21.3], and $K_{Y/G}$ is $\mathbb{Q}$-Cartier since K_Y is Cartier [KM98, Proposition 5.20(1)]. Since $X \ni x$ and $Y/G \ni x'$ are analytically isomorphic, they have a common étale neighborhood (Lemma 2.5(1)). Since the klt property is preserved and reflected by étale morphisms [Kol13, 2.14], $Y/G \ni x'$ is klt, and hence $Y \ni y$ is klt by [KM98, Proposition 5.20(4)]. By Proposition 2.7(1)(2),

$$\widehat{\mathrm{vol}}(Y \ni y) = \deg \pi \cdot \widehat{\mathrm{vol}}(Y/G \ni x') = \deg \pi \cdot V.$$

The Reynolds operator $\frac{1}{|G|} \sum_{g \in G} g^*$ exhibits $\mathcal{O}_{Y/G,x'}$ as a direct summand of the finite $\mathcal{O}_{Y/G,x'}$-module $(\pi_* \mathcal{O}_Y)_{x'}$, completion is exact on finite modules [Mat86, Theorems 8.7 and 8.8], and the completion of $(\pi_* \mathcal{O}_Y)_{x'}$ is $\widehat{\mathcal{O}}_{Y,y}$ since $\pi^{-1}(x') = \{y\}$. Thus

$$\widehat{\mathcal{O}}_{X,x} \cong \widehat{\mathcal{O}}_{Y/G,x'} = (\widehat{\mathcal{O}}_{Y,y})^G.$$

If $\deg \pi = 1$, then $\widehat{\mathcal{O}}_{X,x} \cong \widehat{\mathcal{O}}_{Y,y}$ is a complete intersection, so $X \ni x$ is a local complete intersection [BH98, Definition 2.3.1 and Theorem 2.3.3], a contradiction. Thus $\deg \pi \geq 2$.

By Theorem 2.17(1), $\deg \pi \cdot V \leq 27$. Since $4V > \frac{500}{18} > 27$, we have $\deg \pi \in \{2, 3\}$. If $\deg \pi = 3$, then $\widehat{\mathrm{vol}}(Y \ni y) = 3V > \frac{125}{6} > 16$, so $Y \ni y$ is smooth and $3V = 27$ by Theorem 2.17(1), which contradicts $V < 9$. Thus $\deg \pi = 2$ and $\widehat{\mathrm{vol}}(Y \ni y) = 2V > \frac{125}{9}$. By Lemma 5.2, $Y \ni y$ is smooth or an ordinary double point. If $Y \ni y$ is smooth, then $V = \frac{27}{2} > 9$, a contradiction. Therefore $Y \ni y$ is an ordinary double point, $V = \frac{16}{2} = 8$ by Theorem 2.17(1), and $G = \langle \iota \rangle$ for an involution ι of Y fixing y.

Step 2. In this step, we show that the pair $(\widehat{\mathcal{O}}_{Y,y}, \iota^*)$ is equivariantly isomorphic to

$$\mathbb{C}[[z_1, \dots, z_4]]/(z_1^2 + z_2^2 + z_3^2 + z_4^2)$$

with ι_3^* or ι_4^*, which proves (1).

Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{Y,y}$, and let k be the dimension of the (-1)-eigenspace of ι^* on $\mathfrak{m}/\mathfrak{m}^2$. Since $Y \ni y$ is an ordinary double point of a threefold, $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 = 4$. Choose an eigenbasis of $\mathfrak{m}/\mathfrak{m}^2$, lift it to $\mathfrak{m}$, and apply the Reynolds operators $\frac{1}{2}(1 \pm \iota^*)$ to the lifts. We obtain $z_1, \dots, z_4 \in \mathfrak{m}$ with $\iota^* z_i = -z_i$ for $i \leq k$ and $\iota^* z_i = z_i$ for $i > k$, which induce an equivariant surjection

$$\mathbb{C}[[z_1, \dots, z_4]] \to \widehat{\mathcal{O}}_{Y,y},$$

where ι acts on $\mathbb{C}[[z_1, \dots, z_4]]$ by ι_k^*, and ι_k changes the signs of $z_1, \dots, z_k$. Since $Y \ni y$ is an ordinary double point, the kernel I is principal, and the quadratic part of any generator of I is non-degenerate. The involution acts on the one-dimensional space $I/(z_1, \dots, z_4)I$ by a sign $\epsilon \in \{\pm 1\}$. For a generator φ' of I, the element $\varphi := \frac{1}{2}(\varphi' + \epsilon \iota_k^* \varphi')$ satisfies $\varphi \equiv \varphi' \mod (z_1, \dots, z_4)I$, so φ generates I by Nakayama's lemma, and $\iota_k^* \varphi = \epsilon \varphi$. Let φ_2 be the quadratic part of φ. Then $\iota_k^* \varphi_2 = \epsilon \varphi_2$.

Let $\mathrm{Fix}(\iota) \subset Y$ be the fixed locus of ι. Since G acts freely in codimension one, $\dim_y \mathrm{Fix}(\iota) \leq 1$, and the fixed locus of ι on $\operatorname{Spec} \widehat{\mathcal{O}}_{Y,y}$, which is the completion of $\mathrm{Fix}(\iota)$ at y, is of dimension at most one.

Suppose that $\epsilon = -1$. Then any monomial of φ_2 is of the form $z_i z_j$ with $i \leq k < j$, so the matrix of φ_2 is of the form $\begin{pmatrix} 0 & C \\ C^T & 0 \end{pmatrix}$ with C a $k \times (4-k)$ matrix, and non-degeneracy forces $k = 2$. Since $\iota_k^* \varphi = -\varphi$, any monomial of φ contains an odd number of the variables z_1, z_2, so $\varphi \in (z_1, z_2)$. Thus $\{z_1 = z_2 = 0\}$ is contained in $\operatorname{Spec} \widehat{\mathcal{O}}_{Y,y}$ and is fixed by ι, and $\dim\{z_1 = z_2 = 0\} = 2$, a contradiction. Thus $\epsilon = 1$.

Since $\epsilon = 1$, φ_2 contains no monomial $z_i z_j$ with $i \le k < j$, so φ_2 is the sum of non-degenerate quadratic forms in $z_1, \dots, z_k$ and in $z_{k+1}, \dots, z_4$. After a linear change of coordinates within each eigenspace, $\varphi_2 = g$, where

$$g := z_1^2 + z_2^2 + z_3^2 + z_4^2.$$

We remove the higher terms of φ equivariantly. Assume that $\varphi \equiv g \mod (z_1, \dots, z_4)^d$ for some $d \ge 3$, and let H be the homogeneous part of degree d of φ. Then $\iota_k^* H = H$, so $\iota_k^*(\partial H/\partial z_i) = -\partial H/\partial z_i$ for $i \le k$ and $\iota_k^*(\partial H/\partial z_i) = \partial H/\partial z_i$ for $i > k$. Consider the equivariant automorphism ψ_d of $\mathbb{C}[[z_1, \dots, z_4]]$ defined by

$$\psi_d(z_i) = z_i - \frac{1}{2d}\frac{\partial H}{\partial z_i} \quad (1 \le i \le 4).$$

Since $\sum_i z_i \partial H/\partial z_i = dH$ and $2d - 2 \ge d + 1$, we have $\psi_d(g) \equiv g - H$ and $\psi_d(\varphi) \equiv g \mod (z_1, \dots, z_4)^{d+1}$. Since ψ_d is the identity modulo $(z_1, \dots, z_4)^{d-1}$, the composition ψ of $\psi_3, \psi_4, \dots$ converges, and $\psi(\varphi) = g$. Therefore

$$\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[z_1, \dots, z_4]]/(g)$$

equivariantly, with ι acting by ι_k^*.

Since ι is non-trivial, $k \ne 0$. If $k = 1$, then the fixed locus of ι_1 on $\operatorname{Spec} \mathbb{C}[[z_1, \dots, z_4]]/(g)$ is $\{z_1 = 0,\ z_2^2 + z_3^2 + z_4^2 = 0\}$, which is of dimension two, a contradiction. Suppose that $k = 2$. Let $u := z_1 + \sqrt{-1} z_2$, $v := z_1 - \sqrt{-1} z_2$, $y_1 := z_3$, and $y_2 := z_4$, so that $g = uv + y_1^2 + y_2^2$ and ι_2^* changes the signs of u and v. Since the Reynolds operator is exact, the invariant ring of $\mathbb{C}[[u, v, y_1, y_2]]/(g)$ is generated by u^2, uv, v^2, y_1, and y_2, subject to the relations $u^2 \cdot v^2 = (uv)^2$ and $uv = -(y_1^2 + y_2^2)$. Writing $w_1 := u^2$, $w_2 := uv$, and $w_3 := v^2$, and eliminating $w_2 = -(y_1^2 + y_2^2)$, we obtain

$$\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[w_1, w_3, y_1, y_2]]/\big(w_1 w_3 - (y_1^2 + y_2^2)^2\big),$$

so $X \ni x$ is a local complete intersection [BH98, Definition 2.3.1 and Theorem 2.3.3], a contradiction. Thus $k \in \{3, 4\}$.

For $k \in \{3, 4\}$, the fixed locus of ι_k on Q is $\{z_1 = \cdots = z_k = 0\} \cap Q$, which is supported at the origin. Thus $Q \backslash \{0\} \to (Q \backslash \{0\})/\langle \iota_k \rangle$ is étale. By **Step 1** and the Reynolds argument there,

$$\widehat{\mathcal{O}}_{X,x} \cong \big(\mathbb{C}[[z_1, \dots, z_4]]/(g)\big)^{\iota_k^*} \cong \widehat{\mathcal{O}}_{Q/\langle \iota_k \rangle, o},$$

where o is the image of the origin. Together with $\widehat{\operatorname{vol}}(X \ni x) = 8$ from **Step 1**, this proves (1).

Step 3. In this step, we lift the punctured double cover of $\mathcal{X}_0 \ni \sigma(0)$ to a finite cover of the completed local ring of $\mathcal{X}$ at $\sigma(0)$.

Let $(T, 0)$, f, and σ be as in (2), and fix $k \in \{3, 4\}$ such that $\mathcal{X}_0 \ni \sigma(0)$ is analytically isomorphic to the germ of $Q/\langle \iota_k \rangle$ at the image of the origin. Possibly shrinking T around 0, there exists $\tau \in \mathcal{O}(T)$ with $\tau(0) = 0$ such that $\tau : T \to \mathbb{A}^1$ is étale. Let $B := \widehat{\mathcal{O}}_{\mathcal{X}, \sigma(0)}$, a complete local $\mathbb{C}[[\tau]]$-algebra. Since f is flat, τ is a nonzerodivisor on B, and $B/\tau B = \widehat{\mathcal{O}}_{\mathcal{X}_0, \sigma(0)}$. Let

$$A_0 := \mathbb{C}[[z_1, \dots, z_4]]/(g), \quad B_0 := A_0^{\iota_k^*}.$$

By **Step 2**, $B_0 \cong \widehat{\mathcal{O}}_{Q/\langle \iota_k \rangle, o}$, and we identify $B/\tau B$ with B_0 via a fixed isomorphism. For a Noetherian local ring R with maximal ideal $\mathfrak{m}_R$, write $(\operatorname{Spec} R)^\circ := \operatorname{Spec} R \backslash \{\mathfrak{m}_R\}$. For $n \ge 0$, let $\Lambda_n := \mathbb{C}[\tau]/(\tau^{n+1})$, and for $n \ge 1$, let $B_n := B/\tau^{n+1} B$. Thus $B_n/\tau B_n = B_0$ for any n.

The ring A_0 is a three-dimensional hypersurface, hence Cohen–Macaulay, and B_0 is Cohen–Macaulay [BH98, Theorem 6.4.5]. Since $\dim A_0 = \dim B_0 = 3$, local cohomology implies that

$$\Gamma((\operatorname{Spec} A_0)^\circ, \mathcal{O}) = A_0, \quad H^1((\operatorname{Spec} A_0)^\circ, \mathcal{O}) = 0, \quad \Gamma((\operatorname{Spec} B_0)^\circ, \mathcal{O}) = B_0$$

[Har67, Proposition 2.2 and Theorem 3.8], see also [Har77, Exercises III.2.3 and III.3.4]. Since the ideal $(z_1, \dots, z_k, g)$ of $\mathbb{C}[[z_1, \dots, z_4]]$ is primary to the maximal ideal, the inertia of ι_k is

trivial on $(\operatorname{Spec} A_0)^\circ$, so $(\operatorname{Spec} A_0)^\circ \to (\operatorname{Spec} B_0)^\circ$ is a finite étale μ_2-torsor [Gro71, Exposé V, Proposition 2.6]. Since B is flat over $\mathbb{C}[[\tau]]$, the sequences

$$0 \to B_{n-1} \xrightarrow{\tau} B_n \to B_0 \to 0$$

are exact, so by induction on n and the long exact sequence of local cohomology,

$$\Gamma((\operatorname{Spec} B_n)^\circ, \mathcal{O}) = B_n \quad \text{for any } n.$$

The scheme $(\operatorname{Spec} B_{n+1})^\circ$ is a nilpotent thickening of $(\operatorname{Spec} B_n)^\circ$. By the equivalence of étale schemes under nilpotent thickenings [Sta, Tag 039R], which preserves finite morphisms and μ_2-torsor structures (cf. [Gro71, Exposé I, 8.3–8.4]), there exist finite étale μ_2-torsors $W_n \to (\operatorname{Spec} B_n)^\circ$ with $W_0 = (\operatorname{Spec} A_0)^\circ$ and

$$W_{n+1} \times_{(\operatorname{Spec} B_{n+1})^\circ} (\operatorname{Spec} B_n)^\circ \cong W_n$$

for any n, compatible with the involutions. Let $A_n := \Gamma(W_n, \mathcal{O}_{W_n})$. Since W_n is étale over the Λ_n-flat scheme $(\operatorname{Spec} B_n)^\circ$, $\mathcal{O}_{W_n}$ is flat over Λ_n, so the sequence of sheaves

$$0 \to \mathcal{O}_{W_{n-1}} \xrightarrow{\tau} \mathcal{O}_{W_n} \to \mathcal{O}_{W_0} \to 0$$

is exact [Mat86, Theorem 22.3]. Since $H^1(W_0, \mathcal{O}_{W_0}) = 0$, induction on n shows that $H^1(W_n, \mathcal{O}_{W_n}) = 0$ and that

$$0 \to A_{n-1} \xrightarrow{\tau} A_n \to A_0 \to 0$$

is exact. By induction on n and the snake lemma, applied to the restriction maps between the exact sequences for n and $n-1$, the restriction map $A_n \to A_{n-1}$ is surjective, and hence $A_n/\tau A_n = A_0$. Since $W_n \to (\operatorname{Spec} B_n)^\circ$ is a μ_2-torsor, $A_n^\iota = \Gamma((\operatorname{Spec} B_n)^\circ, \mathcal{O}) = B_n$. Since τ is nilpotent on A_n, lifts to A_n of finitely many generators of the B_0-module A_0 generate A_n over B_n by Nakayama's lemma.

Let $A := \varprojlim_n A_n$. Using also the exact sequences

$$0 \to A_{n-m-1} \xrightarrow{\tau^{m+1}} A_n \to A_m \to 0 \quad (0 \le m < n),$$

which follow from the flatness of $\mathcal{O}_{W_n}$ over Λ_n and $H^1(W_{n-m-1}, \mathcal{O}_{W_{n-m-1}}) = 0$, a Mittag-Leffler argument [Har77, Proposition II.9.1] gives $A/\tau^{n+1}A = A_n$, $A/\tau A = A_0$, and τ is a nonzerodivisor on A. Since B is τ-adically complete, lifts to A of generators of the B_0-module A_0 generate A over B. Thus A is a finite B-algebra with an involution ι such that $A^\iota = \varprojlim_n B_n = B$, and A is local since any maximal ideal of A contains τ and $A/\tau A = A_0$ is local.

Step 4. In this step, we compute A and B, and we compare $\mathcal{X}$ with an explicit algebraic family near $\sigma(0)$.

Applying the Reynolds operators $\frac{1}{2}(1 \pm \iota^*)$ to lifts of $z_1, \dots, z_4 \in A_0$, we obtain lifts $z_1, \dots, z_4 \in A$ with $\iota^* z_i = -z_i$ for $i \le k$ and $\iota^* z_i = z_i$ for $i > k$. Since A is a complete local ring with residue field $\mathbb{C}$ whose maximal ideal is generated by $\tau, z_1, \dots, z_4$, we obtain an equivariant surjection $\mathbb{C}[[\tau, z_1, \dots, z_4]] \to A$, where ι acts on $\mathbb{C}[[\tau, z_1, \dots, z_4]]$ by ι_k^* and fixes τ. Let I be the kernel. Since τ is a nonzerodivisor on A, $I \cap \tau\mathbb{C}[[\tau, z_1, \dots, z_4]] = \tau I$, and since $A/\tau A = A_0$, the image of I in $\mathbb{C}[[z_1, \dots, z_4]]$ is (g). Choose $\Phi' \in I$ with $\Phi'(0, z) = g$, and let $\Phi := \frac{1}{2}(\Phi' + \iota_k^* \Phi')$. Then $\Phi \in I$ is invariant, $\Phi(0, z) = g$, and $I = (\Phi) + \tau I$, so $I = (\Phi)$ by Nakayama's lemma. Thus $A \cong \mathbb{C}[[\tau, z_1, \dots, z_4]]/(\Phi)$ equivariantly.

The Hessian of Φ in z at $\tau = z = 0$ is twice the identity matrix. By the formal implicit function theorem, there exists a unique $b(\tau) = (b_1(\tau), \dots, b_4(\tau)) \in (\tau\mathbb{C}[[\tau]])^4$ with $(\partial\Phi/\partial z_i)(\tau, b(\tau)) = 0$ for $1 \le i \le 4$. Since Φ is invariant, $(\partial\Phi/\partial z_i)(\tau, \iota_k(b(\tau))) = 0$ for $1 \le i \le 4$, so $b_i = 0$ for $i \le k$ by uniqueness. After replacing z_i with $z_i + b_i(\tau)$, which is equivariant, we may write

$$\Phi = a(\tau) + \sum_{i,j} c_{ij}(\tau) z_i z_j + (\text{terms of degree at least three in } z),$$

where $a(\tau) \in \tau\mathbb{C}[[\tau]]$, $(c_{ij}(\tau))$ is symmetric and congruent to the identity matrix modulo τ, and $c_{ij} = 0$ for $i \le k < j$. After a block-diagonal linear change of $z_1, \dots, z_4$ over $\mathbb{C}[[\tau]]$, we may

assume that $\sum_{i,j} c_{ij}(\tau) z_i z_j = g$. Running the argument of **Step 2** over the base ring $\mathbb{C}[[\tau]]$ in place of $\mathbb{C}$, we obtain an equivariant $\mathbb{C}[[\tau]]$-automorphism of $\mathbb{C}[[\tau, z_1, \dots, z_4]]$ which maps Φ to $g + a(\tau)$. Thus

$$A \cong \mathbb{C}[[\tau, z_1, \dots, z_4]]/(g + a(\tau))$$

equivariantly. If $a(\tau) \neq 0$, write $a(\tau) = \tau^\ell u(\tau)$ with $\ell \geq 1$ and $u(\tau) \in \mathbb{C}[[\tau]]$ a unit, and let $s(\tau) \in \mathbb{C}[[\tau]]$ with $s(\tau)^2 = u(\tau)$. Replacing z_i with $s(\tau) z_i$ for all i simultaneously replaces $g + a(\tau)$ with $u(\tau)(g + \tau^\ell)$, and the involution and τ are not changed.

Let $\mathcal{Z} := Q \times \mathbb{A}^1$ if $a(\tau) = 0$, and $\mathcal{Z} := \{g + \tau^\ell = 0\} \subset \mathbb{A}^4 \times \mathbb{A}^1$ otherwise, where τ is the coordinate of $\mathbb{A}^1$ and ι_k acts on the factor $\mathbb{A}^4$. Let $o' \in \mathcal{Z}/\langle \iota_k \rangle$ be the image of $(0, 0)$. By the Reynolds argument of **Step 1**,

$$B = A^\iota \cong \widehat{\mathcal{O}}_{\mathcal{Z}/\langle \iota_k \rangle, o'}$$

as $\mathbb{C}[[\tau]]$-algebras. Since $\mathcal{X}$ and $\mathcal{Z}/\langle \iota_k \rangle$ are of finite type over $\mathbb{A}^1$, [Art69, Corollary 2.6] implies that $\mathcal{X}$ at $\sigma(0)$ and $\mathcal{Z}/\langle \iota_k \rangle$ at o' have a common étale neighborhood over $\mathbb{A}^1$. Since étale morphisms induce local isomorphisms of the associated complex analytic spaces [Gro71, Exposé XII, Proposition 3.1] and $\tau : T \to \mathbb{A}^1$ is étale at 0, after shrinking T around 0 in the analytic topology, there exist an analytic neighborhood of $\sigma(T)$ in $\mathcal{X}$ and an analytic open subset of $\mathcal{Z}/\langle \iota_k \rangle$ containing o' which are isomorphic over $\mathbb{A}^1$. In particular, for any $t \in T$, the analytic germ of $\mathcal{X}_t$ at $\sigma(t)$ is isomorphic to the analytic germ of the fiber of $\mathcal{Z}/\langle \iota_k \rangle$ over $\tau(t)$ at some point close to o'.

Step 5. In this step, we determine the germs of the fibers of $\mathcal{Z}/\langle \iota_k \rangle$ near o', which proves (2).

For $c \in \mathbb{A}^1$, the fiber $(\mathcal{Z}/\langle \iota_k \rangle)_c$ is $\mathcal{Z}_c/\langle \iota_k \rangle$, since taking invariants commutes with base change along $\tau - c$ by the exactness of the Reynolds operator.

Suppose that $\mathcal{Z} = \{g + \tau^\ell = 0\}$ and $c \in \mathbb{A}^1 \backslash \{0\}$. The fiber $\mathcal{Z}_c = \{g + c^\ell = 0\} \subset \mathbb{A}^4$ is smooth, since the gradient $(2z_1, \dots, 2z_4)$ of g vanishes only at $z = 0$, which is not on $\mathcal{Z}_c$. If $k = 4$, then ι_4 has no fixed point on $\mathcal{Z}_c$, so $\mathcal{Z}_c/\langle \iota_4 \rangle$ is smooth. If $k = 3$, then the fixed points of ι_3 on $\mathcal{Z}_c$ are $q_\pm := (0, 0, 0, \pm\sqrt{-c^\ell})$. At q_+ and at q_-, $\partial(g + c^\ell)/\partial z_4 = 2z_4 \neq 0$, so z_1, z_2, z_3 are local analytic coordinates of $\mathcal{Z}_c$ on which ι_3 acts by -1, and the germ of $\mathcal{Z}_c/\langle \iota_3 \rangle$ at the image is analytically isomorphic to the germ of $\mathbb{A}^3/\{\pm 1\}$ at the origin, since completions of quotients at fixed points are invariants of completions (**Step 1**). Away from the images of $q_\pm$, $\mathcal{Z}_c/\langle \iota_3 \rangle$ is smooth. Suppose that $\mathcal{Z} = Q \times \mathbb{A}^1$. Then any fiber is $Q/\langle \iota_k \rangle$, whose germs are smooth except at the image of the origin, since Q is smooth outside the origin and ι_k acts freely on $Q \backslash \{0\}$ by **Step 2**.

By **Step 4** and Lemma 2.5(2), after shrinking T around 0 in the analytic topology, so that $\tau(t) \neq 0$ for $t \neq 0$ since τ is étale at 0, any germ $\mathcal{X}_t \ni \sigma(t)$ with $t \neq 0$ is analytically isomorphic to $\mathcal{X}_0 \ni \sigma(0)$, to a smooth germ, or to the germ of $\mathbb{A}^3/\{\pm 1\}$ at the origin, and the last case occurs only for $k = 3$.

The ordinary double point $Q \ni 0$ is canonical, so $Q/\langle \iota_k \rangle$ is klt [KM98, Proposition 5.20(4)]. By Theorem 2.17(1) and Proposition 2.7(2) applied to the quotient morphisms by ι_k and by $\{\pm 1\}$, which are Galois of degree two, quasi-étale, and totally ramified over the image of the origin,

$$\widehat{\mathrm{vol}}(Q/\langle \iota_k \rangle \ni o) = \frac{16}{2} = 8, \quad \widehat{\mathrm{vol}}(\mathbb{A}^3/\{\pm 1\} \ni 0) = \frac{27}{2},$$

and a smooth threefold germ has local volume 27. In each case, $\mathcal{X}_t$ is normal and klt near $\sigma(t)$, being analytically isomorphic to a klt germ. Since local volumes are invariant under analytic isomorphisms (Proposition 2.7(1)), if $\widehat{\mathrm{vol}}(\mathcal{X}_t \ni \sigma(t)) = 8$, then $\mathcal{X}_t \ni \sigma(t)$ is analytically isomorphic to $\mathcal{X}_0 \ni \sigma(0)$. Since Q is a normal hypersurface and ι_k acts on Q freely in codimension one, $\mathcal{X}_t \ni \sigma(t)$ is LCIQ by Definition 2.13. This proves (2).

Step 6. In this step, we prove (3).

Possibly shrinking T around 0, let $\tau \in \mathcal{O}(T)$ be as in **Step 3**, and let $V := \widehat{\mathrm{vol}}(\mathcal{X}_0 \ni \sigma(0))$. If $V \geq 9$, then by Theorem 2.17(3), any germ $\mathcal{X}_t \ni \sigma(t)$ with $t \neq 0$ close to 0 is a hypersurface germ

or a cyclic quotient singularity of type $\frac{1}{2}(1,1,1)$, $\frac{1}{3}(1,1,0)$, $\frac{1}{3}(1,1,1)$, or $\frac{1}{3}(1,1,2)$. In any case, $\mathcal{X}_t \ni \sigma(t)$ is LCIQ, since the corresponding cyclic groups act on $\mathbb{A}^3$ freely in codimension one.

Assume that $\frac{125}{18} < V < 9$, and that $\mathcal{X}_0 \ni \sigma(0)$ is a local complete intersection. Let $p := \sigma(0)$. Possibly shrinking $\mathcal{X}$ around p and T around 0, we may assume that T and $\mathcal{X}$ are affine, so that $\mathcal{X}$ is a closed subscheme of $M := T \times \mathbb{A}^N$ for some N. Let $J \subset \mathcal{O}_{M,p}$ be the ideal of $\mathcal{X}$. Since $\mathcal{X}_0$ is a local complete intersection at p in the smooth variety $\{0\} \times \mathbb{A}^N$ of dimension N, the image of J in $\mathcal{O}_{M,p}/(\tau)$ is generated by $N-3$ elements, and we lift them to $h_1, \dots, h_{N-3} \in J$. Since τ is a nonzerodivisor on $\mathcal{O}_{\mathcal{X},p}$, $J \cap \tau\mathcal{O}_{M,p} = \tau J$, so $J = (h_1, \dots, h_{N-3}) + \tau J$, and $J = (h_1, \dots, h_{N-3})$ by Nakayama's lemma. Since $\mathcal{O}_{\mathcal{X}_0,p}$ is a domain and τ is a nonzerodivisor on $\mathcal{O}_{\mathcal{X},p}$, $\mathcal{O}_{\mathcal{X},p}$ is a domain of dimension 4. Thus, possibly shrinking $\mathcal{X}$ around p, $h_1, \dots, h_{N-3}$ generate the ideal of $\mathcal{X}$ in M, $\mathcal{X}$ is irreducible of dimension 4, and the ideal of $\mathcal{X}$ has height $N-3$ at any point $p' \in \mathcal{X}$. Hence $h_1, \dots, h_{N-3}$ form a regular sequence in $\mathcal{O}_{M,p'}$ [Mat86, Theorem 17.4(iii)]. Since $\mathcal{X}_t$ is cut out in $\mathcal{X}$ near p' by the nonzerodivisor $\tau - \tau(t)$, where $t := f(p')$, taking $p' = \sigma(t)$ shows that $\mathcal{X}_t \ni \sigma(t)$ is a local complete intersection, hence LCIQ, for any t close to 0.

Assume that $\frac{125}{18} < V < 9$, and that $\mathcal{X}_0 \ni \sigma(0)$ is not a local complete intersection. By (1), $V = 8$, and $\mathcal{X}_0 \ni \sigma(0)$ is analytically isomorphic to the germ of $Q/\langle \iota_3 \rangle$ or of $Q/\langle \iota_4 \rangle$ at the image of the origin. Since $\widehat{\mathrm{vol}}(\mathcal{X}_t \ni \sigma(t)) = V = 8$ for any $t \neq 0$ close to 0, (2) implies that $\mathcal{X}_t \ni \sigma(t)$ is LCIQ. This proves (3). □

5.3. **Associated graded rings of minimizers.** In this subsection, we prove that the associated graded ring of a minimizer over a klt germ with Cartier canonical divisor is a Gorenstein domain whose canonical module is generated by one element homogeneous with respect to the value group (Proposition 5.5). The cone theorems of Section 7 and Section 10 use that the minimizing cone is Gorenstein and that its canonical module has a generator homogeneous with respect to the value group. Since the Gorenstein property of the special fiber of a flat family can fail, we use the structure of a Rees family, and we first prove a criterion for algebras of Rees type (Lemma 5.4). For the stable degeneration, the Gorenstein property of the minimizing cone also follows from [NO25, Proposition 4.1], see also [XZ26b, Lemma 2.8]. The new point of Proposition 5.5 is the generator of the canonical module homogeneous with respect to the value group.

Lemma 5.4. *Let R' be a normal Cohen–Macaulay domain essentially of finite type over $\mathbb{C}$ such that $\omega_{R'} \cong R'$, and let $\mathcal{R} \subset R'[t,t^{-1}]$ be a finitely generated $R'[t]$-subalgebra such that $\mathcal{R}[t^{-1}] = R'[t,t^{-1}]$. Assume that $\mathcal{R}/t\mathcal{R}$ is a non-zero normal Cohen–Macaulay domain. Then $\mathcal{R}$ is normal and Gorenstein, $\mathcal{R}/t\mathcal{R}$ is Gorenstein, $\omega_{\mathcal{R}} \cong \mathcal{R}$, and $\omega_{\mathcal{R}/t\mathcal{R}} \cong \mathcal{R}/t\mathcal{R}$.*

Proof. Since $\mathcal{R}$ is a subring of the domain $R'[t,t^{-1}]$, $\mathcal{R}$ is a domain and t is a nonzerodivisor on $\mathcal{R}$. Since $\mathcal{R}$ is finitely generated over $R'[t]$, $\mathcal{R}$ is Noetherian and essentially of finite type over $\mathbb{C}$.

We show that $\mathcal{R}$ is Cohen–Macaulay. Let $P \subset \mathcal{R}$ be a prime ideal. If $t \notin P$, then $\mathcal{R}_P$ is a localization of $\mathcal{R}[t^{-1}] = R'[t,t^{-1}]$. The homomorphism $R' \to R'[t,t^{-1}]$ is flat, and the fibers $k(\mathfrak{p})[t,t^{-1}]$, for prime ideals $\mathfrak{p} \subset R'$, are normal, so $R'[t,t^{-1}]$ is normal [BH98, Corollary 2.2.23(b)]. Since $R'[t]$ is Cohen–Macaulay [Mat86, Theorem 17.7] and $R'[t,t^{-1}]$ is a localization of $R'[t]$, $R'[t,t^{-1}]$ is Cohen–Macaulay. Thus $\mathcal{R}_P$ is normal and Cohen–Macaulay. If $t \in P$, then $\mathcal{R}_P/t\mathcal{R}_P$ is a localization of $\mathcal{R}/t\mathcal{R}$, hence Cohen–Macaulay. Moreover, t is a non-zero element of the maximal ideal of the Noetherian local domain $\mathcal{R}_P$, so t is $\mathcal{R}_P$-regular. Thus t is part of a system of parameters of $\mathcal{R}_P$ [BH98, Proposition 1.2.12], and

$$\dim \mathcal{R}_P/t\mathcal{R}_P = \dim \mathcal{R}_P - 1$$

by [BH98, Proposition A.4]. By [BH98, Proposition 1.2.10(d)],

$$\operatorname{depth} \mathcal{R}_P = \operatorname{depth} \mathcal{R}_P/t\mathcal{R}_P + 1 = \dim \mathcal{R}_P/t\mathcal{R}_P + 1 = \dim \mathcal{R}_P.$$

Thus $\mathcal{R}$ is Cohen–Macaulay. In particular, $\mathcal{R}$ satisfies Serre's condition (S_2).

We show that $\mathcal{R}$ satisfies Serre's condition (R_1). The height-one prime ideals not containing t were treated above. Since $\mathcal{R}/t\mathcal{R}$ is a domain, $t\mathcal{R}$ is a prime ideal, and $t\mathcal{R}$ has height one by

Krull's principal ideal theorem [Har77, Theorem I.1.11A]. Thus $t\mathcal{R}$ is the only height-one prime ideal of $\mathcal{R}$ containing t. The maximal ideal of the one-dimensional local ring $\mathcal{R}_{(t)}$ is generated by t, so $\mathcal{R}_{(t)}$ is a discrete valuation ring with uniformizer t. By Serre's criterion for normality [Mat86, Theorem 23.8], $\mathcal{R}$ is normal. Moreover, $V(t) \subset \operatorname{Spec} \mathcal{R}$ is a prime divisor and, as Weil divisors on $\operatorname{Spec} \mathcal{R}$,

$$\operatorname{div}(t) = V(t).$$

We compute $\omega_{\mathcal{R}}$. Recall that for a normal Cohen–Macaulay domain A essentially of finite type over $\mathbb{C}$, the canonical module ω_A is the rank one reflexive module of top differential forms on $\operatorname{Spec} A$ that are regular in codimension one, by [KM98, Proposition 5.75] applied to a normal projective compactification of an affine variety of which $\operatorname{Spec} A$ is a localization and localized, since the stalk at a Cohen–Macaulay point of the dualizing sheaf of a projective variety is the canonical module of the local ring [Har77, Proposition III.7.5], [BH98, Theorems 3.3.7(b) and 3.3.5(b)]. Let Ω be a generator of $\omega_{R'}$. Since $\operatorname{Spec} R'[t,t^{-1}] = \operatorname{Spec} R' \times \operatorname{Spec} \mathbb{C}[t,t^{-1}]$, the form $\Omega \wedge dt$ generates $\omega_{R'[t,t^{-1}]}$: it suffices to check that $\Omega \wedge dt$ generates $\omega_{R'[t,t^{-1}]}$ on the smooth locus of $\operatorname{Spec} R'[t,t^{-1}]$, whose complement has codimension at least two. Since $\mathcal{R}[t^{-1}] = R'[t,t^{-1}]$, the divisor of $\Omega \wedge dt$ on $\operatorname{Spec} \mathcal{R}$ is supported on $V(t)$, that is,

$$\operatorname{div}_{\mathcal{R}}(\Omega \wedge dt) = cV(t)$$

for some integer c. Then $\operatorname{div}_{\mathcal{R}}(t^{-c}\Omega \wedge dt) = 0$. For h in the fraction field of $\mathcal{R}$, we have $ht^{-c}\Omega \wedge dt \in \omega_{\mathcal{R}}$ if and only if $\operatorname{div}(h) \geq 0$, that is, if and only if $h \in \mathcal{R}$, because $\mathcal{R}$ is normal. Thus

$$\omega_{\mathcal{R}} = \mathcal{R} \cdot t^{-c}\Omega \wedge dt \cong \mathcal{R}.$$

Since $\omega_{\mathcal{R}_{\mathfrak{p}}} \cong (\omega_{\mathcal{R}})_{\mathfrak{p}} \cong \mathcal{R}_{\mathfrak{p}}$ for any prime ideal $\mathfrak{p}$ of $\mathcal{R}$, any localization of $\mathcal{R}$ at a prime ideal is Gorenstein by [BH98, Theorem 3.3.7(a)], so $\mathcal{R}$ is Gorenstein. Finally, the principal prime divisor $V(t) = \operatorname{div}(t) \geq 0$ is a Cartier divisor on the Cohen–Macaulay scheme $\operatorname{Spec} \mathcal{R}$, and $\mathcal{O}(V(t)) = t^{-1}\mathcal{R} \cong \mathcal{R}$, so by adjunction [Kol13, (4.1.3)],

$$\omega_{\mathcal{R}/t\mathcal{R}} \cong (\omega_{\mathcal{R}} \otimes_{\mathcal{R}} t^{-1}\mathcal{R})/t(\omega_{\mathcal{R}} \otimes_{\mathcal{R}} t^{-1}\mathcal{R}) \cong \mathcal{R}/t\mathcal{R}.$$

Localizing at the prime ideals of $\mathcal{R}/t\mathcal{R}$, any localization of $\mathcal{R}/t\mathcal{R}$ at a prime ideal is Gorenstein by [BH98, Theorem 3.3.7(a)], so $\mathcal{R}/t\mathcal{R}$ is Gorenstein. □

Proposition 5.5. *Let $X \ni x$ be a klt germ of dimension $n \geq 2$ such that K_X is Cartier, let v be a minimizer of $\widehat{\operatorname{vol}}$ on $\operatorname{Val}_{X,x}$, and let $X_0 := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$. Let $M \subset \mathbb{R}$ be the subgroup generated by the degrees of the non-zero homogeneous elements of $\operatorname{gr}_v \mathcal{O}_{X,x}$. Then the degree zero part of $\operatorname{gr}_v \mathcal{O}_{X,x}$ is $\mathbb{C}$, M is a lattice, $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a finitely generated normal Gorenstein domain, X_0 has canonical singularities, and the canonical module of $\operatorname{gr}_v \mathcal{O}_{X,x}$ is generated by one M-homogeneous element.*

Proof. We write ω_{X_0} for the canonical module of $\operatorname{gr}_v \mathcal{O}_{X,x}$. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a finitely generated normal Cohen–Macaulay domain with degree zero part $\mathbb{C}$, and we reduce to the case when X is affine with trivial canonical module.

By Theorem 2.12(2), $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a finitely generated $\mathbb{C}$-algebra and X_0 is a normal klt variety. Thus $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a normal domain, and $\operatorname{gr}_v \mathcal{O}_{X,x}$ is Cohen–Macaulay by [KM98, Theorems 5.22 and 5.10]. Since v is centered at x, $\mathfrak{a}_{>0}(v) = \mathfrak{m}_x$, so the degree zero part of $\operatorname{gr}_v \mathcal{O}_{X,x}$ is $\mathcal{O}_{X,x}/\mathfrak{m}_x = \mathbb{C}$.

Since K_X is Cartier, possibly shrinking X around x, we may assume that $X = \operatorname{Spec} R'$ is affine and $K_X \sim 0$. Then R' is a normal Cohen–Macaulay domain of finite type over $\mathbb{C}$ [KM98, Theorems 5.22 and 5.10], and $\omega_{R'} \cong \mathcal{O}_X(K_X) \cong R'$ by [KM98, Proposition 5.75] applied to a normal projective compactification of X and restricted to X, since the stalk at a Cohen–Macaulay

point of the dualizing sheaf of a projective variety is the canonical module of the local ring [Har77, Proposition III.7.5], [BH98, Theorems 3.3.7(b) and 3.3.5(b)].

Step 2. In this step, we show that $\operatorname{gr}_v \mathcal{O}_{X,x}$ is Gorenstein and that $\omega_{X_0} \cong \operatorname{gr}_v \mathcal{O}_{X,x}$ as ungraded modules.

As v is centered at x, $\operatorname{gr}_v R' = \operatorname{gr}_v \mathcal{O}_{X,x}$, which is finitely generated with X_0 klt by **Step 1**. By Theorem 2.12(1), v is quasi-monomial. Thus by [XZ25, Theorem 4.1, (1)⇒(3)], there exists a Kollár model $\pi : (Y, E) \to X$ [XZ25, Definition 3.7] with $v \in \mathrm{QM}(Y, E)$ [XZ25, Theorem 4.1(3)]. In particular, π is a proper birational morphism which is an isomorphism away from x, $E = \operatorname{Exc}(\pi)$ is a reduced divisor on Y, and v is the monomial valuation with weights $(\gamma_1, \dots, \gamma_s) \in \mathbb{R}^s_{>0}$ with respect to some components $E_1, \dots, E_s$ of E at the generic point of $V := E_1 \cap \cdots \cap E_s$, which is irreducible by [XZ25, Proposition 3.10]. Near the generic point of V, the pair $(Y, E_1 + \cdots + E_s)$ is the quotient of a simple normal crossing pair by a finite abelian group, and monomial valuations at the generic point of V are defined through the simple normal crossing pair, with $v(E_i) = \gamma_i$ for any i [XZ25, Definition 2.8 and the paragraphs following it].

Pick $(\gamma'_1, \dots, \gamma'_s) \in \mathbb{Q}^s_{>0}$, and let w be the monomial valuation with weights $(\gamma'_1, \dots, \gamma'_s)$ with respect to $E_1, \dots, E_s$ at the generic point of V. Let r be the number of components of E, and let Φ be the face of the orthant $\mathbb{R}^r_{\geq 0}$ of non-negative weights consisting of the vectors whose entries at the components of E other than $E_1, \dots, E_s$ are zero. Since $\gamma_i > 0$ and $\gamma'_i > 0$ for any i, the weight vectors of v and w, extended by zeros to all components of E, are in the relative interior of Φ. As $\pi(V) = \{x\}$, w is centered at x, so $\operatorname{gr}_w R' = \operatorname{gr}_w \mathcal{O}_{X,x}$. Thus [XZ25, Corollary 4.10] implies that

$$\operatorname{gr}_w \mathcal{O}_{X,x} \cong \operatorname{gr}_v \mathcal{O}_{X,x}$$

as $\mathbb{C}$-algebras. Since the weights of w are rational, a positive rational multiple of w is ord_F for a prime divisor F over X [XZ25, Definition 2.8 and the paragraphs following it]. Since rescaling a valuation only rescales the grading of the associated graded ring,

$$\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{X,x} \cong \operatorname{gr}_w \mathcal{O}_{X,x} \cong \operatorname{gr}_v \mathcal{O}_{X,x}$$

as $\mathbb{C}$-algebras. The centers of v, w, and ord_F on Y are all equal to V. Since $\operatorname{center}_X v = \{x\}$, we obtain $\operatorname{center}_X F = \{x\}$.

For $j \in \mathbb{Z}$, let $\mathfrak{a}_j := \{g \in R' \mid \operatorname{ord}_F(g) \geq j\}$, so that $\mathfrak{a}_j = R'$ for $j \leq 0$, and let

$$\mathcal{R} := \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j t^{-j} \subset R'[t, t^{-1}].$$

Since $\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{X,x}$ is finitely generated, $\mathcal{R}$ is a finitely generated $R'[t]$-algebra [XZ25, Lemma 2.14 and (2.1)]. Since $\mathfrak{a}_j = R'$ for $j \leq 0$, we have $\mathcal{R}[t^{-1}] = R'[t, t^{-1}]$ and

$$\mathcal{R}/t\mathcal{R} \cong \bigoplus_{j \geq 0} \mathfrak{a}_j / \mathfrak{a}_{j+1}.$$

Since $\operatorname{center}_X F = \{x\}$, $\mathfrak{a}_1$ is the maximal ideal of R' at x and $\mathfrak{a}_j$ is primary to $\mathfrak{a}_1$ for any $j \geq 1$, so $\mathfrak{a}_j/\mathfrak{a}_{j+1} \cong \mathfrak{a}_j \mathcal{O}_{X,x}/\mathfrak{a}_{j+1}\mathcal{O}_{X,x}$ for any $j \geq 0$. Thus

$$\mathcal{R}/t\mathcal{R} \cong \operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{X,x} \cong \operatorname{gr}_v \mathcal{O}_{X,x},$$

which is a non-zero normal Cohen–Macaulay domain by **Step 1**. By Lemma 5.4 and **Step 1**, $\mathcal{R}/t\mathcal{R}$ is Gorenstein and $\omega_{\mathcal{R}/t\mathcal{R}} \cong \mathcal{R}/t\mathcal{R}$. Hence $\operatorname{gr}_v \mathcal{O}_{X,x}$ is Gorenstein and $\omega_{X_0} \cong \operatorname{gr}_v \mathcal{O}_{X,x}$ as ungraded modules.

Step 3. In this step, we show that M is a lattice and that ω_{X_0} is generated by one M-homogeneous element.

By **Step 1**, there exist non-zero homogeneous elements $b_1, \dots, b_N$ of positive degrees which generate $\operatorname{gr}_v \mathcal{O}_{X,x}$ as a $\mathbb{C}$-algebra. The degrees $\deg b_1, \dots, \deg b_N$ generate M, so M is a finitely generated torsion-free abelian group, that is, a lattice. Choose a group homomorphism $\ell : M \to \mathbb{Z}$ such that $\ell(\deg b_i) > 0$ for any i. Such an ℓ exists because, after choosing a basis of M, the

inclusion $M \subset \mathbb{R}$ is a real linear function satisfying finitely many strict linear inequalities, which a rational linear function close to it also satisfies, and ℓ is that rational linear function with denominators cleared. Let $\mathfrak{m}_{x_0} := \bigoplus_{\lambda>0} \mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v)$ be the ideal of the vertex x_0. Any non-zero M-homogeneous element of $\mathfrak{m}_{x_0}$ of degree $\lambda > 0$ is a linear combination of non-constant monomials in $b_1, \dots, b_N$ of degree λ, so $\ell(\lambda) > 0$, that is, the element has positive ℓ-degree.

The action of the torus $\operatorname{Spec} \mathbb{C}[M]$ on X_0 (Theorem 2.12(2)) induces an M-grading of ω_{X_0}, compatible with the M-grading of $\operatorname{gr}_v \mathcal{O}_{X,x}$. Through ℓ, $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a positively graded ring with degree zero part $\mathbb{C}$. By **Step 2**, ω_{X_0} is a finitely generated module, so ω_{X_0} is generated by finitely many M-homogeneous elements, and the ℓ-degrees of the non-zero M-homogeneous elements of ω_{X_0} are bounded from below.

By **Step 2**, $\omega_{X_0}/\mathfrak{m}_{x_0}\omega_{X_0} \cong \operatorname{gr}_v \mathcal{O}_{X,x}/\mathfrak{m}_{x_0} = \mathbb{C}$. Since $\mathfrak{m}_{x_0}$ is M-homogeneous, the one-dimensional vector space $\omega_{X_0}/\mathfrak{m}_{x_0}\omega_{X_0}$ is concentrated in one degree $u \in M$. Pick an M-homogeneous element $\eta \in \omega_{X_0}$ of degree u whose image in $\omega_{X_0}/\mathfrak{m}_{x_0}\omega_{X_0}$ is non-zero, and let $Q := \omega_{X_0}/(\operatorname{gr}_v \mathcal{O}_{X,x})\eta$. Then Q is an M-graded module and $Q = \mathfrak{m}_{x_0}Q$. Suppose that $Q \neq 0$. Let k be the least integer such that Q has a non-zero M-homogeneous element of ℓ-degree k. Since any non-zero M-homogeneous element of $\mathfrak{m}_{x_0}$ has positive ℓ-degree, $Q = \mathfrak{m}_{x_0}Q$ implies that any M-homogeneous element of Q of ℓ-degree k is a sum of products of elements of $\mathfrak{m}_{x_0}$ with elements of Q of ℓ-degree less than k, hence is zero, a contradiction. Thus $Q = 0$, that is, $\omega_{X_0} = (\operatorname{gr}_v \mathcal{O}_{X,x})\eta$. Since ω_{X_0} is torsion-free of rank one and $\eta \neq 0$, the map $\operatorname{gr}_v \mathcal{O}_{X,x} \to \omega_{X_0}$, $g \to g\eta$ is an isomorphism. Thus ω_{X_0} is generated by the M-homogeneous element η.

Step 4. In this step, we conclude the proof.

By **Step 3**, K_{X_0} is Cartier. Thus the discrepancies of the exceptional divisors over X_0 are integers, and since X_0 is klt (**Step 1**), they are greater than -1, hence non-negative [KM98, Definition 2.34]. Thus X_0 has canonical singularities. Together with **Steps 1–3**, this proves the proposition. □

5.4. **An inequality for minimizers.** In this subsection, we prove that a minimizer of $\widehat{\mathrm{vol}}$ bounds log canonical thresholds of ideals from below, and we apply the bound to homogeneous elements of graded rings whose grading valuation is a minimizer.

In Section 7 and Section 10, we need to bound the order of vanishing of homogeneous elements along prime divisors E over the cone whose centers contain the vertex but are not equal to it: the Cox monomials in Section 7, and the divisors used in Section 10. The inequality of [XZ21, Theorem 3.10] concerns valuations centered at x. The second sentence of Proposition 5.6(1) extends the bound to such divisors, and Proposition 5.6(3) bounds the order of vanishing of a homogeneous element in terms of the degree, the log discrepancy of the grading valuation, and the dimension n, which we later apply with $n = 3$.

Proposition 5.6. *Let $(X \ni x, B)$ be a klt pair of dimension $n \geq 2$ such that B is a $\mathbb{Q}$-divisor, let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, and let $a_v := a(v, X, B)$. Let $\mathfrak{b} \subset \mathcal{O}_X$ be a non-zero ideal such that $\mathfrak{b}\mathcal{O}_{X,x} \subset \mathfrak{m}_x$. For a valuation w on X that is non-negative on $\mathcal{O}_{X,x}$, let $w(\mathfrak{b}) := \min_{g\in\mathfrak{b}} w(g)$. For a prime divisor E over X, we write ord_E for the corresponding divisorial valuation.*

(1) For any $w \in \mathrm{Val}_{X,x}$ with $a(w, X, B) < +\infty$, we have $nv(\mathfrak{b})a(w, X, B) \geq a_v w(\mathfrak{b})$. Moreover, the inequality holds for $w = \mathrm{ord}_E$, where E is any prime divisor over X with $x \in \mathrm{center}_X E$. In particular, $\mathrm{lct}(X \ni x, B; \mathfrak{b}) \geq \frac{a_v}{nv(\mathfrak{b})}$.

(2) For any $0 \neq g \in \mathfrak{m}_x$ and any prime divisor D on X through x, we have

$$nv(g)(1 - \mathrm{mult}_D B) \geq a_v \,\mathrm{ord}_D(g).$$

(3) Assume moreover that $B = 0$, that $X = \operatorname{Spec} R$ for a normal positively graded $\mathbb{C}$-domain $R = \bigoplus_{j\geq 0} R_j$ with $R_0 = \mathbb{C}$ and $\gcd\{j \mid R_j \neq 0\} = 1$, that x is the vertex, and that v is the grading valuation, that is, $v(g) = \min\{j \mid g_j \neq 0\}$ for $0 \neq g = \sum_j g_j \in R$ with $g_j \in R_j$. Let $\alpha := a(v, X)$. Then any non-zero homogeneous element $g \in R$ of degree $d > 0$ satisfies

$d \geq \frac{\alpha}{n} \operatorname{ord}_D(g)$ for any prime divisor D on X through x, and $\operatorname{lct}(X \ni x; \operatorname{div}(g)) \geq \frac{\alpha}{nd}$. In particular, $R_d = 0$ for $0 < d < \frac{\alpha}{n}$, and $\operatorname{div}(g)$ is reduced near x if $\frac{\alpha}{n} \leq d < \frac{2\alpha}{n}$.

Proof. For $s \leq 0$, let $\mathfrak{a}_s(v) := \mathcal{O}_{X,x}$. For $w \in \operatorname{Val}_{X,x}$ with $a(w, X, B) < +\infty$ and for $t > 0$, recall from [XZ21, Lemma-Definition 3.1 and p. 158] that

$$\operatorname{vol}(v; w/t) := \lim_{m \to +\infty} \frac{\dim_{\mathbb{C}} \big(\mathfrak{a}_{tm}(w) + \mathfrak{a}_m(v)\big)/\mathfrak{a}_m(v)}{m^n/n!}$$

and

$$S(v; w) := \frac{n+1}{n} \cdot \frac{a_v}{\operatorname{vol}(v)} \int_0^{+\infty} \operatorname{vol}(v; w/t) dt.$$

Since v is a minimizer, $a(w, X, B) \geq S(v; w)$ for any such w [XZ21, Definition 3.4 and Theorem 3.10]. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that for any $0 \neq g \in \mathfrak{m}_x$ and any $w \in \operatorname{Val}_{X,x}$ with $a(w, X, B) < +\infty$,

$$nv(g)a(w, X, B) \geq a_v w(g). \tag{5.1}$$

Let $c := v(g)$ and $e := w(g)$. Since v and w are centered at x, we have $c > 0$ and $e > 0$. Fix a real number $0 < t < \frac{e}{c}$. For any positive integer m, let $k_m := \lceil \frac{tm}{e} \rceil$ and $s_m := m - k_m c$. Then $\lim_{m \to +\infty} \frac{s_m}{m} = 1 - \frac{tc}{e} > 0$. Since $v(g^{k_m} h) = k_m c + v(h)$ for any $h \in \mathcal{O}_{X,x}$, multiplication by g^{k_m} induces an injective map

$$\mathcal{O}_{X,x}/\mathfrak{a}_{s_m}(v) \to \mathcal{O}_{X,x}/\mathfrak{a}_m(v),$$

and since $w(g^{k_m} h) \geq k_m e \geq tm$ for any $h \in \mathcal{O}_{X,x}$, the image of the map is contained in $\big(\mathfrak{a}_{tm}(w) + \mathfrak{a}_m(v)\big)/\mathfrak{a}_m(v)$. Since $\mathfrak{a}_{\lceil s \rceil}(v) \subset \mathfrak{a}_s(v) \subset \mathfrak{a}_{\lfloor s \rfloor}(v)$ for any real number s, the limit defining $\operatorname{vol}(v)$ (Definition 2.6) also holds when m runs over real numbers tending to $+\infty$. Thus

$$\operatorname{vol}(v; w/t) \geq \lim_{m \to +\infty} \frac{\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{a}_{s_m}(v)}{m^n/n!} = \operatorname{vol}(v)\Big(1 - \frac{tc}{e}\Big)^n.$$

Since $\operatorname{vol}(v; w/t) \geq 0$ for any $t > 0$, we have

$$S(v; w) \geq \frac{n+1}{n} \cdot \frac{a_v}{\operatorname{vol}(v)} \int_0^{e/c} \operatorname{vol}(v)\Big(1 - \frac{tc}{e}\Big)^n dt = \frac{a_v e}{nc}.$$

Since $a(w, X, B) \geq S(v; w)$, (5.1) holds.

Step 2. In this step, we show that for any $0 \neq g \in \mathfrak{m}_x$ and any prime divisor E over X such that $x \in \operatorname{center}_X E$,

$$nv(g)a(E, X, B) \geq a_v \operatorname{ord}_E(g). \tag{5.2}$$

If $\operatorname{center}_X E = \{x\}$, then $\operatorname{ord}_E \in \operatorname{Val}_{X,x}$ and $a(\operatorname{ord}_E, X, B) = a(E, X, B) < +\infty$, so (5.2) follows from (5.1). Thus we may assume that $\operatorname{center}_X E \neq \{x\}$. Possibly shrinking X around x, we may assume that g is a regular function on X. Let $\pi : Y \to X$ be a projective log resolution of $(X, B + \operatorname{div}(g))$ such that E is a divisor on Y, $\mathfrak{m}_x \mathcal{O}_Y$ is an invertible ideal sheaf whose zero locus $\pi^{-1}(x)$ is a divisor, and

$$\Sigma := \operatorname{Exc}(\pi) \cup E \cup \operatorname{Supp} \pi_*^{-1} B \cup \operatorname{Supp} \operatorname{div}(\pi^* g) \cup \pi^{-1}(x)$$

is a simple normal crossing divisor (such π exists: by [KM98, Lemma 2.45], there exists a projective birational morphism $X' \to X$, a composite of finitely many blow-ups, such that the center of ord_E on X' is a prime divisor, and [KM98, Theorem 0.2] for X' provides π). Since $\pi(E) = \operatorname{center}_X E$ contains x, $E \cap \pi^{-1}(x) \neq \emptyset$. Let F be an irreducible component of $\pi^{-1}(x)$ such that $E \cap F \neq \emptyset$. Since $\operatorname{center}_X E \neq \{x\}$, $F \neq E$. Let V be an irreducible component of $E \cap F$. Since Σ is simple normal crossing, $\operatorname{codim}_Y V = 2$, the generic point η_V of V is not contained in any other irreducible component of Σ, and local equations z_E and z_F of E and F at η_V form a regular system of parameters of $\mathcal{O}_{Y,\eta_V}$.

For a real number $\epsilon > 0$, let w_ϵ be the monomial valuation in z_E, z_F with weights 1 and ϵ. The center of w_ϵ on Y is V, so $\operatorname{center}_X(w_\epsilon) = \pi(V) = \{x\}$ and $w_\epsilon \in \operatorname{Val}_{X,x}$. Since $(Y, \operatorname{Supp}(E+F+\operatorname{Exc}(\pi)+\pi_*^{-1}B))$ is simple normal crossing, $(Y, E+F)$ is a log smooth model of (X, B) in the sense of [XZ25, Section 2.2.2], and η_V is the generic point of a connected component of $E \cap F$. Thus, by [XZ25, Definition 2.12(2)],

$$a(w_\epsilon, X, B) = a(E, X, B) + \epsilon a(F, X, B) < +\infty.$$

Near η_V, we have $\operatorname{div}(\pi^* g) = \operatorname{ord}_E(g)E + \operatorname{ord}_F(g)F$, so

$$w_\epsilon(g) = \operatorname{ord}_E(g) + \epsilon \operatorname{ord}_F(g).$$

By (5.1) applied to w_ϵ,

$$nv(g)\big(a(E, X, B) + \epsilon a(F, X, B)\big) \geq a_v\big(\operatorname{ord}_E(g) + \epsilon \operatorname{ord}_F(g)\big) \geq a_v \operatorname{ord}_E(g).$$

Letting $\epsilon \to 0$, we obtain (5.2).

Step 3. In this step, we prove (1).

Since $\mathfrak{b} \neq 0$, there exists $g \in \mathfrak{b}$ such that $v(g) = v(\mathfrak{b}) < +\infty$. Then $g \neq 0$ and $g \in \mathfrak{b}\mathcal{O}_{X,x} \subset \mathfrak{m}_x$. For any valuation w on X that is non-negative on $\mathcal{O}_{X,x}$, we have $w(g) \geq w(\mathfrak{b})$. If E is a prime divisor over X with $x \in \operatorname{center}_X E$, then ord_E is non-negative on $\mathcal{O}_{X,x}$. By (5.1) and (5.2), for any $w \in \operatorname{Val}_{X,x}$ with $a(w, X, B) < +\infty$ and for $w = \operatorname{ord}_E$ with $x \in \operatorname{center}_X E$,

$$nv(\mathfrak{b})a(w, X, B) = nv(g)a(w, X, B) \geq a_v w(g) \geq a_v w(\mathfrak{b}).$$

This proves the first two sentences of (1). Let E be a prime divisor over X such that $x \in \operatorname{center}_X E$ and $\operatorname{mult}_E \mathfrak{b} > 0$. Since $\operatorname{ord}_E(\mathfrak{b}) = \operatorname{mult}_E \mathfrak{b}$, we have

$$\frac{a(E, X, B)}{\operatorname{mult}_E \mathfrak{b}} \geq \frac{a_v}{nv(\mathfrak{b})}.$$

Taking the infimum over all such E, Definition 2.3 implies that $\operatorname{lct}(X \ni x, B; \mathfrak{b}) \geq \frac{a_v}{nv(\mathfrak{b})}$.

Step 4. In this step, we prove (2).

Possibly shrinking X around x, we may assume that g is a regular function on X. Let $\mathfrak{b} := g\mathcal{O}_X$. Then $\mathfrak{b}\mathcal{O}_{X,x} \subset \mathfrak{m}_x$, $v(\mathfrak{b}) = v(g)$, and $\operatorname{ord}_D(\mathfrak{b}) = \operatorname{ord}_D(g)$. Since D is a prime divisor on X, $a(D, X, B) = 1 - \operatorname{mult}_D B$. Thus (2) follows from (1) applied to $w = \operatorname{ord}_D$.

Step 5. In this step, we prove (3).

Since x is the vertex, $\mathfrak{m}_x \cap R = \bigoplus_{j>0} R_j$, so $g \in \mathfrak{m}_x$, $v(g) = d$, and $a_v = \alpha$. By (2) with $B = 0$, $nd \geq \alpha \operatorname{ord}_D(g)$ for any prime divisor D on X through x. Since $\operatorname{mult}_E \operatorname{div}(g) = \operatorname{ord}_E(g) = \operatorname{mult}_E(g\mathcal{O}_X)$ for any prime divisor E over X, (1) applied to $\mathfrak{b} = g\mathcal{O}_X$ implies that

$$\operatorname{lct}(X \ni x; \operatorname{div}(g)) = \operatorname{lct}(X \ni x; g\mathcal{O}_X) \geq \frac{\alpha}{nv(g)} = \frac{\alpha}{nd}.$$

Since g is a non-zero nonunit of the Noetherian normal local domain $\mathcal{O}_{X,x}$, a minimal prime ideal of $g\mathcal{O}_{X,x}$ has height one by Krull's principal ideal theorem [Har77, Theorem I.1.11A]. Thus there exists a prime divisor D on X through x such that $\operatorname{ord}_D(g) \geq 1$, and so $d \geq \frac{\alpha}{n} \operatorname{ord}_D(g) \geq \frac{\alpha}{n}$. In particular, $R_d = 0$ for $0 < d < \frac{\alpha}{n}$.

Assume that $\frac{\alpha}{n} \leq d < \frac{2\alpha}{n}$. For any prime divisor D on X through x with $\operatorname{ord}_D(g) > 0$, we have $\operatorname{ord}_D(g) \leq \frac{nd}{\alpha} < 2$, so $\operatorname{ord}_D(g) = 1$. Since X is klt, $\mathcal{O}_{X,x}$ is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10], so $\mathcal{O}_{X,x}/(g)$ is Cohen–Macaulay and has no embedded associated primes. Since $\operatorname{ord}_D(g) = 1$ for any component D of $\operatorname{div}(g)$ through x, $\mathcal{O}_{X,x}/(g)$ is reduced at the minimal primes, hence reduced by [Mat86, §23, p. 183]. Since the non-reduced locus of $\operatorname{div}(g)$ is closed, $\operatorname{div}(g)$ is reduced near x. □

5.5. **Degeneration to the minimizing cone.** In this subsection, we degenerate a klt threefold germ $X \ni x$ which is not LCIQ and satisfies $\widehat{\mathrm{vol}}(X \ni x) \geq q$ to the cone defined by a minimizer of $\widehat{\mathrm{vol}}$.

Proposition 5.8 provides a Gorenstein, K-semistable, and non-LCIQ cone $X_0 \ni x_0$ whose local volume equals that of $X \ni x$, and a flat family of finite type over $\mathbb{A}^1$ which degenerates $X \ni x$ to $X_0 \ni x_0$. Section 11 uses the family to return from $X_0 \ni x_0$ to $X \ni x$ when the local volume of $X \ni x$ equals q. Since the minimizer v may have rational rank greater than one, we construct the family from the extended Rees algebra of a divisorial valuation w such that v and w belong to a common face of a Kollár model [XZ25, Corollary 4.10].

Remark 5.7. In **Step 3** of the proof of Proposition 5.8 below, we use a property of the isomorphisms of [XZ25, Corollary 4.10] which is not contained in the statement of [XZ25, Corollary 4.10]: in the setting of [XZ25, Corollary 4.10], if s and s' belong to the interior of one face of $\mathbb{R}^r_{\geq 0}$, then there exists an isomorphism $\mathrm{gr}_{v_s} R \cong \mathrm{gr}_{v_{s'}} R$ of $\mathbb{C}$-algebras which maps the ideal of the elements of positive degree onto the ideal of the elements of positive degree. The reason is as follows. If s and s' are rational, the proof of [XZ25, Corollary 4.10] realizes $\mathrm{gr}_{v_s} R$ and $\mathrm{gr}_{v_{s'}} R$ as one ring A, the coordinate ring of one fiber of the multigraded Rees family

$$\mathrm{Spec} \bigoplus_{i \in \mathbb{Z}^r} \pi_* \mathcal{O}_Y\Big(-\sum_k i_k E_k\Big) t_1^{-i_1} \cdots t_r^{-i_r} \to \mathbb{A}^r,$$

with the gradings $\langle s, \cdot \rangle$ and $\langle s', \cdot \rangle$. The multidegrees of the non-zero graded pieces of A have non-negative entries, since $\pi_* \mathcal{O}_Y(-\sum_k i_k E_k)$ depends only on the numbers $\max(i_k, 0)$, and the degree zero piece of A is $\mathbb{C}$ for both gradings, since v_s and $v_{s'}$ are centered at x. Thus, for both gradings, the ideal of the elements of positive degree is the sum of the graded pieces of A of non-zero multidegree. If s is irrational, the proof of [XZ25, Corollary 4.10] composes with an isomorphism $\mathrm{gr}_{v_s} R \cong \mathrm{gr}_{v_u} R$ of [LX18, Lemma 2.10] for a rational vector u such that s and u belong to the interior of one face, which sends homogeneous generators of positive degree to homogeneous generators of positive degree, and similarly if s' is irrational.

Proposition 5.8. *Let $X \ni x$ be a klt threefold germ which is not LCIQ, and assume that $V := \widehat{\mathrm{vol}}(X \ni x) \geq q$, where $q = \frac{46+13\sqrt{13}}{12}$. Let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, let $X_0 := \mathrm{Spec}\, \mathrm{gr}_v \mathcal{O}_{X,x}$, let $x_0 \in X_0$ be the vertex, and let $M \subset \mathbb{R}$ be the subgroup generated by the degrees of $\mathrm{gr}_v \mathcal{O}_{X,x}$ (Theorem 2.12(2)). Then the following hold.*

1. *$X \ni x$ is Gorenstein canonical, and $V < 9$.*
2. *X_0 is a normal Gorenstein canonical affine threefold, M is a lattice, the action of the torus $\mathrm{Spec}\, \mathbb{C}[M]$ on X_0 is faithful and good (Definition 2.8), and the inclusion $M \subset \mathbb{R}$ defines a Reeb vector ξ of X_0 such that (X_0, ξ) is a K-semistable Fano cone with $\widehat{\mathrm{vol}}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(X_0 \ni x_0) = V$. The canonical module of X_0 is generated by one homogeneous element.*
3. *$X_0 \ni x_0$ is not LCIQ.*
4. *There exist an affine open neighborhood $U = \mathrm{Spec}\, R$ of x in X such that U is klt and $\omega_U \cong \mathcal{O}_U$, and a prime divisor F over X with $\mathrm{center}_X F = \{x\}$, such that the following hold for $w := \mathrm{ord}_F$: we have $\mathrm{gr}_w \mathcal{O}_{X,x} \cong \mathrm{gr}_v \mathcal{O}_{X,x}$ as $\mathbb{C}$-algebras; the morphism*

$$f : \mathcal{X} := \mathrm{Spec} \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j(w) t^{-j} \to \mathbb{A}^1 = \mathrm{Spec}\, \mathbb{C}[t]$$

defined by the extended Rees algebra *of w, where $\mathfrak{a}_j(w) := \{g \in R \mid w(g) \geq j\}$ for $j > 0$ and $\mathfrak{a}_j(w) := R$ for $j \leq 0$, is flat, affine, and of finite type; $\mathcal{X}$ is normal and Gorenstein with $\omega_{\mathcal{X}} \cong \mathcal{O}_{\mathcal{X}}$; and f has a section σ such that $\mathcal{X}_0 \ni \sigma(0)$ is isomorphic to $X_0 \ni x_0$, and $\mathcal{X}_t \ni \sigma(t)$ is isomorphic to $U \ni x$ for any $t \neq 0$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

By Lemma 4.3, $q > 7$. In particular, $q > \frac{9}{2}$ and $q > \frac{125}{18}$. Since $X \ni x$ is not LCIQ and $V \geq q > \frac{9}{2}$, by Lemma 5.1(4), $X \ni x$ is Gorenstein canonical and $V < 9$.

Step 2. In this step, we prove (2), except for the equality $\dim X_0 = 3$, which we prove in **Step 5**.

By (1), K_X is Cartier near x. By Theorem 2.12(1)(2), v is quasi-monomial, $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a finitely generated $\mathbb{C}$-algebra, X_0 is a normal klt variety with a good action of the torus $\operatorname{Spec}\mathbb{C}[M]$, and the inclusion $M \subset \mathbb{R}$ defines a Reeb vector ξ of X_0. By Proposition 5.5, the degree zero part of $\operatorname{gr}_v \mathcal{O}_{X,x}$ is $\mathbb{C}$, $\operatorname{gr}_v \mathcal{O}_{X,x}$ is a Gorenstein domain, X_0 has canonical singularities, the canonical module of X_0 is generated by one M-homogeneous element, and M is a lattice.

The weights of the action of $\operatorname{Spec}\mathbb{C}[M]$ on X_0 are the degrees of the non-zero homogeneous elements of $\operatorname{gr}_v \mathcal{O}_{X,x}$, and they generate M. Hence the action is faithful. By Theorem 2.12(3), (X_0, ξ) is a K-semistable Fano cone and

$$\widehat{\operatorname{vol}}(\operatorname{wt}_\xi) = \widehat{\operatorname{vol}}(X_0 \ni x_0) = \widehat{\operatorname{vol}}(X \ni x) = V.$$

Step 3. In this step, we construct U, F, and w, and we prove that $\operatorname{gr}_w \mathcal{O}_{X,x} \cong \operatorname{gr}_v \mathcal{O}_{X,x}$.

Since $X \ni x$ is klt and Gorenstein, there exists an affine open neighborhood $U = \operatorname{Spec} R$ of x in X such that U is klt and $\omega_U \cong \mathcal{O}_U$. For a valuation μ on $K(X)$ with center x on X, let

$$\operatorname{gr}_\mu R := \bigoplus_{\lambda \geq 0} \{g \in R \mid \mu(g) \geq \lambda\}/\{g \in R \mid \mu(g) > \lambda\}.$$

We show that the natural homomorphism $\operatorname{gr}_\mu R \to \operatorname{gr}_\mu \mathcal{O}_{X,x}$ is an isomorphism for any such μ. It is injective, since the filtration of R defined by μ is the restriction of the filtration of $\mathcal{O}_{X,x}$ defined by μ. Let $g \in \mathcal{O}_{X,x}$ be non-zero, and write $g = r/h$ with $r, h \in R$ and $h(x) \neq 0$. Since μ is centered at x, we have $\mu(h) = 0$ and $\mu(h(x) - h) > 0$. A direct calculation shows that

$$\mu\Big(g - \frac{r}{h(x)}\Big) = \mu\Big(\frac{r(h(x) - h)}{h(x)h}\Big) = \mu(r) + \mu(h(x) - h) > \mu(r) = \mu(g).$$

Thus the image of g in degree $\mu(g)$ equals the image of $r/h(x) \in R$, and the homomorphism is surjective.

Since v is quasi-monomial, $\operatorname{gr}_v R \cong \operatorname{gr}_v \mathcal{O}_{X,x}$ is finitely generated, and $\operatorname{Spec}\operatorname{gr}_v R \cong X_0$ is klt (**Step 2**), by the implication (1)⇒(3) of [XZ25, Theorem 4.1] there exists a Kollár model $\pi : (Y, E) \to U$ of $x \in U$ [XZ25, Definition 3.7] such that $v \in \operatorname{QM}(Y, E)$. By [XZ25, Definition 2.8 and the paragraphs following it], there exist components $E_1, \dots, E_r$ of E, the generic point η of an irreducible component of $E_1 \cap \cdots \cap E_r$, and a vector $s = (s_1, \dots, s_r) \in \mathbb{R}^r_{\geq 0} \backslash \{0\}$ such that $v = v_s$, where for $s' \in \mathbb{R}^r_{\geq 0} \backslash \{0\}$ we denote by $v_{s'}$ the quasi-monomial valuation with weights s' along $E_1, \dots, E_r$ at η. By definition, $v_{\lambda s'} = \lambda v_{s'}$ for any real number $\lambda > 0$.

Let $\tau \subset \mathbb{R}^r_{\geq 0}$ be the face consisting of the vectors s' with $s'_i = 0$ whenever $s_i = 0$, and let τ° be the relative interior of τ, that is, the set of vectors $s' \in \tau$ with $s'_i > 0$ whenever $s_i > 0$. Then $s \in \tau^\circ$. Since τ° is a non-empty open subset of the linear subspace $\{s' \in \mathbb{R}^r \mid s'_i = 0 \text{ whenever } s_i = 0\}$, which is defined over $\mathbb{Q}$, there exists a vector $b \in \tau^\circ \cap \mathbb{Z}^r$. By [XZ25, Definition 2.8 and the paragraphs following it], v_b is given by the toroidal divisor of the weighted blow-up with weight b: there exist a prime divisor F over X and a real number $c > 0$ such that $v_b = c \operatorname{ord}_F$. Since b is integral, v_b takes values in $\mathbb{Z}$ on $K(X)^*$ (in the toroidal case, v_b is the restriction of the monomial valuation with the integral weights $(n_1 b_1, \dots, n_r b_r)$ on an SNC abelian cover, where E_i pulls back to $n_i E'_i$ [XZ25, p. 593], so v_b is integer-valued). Since $\operatorname{ord}_F(g) = 1$ for some $g \in K(X)^*$, we have $c = v_b(g) \in \mathbb{N}^+$. Let $w := \operatorname{ord}_F = v_{b/c}$. Then $b/c \in \tau^\circ$.

Let ζ be the generic point of the irreducible component of $\bigcap_{s_i > 0} E_i$ which contains η, let $z_1, \dots, z_r$ be local equations of $E_1, \dots, E_r$ at η, and let $s' \in \tau^\circ$. By the formula in [XZ25, Definition 2.8(3)], computed in the toroidal case on the SNC abelian cover described in the paragraph following it, for $g \in \mathcal{O}_{Y,\eta}$ we have $v_{s'}(g) > 0$ if and only if every monomial of g with

non-zero coefficient contains some z_i with $s'_i > 0$, that is, g vanishes along the closure of ζ. Thus the center of $v_{s'}$ on Y is ζ. In particular, the centers of v and w on Y are both ζ. Since $E = \mathrm{Exc}(\pi)$ and π is an isomorphism over $U\backslash\{x\}$ [XZ25, Definition 3.7], we have $\pi(\zeta) = x$. Thus $\mathrm{center}_X F = \{x\}$. Since s and b/c belong to τ° (extended by zeros to all components of E, which does not change v_s and $v_{b/c}$ by [XZ25, Proposition 3.10]), by [XZ25, Corollary 4.10],

$$\mathrm{gr}_w R \cong \mathrm{gr}_v R \tag{5.3}$$

as $\mathbb{C}$-algebras, and by Remark 5.7, we may take the isomorphism (5.3) to map the ideal of the elements of positive degree onto the ideal of the elements of positive degree, that is, the vertex of $\mathrm{Spec}\,\mathrm{gr}_w R$ to the vertex of $\mathrm{Spec}\,\mathrm{gr}_v R$. Since w is centered at x, the natural homomorphism $\mathrm{gr}_w R \to \mathrm{gr}_w \mathcal{O}_{X,x}$ is an isomorphism, as shown above. Together with (5.3), we obtain

$$\mathrm{gr}_w \mathcal{O}_{X,x} \cong \mathrm{gr}_w R \cong \mathrm{gr}_v R \cong \mathrm{gr}_v \mathcal{O}_{X,x}$$

as $\mathbb{C}$-algebras, and the composite isomorphism maps the vertex to the vertex.

Step 4. In this step, we construct f and σ, and we prove the assertions on the fibers of f in (4).

Let

$$\mathcal{R} := \bigoplus_{j\in\mathbb{Z}} \mathfrak{a}_j(w)t^{-j} \subset R[t,t^{-1}].$$

Since w is a valuation, $\mathfrak{a}_j(w)\mathfrak{a}_k(w) \subset \mathfrak{a}_{j+k}(w)$ for any $j,k \in \mathbb{Z}$, so $\mathcal{R}$ is a subring of $R[t,t^{-1}]$. Since $\mathfrak{a}_j(w) = R$ for $j \le 0$, we have $R[t] \subset \mathcal{R}$, and hence $\mathcal{R}[t^{-1}] = R[t,t^{-1}]$. Since $\mathrm{gr}_w R$ is finitely generated (**Step 3**), by [XZ25, Lemma 2.14 and (2.1)], $\mathcal{R}$ is a finitely generated $R[t]$-algebra, hence a finitely generated $\mathbb{C}[t]$-algebra. Since $w = \mathrm{ord}_F$ takes values in $\mathbb{Z}$, we have $\{g \in R \mid w(g) > j\} = \mathfrak{a}_{j+1}(w)$ for any integer $j \ge 0$. Since $t\cdot\mathfrak{a}_{j+1}(w)t^{-j-1} = \mathfrak{a}_{j+1}(w)t^{-j}$ and $\mathfrak{a}_j(w) = \mathfrak{a}_{j+1}(w)$ for $j < 0$, we obtain

$$\mathcal{R}/t\mathcal{R} = \bigoplus_{j\ge 0} \big(\mathfrak{a}_j(w)/\mathfrak{a}_{j+1}(w)\big)t^{-j} \cong \mathrm{gr}_w R.$$

Let $\mathcal{X} := \mathrm{Spec}\,\mathcal{R}$ and let $f : \mathcal{X} \to \mathbb{A}^1$ be the morphism defined by $\mathbb{C}[t] \subset \mathcal{R}$. Then f is affine and of finite type. Since $\mathcal{R} \subset R[t,t^{-1}]$, $\mathcal{R}$ is torsion-free over the principal ideal domain $\mathbb{C}[t]$, so f is flat [Har77, Chapter III, Example 9.1.3].

Let $\mathfrak{m}_x \subset R$ be the maximal ideal of x. If $g \in \mathfrak{m}_x$, then $w(g) > 0$ since $\mathrm{center}_X F = \{x\}$, and hence $w(g) \ge 1$. If $g \in R\backslash\mathfrak{m}_x$, then $w(g) = 0$. Thus $\mathfrak{a}_1(w) = \mathfrak{m}_x$, and $\mathfrak{a}_j(w) \subset \mathfrak{m}_x$ for any $j > 0$. Consider the evaluation map at x,

$$\epsilon : R[t,t^{-1}] \to \mathbb{C}[t,t^{-1}], \quad gt^k \to g(x)t^k.$$

For $j > 0$ and $g \in \mathfrak{a}_j(w)$, we have $\epsilon(gt^{-j}) = 0$. Hence $\epsilon(\mathcal{R}) \subset \mathbb{C}[t]$, and $\epsilon|_{\mathcal{R}} : \mathcal{R} \to \mathbb{C}[t]$ is a homomorphism of $\mathbb{C}[t]$-algebras which is the identity on $\mathbb{C}[t]$. Let $\sigma : \mathbb{A}^1 \to \mathcal{X}$ be the section of f defined by $\epsilon|_{\mathcal{R}}$.

Let $t_0 \in \mathbb{C}\backslash\{0\}$. Since t is a unit in $\mathcal{R}/(t-t_0)\mathcal{R}$, we have

$$\mathcal{R}/(t-t_0)\mathcal{R} \cong \mathcal{R}[t^{-1}]/(t-t_0)\mathcal{R}[t^{-1}] = R[t,t^{-1}]/(t-t_0)R[t,t^{-1}] \cong R,$$

where the last isomorphism maps gt^k to $t_0^k g$. Under the composite isomorphism, the homomorphism induced by $\epsilon|_{\mathcal{R}}$ from $\mathcal{R}/(t-t_0)\mathcal{R}$ to $\mathbb{C}$ corresponds to the evaluation $g \to g(x)$ on R. Thus $\mathcal{X}_{t_0} \ni \sigma(t_0)$ is isomorphic to $U \ni x$.

The homomorphism $\mathcal{R}/t\mathcal{R} \to \mathbb{C}$ induced by $\epsilon|_{\mathcal{R}}$ vanishes on $\mathfrak{a}_j(w)/\mathfrak{a}_{j+1}(w)$ for $j > 0$, and it is the isomorphism $R/\mathfrak{m}_x \to \mathbb{C}$ on $\mathfrak{a}_0(w)/\mathfrak{a}_1(w) = R/\mathfrak{m}_x$. Thus $\sigma(0)$ is the vertex of $\mathcal{X}_0 = \mathrm{Spec}\,\mathrm{gr}_w R$, that is, the point defined by the ideal of elements of positive degree. By **Step 3**, $\mathcal{X}_0 \ni \sigma(0)$ is isomorphic to $X_0 \ni x_0$.

Step 5. In this step, we prove that $\mathcal{X}$ is normal and Gorenstein with $\omega_{\mathcal{X}} \cong \mathcal{O}_{\mathcal{X}}$, and that $\dim X_0 = 3$.

By **Steps 2–4**, $\mathcal{R}/t\mathcal{R} \cong \mathrm{gr}_w R \cong \mathrm{gr}_v \mathcal{O}_{X,x}$ is a non-zero normal Gorenstein domain. Since U is klt, R is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. Thus R is a normal Cohen–Macaulay

domain of finite type over $\mathbb{C}$ with $\omega_R \cong R$. By **Step 4**, $\mathcal{R} \subset R[t,t^{-1}]$ is a finitely generated $R[t]$-subalgebra with $\mathcal{R}[t^{-1}] = R[t,t^{-1}]$, and $\mathcal{R}/t\mathcal{R}$ is a non-zero normal Cohen–Macaulay domain, since Gorenstein rings are Cohen–Macaulay [BH98, Proposition 3.1.20]. By Lemma 5.4 with $R' := R$, $\mathcal{R}$ is normal and Gorenstein, and $\omega_{\mathcal{R}} \cong \mathcal{R}$. Thus $\mathcal{X}$ is normal and Gorenstein, and $\omega_{\mathcal{X}} \cong \mathcal{O}_{\mathcal{X}}$.

Since $R[t] \subset \mathcal{R} \subset R[t,t^{-1}]$, the fraction field of $\mathcal{R}$ is $K(X)(t)$. Thus $\mathcal{X}$ is an integral scheme of dimension 4. Since f is flat and of finite type, by [Har77, Chapter III, Corollary 9.6], any irreducible component of $\mathcal{X}_0$ is of dimension 3. Since $\mathcal{X}_0 \cong X_0$ (**Step 4**), $\dim X_0 = 3$. Together with **Step 2**, this proves (2), and together with **Steps 3 and 4**, this proves (4).

Step 6. In this step, we prove (3).

By **Step 4** and **Step 5**, $f : \mathcal{X} \to \mathbb{A}^1$ is a flat morphism of finite type with a section σ, the germ $\mathcal{X}_0 \ni \sigma(0)$ is isomorphic to the klt threefold germ $X_0 \ni x_0$, and the germ $\mathcal{X}_t \ni \sigma(t)$ is isomorphic to the klt threefold germ $U \ni x$ for any $t \neq 0$. By (2) and **Step 1**,

$$\widehat{\mathrm{vol}}(\mathcal{X}_t \ni \sigma(t)) = \widehat{\mathrm{vol}}(X \ni x) = V = \widehat{\mathrm{vol}}(X_0 \ni x_0) = \widehat{\mathrm{vol}}(\mathcal{X}_0 \ni \sigma(0)) \geq q > \frac{125}{18}$$

for any $t \neq 0$. Suppose that $X_0 \ni x_0$ is LCIQ. Then $\mathcal{X}_0 \ni \sigma(0)$ is LCIQ, and by Proposition 5.3(3), $\mathcal{X}_t \ni \sigma(t)$ is LCIQ for any $t \neq 0$ close to 0. Thus $X \ni x$ is LCIQ, a contradiction. Hence $X_0 \ni x_0$ is not LCIQ. □

5.6. **From formal to analytic isomorphisms.** In this subsection, we show that an isomorphism of completed local rings of a family over a curve and of a product family implies an analytic isomorphism between neighborhoods of the given points over the base.

An isomorphism of completed local rings is only formal, and it need not preserve the section. Relative Artin approximation implies an analytic product structure over the base, and, since the singularity of the central germ is isolated, for t close to 0 the product structure sends $\sigma(t)$ to (x',t) whenever $\mathcal{X}_t$ is singular at $\sigma(t)$.

Proposition 5.9. *Let $(T,0)$ be a smooth pointed curve, $f : \mathcal{X} \to T$ a flat morphism of finite type, and σ a section of f. Let $X' \ni x'$ be a germ with an isolated singularity at x' such that $\mathcal{X}_0 \ni \sigma(0)$ is analytically isomorphic to $X' \ni x'$. Assume that $\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} \cong \widehat{\mathcal{O}}_{X'\times T,(x',0)}$ as $\widehat{\mathcal{O}}_{T,0}$-algebras. Then there exist an open neighborhood U of $\sigma(0)$ in $\mathcal{X}$ and an open neighborhood U' of $(x',0)$ in $X' \times T$, both in the analytic topology, and an isomorphism of complex analytic spaces $\phi : U \to U'$ such that $\phi(\sigma(0)) = (x',0)$ and $\mathrm{pr}_T \circ \phi = f|_U$, where $\mathrm{pr}_T : X' \times T \to T$ is the projection. Moreover, after shrinking T around 0 in the analytic topology, any singular germ $\mathcal{X}_t \ni \sigma(t)$ is analytically isomorphic to $X' \ni x'$. In particular, if $\mathcal{X}_t \ni \sigma(t)$ is analytically isomorphic to a singular germ $X \ni x$ for any $t \neq 0$ close to 0, then $X \ni x$ is analytically isomorphic to $X' \ni x'$.*

Proof. Let $p := \sigma(0)$. The structure homomorphisms $\mathcal{O}_{T,0} \to \widehat{\mathcal{O}}_{\mathcal{X},p}$ and $\mathcal{O}_{T,0} \to \widehat{\mathcal{O}}_{X'\times T,(x',0)}$ factor through $\mathcal{O}_{T,0} \to \widehat{\mathcal{O}}_{T,0}$, so the isomorphism of $\widehat{\mathcal{O}}_{T,0}$-algebras in the hypothesis is an isomorphism of $\mathcal{O}_{T,0}$-algebras. The curve T is of finite type over $\mathbb{C}$, and $\mathcal{X}$ and $X' \times T$ are of finite type over T. By [Art69, Corollary 2.6], $\mathcal{X}$ and $X' \times T$ have a common étale neighborhood over T: there exist a scheme W of finite type over T, a point $y \in W$, and étale T-morphisms $g : W \to \mathcal{X}$ and $h : W \to X' \times T$ such that $g(y) = p$ and $h(y) = (x',0)$.

An étale morphism induces local isomorphisms of the associated complex analytic spaces [Gro71, Exposé XII, Proposition 3.1]. Thus there exist an open neighborhood W_1 of y in W^{an}, an open neighborhood U of p in $\mathcal{X}^{\mathrm{an}}$, and an open neighborhood U' of $(x',0)$ in $(X' \times T)^{\mathrm{an}}$ such that g^{an} restricts to an isomorphism $W_1 \to U$ and h^{an} restricts to an isomorphism $W_1 \to U'$. Let

$$\phi := h^{\mathrm{an}}|_{W_1} \circ \left(g^{\mathrm{an}}|_{W_1}\right)^{-1} : U \to U'.$$

Then ϕ is an isomorphism of complex analytic spaces and $\phi(p) = (x',0)$. Since g and h are T-morphisms, $\mathrm{pr}_T \circ h = f \circ g$, and hence $\mathrm{pr}_T \circ \phi = f|_U$. This proves the existence of ϕ.

Since $X' \ni x'$ has an isolated singularity at x' and the singular locus of X' is closed [Har77, Theorem I.5.3], there exists a Zariski open neighborhood X'_1 of x' in X' such that X'_1 is smooth outside x'. Possibly replacing U' with $U' \cap (X'_1 \times T)^{\mathrm{an}}$ and U with $\phi^{-1}(U' \cap (X'_1 \times T)^{\mathrm{an}})$, we may assume that $U' \subset (X'_1 \times T)^{\mathrm{an}}$. Since σ^{an} is continuous and $\sigma(0) = p \in U$, there exists an open neighborhood Δ of 0 in T^{an} such that $\sigma(\Delta) \subset U$. We replace T with Δ.

Let $t \in \Delta$. Since $f(\sigma(t)) = t$ and $\mathrm{pr}_T \circ \phi = f|_U$, we have $\phi(\sigma(t)) = (z_t, t)$ for some $z_t \in X'_1$. Since $\mathrm{pr}_T \circ \phi = f|_U$, ϕ restricts to an isomorphism from $U \cap (f^{\mathrm{an}})^{-1}(t)$ onto $U' \cap (X'^{\mathrm{an}} \times \{t\})$. Thus the complex analytic germ of $\mathcal{X}_t$ at $\sigma(t)$ is isomorphic to the complex analytic germ of X' at z_t, and hence

$$\widehat{\mathcal{O}}_{\mathcal{X}_t, \sigma(t)} \cong \widehat{\mathcal{O}}_{X', z_t}$$

(Lemma 2.5(2)). A Noetherian local ring is regular if and only if the completion is regular [Har77, Theorem I.5.4A(d)]. Hence, if $\mathcal{X}_t \ni \sigma(t)$ is singular, then X' is singular at z_t, so $z_t = x'$, and $\mathcal{X}_t \ni \sigma(t)$ is analytically isomorphic to $X' \ni x'$. This proves the second statement of the proposition.

Finally, assume that $\mathcal{X}_t \ni \sigma(t)$ is analytically isomorphic to a singular germ $X \ni x$ for any $t \neq 0$ close to 0, and fix such a $t \in \Delta \backslash \{0\}$. Then $\widehat{\mathcal{O}}_{\mathcal{X}_t, \sigma(t)} \cong \widehat{\mathcal{O}}_{X,x}$ is not regular [Har77, Theorem I.5.4A(d)], so $\mathcal{X}_t \ni \sigma(t)$ is singular. By the second statement of the proposition,

$$\widehat{\mathcal{O}}_{X,x} \cong \widehat{\mathcal{O}}_{\mathcal{X}_t, \sigma(t)} \cong \widehat{\mathcal{O}}_{X', x'}.$$

Thus $X \ni x$ is analytically isomorphic to $X' \ni x'$. □

6. Toric singularities

In this section, we prove Theorem 6.1 below and Proposition 6.4. Both rely on the list of lattice polygons in Lemma 6.3(3). Theorem 6.1 is used in Section 7 and Section 11, and Proposition 6.4 in Section 11. The toric case is the only case of Theorem 1.4 in which the bound q is attained, and the equality case of Theorem 1.4 is reduced to the equality case of Theorem 6.1.

Theorem 6.1. *Let N be a lattice of rank three and $\sigma \subset N_{\mathbb{R}}$ a strongly convex rational polyhedral cone with $\dim \sigma = 3$ such that the affine toric variety X_σ is $\mathbb{Q}$-Gorenstein and klt, and let $x \in X_\sigma$ be the torus-fixed point. Assume that $X_\sigma \ni x$ is not LCIQ. Then $\widehat{\mathrm{vol}}(X_\sigma \ni x) \leq q$. Moreover, equality holds if and only if (σ, N) is isomorphic to $(\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$, where $\sigma_{\mathbb{F}_1}$ is as in Construction-Proposition 4.2, that is, if and only if $X_\sigma \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$.*

6.1. **Lattice polygons.** In this subsection, we express the local volume of a Gorenstein toric threefold germ through areas of polar polygons and bound it in terms of the area of the corresponding lattice polygon (Lemma 6.3(2)). When the local volume is at least q, only finitely many lattice polygons remain up to isomorphism, and we describe them in Lemma 6.3(3).

In the following lemma, we describe the finite quasi-étale covers of toric germs used in Lemma 6.3(1) and in the proof of Theorem 6.1.

Lemma 6.2. *Let $N' \subset N$ be lattices of rank d such that N/N' is finite, let $M := \mathrm{Hom}(N, \mathbb{Z})$ and $M' := \mathrm{Hom}(N', \mathbb{Z})$, and let $\sigma \subset N_{\mathbb{R}}$ be a strongly convex rational polyhedral cone. Assume that the primitive generator in N of any ray of σ belongs to N'. Then the finite group $G := N/N'$ acts on $X'_\sigma := \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M']$, the toric morphism $X'_\sigma \to X_\sigma := \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M]$ is the quotient morphism by G, and G acts freely on X'_σ outside a closed subset of codimension at least two. In particular, $X'_\sigma \to X_\sigma$ is quasi-étale.*

Proof. By [CLS11, Proposition 1.3.18], G acts on X'_σ, the toric morphism $X'_\sigma \to X_\sigma$ is the quotient morphism by G, and $G \cong \mathrm{Hom}(M'/M, \mathbb{C}^*)$, where the class of $u \in N$ acts on the character χ^m, $m \in M'$, by the scalar $e^{2\pi\sqrt{-1}\langle m, u\rangle}$. By [CLS11, Theorem 3.2.6], X'_σ is the union of the torus orbits $O(\tau)$ for the faces τ of σ, and $\dim O(\tau) = d - \dim \tau$. For a ray ρ of σ, the open subset $X'_\rho := \operatorname{Spec} \mathbb{C}[\rho^\vee \cap M']$ of X'_σ is G-stable and is the union of the dense orbit and $O(\rho)$. If $\sigma = \{0\}$, then X'_σ is the torus $T_{N'} := \operatorname{Spec} \mathbb{C}[M']$, and $G \subset T_{N'}$ acts on X'_σ by translations,

hence freely. Otherwise, σ has a ray, any X'_ρ contains the dense orbit, and the complement of $\bigcup_\rho X'_\rho$ in X'_σ is the union of the orbits $O(\tau)$ with $\dim\tau \geq 2$, which has codimension at least two. Thus it suffices to show that G acts freely on X'_ρ for any ray ρ of σ.

Let $e \in N$ be the primitive generator of ρ. Since $e \in N' \subset N$, e is primitive in N', so e extends to a basis $e = f_1, f_2, \dots, f_d$ of N'. Let $f_1^*, \dots, f_d^*$ be the dual basis of M'. Then $X'_\rho \cong \mathbb{A}^1 \times \mathbb{G}_m^{d-1}$ with coordinates $x_i := \chi^{f_i^*}$, and the class of $u = \sum_i c_i f_i \in N$, $c_i \in \mathbb{Q}$, acts by $x_i \to e^{2\pi\sqrt{-1}c_i}x_i$. Suppose that the class of u fixes a point of X'_ρ. Since $x_2, \dots, x_d$ do not vanish on X'_ρ, we have $c_2, \dots, c_d \in \mathbb{Z}$. Hence

$$c_1 e = u - \sum_{i\geq 2} c_i f_i \in N \cap \mathbb{Q}e = \mathbb{Z}e,$$

so $c_1 \in \mathbb{Z}$ and $u \in N'$. Thus G acts freely on X'_ρ. Since the quotient morphism by a finite group is étale over the image of the locus where the group acts freely [Gro71, Exposé V, Proposition 2.2(ii)], $X'_\sigma \to X_\sigma$ is quasi-étale. □

Lemma 6.3. *Let N be a lattice of rank three, $M := \mathrm{Hom}(N, \mathbb{Z})$, and $\sigma \subset N_\mathbb{R}$ a strongly convex rational polyhedral cone with $\dim\sigma = 3$ such that $X_\sigma := \operatorname{Spec}\mathbb{C}[\sigma^\vee \cap M]$ is $\mathbb{Q}$-Gorenstein and klt. Let $x \in X_\sigma$ be the torus-fixed point, and let $m_0 \in M_\mathbb{Q}$ be the element with $\langle m_0, e\rangle = 1$ for any primitive generator e of a ray of σ. Then the following hold.*

(1) Let $N' := \{n \in N \mid \langle m_0, n\rangle \in \mathbb{Z}\}$ and $r := [N : N']$, and let x' be the torus-fixed point of the affine toric variety X'_σ of σ with respect to N'. Then the toric morphism $X'_\sigma \to X_\sigma$ is a cyclic quasi-étale quotient of degree r, the germ $X'_\sigma \ni x'$ is Gorenstein canonical, and $\widehat{\mathrm{vol}}(X'_\sigma \ni x') = r\widehat{\mathrm{vol}}(X_\sigma \ni x)$. Moreover, for any $\xi \in \mathrm{Int}\,\sigma$, the Reeb valuations wt_ξ on X_σ and on X'_σ satisfy $a(\mathrm{wt}_\xi, X'_\sigma) = a(\mathrm{wt}_\xi, X_\sigma)$ and $\mathrm{vol}_{X'_\sigma}(\mathrm{wt}_\xi) = r\,\mathrm{vol}_{X_\sigma}(\mathrm{wt}_\xi)$; in particular, wt_ξ computes $\widehat{\mathrm{vol}}(X_\sigma \ni x)$ if and only if it computes $\widehat{\mathrm{vol}}(X'_\sigma \ni x')$, and the rational rank $\dim_\mathbb{Q}\left(\mathbb{Q} \otimes_\mathbb{Z} \mathrm{wt}_\xi(K(X_\sigma)^*)\right)$ of wt_ξ on X_σ is equal to the rational rank of wt_ξ on X'_σ.

(2) Assume that X_σ is Gorenstein. Choose a splitting $N = \mathbb{Z}^2 \oplus \mathbb{Z}$ such that m_0 is the last coordinate, and let $Q \subset \mathbb{R}^2$ be the lattice polygon with $\sigma = \mathrm{cone}(Q \times \{1\})$. Then

$$\widehat{\mathrm{vol}}(X_\sigma \ni x) = 2 \inf_{w \in \mathrm{Int}\,Q} \mathrm{Area}\left((Q-w)^\circ\right) \leq \frac{2\pi^2}{\mathrm{Area}(Q)},$$

where, for $w \in \mathrm{Int}\,Q$, the polar polygon of $Q - w$ is

$$(Q-w)^\circ := \{y \in \mathbb{R}^2 \mid \langle y, z\rangle \leq 1 \text{ for any } z \in Q - w\}.$$

(3) Assume that X_σ is Gorenstein and that $\widehat{\mathrm{vol}}(X_\sigma \ni x) \geq q$. Then $\mathrm{Area}(Q) < 3$, and, up to an affine lattice isomorphism of $\mathbb{Z}^2$, the polygon Q is a triangle, the Lawrence trapezoid $\mathrm{conv}((0,0),(a,0),(0,1),(b,1))$ for integers $a, b \geq 0$ with $a + b > 0$, the polygon $Q_0 := \mathrm{conv}((\pm 1, 0), (0, \pm 1))$, or the polygon $Q_1 := \mathrm{conv}((1,0),(1,1),(0,1),(-1,-1))$.

(4) Assume that X_σ is Gorenstein and that, with respect to the splitting in (2),

$$Q = \mathrm{conv}((0,0),(a,0),(0,1),(b,1))$$

for integers $a, b \geq 0$ with $a + b > 0$, and identify M with $\mathbb{Z}^3$ by the dual coordinates. Then the semigroup $\sigma^\vee \cap M$ is generated by

$$g_1 := (1,0,0), \quad g_2 := (-1, b-a, a), \quad g_3 := (0,1,0), \quad g_4 := (0,-1,1),$$

and

$$X_\sigma \cong \operatorname{Spec}\mathbb{C}[x_1, x_2, x_3, x_4]/(x_1x_2 - x_3^b x_4^a).$$

Moreover, the linear map $M \to M$ which interchanges g_1 and g_2 and fixes g_3 and g_4 is a lattice involution, preserves $\sigma^\vee \cap M$, and fixes $m_0 = g_3 + g_4$.

Proof. Such an element m_0 exists since $K_{X_\sigma} = -\sum_\rho D_\rho$ [CLS11, Theorem 8.2.3] is $\mathbb{Q}$-Cartier, and it is unique since the rays of σ span $N_\mathbb{R}$. The torus $\operatorname{Spec}\mathbb{C}[M]$ acts on X_σ with weights $\sigma^\vee \cap M$. Since $\dim\sigma = 3$, the cone $\sigma^\vee$ is strongly convex, so the action is good, and the Reeb cone is $\operatorname{Int}\sigma$ (Definition 2.8). The toric valuations centered at x in [Blu18, Section 8] are the Reeb valuations wt_ξ with $\xi \in \operatorname{Int}\sigma$. Let Vol_M be the Lebesgue measure on $M_\mathbb{R}$ for which a fundamental domain of M has volume one. By [Blu18, Theorem 1.4 and Section 8], for any $\xi \in \operatorname{Int}\sigma$,

$$a(\mathrm{wt}_\xi, X_\sigma) = \langle m_0, \xi\rangle \quad \text{and} \quad \mathrm{vol}_{X_\sigma}(\mathrm{wt}_\xi) = 3!\,\mathrm{Vol}_M\big(\{u \in \sigma^\vee \mid \langle u, \xi\rangle < 1\}\big), \tag{6.1}$$

and

$$\widehat{\mathrm{vol}}(X_\sigma \ni x) = \inf_{\xi \in \operatorname{Int}\sigma} \widehat{\mathrm{vol}}(\mathrm{wt}_\xi).$$

(1) The homomorphism $N \to \mathbb{Q}/\mathbb{Z}$, $n \to \langle m_0, n\rangle + \mathbb{Z}$, has kernel N', and the image is a finitely generated subgroup of $\mathbb{Q}/\mathbb{Z}$, hence finite and cyclic. Thus N/N' is cyclic of order r. For any primitive generator e of a ray of σ, we have $\langle m_0, e\rangle = 1$, so $e \in N'$, and e is primitive in N' since $N' \subset N$. By Lemma 6.2, the toric morphism $X'_\sigma \to X_\sigma$ is the quotient by the cyclic group N/N', which acts freely in codimension one, and $X'_\sigma \to X_\sigma$ is quasi-étale of degree r. Since the torus is connected, the preimage of x, which is finite and torus-invariant, consists of torus-fixed points, so it is $\{x'\}$.

Let $M' := \operatorname{Hom}(N', \mathbb{Z})$. Then $M \subset M'$ and $[M' : M] = [N : N'] = r$. Since $\langle m_0, N'\rangle \subset \mathbb{Z}$, we have $m_0 \in M'$. Thus $K_{X'_\sigma}$ is Cartier [CLS11, Proposition 8.2.12], so X'_σ is Gorenstein since normal toric varieties are Cohen–Macaulay [CLS11, Theorem 9.2.9], and X'_σ is canonical [CLS11, Proposition 11.4.11].

Fix $\xi \in \operatorname{Int}\sigma$. The primitive generators of the rays of σ with respect to N' coincide with those with respect to N, so (6.1) for X'_σ and (6.1) for X_σ both involve the element m_0 and the region $\{u \in \sigma^\vee \mid \langle u, \xi\rangle < 1\}$. Since $[M' : M] = r$, we have $\mathrm{Vol}_{M'} = r\,\mathrm{Vol}_M$. Thus

$$a(\mathrm{wt}_\xi, X'_\sigma) = a(\mathrm{wt}_\xi, X_\sigma) \quad \text{and} \quad \mathrm{vol}_{X'_\sigma}(\mathrm{wt}_\xi) = r\,\mathrm{vol}_{X_\sigma}(\mathrm{wt}_\xi),$$

so the normalized volume of wt_ξ on X'_σ is r times the normalized volume of wt_ξ on X_σ. Taking the infimum over $\xi \in \operatorname{Int}\sigma$, we obtain $\widehat{\mathrm{vol}}(X'_\sigma \ni x') = r\widehat{\mathrm{vol}}(X_\sigma \ni x)$, and wt_ξ computes $\widehat{\mathrm{vol}}(X_\sigma \ni x)$ if and only if it computes $\widehat{\mathrm{vol}}(X'_\sigma \ni x')$.

For any non-zero $f \in \mathbb{C}[\sigma^\vee \cap M]$, we have $\mathrm{wt}_\xi(f) = \langle u, \xi\rangle$ for some $u \in \sigma^\vee \cap M$. Since $K(X_\sigma)$ is the fraction field of $\mathbb{C}[\sigma^\vee \cap M]$ and $\sigma^\vee \cap M$ generates M as a group, the value group of wt_ξ on X_σ is $\{\langle u, \xi\rangle \mid u \in M\}$. Similarly, the value group of wt_ξ on X'_σ is $\{\langle u, \xi\rangle \mid u \in M'\}$. Since $rM' \subset M \subset M'$, both value groups span the $\mathbb{Q}$-vector space $\{\langle u, \xi\rangle \mid u \in M_\mathbb{Q}\}$, which implies (1).
(2) Since X_σ is Gorenstein, $m_0 \in M$ [CLS11, Proposition 8.2.12], and m_0 is primitive in M since $\langle m_0, e\rangle = 1$ for a primitive generator e of a ray. Thus a splitting as in (2) exists, the primitive generators of the rays of σ are $(v, 1)$ for the vertices $v \in \mathbb{Z}^2$ of Q, and $\dim Q = 2$ since $\dim\sigma = 3$. We have $\operatorname{Int}\sigma = \mathbb{R}_{>0} \cdot (\operatorname{Int} Q \times \{1\})$. For $c > 0$, we have $a(\mathrm{wt}_{c\xi}, X_\sigma) = ca(\mathrm{wt}_\xi, X_\sigma)$ and $\mathrm{vol}(\mathrm{wt}_{c\xi}) = c^{-3}\,\mathrm{vol}(\mathrm{wt}_\xi)$, so $\widehat{\mathrm{vol}}(\mathrm{wt}_{c\xi}) = \widehat{\mathrm{vol}}(\mathrm{wt}_\xi)$. Thus

$$\widehat{\mathrm{vol}}(X_\sigma \ni x) = \inf_{w \in \operatorname{Int} Q} \widehat{\mathrm{vol}}(\mathrm{wt}_{(w,1)}).$$

Fix $w \in \operatorname{Int} Q$, and write elements of $M_\mathbb{R} = \mathbb{R}^2 \times \mathbb{R}$ as (y, s). By (6.1),

$$a(\mathrm{wt}_{(w,1)}, X_\sigma) = \langle m_0, (w, 1)\rangle = 1.$$

We have

$$\sigma^\vee = \{(y, s) \mid \langle y, z\rangle + s \geq 0 \text{ for any } z \in Q\} \quad \text{and} \quad \langle (y, s), (w, 1)\rangle = \langle y, w\rangle + s.$$

The linear map $(y, s) \to (y, s')$ with $s' := s + \langle y, w\rangle$ has determinant one, so it preserves Vol_M. In the coordinates (y, s'), the region $\{u \in \sigma^\vee \mid \langle u, (w, 1)\rangle < 1\}$ is

$$\{(y, s') \mid \langle y, z - w\rangle + s' \geq 0 \text{ for any } z \in Q,\ s' < 1\}.$$

Taking $z = w$, we obtain $s' \geq 0$ on the region. For $s' = 0$, the condition $\langle y, z - w\rangle \geq 0$ for any $z \in Q$ implies $y = 0$, since $Q - w$ contains a neighborhood of the origin. For $0 < s' < 1$, the slice of the region at s' is $-s'(Q-w)^\circ$. Since $w \in \operatorname{Int} Q$, the polygon $(Q-w)^\circ$ is bounded. Thus

$$\widehat{\operatorname{vol}}(\operatorname{wt}_{(w,1)}) = 3! \int_0^1 s'^2 \operatorname{Area}\big((Q-w)^\circ\big) ds' = 2 \operatorname{Area}\big((Q-w)^\circ\big),$$

which implies the equality in (2).

By the nonsymmetric Blaschke–Santaló inequality [FM06, Introduction, p. 2], there exists $w_0 \in \operatorname{Int} Q$ such that

$$\operatorname{Area}(Q) \cdot \operatorname{Area}\big((Q-w_0)^\circ\big) \leq \pi^2.$$

The polar body of Q with respect to w_0 in [FM06] is $w_0 + (Q-w_0)^\circ$, a translate of $(Q-w_0)^\circ$. Thus

$$\widehat{\operatorname{vol}}(X_\sigma \ni x) \leq 2 \operatorname{Area}\big((Q-w_0)^\circ\big) \leq \frac{2\pi^2}{\operatorname{Area}(Q)},$$

which implies (2).

(3) An affine lattice isomorphism $z \to \varphi(z) + c$ of $\mathbb{Z}^2$ with $\varphi \in \mathrm{GL}_2(\mathbb{Z})$ and $c \in \mathbb{Z}^2$ induces the lattice automorphism $(z,t) \to (\varphi(z) + tc, t)$ of $N = \mathbb{Z}^2 \oplus \mathbb{Z}$, which preserves the last coordinate and maps σ onto the cone over the image of Q. Thus we may replace Q with the image of Q under any affine lattice isomorphism of $\mathbb{Z}^2$.

By Lemma 4.3, $q > 7$. Since $\pi < \frac{22}{7}$, we have $\frac{2\pi^2}{3} < \frac{968}{147} < 7$. If $\operatorname{Area}(Q) \geq 3$, then $\widehat{\operatorname{vol}}(X_\sigma \ni x) \leq \frac{2\pi^2}{3} < 7 < q$ by (2), a contradiction. Thus $\operatorname{Area}(Q) < 3$.

Suppose that Q has no interior lattice point. By the classification of lattice polygons without interior lattice points [BN06, Theorem 2.5], up to an affine lattice isomorphism, Q is a Lawrence trapezoid $\operatorname{conv}((0,0),(a,0),(0,1),(b,1))$ for integers $a, b \geq 0$ with $a + b > 0$, or the triangle $\operatorname{conv}((0,0),(2,0),(0,2))$.

Suppose that Q is not a triangle and that Q has an interior lattice point. Then Q has at least four vertices, so $\#(\partial Q \cap \mathbb{Z}^2) \geq 4$. By Pick's theorem [Pic99],

$$\operatorname{Area}(Q) = \#(\operatorname{Int} Q \cap \mathbb{Z}^2) + \frac{1}{2}\#(\partial Q \cap \mathbb{Z}^2) - 1 < 3.$$

If $\#(\operatorname{Int} Q \cap \mathbb{Z}^2) \geq 2$, then $\operatorname{Area}(Q) \geq 2 + 2 - 1 = 3$, a contradiction. Thus $\#(\operatorname{Int} Q \cap \mathbb{Z}^2) = 1$ and $\#(\partial Q \cap \mathbb{Z}^2) \in \{4, 5\}$. Possibly replacing Q with a lattice translate, we may assume that $\operatorname{Int} Q \cap \mathbb{Z}^2 = \{0\}$.

Let $p, p' \in \partial Q \cap \mathbb{Z}^2$ be two consecutive boundary lattice points. Any point of $\operatorname{conv}(0, p, p')$ outside the segment $[p, p']$ is a convex combination with a positive weight on the interior point 0, hence is contained in $\operatorname{Int} Q$. Thus the only lattice points of $\operatorname{conv}(0, p, p')$ are 0, p, and p', and Pick's theorem [Pic99] implies $\operatorname{Area}(\operatorname{conv}(0, p, p')) = \frac{1}{2}$, that is, $|\det(p, p')| = 1$. Hence, for any edge E of Q, the matrix with columns p, p' for two consecutive boundary lattice points $p, p' \in E$ is in $\mathrm{GL}_2(\mathbb{Z})$, and there exists $y_E \in \mathbb{Z}^2$ with $\langle y_E, z\rangle = 1$ for any $z \in E$. Since $Q = \{z \in \mathbb{R}^2 \mid \langle y_E, z\rangle \leq 1 \text{ for any edge } E\}$, the polar polygon Q° is the convex hull of the elements y_E, so Q° is a lattice polygon. If $y \in \operatorname{Int} Q^\circ \cap \mathbb{Z}^2$, then $\langle y, v\rangle < 1$, hence $\langle y, v\rangle \leq 0$, for any vertex v of Q, so $\langle y, z\rangle \leq 0$ for any $z \in Q$, and $y = 0$ since $0 \in \operatorname{Int} Q$. Thus $\operatorname{Int} Q^\circ \cap \mathbb{Z}^2 = \{0\}$.

By [PRV00, Theorem 1], $\#(\partial Q \cap \mathbb{Z}^2) + \#(\partial Q^\circ \cap \mathbb{Z}^2) = 12$. The dual polygon in [PRV00] is obtained by rotating by 90° the differences of consecutive boundary lattice points of Q, so it differs from Q° by a unimodular transformation and has exactly $\#(\partial Q^\circ \cap \mathbb{Z}^2)$ boundary lattice points. By Pick's theorem [Pic99] for Q°, we obtain

$$2 \operatorname{Area}(Q^\circ) = \#(\partial Q^\circ \cap \mathbb{Z}^2) = 12 - \#(\partial Q \cap \mathbb{Z}^2).$$

By (2) with $w = 0$,

$$\widehat{\operatorname{vol}}(X_\sigma \ni x) \leq 2 \operatorname{Area}(Q^\circ) = 12 - \#(\partial Q \cap \mathbb{Z}^2).$$

If $\#(\partial Q \cap \mathbb{Z}^2) = 5$, then $\widehat{\operatorname{vol}}(X_\sigma \ni x) \leq 7 < q$, a contradiction. Thus $\#(\partial Q \cap \mathbb{Z}^2) = 4$.

Since Q has at least four vertices, Q has exactly four vertices v_1, v_2, v_3, v_4, ordered counterclockwise, and they are the only lattice points of ∂Q. Thus $\det(v_i, v_{i+1}) = 1$ for $i = 1, 2, 3, 4$, where $v_5 := v_1$, and the sign is positive since the order is counterclockwise and $0 \in \operatorname{Int} Q$. Possibly replacing Q with the image of Q under an element of $\mathrm{GL}_2(\mathbb{Z})$, we may assume that $v_1 = (1, 0)$ and $v_2 = (0, 1)$. The conditions $\det(v_2, v_3) = 1$ and $\det(v_4, v_1) = 1$ imply $v_3 = (-1, c)$ and $v_4 = (d, -1)$ for some $c, d \in \mathbb{Z}$, and $\det(v_3, v_4) = 1 - cd = 1$ implies $cd = 0$. If $c = 0$, then the reflection $(y_1, y_2) \to (y_2, y_1)$ maps Q onto $\operatorname{conv}((1, 0), (0, 1), (-1, d), (0, -1))$, which is the polygon $\operatorname{conv}((1, 0), (0, 1), (-1, c), (d, -1))$ with (c, d) replaced by $(d, 0)$. Thus we may assume that $d = 0$, that is,

$$Q = \operatorname{conv}((1, 0), (0, 1), (-1, c), (0, -1)).$$

Strict convexity of Q at $(0, 1)$ and at $(0, -1)$ implies

$$\det\big((0, 1) - (1, 0), (-1, c) - (0, 1)\big) = 2 - c > 0,$$
$$\det\big((0, -1) - (-1, c), (1, 0) - (0, -1)\big) = 2 + c > 0,$$

so $c \in \{-1, 0, 1\}$. For $c = 0$, we have $Q = Q_0$. The reflection $(y_1, y_2) \to (y_1, -y_2)$ maps the polygon for $c = 1$ onto the polygon for $c = -1$, and the linear map $(y_1, y_2) \to (-y_2, y_1 - y_2)$, which is in $\mathrm{GL}_2(\mathbb{Z})$, maps $(1, 0), (0, 1), (-1, -1), (0, -1)$ to $(0, 1), (-1, -1), (1, 0), (1, 1)$, respectively. Thus for $c = \pm 1$, the polygon Q is isomorphic to Q_1, which implies (3).
(4) The primitive generators of the rays of σ are contained in $\{(0, 0, 1), (a, 0, 1), (0, 1, 1), (b, 1, 1)\}$. Thus

$$\sigma^\vee \cap M = \{(u_1, u_2, u_3) \in \mathbb{Z}^3 \mid u_3 \geq 0,\ au_1 + u_3 \geq 0,\ u_2 + u_3 \geq 0,\ bu_1 + u_2 + u_3 \geq 0\}.$$

The four elements g_1, g_2, g_3, g_4 satisfy the four inequalities. Let $u = (u_1, u_2, u_3) \in \sigma^\vee \cap M$. If $u_1 \geq 0$, then

$$u = u_1 g_1 + (u_2 + u_3) g_3 + u_3 g_4$$

with $u_1, u_2 + u_3, u_3 \geq 0$. If $u_1 = -k < 0$, then $u - kg_2 = (0, u_2 - k(b - a), u_3 - ka)$, and

$$u_3 - ka = au_1 + u_3 \geq 0 \quad \text{and} \quad (u_2 - k(b - a)) + (u_3 - ka) = bu_1 + u_2 + u_3 \geq 0,$$

so $u - kg_2$ is a non-negative integral combination of g_3 and g_4 by the case $u_1 \geq 0$. Thus g_1, g_2, g_3, g_4 generate $\sigma^\vee \cap M$, and they satisfy

$$g_1 + g_2 = bg_3 + ag_4.$$

Hence the homomorphism $\mathbb{C}[x_1, x_2, x_3, x_4] \to \mathbb{C}[\sigma^\vee \cap M]$, $x_i \to \chi^{g_i}$, is surjective, and the kernel contains $x_1x_2 - x_3^b x_4^a$. The kernel is a prime ideal of height one since $\dim X_\sigma = 3$, and $x_1x_2 - x_3^b x_4^a$ is irreducible since it has degree one in x_1 and x_2 is coprime to $x_3^b x_4^a$. Since $\mathbb{C}[x_1, x_2, x_3, x_4]$ is a unique factorization domain, the kernel is generated by $x_1x_2 - x_3^b x_4^a$, so $X_\sigma \cong \operatorname{Spec} \mathbb{C}[x_1, x_2, x_3, x_4]/(x_1x_2 - x_3^b x_4^a)$.

The determinant of the matrix with rows g_1, g_3, g_4 is 1, so g_1, g_3, g_4 is a basis of M, and $g_2 = -g_1 + bg_3 + ag_4$. Let $\iota : M \to M$ be the linear map with $\iota(g_1) = g_2$, $\iota(g_3) = g_3$, and $\iota(g_4) = g_4$. Then $\iota(g_2) = -g_2 + bg_3 + ag_4 = g_1$, so $\iota^2 = \mathrm{id}_M$ and ι is a lattice involution. Since ι permutes g_1, g_2, g_3, g_4, it preserves $\sigma^\vee \cap M$. Finally, $m_0 = (0, 0, 1) = g_3 + g_4$ is fixed by ι, which implies (4). □

6.2. **The toric bound and the rational rank.** In this subsection, we prove Theorem 6.1 and Proposition 6.4.

Proof of Theorem 6.1. Let $V := \widehat{\mathrm{vol}}(X_\sigma \ni x)$, and let $m_0 \in M_\mathbb{Q}$ be the element with $\langle m_0, e\rangle = 1$ for any primitive generator e of a ray of σ. Since x is contained in the closure of any torus orbit of X_σ [CLS11, Theorem 3.2.6] and $K_{X_\sigma} = -\sum_\rho D_\rho$ is torus-invariant [CLS11, Theorem 8.2.3], K_{X_σ} is Cartier at x if and only if X_σ is Gorenstein. If X_σ is not Gorenstein, then the Cartier index of K_{X_σ} at x is at least two, and $V \leq \frac{9}{2} < q$ by Lemma 5.1(3).

Thus we may assume that X_σ is Gorenstein. Choose a splitting $N = \mathbb{Z}^2 \oplus \mathbb{Z}$ such that m_0 is the last coordinate, and let $Q \subset \mathbb{R}^2$ be the lattice polygon with $\sigma = \operatorname{cone}(Q \times \{1\})$. Suppose that $V \geq q$. By Lemma 6.3(3), after a lattice automorphism of N which preserves the last coordinate, Q is a triangle, a Lawrence trapezoid $\operatorname{conv}((0,0),(a,0),(0,1),(b,1))$ for integers $a, b \geq 0$ with $a + b > 0$, Q_0, or Q_1.

Suppose that Q is a triangle. Let e_1, e_2, e_3 be the primitive generators of the rays of σ, and $N_0 \subset N$ the sublattice generated by e_1, e_2, e_3. Since e_1, e_2, e_3 is a basis of N_0, the affine toric variety of σ with respect to N_0 is $\mathbb{A}^3$ [CLS11, Example 1.3.20]. Since $e_1, e_2, e_3 \in N_0$, Lemma 6.2 implies that the toric morphism $\mathbb{A}^3 \to X_\sigma$ is the quotient by the finite group N/N_0, which acts freely in codimension one. The group N/N_0 fixes the origin. Thus $X_\sigma \ni x$ is LCIQ, a contradiction.

Suppose that Q is a Lawrence trapezoid. Then $X_\sigma \ni x$ is a hypersurface germ by Lemma 6.3(4), hence LCIQ, a contradiction.

Suppose that $Q = Q_0$. The primitive generators of the rays of σ are $(1,0,1)$, $(-1,0,1)$, $(0,1,1)$, and $(0,-1,1)$, and they generate the sublattice $N_0 \subset N$ with basis

$$f_1 := (-1,1,0), \quad f_2 := (-1,-1,0), \quad f_3 := (1,0,1).$$

The determinant of the matrix with rows f_1, f_2, f_3 is 2, so $[N : N_0] = 2$. In the basis f_1, f_2, f_3, the primitive generators $(1,0,1)$, $(-1,0,1)$, $(0,1,1)$, and $(0,-1,1)$ have coordinates $(0,0,1)$, $(1,1,1)$, $(1,0,1)$, and $(0,1,1)$, respectively, and m_0 restricts to the third coordinate. Thus, with respect to N_0, the polygon of σ is the Lawrence trapezoid with $a = b = 1$, and Lemma 6.3(4) implies that the affine toric variety of σ with respect to N_0 is the threefold ordinary double point $\operatorname{Spec} \mathbb{C}[x_1, x_2, x_3, x_4]/(x_1x_2 - x_3x_4)$. The toric morphism from $\operatorname{Spec} \mathbb{C}[x_1, x_2, x_3, x_4]/(x_1x_2 - x_3x_4)$ to X_σ is the quotient by $N/N_0 \cong \mathbb{Z}/2$, which fixes the torus-fixed point and acts freely in codimension one by Lemma 6.2, since the primitive generators of the rays with respect to N belong to N_0. Thus $X_\sigma \ni x$ is LCIQ, a contradiction.

Therefore $Q = Q_1$, and σ is the cone spanned by $(1,0,1)$, $(1,1,1)$, $(0,1,1)$, and $(-1,-1,1)$, that is, $(\sigma, N) \cong (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$. By Construction-Proposition 4.2, $X_\sigma \ni x$ is isomorphic to $C_{\mathbb{F}_1} \ni o$, and $V = q$. Hence $V \geq q$ implies that $(\sigma, N) \cong (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$ and $V = q$. Thus $V \leq q$, and $V = q$ implies $(\sigma, N) \cong (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$. Conversely, if $(\sigma, N) \cong (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$, then $X_\sigma \ni x$ is isomorphic to $C_{\mathbb{F}_1} \ni o$ by Construction-Proposition 4.2, so $V = q$.

Finally, if $X_\sigma \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$, then $V = \widehat{\operatorname{vol}}(C_{\mathbb{F}_1} \ni o) = q$ by Proposition 2.7(1) and Construction-Proposition 4.2, so $(\sigma, N) \cong (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$. Conversely, a lattice isomorphism $(\sigma, N) \cong (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$ induces an isomorphism $X_\sigma \cong C_{\mathbb{F}_1}$ which maps x to o (Construction-Proposition 4.2), so $X_\sigma \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$. □

If a minimizer has rational rank three, then the minimizing cone X_0 is an affine toric threefold. Thus it suffices to show that a Reeb valuation computing the local volume of an affine toric threefold with local volume at least q has rational rank at most two, which we prove by the symmetries of the polygons in Lemma 6.3(3), the inequality of arithmetic and geometric means, and, for Q_1, the explicit minimizer in Construction-Proposition 4.2.

Proposition 6.4. *Let $X \ni x$ be a klt threefold germ with $\widehat{\operatorname{vol}}(X \ni x) \geq q$, and let v be a minimizer of $\widehat{\operatorname{vol}}$ on $\operatorname{Val}_{X,x}$. Then the rational rank $\dim_{\mathbb{Q}} \left(\mathbb{Q} \otimes_{\mathbb{Z}} v(K(X)^*)\right)$ of v is at most two.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove the proposition for Reeb valuations of affine toric threefolds. Let N be a lattice of rank three, $M := \operatorname{Hom}(N, \mathbb{Z})$, and $\sigma \subset N_{\mathbb{R}}$ a strongly convex rational polyhedral cone with $\dim \sigma = 3$ such that $X_\sigma = \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M]$ is $\mathbb{Q}$-Gorenstein and klt, let $x \in X_\sigma$ be the torus-fixed point, and let $\xi \in \operatorname{Int} \sigma$. Assume that $\widehat{\operatorname{vol}}(X_\sigma \ni x) \geq q$ and that wt_ξ computes $\widehat{\operatorname{vol}}(X_\sigma \ni x)$. We show that the rational rank of wt_ξ is at most two.

Since $K(X_\sigma)$ is the fraction field of $\mathbb{C}[\sigma^\vee \cap M]$, $\sigma^\vee \cap M$ generates M as a group, and $\mathrm{wt}_\xi(f) = \langle u, \xi\rangle$ for some $u \in \sigma^\vee \cap M$ for any non-zero $f \in \mathbb{C}[\sigma^\vee \cap M]$, the value group of wt_ξ is $\{\langle u, \xi\rangle \mid u \in M\}$.

By Lemma 6.3(1), replacing N with N', we may assume that X_σ is Gorenstein. This is allowed since X'_σ is Gorenstein canonical (by Lemma 6.3(1) the germ $X'_\sigma \ni x'$ is, and the loci where X'_σ is Gorenstein and canonical are open and torus-invariant, while x' is contained in the closure of any torus orbit [CLS11, Theorem 3.2.6]), $\widehat{\mathrm{vol}}(X'_\sigma \ni x') = r\widehat{\mathrm{vol}}(X_\sigma \ni x) \geq q$, and wt_ξ computes $\widehat{\mathrm{vol}}(X'_\sigma \ni x')$, and the rational rank of wt_ξ on X'_σ is equal to the rational rank of wt_ξ on X_σ, by Lemma 6.3(1). Let $m_0 \in M$ be the element with $\langle m_0, e\rangle = 1$ for any primitive generator e of a ray of σ [CLS11, Proposition 8.2.12]. Choose a splitting $N = \mathbb{Z}^2 \oplus \mathbb{Z}$ such that m_0 is the last coordinate, and let $Q \subset \mathbb{R}^2$ be the lattice polygon with $\sigma = \mathrm{cone}(Q \times \{1\})$. A lattice automorphism g of N preserving the last coordinate does not change the value group, since $\langle u, \xi\rangle = \langle u \circ g^{-1}, g(\xi)\rangle$ for $u \in M$, and $\mathrm{wt}_{g(\xi)}$ computes the local volume of $X_{g(\sigma)}$ at its torus-fixed point, since g induces an isomorphism $X_\sigma \to X_{g(\sigma)}$. Thus, by Lemma 6.3(3), we may assume that Q is a triangle, a Lawrence trapezoid $\mathrm{conv}((0,0),(a,0),(0,1),(b,1))$ for integers $a, b \geq 0$ with $a + b > 0$, Q_0, or Q_1.

We show that $g(\xi) = \xi$ for any lattice automorphism $g : N \to N$ with $g(\sigma) = \sigma$. The dual map $M \to M$, $u \to u \circ g$, preserves $\sigma^\vee \cap M$, so it induces an automorphism α of $\mathbb{C}[\sigma^\vee \cap M]$ with $\alpha(\chi^u) = \chi^{u\circ g}$, and the corresponding automorphism of X_σ fixes x. For any $u \in M$,

$$\mathrm{wt}_\xi(\alpha(\chi^u)) = \langle u \circ g, \xi\rangle = \langle u, g(\xi)\rangle = \mathrm{wt}_{g(\xi)}(\chi^u),$$

so $\mathrm{wt}_{g(\xi)} = \mathrm{wt}_\xi \circ \alpha$. Since log discrepancies and volumes are preserved by automorphisms of X_σ fixing x, $\mathrm{wt}_{g(\xi)}$ computes $\widehat{\mathrm{vol}}(X_\sigma \ni x)$ and $a(\mathrm{wt}_{g(\xi)}, X_\sigma) = a(\mathrm{wt}_\xi, X_\sigma)$. By Theorem 2.12(1), $\mathrm{wt}_{g(\xi)} = c\,\mathrm{wt}_\xi$ for some $c > 0$. Since $a(c\,\mathrm{wt}_\xi, X_\sigma) = ca(\mathrm{wt}_\xi, X_\sigma)$ and $0 < a(\mathrm{wt}_\xi, X_\sigma) < +\infty$, we have $c = 1$. Comparing the values on the characters χ^u, $u \in \sigma^\vee \cap M$, we obtain $g(\xi) = \xi$.

Suppose that Q is a triangle. Let e_1, e_2, e_3 be the primitive generators of the rays of σ, and write $\xi = c_1e_1 + c_2e_2 + c_3e_3$ with $c_1, c_2, c_3 > 0$. Let $e_1^*, e_2^*, e_3^* \in M_\mathbb{R}$ be the dual basis of e_1, e_2, e_3, and $\omega > 0$ the volume with respect to Vol_M of the parallelepiped spanned by e_1^*, e_2^*, e_3^*. Then

$$\{u \in \sigma^\vee \mid \langle u, \xi\rangle < 1\} = \Big\{ \sum_{i=1}^3 s_i e_i^* \;\Big|\; s_i \geq 0,\ \sum_{i=1}^3 c_i s_i < 1 \Big\},$$

and by [Blu18, Theorem 1.4 and Section 8],

$$a(\mathrm{wt}_\xi, X_\sigma) = \langle m_0, \xi\rangle = c_1 + c_2 + c_3 \quad \text{and} \quad \mathrm{vol}(\mathrm{wt}_\xi) = 3! \cdot \frac{\omega}{6c_1c_2c_3} = \frac{\omega}{c_1c_2c_3}.$$

By the inequality of arithmetic and geometric means,

$$\widehat{\mathrm{vol}}(\mathrm{wt}_\xi) = \frac{\omega(c_1 + c_2 + c_3)^3}{c_1c_2c_3} \geq 27\omega,$$

and equality holds if and only if $c_1 = c_2 = c_3$. Since $\mathrm{wt}_{e_1+e_2+e_3}$ attains 27ω and wt_ξ computes $\widehat{\mathrm{vol}}(X_\sigma \ni x)$, we have $c_1 = c_2 = c_3$. Thus $\xi = c_1(e_1 + e_2 + e_3)$ with $e_1 + e_2 + e_3 \in N$, so the value group $\{\langle u, \xi\rangle \mid u \in M\}$ is contained in $c_1\mathbb{Z}$, and the rational rank of wt_ξ is one.

Suppose that Q is a Lawrence trapezoid. Let $\iota : M \to M$ be the lattice involution in Lemma 6.3(4), and $g : N \to N$ the dual map, so that $\langle u, g(n)\rangle = \langle \iota(u), n\rangle$ for $u \in M$ and $n \in N$. Since ι preserves $\sigma^\vee \cap M$, we have $g(\sigma) = \sigma$, so $g(\xi) = \xi$. Thus

$$\langle g_1, \xi\rangle = \langle g_1, g(\xi)\rangle = \langle \iota(g_1), \xi\rangle = \langle g_2, \xi\rangle.$$

Since $g_1 - g_2 = (2, a-b, -a)$ is a non-zero element of M with $\langle g_1 - g_2, \xi\rangle = 0$, the homomorphism $M \to \mathbb{R}$, $u \to \langle u, \xi\rangle$, has a non-zero kernel. Thus the rational rank of wt_ξ is at most two.

Suppose that $Q = Q_0$. The lattice automorphisms

$$(y_1, y_2, t) \to (-y_1, y_2, t) \quad \text{and} \quad (y_1, y_2, t) \to (y_1, -y_2, t)$$

of N map σ onto σ, so they fix ξ. Thus ξ is a positive multiple of $(0,0,1) \in N$, and the rational rank of wt_ξ is one.

Suppose that $Q = Q_1$. Then $(\sigma, N) = (\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$, and $\mathrm{wt}_{\xi_{\mathbb{F}_1}}$ computes $\widehat{\mathrm{vol}}(X_\sigma \ni x)$ by Construction-Proposition 4.2. By Theorem 2.12(1), $\mathrm{wt}_\xi = c\,\mathrm{wt}_{\xi_{\mathbb{F}_1}}$ for some $c > 0$, so $\xi = c\xi_{\mathbb{F}_1}$. Let $t := \frac{4-\sqrt{13}}{3}$. Then the value group of wt_ξ is

$$\{c((u_1+u_2)t+u_3) \mid (u_1,u_2,u_3) \in \mathbb{Z}^3\} = c(\mathbb{Z}t+\mathbb{Z}).$$

Since $t \notin \mathbb{Q}$, the rational rank of wt_ξ is two.

Step 2. In this step, we conclude the proof. By Theorem 2.12(1), v is quasi-monomial. By the Abhyankar inequality [Abh56, Theorem 1(1)] applied to $\mathcal{O}_{X,x}$, the rational rank of v is at most $\dim X = 3$. Suppose that the rational rank of v is three.

Let $\Gamma \subset \mathbb{R}$ be the subgroup generated by the degrees λ with $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$. For any non-zero $f \in \mathcal{O}_{X,x}$, the image of f in $\mathfrak{a}_{v(f)}(v)/\mathfrak{a}_{>v(f)}(v)$ is non-zero, and conversely any λ with $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$ is $v(f)$ for some $f \in \mathcal{O}_{X,x}$. Since $K(X)$ is the fraction field of $\mathcal{O}_{X,x}$, we have $\Gamma = v(K(X)^*)$. By Theorem 2.12(2), $\mathrm{gr}_v \mathcal{O}_{X,x}$ is generated by finitely many non-zero homogeneous elements $h_1, \dots, h_k$, of degrees $\gamma_1, \dots, \gamma_k$, and any degree with a non-zero graded piece is a non-negative integral combination of $\gamma_1, \dots, \gamma_k$. Thus Γ is generated by $\gamma_1, \dots, \gamma_k$, and $\Gamma \cong \mathbb{Z}^3$ since $\Gamma \subset \mathbb{R}$ is torsion free of rational rank three.

By Theorem 2.12(2), $X_0 = \operatorname{Spec} \mathrm{gr}_v \mathcal{O}_{X,x}$ is a normal klt variety with a good action of the torus $T := \operatorname{Spec} \mathbb{C}[\Gamma]$, and the inclusion $\Gamma \subset \mathbb{R}$ defines a Reeb vector ξ of X_0. By Theorem 2.12 and [XZ25, Theorem 1.2(1)], X_0 is a degeneration of $X \ni x$, so $\dim X_0 = 3$. Let $x_1 \in X_0$ be a point at which $h_1, \dots, h_k$ are non-zero. An element of $T = \operatorname{Hom}(\Gamma, \mathbb{C}^*)$ fixing x_1 is trivial on $\gamma_1, \dots, \gamma_k$, hence trivial. Thus $\dim(T \cdot x_1) = 3$, so the orbit $T \cdot x_1$ is open and dense in X_0. Since the stabilizer of x_1 is trivial and $\operatorname{char} \mathbb{C} = 0$, the orbit map $T \to T \cdot x_1$ is an isomorphism onto the open orbit [Bor91, Proposition II.6.7]. Since X_0 is normal and affine, X_0 is an affine toric variety with torus T [CLS11, Definition 1.1.3 and Theorem 1.3.5], that is, $X_0 \cong \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap \Gamma]$ for a strongly convex rational polyhedral cone $\sigma \subset \operatorname{Hom}(\Gamma, \mathbb{R})$, where the grading of $\mathrm{gr}_v \mathcal{O}_{X,x}$ by Γ is the weight decomposition. Since $(\mathrm{gr}_v \mathcal{O}_{X,x})_0 = \mathbb{C}$ and the weight cone $\sigma^\vee$ is strictly convex, we have $\dim \sigma = 3$, and the vertex x_0 is the torus-fixed point. The Reeb vector ξ is contained in the Reeb cone $\operatorname{Int} \sigma$, and $\langle u, \xi \rangle = u$ for $u \in \Gamma$. Thus the value group of wt_ξ is $\{\langle u, \xi \rangle \mid u \in \Gamma\} = \Gamma$, which has rational rank three.

On the other hand, X_0 is $\mathbb{Q}$-Gorenstein and klt, and by Theorem 2.12(3), wt_ξ computes $\widehat{\mathrm{vol}}(X_0 \ni x_0)$ and $\widehat{\mathrm{vol}}(X_0 \ni x_0) = \widehat{\mathrm{vol}}(X \ni x) \geq q$. By **Step 1**, the rational rank of wt_ξ is at most two, a contradiction. Thus the rational rank of v is at most two. □

7. Fano cones of complexity one

In this section, we prove an upper bound for the local volumes of non-LCIQ Gorenstein Fano cones of dimension three with a faithful action of a two-dimensional torus (Theorem 7.3). Such cones arise as minimizing cones: by Proposition 6.4, a minimizer over a klt threefold germ of local volume at least q has rational rank at most two, and in Section 11 we show that the minimizing cone of a minimizer of rational rank two satisfies the conditions of Set-up 7.1 and apply the results of this section to it. The cones of Set-up 7.1 do not form a finite list, and the condition that $X \ni o$ is not LCIQ concerns all finite quasi-étale covers of $X \ni o$. We first translate the failure of the LCIQ property into conditions on the Cox ring of X, and then bound $\widehat{\mathrm{vol}}(X \ni o)$ by the normalized volumes of explicit Reeb valuations.

Set-up 7.1. Let X, T, M, and o be as follows.

(1) X is a normal affine threefold with klt singularities.
(2) $T \cong (\mathbb{C}^*)^2$ is a torus with character lattice M which acts faithfully on X.
(3) A one-parameter subgroup of T contracts X to a closed point $o \in X$.

Definition-Lemma 7.2. Notation and conditions as in Set-up 7.1. Then the action of T on X is good, and o is the vertex. In particular, (X, ξ) is a Fano cone for any Reeb vector ξ of X. If moreover X is Gorenstein, then the canonical module of $\mathbb{C}[X]$ is generated by a T-homogeneous element, and the weight $\kappa \in M$ of such a generator, with respect to the grading in which dg and g have equal weights for any homogeneous $g \in \mathbb{C}[X]$, does not depend on the generator and satisfies $a(\mathrm{wt}_\xi, X) = \langle \kappa, \xi \rangle$ for any Reeb vector ξ. We call κ the *canonical character* of X, and write $\kappa = \zeta\theta$ with $\theta \in M$ primitive and ζ a positive integer. We say that X is *toric* if X is isomorphic to an affine toric variety.

Proof. Let $\lambda \in \mathrm{Hom}(M, \mathbb{Z})$ be a one-parameter subgroup of T which contracts X to o, that is, $\lim_{t\to 0} \lambda(t) \cdot x = o$ for any closed point $x \in X$. For any $u \in M$, any $g \in \mathbb{C}[X]_u$, and any closed point $x \in X$, we have

$$g(\lambda(t) \cdot x) = t^{\langle u, \lambda \rangle} g(x) \quad \text{for any } t \in \mathbb{C}^*,$$

and the left-hand side converges to $g(o)$ as $t \to 0$. Thus if $g \neq 0$, then $\langle u, \lambda \rangle \geq 0$. If moreover $\langle u, \lambda \rangle = 0$, then $g(x) = g(o)$ for any $x \in X$, so g is a non-zero constant and $u = 0$. Hence $\mathbb{C}[X]_0 = \mathbb{C}$, and

$$\langle u, \lambda \rangle > 0 \quad \text{for any } u \neq 0 \text{ with } \mathbb{C}[X]_u \neq 0.$$

Let w be an element such that w and $-w$ belong to the cone generated by the weights u with $\mathbb{C}[X]_u \neq 0$. Write $w = \sum_j c_j u_j$ with $c_j \geq 0$ and $u_j \neq 0$ such weights. Since $\langle -w, \lambda \rangle \geq 0$, we have

$$0 \geq \langle w, \lambda \rangle = \sum_j c_j \langle u_j, \lambda \rangle,$$

so $c_j = 0$ for any j and $w = 0$. Thus the cone generated by the weights is strictly convex, the action of T is good, and λ is a Reeb vector.

For any $u \neq 0$, any $g \in \mathbb{C}[X]_u$, and any closed point $x \in X$, we have $g(o) = \lim_{t\to 0} t^{\langle u, \lambda \rangle} g(x) = 0$. Hence o is the closed point of the T-stable maximal ideal

$$\mathfrak{m} := \bigoplus_{u \neq 0} \mathbb{C}[X]_u,$$

and o is T-fixed. Any T-fixed closed point x satisfies $x = \lim_{t\to 0} \lambda(t) \cdot x = o$, so o is the unique T-fixed point, that is, the vertex. Since X is klt, (X, ξ) is a Fano cone for any Reeb vector ξ.

Assume that X is Gorenstein, and let ω be the canonical module of $\mathbb{C}[X]$ with the M-grading in the statement. The pairing with λ coarsens the M-gradings of $\mathbb{C}[X]$ and ω to $\mathbb{Z}$-gradings, in which a homogeneous element of weight u has degree $\langle u, \lambda \rangle$. Then

$$\mathbb{C}[X] = \bigoplus_{k \geq 0} \mathbb{C}[X]_{(k)}, \quad \mathbb{C}[X]_{(0)} = \mathbb{C}, \quad \text{and} \quad \mathfrak{m} = \bigoplus_{k>0} \mathbb{C}[X]_{(k)}.$$

Since X is Gorenstein, ω is an invertible $\mathbb{C}[X]$-module, so $\dim_{\mathbb{C}} \omega/\mathfrak{m}\omega = 1$. Since $\mathfrak{m}\omega$ is an M-graded submodule of ω, the vector space $\omega/\mathfrak{m}\omega$ is concentrated in a single weight $\kappa \in M$. Let $s \in \omega$ be a homogeneous element of weight κ whose image in $\omega/\mathfrak{m}\omega$ is non-zero. By the graded Nakayama lemma [BH98, Exercise 1.5.24(a)], applied to the submodule $\mathbb{C}[X]s$ of ω and the $\mathbb{Z}$-grading, s generates ω.

Let s' be another homogeneous generator of ω. Then $s' = bs$ and $s = b's'$ for some $b, b' \in \mathbb{C}[X]$, and $bb' = 1$ since ω is torsion free of rank one. In the domain $\mathbb{C}[X]$, the highest degree component of bb' is the product of the highest degree components of b and b', so $b, b' \in \mathbb{C}[X]_{(0)} = \mathbb{C}$. Thus s' has weight κ, and κ does not depend on the generator. By Lemma 2.9, $a(\mathrm{wt}_\xi, X) = \langle \kappa, \xi \rangle$ for any Reeb vector ξ. Since X is klt, $a(\mathrm{wt}_\xi, X) > 0$. Hence $\kappa \neq 0$, and $\kappa = \zeta\theta$ for a unique primitive element $\theta \in M$ and a unique positive integer ζ. □

We treat the cases $\zeta = 1$ and $\zeta \geq 2$ separately, since $\langle \kappa, \cdot \rangle$ is a lattice coordinate on $N := \mathrm{Hom}(M, \mathbb{Z})$ if and only if $\zeta = 1$. The case $\zeta = 1$ is Proposition 7.12, and the case $\zeta \geq 2$ is treated in Subsection 7.4.

Theorem 7.3. *Notation and conditions as in Set-up 7.1. Assume that X is Gorenstein, that $X \ni o$ is not LCIQ, and that the Reeb valuation wt_ξ of a Reeb vector ξ computes $\widehat{\mathrm{vol}}(X \ni o)$. Then*

$$\widehat{\mathrm{vol}}(X \ni o) \leq q.$$

Moreover, if X is not toric and the canonical character of X is not primitive in M, then $\widehat{\mathrm{vol}}(X \ni o) \leq \frac{27}{4}$.

We prove Theorem 7.3 in Subsection 7.4.

7.1. **Cox covers.** In this subsection, we construct finite quasi-étale covers of the cones of Set-up 7.1 from their Cox rings. We use the Cox rings of rational T-varieties of complexity one and their presentations of Type 2 in the sense of [ABHW18, Constructions 2.1, 2.2, and 2.6].

Lemma 7.4. *Notation and conditions as in Set-up 7.1. Then the following hold.*

1. *X is rational, $\mathbb{C}[X]^* = \mathbb{C}^*$, and X is T-equivariantly isomorphic to an affine variety $X(A, P)$ of Type 2 in the sense of* [ABHW18, Constructions 2.1, 2.2, and 2.6]. *In particular, the Cox ring $\mathcal{R}$ of X has a presentation of Type 2, and $X = \operatorname{Spec} \mathcal{R}^H$ for the characteristic quasitorus H.*
2. *Let $H' \subset H$ be a closed subgroup such that $G := H/H'$ is finite. Then $Y := \operatorname{Spec} \mathcal{R}^{H'}$ is a normal affine variety, the induced morphism $Y \to X$ is the quotient morphism by G and is quasi-étale, G acts freely on Y outside a closed subset of codimension at least two, and the fiber of $Y \to X$ over o is one G-fixed point, namely the image of the origin of* $\operatorname{Spec} \mathcal{R}$.

Proof. We prove (1). Let λ be the one-parameter subgroup of T in Set-up 7.1(3). By [LS10, Section 1] (see also [LMS26, Construction 2.22 and Theorem 2.24]), X is T-equivariantly isomorphic to the affine T-variety of a proper polyhedral divisor on a smooth semiprojective curve Z, with $\mathbb{C}[X]_0 = H^0(Z, \mathcal{O}_Z)$, and the function field $\mathbb{C}(X)$ is the fraction field of $\mathbb{C}(Z)[M]$. By Definition-Lemma 7.2, $H^0(Z, \mathcal{O}_Z) = \mathbb{C}[X]_0 = \mathbb{C}$, and since Z is projective over an affine variety [LS10, Section 1] and the structure morphism to it factors through $\operatorname{Spec} H^0(Z, \mathcal{O}_Z)$, which is a point, Z is a projective curve. Since X is klt, $Z \cong \mathbb{P}^1$ by [LS10, Corollary 5.8] (see also [LS10, Proposition 5.5]). Thus $\mathbb{C}(X)$ is purely transcendental of degree three over $\mathbb{C}$, and X is rational.

The T-invariant functions on X are the elements of $\mathbb{C}[X]_0 = \mathbb{C}$. With respect to the $\mathbb{Z}$-grading of $\mathbb{C}[X]$ defined by λ, the degree-zero part is $\mathbb{C}[X]_{(0)} = \mathbb{C}[X]_0 = \mathbb{C}$ and $\mathbb{C}[X]_{(d)} = 0$ for $d < 0$ ($\mathbb{C}[X]_0 = \mathbb{C}$ by Definition-Lemma 7.2, and $\langle u, \lambda \rangle > 0$ for any weight $u \neq 0$ of $\mathbb{C}[X]$, since λ contracts X to o). Comparing the highest degree components of a unit and of the inverse of the unit in the domain $\mathbb{C}[X]$ shows that the invertible functions on X are constant. Hence X is of Type 2, since varieties of Type 1 have non-constant T-invariant functions [ABHW18, Introduction]. Thus $X \cong X(A, P)$ for data A and P as in [ABHW18, Construction 2.2] by [ABHW18, Section 2] (the paragraph after Remark 2.7 there). The maximality property with respect to embeddings into toric varieties in [ABHW18, Section 2, the paragraph after Remark 2.7] and [HW17, Theorem 1.8] concerns open embeddings $\iota : X \to X'$ into normal varieties X' with $\operatorname{codim}_{X'}(X' \setminus \iota(X)) \geq 2$, and X itself, being affine, embeds into an affine space, which is a toric variety. We show that any such ι with X' normal and separated satisfies $\iota(X) = X'$. The argument does not use the T-action, so X has the maximality property. Let $\iota : X \to X'$ be such an open embedding. Since X' is normal and $\operatorname{codim}_{X'}(X' \setminus \iota(X)) \geq 2$, the restriction map $H^0(X', \mathcal{O}_{X'}) \to H^0(X, \mathcal{O}_X) = \mathbb{C}[X]$ is an isomorphism by Hartogs' theorem [Har80, Proposition 1.6], so there exists a morphism $\pi : X' \to \operatorname{Spec} \mathbb{C}[X] = X$ with $\pi \circ \iota = \mathrm{id}_X$. The morphisms $\iota \circ \pi$ and $\mathrm{id}_{X'}$ coincide on the dense open subset $\iota(X)$ of the reduced separated scheme X', so $\iota \circ \pi = \mathrm{id}_{X'}$ and $X' = \iota(X)$. Let $\mathcal{R}$ be the Cox ring of X and H the characteristic quasitorus, so that $X = \operatorname{Spec} \mathcal{R}^H$ [ABHW18, Construction 2.6, Remark 2.7, Proposition 2.12].

We prove (2). By [ABHW18, Construction 2.1], $\mathcal{R}$ is a normal domain. Since X is affine and of Type 2, the characteristic space of X is $\operatorname{Spec} \mathcal{R}$, and the morphism $p : \operatorname{Spec} \mathcal{R} \to X$ induced by $\mathcal{R}^H \subset \mathcal{R}$ is the good quotient by H [ABHW18, Remark 2.7, Proposition 2.12].

Since H' is reductive, $\mathcal{R}^{H'}$ is a finitely generated $\mathbb{C}$-algebra, and $\mathcal{R}^{H'}$ is a domain. If an element of the fraction field of $\mathcal{R}^{H'}$ is integral over $\mathcal{R}^{H'}$, then it is integral over $\mathcal{R}$, so it belongs to $\mathcal{R}$ by the normality of $\mathcal{R}$, and it is H'-invariant. Thus $\mathcal{R}^{H'}$ is normal. The finite group G acts on $\mathcal{R}^{H'}$ with $(\mathcal{R}^{H'})^G = \mathcal{R}^H = \mathbb{C}[X]$, and any $b \in \mathcal{R}^{H'}$ is a root of the monic polynomial $\prod_{\gamma\in G}(z - \gamma \cdot b) \in \mathbb{C}[X][z]$. Hence $\mathcal{R}^{H'}$ is finite over $\mathbb{C}[X]$, and the morphism $f : Y \to X$ induced by $\mathbb{C}[X] \subset \mathcal{R}^{H'}$ is the quotient morphism by G. In particular, $\dim Y = \dim X = 3$.

Let X_{reg} be the smooth locus of X. By [ADHL15, Proposition 1.6.1.6(i)], $p^{-1}(X_{\mathrm{reg}}) \to X_{\mathrm{reg}}$ is an H-torsor. Since $\operatorname{Spec}\mathcal{R} \to Y$ is the good quotient by H', the quotient of $p^{-1}(X_{\mathrm{reg}})$ by H' is $f^{-1}(X_{\mathrm{reg}})$, so $f^{-1}(X_{\mathrm{reg}}) \to X_{\mathrm{reg}}$ is a G-torsor. In particular, G acts freely on $f^{-1}(X_{\mathrm{reg}})$, and f is étale over X_{reg}. Since X is normal, $\operatorname{codim}_X(X\backslash X_{\mathrm{reg}}) \geq 2$. Since f is finite, $\operatorname{codim}_Y f^{-1}(X\backslash X_{\mathrm{reg}}) \geq 2$. Thus f is quasi-étale, and G acts freely on Y outside the closed subset $f^{-1}(X\backslash X_{\mathrm{reg}})$ of codimension at least two.

By [ABHW18, Section 2], H is a closed subgroup of a quasitorus H_0 which acts diagonally on $\operatorname{Spec}\mathcal{R}$, and the action of T on X is induced by $T = H_0/H$. The origin $0 \in \operatorname{Spec}\mathcal{R}$, where all the variables of the presentation vanish, is H_0-fixed. Hence the image $\bar{0} \in Y$ of 0 is G-fixed, and $p(0)$ is T-fixed. By Definition-Lemma 7.2, o is the unique T-fixed point of X, so $f(\bar{0}) = p(0) = o$. Since the fibers of the quotient morphism f by the finite group G over closed points are G-orbits, we have $f^{-1}(o) = \{\bar{0}\}$. □

Each of two conditions on X yields an explicit finite quasi-étale cover of X by a complete intersection: $\mathbb{Q}$-factoriality at o (Proposition 7.5), and a Cox presentation without free variables and with constant exponents within any block (Proposition 7.6). Thus if $X \ni o$ is not LCIQ, then neither condition holds. The failure of $\mathbb{Q}$-factoriality is the input of Proposition 7.12, and a block with unequal exponents is the input of the non-primitive case of Theorem 7.3.

Proposition 7.5. *Notation and conditions as in Set-up 7.1. Assume that X is $\mathbb{Q}$-factorial at o. Then there exist a normal affine complete intersection threefold Y and a finite abelian group G acting on Y such that $X \cong Y/G$, the quotient morphism $Y \to X$ is quasi-étale, and the fiber of $Y \to X$ over o is one G-fixed point. In particular, $X \ni o$ is LCIQ.*

Proof. By Lemma 7.4(1), X is rational and of Type 2. Let $\mathcal{R}$ be the Cox ring of X, P the defining matrix of a presentation of Type 2 of $\mathcal{R}$ (Lemma 7.4(1)), k the number of columns of P, and H the characteristic quasitorus, so that $X = \operatorname{Spec}\mathcal{R}^H$. Let λ be the one-parameter subgroup of T in Set-up 7.1(3). We show that X is $\mathbb{Q}$-factorial. By the paragraph after [ABHW18, Construction 2.6] (which recalls [ADHL15, Theorem 3.4.3.7]), $\mathrm{Cl}(X)$ is isomorphic to the cokernel K of P^t, where the class of the T-invariant prime divisor D_j of X corresponding to the j-th column of P is the degree in K of the j-th variable of $\mathcal{R}$ $(1 \leq j \leq k)$. Since the degrees of the variables of $\mathcal{R}$ generate K, $\mathrm{Cl}(X)$ is generated by the classes of $D_1, \ldots, D_k$. Fix $1 \leq j \leq k$, and let c be a positive integer such that cD_j is Cartier near o. The locus $V_j \subset X$ where cD_j is not Cartier is closed and T-invariant, and $o \notin V_j$. If $x \in V_j$ is a closed point, then $o = \lim_{t\to 0} \lambda(t) \cdot x \in V_j$, a contradiction. Thus $V_j = \emptyset$, and cD_j is Cartier. Since j is arbitrary, X is $\mathbb{Q}$-factorial.

By [ABHW18, Corollary 2.13], the columns of P are linearly independent. By [ABHW18, Constructions 2.1 and 2.2], P is an $(r+2) \times k$ integer matrix with $\operatorname{rank} P = r + 2$ for an integer $r \geq 1$, and $\mathcal{R}$ is the quotient of a polynomial ring in k variables by $r - 1$ trinomials, which is a normal complete intersection domain with

$$\dim \mathcal{R} = k + 1 - r.$$

Thus $k = r + 2$, P is a square matrix with $\det P \neq 0$, and $\mathrm{Cl}(X) = \mathbb{Z}^{r+2}/P^t(\mathbb{Z}^{r+2})$ is finite. Hence $H = \operatorname{Hom}(\mathrm{Cl}(X), \mathbb{C}^*)$ is a finite abelian group, and we let $G := H$. Moreover, $Y := \operatorname{Spec}\mathcal{R}$ is a normal affine complete intersection with $\dim Y = 3$. By Lemma 7.4(2) with $H' = \{1\}$, we have $X \cong Y/G$, the quotient morphism $Y \to X$ is quasi-étale, and the fiber of $Y \to X$ over o is one G-fixed point y.

By Lemma 7.4(2), G acts freely on Y outside a closed subset of codimension at least two. Moreover, $Y \ni y$ is a normal local complete intersection germ, and $X \ni o$ is isomorphic to the germ of Y/G at the image of y. Thus $X \ni o$ is LCIQ. □

Proposition 7.6. *Notation and conditions as in Set-up 7.1. Assume that X has a Cox presentation of Type 2 in the sense of [ABHW18] without free variables and with constant exponents within any block. That is, the Cox ring of X is*

$$\mathcal{R} = \mathbb{C}[y_{ij} \mid 0 \le i \le r,\ 1 \le j \le n_i]/(g_0, \dots, g_{r-2})$$

for an integer $r \ge 1$, where p_i and n_i are positive integers, $y_i := \prod_{j=1}^{n_i} y_{ij}$, the vectors $c_0, \dots, c_r \in \mathbb{C}^2$ are pairwise linearly independent, and g_i is the determinant of the 3×3 matrix with first row $(y_i^{p_i}, y_{i+1}^{p_{i+1}}, y_{i+2}^{p_{i+2}})$ and with the columns c_i, c_{i+1}, and c_{i+2} below it (for $r = 1$ the Cox ring is a polynomial ring). Then $X \ni o$ is LCIQ. More precisely, $n_i \in \{1, 2\}$ for any i, and there exist integers $d_i \ge 0$ with $d_i = 0$ if and only if $n_i = 1$, at least one of them positive, and a finite abelian group G acting on the normal affine complete intersection threefold $Y \subset \mathbb{A}^{r+3}$ with coordinates $z_0, \dots, z_r, U, V$ defined by

$$UV = \prod_{i=0}^{r} z_i^{d_i} \quad \text{and} \quad \det \begin{pmatrix} z_i^{p_i} & z_{i+1}^{p_{i+1}} & z_{i+2}^{p_{i+2}} \\ c_i & c_{i+1} & c_{i+2} \end{pmatrix} = 0 \quad (0 \le i \le r-2),$$

such that $X \cong Y/G$, the quotient morphism is quasi-étale, and the fiber over o is one G-fixed point.

Proof. The proof is long, so we divide it into several steps. Let P be the defining matrix of the presentation and H the characteristic quasitorus. By [ABHW18, Construction 2.2] and the paragraph after [ABHW18, Construction 2.6] (which recalls [ADHL15, Theorem 3.4.3.7]), $\mathrm{Cl}(X)$ is isomorphic to the cokernel of P^t, and $X = \operatorname{Spec} \mathcal{R}^H$ by Lemma 7.4(1). Let $n := \sum_{i=0}^{r} n_i$, so that P is an $(r+2) \times n$ integer matrix whose columns v_{ij} correspond to the variables y_{ij}.

Step 1. In this step, we bring P into a normal form, show that $n_i \in \{1, 2\}$, and construct a sublattice $N' \subset \mathbb{Z}^{r+2}$ of finite index which contains all the columns of P.

By [ABHW18, Construction 2.2], the columns of P are pairwise different primitive vectors in $\mathbb{Z}^{r+2}$, and $\operatorname{rank} P = r + 2$. Since X is affine, the columns of P generate the extremal rays of a pointed cone of dimension $r + 2$ by [ABHW18, Proposition 2.12].

Since X is klt, X is $\mathbb{Q}$-Gorenstein. Let $\imath_X$ be the Gorenstein index of X, and let ζ_X be the canonical multiplicity of X [ABHW18, Definition 4.2, Proposition 4.3]. Here X need not be Gorenstein, and ζ_X is defined from P. The integer ζ_X is compared with the integer ζ of Definition-Lemma 7.2 in (7.11). By [ABHW18, Proposition 4.3], $\zeta_X > 0$, and after elementary row operations among the last two rows of P [ABHW18, Proposition 4.3], which are admissible operations of type (iv) in [ABHW18, Remark 2.3] and change neither $\mathcal{R}$ nor the image of P^t, hence neither $\mathrm{Cl}(X)$ nor X, we have

$$v_{ij} = (p_i e_i, \gamma_{ij}, \nu_i) \in \mathbb{Z}^r \oplus \mathbb{Z} \oplus \mathbb{Z} = \mathbb{Z}^{r+2}, \quad \nu_i = \frac{\imath_X - s_i p_i}{\zeta_X},$$

where $e_1, \dots, e_r$ is the standard basis of $\mathbb{Z}^r$, $e_0 := -(e_1 + \cdots + e_r)$, and γ_{ij}, ν_i, and s_i are integers with

$$\sum_{i=0}^{r} s_i = \imath_X (r-1).$$

The last coordinate ν_i of v_{ij} depends only on i since the exponents are constant within any block.

Since X is klt, $\mathcal{R}$ is platonic [ABHW18, Theorem 1.3]. Since the exponents are constant within any block, $\mathcal{R}$ is platonic if and only if $(p_0, \dots, p_r)$ is a platonic tuple, that is, after reordering, $p_i = 1$ for any $i \ge 3$ and $\frac{1}{p_0} + \frac{1}{p_1} + \frac{1}{p_2} > 1$, where we set $p_i := 1$ for $i > r$. Since $1 - \frac{1}{p_i} \ge \frac{1}{2}$

whenever $p_i \geq 2$, $(p_0, \dots, p_r)$ is a platonic tuple if and only if $\sum_{i=0}^r (1 - \frac{1}{p_i}) < 2$, that is,

$$\sum_{i=0}^{r} \frac{1}{p_i} > r - 1. \tag{7.1}$$

Let $w \in \mathbb{Z}^{r+2}$ be the penultimate coordinate vector. Then $v_{ij} - v_{ij'} = (\gamma_{ij} - \gamma_{ij'})w$, and the integers $\gamma_{i1}, \dots, \gamma_{in_i}$ are pairwise different since the columns of P are pairwise different. Suppose that $n_i \geq 3$ for some i, and renumber so that $\gamma_{i1} < \gamma_{i2} < \gamma_{i3}$. Then

$$v_{i2} = \frac{\gamma_{i3} - \gamma_{i2}}{\gamma_{i3} - \gamma_{i1}} v_{i1} + \frac{\gamma_{i2} - \gamma_{i1}}{\gamma_{i3} - \gamma_{i1}} v_{i3}$$

with positive coefficients. Since v_{i2} generates an extremal ray of the cone generated by the columns of P, the vectors v_{i1} and v_{i3} belong to the ray $\mathbb{R}_{\geq 0} v_{i2}$, so $v_{i1} = v_{i2} = v_{i3}$ since they are primitive, a contradiction. Thus $n_i \in \{1, 2\}$ for any i. If $n_i = 2$, then we renumber so that $\gamma_{i1} < \gamma_{i2}$ and let $d_i := \gamma_{i2} - \gamma_{i1} > 0$. If $n_i = 1$, then we let $d_i := 0$. If $d_i = 0$ for any i, then P has $r + 1$ columns, which contradicts $\operatorname{rank} P = r + 2$. Thus $d_i > 0$ for some i.

We show that $v_{01}, \dots, v_{r1}, w$ are linearly independent over $\mathbb{Q}$. Modulo $\mathbb{Q}w$, the vector v_{i1} becomes $(p_i e_i, \nu_i) \in \mathbb{Q}^r \oplus \mathbb{Q}$. Let $\beta_0, \dots, \beta_r \in \mathbb{Q}$ such that $\sum_i \beta_i (p_i e_i, \nu_i) = 0$. Comparing the coefficients of $e_1, \dots, e_r$, we have $\beta_i p_i = \beta_0 p_0 =: c$ for any i. Comparing the last coordinates, we have

$$0 = \sum_{i=0}^{r} \beta_i \nu_i = c \sum_{i=0}^{r} \frac{\nu_i}{p_i} = \frac{c}{\zeta_X} \Big(\iota_X \sum_{i=0}^{r} \frac{1}{p_i} - \sum_{i=0}^{r} s_i \Big) = \frac{c \iota_X}{\zeta_X} \Big(\sum_{i=0}^{r} \frac{1}{p_i} - (r-1) \Big).$$

By (7.1), $c = 0$, so $\beta_i = 0$ for any i. Thus $v_{01}, \dots, v_{r1}, w$ are linearly independent, and

$$N' := \bigoplus_{i=0}^{r} \mathbb{Z} v_{i1} \oplus \mathbb{Z} w \subset \mathbb{Z}^{r+2}$$

is a sublattice of finite index. Since $v_{i2} = v_{i1} + d_i w$, all the columns of P belong to N'. Let $f_0, \dots, f_r$ be the standard basis of $\mathbb{Z}^{r+1}$, and identify N' with $\mathbb{Z}^{r+1} \oplus \mathbb{Z}$ via the basis $v_{01}, \dots, v_{r1}, w$. Then the columns of P are $v_{i1} = (f_i, 0)$ and, if $n_i = 2$, $v_{i2} = (f_i, d_i)$.

Step 2. In this step, we construct a closed subgroup $H' \subset H$ with H/H' finite, and use it in Lemma 7.4.

Let $(\mathbb{Z}^{r+2})^\vee$ and $(N')^\vee$ be the dual lattices, and let

$$P'^t : (N')^\vee \to \mathbb{Z}^n, \quad m' \to (\langle m', v_{ij} \rangle)_{i,j}.$$

Since the columns of P span $N' \otimes \mathbb{Q}$, P'^t is injective. Since all the columns of P belong to N', we have $P^t(m) = P'^t(m|_{N'})$ for any $m \in (\mathbb{Z}^{r+2})^\vee$, and the restriction $(\mathbb{Z}^{r+2})^\vee \to (N')^\vee$ is injective with finite cokernel. Thus $P^t((\mathbb{Z}^{r+2})^\vee)$ is a subgroup of finite index of $P'^t((N')^\vee)$, and the natural surjection

$$\operatorname{Cl}(X) = \mathbb{Z}^n / P^t((\mathbb{Z}^{r+2})^\vee) \to \Gamma' := \mathbb{Z}^n / P'^t((N')^\vee)$$

has finite kernel. Let $H' := \operatorname{Spec} \mathbb{C}[\Gamma']$, which is a closed subgroup of $H = \operatorname{Spec} \mathbb{C}[\operatorname{Cl}(X)]$. Then $G := H/H'$ is a finite abelian group. Let $S := \mathbb{C}[y_{ij}]$ be the polynomial ring. The quasitorus H acts on $\operatorname{Spec} S$ through the $\operatorname{Cl}(X)$-grading of S in which y_{ij} has the class of the corresponding standard basis vector of $\mathbb{Z}^n$, and H' acts through the induced Γ'-grading. By Lemma 7.4(2), $Y := \operatorname{Spec} \mathcal{R}^{H'}$ is a normal affine variety, the morphism $Y \to X$ is the quotient morphism by G and is quasi-étale, and the fiber of $Y \to X$ over o is one G-fixed point, namely the image of the origin of $\operatorname{Spec} \mathcal{R}$. In particular, $\dim Y = \dim X = 3$.

Step 3. In this step, we show that $\mathcal{R}^{H'}$ is the coordinate ring of the complete intersection in the statement.

For $a \in \mathbb{Z}^n_{\geq 0}$, the monomial $y^a := \prod_{i,j} y_{ij}^{a_{ij}}$ is H'-invariant if and only if $a = P'^t(m')$ for some $m' \in (N')^\vee$. Write $m' = (\alpha_0, \dots, \alpha_r, t)$ in the basis of $(N')^\vee$ dual to $v_{01}, \dots, v_{r1}, w$. Then $P'^t(m')$

has the entry α_i at $(i,1)$ and, if $n_i = 2$, the entry $\alpha_i + d_i t$ at $(i,2)$. Hence $y^{P'^t(m')}$ is a monomial if and only if

$$\alpha_i \geq 0 \text{ for any } i, \quad \text{and} \quad \alpha_i + d_i t \geq 0 \text{ for any } i \text{ with } n_i = 2.$$

Let $A := \mathbb{C}[z_0, \dots, z_r, U, V]$ be the polynomial ring, and let $\phi : A \to S$ be the homomorphism with

$$\phi(z_i) = y_i, \quad \phi(U) = \prod_{n_i=2} y_{i2}^{d_i}, \quad \text{and} \quad \phi(V) = \prod_{n_i=2} y_{i1}^{d_i}.$$

Then $\phi(UV) = \prod_{i=0}^r y_i^{d_i} = \phi(\prod_i z_i^{d_i})$. Let $\alpha := (\alpha_0, \dots, \alpha_r)$ and $d := (d_0, \dots, d_r)$, and write $z^\alpha := \prod_i z_i^{\alpha_i}$. If $t \geq 0$, then

$$y^{P'^t(m')} = \prod_{i=0}^r y_i^{\alpha_i} \prod_{n_i=2} y_{i2}^{d_i t} = \phi(z^\alpha U^t).$$

If $t = -l < 0$, then $\alpha_i \geq l d_i$ for any i, and

$$y^{P'^t(m')} = \prod_{i=0}^r y_i^{\alpha_i - l d_i} \prod_{n_i=2} y_{i1}^{l d_i} = \phi(z^{\alpha - l d} V^l).$$

Since H' is diagonalizable, $S^{H'}$ is spanned by the H'-invariant monomials, so $\phi(A) = S^{H'}$. The quotient $A/(UV - \prod_i z_i^{d_i})$ is spanned by the images of the monomials $z^\alpha U^l$ and $z^\alpha V^l$ with $\alpha \in \mathbb{Z}_{\geq 0}^{r+1}$ and $l \geq 0$. By the formulas for $y^{P'^t(m')}$ above, ϕ sends $z^\alpha U^l$ to $y^{P'^t(\alpha, l)}$ and $z^\alpha V^l$ to $y^{P'^t(\alpha + l d, -l)}$. Since P'^t is injective, the different monomials $z^\alpha U^l$ and $z^\alpha V^l$ with $l \geq 0$ are sent to pairwise different monomials of S, which are linearly independent. Thus

$$S^{H'} \cong A/(UV - \prod_{i=0}^r z_i^{d_i}).$$

For any i, the monomial $y_i^{p_i} = \phi(z_i^{p_i})$ is H'-invariant. Let

$$\tilde{g}_i := \det \begin{pmatrix} z_i^{p_i} & z_{i+1}^{p_{i+1}} & z_{i+2}^{p_{i+2}} \\ c_i & c_{i+1} & c_{i+2} \end{pmatrix} \in A \quad (0 \leq i \leq r-2),$$

so that $g_i = \phi(\tilde{g}_i) \in S^{H'}$. Let $I := (g_0, \dots, g_{r-2})S$. The projection $\rho : S \to S^{H'}$ onto the degree-zero part of the Γ'-grading is $S^{H'}$-linear. If $s \in I \cap S^{H'}$ and $s = \sum_i h_i g_i$ with $h_i \in S$, then $s = \rho(s) = \sum_i \rho(h_i) g_i$. Thus $I \cap S^{H'} = (g_0, \dots, g_{r-2}) S^{H'}$. Since I is Γ'-homogeneous, we have $\mathcal{R}^{H'} = S^{H'}/(I \cap S^{H'})$, and hence

$$\mathcal{R}^{H'} \cong A/J, \quad J := \Big(UV - \prod_{i=0}^r z_i^{d_i}, \tilde{g}_0, \dots, \tilde{g}_{r-2}\Big).$$

By **Step 2**, A/J is a domain with $\dim A/J = 3$, so J is a prime ideal of height r in the polynomial ring A in $r+3$ variables. For any prime ideal $\mathfrak{p} \supset J$ of A, the ideal $JA_\mathfrak{p}$ of the Cohen–Macaulay local ring $A_\mathfrak{p}$ has height r and is generated by r elements, so the r generators form a regular sequence by [Mat86, Theorem 17.4(iii)]. Thus $Y \subset \mathbb{A}^{r+3}$ is a normal affine complete intersection threefold defined by the equations in the statement.

Step 4. In this step, we conclude the proof.

By **Steps 2 and 3**, $X \cong Y/G$, the quotient morphism $Y \to X$ is quasi-étale, and the fiber of $Y \to X$ over o is one G-fixed point, namely the image of the origin of $\operatorname{Spec} \mathcal{R}$. Under the isomorphism of **Step 3**, the coordinates z_i, U, and V correspond to y_i, $\prod_{n_i=2} y_{i2}^{d_i}$, and $\prod_{n_i=2} y_{i1}^{d_i}$, which all vanish at the origin of $\operatorname{Spec} \mathcal{R}$ since $d_i > 0$ for some i (**Step 1**). Hence the fiber over o is the origin $\bar{0}$ of $\mathbb{A}^{r+3}$. By Lemma 7.4(2), G acts freely on Y outside a closed subset of codimension at least two. Moreover, $Y \ni \bar{0}$ is a normal local complete intersection germ, and $X \ni o$ is isomorphic to the germ of Y/G at the image of $\bar{0}$. Thus $X \ni o$ is LCIQ. □

7.2. **Polyhedral divisors and volumes.** In this subsection, we describe the cones of Set-up 7.1 by polyhedral divisors on $\mathbb{P}^1$, and we express the volumes of their Reeb valuations in terms of the corresponding polyhedral divisors.

Construction-Proposition 7.7. Notation and conditions as in Set-up 7.1, and let $N := \mathrm{Hom}(M, \mathbb{Z})$. Then there exist a strongly convex rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$ with $\dim \sigma = 2$ and σ-polyhedra $\Delta_y \subset N_{\mathbb{Q}}$ for $y \in \mathbb{P}^1$, with $\Delta_y = \sigma$ for all but finitely many y, such that $Q := \sum_y \Delta_y$ is a proper subset of σ and

$$\mathbb{C}[X] \cong \bigoplus_{m \in \sigma^\vee \cap M} H^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor \mathcal{D}(m) \rfloor)\big)\chi^m$$

as M-graded $\mathbb{C}$-algebras, where $\mathcal{D}(m) := \sum_y \big(\min_{v \in \Delta_y} \langle m, v\rangle\big)[y]$. We fix such data, and call $\mathcal{D} := \sum_y \Delta_y \cdot [y]$ the *p-divisor* of X, σ the *tail cone* of $\mathcal{D}$, and Q the *degree polyhedron* of $\mathcal{D}$. We write $\deg \mathcal{D}(m) := \min_{z \in Q} \langle m, z\rangle$ for $m \in \sigma^\vee$. For a vertex v of Δ_y, let $\mu(v)$ be the smallest positive integer with $\mu(v)v \in N$, and let μ_y be the maximum of $\mu(v)$ over the vertices v of Δ_y. For $y \in \mathbb{P}^1$ and a vertex v of Δ_y, let $D_{y,v}$ be the T-invariant prime divisor of X corresponding to (y, v), and for a ray ρ of σ disjoint from Q, let D_ρ be the T-invariant prime divisor of X corresponding to ρ [LMS26, Section 5], and let $n_\rho \in N$ be the primitive generator of ρ. Then the following hold.

(1) $0 \notin Q$, and the Reeb cone of X is $\mathrm{Int}\, \sigma$.
(2) Assume that X is Gorenstein, and let κ be the canonical character of X. Then there exist integers b_y for $y \in \mathbb{P}^1$, zero for all but finitely many y, with $\sum_y b_y = 2$, such that $\langle \kappa, v\rangle = b_y - 1 + \frac{1}{\mu(v)}$ for any vertex v of Δ_y, and $\langle \kappa, n_\rho\rangle = 1$ for any ray ρ of σ disjoint from Q.
(3) The T-invariant prime divisors of X are exactly the $D_{y,v}$ and the D_ρ, and for any $f \in \mathbb{C}(\mathbb{P}^1)^*$ and $m \in M$,

$$\mathrm{div}(f\chi^m) = \sum_\rho \langle m, n_\rho\rangle D_\rho + \sum_{y,v} \mu(v)\big(\mathrm{ord}_y\, f + \langle m, v\rangle\big) D_{y,v},$$

where ρ runs over the rays of σ disjoint from Q, and (y, v) runs over the points $y \in \mathbb{P}^1$ and the vertices v of Δ_y.

Proof. By Definition-Lemma 7.2, the action of T on X is good with vertex o, and (X, ξ) is a Fano cone for any Reeb vector ξ of X. Since $\dim X - \dim T = 1$, by [LMS26, Construction 2.22, Theorem 5.1], X is T-equivariantly isomorphic to the affine variety of a proper polyhedral divisor on $(\mathbb{P}^1, N_{\mathbb{Q}})$ with complete locus, see also [LS10, Section 1]. In other words, there exist a strongly convex rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$ and σ-polyhedra $\Delta_y \subset N_{\mathbb{Q}}$, with $\Delta_y = \sigma$ for all but finitely many y and with $Q \subsetneq \sigma$ by properness, such that

$$\mathbb{C}[X] \cong \bigoplus_{m \in \sigma^\vee \cap M} H^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor \mathcal{D}(m) \rfloor)\big)\chi^m$$

as M-graded $\mathbb{C}$-algebras. We prove $\dim \sigma = 2$ together with (1).

(1) For $m \in \sigma^\vee \cap M$, the weight-m part $\mathbb{C}[X]_m$ is isomorphic to $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor \mathcal{D}(m) \rfloor))$, and $\mathbb{C}[X]_m = 0$ for $m \in M \backslash \sigma^\vee$. Since $Q = \sum_y \Delta_y$, the degree of the $\mathbb{Q}$-divisor $\mathcal{D}(m)$ is

$$\sum_y \min_{v \in \Delta_y} \langle m, v\rangle = \min_{z \in Q} \langle m, z\rangle = \deg \mathcal{D}(m).$$

Since Q is a σ-polyhedron and $Q \subsetneq \sigma$, we have $0 \notin Q$, for otherwise $\sigma = 0 + \sigma \subset Q$. Thus

$$\deg \mathcal{D}(m) > 0 \quad \text{for any } m \in \mathrm{Int}\, \sigma^\vee.$$

Let s be the number of points $y \in \mathbb{P}^1$ with $\Delta_y \neq \sigma$. Since $\min_{v \in \sigma} \langle m, v\rangle = 0$ for $m \in \sigma^\vee$, we have

$$\deg \lfloor \mathcal{D}(m) \rfloor \geq \deg \mathcal{D}(m) - s \quad \text{for any } m \in \sigma^\vee.$$

Hence, for any $m \in \operatorname{Int}\sigma^\vee \cap M$ and any integer $k > \frac{s}{\deg \mathcal{D}(m)}$, we have

$$\deg\lfloor \mathcal{D}(km)\rfloor \geq k \deg \mathcal{D}(m) - s > 0,$$

so $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor \mathcal{D}(km)\rfloor)) \neq 0$.

Let $\omega \subset M_{\mathbb{R}}$ be the cone generated by the weights m with $\mathbb{C}[X]_m \neq 0$. Since $\mathbb{C}[X]$ is generated over $\mathbb{C}$ by finitely many homogeneous elements of non-zero weights $m_1, \dots, m_n$, we have $\omega = \operatorname{Cone}(m_1, \dots, m_n)$, so ω is a closed rational polyhedral cone. Since $km \in \omega$ for any $m \in \operatorname{Int}\sigma^\vee \cap M$ and $k \gg 0$, we have

$$\operatorname{Int}\sigma^\vee \cap M_{\mathbb{Q}} \subset \omega \subset \sigma^\vee.$$

Since ω is closed, $\omega = \sigma^\vee$. Since the action of T is good, $\sigma^\vee$ is strictly convex, so $\dim \sigma = 2$. The weights $u \neq 0$ with $\mathbb{C}[X]_u \neq 0$ are non-zero sums of $m_1, \dots, m_n$, and $\sigma^\vee = \omega = \operatorname{Cone}(m_1, \dots, m_n)$. Hence an element $\xi \in N_{\mathbb{R}}$ is a Reeb vector if and only if $\langle m_i, \xi\rangle > 0$ for $i = 1, \dots, n$, that is, if and only if $\langle m, \xi\rangle > 0$ for any $m \in \sigma^\vee \backslash \{0\}$, that is, if and only if $\xi \in \operatorname{Int}\sigma$.

We prove (3). By [LMS26, Section 5], the T-invariant prime divisors of X are exactly the $D_{y,v}$ and the D_ρ. Here every vertex of Δ_y and every ray of σ disjoint from Q is extremal in the sense of [LS10, Section 1]: a vertex v of Δ_y is extremal, since the condition of [LS10, Section 1] asks that the restriction of $\mathcal{D}(m)$ to the prime divisor $\{y\}$ of $\mathbb{P}^1$ be big for any m in the interior of the normal cone of v, and the restriction of a $\mathbb{Q}$-divisor on $\mathbb{P}^1$ to a point has Iitaka dimension 0, which is the dimension of the point, so it is big, and a ray ρ of σ is extremal if and only if $\rho \cap Q = \emptyset$ [LS10, the sentence before Proposition 5.6]. The formula for $\operatorname{div}(f\chi^m)$ is [LMS26, Section 5, (6)].

(2) We compare the coefficients of a T-invariant canonical divisor of X with the formula of (3). Write $K_{\mathbb{P}^1} = \sum_y c_y[y]$ with $c_y \in \mathbb{Z}$, so that $\sum_y c_y = -2$. By [LS10, (2)],

$$K_X = \sum_{y,v} \big(\mu(v)c_y + \mu(v) - 1\big) D_{y,v} - \sum_\rho D_\rho$$

is a T-invariant canonical divisor of X. Since X is Gorenstein, K_X is Cartier. Thus K_X is principal by [LS10, Lemma 4.1], and we may write $K_X = \operatorname{div}(f\chi^{u'})$ with $f \in \mathbb{C}(\mathbb{P}^1)^*$ and $u' \in M$, see [LS10, Proposition 4.4].

Comparing the coefficients of $D_{y,v}$ in K_X and in $\operatorname{div}(f\chi^{u'})$ by (3), we obtain

$$\mu(v)\big(\operatorname{ord}_y f + \langle u', v\rangle\big) = \mu(v)c_y + \mu(v) - 1.$$

Let $b_y := \operatorname{ord}_y f - c_y \in \mathbb{Z}$. Dividing by $\mu(v)$, we obtain

$$\langle -u', v\rangle = b_y - 1 + \frac{1}{\mu(v)}$$

for any vertex v of Δ_y. Since f has finitely many zeros and poles and $c_y = 0$ for all but finitely many y, we have $b_y = 0$ for all but finitely many y, and

$$\sum_y b_y = \deg \operatorname{div}(f) - \deg K_{\mathbb{P}^1} = 0 - (-2) = 2.$$

Comparing the coefficients of D_ρ by (3), we obtain $\langle u', n_\rho\rangle = -1$, that is,

$$\langle -u', n_\rho\rangle = 1.$$

Thus $(b_y)_y$ and $-u'$ form a solution of the system of linear equations [Süß21, (4.1)] (with $a_y = b_y$), which has a unique solution. By the remark after [Süß21, Definition 4.1], $-u'$ is the weight of a generator of the canonical module of $\mathbb{C}[X]$. By Definition-Lemma 7.2, the weight of a generator of the canonical module of $\mathbb{C}[X]$ does not depend on the generator, so $\kappa = -u'$. Replacing $-u'$ by κ, we obtain $\langle \kappa, v\rangle = b_y - 1 + \frac{1}{\mu(v)}$ for any vertex v of Δ_y, and $\langle \kappa, n_\rho\rangle = 1$. □

Lemma 7.8. *Notation and conditions as in Set-up 7.1, and let $\mathcal{D}$, σ, and Q be as in Construction-Proposition 7.7. Then the following hold.*

(1) *For any* $\xi \in \operatorname{Int}\sigma$, *we have*

$$\operatorname{vol}(\mathrm{wt}_\xi) = \int_{\sigma^\vee} e^{-\langle m,\xi\rangle} \deg \mathcal{D}(m) dm,$$

where dm *is the Lebesgue measure on* $M_{\mathbb{R}}$ *for which* M *has covolume one.*

(2) *Assume that* X *is Gorenstein, and write* $\kappa = \zeta\theta$ *for the canonical character of* X *as in Definition-Lemma 7.2. Let* $U, W \in Q$ *be linearly independent, and let* $\det(U,W)$ *be the determinant of* U *and* W *with respect to a basis of* N. *Then, for any real numbers* $\lambda_1, \lambda_2 > 0$,

$$\widehat{\operatorname{vol}}(X \ni o) \le \frac{(\lambda_1\langle\kappa,U\rangle + \lambda_2\langle\kappa,W\rangle)^3}{|\det(U,W)|\lambda_1\lambda_2(\lambda_1+\lambda_2)}.$$

In particular, if $\langle\theta,U\rangle = \langle\theta,W\rangle > 0$ *and* $W - U = Le$ *for a primitive element* $e \in N$ *and a rational number* $L > 0$, *then* $\widehat{\operatorname{vol}}(X \ni o) \le \frac{4\zeta^3\langle\theta,U\rangle^2}{L}$.

Proof. (1) By Construction-Proposition 7.7(1), ξ is a Reeb vector of X. For $t > 0$, let $\mathfrak{a}_t(\mathrm{wt}_\xi) \subset \mathcal{O}_{X,o}$ be the ideal of elements g with $\mathrm{wt}_\xi(g) \ge t$, so that

$$\operatorname{vol}(\mathrm{wt}_\xi) = \lim_{t\to+\infty} \frac{3!}{t^3} \dim_{\mathbb{C}} \mathcal{O}_{X,o}/\mathfrak{a}_t(\mathrm{wt}_\xi).$$

The ideal $\mathfrak{a}_t := \mathfrak{a}_t(\mathrm{wt}_\xi) \cap \mathbb{C}[X]$ is $\bigoplus_{\langle m,\xi\rangle \ge t} \mathbb{C}[X]_m$. Let c_0 be the minimum of $\langle m,\xi\rangle$ over $m \in \sigma^\vee \cap M\backslash\{0\}$, which exists and is positive since $\xi \in \operatorname{Int}\sigma$. Since $\mathbb{C}[X]_0 = \mathbb{C}$, the maximal ideal $\mathfrak{m}_o$ of o is $\bigoplus_{m\ne 0} \mathbb{C}[X]_m$, so $\mathfrak{m}_o^k \subset \mathfrak{a}_t$ for any integer $k \ge \frac{t}{c_0}$. Thus $\mathcal{O}_{X,o}/\mathfrak{a}_t(\mathrm{wt}_\xi) \cong \mathbb{C}[X]/\mathfrak{a}_t$, and

$$\dim_{\mathbb{C}} \mathbb{C}[X]/\mathfrak{a}_t = \sum_{m\in\sigma^\vee\cap M,\ \langle m,\xi\rangle<t} h^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor \mathcal{D}(m)\rfloor)\big).$$

For any divisor D on $\mathbb{P}^1$, $h^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(D)) = \max(0, 1+\deg D)$. Let s be the number of points $y \in \mathbb{P}^1$ with $\Delta_y \ne \sigma$. For $m \in \sigma^\vee$, we have $\deg\mathcal{D}(m) \ge 0$ since $Q \subset \sigma$, and $\deg\mathcal{D}(m) - s \le \deg\lfloor\mathcal{D}(m)\rfloor \le \deg\mathcal{D}(m)$. Hence

$$\Big|h^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor\mathcal{D}(m)\rfloor)\big) - \deg\mathcal{D}(m)\Big| \le s+1 \quad \text{for any } m \in \sigma^\vee \cap M.$$

Let $\Pi := \{m \in \sigma^\vee \mid \langle m,\xi\rangle \le 1\}$, which is a compact convex polygon since $\xi \in \operatorname{Int}\sigma$. Then $\#(t\Pi \cap M) \le c_1 t^2$ for some constant c_1 and any $t \ge 1$. Since the sum for $\dim_{\mathbb{C}} \mathbb{C}[X]/\mathfrak{a}_t$ runs over a subset of $t\Pi \cap M$ and the summand for m differs from $\deg\mathcal{D}(m)$ by at most $s+1$, the numbers $\dim_{\mathbb{C}} \mathbb{C}[X]/\mathfrak{a}_t$ and $\sum_{m\in\sigma^\vee\cap M,\ \langle m,\xi\rangle<t} \deg\mathcal{D}(m)$ differ by at most $(s+1)c_1t^2$. The function $\deg\mathcal{D}$ is continuous on $\sigma^\vee$ and satisfies $\deg\mathcal{D}(rm) = r\deg\mathcal{D}(m)$ for $r \ge 0$. Thus

$$\sum_{m\in\sigma^\vee\cap M,\ \langle m,\xi\rangle<t} \deg\mathcal{D}(m) = t^3 \cdot \frac{1}{t^2} \sum_{m'\in\frac{1}{t}M\cap\Pi,\ \langle m',\xi\rangle<1} \deg\mathcal{D}(m'),$$

and

$$\frac{1}{t^2} \sum_{m'\in\frac{1}{t}M\cap\Pi,\ \langle m',\xi\rangle<1} \deg\mathcal{D}(m') = \int_\Pi \deg\mathcal{D}(m)dm + o(1)$$

as $t \to +\infty$, because the left-hand side is a Riemann sum of $\deg\mathcal{D}$ over Π for the lattice $\frac{1}{t}M$ of covolume t^{-2}. Since $(s+1)c_1t^2 = o(t^3)$, we obtain

$$\frac{3!}{t^3} \dim_{\mathbb{C}} \mathbb{C}[X]/\mathfrak{a}_t \to 6\int_\Pi \deg\mathcal{D}(m)dm \quad \text{as } t \to +\infty.$$

Therefore

$$\operatorname{vol}(\mathrm{wt}_\xi) = 6\int_\Pi \deg\mathcal{D}(m)dm.$$

By homogeneity, $\int_{r\Pi} \deg \mathcal{D}(m)dm = r^3 \int_\Pi \deg \mathcal{D}(m)dm$ for $r > 0$. Writing $e^{-\langle m,\xi\rangle} = \int_{\langle m,\xi\rangle}^\infty e^{-r}dr$ and applying Fubini's theorem, we obtain

$$\int_{\sigma^\vee} e^{-\langle m,\xi\rangle} \deg \mathcal{D}(m)dm = \int_0^\infty e^{-r}\Big(\int_{r\Pi} \deg \mathcal{D}(m)dm\Big)dr$$

and

$$\int_0^\infty e^{-r}\Big(\int_{r\Pi} \deg \mathcal{D}(m)dm\Big)dr = \int_0^\infty r^3 e^{-r}dr \cdot \int_\Pi \deg \mathcal{D}(m)dm = 6\int_\Pi \deg \mathcal{D}(m)dm.$$

This implies (1). Compare [Süß21, Propositions 4.5 and 4.6], where the volume is normalized as the lattice volume, which differs from the Lebesgue volume by the factor 3! in our case, see the remark after [Süß21, Example 4.7].

(2) Let $\tau := \mathrm{Cone}(U, W)$. Since $U, W \in Q \subset \sigma$ are linearly independent, $\tau \subset \sigma$ and $\dim \tau = 2$. Let $\xi := \lambda_1 U + \lambda_2 W$. Then $\xi \in \mathrm{Int}\,\tau \subset \mathrm{Int}\,\sigma$, so ξ is a Reeb vector of X by Construction-Proposition 7.7(1). By Definition-Lemma 7.2,

$$a(\mathrm{wt}_\xi, X) = \langle \kappa, \xi\rangle = \lambda_1\langle \kappa, U\rangle + \lambda_2\langle \kappa, W\rangle.$$

For $m \in \sigma^\vee$, we have $0 \le \deg \mathcal{D}(m) \le \min(\langle m, U\rangle, \langle m, W\rangle)$ since $U, W \in Q$. Moreover, $\sigma^\vee \subset \tau^\vee$ and $\min(\langle m, U\rangle, \langle m, W\rangle) \ge 0$ on $\tau^\vee$. By (1),

$$\mathrm{vol}(\mathrm{wt}_\xi) \le \int_{\tau^\vee} e^{-\langle m,\xi\rangle} \min(\langle m, U\rangle, \langle m, W\rangle)dm.$$

The linear map $M_\mathbb{R} \to \mathbb{R}^2$, $m \to (\langle m, U\rangle, \langle m, W\rangle)$, sends $\tau^\vee$ onto $\mathbb{R}^2_{\ge 0}$ and multiplies the Lebesgue measure dm by $|\det(U, W)|$. Writing $\min(x, y) = \int_0^\infty \mathbf{1}_{\{x\ge r, y\ge r\}}dr$ for $x, y \ge 0$, we obtain

$$\int_{\tau^\vee} e^{-\langle m,\xi\rangle} \min(\langle m, U\rangle, \langle m, W\rangle)dm = \frac{1}{|\det(U, W)|}\int_0^\infty \Big(\int_r^\infty e^{-\lambda_1 x}dx\Big)\Big(\int_r^\infty e^{-\lambda_2 y}dy\Big)dr,$$

and the right-hand side is

$$\frac{1}{|\det(U, W)|}\int_0^\infty \frac{e^{-(\lambda_1+\lambda_2)r}}{\lambda_1\lambda_2}dr = \frac{1}{|\det(U, W)|\lambda_1\lambda_2(\lambda_1 + \lambda_2)}.$$

Since $\widehat{\mathrm{vol}}(X \ni o) \le a(\mathrm{wt}_\xi, X)^3 \mathrm{vol}(\mathrm{wt}_\xi)$, the first inequality of (2) follows.

Assume that $\langle \theta, U\rangle = \langle \theta, W\rangle > 0$ and $W - U = Le$ as in (2). Then $\langle \theta, e\rangle = 0$. Since θ and e are primitive, there exists $f_0 \in N$ with $\langle \theta, f_0\rangle = 1$, and (e, f_0) is a basis of N. Writing $U = x_U e + \langle \theta, U\rangle f_0$ with $x_U \in \mathbb{Q}$, we obtain

$$|\det(U, W)| = |\det(U, U + Le)| = L|\det(U, e)| = L\langle \theta, U\rangle.$$

Taking $\lambda_1 = \lambda_2 = 1$ and using $\langle \kappa, U\rangle = \langle \kappa, W\rangle = \zeta\langle \theta, U\rangle$, we obtain

$$\widehat{\mathrm{vol}}(X \ni o) \le \frac{8\zeta^3\langle \theta, U\rangle^3}{2L\langle \theta, U\rangle} = \frac{4\zeta^3\langle \theta, U\rangle^2}{L}.$$

□

7.3. **Primitive canonical characters.** In this subsection, we prove Proposition 7.12 below, a bound for cones with primitive canonical characters. We only use the Reeb valuations of Reeb vectors generated by one or two points of the degree polyhedron to bound the local volume from above, and we do not determine which Reeb vector computes the local volume. The role of the primitivity of κ is explained in the paragraph after Definition-Lemma 7.2, and that of the failure of $\mathbb{Q}$-factoriality in the paragraph before Proposition 7.5.

We first prove three lemmas.

Lemma 7.9. *Notation and conditions as in Set-up 7.1, and let $\mathcal{D} = \sum_y \Delta_y \cdot [y]$ and σ be as in Construction-Proposition 7.7. Let $y \in \mathbb{P}^1$, and let $v \ne v'$ be vertices of Δ_y. Then $v - v' \notin \sigma$. In particular, for any $z \in N_\mathbb{R}$ such that $z + v$ and $z + v'$ are non-zero elements of σ, the vectors $z + v$ and $z + v'$ are linearly independent.*

Proof. Suppose that $v - v' = s_0$ for some $s_0 \in \sigma$. Then $s_0 \neq 0$, and $v' + 2s_0 \in \Delta_y$ since $\Delta_y + \sigma \subset \Delta_y$. Thus $v = \frac{1}{2}(v' + (v' + 2s_0))$ is the midpoint of two distinct points of Δ_y, which contradicts that v is a vertex of Δ_y.

Let $U := z + v$ and $W := z + v'$, and suppose that $W = cU$ for some $c \in \mathbb{R}$. Since σ is strongly convex and $U, W \in \sigma\backslash\{0\}$, we have $c > 0$, and $c \neq 1$ since $U \neq W$. Then $v' - v = W - U = (c-1)U$ belongs to σ or to $-\sigma$, which contradicts the first part applied to (v, v') or to (v', v). □

Lemma 7.10. *Notation and conditions as in Set-up 7.1, and let $\mathcal{D} = \sum_y \Delta_y \cdot [y]$, σ, Q, and μ_y be as in Construction-Proposition 7.7. Let $\delta := 2 - \sum_y (1 - \frac{1}{\mu_y})$, and let (p, h) be the coordinates on $N_{\mathbb{R}}$ with respect to a basis of N. Assume that X is Gorenstein, that the canonical character κ of X is primitive in M, and that $\langle \kappa, (p, h)\rangle = h$ for any $(p, h) \in N_{\mathbb{R}}$. Then the following hold.*

(1) $\delta > 0$, $h \geq \delta$ for any $(p, h) \in Q$, and $h > 0$ for any $(p, h) \in \sigma\backslash\{0\}$.
(2) There exist rational numbers $s_- < s_+$ such that $\sigma = \mathrm{Cone}((s_-, 1), (s_+, 1))$.
(3) For any $(P, H) \in Q$, we have

$$\widehat{\mathrm{vol}}(X \ni o) \leq \frac{8H}{s_+ - s_-}.$$

Proof. (1) Since X is Gorenstein and klt, $\delta > 0$ by [LS10, Corollary 5.8], where μ_y is the maximum of $\mu(v)$ over the vertices v of Δ_y as in the definition before [LS10, Theorem 4.7]. For $\xi \in \mathrm{Int}\,\sigma$, ξ is a Reeb vector by Construction-Proposition 7.7(1), so $\langle \kappa, \xi\rangle = a(\mathrm{wt}_\xi, X) > 0$ by Definition-Lemma 7.2 since X is klt. Hence $h \geq 0$ on σ.

Let b_y be the integers of Construction-Proposition 7.7(2). Since $Q = \sum_y \Delta_y$, any point of Q is of the form $z + s_0$, where $s_0 \in \sigma$ and z is a convex combination of points $\sum_y v_y$ with v_y a vertex of Δ_y for any y (and $v_y = 0$ if $\Delta_y = \sigma$). For such vertices v_y, Construction-Proposition 7.7(2) implies

$$\Big\langle \kappa, \sum_y v_y \Big\rangle = \sum_y \Big(b_y - 1 + \frac{1}{\mu(v_y)}\Big) = 2 - \sum_y \Big(1 - \frac{1}{\mu(v_y)}\Big) \geq 2 - \sum_y \Big(1 - \frac{1}{\mu_y}\Big) = \delta,$$

where the sums run over the finitely many y with $\Delta_y \neq \sigma$ or $b_y \neq 0$. Since $\langle \kappa, s_0\rangle \geq 0$, we obtain $h \geq \delta$ on Q.

Let ρ be a ray of σ and $n_\rho \in N$ the primitive generator of ρ, and suppose that $\langle \kappa, n_\rho\rangle = 0$. Then $h = 0$ on ρ, so $\rho \cap Q = \emptyset$. By Construction-Proposition 7.7(2), $\langle \kappa, n_\rho\rangle = 1$, a contradiction. Thus $h > 0$ on $\sigma\backslash\{0\}$.

(2) By (1) and Construction-Proposition 7.7, σ is a two-dimensional cone contained in $\{h > 0\} \cup \{0\}$. Thus each ray of σ contains a unique point $(s, 1)$, and $s \in \mathbb{Q}$ since the rays are rational. This implies (2).

(3) Let $w := s_+ - s_-$, $s := \frac{s_- + s_+}{2}$, and $\xi := (s, 1) \in \mathrm{Int}\,\sigma$. By Definition-Lemma 7.2, $a(\mathrm{wt}_\xi, X) = \langle \kappa, \xi\rangle = 1$. Since $(P, H) \in Q$, we have $0 \leq \deg \mathcal{D}(m) \leq \langle m, (P, H)\rangle$ for $m \in \sigma^\vee$. Let (u, v) be the dual coordinates on $M_{\mathbb{R}}$, so that $\langle (u, v), (p, h)\rangle = up + vh$. Then

$$\sigma^\vee = \{(u, v) \mid v \geq -s_- u \text{ and } v \geq -s_+ u\},$$

and the condition is $v \geq -s_- u$ when $u \geq 0$ and $v \geq -s_+ u$ when $u \leq 0$. By Lemma 7.8(1),

$$\mathrm{vol}(\mathrm{wt}_\xi) \leq \int_{\sigma^\vee} e^{-su-v}(Pu + Hv)dudv.$$

Substituting $v = -s_- u + r$ on $\{u \geq 0\}$, we obtain

$$\int_0^\infty \int_{-s_- u}^\infty e^{-su-v}(Pu + Hv)dvdu = \int_0^\infty e^{-(s-s_-)u}\big(H + (P - Hs_-)u\big)du,$$

which is equal to

$$\frac{H}{s - s_-} + \frac{P - Hs_-}{(s - s_-)^2}.$$

Substituting $u=-u_1$ and $v=s_+u_1+r$ on $\{u\le 0\}$, we obtain

$$\int_{-\infty}^{0}\int_{-s_+u}^{\infty} e^{-su-v}(Pu+Hv)dvdu=\int_0^{\infty} e^{-(s_+-s)u_1}\big(H+(Hs_+-P)u_1\big)du_1,$$

which is equal to

$$\frac{H}{s_+-s}+\frac{Hs_+-P}{(s_+-s)^2}.$$

Since $s-s_-=s_+-s=\frac{w}{2}$ and $(P-Hs_-)+(Hs_+-P)=Hw$, the sum of the integrals over $\{u\ge 0\}$ and $\{u\le 0\}$ is

$$\frac{4H}{w}+\frac{4Hw}{w^2}=\frac{8H}{w}.$$

Thus $\widehat{\mathrm{vol}}(X\ni o)\le a(\mathrm{wt}_\xi,X)^3\,\mathrm{vol}(\mathrm{wt}_\xi)\le\frac{8H}{w}$. □

Lemma 7.11. *Notation and conditions as in Set-up 7.1, and let $\mathcal{D}=\sum_y\Delta_y\cdot[y]$ and σ be as in Construction-Proposition 7.7. Let $y_1,y_2\in\mathbb{P}^1$. Assume that for any $y\in\mathbb{P}^1\backslash\{y_1,y_2\}$, $\Delta_y=v_y+\sigma$ for some $v_y\in N$. Then X is toric.*

Proof. Since $\Delta_y=\sigma$ for all but finitely many y, there exists $y'\in\mathbb{P}^1\backslash\{y_1\}$ with $\Delta_{y'}=\sigma$, and if $y_1=y_2$, the assumption still holds after replacing y_2 with y'. Possibly replacing y_2 with y', we may assume that $y_1\ne y_2$. Let Y_0 be the finite set of $y\in\mathbb{P}^1\backslash\{y_1,y_2\}$ with $\Delta_y\ne\sigma$. For $y\in Y_0$, let $f_y\in\mathbb{C}(\mathbb{P}^1)^*$ with $\mathrm{div}(f_y)=[y_1]-[y]$, and for $m\in M$, let $f_m:=\prod_{y\in Y_0}f_y^{\langle m,v_y\rangle}$. Then $f_{m+m'}=f_mf_{m'}$, and

$$E(m):=\mathrm{div}(f_m)=\sum_{y\in Y_0}\langle m,v_y\rangle\big([y_1]-[y]\big).$$

Let $\Delta'_{y_1}:=\Delta_{y_1}+\sum_{y\in Y_0}v_y$, $\Delta'_{y_2}:=\Delta_{y_2}$, and $\Delta'_y:=\sigma$ for $y\ne y_1,y_2$, and let $\mathcal{D}'(m):=\sum_y\big(\min_{v\in\Delta'_y}\langle m,v\rangle\big)[y]$ for $m\in\sigma^\vee$. Then $\mathcal{D}'(m)=\mathcal{D}(m)+E(m)$, and $E(m)$ is an integral divisor, so $\lfloor\mathcal{D}'(m)\rfloor=\lfloor\mathcal{D}(m)\rfloor+E(m)$. Thus the maps

$$H^0\big(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(\lfloor\mathcal{D}(m)\rfloor)\big)\to H^0\big(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(\lfloor\mathcal{D}'(m)\rfloor)\big),\quad \phi\to\phi f_m^{-1},$$

for $m\in\sigma^\vee\cap M$ define an isomorphism of M-graded $\mathbb{C}$-algebras

$$\mathbb{C}[X]\cong\bigoplus_{m\in\sigma^\vee\cap M}H^0\big(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(\lfloor\mathcal{D}'(m)\rfloor)\big)\chi^m.$$

Choose a coordinate z on $\mathbb{P}^1$ with $y_1=\{z=0\}$ and $y_2=\{z=+\infty\}$. For integers a_0,a_∞, the space $H^0(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(a_0[y_1]+a_\infty[y_2]))$ has the basis z^i, $-a_0\le i\le a_\infty$. Since i is an integer,

$$-\Big\lfloor\min_{v\in\Delta'_{y_1}}\langle m,v\rangle\Big\rfloor\le i\le\Big\lfloor\min_{v\in\Delta'_{y_2}}\langle m,v\rangle\Big\rfloor$$

if and only if $\langle m,v\rangle+i\ge 0$ for any $v\in\Delta'_{y_1}$ and $\langle m,v\rangle-i\ge 0$ for any $v\in\Delta'_{y_2}$. Let $\tau'\subset N_\mathbb{R}\oplus\mathbb{R}$ be the cone generated by $\sigma\times\{0\}$, $\Delta'_{y_1}\times\{1\}$, and $\Delta'_{y_2}\times\{-1\}$, which is a rational polyhedral cone. Then $\mathbb{C}[X]$ is isomorphic to the semigroup algebra of the monoid $\tau'^\vee\cap(M\oplus\mathbb{Z})$, where $z^i\chi^m$ corresponds to (m,i). Since $\dim X=3$, $\tau'^\vee$ is full-dimensional, so τ' is strongly convex. Thus X is an affine toric variety. □

Proposition 7.12. *Notation and conditions as in Set-up 7.1. Assume that X is Gorenstein and not toric, that X is not $\mathbb{Q}$-factorial at o, and that the canonical character of X is primitive in M. Then*

$$\widehat{\mathrm{vol}}(X\ni o)\le\frac{20}{3}.$$

Proof. The proof is long, so we divide it into several steps. Let $\mathcal{D}=\sum_y\Delta_y\cdot[y]$, σ, Q, $\mu(v)$, μ_y, and b_y be as in Construction-Proposition 7.7, and let κ be the canonical character of X. Then $\kappa=\theta$ and $\zeta=1$.

Step 1. In this step, we fix coordinates on $N_\mathbb{R}$ and compute the rank of $\mathrm{Cl}(X)$.

Since κ is primitive in M, there exists a basis (e_0, f_0) of N such that the coordinates (p, h) on $N_{\mathbb{R}}$ with respect to (e_0, f_0) satisfy $\langle \kappa, (p, h)\rangle = h$. We call h the *height* of (p, h), and $\frac{p}{h}$ the *slope* of (p, h) if $h > 0$. Let $\delta := 2 - \sum_y (1 - \frac{1}{\mu_y})$. By Lemma 7.10, $\delta > 0$, any point of Q has height at least δ, and $\sigma = \mathrm{Cone}((s_-, 1), (s_+, 1))$ for rational numbers $s_- < s_+$. Let $w := s_+ - s_-$. Since the slopes of the points of $\sigma\backslash\{0\}$ are in $[s_-, s_+]$, we have

$$w \geq \left|\frac{p}{h} - \frac{p'}{h'}\right| \quad \text{for any } (p, h), (p', h') \in \sigma\backslash\{0\}. \tag{7.2}$$

For a vertex v of Δ_y, we have $v \in \frac{1}{\mu(v)}N$, and the height of v is $b_y - 1 + \frac{1}{\mu(v)}$ by Construction-Proposition 7.7(2). Let ϵ be the number of rays of σ disjoint from Q. For any such ray ρ, the primitive generator of ρ is $(j, 1)$ for some $j \in \mathbb{Z}$ by Construction-Proposition 7.7(2), so the slope of ρ is an integer.

Let $D_{y,v}$ and D_ρ be the T-invariant prime divisors of Construction-Proposition 7.7. By [LS10, Theorem 4.2], $\mathrm{Cl}(X)$ is generated by $\mathrm{Cl}(\mathbb{P}^1)$ and the divisors $D_{y,v}$ and D_ρ, subject to the relations $[y] = \sum_v \mu(v) D_{y,v}$ and $\mathrm{div}(\chi^m) = 0$ for $m \in M$. For $y \in \mathbb{P}^1$, let $F_y := \sum_v \mu(v) D_{y,v}$, where v runs over the vertices of Δ_y. By Construction-Proposition 7.7(3) with $m = 0$,

$$\mathrm{div}(f) = \sum_y \mathrm{ord}_y(f) F_y \quad \text{for any } f \in \mathbb{C}(\mathbb{P}^1)^*.$$

Since $\mathrm{Cl}(\mathbb{P}^1) \cong \mathbb{Z}$ and any two points of $\mathbb{P}^1$ are linearly equivalent, eliminating $\mathrm{Cl}(\mathbb{P}^1)$ by the relations $[y] = F_y$ shows that the classes of the divisors F_y in $\mathrm{Cl}(X)$ are equal to a common class F, and that the remaining relations are $F_y = F$ and $\mathrm{div}(\chi^m) = 0$ for $m \in M$. In $\mathrm{Cl}(X) \otimes \mathbb{Q}$, we eliminate one divisor $D_{y,v}$ for each y by the relations $F_y = F$. Thus $\mathrm{Cl}(X) \otimes \mathbb{Q}$ is generated by F, the remaining $\sum_y (\#\mathrm{vertices}(\Delta_y) - 1)$ divisors $D_{y,v}$, and the ϵ divisors D_ρ, subject to the relations $\mathrm{div}(\chi^{m_1})$ and $\mathrm{div}(\chi^{m_2})$ for a basis m_1, m_2 of M, which are linearly independent by [LS10, Lemma 4.6]. Hence

$$\mathrm{rank}\,\mathrm{Cl}(X) = \sum_y \big(\#\mathrm{vertices}(\Delta_y) - 1\big) + \epsilon - 1. \tag{7.3}$$

If $\mathrm{Cl}(X)$ is a torsion group, then any Weil divisor on X is $\mathbb{Q}$-Cartier. Since X is not $\mathbb{Q}$-factorial at o, we have $\mathrm{rank}\,\mathrm{Cl}(X) > 0$ (cf. [ABHW18, Corollaries 2.13–2.14]).

Step 2. In this step, we prove $\widehat{\mathrm{vol}}(X \ni o) \leq \frac{20}{3}$ unless exactly three of the integers μ_y are larger than one and $\Delta_y = v_y + \sigma$ with $v_y \in N$ for any other y.

Let $y \in \mathbb{P}^1$ with $\mu_y = 1$. Then any vertex v of Δ_y is in N and has height b_y. Since Δ_y is the sum of σ and the convex hull of the vertices of Δ_y, and since $e_0 = (1, 0)$, there exist $v_y \in N$ and an integer $L_y \geq 0$ such that

$$\Delta_y = [v_y, v_y + L_y e_0] + \sigma,$$

where $v_y = 0$ and $L_y = 0$ if $\Delta_y = \sigma$. Let t be the number of points y with $\mu_y > 1$, let k be the number of points y with $\mu_y = 1$ and $L_y > 0$, and let $L := \sum_{\mu_y = 1} L_y$. Then $L \geq k$. For any y with $\mu_y > 1$, let u_y be a vertex of Δ_y with $\mu(u_y) = \mu_y$. Let

$$z_1 := \sum_{\mu_y > 1} u_y + \sum_{\mu_y = 1} v_y.$$

Then $[z_1, z_1 + Le_0] \subset Q$, and the height of z_1 is

$$\sum_{\mu_y > 1} \left(b_y - 1 + \frac{1}{\mu_y}\right) + \sum_{\mu_y = 1} b_y = 2 - \sum_y \left(1 - \frac{1}{\mu_y}\right) = \delta.$$

If $L > 0$, then Lemma 7.8(2) with $U = z_1$, $W = z_1 + Le_0$, and $e = e_0$ implies

$$\widehat{\mathrm{vol}}(X \ni o) \leq \frac{4\delta^2}{L}. \tag{7.4}$$

If $t+k\le 2$, then $\Delta_y = v_y+\sigma$ with $v_y\in N$ for all but at most two points y, so X is toric by Lemma 7.11, a contradiction. Thus $t+k\ge 3$. By [LS10, Remark 5.9(i)], $t\le 3$. We have the following cases.

- If $t=0$, then $\delta=2$ and $L\ge k\ge 3$, so $\widehat{\mathrm{vol}}(X\ni o)\le \frac{16}{3}$ by (7.4).
- If $t=1$, then $\delta\le 1+\frac{1}{2}=\frac{3}{2}$ and $L\ge k\ge 2$, so $\widehat{\mathrm{vol}}(X\ni o)\le \frac{9}{2}$ by (7.4).
- If $t=2$, then $\delta\le \frac{1}{2}+\frac{1}{2}=1$ and $L\ge k\ge 1$, so $\widehat{\mathrm{vol}}(X\ni o)\le 4$ by (7.4).
- If $t=3$ and $L>0$, then $\delta\le \frac{3}{2}-1=\frac{1}{2}$ and $L\ge 1$, so $\widehat{\mathrm{vol}}(X\ni o)\le 1$ by (7.4).

In each case, $\widehat{\mathrm{vol}}(X\ni o)<\frac{20}{3}$. This finishes **Step 2**.

From now on, we assume that $t=3$ and $L=0$. Let y_1,y_2,y_3 be the three points with $\mu_{y_i}>1$, and let n_i be the number of vertices of Δ_{y_i}. For $y\neq y_1,y_2,y_3$, we have $\Delta_y=v_y+\sigma$ with $v_y\in N$, and $v_y=0$ if $\Delta_y=\sigma$. Let $z_0:=\sum_{y\neq y_1,y_2,y_3} v_y\in N$. Then

$$Q = z_0+\Delta_{y_1}+\Delta_{y_2}+\Delta_{y_3},$$

and (7.3) and $\operatorname{rank}\mathrm{Cl}(X)>0$ imply

$$n_1+n_2+n_3+\epsilon-4>0. \tag{7.5}$$

For vertices u_i' of Δ_{y_i}, $i=1,2,3$, the point $z_0+u_1'+u_2'+u_3'\in Q$ is in $\mathrm{lcm}(\mu(u_1'),\mu(u_2'),\mu(u_3'))^{-1}N$, and it has height

$$\sum_{i=1}^{3}\Big(b_{y_i}-1+\frac{1}{\mu(u_i')}\Big)+\sum_{y\neq y_1,y_2,y_3} b_y = \frac{1}{\mu(u_1')}+\frac{1}{\mu(u_2')}+\frac{1}{\mu(u_3')}-1. \tag{7.6}$$

By [LS10, Remark 5.9(i)], up to reordering, $(\mu_{y_1},\mu_{y_2},\mu_{y_3})$ is one of $(2,2,r)$ with $r\ge 2$, $(2,3,3)$, $(2,3,4)$, and $(2,3,5)$. For $i=1,2,3$, let u_i be a vertex of Δ_{y_i} with $\mu(u_i)=\mu_{y_i}$, and let $U:=z_0+u_1+u_2+u_3\in Q$. By (7.6), the height of U is δ.

Step 3. In this step, we treat the case when $(\mu_{y_1},\mu_{y_2},\mu_{y_3})$ is $(2,3,3)$, $(2,3,4)$, or $(2,3,5)$ up to reordering.

Let $d:=\mathrm{lcm}(\mu_{y_1},\mu_{y_2},\mu_{y_3})$. Then $d=6,12,30$ and $\delta=\frac{1}{d}$ respectively. Since $U\in\frac{1}{d}N$ has height $\frac{1}{d}$, we have $U=(\frac{c}{d},\frac{1}{d})$ for some $c\in\mathbb{Z}$, and the slope of U is c.

Suppose that $\epsilon\ge 1$. Then σ has a ray ρ disjoint from Q, and the slope j of ρ is an integer by **Step 1**. Since $U\in Q$, $U\notin\rho$, so $c\neq j$ and $|c-j|\ge 1$. By (7.2), $w\ge 1$. By Lemma 7.10(3), $\widehat{\mathrm{vol}}(X\ni o)\le\frac{8}{dw}\le\frac{8}{d}\le\frac{4}{3}$.

Suppose that $\epsilon=0$. By (7.5), $n_i\ge 2$ for some i. Let $l:=\mu_{y_i}$, let $u_i'\neq u_i$ be a vertex of Δ_{y_i}, and let $l':=\mu(u_i')$, so $1\le l'\le l$. Let $W:=U-u_i+u_i'\in Q$. Let $\{i',i''\}:=\{1,2,3\}\backslash\{i\}$, let $\mu':=\mu_{y_{i'}}$ and $\mu'':=\mu_{y_{i''}}$, let $d':=\mathrm{lcm}(\mu',\mu'',l')$, and let

$$e':=d'\Big(\frac{1}{\mu'}+\frac{1}{\mu''}+\frac{1}{l'}-1\Big).$$

By (7.6), the height of W is $\frac{e'}{d'}$, so e' is an integer with $e'\ge d'\delta>0$, and $W=(\frac{e}{d'},\frac{e'}{d'})$ for some $e\in\mathbb{Z}$. By Lemma 7.9, U and W are linearly independent, so $\frac{e}{e'}\neq c$. Thus $\big|\frac{e}{e'}-c\big|=\frac{|e-ce'|}{e'}\ge\frac{1}{e'}$, and $w\ge\frac{1}{e'}$ by (7.2). By Lemma 7.10(3) applied to U,

$$\widehat{\mathrm{vol}}(X\ni o)\le\frac{8}{dw}\le\frac{8e'}{d}.$$

The values of e' for all possible (l,l') are listed in Table 1. They are obtained by substituting μ',μ'', and l' into the definition of e'. The maximum of e' is 5, 7, and 8 in the three cases, so $\widehat{\mathrm{vol}}(X\ni o)$ is at most $\frac{8\cdot 5}{6}=\frac{20}{3}$, $\frac{8\cdot 7}{12}=\frac{14}{3}$, and $\frac{8\cdot 8}{30}=\frac{32}{15}$ respectively. Thus $\widehat{\mathrm{vol}}(X\ni o)\le\frac{20}{3}$ in **Step 3**.

Step 4. In this step, we treat the case when $(\mu_{y_1},\mu_{y_2},\mu_{y_3})=(2,2,r)$ with $r\ge 2$.

In this case, $\delta=\frac{1}{r}$. We consider three cases.

$(\mu_{y_1},\mu_{y_2},\mu_{y_3})$	l	$l'=1,2,\dots,l$	e'
$(2,3,3)$	2	$1,2$	$2,1$
$(2,3,3)$	3	$1,2,3$	$5,2,1$
$(2,3,4)$	2	$1,2$	$7,1$
$(2,3,4)$	3	$1,2,3$	$3,1,1$
$(2,3,4)$	4	$1,2,3,4$	$5,2,1,1$
$(2,3,5)$	2	$1,2$	$8,1$
$(2,3,5)$	3	$1,2,3$	$7,2,1$
$(2,3,5)$	5	$1,2,3,4,5$	$5,2,1,1,1$

TABLE 1. The integers $e'=d'(\frac{1}{\mu'}+\frac{1}{\mu''}+\frac{1}{l'}-1)$ in **Step 3**, listed in the order of l'.

Case (a): $n_3\geq 2$. Let $u_3'\neq u_3$ be a vertex of Δ_{y_3}, let $l':=\mu(u_3')$, so $1\leq l'\leq r$, and let $W:=z_0+u_1+u_2+u_3'\in Q$. By (7.6), the heights of U and W are $\frac{1}{r}$ and $\frac{1}{l'}$. Since $z_0+u_1+u_2\in\frac{1}{2}N$, writing $z_0+u_1+u_2=(x_0,h_0)$ in the coordinates of **Step 1**, we have $x_0\in\frac{1}{2}\mathbb{Z}$, and

$$U=\Big(x_0+\frac{j_U}{r},\frac{1}{r}\Big),\quad W=\Big(x_0+\frac{j_W}{l'},\frac{1}{l'}\Big)$$

for some $j_U,j_W\in\mathbb{Z}$. Then

$$\det(U,W)=\frac{x_0(r-l')+j_U-j_W}{rl'}.$$

By Lemma 7.9, $\det(U,W)\neq 0$. Since the numerator is in $\frac{1}{2}\mathbb{Z}$, $|\det(U,W)|\geq\frac{1}{2rl'}$. Lemma 7.8(2) with $\lambda_1=r$ and $\lambda_2=l'$ implies

$$\widehat{\mathrm{vol}}(X\ni o)\leq\frac{(r\cdot\frac{1}{r}+l'\cdot\frac{1}{l'})^3}{\frac{1}{2rl'}\cdot rl'(r+l')}=\frac{16}{r+l'}\leq\frac{16}{3}.$$

Suppose that $r=2$. Since the three integers μ_{y_i} are equal to 2, possibly reordering y_1,y_2,y_3, Case (a) covers the case when $n_i\geq 2$ for some i. If $n_1=n_2=n_3=1$, then $\epsilon\geq 2$ by (7.5), so $\epsilon=2$. Thus the slopes s_- and s_+ are distinct integers by **Step 1**, so $w\geq 1$. By Lemma 7.10(3) applied to U, $\widehat{\mathrm{vol}}(X\ni o)\leq\frac{8}{rw}\leq 4$. This finishes the case $r=2$.

From now on, we assume that $r\geq 3$ and $n_3=1$.

Case (b): $r\geq 3$, $n_3=1$, *and* $\epsilon\geq 1$. Then σ has a ray ρ disjoint from Q, and the slope j of ρ is an integer by **Step 1**. Since $U\in\mathrm{lcm}(2,r)^{-1}N$ has height $\frac{1}{r}$, the slope of U is in $r\,\mathrm{lcm}(2,r)^{-1}\mathbb{Z}\subset\frac{1}{2}\mathbb{Z}$. Since $U\in Q$, $U\notin\rho$, so the slope of U is different from j. Thus $w\geq\frac{1}{2}$ by (7.2). By Lemma 7.10(3) applied to U, $\widehat{\mathrm{vol}}(X\ni o)\leq\frac{16}{r}\leq\frac{16}{3}$.

Case (c): $r\geq 3$, $n_3=1$, *and* $\epsilon=0$. By (7.5), $n_1+n_2\geq 4$.

Suppose first that for some $i\in\{1,2\}$, Δ_{y_i} has two vertices $u\neq u'$ with $\mu(u)=\mu(u')=:l$. Then $l\in\{1,2\}$, and $u-u'\in\frac{1}{l}N$ has height 0, so $u-u'=(x,0)$ with $|x|\geq\frac{1}{l}$. Replacing u_i by u and by u' in U, we obtain two points of Q whose difference is $\pm(x,0)$. By (7.6), both points have height

$$H=\frac{1}{l}+\frac{1}{2}+\frac{1}{r}-1=\frac{1}{l}-\frac{1}{2}+\frac{1}{r}.$$

By Lemma 7.8(2) with $e=e_0$ and $L=|x|$, we have $\widehat{\mathrm{vol}}(X\ni o)\leq\frac{4H^2}{|x|}\leq 4lH^2$. If $l=1$, then

$$\widehat{\mathrm{vol}}(X\ni o)\leq 4\Big(\frac{1}{2}+\frac{1}{r}\Big)^2\leq\frac{25}{9}.$$

If $l=2$, then $H=\frac{1}{r}$ and $\widehat{\mathrm{vol}}(X\ni o)\leq\frac{8}{r^2}\leq\frac{8}{9}$.

Suppose next that for $i=1,2$, the integers $\mu(v)$ for the vertices v of Δ_{y_i} are pairwise different. Since $\mu(v)\leq\mu_{y_i}=2$ for any vertex v of Δ_{y_i}, we have $n_1,n_2\leq 2$, so $n_1=n_2=2$ since $n_1+n_2\geq 4$.

Since $\mu_{y_i}=2$, the vertices of Δ_{y_i} are u_i with $\mu(u_i)=2$ and a vertex u_i' with $\mu(u_i')=1$. Let $\eta_i:=u_i'-u_i$. Since $u_i'\in N$, $u_i\in\frac{1}{2}N$, and the heights of u_i' and u_i are b_{y_i} and $b_{y_i}-\frac{1}{2}$, we have

$$\eta_i=\Big(\frac{\gamma_i}{2},\frac{1}{2}\Big)\quad\text{for some }\gamma_i\in\mathbb{Z}.$$

The vertex u_3 is the unique vertex of Δ_{y_3}, and $U=z_0+u_1+u_2+u_3$ has height $\frac{1}{r}$.

If $\gamma_1\neq\gamma_2$, then $U+\eta_1=z_0+u_1'+u_2+u_3$ and $U+\eta_2=z_0+u_1+u_2'+u_3$ are points of Q of height $\frac{1}{2}+\frac{1}{r}$, and $(U+\eta_1)-(U+\eta_2)=(\frac{\gamma_1-\gamma_2}{2},0)$ with $\big|\frac{\gamma_1-\gamma_2}{2}\big|\geq\frac{1}{2}$. By Lemma 7.8(2),

$$\widehat{\mathrm{vol}}(X\ni o)\leq 8\Big(\frac{1}{2}+\frac{1}{r}\Big)^2\leq\frac{50}{9}<\frac{20}{3}.$$

If $\gamma_1=\gamma_2=:\gamma$, then $u_1+u_2=u_1'+u_2'-(\gamma,1)\in N$. Since $z_0\in N$ and $u_3\in\frac{1}{r}N$, we have $U\in\frac{1}{r}N$, so $U=(\frac{c}{r},\frac{1}{r})$ for some $c\in\mathbb{Z}$. Let

$$W:=U+\eta_1+\eta_2=U+(\gamma,1)=z_0+u_1'+u_2'+u_3\in Q,$$

which has height $\frac{r+1}{r}$. Then $\det(U,W)=\det(U,(\gamma,1))=\frac{c-\gamma}{r}$. If $c=\gamma$, then $U=\frac{1}{r}(\gamma,1)$, so $\eta_1=\frac{1}{2}(\gamma,1)=\frac{r}{2}U\in\sigma$, which contradicts Lemma 7.9 since $\eta_1=u_1'-u_1$. Thus $c\neq\gamma$ and $|\det(U,W)|\geq\frac{1}{r}$. Lemma 7.8(2) with $\lambda_1=r+1$ and $\lambda_2=1$ implies

$$\widehat{\mathrm{vol}}(X\ni o)\leq\frac{\big((r+1)\cdot\frac{1}{r}+\frac{r+1}{r}\big)^3}{\frac{1}{r}(r+1)(r+2)}=\frac{8(r+1)^2}{r^2(r+2)}\leq\frac{9}{2},$$

where the last inequality follows from

$$9r^2(r+2)-16(r+1)^2=(r-2)(9r^2+20r+8)\geq 0.$$

Thus $\widehat{\mathrm{vol}}(X\ni o)\leq\frac{20}{3}$ in all cases, and we are done. □

7.4. **Non-primitive canonical characters.** In this subsection, we prove Theorem 7.3. The toric case and the case of a primitive canonical character follow from Theorem 6.1 and Propositions 7.5 and 7.12. For a non-primitive canonical character, we write the Cox ring of X in the normal form of [ABHW18, Proposition 4.7], bound the exponents of the normal form by Proposition 5.6(2) for an explicit Cox monomial, and bound the local volume by Lemma 7.8(2).

Remark 7.13. In **Step 4** of the proof below, we use a property of the normal form of [ABHW18, Proposition 4.7] which is not contained in the statement of [ABHW18, Proposition 4.7]: if the exponents of the defining matrix P satisfy $l_{i1}\geq\cdots\geq l_{in_i}$ and $n_il_{i1}\geq 2$ for any i, then so do the exponents of the matrix to which [ABHW18, Proposition 4.7] brings P. The reason is that the proof of [ABHW18, Proposition 4.7] changes P only by exchanging the data of column blocks, by elementary row operations among the last two rows, and by adding multiples of the upper rows to the last row, that is, by admissible operations of types (ii)–(iv) of [ABHW18, Remark 2.3], and admissible operations of types (ii)–(iv) only permute the blocks of exponents.

Proof of Theorem 7.3. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove the theorem when X is toric or the canonical character of X is primitive, and reduce the remaining case to the inequality $\widehat{\mathrm{vol}}(X\ni o)\leq\frac{27}{4}$.

Let $\kappa=\zeta\theta$ be the canonical character of X, where $\theta\in M$ is primitive and ζ is a positive integer. By Definition-Lemma 7.2, the action of T on X is good with vertex o, and

$$a(\mathrm{wt}_\eta,X)=\langle\kappa,\eta\rangle=\zeta\langle\theta,\eta\rangle\tag{7.7}$$

for any Reeb vector η of X. By Lemma 7.4(1), $\mathbb{C}[X]^*=\mathbb{C}^*$.

Suppose that X is toric, that is, $X\cong X_\tau$ for a lattice N' of rank three and a strongly convex rational polyhedral cone $\tau\subset N'_{\mathbb{R}}$. For any $m\in\tau^\perp\cap\mathrm{Hom}(N',\mathbb{Z})$, the character χ^m is a unit of $\mathbb{C}[X_\tau]$. Since $\mathbb{C}[X]^*=\mathbb{C}^*$, we have $\tau^\perp=0$, that is, $\dim\tau=3$. Let $x\in X_\tau$ be the torus-fixed point. Any point $x'\neq x$ of X_τ belongs to the torus orbit $O(\tau')$ corresponding to a proper face τ' of τ, and $O(\tau')\subset X_{\tau'}\subset X_\tau$ [CLS11, Theorem 3.2.6]. Since $\dim\tau'\leq 2$, the cone τ' is simplicial. Let

$k := \dim \tau'$, let $u_1, \dots, u_k$ be the primitive generators of the rays of τ', let $N'' := \mathrm{Span}(\tau') \cap N'$, and let $N_1 := \mathbb{Z}u_1 + \cdots + \mathbb{Z}u_k$. Since τ' is simplicial, $N_1 \subset N''$ is a sublattice of finite index. Let $X_{\tau',N''}$ be the affine toric variety of τ' with respect to N'', and let x'' be the torus-fixed point of $X_{\tau',N''}$. Since N'/N'' is torsion free, N'' is spanned by a subset of a basis of N', so there exists a toric isomorphism $X_{\tau'} \cong X_{\tau',N''} \times (\mathbb{C}^*)^{3-k}$ [CLS11, Proposition 3.3.11(a)], under which $O(\tau')$ corresponds to $\{x''\} \times (\mathbb{C}^*)^{3-k}$. The affine toric variety of τ' with respect to N_1 is $\mathbb{C}^k$ [CLS11, Example 1.3.20]. Since $u_1, \dots, u_k$ are primitive in N'' and belong to N_1, Lemma 6.2 for $N_1 \subset N''$ and τ' shows that the toric morphism $\mathbb{C}^k \to X_{\tau',N''}$ is the quotient morphism by the finite group $G := N''/N_1$, and that G acts freely on $\mathbb{C}^k$ outside a closed subset of codimension at least two. Moreover, G fixes the origin of $\mathbb{C}^k$, whose image is x''. Thus the germ $X_\tau \ni x'$ is isomorphic to the germ of $(\mathbb{C}^k \times (\mathbb{C}^*)^{3-k})/G$ at the image of a G-fixed point, where G acts on the first factor. In particular, $X_\tau \ni x'$ is LCIQ.

Since $X \ni o$ is not LCIQ, any isomorphism $X \cong X_\tau$ maps o to x. Since X_τ is Gorenstein and klt and $X_\tau \ni x$ is not LCIQ, Theorem 6.1 implies that

$$\widehat{\mathrm{vol}}(X \ni o) = \widehat{\mathrm{vol}}(X_\tau \ni x) \le q.$$

Suppose that X is not toric and that $\zeta = 1$. If X is $\mathbb{Q}$-factorial at o, then $X \ni o$ is LCIQ by Proposition 7.5, which is not the case. Thus X is not $\mathbb{Q}$-factorial at o, and Proposition 7.12 implies that $\widehat{\mathrm{vol}}(X \ni o) \le \frac{20}{3}$.

By Lemma 4.3, $q > 7 > \frac{27}{4} > \frac{20}{3}$. Thus it suffices to show that $\widehat{\mathrm{vol}}(X \ni o) \le \frac{27}{4}$ when X is not toric and $\zeta \ge 2$. From now on, we assume that X is not toric and that $\zeta \ge 2$.

Step 2. In this step, we show that any T-invariant prime divisor of X is of the form $D_{y,v}$, and that the degree polyhedron of the p-divisor of X has a vertex on each ray of the tail cone.

Let $N := \mathrm{Hom}(M, \mathbb{Z})$, and let $\mathcal{D} = \sum_y \Delta_y \cdot [y]$, σ, and Q be the p-divisor, the tail cone, and the degree polyhedron of X (Construction-Proposition 7.7). By the isomorphism of graded algebras in Construction-Proposition 7.7, the field of fractions of $\mathbb{C}[X]$ is $\mathbb{C}(\mathbb{P}^1)(M)$ [LS10, Section 1], which is purely transcendental over $\mathbb{C}(\mathbb{P}^1)$, with T acting on χ^u by the character u. Hence $\mathbb{C}(\mathbb{P}^1)$ is the field of T-invariant rational functions on X, and χ^u is a T-homogeneous rational function of weight u for any $u \in M$. By Construction-Proposition 7.7(3), the T-invariant prime divisors of X are exactly the D_ρ and the $D_{y,v}$, and $\mathrm{ord}_{D_{y,v}}(h) = \mu(v)\,\mathrm{ord}_y(h)$ for any $h \in \mathbb{C}(\mathbb{P}^1)^*$. In particular, any $h \in \mathbb{C}(\mathbb{P}^1)^*$ with $\mathrm{ord}_y(h) > 0$ vanishes along $D_{y,v}$, so the rational quotient $X \dashrightarrow \mathbb{P}^1$ maps $D_{y,v}$ to y.

By Construction-Proposition 7.7(2), $\langle \kappa, n_\rho \rangle = 1$ for any ray ρ of σ with $\rho \cap Q = \emptyset$. Since $\langle \kappa, n_\rho \rangle = \zeta \langle \theta, n_\rho \rangle \in \zeta\mathbb{Z}$ and $\zeta \ge 2$, no such ray exists. Thus any T-invariant prime divisor of X is of the form $D_{y,v}$.

Let ρ_1 and ρ_2 be the rays of σ, with primitive generators $n_{\rho_1}, n_{\rho_2} \in N$, so that $\sigma = \mathbb{R}_{\ge 0} n_{\rho_1} + \mathbb{R}_{\ge 0} n_{\rho_2}$. By Construction-Proposition 7.7(1), $0 \notin Q$. Since no ray of σ is disjoint from Q, there exists a real number $a > 0$ with $a n_{\rho_1} \in Q$. Write $a n_{\rho_1} = \sum_k t_k w'_k + w''$, where w'_k are vertices of Q, $t_k > 0$, $\sum_k t_k = 1$, and $w'' \in \sigma$. The coefficients of n_{ρ_2} in the w'_k and in w'' are non-negative with sum zero, so $w'_k \in \mathbb{R}_{>0} n_{\rho_1}$ for any k. Thus Q has a vertex $w_1 \in \mathbb{R}_{>0} n_{\rho_1}$, and similarly a vertex $w_2 \in \mathbb{R}_{>0} n_{\rho_2}$. In particular, w_1 and w_2 are linearly independent.

Step 3. In this step, we describe the Cox ring of X, express the p-divisor of X through the defining matrix of the Cox ring, and show that $\kappa = (0, \zeta_X)$ in the coordinates of P, where ζ_X is the canonical multiplicity of X.

By Lemma 7.4(1), X is T-equivariantly isomorphic to an affine variety $X(A, P)$ of Type 2. Thus the Cox ring of X is

$$\mathcal{R} = \mathbb{C}[y_{ij}, S_1, \dots, S_m \mid 0 \le i \le r,\ 1 \le j \le n_i]/(g_0, \dots, g_{r-2}),$$

where $m \ge 0$ is an integer, n_i and l_{ij} are positive integers, we write $y_i^{l_i} := \prod_j y_{ij}^{l_{ij}}$, the vectors $c_0, \dots, c_r \in \mathbb{C}^2$ are pairwise linearly independent, and g_i is the determinant of the 3×3 matrix

with first row $(y_i^{l_i}, y_{i+1}^{l_{i+1}}, y_{i+2}^{l_{i+2}})$ and with the columns c_i, c_{i+1}, and c_{i+2} below it. The ring $\mathcal{R}$ is not a polynomial ring: otherwise X would be the quotient of an affine space by a diagonal action of a quasitorus, hence an affine toric variety. Thus, by [ABHW18, Remark 2.4], we may assume that no block of the presentation consists of a single variable with exponent one, that is, $n_i l_{i1} \geq 2$ for any i, and we say that P is *irredundant* [ABHW18, Remark 2.4]. Moreover, by admissible operations of type (i) [ABHW18, Remark 2.3], we may assume that $l_{i1} \geq \cdots \geq l_{in_i}$ for any i [ABHW18, Remark 2.5], and we say that the presentation is *ordered* (this is weaker than being ordered in the sense of [ABHW18, Remark 6.1], which also asks that $l_{01} \geq \cdots \geq l_{r1}$). Since $\mathcal{R}$ is not a polynomial ring, $r \geq 2$. Moreover, $X = \operatorname{Spec} \mathcal{R}^H$ for the characteristic quasitorus H, the class group of X is the cokernel of P^t, the columns of the defining matrix P corresponding to the variables y_{ij} are the pairwise different primitive vectors

$$v_{ij} = (l_{ij} e_i, \bar{v}_{ij}) \quad (0 \leq i \leq r)$$

in $\mathbb{Z}^r \times N$, where $\bar{v}_{ij} \in N$, $e_i \in \mathbb{Z}^r$ is the i-th standard basis vector for $1 \leq i \leq r$, and $e_0 := -(e_1 + \cdots + e_r)$, and the first r coordinates of the columns corresponding to $S_1, \ldots, S_m$ are zero [ABHW18, Constructions 2.1, 2.2, and 2.6]. We write elements of N in a basis of N, and elements of M in the dual basis.

Since X is $\mathbb{Q}$-Gorenstein, after admissible operations [ABHW18, Remark 2.3], which keep the form above, we may assume that P is in the shape of [ABHW18, Proposition 4.3]. Since X is Gorenstein, the Gorenstein index $\imath_X$ of X is one, so by [ABHW18, Proposition 4.3 and Remark 4.5] there exist integers $s_0, \ldots, s_r$ with $s_0 + \cdots + s_r = r - 1$ such that

$$\langle \kappa_0, \bar{v}_{ij} \rangle = 1 - s_i l_{ij} \quad \text{for any } i \text{ and } j, \tag{7.8}$$

where $\kappa_0 := (0, \zeta_X) \in M$ and ζ_X is the canonical multiplicity of X [ABHW18, Remark 4.1, Definition 4.2, and Remark 4.4].

Since c_0 and c_1 are linearly independent, there exists a pair $\varphi = (\varphi_1, \varphi_2)$ of regular functions on $\operatorname{Spec} \mathcal{R}$ with $y_i^{l_i} = \det(c_i, \varphi)$ for $i = 0, 1$. For $0 \leq i \leq r-2$, the relation $g_i = 0$ says that the row $(y_i^{l_i}, y_{i+1}^{l_{i+1}}, y_{i+2}^{l_{i+2}})$ is a linear combination of the rows of the 2×3 matrix with the columns c_i, c_{i+1}, and c_{i+2}, so induction on i shows that $y_i^{l_i} = \det(c_i, \varphi)$ for $0 \leq i \leq r$. Let $z_i := [c_i] \in \mathbb{P}^1$. Then $z_0, \ldots, z_r$ are pairwise distinct. For $1 \leq i \leq r$, let $h_i \in \mathbb{C}(\mathbb{P}^1)$ be the function $h_i([a]) := \det(c_i, a)/\det(c_0, a)$, which has a simple zero at z_i, a simple pole at z_0, and no other zeros or poles. Let $a \in \mathbb{C}^2$ with $\det(c_i, a) \neq 0$ for $0 \leq i \leq r$. Since $\det(c, a)$ is linear in $c \in \mathbb{C}^2$, the polynomials $g_0, \ldots, g_{r-2}$ vanish at any point p of the affine space with coordinates y_{ij} and $S_1, \ldots, S_m$ such that $y_i^{l_i}(p) = \det(c_i, a)$ for $0 \leq i \leq r$. Thus such points p, with arbitrary values $S_1(p), \ldots, S_m(p)$, are points of $\operatorname{Spec} \mathcal{R}$, and $\varphi(p) = a$.

By [ABHW18, Construction 2.1], $H_0 = \operatorname{Spec} \mathbb{C}[K_0]$, where K_0 is the cokernel of the transpose of the matrix P_0 formed by the first r rows of P, and $T = H_0/H$ [ABHW18, Construction 2.6]. Since the i-th row of P_0 is the exponent vector of $y_i^{l_i}/y_0^{l_0}$, the quasitorus H_0 is the subgroup of the torus acting diagonally on the coordinates y_{ij} and $S_1, \ldots, S_m$ which consists of the elements t with

$$\prod_j t_{0j}^{l_{0j}} = \prod_j t_{1j}^{l_{1j}} = \cdots = \prod_j t_{rj}^{l_{rj}},$$

where t_{ij} is the coordinate of t acting on y_{ij}, and the coordinates of t acting on $S_1, \ldots, S_m$ are arbitrary. In particular, $y_i^{l_i}/y_0^{l_0}$ is H_0-invariant, so it is a T-invariant rational function on X, and $[\varphi]$ induces a T-invariant rational map $[\varphi] : X \dashrightarrow \mathbb{P}^1$ with $h_i \circ [\varphi] = y_i^{l_i}/y_0^{l_0}$ for $1 \leq i \leq r$. By the previous paragraph, $[\varphi]$ is dominant.

We show that $[\varphi]$ is the rational quotient of X. Let p and p' be points of $\operatorname{Spec} \mathcal{R}$ with all coordinates non-zero and with $[\varphi(p)] = [\varphi(p')]$. Then $\varphi(p') = \epsilon \varphi(p)$ for some $\epsilon \in \mathbb{C}^*$, so $y_i^{l_i}(p') = \epsilon y_i^{l_i}(p)$ for $0 \leq i \leq r$. Thus the element t of the diagonal torus with coordinates $y_{ij}(p')/y_{ij}(p)$ and $S_1(p')/S_1(p), \ldots, S_m(p')/S_m(p)$ belongs to H_0, and $p' = t \cdot p$. By Chevalley's theorem, the image in X of the set of points of $\operatorname{Spec} \mathcal{R}$ with all coordinates non-zero contains a

dense open subset X° of X, and we may assume that $[\varphi]$ is defined on X°. Thus any fiber of $[\varphi]|_{X^\circ}$ is contained in one T-orbit. By **Step 2**, the field of T-invariant rational functions on X is $\mathbb{C}(\mathbb{P}^1)$, that is, it is the pullback of $\mathbb{C}(\mathbb{P}^1)$ by the rational quotient $\pi : X \dashrightarrow \mathbb{P}^1$ [LS10, Section 1]. Since $[\varphi]$ is T-invariant and dominant, $[\varphi] = \psi \circ \pi$ for a finite morphism $\psi : \mathbb{P}^1 \to \mathbb{P}^1$. Suppose that $\deg \psi \geq 2$. For a general point $z \in \mathbb{P}^1$, there exist two different points $w, w' \in \psi^{-1}(z)$ and points $x, x' \in X^\circ$ at which π is defined with $\pi(x) = w$ and $\pi(x') = w'$. Since $[\varphi](x) = [\varphi](x') = z$, we have $x' \in T \cdot x$, so $\pi(x) = \pi(x')$ since π is T-invariant, a contradiction. Thus $\deg \psi = 1$, that is, $[\varphi]$ is the rational quotient of X, and we regard $\mathbb{P}^1$ in Construction-Proposition 7.7 as the codomain of $[\varphi]$.

We show that $m = 0$. Suppose that $m \geq 1$, and let $E \subset X$ be the prime divisor whose preimage in the characteristic space is $V(S_1)$. Since $V(S_1)$ is H_0-invariant, E is T-invariant. Since S_1 does not occur in $g_0, \dots, g_{r-2}$, the ring $\mathcal{R}/(S_1)$ is a polynomial ring in $S_2, \dots, S_m$ over the domain $\mathbb{C}[y_{ij} \mid i, j]/(g_0, \dots, g_{r-2})$, so $V(S_1)$ is irreducible. By the points p with $S_1(p) = 0$ constructed above, $h_1 \circ [\varphi] = y_1^{l_1}/y_0^{l_0}$ is not constant on $V(S_1)$, hence not constant on E. Thus $[\varphi]$ maps E onto a dense subset of $\mathbb{P}^1$. On the other hand, by **Step 2**, $E = D_{y,v}$ for some $y \in \mathbb{P}^1$ and some vertex v of Δ_y, and $[\varphi]$ maps $D_{y,v}$ to the point y, a contradiction. Thus $m = 0$, that is,

$$\mathcal{R} = \mathbb{C}[y_{ij} \mid 0 \leq i \leq r,\ 1 \leq j \leq n_i]/(g_0, \dots, g_{r-2}).$$

For $\lambda \in \mathbb{Z}^r \times \mathbb{Z}^2$, the Laurent monomial $y^{P^t\lambda} := \prod_{i,j} y_{ij}^{\langle \lambda, v_{ij} \rangle}$ has trivial H-degree, so it is a rational function on X. The variety X is a T-equivariant closed subvariety of the minimal toric ambient variety Z of X, on which T acts as the subtorus of the big torus of Z corresponding to $0 \times \mathbb{Z}^2 \subset \mathbb{Z}^r \times \mathbb{Z}^2$, and $y^{P^t\lambda}$ is the restriction to X of the character χ^λ of the big torus of Z [ABHW18, Construction 2.9]. Thus $y^{P^t\lambda}$ is T-homogeneous of weight $(\lambda_{r+1}, \lambda_{r+2}) \in M$. Let $D_{ij} \subset X$ be the prime divisor whose preimage in the characteristic space is $V(y_{ij})$. Over the smooth locus of X the characteristic space is an H-torsor [ADHL15, Proposition 1.6.1.6(i)], and the generic point of D_{ij} is a smooth point of X since X is normal. Thus

$$\operatorname{div}(y^{P^t\lambda}) = \sum_{i,j} \langle \lambda, v_{ij} \rangle D_{ij}. \tag{7.9}$$

For $1 \leq i \leq r$, by (7.9) with $\lambda = (e_i, 0)$,

$$\operatorname{div}\left(y_i^{l_i}/y_0^{l_0}\right) = \sum_j l_{ij} D_{ij} - \sum_j l_{0j} D_{0j},$$

while by the formula of Construction-Proposition 7.7(3) for $\operatorname{div}(h_i)$,

$$\operatorname{div}\left(y_i^{l_i}/y_0^{l_0}\right) = \sum_v \mu(v) D_{z_i,v} - \sum_v \mu(v) D_{z_0,v}.$$

Thus, for $0 \leq i \leq r$, the prime divisors $D_{z_i,v}$ for the vertices v of Δ_{z_i} are exactly $D_{i1}, \dots, D_{in_i}$, and $\mu(v) = l_{ij}$ if $D_{ij} = D_{z_i,v}$. Moreover, for $y \notin \{z_0, \dots, z_r\}$, no divisor $D_{y,v}$ is of the form D_{ij}.

For $u \in M$, both $y^{P^t(0,u)}$ and χ^u are T-homogeneous of weight u, so $y^{P^t(0,u)} = h'_u \chi^u$ for some $h'_u \in \mathbb{C}(\mathbb{P}^1)^*$, and $u \to h'_u$ is a group homomorphism $M \to \mathbb{C}(\mathbb{P}^1)^*$. Hence there exist $k_y \in N$ for $y \in \mathbb{P}^1$, zero for all but finitely many y, such that $\operatorname{ord}_y(h'_u) = \langle u, k_y \rangle$ for any $u \in M$, and $\sum_y k_y = 0$ since a principal divisor on $\mathbb{P}^1$ has degree zero. If $D_{ij} = D_{z_i,v}$, then comparing the coefficients of D_{ij} in (7.9) with $\lambda = (0, u)$ and in the formula of Construction-Proposition 7.7(3) for $\operatorname{div}(h'_u \chi^u)$ shows that

$$\langle u, \bar{v}_{ij} \rangle = l_{ij}\big(\langle u, k_{z_i} \rangle + \langle u, v \rangle\big) \quad \text{for any } u \in M,$$

that is, $v = \frac{1}{l_{ij}} \bar{v}_{ij} - k_{z_i}$. If $y \notin \{z_0, \dots, z_r\}$ and v is a vertex of Δ_y, then the coefficient of $D_{y,v}$ in $\operatorname{div}(y^{P^t(0,u)})$ is zero, so $\mu(v)(\langle u, k_y \rangle + \langle u, v \rangle) = 0$ for any $u \in M$, that is, $v = -k_y$. Thus, for

$0 \le i \le r$, the vertices of Δ_{z_i} are the points $\frac{1}{l_{ij}}\bar{v}_{ij} - k_{z_i}$ for $1 \le j \le n_i$, and

$$\mu\Big(\frac{1}{l_{ij}}\bar{v}_{ij} - k_{z_i}\Big) = l_{ij} \quad \text{and} \quad \Delta_y = -k_y + \sigma \ \text{ for } y \notin \{z_0, \dots, z_r\}. \tag{7.10}$$

We show that $\kappa = \kappa_0$. Let b_y be the integers of Construction-Proposition 7.7(2), so that $\sum_y b_y = 2$ and $\langle \kappa, v\rangle = b_y - 1 + \frac{1}{\mu(v)}$ for any vertex v of Δ_y. Let $\delta := \kappa - \kappa_0$. By (7.8), (7.10), and the description of the vertices of Δ_{z_i}, for any vertex $v = \frac{1}{l_{ij}}\bar{v}_{ij} - k_{z_i}$ of Δ_{z_i},

$$\langle \delta, v\rangle = \Big(b_{z_i} - 1 + \frac{1}{l_{ij}}\Big) - \Big(\frac{1}{l_{ij}} - s_i - \langle \kappa_0, k_{z_i}\rangle\Big) = b_{z_i} - 1 + s_i + \langle \kappa_0, k_{z_i}\rangle,$$

and for $y \notin \{z_0, \dots, z_r\}$, the only vertex $-k_y$ of Δ_y satisfies $\langle \delta, -k_y\rangle = b_y + \langle \kappa_0, k_y\rangle$. Thus, for any y, the function $\langle \delta, \cdot\rangle$ takes a constant value d_y on the vertices of Δ_y, and

$$\sum_{y\in\mathbb{P}^1} d_y = \sum_y b_y - (r+1) + \sum_{i=0}^{r} s_i + \Big\langle \kappa_0, \sum_y k_y\Big\rangle = 2 - (r+1) + (r-1) + 0 = 0.$$

Any vertex of Q is a sum $\sum_y v_y$ of vertices v_y of Δ_y (a linear function minimized only at a given vertex of Q is minimized on each Δ_y only at a vertex v_y, and the vertex equals $\sum_y v_y$), so $\langle \delta, \cdot\rangle$ vanishes on the vertices of Q. In particular, $\langle \delta, w_1\rangle = \langle \delta, w_2\rangle = 0$, where w_1 and w_2 are the linearly independent vertices of Q found in **Step 2**. Thus $\delta = 0$, that is,

$$\kappa = (0, \zeta_X). \tag{7.11}$$

Since $(0,1) \in M$ is primitive, $\zeta = \zeta_X$ and $\theta = (0,1)$. In other words, $\langle \theta, \cdot\rangle$ is the second coordinate function on N.

Step 4. In this step, we bring P into the normal form of case (vi) of [ABHW18, Proposition 4.7] and derive congruences for the exponents l_{0j} and l_{1j}.

By **Step 3**, $\zeta_X = \zeta > 1$. Since X is klt and $\imath_X = 1$, by [ABHW18, Proposition 4.7], after admissible operations P is in the shape of [ABHW18, Proposition 4.3] and belongs to one of the cases (i)–(vi) of [ABHW18, Proposition 4.7]. In each of the cases (i)–(v), the Gorenstein index $\imath_X$ is divisible by 2 or 3, so case (vi) holds. In case (vi), $l_{21} = 1$ and $l_{ij} = 1$ for $3 \le i \le r$. Since the matrix of **Step 3** is ordered and irredundant, so is the matrix of case (vi) by Remark 7.13. Since the presentation is ordered, (l_{01}, l_{11}, l_{21}) in case (vi) are the largest exponents of the blocks 0, 1, and 2, so $l_{21} = 1$ implies $l_{2j} = 1$ for any j. Thus $l_{ij} = 1$ for $2 \le i \le r$, and there exists an integer s such that

$$\begin{aligned} v_{0j} &= \Big(-l_{0j}, \dots, -l_{0j}, \gamma_{0j}, \frac{1 - sl_{0j}}{\zeta}\Big), \\ v_{1j} &= \Big(l_{1j}, 0, \dots, 0, \gamma_{1j}, \frac{1 + sl_{1j}}{\zeta}\Big), \\ v_{ij} &= (e_i, \gamma_{ij}, 0) \quad (2 \le i \le r) \end{aligned} \tag{7.12}$$

for some integers γ_{ij} [ABHW18, Proposition 4.7]. Since **Step 3** used only that P has the form above and is in the shape of [ABHW18, Proposition 4.3], its conclusions hold for the new matrix P. In particular, $\theta = (0,1)$, and (7.8) holds with $s_0 = s$, $s_1 = -s$, and $s_i = 1$ for $2 \le i \le r$.

Let $\ell_0 := \max_j l_{0j}$ and $\ell_1 := \max_j l_{1j}$. Exchanging the data of the blocks 0 and 1 is an admissible operation [ABHW18, Remark 2.3(ii)], and by the criterion of [ABHW18, Remark 4.5], the new matrix is again of the form (7.12) with s replaced by $-s$. Possibly exchanging the blocks 0 and 1, we may assume that $\ell_0 \ge \ell_1$, and the conclusions of **Step 3** hold for the new matrix, since it is of the form (7.12) and, by the criterion of [ABHW18, Remark 4.5], in the shape of [ABHW18, Proposition 4.3].

Since the last coordinates in (7.12) are integers, we have

$$sl_{0j} \equiv 1 \pmod{\zeta} \quad \text{and} \quad sl_{1j} \equiv -1 \pmod{\zeta} \quad \text{for any } j. \tag{7.13}$$

Thus s is invertible modulo ζ, any two exponents l_{0j} and $l_{0j'}$ are congruent modulo ζ, any two exponents l_{1j} and $l_{1j'}$ are congruent modulo ζ, and $\gcd(\ell_1, \zeta) = 1$. Moreover, $\zeta \mid s(\ell_0 + \ell_1)$ by (7.13), so

$$\zeta \mid \ell_0 + \ell_1. \tag{7.14}$$

Step 5. In this step, we construct a T-homogeneous regular function f on X vanishing along a prime divisor D_{0j}, and show by Proposition 5.6(2) with $g = f$ that $\ell_0 \le 2\ell_1$. **Step 5** is the only step of the proof in which we use that wt_ξ computes $\widehat{\mathrm{vol}}(X \ni o)$.

Let

$$\lambda := \Big(-\frac{1 + s\ell_1}{\zeta}, 0, \dots, 0, 0, \ell_1 \Big) \in \mathbb{Z}^r \times \mathbb{Z}^2,$$

which is integral by (7.13). By (7.12),

$$\langle \lambda, v_{0j} \rangle = \frac{l_{0j} + \ell_1}{\zeta}, \qquad \langle \lambda, v_{1j} \rangle = \frac{\ell_1 - l_{1j}}{\zeta}, \qquad \langle \lambda, v_{ij} \rangle = 0 \quad (2 \le i \le r).$$

By (7.13), $s(l_{0j} + \ell_1) \equiv 0$ and $s(\ell_1 - l_{1j}) \equiv 0 \pmod{\zeta}$, so the pairings $\langle \lambda, v_{ij} \rangle$ are integers since s is invertible modulo ζ. They are non-negative since $\ell_1 = \max_j l_{1j}$. Thus

$$F := y^{P^t \lambda} = \prod_j y_{0j}^{(l_{0j} + \ell_1)/\zeta} \prod_j y_{1j}^{(\ell_1 - l_{1j})/\zeta}$$

is a non-zero element of $\mathcal{R}$ with trivial H-degree, so $F \in \mathcal{R}^H$ defines a non-zero regular function f on X. By **Step 3** and $\theta = (0, 1)$, f is T-homogeneous of weight $(0, \ell_1) = \ell_1 \theta$. Fix j_0 with $l_{0j_0} = \ell_0$. By (7.9),

$$\mathrm{ord}_{D_{0j_0}}(f) = \frac{\ell_0 + \ell_1}{\zeta}. \tag{7.15}$$

Since a one-parameter subgroup of T contracts X to o, the point o is contained in the closure of any T-orbit, hence in the T-invariant prime divisor D_{0j_0}. Since f vanishes along D_{0j_0}, we have $0 \ne f \in \mathfrak{m}_o$.

Since f is a non-constant homogeneous element of weight $\ell_1 \theta$ and ξ is a Reeb vector, $\mathrm{wt}_\xi(f) = \ell_1 \langle \theta, \xi \rangle > 0$. In particular, $\langle \theta, \xi \rangle > 0$. Since wt_ξ computes $\widehat{\mathrm{vol}}(X \ni o)$, it is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,o}$. By Proposition 5.6(2) with $n = 3$, $B = 0$, $v = \mathrm{wt}_\xi$, $g = f$, and $D = D_{0j_0}$, together with (7.7) and (7.15), we have

$$3\ell_1 \langle \theta, \xi \rangle = 3\,\mathrm{wt}_\xi(f) \ge a(\mathrm{wt}_\xi, X)\,\mathrm{ord}_{D_{0j_0}}(f) = \zeta \langle \theta, \xi \rangle \cdot \frac{\ell_0 + \ell_1}{\zeta} = (\ell_0 + \ell_1) \langle \theta, \xi \rangle.$$

Since $\langle \theta, \xi \rangle > 0$, we have

$$\ell_0 \le 2\ell_1. \tag{7.16}$$

Step 6. In this step, we show that block 0 or block 1 contains two different exponents, and deduce that $\ell_0 \ge \zeta + 1$ and $\zeta \le \frac{\ell_0 + \ell_1}{2}$.

The Cox presentation of X of Type 2 has no free variables by **Step 3**. If the exponents l_{ij} were constant within any block, then $X \ni o$ would be LCIQ by Proposition 7.6, which is not the case. Since $l_{ij} = 1$ for $2 \le i \le r$, the exponents are not constant within block 0 or within block 1. By (7.13), two different exponents in block 0 or in block 1 differ by a positive multiple of ζ, so the larger exponent is at least $\zeta + 1$. Since $\ell_0 = \max\{l_{0j}, l_{1j} \mid j\}$, we have $\ell_0 \ge \zeta + 1$. Thus $\ell_0 + \ell_1 > \zeta$, and (7.14) implies $\ell_0 + \ell_1 \ge 2\zeta$. Therefore,

$$\ell_0 \ge \zeta + 1 \quad \text{and} \quad \zeta \le \frac{\ell_0 + \ell_1}{2}. \tag{7.17}$$

Step 7. In this step, we show that

$$\widehat{\mathrm{vol}}(X \ni o) \le 4\zeta \Big(\frac{1}{\ell_0} + \frac{1}{\ell_1} \Big)^2. \tag{7.18}$$

Fix j_1 with $l_{1j_1} = \ell_1$, and recall j_0 from **Step 5**. Since the operations in **Step 4** keep irredundancy, we have $n_2 l_{21} \geq 2$. Since $l_{21} = 1$, we have $n_2 \geq 2$. Let

$$U := \frac{1}{\ell_0}\bar{v}_{0j_0} + \frac{1}{\ell_1}\bar{v}_{1j_1} + \sum_{i=2}^{r} \bar{v}_{i1} \quad \text{and} \quad W := U + \bar{v}_{22} - \bar{v}_{21}.$$

Recall that $k_y \in N$ in (7.10) satisfy $\sum_y k_y = 0$. Thus

$$U = \Big(\frac{1}{\ell_0}\bar{v}_{0j_0} - k_{z_0}\Big) + \Big(\frac{1}{\ell_1}\bar{v}_{1j_1} - k_{z_1}\Big) + \sum_{i=2}^{r} \big(\bar{v}_{i1} - k_{z_i}\big) + \sum_{y \notin \{z_0,\dots,z_r\}} (-k_y)$$

is a sum of vertices of the polyhedra Δ_y by (7.10), the description of the vertices of Δ_{z_i} before it, and $l_{i1} = 1$ for $i \geq 2$, so $U \in Q$. Replacing $\bar{v}_{21} - k_{z_2}$ with $\bar{v}_{22} - k_{z_2}$ in the sum shows that $W \in Q$. By (7.12) and $\theta = (0, 1)$,

$$\langle \theta, U\rangle = \langle \theta, W\rangle = \frac{1 - s\ell_0}{\zeta\ell_0} + \frac{1 + s\ell_1}{\zeta\ell_1} = \frac{1}{\zeta}\Big(\frac{1}{\ell_0} + \frac{1}{\ell_1}\Big) > 0.$$

Since $v_{21} = (e_2, \gamma_{21}, 0)$ and $v_{22} = (e_2, \gamma_{22}, 0)$ are different columns of P, we have $\gamma_{21} \neq \gamma_{22}$. Thus $W - U = (\gamma_{22} - \gamma_{21}, 0) = Le$, where $L := |\gamma_{22} - \gamma_{21}| \geq 1$ and $e := \pm(1, 0)$ is primitive in N. Since $U \neq W$ and $\langle \theta, U\rangle = \langle \theta, W\rangle > 0$, the vectors U and W are linearly independent. Thus Lemma 7.8(2) implies that

$$\widehat{\mathrm{vol}}(X \ni o) \leq \frac{4\zeta^3\langle \theta, U\rangle^2}{L} = \frac{4\zeta}{L}\Big(\frac{1}{\ell_0} + \frac{1}{\ell_1}\Big)^2 \leq 4\zeta\Big(\frac{1}{\ell_0} + \frac{1}{\ell_1}\Big)^2.$$

Step 8. In this step, we conclude the proof by going through the possible values of ℓ_1.

By **Steps 4–6**, the integers ℓ_0, ℓ_1, and ζ satisfy

$$1 \leq \ell_1 \leq \ell_0 \leq 2\ell_1, \quad \zeta \geq 2, \quad \ell_0 \geq \zeta + 1,$$

$$\gcd(\ell_1, \zeta) = 1, \quad \zeta \mid \ell_0 + \ell_1, \quad \text{and} \quad \zeta \leq \frac{\ell_0 + \ell_1}{2}.$$

Case 1. $\ell_1 = 1$. Then $\ell_0 \leq 2$ by (7.16), which contradicts $\ell_0 \geq \zeta + 1 \geq 3$.

Case 2. $\ell_1 = 2$. Then $\ell_0 \leq 4$. Since $\gcd(2, \zeta) = 1$ and $\zeta \geq 2$, ζ is odd and $\zeta \geq 3$, so $4 \geq \ell_0 \geq \zeta + 1 \geq 4$. Thus $\ell_0 = 4$ and $\zeta = 3$, and (7.18) implies that

$$\widehat{\mathrm{vol}}(X \ni o) \leq 4 \cdot 3 \cdot \Big(\frac{1}{4} + \frac{1}{2}\Big)^2 = \frac{27}{4}.$$

Case 3. $\ell_1 \geq 3$. By (7.18) and (7.17),

$$\widehat{\mathrm{vol}}(X \ni o) \leq 4\zeta\frac{(\ell_0 + \ell_1)^2}{\ell_0^2\ell_1^2} \leq \frac{2(\ell_0 + \ell_1)^3}{\ell_0^2\ell_1^2} = \frac{2}{\ell_1} \cdot \frac{(t+1)^3}{t^2},$$

where $t := \ell_0/\ell_1 \in [1, 2]$ by (7.16). Since

$$\frac{d}{dt}\frac{(t+1)^3}{t^2} = \frac{(t+1)^2(t-2)}{t^3} \leq 0 \quad \text{for } 1 \leq t \leq 2,$$

we have $\frac{(t+1)^3}{t^2} \leq \frac{(1+1)^3}{1^2} = 8$. Thus

$$\widehat{\mathrm{vol}}(X \ni o) \leq \frac{16}{\ell_1} \leq \frac{16}{3} < \frac{27}{4}.$$

Thus $\widehat{\mathrm{vol}}(X \ni o) \leq \frac{27}{4}$ when X is not toric and $\zeta \geq 2$. By **Step 1**, Theorem 7.3 follows. □

8. Singularities with one crepant divisor

In this section, we prove Theorem 8.1, which classifies the Gorenstein canonical threefold germs with exactly one crepant divisor and local volume greater than 7. Theorem 8.1 is used in Section 9, for the vertex of an elliptic cone (Propositions 9.20 and 9.21) and for the last sentence of Theorem 9.1, which is used in Section 10 through Proposition 10.14. Theorem 8.1 is also used directly in Section 10, for Proposition 10.13(3).

Theorem 8.1. *Let $X \ni x$ be a Gorenstein canonical threefold germ with $\operatorname{crex}(X \ni x) = 1$ and $\widehat{\operatorname{vol}}(X \ni x) > 7$. Then $X \ni x$ is analytically isomorphic to exactly one of the following germs.*

(1) $C_{\mathbb{P}^2} \ni o$, which is the quotient of $\mathbb{A}^3$ by μ_3 acting by scalars, is LCIQ, and has local volume 9.
(2) $C_{\mathbb{P}^1\times\mathbb{P}^1} \ni o$, which is the quotient of the threefold ordinary double point by -1, is LCIQ, and has local volume 8.
(3) $C_{\mathbb{F}_1} \ni o$, which is not LCIQ and has local volume q.

Conversely, each of the germs in (1)–(3) has $\operatorname{crex} = 1$. In particular, if $X \ni x$ is a Gorenstein canonical threefold germ with $\operatorname{crex}(X \ni x) = 1$ which is not LCIQ, then $\widehat{\operatorname{vol}}(X \ni x) \le q$, and equality holds if and only if $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$.

We prove Theorem 8.1 in Subsection 8.4.

8.1. Crepant extractions.

In this subsection, we fix the set-up for a crepant extraction over a Gorenstein canonical threefold germ whose reduced exceptional fiber is a Cartier divisor.

Set-up 8.2. Let $X \ni x$, $f : Y \to X$, and E be as follows.

(1) $X \ni x$ is a Gorenstein canonical threefold germ.
(2) $f : Y \to X$ is a projective birational morphism from a normal variety Y with $K_Y = f^*K_X$, and f is an isomorphism over $X\setminus\{x\}$.
(3) $E := f^{-1}(x)_{\mathrm{red}}$ is a Cartier divisor on Y, and $-E$ is f-ample.

Remark 8.3. Condition (3) of Set-up 8.2 asks E itself to be Cartier. Thus E is Gorenstein by Lemma 2.19(2), $\omega_E \cong \mathcal{O}_E(E)$ by adjunction, and $\mathcal{O}_E(-E) \cong \omega_E^{-1}$ is ample, so E, which is projective and connected [Har77, Corollary III.11.4], together with $\mathcal{O}_E(-E)$ is a *Gorenstein del Pezzo surface*, that is, a connected projective Gorenstein scheme S with $\dim S = 2$ together with an ample line bundle $\mathcal{O}_S(1)$ such that $\omega_S \cong \mathcal{O}_S(-1)$ [Rei94, §0.1]. In particular, the classification of non-normal Gorenstein del Pezzo surfaces [Rei94, Theorem 1.5] and that of normal ones [HW81, Theorems 2.2 and 3.4], [Bre80, Theorem 1] apply to E, and Lemma 2.19(2) implies

$$\widehat{\operatorname{vol}}(X \ni x) \le (K_E^2).$$

By Proposition 8.7, for any Gorenstein canonical threefold germ $X \ni x$ with $\operatorname{crex}(X \ni x) = 1$, there exist f and E as in Set-up 8.2.

In Set-up 8.2, if E is isomorphic to $\mathbb{F}_1$, $\mathbb{P}^1\times\mathbb{P}^1$, or $\mathbb{P}(1,1,2)$, then $(K_E^2) = 8$, and Lemma 2.19(2) only bounds $\widehat{\operatorname{vol}}(X \ni x)$ by 8, while $8 > q$ by Lemma 4.3. Thus, for $E \cong \mathbb{F}_1$, $\mathbb{P}^1\times\mathbb{P}^1$, or $\mathbb{P}(1,1,2)$, we do not estimate the local volume, but determine the germ $X \ni x$ itself (Propositions 8.9 and 8.5).

8.2. Cyclic covers and quadric surfaces.

In this subsection, we construct cyclic covers of $X \ni x$ from roots of $\mathcal{O}_E(-E)$, and we determine $X \ni x$ when E is a quadric surface.

Lemma 8.4. *Notation and conditions as in Set-up 8.2. Assume that (Y, E) is lc. Let $r \ge 2$ be an integer, and assume that there exists a line bundle $\mathcal{L}_E$ on E with $\mathcal{L}_E^{\otimes r} \cong \mathcal{O}_E(-E)$. Then, after replacing $X \ni x$ by a pointed étale neighborhood with residue field $\mathbb{C}$, there exists a line bundle $\mathcal{L}$ on Y with $\mathcal{L}|_E \cong \mathcal{L}_E$ and $\mathcal{L}^{\otimes r} \cong \mathcal{O}_Y(-E)$, and for any such $\mathcal{L}$ the following hold.*

(1) *The cyclic cover* $\pi : \tilde{Y} := \operatorname{Spec}_Y \bigoplus_{i=0}^{r-1} \mathcal{L}^{\otimes i} \to Y$ *defined by* $\mathcal{L}^{\otimes r} \cong \mathcal{O}_Y(-E) \subset \mathcal{O}_Y$ *is normal and connected, and* $\tilde{E} := \pi^{-1}(E)_{\mathrm{red}}$ *is a Cartier divisor on* $\tilde{Y}$ *such that* $\pi|_{\tilde{E}} : \tilde{E} \to E$ *is an isomorphism,* $\mathcal{O}_{\tilde{E}}(-\tilde{E}) \cong (\pi|_{\tilde{E}})^*\mathcal{L}_E$*, and* μ_r *acts on* $\mathcal{O}_{\tilde{E}}(-\tilde{E})$ *by a primitive character.*

(2) *For the Stein factorization* $\tilde{Y} \xrightarrow{\tilde{f}} W \xrightarrow{p} X$ *of* $f \circ \pi$*, the morphism* $p : (W \ni w) \to (X \ni x)$ *is a connected normal cyclic cover of degree* r *with* $p^{-1}(x) = \{w\}$ *which is étale over* $X \backslash \{x\}$*, and* $\tilde{f}$ *is a projective birational morphism which is an isomorphism over* $W \backslash \{w\}$ *with* $\tilde{f}^{-1}(w)_{\mathrm{red}} = \tilde{E}$*. In particular,* p *is quasi-étale and* W *is klt.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that Y is Cohen–Macaulay, that E is connected and slc with $\omega_E \cong \mathcal{O}_E(E)$, and that

$$H^i(E, \mathcal{O}_E(-kE)) = 0 \quad \text{for any integers } i > 0 \text{ and } k \geq 0.$$

Since $K_Y = f^*K_X$, the divisor K_Y is Cartier, and $a(F, Y) = a(F, X) \geq 1$ for any prime divisor F which is exceptional over Y [KM98, Lemma 2.30]. Thus Y is canonical and K_Y is Cartier, hence Y is klt, so Y has rational singularities [KM98, Theorem 5.22] and is Cohen–Macaulay [KM98, Theorem 5.10]. Since K_Y is Cartier, Y is Gorenstein. By Zariski's Main Theorem [Har77, Corollary III.11.4], $f^{-1}(x)$ is connected. Since $E_{\mathrm{red}} = f^{-1}(x)_{\mathrm{red}}$, E is connected.

Shrinking X around x, we may assume that $\omega_X \cong \mathcal{O}_X$, so $\omega_Y \cong f^*\omega_X \cong \mathcal{O}_Y$. Since E is a Cartier divisor on the Cohen–Macaulay variety Y, adjunction [Kol13, (4.1.3)] implies

$$\omega_E \cong \omega_Y(E)|_E \cong \mathcal{O}_E(E).$$

In particular, K_E is Cartier, and $\mathcal{O}_E(-E) \cong \omega_E^{-1}$.

Next we show that E is slc. Locally E is defined by one nonzerodivisor in a Cohen–Macaulay local ring, so E is Cohen–Macaulay [Har77, Theorem II.8.21A(d)], and in particular E is S_2. Since (Y, E) is lc, E is regular or has a node at any codimension two point of Y which is contained in E [Kol13, Corollary 2.32]. Thus E is demi-normal [Kol13, Definition 5.1]. Let $\bar{E} \to E$ be the normalization, and let $\bar{D}$ be the conductor on $\bar{E}$. Since E is Cartier, $\operatorname{Diff}_E(0) = 0$, so $\operatorname{Diff}_{\bar{E}}(0) = \bar{D}$ [Kol13, Proposition 4.5(4)(6)], and $(\bar{E}, \bar{D})$ is lc [Kol13, Theorem 4.9]. Since K_E is Cartier, E is slc [Kol13, Definition–Lemma 5.10].

Finally, $-E$ is f-ample and E is a closed subscheme of $f^{-1}(x)$, so $\mathcal{O}_E(-E)$ is ample. For any integer $k \geq 0$, the divisor $-kE|_E$ is Cartier, and

$$-kE|_E - K_E \sim -(k+1)E|_E$$

is ample. Thus [Fuj14, Theorem 1.7], applied to the slc pair $(E, 0)$ and the structure morphism $E \to \operatorname{Spec} \mathbb{C}$, implies that $H^i(E, \mathcal{O}_E(-kE)) = 0$ for any integers $i > 0$ and $k \geq 0$.

Step 2. In this step, after a pointed étale base change, we extend $\mathcal{L}_E$ to a line bundle $\mathcal{L}$ on Y and show that $\mathcal{L}^{\otimes r} \cong \mathcal{O}_Y(-E)$ after shrinking X around x.

Since $K_Y = f^*K_X$ and $-E$ is f-ample, $-(K_Y + E)$ is f-ample. Moreover, $E = f^{-1}(x)_{\mathrm{red}}$, f is an isomorphism over $X \backslash \{x\}$, and (Y, E) is lc. Thus $f : (Y, E) \to X$ is an lc blow-up in the sense of [Liu25, Definition 2.4]. By [Liu25, Proposition 3.1], there exist an étale morphism $(X' \ni x') \to (X \ni x)$, where x' has residue field $\mathbb{C}$, and a line bundle $\mathcal{L}'$ on $Y' := Y \times_X X'$ such that $\mathcal{L}'|_{E'} \cong \mathcal{L}_E$, where $E' \cong E$ is the reduced fiber of $Y' \to X'$ over x'. Shrinking X' around x', we may assume that x' is the only point of X' over x. Since $X' \to X$ is étale, the germ $X' \ni x'$, the morphism $Y' \to X'$, and E' satisfy the conditions of Set-up 8.2, and (Y', E') is lc [Kol13, 2.14]. Possibly replacing $X \ni x$ with $X' \ni x'$, we may assume that there exists a line bundle $\mathcal{L}$ on Y with $\mathcal{L}|_E \cong \mathcal{L}_E$.

Let

$$\mathcal{N} := \mathcal{L}^{\otimes r} \otimes \mathcal{O}_Y(E).$$

Then $\mathcal{N}|_E \cong \mathcal{L}_E^{\otimes r} \otimes \mathcal{O}_E(E) \cong \mathcal{O}_E$. We show that $\mathcal{N} \cong \mathcal{O}_Y$ after shrinking X around x. Let $\mathcal{N}(-E) := \mathcal{N} \otimes \mathcal{O}_Y(-E)$, and for any integer $n \geq 1$, let E_n be the closed subscheme of Y defined by $\mathcal{O}_Y(-nE)$. The images of $\mathcal{O}_Y(-kE) \otimes \mathcal{N}(-E)$ for $0 \leq k \leq n$ define a filtration of $\mathcal{N}(-E)|_{E_n}$ with graded pieces

$$\mathcal{N}|_E \otimes \mathcal{O}_E(-(k+1)E) \cong \mathcal{O}_E(-(k+1)E), \quad 0 \leq k < n.$$

By **Step 1**, the first cohomology of each graded piece vanishes. By the long exact sequences of cohomology,

$$H^1(E_n, \mathcal{N}(-E)|_{E_n}) = 0 \quad \text{for any integer } n \geq 1.$$

We may assume that X is affine. Let $\mathfrak{m}_x \subset \mathcal{O}_X$ be the ideal sheaf of x. Since E is reduced and $E = f^{-1}(x)_{\mathrm{red}}$, the ideal sheaf $\mathcal{O}_Y(-E)$ is the radical of $\mathfrak{m}_x\mathcal{O}_Y$. Since Y is Noetherian, there exists an integer $c \geq 1$ such that

$$\mathcal{O}_Y(-cE) \subset \mathfrak{m}_x\mathcal{O}_Y \subset \mathcal{O}_Y(-E).$$

Thus the thickenings E_n and the closed subschemes $Y \times_X \operatorname{Spec}(\mathcal{O}_{X,x}/\mathfrak{m}_x^n)$ form cofinal systems, and the theorem on formal functions [Har77, Theorem III.11.1] implies that the completion of the $\mathcal{O}_{X,x}$-module $(R^1 f_* \mathcal{N}(-E))_x$ is

$$\varprojlim_n H^1(E_n, \mathcal{N}(-E)|_{E_n}) = 0.$$

The module $(R^1 f_* \mathcal{N}(-E))_x$ is finitely generated [Har77, Theorem III.8.8(b)], and the completion of a finitely generated $\mathcal{O}_{X,x}$-module is the tensor product with $\widehat{\mathcal{O}}_{X,x}$ [Mat86, Theorem 8.7]. Hence $(R^1 f_* \mathcal{N}(-E))_x$ is zero modulo $\mathfrak{m}_x$, so $(R^1 f_* \mathcal{N}(-E))_x = 0$ by Nakayama's lemma. Shrinking X around x, we may assume that $R^1 f_* \mathcal{N}(-E) = 0$. Applying f_* to the exact sequence

$$0 \to \mathcal{N}(-E) \to \mathcal{N} \to \mathcal{N}|_E \to 0,$$

we obtain a surjection $H^0(Y, \mathcal{N}) \to H^0(E, \mathcal{N}|_E) \cong H^0(E, \mathcal{O}_E)$. Let $\sigma \in H^0(Y, \mathcal{N})$ be a lift of 1. Then σ does not vanish at any point of E. Since f is proper, $f(\{\sigma = 0\})$ is a closed subset of X which does not contain x. Replacing X by $X \backslash f(\{\sigma = 0\})$, σ is a nowhere vanishing section of $\mathcal{N}$, so $\mathcal{N} \cong \mathcal{O}_Y$, that is, $\mathcal{L}^{\otimes r} \cong \mathcal{O}_Y(-E)$.

From now on, let $\mathcal{L}$ be any line bundle on Y with $\mathcal{L}|_E \cong \mathcal{L}_E$ and $\mathcal{L}^{\otimes r} \cong \mathcal{O}_Y(-E)$, as in the statement, and fix an isomorphism $\mathcal{L}^{\otimes r} \cong \mathcal{O}_Y(-E)$.

Step 3. In this step, we show that $\tilde{Y}$ is normal and integral, and we prove the properties of $\tilde{E}$ in (1).

The multiplication of $\bigoplus_{i=0}^{r-1} \mathcal{L}^{\otimes i}$ is the tensor product $\mathcal{L}^{\otimes i} \otimes \mathcal{L}^{\otimes j} \to \mathcal{L}^{\otimes (i+j)}$ when $i + j < r$, and is the composition

$$\mathcal{L}^{\otimes i} \otimes \mathcal{L}^{\otimes j} \to \mathcal{L}^{\otimes (i+j-r)} \otimes \mathcal{L}^{\otimes r} \cong \mathcal{L}^{\otimes (i+j-r)} \otimes \mathcal{O}_Y(-E) \to \mathcal{L}^{\otimes (i+j-r)}$$

when $i + j \geq r$. Let $V \subset Y$ be an affine open subset on which $\mathcal{L}$ has a generator ℓ, and let $s_E \in \mathcal{O}_Y(V)$ be the image of $\ell^{\otimes r}$. Then s_E is a local equation of E on V, and

$$\pi^{-1}(V) \cong \operatorname{Spec} \mathcal{O}_Y(V)[\tau]/(\tau^r - s_E),$$

where τ corresponds to ℓ. In particular, π is finite and locally free of degree r. The group μ_r acts on $\tilde{Y}$ over Y, where $\zeta \in \mu_r$ acts on $\mathcal{L}^{\otimes i}$ by ζ^i, that is, $\tau \to \zeta\tau$, and the invariant subalgebra is $\mathcal{O}_Y$. The elements s_E and τ are units over $Y \backslash E$, and the derivative $r\tau^{r-1}$ of $\tau^r - s_E$ is a unit, so π is étale over $Y \backslash E$.

We show that $\tilde{Y}$ is integral. Let η be the generic point of an irreducible component of E. Then $\mathcal{O}_{Y,\eta}$ is a DVR, and s_E is a uniformizer of $\mathcal{O}_{Y,\eta}$ because E is reduced. Thus $T^r - s_E$ is an Eisenstein polynomial over $\mathcal{O}_{Y,\eta}$, hence irreducible over the function field $K(Y)$, and the generic fiber $K(Y)[T]/(T^r - s_E)$ of π is a field. Since π is flat, any irreducible component of $\tilde{Y}$ dominates Y, and since π is locally free, $\mathcal{O}_{\tilde{Y}}$ is a subsheaf of the constant sheaf $K(Y)[T]/(T^r - s_E)$. Thus $\tilde{Y}$ is integral, and in particular connected.

We show that $\tilde{Y}$ is normal. Locally $\tilde{Y}$ is defined in the polynomial ring $\mathcal{O}_Y(V)[\tau]$, which is Cohen–Macaulay [Mat86, Theorem 17.7], by the monic polynomial $\tau^r - s_E$, which is a nonzerodivisor. Thus $\tilde{Y}$ is Cohen–Macaulay [Har77, Theorem II.8.21A(b)(d)], and in particular S_2. Let $\tilde{y} \in \tilde{Y}$ be a codimension one point. Since π is finite and flat, $y := \pi(\tilde{y})$ is a codimension one point of Y [Mat86, Theorem 15.1(ii)], so $\mathcal{O}_{Y,y}$ is a DVR. If $y \notin E$, then π is étale at $\tilde{y}$, so $\mathcal{O}_{\tilde{Y},\tilde{y}}$ is regular. If $y \in E$, then y is the generic point of an irreducible component of E, and $\mathcal{O}_{Y,y}[\tau]/(\tau^r - s_E)$ is a DVR with uniformizer τ, since $\tau^r - s_E$ is Eisenstein. Thus $\tilde{Y}$ is regular in codimension one, and $\tilde{Y}$ is normal by Serre's criterion [Har77, Theorem II.8.22A].

We show the properties of $\tilde{E}$. Over V, the scheme $\pi^{-1}(E)$ is $\operatorname{Spec}(\mathcal{O}_Y(V)/(s_E))[\tau]/(\tau^r)$. Since E is reduced, $\pi^{-1}(E)_{\mathrm{red}}$ is defined by $\tau = 0$ over V. The ideal generated by τ is the image of the multiplication map

$$\mathcal{L} \otimes \bigoplus_{i=0}^{r-1} \mathcal{L}^{\otimes i} \to \bigoplus_{i=0}^{r-1} \mathcal{L}^{\otimes i},$$

which does not depend on the choice of ℓ. Since $\tilde{Y}$ is integral, τ is a nonzerodivisor, and the multiplication map induces an isomorphism

$$\pi^*\mathcal{L} \cong \mathcal{O}_{\tilde{Y}}(-\tilde{E}). \tag{8.1}$$

Thus $\tilde{E}$ is a Cartier divisor on $\tilde{Y}$. Restricting (8.1) to $\tilde{E}$, we obtain

$$\mathcal{O}_{\tilde{E}}(-\tilde{E}) \cong (\pi|_{\tilde{E}})^*(\mathcal{L}|_E) \cong (\pi|_{\tilde{E}})^*\mathcal{L}_E.$$

Moreover, $\pi|_{\tilde{E}} : \tilde{E} \to E$ is an isomorphism, since over V

$$\mathcal{O}_{\tilde{Y}}(V)/(\tau) \cong \mathcal{O}_Y(V)/(s_E).$$

The conormal sheaf $\mathcal{O}_{\tilde{E}}(-\tilde{E})$ is generated over V by the class of τ, and $\zeta \in \mu_r$ maps τ to $\zeta\tau$. Thus μ_r acts on $\mathcal{O}_{\tilde{E}}(-\tilde{E})$ by the character $\zeta \to \zeta$, which is primitive. This proves (1).

Step 4. In this step, we study the Stein factorization of $f \circ \pi$ and prove (2).

Since π is finite and f is projective, $f \circ \pi$ is a proper morphism between quasi-projective varieties, hence projective, so $(f \circ \pi)_*\mathcal{O}_{\tilde{Y}}$ is a coherent sheaf of $\mathcal{O}_X$-algebras [Har77, Theorem III.8.8(b)]. Let

$$W := \operatorname{Spec}_X (f \circ \pi)_*\mathcal{O}_{\tilde{Y}},$$

let $p : W \to X$ be the induced finite morphism, and let $\tilde{f} : \tilde{Y} \to W$ be the morphism over X induced by the identity of $(f \circ \pi)_*\mathcal{O}_{\tilde{Y}}$, so that $f \circ \pi = p \circ \tilde{f}$ is the Stein factorization of $f \circ \pi$. Since p is affine and $p_*\tilde{f}_*\mathcal{O}_{\tilde{Y}} = (f \circ \pi)_*\mathcal{O}_{\tilde{Y}} = p_*\mathcal{O}_W$, we have $\tilde{f}_*\mathcal{O}_{\tilde{Y}} = \mathcal{O}_W$. Since p is separated, $\tilde{f}$ is projective [Har77, Exercises II.4.8(e) and II.4.9]. For any open subset $U \subset X$, the ring $\mathcal{O}_{\tilde{Y}}((f \circ \pi)^{-1}(U))$ is contained in the function field $K(\tilde{Y})$, and any element of $K(\tilde{Y})$ which is integral over $\mathcal{O}_{\tilde{Y}}((f \circ \pi)^{-1}(U))$ is contained in $\mathcal{O}_{\tilde{Y},\tilde{y}}$ for any $\tilde{y} \in (f \circ \pi)^{-1}(U)$ because $\tilde{Y}$ is normal, hence is a section over $(f \circ \pi)^{-1}(U)$. Thus W is normal and integral.

Over $X \backslash \{x\}$, f is an isomorphism, so $f \circ \pi$ is finite over $X \backslash \{x\}$. Thus $\tilde{f}$ is an isomorphism over $p^{-1}(X \backslash \{x\})$, and p is étale over $X \backslash \{x\}$ because π is étale over $Y \backslash E$. In particular, $\tilde{f}$ is birational, $K(W) = K(\tilde{Y})$, and $\deg p = [K(\tilde{Y}) : K(Y)] = r$.

As a set, $p^{-1}(x)$ is the image under $\tilde{f}$ of $(f \circ \pi)^{-1}(x) = \pi^{-1}(E)$, and $\pi^{-1}(E)_{\mathrm{red}} = \tilde{E} \cong E$. By **Step 1**, E is connected, so $p^{-1}(x)$ is a connected finite set of closed points. Thus $p^{-1}(x) = \{w\}$ for a closed point $w \in W$ with residue field $\mathbb{C}$. In particular, $\tilde{f}^{-1}(w) = (f \circ \pi)^{-1}(x)$ as sets, so $\tilde{f}^{-1}(w)_{\mathrm{red}} = \tilde{E}$, and $\tilde{f}$ is an isomorphism over $W \backslash \{w\}$.

The group μ_r acts on $\tilde{Y}$ over X, hence on $(f \circ \pi)_*\mathcal{O}_{\tilde{Y}}$, and the invariant subalgebra is

$$\big((f \circ \pi)_*\mathcal{O}_{\tilde{Y}}\big)^{\mu_r} = f_*\big((\pi_*\mathcal{O}_{\tilde{Y}})^{\mu_r}\big) = f_*\mathcal{O}_Y = \mathcal{O}_X,$$

where $f_*\mathcal{O}_Y = \mathcal{O}_X$ because $f_*\mathcal{O}_Y$ is a coherent $\mathcal{O}_X$-algebra [Har77, Theorem III.8.8(b)] contained in $K(X)$ and X is normal. Thus $p : W \to X$ is the quotient by μ_r, and p is a connected normal

cyclic cover of degree r with group μ_r. Since p is étale over $X\backslash\{x\}$ and $\dim X = 3$, p is quasi-étale. Thus $K_W = p^*K_X$, and W is klt [KM98, Proposition 5.20(4)]. This proves (2). □

Proposition 8.5(2) is used in the proof of Theorem 8.1 to exclude $E \cong \mathbb{P}(1,1,2)$, and Proposition 8.5(3) is used in the proof of Proposition 9.10. In Set-up 8.2, if (Y,E) is lc and $E \cong \mathbb{P}^1 \times \mathbb{P}^1$, then Proposition 8.5(1) gives the description of $X \ni x$ in the last sentence of Proposition 8.9 by a second argument (embedded in $\mathbb{P}^3$ by $|\mathcal{O}_{\mathbb{P}^1\times\mathbb{P}^1}(1,1)|$, E is a quadric surface of rank 4, and $\mathcal{O}_E(-E) \cong \omega_E^{-1} \cong \mathcal{O}_E(2)$ by the remark after Set-up 8.2 and [Har77, Example II.8.20.3]). The double cover in Proposition 8.5 is given by Lemma 8.4 applied to the square root $\mathcal{O}_E(1)$ of $\mathcal{O}_E(-E)$.

Proposition 8.5. *Notation and conditions as in Set-up 8.2. Assume that (Y,E) is lc, that $E \subset \mathbb{P}^3$ is a reduced quadric surface of rank $r \in \{2,3,4\}$ with hyperplane bundle $\mathcal{O}_E(1)$, and that $\mathcal{O}_E(-E) \cong \mathcal{O}_E(2)$. Then, after replacing $X \ni x$ by a pointed étale neighborhood, there exists a finite quasi-étale morphism $(W \ni w) \to (X \ni x)$ of degree two such that $W \ni w$ is a hypersurface germ whose tangent cone is the affine quadric cone of rank r and whose deck involution acts by -1 on the Zariski tangent space of W at w. In particular, $X \ni x$ is LCIQ. Moreover, the following hold.*

(1) If $r = 4$, then $X \ni x$ is analytically isomorphic to the quotient of the threefold ordinary double point $\{u_1^2+u_2^2+u_3^2+u_4^2=0\}$ by the involution

$$(u_1,u_2,u_3,u_4) \to (-u_1,-u_2,-u_3,-u_4),$$

and $\widehat{\mathrm{vol}}(X \ni x) = 8$.

(2) If $r = 3$, then $E \cong \mathbb{P}(1,1,2)$, and $X \ni x$ is analytically isomorphic to the quotient of $\{u_1^2+u_2^2+u_3^2+t^{2k}=0\}$ for an integer $k \geq 2$, or of $\{u_1^2+u_2^2+u_3^2=0\}$, by the involution $(u_1,u_2,u_3,t) \to (-u_1,-u_2,-u_3,-t)$. In both cases $\widehat{\mathrm{vol}}(X \ni x) = \frac{27}{4}$.

(3) If $r = 2$, then E is the union of two planes intersecting along a line,

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[u,v,z,t]]/(uv + h(z,t)),$$

where $h(-z,-t) = h(z,t)$ and $\operatorname{ord} h \geq 4$, and $\widehat{\mathrm{vol}}(X \ni x) \leq 4$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct the double cover and describe E.

Since $\mathcal{O}_E(1)^{\otimes 2} \cong \mathcal{O}_E(2) \cong \mathcal{O}_E(-E)$, we may use Lemma 8.4 with $\mathcal{L}_E = \mathcal{O}_E(1)$ and the integer 2. A pointed étale neighborhood of $X \ni x$ with residue field $\mathbb{C}$ has completed local ring isomorphic to $\widehat{\mathcal{O}}_{X,x}$ [Har77, Exercise III.10.4], so replacing $X \ni x$ by such a neighborhood changes neither the analytic isomorphism class of $X \ni x$, nor $\widehat{\mathrm{vol}}(X \ni x)$ (Proposition 2.7(1)), nor the LCIQ property. By Lemma 8.4, after replacing $X \ni x$ by such a neighborhood, there exists a line bundle $\mathcal{L}$ on Y with $\mathcal{L}|_E \cong \mathcal{O}_E(1)$ and $\mathcal{L}^{\otimes 2} \cong \mathcal{O}_Y(-E)$. Let $\pi : \tilde{Y} := \operatorname{Spec}_Y(\mathcal{O}_Y \oplus \mathcal{L}) \to Y$ be the double cover defined by $\mathcal{L}^{\otimes 2} \cong \mathcal{O}_Y(-E) \subset \mathcal{O}_Y$, let $\tilde{E} := \pi^{-1}(E)_{\mathrm{red}}$, and let $\tilde{Y} \xrightarrow{\tilde{f}} W \xrightarrow{p} X$ be the Stein factorization of $f \circ \pi$. Let ι be the non-trivial element of μ_2, acting on $\tilde{Y}$ and on W. By Lemma 8.4(2), $p^{-1}(x)$ is a single point w, $p : (W \ni w) \to (X \ni x)$ is quasi-étale of degree two, $X = W/\iota$, and W is klt.

Since $\pi|_{\tilde{E}} : \tilde{E} \to E$ is an isomorphism and $\pi \circ \iota = \pi$, the involution ι acts trivially on $\tilde{E}$. We identify $\tilde{E}$ with E by $\pi|_{\tilde{E}}$. By Lemma 8.4(1), $\mathcal{O}_{\tilde{E}}(-\tilde{E}) \cong \mathcal{O}_E(1)$, and ι acts on $\mathcal{O}_{\tilde{E}}(-\tilde{E})$ by -1, the only primitive character of μ_2.

Let $\mathbb{P}^3 = \operatorname{Proj}\mathbb{C}[y_1,y_2,y_3,y_4]$, and let $Q_r \in \mathbb{C}[y_1,y_2,y_3,y_4]$ be a quadratic form of rank r defining E. If $r = 3$, then after a linear change of coordinates, $Q_3 = y_1y_3 - y_2^2$. Let $\mathbb{C}[c_1,c_2,c_3]$ be graded by $\deg c_1 = \deg c_2 = 1$ and $\deg c_3 = 2$. The subring of $\mathbb{C}[c_1,c_2,c_3]$ of elements of even degree is generated by c_1^2, c_1c_2, c_2^2, and c_3, with the single relation $c_1^2 \cdot c_2^2 = (c_1c_2)^2$ (the surjection from $\mathbb{C}[y_1,y_2,y_3,y_4]/(Q_3)$ onto $\bigoplus_{n\geq 0}\mathbb{C}[c_1,c_2,c_3]_{2n}$ sending y_1,y_2,y_3,y_4 to $c_1^2, c_1c_2, c_2^2, c_3$ is an isomorphism, since both graded rings have Hilbert function $(n+1)^2$). Since Proj of the subring

of elements of even degree is isomorphic to $\operatorname{Proj}\mathbb{C}[c_1,c_2,c_3]$ [Gro61, Proposition 2.4.7(i)], we obtain $E \cong \operatorname{Proj}\mathbb{C}[c_1,c_2,c_3] = \mathbb{P}(1,1,2)$. If $r=2$, then after a linear change of coordinates, $Q_2 = y_1y_2$, and E is the union of the planes $\{y_1=0\}$ and $\{y_2=0\}$, which intersect along the line $\{y_1=y_2=0\}$.

Step 2. In this step, we show that $H^1(E,\mathcal{O}_E(n))=0$ for any integer n, and that

$$\bigoplus_{n\geq 0} H^0(E,\mathcal{O}_E(n)) \cong \mathbb{C}[y_1,y_2,y_3,y_4]/(Q_r)$$

as graded rings.

For any integer n, consider the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^3}(n-2) \xrightarrow{Q_r} \mathcal{O}_{\mathbb{P}^3}(n) \to \mathcal{O}_E(n) \to 0.$$

Since $H^i(\mathbb{P}^3,\mathcal{O}_{\mathbb{P}^3}(j))=0$ for $i=1,2$ and any integer j [Har77, Theorem III.5.1(b)], we have $H^1(E,\mathcal{O}_E(n))=0$, and the restriction map

$$H^0(\mathbb{P}^3,\mathcal{O}_{\mathbb{P}^3}(n)) \to H^0(E,\mathcal{O}_E(n))$$

is surjective with kernel $Q_r\cdot H^0(\mathbb{P}^3,\mathcal{O}_{\mathbb{P}^3}(n-2))$. This implies the isomorphism of graded rings. In particular, $H^0(E,\mathcal{O}_E)=\mathbb{C}$, and $H^0(E,\mathcal{O}_E(1))$ has basis y_1,y_2,y_3,y_4.

Step 3. In this step, we construct a filtration $(\mathfrak{a}_n)_{n\geq 0}$ of $\widehat{\mathcal{O}}_{W,w}$ by ideals and compute the associated graded ring.

Shrinking X around x, we may assume that X is affine, so W is affine. Let $\mathfrak{m}_w \subset \mathcal{O}_W$ be the ideal sheaf of w, let $I := \mathcal{O}_{\tilde{Y}}(-\tilde{E})$, and for any integer $n\geq 1$, let $\tilde{E}_n$ be the closed subscheme of $\tilde{Y}$ defined by I^n. By Lemma 8.4(2), $\tilde{f}^{-1}(w)_{\mathrm{red}} = \tilde{E}$. Since $\tilde{E}$ is reduced, I is the radical of $\mathfrak{m}_w\mathcal{O}_{\tilde{Y}}$, so there exists an integer $c\geq 1$ such that

$$I^c \subset \mathfrak{m}_w\mathcal{O}_{\tilde{Y}} \subset I.$$

Since $\tilde{f}_*\mathcal{O}_{\tilde{Y}} = \mathcal{O}_W$, the theorem on formal functions [Har77, Theorem III.11.1], applied to the projective morphism $\tilde{f}$ and $\mathcal{O}_{\tilde{Y}}$, together with the cofinality of the systems $(I^n)_n$ and $(\mathfrak{m}_w^n\mathcal{O}_{\tilde{Y}})_n$, implies a ring isomorphism

$$\widehat{\mathcal{O}}_{W,w} \cong \varprojlim_n H^0(\tilde{E}_n,\mathcal{O}_{\tilde{E}_n}),$$

which is compatible with ι.

Since I is invertible, $I^n/I^{n+1} \cong \mathcal{O}_{\tilde{E}}(-n\tilde{E}) \cong \mathcal{O}_E(n)$, and ι acts on I^n/I^{n+1} by $(-1)^n$. By **Step 2**, the exact sequence

$$0 \to \mathcal{O}_E(n) \to \mathcal{O}_{\tilde{E}_{n+1}} \to \mathcal{O}_{\tilde{E}_n} \to 0$$

induces a surjection

$$H^0(\tilde{E}_{n+1},\mathcal{O}_{\tilde{E}_{n+1}}) \to H^0(\tilde{E}_n,\mathcal{O}_{\tilde{E}_n})$$

with kernel $H^0(E,\mathcal{O}_E(n))$. Thus the natural map $\widehat{\mathcal{O}}_{W,w} \to H^0(\tilde{E}_n,\mathcal{O}_{\tilde{E}_n})$ is surjective for any $n\geq 1$. Let $\mathfrak{a}_n$ be the kernel of $\widehat{\mathcal{O}}_{W,w} \to H^0(\tilde{E}_n,\mathcal{O}_{\tilde{E}_n})$ for $n\geq 1$, and let $\mathfrak{a}_0 := \widehat{\mathcal{O}}_{W,w}$. Then the following hold, by the isomorphism and the exact sequence above.

(i) $\bigcap_{n\geq 0}\mathfrak{a}_n = 0$ by the isomorphism $\widehat{\mathcal{O}}_{W,w} \cong \varprojlim_n H^0(\tilde{E}_n,\mathcal{O}_{\tilde{E}_n})$, and $\mathfrak{a}_m\mathfrak{a}_n \subset \mathfrak{a}_{m+n}$ for any integers $m,n\geq 0$, since for $k\geq n$ the image of $\mathfrak{a}_n$ in $H^0(\tilde{E}_k,\mathcal{O}_{\tilde{E}_k})$ is contained in $H^0(\tilde{E}_k, I^n/I^k)$ and $I^mI^n \subset I^{m+n}$.

(ii) Since $\widehat{\mathcal{O}}_{W,w} \to H^0(\tilde{E}_{n+1},\mathcal{O}_{\tilde{E}_{n+1}})$ is surjective, $\mathfrak{a}_n/\mathfrak{a}_{n+1}$ is the kernel $H^0(E,\mathcal{O}_E(n))$ of the surjection above, and the multiplication corresponds to $I^m/I^{m+1}\otimes I^n/I^{n+1} \to$

I^{m+n}/I^{m+n+1}, that is, to $\mathcal{O}_E(m) \otimes \mathcal{O}_E(n) \to \mathcal{O}_E(m+n)$. Thus there exists an isomorphism of graded rings

$$\bigoplus_{n\geq 0} \mathfrak{a}_n/\mathfrak{a}_{n+1} \cong \bigoplus_{n\geq 0} H^0(E, \mathcal{O}_E(n)) \cong \mathbb{C}[y_1, y_2, y_3, y_4]/(Q_r),$$

and ι acts on $\mathfrak{a}_n/\mathfrak{a}_{n+1}$ by $(-1)^n$, as on I^n/I^{n+1}.

Since $\mathfrak{a}_0/\mathfrak{a}_1 \cong H^0(E, \mathcal{O}_E) = \mathbb{C}$, the ideal $\mathfrak{a}_1$ is the maximal ideal $\widehat{\mathfrak{m}}$ of $\widehat{\mathcal{O}}_{W,w}$.

Step 4. In this step, we show that $\mathfrak{a}_n = \widehat{\mathfrak{m}}^n$, and we write $\widehat{\mathcal{O}}_{W,w}$ as the quotient of a power series ring by an even series.

Choose $s_1, s_2, s_3, s_4 \in \mathfrak{a}_1$ whose classes in $\mathfrak{a}_1/\mathfrak{a}_2$ are y_1, y_2, y_3, y_4 under the isomorphism in (ii). Since ι acts on $\mathfrak{a}_1/\mathfrak{a}_2$ by -1, the class of $\frac{1}{2}(s_i - \iota(s_i))$ in $\mathfrak{a}_1/\mathfrak{a}_2$ is y_i. Replace s_i by $\frac{1}{2}(s_i - \iota(s_i))$ for $1 \leq i \leq 4$. Then $\iota(s_i) = -s_i$.

Let $n \geq 0$ be an integer and $g \in \mathfrak{a}_n$. By (ii), there exists a homogeneous polynomial g_n of degree n such that $g - g_n(s_1, \dots, s_4) \in \mathfrak{a}_{n+1}$. Repeating this, we obtain homogeneous polynomials g_j of degree j for any $j \geq n$ such that

$$g - \sum_{j=n}^{m} g_j(s_1, \dots, s_4) \in \mathfrak{a}_{m+1} \quad \text{for any integer } m \geq n.$$

Since $g_j(s_1, \dots, s_4) \in \widehat{\mathfrak{m}}^j$ and $\widehat{\mathcal{O}}_{W,w}$ is complete, the series $g' := \sum_{j\geq n} g_j(s_1, \dots, s_4)$ converges, and $g' \in \widehat{\mathfrak{m}}^n$. For any integer $m \geq n$, we have

$$\sum_{j>m} g_j(s_1, \dots, s_4) \in \widehat{\mathfrak{m}}^{m+1} = \mathfrak{a}_1^{m+1} \subset \mathfrak{a}_{m+1},$$

so $g - g' \in \mathfrak{a}_{m+1}$. By (i), $g = g'$. Thus $\mathfrak{a}_n \subset \widehat{\mathfrak{m}}^n$, and $\widehat{\mathfrak{m}}^n = \mathfrak{a}_1^n \subset \mathfrak{a}_n$ by (i), so $\mathfrak{a}_n = \widehat{\mathfrak{m}}^n$. In particular,

$$\operatorname{gr}_{\widehat{\mathfrak{m}}} \widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[y_1, y_2, y_3, y_4]/(Q_r),$$

where y_i is the class of s_i. Moreover, the case $n = 0$ shows that the continuous homomorphism

$$\Phi : \mathbb{C}[[y_1, y_2, y_3, y_4]] \to \widehat{\mathcal{O}}_{W,w}, \quad y_i \to s_i,$$

is surjective. Let ι act on $\mathbb{C}[[y_1, y_2, y_3, y_4]]$ by $y_i \to -y_i$. Then Φ is ι-equivariant.

Since $\mathcal{O}_{W,w}$ is a normal local ring of a variety, $\widehat{\mathcal{O}}_{W,w}$ is normal [Gro65, Scholie 7.8.3(ii)(iii)(v)], hence a domain, and $\dim \widehat{\mathcal{O}}_{W,w} = 3$. Thus $\operatorname{Ker} \Phi$ is a prime ideal of height one in the regular local ring $\mathbb{C}[[y_1, y_2, y_3, y_4]]$, which is a unique factorization domain [Mat86, Theorem 20.3], so $\operatorname{Ker} \Phi = (\phi)$ for some ϕ [Mat86, Theorem 20.1]. For a non-zero series $g \in \mathbb{C}[[y_1, y_2, y_3, y_4]]$, the *leading form* of g is the non-zero homogeneous part of g of lowest degree. Since $\mathbb{C}[y_1, y_2, y_3, y_4]$ is a domain, the leading forms of the non-zero elements of (ϕ) are the multiples of the leading form of ϕ, so

$$\operatorname{gr}_{\widehat{\mathfrak{m}}} \widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[y_1, y_2, y_3, y_4]/(\text{leading form of } \phi).$$

Comparing with the presentation $\mathbb{C}[y_1, y_2, y_3, y_4]/(Q_r)$ of $\operatorname{gr}_{\widehat{\mathfrak{m}}} \widehat{\mathcal{O}}_{W,w}$ above, the leading form of ϕ is a non-zero scalar multiple of Q_r. Rescaling ϕ, we may assume that the leading form of ϕ is Q_r.

Since Φ is ι-equivariant, (ϕ) is ι-stable, so $\iota(\phi) = U\phi$ for a unit U. Since $\iota(Q_r) = Q_r$, the constant term of U is 1. Thus

$$\frac{1}{2}(\phi + \iota(\phi)) = \frac{1+U}{2}\phi$$

is again a generator of $\operatorname{Ker} \Phi$, with leading form Q_r. Replacing ϕ with $\frac{1}{2}(\phi + \iota(\phi))$, we may assume that ϕ is even, that is, $\iota(\phi) = \phi$. Hence

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[y_1, y_2, y_3, y_4]]/(\phi),$$

ι-equivariantly. In particular, $W \ni w$ is a hypersurface germ whose tangent cone

$$\operatorname{Spec} \mathbb{C}[y_1, y_2, y_3, y_4]/(Q_r)$$

is the affine quadric cone of rank r, and ι acts by -1 on $\widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2$, hence on the Zariski tangent space of W at w.

Step 5. In this step, we find ι-equivariant coordinates in which ϕ splits as a sum of squares and a series in the remaining variables.

After a linear change of coordinates, which commutes with ι, we may write the coordinates as $u = (u_1, \dots, u_r)$ and $t = (t_1, \dots, t_{4-r})$, where t is empty when $r = 4$, so that $Q_r = \sum_{i=1}^r u_i^2$. Since $\frac{\partial \phi}{\partial u_i}$ is $2u_i$ plus a series of order at least 2, the equations

$$\frac{\partial \phi}{\partial u_i}(u, t) = 0, \quad 1 \le i \le r,$$

have a unique solution $u = \psi(t)$ with $\psi \in \mathbb{C}[[t]]^r$ and $\psi(0) = 0$, determined degree by degree, and $\operatorname{ord} \psi \ge 2$. Since $\phi(-u, -t) = \phi(u, t)$, we have

$$\frac{\partial \phi}{\partial u_i}(-u, -t) = -\frac{\partial \phi}{\partial u_i}(u, t),$$

so $u = -\psi(-t)$ is also a solution. By uniqueness, $\psi(-t) = -\psi(t)$. Thus the coordinate change $v := u - \psi(t)$ commutes with ι. In the coordinates (v, t), the series ϕ is even, has leading form $\sum_{i=1}^r v_i^2$, and satisfies $\frac{\partial \phi}{\partial v_i}(0, t) = 0$ for $1 \le i \le r$. Let $h(t) := \phi(0, t)$. By the Taylor formula with integral remainder in v,

$$\phi(v, t) = h(t) + v^T S(v, t) v, \quad S(v, t) := \int_0^1 (1 - \lambda) \operatorname{Hess}_v \phi(\lambda v, t) d\lambda,$$

where $\operatorname{Hess}_v \phi$ is the $r \times r$ matrix of second partial derivatives of ϕ in v, and the integral is taken monomial by monomial, which only divides coefficients by positive integers. The matrix $S(v, t)$ is symmetric, $S(0, 0) = I_r$, and $S(-v, -t) = S(v, t)$ because second partial derivatives of an even series are even. The entries of $S - I_r$ are contained in the maximal ideal, so the binomial series

$$P(v, t) := \big(I_r + (S(v, t) - I_r)\big)^{1/2} = \sum_{j \ge 0} \binom{1/2}{j} \big(S(v, t) - I_r\big)^j$$

converges. The matrix $P(v, t)$ is symmetric and even, and $P(v, t)^2 = S(v, t)$, so

$$v^T S(v, t) v = (P(v, t) v)^T (P(v, t) v).$$

The substitution $v \to P(v, t) v$ has identity linear part and maps odd series to odd series, so it is a coordinate change which commutes with ι. This is the splitting lemma [AGV85, Section 11.1], with the involution taken into account. Renaming the new coordinates $u_1, \dots, u_r$, we obtain

$$\phi = u_1^2 + \cdots + u_r^2 + h(t),$$

where ι acts by -1 on $u_1, \dots, u_r, t$. The series h is even. Moreover, $h(t) = \phi(\psi(t), t)$, where ϕ is written in the coordinates (u, t) before the substitution $v = u - \psi(t)$, so that $\phi - \sum_{i=1}^r u_i^2$ has order at least 3. Since $\operatorname{ord} \psi \ge 2$, we have $\operatorname{ord} h \ge 3$. Since h is even, all monomials of h have even degree, so $\operatorname{ord} h \ge 4$.

If $r = 4$, then t is empty and $h = 0$. If $r = 3$, then t is one variable, and either $h = 0$ or $h = t^{2k} U(t)$ for an integer $k \ge 2$ and an even unit $U \in \mathbb{C}[[t]]$. In the second case, choose $c \in \mathbb{C}$ with $c^{2k} = U(0)$, and let

$$\rho(t) := c \sum_{j \ge 0} \binom{1/(2k)}{j} \Big(\frac{U(t)}{U(0)} - 1\Big)^j.$$

Then ρ is even and $\rho^{2k} = U$, so $T := t\rho(t)$ is an odd coordinate with $h = T^{2k}$. Renaming T as t, we obtain $\phi = u_1^2 + u_2^2 + u_3^2 + t^{2k}$ or $\phi = u_1^2 + u_2^2 + u_3^2$. If $r = 2$, then write (z, t) for (t_1, t_2), and replace (u_1, u_2) by $(u_1 + \sqrt{-1} u_2, u_1 - \sqrt{-1} u_2)$, which we rename (u, v). Then $\phi = uv + h(z, t)$ with $h(-z, -t) = h(z, t)$ and $\operatorname{ord} h \ge 4$.

Step 6. In this step, we show that $X \ni x$ is LCIQ, and we prove (1)–(3).

We first show that taking invariants commutes with completion. Let Z be an affine variety with an involution ι_Z, and let $\xi \in Z$ be a closed point which is the only point of Z over the image $\xi' \in Z/\iota_Z$ of ξ. Then $\mathcal{O}_{Z,\xi}$ is a finite $\mathcal{O}_{Z/\iota_Z,\xi'}$-module, and $\mathfrak{m}_{\xi'}\mathcal{O}_{Z,\xi}$ is $\mathfrak{m}_\xi$-primary, so $\widehat{\mathcal{O}}_{Z,\xi} = \mathcal{O}_{Z,\xi} \otimes \widehat{\mathcal{O}}_{Z/\iota_Z,\xi'}$ [Mat86, Theorem 8.7]. The ring $\mathcal{O}_{Z/\iota_Z,\xi'}$ is the kernel of $1-\iota_Z$ on $\mathcal{O}_{Z,\xi}$, and $\widehat{\mathcal{O}}_{Z/\iota_Z,\xi'}$ is flat over $\mathcal{O}_{Z/\iota_Z,\xi'}$ [Mat86, Theorem 8.8], so

$$\widehat{\mathcal{O}}_{Z/\iota_Z,\xi'} = \left(\widehat{\mathcal{O}}_{Z,\xi}\right)^{\iota_Z}. \tag{8.2}$$

Taking $Z = W$, $\iota_Z = \iota$, and $\xi = w$, we obtain $\widehat{\mathcal{O}}_{X,x} = (\widehat{\mathcal{O}}_{W,w})^\iota$.

By **Step 4**, $\widehat{\mathcal{O}}_{W,w}$ is a complete intersection, so $\mathcal{O}_{W,w}$ is a complete intersection [BH98, Definition 2.3.1 and Theorem 2.3.3]. The involution ι fixes w, and acts freely on $W\backslash\{w\}$ because p is étale of degree two over $X\backslash\{x\}$. Since $X = W/\iota$, the germ $X \ni x$ is LCIQ (Definition 2.13). By **Step 1**, p is a finite Galois morphism of degree two between klt germs with $p^{-1}(x) = \{w\}$, and $K_W = p^*K_X$ since p is quasi-étale. Thus Proposition 2.7(2) implies

$$\widehat{\mathrm{vol}}(W \ni w) = 2\widehat{\mathrm{vol}}(X \ni x).$$

Assume that $r = 4$ or $r = 3$. Let $V \subset \mathbb{A}^4$ be the hypersurface defined by the polynomial ϕ of **Step 5**, that is,

$$V = \begin{cases} \{u_1^2 + u_2^2 + u_3^2 + u_4^2 = 0\} & \text{if } r = 4, \\ \{u_1^2 + u_2^2 + u_3^2 + t^{2k} = 0\} \text{ or } \{u_1^2 + u_2^2 + u_3^2 = 0\} & \text{if } r = 3. \end{cases}$$

Let ι_V be the involution of V given by -1 on the coordinates of $\mathbb{A}^4$. The origin 0 is the only fixed point of -1 on $\mathbb{A}^4$. By **Step 5**, $\widehat{\mathcal{O}}_{V,0} \cong \widehat{\mathcal{O}}_{W,w}$ compatibly with the involutions, so by (8.2) applied to $Z = V$ and $\xi = 0$,

$$\widehat{\mathcal{O}}_{X,x} = (\widehat{\mathcal{O}}_{W,w})^\iota \cong (\widehat{\mathcal{O}}_{V,0})^{\iota_V} = \widehat{\mathcal{O}}_{V/\iota_V,0'},$$

where $0'$ is the image of 0. Thus $X \ni x$ is analytically isomorphic to $V/\iota_V \ni 0'$, which is the germ in (1) or (2).

If $r = 4$, then after a linear change of coordinates, $\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[u,v,z,t]]/(uv + z^2 + t^2)$, so $\widehat{\mathrm{vol}}(W \ni w) = 16$ by Proposition 3.1(1). Thus $\widehat{\mathrm{vol}}(X \ni x) = 8$, and (1) holds.

If $r = 3$, then $E \cong \mathbb{P}(1,1,2)$ by **Step 1**. After a linear change of coordinates, $\widehat{\mathcal{O}}_{W,w}$ is isomorphic to $\mathbb{C}[[u,v,z,t]]/(uv + z^2 + t^{2k})$, which is the ring in Proposition 3.1(1) with k replaced by $2k-1 \geq 3$, or to $\mathbb{C}[[u,v,z,t]]/(uv + z^2)$, which is the ring in Proposition 3.1(2). By Proposition 3.1(1)(2), $\widehat{\mathrm{vol}}(W \ni w) = \frac{27}{2}$ in both cases. Thus $\widehat{\mathrm{vol}}(X \ni x) = \frac{27}{4}$, and (2) holds.

If $r = 2$, then E is the union of two planes intersecting along a line by **Step 1**, and by **Step 5**,

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[u,v,z,t]]/(uv + h(z,t)),$$

where $h(-z,-t) = h(z,t)$ and $\operatorname{ord} h \geq 4$. Let $v_{(2,2,1,1)}$ be the monomial valuation on $\mathbb{C}[[u,v,z,t]]$ with weights $(2,2,1,1)$. Then $v_{(2,2,1,1)}(uv) = 4$, and any monomial of h has weight at least 4, so $v_{(2,2,1,1)}(uv + h) = 4$. Since W is klt, Proposition 2.20 implies

$$\widehat{\mathrm{vol}}(W \ni w) \leq (2+2+1+1-4)^3 \cdot \frac{4}{2 \cdot 2 \cdot 1 \cdot 1} = 8.$$

Thus $\widehat{\mathrm{vol}}(X \ni x) \leq 4$, and (3) holds. □

8.3. **Non-normal exceptional surfaces.** In this subsection, we show that if $\mathrm{crex}(X \ni x) = 1$, then the crepant divisor over $X \ni x$ can be extracted as a Cartier divisor which is an integral Gorenstein del Pezzo surface, and that $\widehat{\mathrm{vol}}(X \ni x) \leq 7$ if the extracted surface is not normal. We first treat non-normal surfaces whose normalization is a Hirzebruch surface with a double conductor curve.

Lemma 8.6 treats the surfaces of type (e) in [Rei94, Theorem 1.1]. It is used in **Step 5** of the proof of Proposition 8.7.

Lemma 8.6. *Notation and conditions as in Set-up 8.2. Assume that E is integral, Y is locally a hypersurface along E, the normalization $\bar{E}$ of E is isomorphic to $\mathbb{F}_a$ for an integer $a \geq 0$, the conductor on $\bar{E}$ is the negative section B of $\mathbb{F}_a$, and the conductor C on E is a smooth rational curve. Assume moreover that $B \to C$ is finite of degree two, that E has only double crossings at the points of C over which $B \to C$ is unramified and pinch points at the branch points, and that Y is smooth at the generic point of C. For $p \in C$, choose local equations $\phi(s_1, s_2, s_3) = 0$ of E in a smooth threefold germ with $C = \{s_1 = s_2 = 0\}$ and $\phi(s_1, s_2, s_3) + s_4 g(s_1, s_2, s_3, s_4) = 0$ of Y in a smooth fourfold germ with $E = \{s_4 = 0\}$, and let $\operatorname{mult}_p D := \operatorname{ord}_p(g(0, 0, s_3, 0))$. Then $D \geq 0$ is a well-defined divisor on C, and*

$$(K_E^2) = a + 4 = 7 - \deg D \leq 7.$$

In particular, $\widehat{\operatorname{vol}}(X \ni x) \leq 7$.

Proof. The proof is long, so we divide it into several steps. Let $\nu : \bar{E} \to E$ be the normalization, and fix an isomorphism $\bar{E} \cong \mathbb{F}_a$, so that $\nu|_B : B \to C$ is the finite morphism of degree two. Let $\mathcal{I}$ be the ideal sheaf of C in Y. We compute in analytic local coordinates. Since $\mathcal{O}_{C,p} \to \mathcal{O}_{C,p}^{an}$ is faithfully flat for any $p \in C$, lengths and orders of vanishing do not change. For local equations as in the statement, we write

$$g_C := g(0, 0, s_3, 0).$$

Step 1. In this step, we show that local equations as in the statement exist, that D is a well-defined divisor with $D \geq 0$, and that $\mathcal{N} := \operatorname{Hom}_{\mathcal{O}_C}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_C)$ is locally free of rank two.

Let $p \in C$, and let $\phi(s_1, s_2, s_3) = 0$ be a local equation of E in a smooth threefold germ with $C = \{s_1 = s_2 = 0\}$, for example $\phi = s_1 s_2$ at a double crossing and $\phi = s_1^2 - s_2^2 s_3$ at a pinch point. Since E is singular along C, the function ϕ and the derivatives $\partial\phi/\partial s_1$, $\partial\phi/\partial s_2$ vanish along C, so

$$\phi \in (s_1, s_2)^2.$$

Since E is a Cartier divisor on Y, the ideal of E in $\mathcal{O}_{Y,p}$ is generated by one element h. If Y is smooth at p, then lifts of s_1, s_2, s_3 to $\mathcal{O}_{Y,p}$ are local coordinates of Y, because the embedding dimension of E at p is $3 = \dim Y$. Thus h is a unit times ϕ, and Y is the germ $\{\phi - s_4 = 0\}$ with $E = \{s_4 = 0\}$. If Y is singular at p, then the embedding dimension of Y at p is 4, since Y is locally a hypersurface along E. Since

$$\mathfrak{m}_{E,p}/\mathfrak{m}_{E,p}^2 \cong \mathfrak{m}_{Y,p}/(\mathfrak{m}_{Y,p}^2 + (h))$$

and the embedding dimension of E at p is 3, we have $h \notin \mathfrak{m}_{Y,p}^2$. Let $s_4 := h$, and lift s_1, s_2, s_3 to $\mathcal{O}_{Y,p}$. Then Y is a hypersurface $\{F = 0\}$ in a smooth fourfold germ with coordinates $s_1, \dots, s_4$, and $F(s_1, s_2, s_3, 0) = c\phi$ for a unit c. Replacing F with $c^{-1}F$, we may write $F = \phi + s_4 g$. Thus local equations as in the statement exist.

Now let $F = \phi + s_4 g$ be any local equation of Y as in the statement, and let $J := (s_1, s_2, s_4)$ in the local ring of the smooth fourfold germ. Then $\mathcal{I}/\mathcal{I}^2$ is locally $J/(J^2 + (F))$. Since $\phi \in J^2$ and $s_4 g \equiv g_C s_4$ modulo J^2, we have

$$\mathcal{I}/\mathcal{I}^2 \cong (\mathcal{O}_C s_1 \oplus \mathcal{O}_C s_2 \oplus \mathcal{O}_C s_4)/(g_C s_4)$$

near p. If $g_C = 0$, then $F \in J^2$, all partial derivatives of F vanish along C, and Y is singular along C, a contradiction. Thus $g_C \neq 0$. Hence the torsion subsheaf of $\mathcal{I}/\mathcal{I}^2$ is generated near p by s_4 and is isomorphic to $\mathcal{O}_C/(g_C)$, of length $\operatorname{ord}_p(g_C)$ at p. Thus D is the divisor of lengths of the torsion subsheaf of $\mathcal{I}/\mathcal{I}^2$, which does not depend on the local equations. In particular, $D \geq 0$. Since $\partial F/\partial s_j \in (s_1, s_2, s_4)$ for $j = 1, 2, 3$ and $(\partial F/\partial s_4)|_C = g_C$, Y is singular at p if and only if $\operatorname{mult}_p D > 0$. Since Y is smooth at the generic point of C, $\operatorname{Supp} D = \operatorname{Sing} Y \cap C$ is finite, so D is a divisor on C.

Moreover, any homomorphism $\mathcal{I}/\mathcal{I}^2 \to \mathcal{O}_C$ maps the torsion element s_4 to 0. Thus $\mathcal{N}$ is locally free of rank two, with local basis e_1, e_2 dual to s_1, s_2.

Step 2. In this step, we show that

$$\deg \mathcal{N} = 1 - a.$$

Since $\nu(B) = C$, we have $\mathcal{I} \cdot \mathcal{O}_{\bar{E}} \subset \mathcal{O}_{\bar{E}}(-B)$, where $\mathcal{O}_{\bar{E}}(-B)$ is the ideal sheaf of B. Restricting to B, we obtain a homomorphism $(\nu|_B)^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{O}_B(-B)$, hence by duality a homomorphism

$$\mathcal{O}_B(B) \to \operatorname{Hom}_{\mathcal{O}_B}((\nu|_B)^*(\mathcal{I}/\mathcal{I}^2), \mathcal{O}_B).$$

Since $(\nu|_B)^* g_C \neq 0$ on the integral curve B, any homomorphism $(\nu|_B)^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{O}_B$ maps the pullback of the torsion subsheaf to 0. Thus the right-hand side is $(\nu|_B)^*\mathcal{N}$. Pushing forward to C and composing with the trace $\operatorname{Tr} : (\nu|_B)_*\mathcal{O}_B \to \mathcal{O}_C$, we obtain a homomorphism

$$T : (\nu|_B)_*\mathcal{O}_B(B) \to (\nu|_B)_*(\nu|_B)^*\mathcal{N} \cong \mathcal{N} \otimes (\nu|_B)_*\mathcal{O}_B \xrightarrow{\mathrm{id}_{\mathcal{N}} \otimes \operatorname{Tr}} \mathcal{N}$$

between locally free sheaves of rank two on C.

Let $p \in C$ be a point over which $\nu|_B$ is unramified. Near p, $\phi = s_1 s_2$, and $\bar{E}$ consists of two smooth branches, $\{s_2 = 0\}$ with coordinates (s_1, s_3) and $\{s_1 = 0\}$ with coordinates (s_2, s_3), where $B = \{s_1 = 0\}$ on the first branch and $B = \{s_2 = 0\}$ on the second branch. The normal vectors $\partial/\partial s_1$ and $\partial/\partial s_2$ of B on the first and second branches are mapped to e_1 and e_2 respectively, and Tr is the sum over both branches. Thus T is an isomorphism near p.

Let $p \in C$ be a branch point. Near p, $\phi = s_1^2 - s_2^2 s_3$, and ν is given by

$$(s_1, s_2, s_3) = (st, t, s^2), \quad B = \{t = 0\},$$

where (s, t) are local coordinates of $\bar{E}$ (the map $(s, t) \to (st, t, s^2)$ from a smooth germ is finite and invertible on $s_2 \neq 0$, where $s = s_1/s_2$, hence it is the normalization of $\{s_1^2 = s_2^2 s_3\}$). The pullbacks of s_1, s_2, s_4 to $\mathcal{O}_B(-B)$ are st, t, and 0. Thus the normal vector $\partial/\partial t$ of B is mapped to $se_1 + e_2$. Since $(\nu|_B)_*\mathcal{O}_B$ is free over $\mathcal{O}_C$ with basis $1, s$ near p, and

$$\operatorname{Tr}(1) = 2, \quad \operatorname{Tr}(s) = 0, \quad \operatorname{Tr}(s^2) = 2s_3,$$

we have $T(\partial/\partial t) = 2e_2$ and $T(s\partial/\partial t) = 2s_3 e_1$. Thus the matrix of T in the bases $\partial/\partial t, s\partial/\partial t$ and e_1, e_2 is

$$\begin{pmatrix} 0 & 2s_3 \\ 2 & 0 \end{pmatrix},$$

whose determinant $-4s_3$ vanishes to order one at p.

Since $B \cong C \cong \mathbb{P}^1$ and $\deg(\nu|_B) = 2$, the Hurwitz formula [Har77, Corollary IV.2.4] implies that the ramification divisor R of $\nu|_B$ satisfies

$$-2 = 2 \cdot (-2) + \deg R,$$

so $\deg R = 2$. Since any ramification index of $\nu|_B$ is at most 2, $\nu|_B$ has exactly two ramification points, and since any fiber of $\nu|_B$ has degree 2, they are mapped to two distinct branch points. Thus $\det T$ is a section of $\det \mathcal{N} \otimes (\det(\nu|_B)_*\mathcal{O}_B(B))^{-1}$ with exactly two zeros, each of order one, and

$$\deg \mathcal{N} = \deg(\nu|_B)_*\mathcal{O}_B(B) + 2.$$

Since $\nu|_B$ is finite, $\chi(C, (\nu|_B)_*\mathcal{O}_B(B)) = \chi(B, \mathcal{O}_B(B))$ [Har77, Exercise III.4.1]. By the Riemann–Roch theorem on $\mathbb{P}^1$ [Har77, Theorem IV.1.3], applied to $\mathcal{O}_B(B)$ and to the line bundles in a splitting of $(\nu|_B)_*\mathcal{O}_B(B)$ [Har77, Exercise V.2.6], we have

$$\deg(\nu|_B)_*\mathcal{O}_B(B) + 2 = \chi(C, (\nu|_B)_*\mathcal{O}_B(B)) = \chi(B, \mathcal{O}_B(B)) = 1 + (B^2) = 1 - a.$$

Thus $\deg(\nu|_B)_*\mathcal{O}_B(B) = -a - 1$, and $\deg \mathcal{N} = 1 - a$.

Step 3. In this step, we show that

$$\deg \mathcal{N} = \deg D - 2.$$

Let $C^\circ := C \setminus \operatorname{Sing} Y$. By the adjunction formula [Har77, Proposition II.8.20], there exists an isomorphism

$$\omega_Y|_{C^\circ} \cong (\omega_C \otimes (\det \mathcal{N})^{-1})|_{C^\circ}.$$

Let $p \in C$, and take local equations as in the statement. By adjunction [Kol13, (4.1.3)], ω_Y is generated near p by the residue of $ds_1 \wedge ds_2 \wedge ds_3 \wedge ds_4/F$, which is equal, up to sign, to

$$\eta := \frac{ds_1 \wedge ds_2 \wedge ds_3}{\partial F/\partial s_4}$$

where $\partial F/\partial s_4 \neq 0$, and $(\partial F/\partial s_4)|_C = g_C$ since $s_4 = 0$ on C. At the points of C° near p, we have $g_C \neq 0$, so (s_1, s_2, s_3) are local coordinates of Y, and the isomorphism $\omega_Y|_{C^\circ} \cong (\omega_C \otimes (\det \mathcal{N})^{-1})|_{C^\circ}$ maps $\eta|_C$ to

$$g_C^{-1} ds_3 \otimes (e_1 \wedge e_2)^{-1}$$

up to sign. Since $ds_3 \otimes (e_1 \wedge e_2)^{-1}$ generates $\omega_C \otimes (\det \mathcal{N})^{-1}$ at p, and g_C^{-1} generates $\mathcal{O}_C(D)$ at p, the isomorphism over C° extends to an isomorphism

$$\omega_Y|_C \cong \omega_C \otimes (\det \mathcal{N})^{-1} \otimes \mathcal{O}_C(D).$$

Since $K_Y = f^*K_X$, $f(C) = x$, and K_X is Cartier, $\omega_Y|_C \cong \mathcal{O}_C$. Taking degrees, we obtain $0 = -2 - \deg \mathcal{N} + \deg D$, hence $\deg \mathcal{N} = \deg D - 2$.

Step 4. In this step, we compute (K_E^2) and conclude the proof.

By **Steps 2 and 3**, $1 - a = \deg D - 2$, so $a = 3 - \deg D$. Since $K_Y = f^*K_X$ is trivial over the germ $X \ni x$, we have $K_E = (K_Y + E)|_E = E|_E$ [Kol13, (4.1.3)], so $(E^3) = (K_E^2)$. By [Rei94, Proposition 2.3],

$$\nu^* K_E = K_{\mathbb{F}_a} + B.$$

Since ν is finite and birational, the projection formula [Laz04, Remark 1.1.13] implies that $(K_E^2) = ((K_{\mathbb{F}_a} + B)^2)$. Let A be a fiber of $\mathbb{F}_a$. Since $(A^2) = 0$, $(A \cdot B) = 1$, $(B^2) = -a$, and $K_{\mathbb{F}_a} \sim -2B - (a+2)A$ [Har77, Proposition V.2.9 and Lemma V.2.10], we have

$$(K_E^2) = ((B + (a+2)A)^2) = -a + 2(a+2) = a + 4 = 7 - \deg D \leq 7.$$

Since E is irreducible, Lemma 2.19(2) implies that

$$\widehat{\mathrm{vol}}(X \ni x) \leq \widehat{\mathrm{vol}}(\mathrm{ord}_E) = (K_E^2) \leq 7. \qquad \square$$

The Cartier property of E in the following proposition reduces the study of germs with $\mathrm{crex}(X \ni x) = 1$ to integral Gorenstein del Pezzo surfaces, to which the classification of [Rei94, Theorems 1.1 and 1.5] can be applied. For non-normal E, the bound 7 improves the bound 9 of [Liu25, Theorem 1.1], which holds for any Gorenstein canonical threefold germ that is not a hypersurface germ. For $\mathrm{crex}(X \ni x) = 1$, the bound 9 is [Liu25, Proposition 4.5].

Proposition 8.7. *Let $X \ni x$ be a Gorenstein canonical threefold germ with $\mathrm{crex}(X \ni x) = 1$, and let E be the crepant divisor over $X \ni x$. Then there exists a projective birational morphism $f : Y \to X$ with $K_Y = f^*K_X$ which is an isomorphism over $X \backslash \{x\}$, such that $f^{-1}(x)_{\mathrm{red}} = E$, E is Cartier, $-E$ is f-ample, Y is locally a hypersurface along E, and $\mathrm{Sing}\, Y \cap E$ is finite. In particular, E is an integral Gorenstein del Pezzo surface, and*

$$\widehat{\mathrm{vol}}(X \ni x) \leq \widehat{\mathrm{vol}}(\mathrm{ord}_E) = (E^3) = (K_E^2).$$

If moreover E is not normal, then $(K_E^2) \leq 7$, and hence $\widehat{\mathrm{vol}}(X \ni x) \leq 7$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we extract E and show that Y is locally a hypersurface along E and that $\mathrm{Sing}\, Y \cap E$ is finite.

Since X is klt and $a(E, X) = 1$, by [Liu25, Proposition 2.17], there exists a projective birational morphism $f : Y \to X$ from a normal variety Y which is an isomorphism over $X \backslash \{x\}$, such that $f^{-1}(x)_{\mathrm{red}} = E$ and $-E$ is $\mathbb{Q}$-Cartier and f-ample. Since E is the only exceptional divisor of f and $a(E, X) = 1$, we have $K_Y = f^*K_X$. Thus K_Y is Cartier, and $a(F, Y) = a(F, X)$ for any prime divisor F over Y [KM98, Lemma 2.30]. In particular, Y is Gorenstein canonical.

Let $y \in E$, and let F be a crepant divisor over $Y \ni y$. Then $a(F, X) = a(F, Y) = 1$, $\operatorname{center}_X F = \{f(y)\} = \{x\}$, and $F \neq E$ since $\operatorname{center}_Y F = \{y\}$. This contradicts $\operatorname{crex}(X \ni x) = 1$. Thus $\operatorname{crex}(Y \ni y) = 0$, and by Theorem 2.15(1), $Y \ni y$ is a cDV hypersurface germ. Hence Y is locally a hypersurface along E.

Suppose that $\operatorname{Sing} Y \cap E$ contains a curve Γ. Let W be the complement in Y of the union of the irreducible components of $\operatorname{Sing} Y$ other than Γ. Then W is open, $W \cap \Gamma \neq \emptyset$, and $\operatorname{Sing} W \subset \Gamma$. Since W is singular in codimension two, W is not terminal [KM98, Corollary 5.18]. Since W is canonical, there exists an exceptional divisor F over W with $a(F, W) = 1$. Since any exceptional divisor over a smooth variety has log discrepancy at least 2 [KM98, Corollary 2.31(2)], we have $\operatorname{center}_W F \subset \operatorname{Sing} W \subset \Gamma \subset E$. Thus $a(F, X) = a(F, W) = 1$, $\operatorname{center}_X F = \{x\}$, and $F \neq E$ since F is exceptional over Y. This contradicts $\operatorname{crex}(X \ni x) = 1$. Thus $\operatorname{Sing} Y \cap E$ is finite.

Step 2. In this step, we show that E is Cartier.

Since any Weil divisor is Cartier near a smooth point, the locus where E is not Cartier is contained in $\operatorname{Sing} Y \cap E$, which is finite by **Step 1**. Let $p \in \operatorname{Sing} Y \cap E$, let $R := \mathcal{O}_{Y,p}$, let $U := \operatorname{Spec} R \backslash \{p\}$, and let $j : U \to \operatorname{Spec} R$ be the inclusion. Then $\mathcal{O}_Y(E)|_U$ is invertible. Since E is $\mathbb{Q}$-Cartier, there exists an integer $r > 0$ such that rE is Cartier, so $(\mathcal{O}_Y(E)|_U)^{\otimes r} \cong \mathcal{O}_Y(rE)|_U \cong \mathcal{O}_U$. Thus $\mathcal{O}_Y(E)|_U$ is a torsion element of $\operatorname{Pic}(U)$. By **Step 1**, R is a hypersurface of dimension 3, that is, the completion of R is a complete regular local ring modulo one element. Thus $\operatorname{Pic}(U)$ is torsion-free [Dao12, Corollary 3.5], and $\mathcal{O}_Y(E)|_U \cong \mathcal{O}_U$. Since R is normal and $\mathcal{O}_Y(E)$ is reflexive, we have [Har80, Proposition 1.6]

$$\mathcal{O}_Y(E)_p \cong j_*(\mathcal{O}_Y(E)|_U) \cong j_*\mathcal{O}_U = R.$$

Thus E is Cartier at p. Since $p \in \operatorname{Sing} Y \cap E$ is arbitrary, E is Cartier.

Step 3. In this step, we show that E is an integral Gorenstein del Pezzo surface and prove the inequality for $\widehat{\operatorname{vol}}(X \ni x)$.

By **Steps 1 and 2**, $X \ni x$, f, and E satisfy the conditions of Set-up 8.2. Since E is irreducible, Lemma 2.19(2) implies that E is Gorenstein and that

$$\widehat{\operatorname{vol}}(X \ni x) \leq \widehat{\operatorname{vol}}(\operatorname{ord}_E) = (E^3) = (K_E^2).$$

Since E is a prime divisor, E is integral. Since f is projective and $E = f^{-1}(x)_{\mathrm{red}}$, E is projective, and E is connected [Har77, Corollary III.11.4]. Since $K_Y = f^*K_X$ is trivial over the germ $X \ni x$, we have $\omega_E \cong \mathcal{O}_E(E)$ [Kol13, (4.1.3)]. Since $-E$ is f-ample and $E \subset f^{-1}(x)$, $\mathcal{O}_E(-E)$ is ample. Thus E, together with $\mathcal{O}_E(1) := \mathcal{O}_E(-E)$, is an integral Gorenstein del Pezzo surface. Moreover, since $E \subset Y$ and Y is locally a hypersurface along E, the embedding dimension of E at any point is at most 4.

Step 4. In this step, we assume that E is not normal and that $m := (K_E^2) \geq 8$, and we reduce to two types of projected scrolls.

For integers $a, k \geq 0$, let $\mathbb{F}_{a;k}$ be $\mathbb{F}_a$ embedded by $\mathcal{O}_{\mathbb{F}_a}((a+k)A + B)$, a scroll of degree $a + 2k$, where for $k = 0$, $\mathbb{F}_{a;0}$ is the cone over a rational normal curve of degree a [Rei94, Theorem 1.1]. Since E is an integral non-normal Gorenstein del Pezzo surface, by [Rei94, Theorems 1.1 and 1.5, Section 1.3], E is obtained by the gluing constructions of [Rei94, Section 1.3] from one of the pairs of types (a1)–(e) in [Rei94, Theorem 1.1]. The types (a1), (a2), (a3), (b), (c0), and (d0) have degree at most 4, so they do not occur since $m \geq 8$. Thus one of the following holds.

(1) E is of type (c1) or (c2), and E is the projective cone over a nodal or cuspidal rational curve $\Gamma \subset \mathbb{P}^{m-1}$ of degree m which spans $\mathbb{P}^{m-1}$.
(2) E is of type (e), and E is a projection of $\mathbb{F}_{m-4;2}$.
(3) E is of type (d1), and E is a projection of $\mathbb{F}_{m-2;1}$.

In case (1), the local ring of E at the vertex is the localization of the homogeneous coordinate ring of Γ at the irrelevant ideal. Since Γ spans $\mathbb{P}^{m-1}$, the homogeneous ideal of Γ contains no linear form, so the embedding dimension of E at the vertex is $m \geq 8$. This contradicts **Step 3**. In the next step, we exclude cases (2) and (3).

Step 5. In this step, we show that cases (2) and (3) of **Step 4** do not occur. This implies that $(K_E^2) \leq 7$ if E is not normal, and hence $\widehat{\mathrm{vol}}(X \ni x) \leq 7$ by **Step 3**.

In both cases, let $\nu : \bar{E} \to E$ be the normalization, and let $C \subset E$ be the conductor. Since $\operatorname{Sing} Y \cap E$ is finite, Y is smooth at the generic point of C.

Case (2). E is a projection of $\mathbb{F}_{m-4;2}$. Let $a := m-4 \geq 4$. By [Rei94, Section 1.3], $\bar{E} \cong \mathbb{F}_a$, the conductor on $\bar{E}$ is the negative section B, the conductor C on E is a line, $\nu|_B : B \to C$ is finite of degree two, and E is the gluing of $\bar{E}$ along the involution of B over C. Thus near a point of C over which $\nu|_B$ is unramified, E is two smooth branches glued transversally along C, that is, a double crossing. Near a branch point, there exist local coordinates (s,t) of $\bar{E}$ with $B = \{t = 0\}$ such that $s_3 := s^2$ is a coordinate of C ($B \to C$ is simply ramified at the branch point, so $\nu|_B$ is $s \to s^2$ in suitable coordinates, and s extends to $\bar{E}$), and the local ring of E consists of the power series $u(s,t)$ with $u(s,0)$ even in s. The local ring of E at the branch point is generated by $s_1 := st$, $s_2 := t$, and s_3, with the relation

$$s_1^2 - s_2^2 s_3 = 0.$$

For such u, $u(s,0) = \bar{u}(s_3)$ for a series $\bar{u}$, and $u - \bar{u}(s_3) \in (t)$. Since $\mathbb{C}[[s,t]]$ is generated by 1 and s as a module over $\mathbb{C}[[s^2,t]] = \mathbb{C}[[s_3,s_2]]$, we get $t \cdot \mathbb{C}[[s,t]] = s_2\mathbb{C}[[s_2,s_3]] + s_1\mathbb{C}[[s_2,s_3]]$, so $u \in \mathbb{C}[[s_1,s_2,s_3]]$. The series $s_1^2 - s_2^2 s_3$ is irreducible, since $s_2^2 s_3$ is not a square in $\mathbb{C}[[s_2,s_3]]$. Since both $\mathbb{C}[[s_1,s_2,s_3]]/(s_1^2 - s_2^2 s_3)$ and the local ring of E are two-dimensional domains, $s_1^2 - s_2^2 s_3$ generates the kernel. Thus E has a pinch point at any branch point. Hence E, f, and Y satisfy all the conditions of Lemma 8.6, which implies that $m = (K_E^2) \leq 7$. This contradicts $m \geq 8$.

Case (3). E is a projection of $\mathbb{F}_{m-2;1}$. The following computation is taken from [Liu25, Proposition 4.5]. Let A be a fiber and B the negative section of $\bar{E} \cong \mathbb{F}_{m-2}$. By [Rei94, Section 1.3], E is the gluing of $\bar{E}$ along an isomorphism $A \to B$ which fixes the point $A \cap B$, and $C = \nu(A) = \nu(B)$. Let $p := \nu(A \cap B)$. The description of the local ring of E at p below goes back to [LX19, Section 3.1]. Away from p, two smooth branches of $\bar{E}$ are glued transversally along C, so E has double crossings along $C \backslash \{p\}$.

Let (s,t) be local coordinates of $\bar{E}$ at $A \cap B$ with $A = \{s = 0\}$ and $B = \{t = 0\}$ such that the gluing is $(0,\tau) \sim (\tau,0)$ (A and B meet transversally since $(A \cdot B) = 1$, the gluing fixes the intersection point, and reparametrizing s along B gives the gluing $(0,\tau) \sim (\tau,0)$). Then the local ring of E at p consists of the power series $u(s,t)$ with $u(0,\tau) = u(\tau,0)$. The local ring of E at p is generated by

$$s_1 := st, \quad s_2 := s^2 t, \quad s_3 := s + t,$$

with the relation

$$\phi := s_2^2 + s_1^3 - s_1 s_2 s_3 = 0.$$

For such u, let $\bar{u}(\tau) := u(0,\tau) = u(\tau,0)$. Then $u - \bar{u}(s_3)$ vanishes on $\{s = 0\}$ and on $\{t = 0\}$, so $u - \bar{u}(s_3) \in (st)$. Since $t = s_3 - s$ and $s^2 = s_3 s - s_1$, Weierstrass division shows that $\mathbb{C}[[s,t]]$ is generated by 1 and s as a module over $\mathbb{C}[[s_1,s_3]]$. Since $s \cdot st = s_2$, we get $st \cdot \mathbb{C}[[s,t]] = s_1\mathbb{C}[[s_1,s_3]] + s_2\mathbb{C}[[s_1,s_3]]$, so $u \in \mathbb{C}[[s_1,s_2,s_3]]$. The series ϕ is irreducible, since $\phi \equiv s_2^2 + s_1^3$ modulo s_3, so ϕ generates the kernel as in Case (2). Here $C = \{s_1 = s_2 = 0\}$, s_3 is a coordinate of C, and $\phi \in (s_1,s_2)^2$.

Let $c \in C$, and let $\phi(s_1,s_2,s_3) = 0$ be a local equation of E in a smooth threefold germ with $C = \{s_1 = s_2 = 0\}$, where we work in analytic local coordinates. Since E is singular along C, $\phi \in (s_1,s_2)^2$. At $c = p$, we take the series ϕ above. Since E is Cartier and Y is a hypersurface germ at c, there exists a local equation $F = \phi + s_4 g = 0$ of Y in a smooth fourfold germ with $E = \{s_4 = 0\}$. If Y is singular at c, then a generator s_4 of the ideal of E in $\mathcal{O}_{Y,c}$ is not in $\mathfrak{m}_{Y,c}^2$, since the embedding dimensions of Y and E at c are 4 and 3 (so lifts of s_1, s_2, s_3 to $\mathcal{O}_{Y,c}$, together with s_4, are coordinates of a smooth fourfold germ containing the germ of Y at c, and Y is defined in it by a series F with $F(s_1,s_2,s_3,0) = \phi$ after dividing by a unit, since $F(s_1,s_2,s_3,0)$ and ϕ both define E in $\{s_4 = 0\}$). If Y is smooth at c, then we may take $F = \phi - s_4$. We write $g_C := g(0,0,s_3,0)$.

Let $\Theta := \mathrm{Hom}_{\mathcal{O}_C}(\Omega_Y|_C, \mathcal{O}_C)$, which is $T_Y|_C$ over the smooth locus of Y. Near any point of C, $\Omega_Y|_C$ is the quotient of $\bigoplus_{j=1}^4 \mathcal{O}_C ds_j$ by

$$dF|_C = g_C ds_4,$$

since $\partial F/\partial s_j \in (s_1, s_2, s_4)$ for $j = 1, 2, 3$. Since Y is smooth at the generic point of C, $g_C \neq 0$. Thus Θ is locally free of rank three, with local basis $\partial_1, \partial_2, \partial_3$ dual to ds_1, ds_2, ds_3. The differential of ν, restricted to A and B, induces a homomorphism

$$T_{\bar{E}}|_A \oplus T_{\bar{E}}|_B \to \Theta$$

of sheaves on C, where we view sheaves on A and B as sheaves on C via ν. Let T be the image. Then T is torsion-free, hence locally free.

Near p, the images of ∂_s, ∂_t on A are $\partial_3 + s_3\partial_1$ and ∂_3, and the images of ∂_s, ∂_t on B are ∂_3 and $\partial_3 + s_3\partial_1 + s_3^2\partial_2$. Thus T is generated near p by

$$\partial_3, \quad s_3\partial_1, \quad s_3^2\partial_2,$$

and $\det T$ is generated at p by $s_3^3 \partial_1 \wedge \partial_2 \wedge \partial_3$. Near a point of $C\backslash\{p\}$, $\phi = s_1 s_2$, the images of the tangent vectors of both branches are ∂_1, ∂_3 and ∂_2, ∂_3, and T is generated by $\partial_1, \partial_2, \partial_3$. In particular, T has rank three. The morphisms $\nu|_A : A \to C$ and $\nu|_B : B \to C$ are isomorphisms. For a local section v of T_C, let v_A and v_B be the corresponding sections of T_A and T_B. Since v_A and v_B both map to v in Θ, the image K' of T_C under $v \to (v_A, -v_B)$ is contained in the kernel K of $T_{\bar{E}}|_A \oplus T_{\bar{E}}|_B \to T$. Both K and K' have rank one, and K' is a subbundle since $T_A \subset T_{\bar{E}}|_A$ is a subbundle. Thus K/K' is a torsion subsheaf of the locally free sheaf $(T_{\bar{E}}|_A \oplus T_{\bar{E}}|_B)/K'$, so $K = K' \cong T_C$. Since $(A^2) = 0$ and $(B^2) = -(m-2)$, [Har77, Proposition V.1.5] implies that

$$\deg T = \deg(T_{\bar{E}}|_A) + \deg(T_{\bar{E}}|_B) - \deg T_C = (2+0) + (2-(m-2)) - 2 = 4 - m.$$

Now we compare $\det T$ with $\omega_Y^\vee|_C$. Near any point of C, by adjunction [Kol13, (4.1.3)], ω_Y is generated by the residue of $ds_1 \wedge ds_2 \wedge ds_3 \wedge ds_4/F$, which is equal, up to sign, to

$$\eta := \frac{ds_1 \wedge ds_2 \wedge ds_3}{\partial F/\partial s_4}$$

where $\partial F/\partial s_4 \neq 0$, and $(\partial F/\partial s_4)|_C = g_C$. At the points of C where $g_C \neq 0$, Y is smooth with local coordinates (s_1, s_2, s_3), $\det\Theta \cong \omega_Y^\vee|_C$, and

$$\langle \partial_1 \wedge \partial_2 \wedge \partial_3, \eta \rangle = \frac{1}{g_C}.$$

Thus the isomorphism $\det\Theta \cong \omega_Y^\vee|_C$ over $C \setminus \mathrm{Sing}\, Y$ induces a rational homomorphism $\det T \to \omega_Y^\vee|_C$, whose order at a point $q \in C$ is the order at q of $\langle \sigma, \eta \rangle$ for a local generator σ of $\det T$ at q. Let $l := \mathrm{ord}_p(g_C)$, which is finite since $g_C \neq 0$. Let $p_1, \ldots, p_n$ be the points of $\mathrm{Sing}\, Y \cap C$ other than p, and let $l_i := \mathrm{ord}_{p_i}(g_{i,C})$, where $g_{i,C}$ is g_C for local equations at p_i. Since Y is singular at p_i and $\partial F/\partial s_j \in (s_1, s_2, s_4)$ for $j = 1, 2, 3$, we have $l_i > 0$. Then:

- at p, $\langle s_3^3 \partial_1 \wedge \partial_2 \wedge \partial_3, \eta \rangle = s_3^3/g_C$ has order $3 - l$;
- at p_i, $\langle \partial_1 \wedge \partial_2 \wedge \partial_3, \eta \rangle = 1/g_{i,C}$ has order $-l_i$;
- at any other point of C, g_C is a unit and $\langle \partial_1 \wedge \partial_2 \wedge \partial_3, \eta \rangle$ has order 0.

Hence

$$\det T \otimes \mathcal{O}_C\Big((3-l)p - \sum_{i=1}^n l_i p_i\Big) \cong \omega_Y^\vee|_C.$$

Since $K_Y = f^* K_X$, $f(C) = x$, and K_X is Cartier, $\deg \omega_Y|_C = 0$. Since $\deg T = 4 - m$, taking degrees, we obtain

$$m = 7 - l - \sum_{i=1}^n l_i \leq 7.$$

This contradicts $m \geq 8$.

Thus neither case (2) nor case (3) of **Step 4** occurs, and Proposition 8.7 follows. □

8.4. **Classification above volume seven.** In this subsection, we prove Theorem 8.1. We first show that if E is smooth and $H^1(E, T_E \otimes \omega_E^{\otimes(-k)}) = H^1(E, \omega_E^{\otimes(1-k)}) = 0$ for any integer $k \geq 1$, then E determines the germ.

Lemma 8.8. *Let $X \ni x$ be a Gorenstein canonical threefold germ, and let $f : Y \to X$ be a projective birational morphism from a normal variety Y with $K_Y = f^*K_X$ which is an isomorphism over $X\backslash\{x\}$, such that $E := f^{-1}(x)_{\mathrm{red}}$ is a smooth Cartier divisor and $-K_E$ is ample. Assume that $H^1(E, T_E \otimes \omega_E^{\otimes(-k)}) = 0$ and $H^1(E, \omega_E^{\otimes(1-k)}) = 0$ for any integer $k \geq 1$. Then Y is smooth along E, and there exists a biholomorphism from a neighborhood of E in the complex analytic space Y^{an} onto a neighborhood of the zero section in the total space of the line bundle ω_E which induces an isomorphism between the complex analytic germs $X \ni x$ and $C_E \ni o$.*

Proof. Let $y \in E$ be a closed point, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{Y,y}$, and let $u \in \mathfrak{m}$ be a local equation of E at y. Since $\mathcal{O}_{Y,y}/(u) \cong \mathcal{O}_{E,y}$ is a regular local ring of dimension 2, the ideal $\mathfrak{m}$ is generated by u and two lifts of a regular system of parameters of $\mathcal{O}_{E,y}$. Thus $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 \leq 3 = \dim \mathcal{O}_{Y,y}$, and $\mathcal{O}_{Y,y}$ is regular. Since the smooth locus of Y is open, Y is smooth along E.

Since $f(E) = \{x\}$ and K_X is Cartier, $\mathcal{O}_Y(K_Y)|_E \cong \mathcal{O}_Y(f^*K_X)|_E \cong \mathcal{O}_E$. Thus the adjunction formula [Kol13, (4.1.3)] implies

$$\mathcal{O}_E(E) \cong \mathcal{O}_Y(K_Y + E)|_E \cong \omega_E.$$

In particular, the dual of the normal bundle $\mathcal{O}_E(E)$ is the ample line bundle $\mathcal{O}_E(-K_E)$. Choose $l \geq 1$ such that $\mathcal{O}_E(-lK_E)$ is very ample. The pullback of the Fubini–Study form under the corresponding embedding represents the Chern class of $\mathcal{O}_E(-lK_E)$ and is positive definite, so $\mathcal{O}_E(E)^{\otimes l}$ is negative in the sense of Kodaira used in [Gra62, §3, p. 341], and hence $\mathcal{O}_E(E)$ is negative [Gra62, §3, (∗) on p. 343].

Let $Y' := \operatorname{Spec}_E \bigoplus_{m\geq 0} \mathcal{O}_E(-mK_E)$ be the total space of ω_E, let $\pi : Y' \to E$ be the projection, and let $E' \subset Y'$ be the zero section, so that $\mathcal{O}_{E'}(E') \cong \omega_E$ under the isomorphism $\pi|_{E'} : E' \to E$. Since $(\mathcal{O}_E(E)^*)^{\otimes k} \cong \omega_E^{\otimes(-k)}$ for any integer k, our assumption says that

$$H^1\big(E, T_E \otimes (\mathcal{O}_E(E)^*)^{\otimes k}\big) = 0 \quad \text{and} \quad H^1\big(E, (\mathcal{O}_E(E)^*)^{\otimes(k-1)}\big) = 0$$

for any integer $k \geq 1$. By [Gra62, §4, Satz 7 and Corollar] (cf. [ABT09, Theorem 4.1 (p. 645), Remarks 4.2–4.3 and Corollary 4.3 (pp. 649–650)]), there exist an open neighborhood U of E in Y^{an}, an open neighborhood U' of E' in Y'^{an}, and a biholomorphism $\varphi : U \to U'$ with $\varphi(E) = E'$.

Let $R := \bigoplus_{m\geq 0} H^0(E, \mathcal{O}_E(-mK_E))$, so that $C_E = \operatorname{Spec} R$. Since π is affine, $H^0(Y', \mathcal{O}_{Y'}) = R$, and we let $g : Y' \to C_E$ be the induced morphism. Since $\mathcal{O}_E(-K_E)$ is ample, C_E is normal, g is proper, $g^{-1}(o) = E'$, and g is an isomorphism over $C_E\backslash\{o\}$ [Kol13, 3.8].

We show that $f_*\mathcal{O}_{Y^{\mathrm{an}}} = \mathcal{O}_{X^{\mathrm{an}}}$. Let V be an open subset of X^{an} containing x, and let s be a holomorphic function on $f^{-1}(V)$. Since f is an isomorphism over $X\backslash\{x\}$, s induces a holomorphic function on $V\backslash\{x\}$, which is locally bounded near x because f is proper. Since X is normal, X^{an} is normal [Gro71, Exposé XII, Proposition 2.1], so the function induced by s extends uniquely to a holomorphic function on V by the Riemann extension theorem [GR84, Chapter 7, §4.2]. Since C_E is normal and g is proper, the Riemann extension argument above with C_E and g in place of X and f shows that $g_*\mathcal{O}_{Y'^{\mathrm{an}}} = \mathcal{O}_{C_E^{\mathrm{an}}}$.

Since f is proper and $x \notin f(Y^{\mathrm{an}}\backslash U)$, there exists an open neighborhood V of x in X^{an} such that $f^{-1}(V) \subset U$. Fix generators $r_1, \ldots, r_n$ of the $\mathbb{C}$-algebra R, which define a closed embedding $C_E \subset \mathbb{A}^n$. Since $f_*\mathcal{O}_{Y^{\mathrm{an}}} = \mathcal{O}_{X^{\mathrm{an}}}$, the functions $r_i \circ g \circ \varphi$ on $f^{-1}(V)$ descend to holomorphic functions on V. Thus we obtain a holomorphic map $\alpha : V \to \mathbb{C}^n$ such that $\alpha \circ f = g \circ \varphi$ on $f^{-1}(V)$. Since $\alpha(V\backslash\{x\}) \subset C_E^{\mathrm{an}}$ and C_E^{an} is closed in $\mathbb{C}^n$, we have $\alpha(V) \subset C_E^{\mathrm{an}}$, and $\alpha(x) = g(E') = o$. Next, fix a closed embedding of an affine neighborhood of x in X into an affine space. Since g is proper and $f(\varphi^{-1}(E')) = \{x\}$, there exists an open neighborhood V' of o in C_E^{an} such that $g^{-1}(V') \subset U'$ and $f(\varphi^{-1}(g^{-1}(V')))$ is contained in that affine neighborhood. Since $g_*\mathcal{O}_{Y'^{\mathrm{an}}} = \mathcal{O}_{C_E^{\mathrm{an}}}$, the

coordinate functions of $f \circ \varphi^{-1}$ on $g^{-1}(V')$ descend to V'. As for α, the image of $V'\backslash\{o\}$ is contained in the closed analytic subset given by that affine neighborhood, hence so is the image of V', and we obtain a holomorphic map $\beta : V' \to X^{\mathrm{an}}$ such that $\beta \circ g = f \circ \varphi^{-1}$ on $g^{-1}(V')$ and $\beta(o) = x$.

Since f and g are isomorphisms over $X\backslash\{x\}$ and $C_E\backslash\{o\}$ respectively, we have

$$\alpha = g \circ \varphi \circ f^{-1} \text{ on } V\backslash\{x\} \quad \text{and} \quad \beta = f \circ \varphi^{-1} \circ g^{-1} \text{ on } V'\backslash\{o\}.$$

Hence $\beta \circ \alpha$ is the identity on $\alpha^{-1}(V')\backslash\{x\}$ and $\alpha \circ \beta$ is the identity on $\beta^{-1}(V)\backslash\{o\}$. Since X^{an} and C_E^{an} are reduced, $\beta \circ \alpha$ and $\alpha \circ \beta$ are the identity on $\alpha^{-1}(V')$ and $\beta^{-1}(V)$ respectively. Therefore α, which is induced by φ through $\alpha \circ f = g \circ \varphi$, induces an isomorphism between the complex analytic germs $X \ni x$ and $C_E \ni o$. □

As explained in the remark after Set-up 8.2, for $E \cong \mathbb{F}_1$ or $E \cong \mathbb{P}^1 \times \mathbb{P}^1$ we do not estimate the local volume but determine the germ itself: we show that the cohomology groups in Lemma 8.8 vanish.

Proposition 8.9. *Let $X \ni x$ be a Gorenstein canonical threefold germ, and let $f : Y \to X$ be a projective birational morphism from a normal variety Y with $K_Y = f^*K_X$ which is an isomorphism over $X\backslash\{x\}$, such that $E := f^{-1}(x)_{\mathrm{red}}$ is a Cartier divisor. Assume that $E \cong \mathbb{F}_1$ or $E \cong \mathbb{P}^1 \times \mathbb{P}^1$. Then a neighborhood of E in the complex analytic space Y^{an} is biholomorphic to a neighborhood of the zero section in the total space of the line bundle ω_E, and the biholomorphism induces an isomorphism between the complex analytic germs $X \ni x$ and $C_E \ni o$. In particular, $X \ni x$ and $C_E \ni o$ are analytically isomorphic. If $E \cong \mathbb{F}_1$, then $\widehat{\mathrm{vol}}(X \ni x) = q$ and $X \ni x$ is not LCIQ. If $E \cong \mathbb{P}^1 \times \mathbb{P}^1$, then $X \ni x$ is the quotient of the threefold ordinary double point by -1, $X \ni x$ is LCIQ, and $\widehat{\mathrm{vol}}(X \ni x) = 8$.*

Proof. Write $E \cong \mathbb{F}_a$ with $a \in \{0, 1\}$, where $\mathbb{F}_0 = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}) \cong \mathbb{P}^1 \times \mathbb{P}^1$. Let $\pi : E \to \mathbb{P}^1$ be the ruling, let A be a fiber, and let B be a section with $(B^2) = -a$. Then $(A^2) = 0$, $(A \cdot B) = 1$, and

$$K_E \sim -2B - (a+2)A$$

[Har77, Lemma V.2.10]. Since E is a smooth Cartier divisor and $-K_E$ is ample, by Lemma 8.8, it suffices to show that $H^1(E, T_E \otimes \omega_E^{\otimes(-k)}) = 0$ and $H^1(E, \omega_E^{\otimes(1-k)}) = 0$ for any integer $k \geq 1$.

We first show that for any integers $m \geq 0$ and n with $n - ma \geq -1$,

$$H^1(E, \mathcal{O}_E(mB + nA)) = 0. \tag{8.3}$$

Since $\mathcal{O}_E(B)$ is the tautological line bundle of $\mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(-a))$ [Har77, Proposition V.2.8], by [Har77, Exercise III.8.4] and the projection formula,

$$\pi_*\mathcal{O}_E(mB + nA) \cong \bigoplus_{j=0}^{m} \mathcal{O}_{\mathbb{P}^1}(n - ja) \quad \text{and} \quad R^1\pi_*\mathcal{O}_E(mB + nA) = 0.$$

By [Har77, Exercise III.8.1],

$$H^1(E, \mathcal{O}_E(mB + nA)) \cong \bigoplus_{j=0}^{m} H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(n - ja)),$$

which is zero by [Har77, Theorem III.5.1] since $n - ja \geq n - ma \geq -1$ for $0 \leq j \leq m$.

Since π is smooth, we have the exact sequence $0 \to T_{E/\mathbb{P}^1} \to T_E \to \pi^*T_{\mathbb{P}^1} \to 0$, where

$$T_{E/\mathbb{P}^1} \cong \mathcal{O}_E(-K_E + \pi^*K_{\mathbb{P}^1}) \cong \mathcal{O}_E(2B + aA) \quad \text{and} \quad \pi^*T_{\mathbb{P}^1} \cong \mathcal{O}_E(2A).$$

Let $k \geq 1$ be an integer. Since $\omega_E^{\otimes(-k)} \cong \mathcal{O}_E(2kB + k(a+2)A)$, we obtain the exact sequence

$$0 \to \mathcal{O}_E\big((2k+2)B + (k(a+2)+a)A\big) \to T_E \otimes \omega_E^{\otimes(-k)} \to \mathcal{O}_E\big(2kB + (k(a+2)+2)A\big) \to 0.$$

Since $a \in \{0, 1\}$, we have

$$\big(k(a+2)+a\big) - (2k+2)a = k(2-a) - a \geq 0 \quad \text{and} \quad \big(k(a+2)+2\big) - 2ka = k(2-a) + 2 \geq 3.$$

Thus by (8.3), the first cohomology groups of both line bundles in the exact sequence vanish, and hence $H^1(E, T_E \otimes \omega_E^{\otimes(-k)}) = 0$. Moreover, $\omega_E^{\otimes(1-k)} \cong \mathcal{O}_E(2(k-1)B + (k-1)(a+2)A)$ and

$$(k-1)(a+2) - 2(k-1)a = (k-1)(2-a) \geq 0,$$

so $H^1(E, \omega_E^{\otimes(1-k)}) = 0$ by (8.3).

Thus Lemma 8.8 implies the first sentence of the conclusion, and Lemma 2.5(2) implies that $X \ni x$ and $C_E \ni o$ are analytically isomorphic. Hence $X \ni x$ is LCIQ if and only if $C_E \ni o$ is LCIQ.

Suppose that $E \cong \mathbb{F}_1$. By Proposition 2.7(1) and Construction-Proposition 4.2, $\widehat{\mathrm{vol}}(X \ni x) = \widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q$, and $X \ni x$ is not LCIQ since $C_{\mathbb{F}_1} \ni o$ is not LCIQ.

Suppose that $E \cong \mathbb{P}^1 \times \mathbb{P}^1$, that is, $a = 0$. Let $H := A + B$ and $R_H := \bigoplus_{m \geq 0} H^0(E, \mathcal{O}_E(mH))$. Since R_H is generated by $H^0(E, \mathcal{O}_E(H))$ [CLS11, Proposition 4.3.3 and Corollary 2.2.13] and $\mathcal{O}_E(H)$ defines the Segre embedding of E as the quadric $\{u_1u_4 = u_2u_3\} \subset \mathbb{P}^3$, we have

$$R_H \cong \mathbb{C}[u_1, u_2, u_3, u_4]/(u_1u_4 - u_2u_3)$$

with $u_1, \dots, u_4$ of degree one. Let $Q := \operatorname{Spec} R_H \subset \mathbb{A}^4$, and let ι be the involution of Q given by $u_i \to -u_i$ for $1 \leq i \leq 4$. Since $-K_E \sim 2H$, the ring $\bigoplus_{m \geq 0} H^0(E, \mathcal{O}_E(-mK_E))$ is the even-degree part of R_H, which is the ring of ι-invariants of R_H. Thus $C_E = Q/\langle \iota \rangle$, and we let $p : Q \to C_E$ be the quotient morphism. Since ι acts freely on $Q \backslash \{0\}$, p is quasi-étale of degree 2 and $p^{-1}(o) = \{0\}$. Since $u_1u_4 - u_2u_3$ is a non-degenerate quadratic form, after a linear change of coordinates, which commutes with ι, $Q = \{u_1^2 + u_2^2 + u_3^2 + u_4^2 = 0\}$ is the threefold ordinary double point and ι is -1. Thus $X \ni x$ is the quotient of the threefold ordinary double point by -1. Since $Q \ni 0$ is a hypersurface germ, it is a local complete intersection, so $C_E \ni o$ is LCIQ, and hence $X \ni x$ is LCIQ. Since $Q \ni 0$ is klt and p is quasi-étale, $K_Q = p^*K_{C_E}$ and $C_E \ni o$ is klt [KM98, Proposition 5.20(4)]. By Theorem 2.17(1), $\widehat{\mathrm{vol}}(Q \ni 0) = 16$. Thus Proposition 2.7(2) implies $\widehat{\mathrm{vol}}(C_E \ni o) = 8$, and Proposition 2.7(1) implies $\widehat{\mathrm{vol}}(X \ni x) = 8$. □

Proof of Theorem 8.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that for $S \in \{\mathbb{P}^2, \mathbb{P}^1 \times \mathbb{P}^1, \mathbb{F}_1\}$, the germ $C_S \ni o$ is Gorenstein canonical, $\operatorname{crex}(C_S \ni o) = 1$, and $C_S \ni o$ satisfies the properties listed in (1)–(3).

Let $Y' := \operatorname{Spec}_S \bigoplus_{m \geq 0} \mathcal{O}_S(-mK_S)$ be the total space of ω_S, let $\pi : Y' \to S$ be the projection, let $E' \subset Y'$ be the zero section, and let $g : Y' \to C_S$ be the morphism induced by $H^0(Y', \mathcal{O}_{Y'}) = \bigoplus_{m \geq 0} H^0(S, \mathcal{O}_S(-mK_S))$. Since $\mathcal{O}_S(-K_S)$ is ample, C_S is normal, g is proper, $g^{-1}(o) = E'$, and g is an isomorphism over $C_S \backslash \{o\}$ [Kol13, 3.8]. Since Y' is quasi-projective, g is projective. Moreover, Y' is smooth and $E' \cong S$ is a Cartier divisor on Y'.

Since Y' is the total space of ω_S, we have the exact sequence

$$0 \to \pi^*\omega_S \to T_{Y'} \to \pi^*T_S \to 0.$$

Taking determinants, $\omega_{Y'}^{-1} \cong \pi^*\omega_S \otimes \pi^*\omega_S^{-1} \cong \mathcal{O}_{Y'}$. Let σ be a nowhere vanishing section of $\omega_{Y'}$. Since g is an isomorphism over $C_S \backslash \{o\}$ and $\operatorname{codim}_{C_S}\{o\} = 3$, σ induces a nowhere vanishing section of $\mathcal{O}_{C_S}(K_{C_S})$. Thus $K_{C_S} \sim 0$ is Cartier, and taking $K_{Y'}$ and K_{C_S} to be the divisors of σ, we have $K_{Y'} = 0 = g^*K_{C_S}$. Since Y' is smooth, for any prime divisor F over C_S, $a(F, C_S) = a(F, Y') \geq 1$ [KM98, Lemma 2.30], so $C_S \ni o$ is canonical, and hence Gorenstein canonical [KM98, Theorems 5.22 and 5.10]. Let $F \neq E'$ be a prime divisor over C_S with $\operatorname{center}_{C_S} F = \{o\}$. Since $g^{-1}(o) = E'$, the center of F on Y' is a proper closed subset of E', so F is exceptional over Y', and

$$a(F, C_S) = a(F, Y') \geq 2$$

by [KM98, Lemma 2.30 and Corollary 2.31(2)]. Since $a(E', C_S) = a(E', Y') = 1$, we have $\operatorname{crex}(C_S \ni o) = 1$.

Suppose that $S = \mathbb{P}^2$. Then $\bigoplus_{m \geq 0} H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3m))$ is the subring of $\mathbb{C}[z_1, z_2, z_3]$ spanned by the monomials of degree divisible by 3, which is the ring of invariants of μ_3 acting on $\mathbb{A}^3$ by

scalars. Thus $C_{\mathbb{P}^2} = \mathbb{A}^3/\mu_3$. Since μ_3 acts freely on $\mathbb{A}^3\backslash\{0\}$, the quotient morphism $\mathbb{A}^3 \to C_{\mathbb{P}^2}$ is quasi-étale of degree 3, the preimage of o is $\{0\}$, and $K_{\mathbb{A}^3}$ is the pullback of $K_{C_{\mathbb{P}^2}}$. Since $\mathbb{A}^3$ is smooth, $C_{\mathbb{P}^2} \ni o$ is LCIQ. By Theorem 2.17(1), $\widehat{\mathrm{vol}}(\mathbb{A}^3 \ni 0) = 27$, so Proposition 2.7(2) implies $\widehat{\mathrm{vol}}(C_{\mathbb{P}^2} \ni o) = 9$.

Suppose that $S = \mathbb{P}^1 \times \mathbb{P}^1$. Then $C_S \ni o$ is a Gorenstein canonical threefold germ, g is a projective birational morphism from the smooth variety Y' with $K_{Y'} = g^*K_{C_S}$ which is an isomorphism over $C_S\backslash\{o\}$, and $g^{-1}(o)_{\mathrm{red}} = E' \cong \mathbb{P}^1 \times \mathbb{P}^1$ is a Cartier divisor. By Proposition 8.9, $C_S \ni o$ is the quotient of the threefold ordinary double point by -1, $C_S \ni o$ is LCIQ, and $\widehat{\mathrm{vol}}(C_S \ni o) = 8$.

Suppose that $S = \mathbb{F}_1$. By Construction-Proposition 4.2, $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q$ and $C_{\mathbb{F}_1} \ni o$ is not LCIQ.

Step 2. In this step, we show that $X \ni x$ admits a projective birational morphism $f : Y \to X$ with $K_Y = f^*K_X$ such that $E := f^{-1}(x)_{\mathrm{red}}$ is isomorphic to $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, $\mathbb{F}_1$, or a quadric surface of rank 3 in $\mathbb{P}^3$.

Let $X \ni x$ be a Gorenstein canonical threefold germ with $\mathrm{crex}(X \ni x) = 1$ and $\widehat{\mathrm{vol}}(X \ni x) > 7$. By Proposition 8.7, there exists a projective birational morphism $f : Y \to X$ with $K_Y = f^*K_X$ which is an isomorphism over $X\backslash\{x\}$, such that $E := f^{-1}(x)_{\mathrm{red}}$ is the crepant divisor over $X \ni x$, E is Cartier, $-E$ is f-ample, Y is locally a hypersurface along E, and E is an integral Gorenstein del Pezzo surface with

$$7 < \widehat{\mathrm{vol}}(X \ni x) \le d := (K_E^2).$$

Since $\widehat{\mathrm{vol}}(X \ni x) > 7$, Proposition 8.7 also implies that E is normal.

Suppose that E is not rational. By [HW81, Theorem 2.2] and [Bre80, Theorem 1], the minimal resolution $\tau : \tilde{E} \to E$ is $\mathbb{P}_C(\mathcal{O}_C \oplus \mathcal{L})$ for an elliptic curve C and a line bundle $\mathcal{L}$ on C with $\deg \mathcal{L} > 0$, and τ contracts the minimal section C_0, where $(C_0^2) = -\deg \mathcal{L}$. That is, E is the projective cone over C with conormal bundle $\mathcal{L}$. Since C_0 has genus one, the adjunction formula [Har77, Proposition V.1.5] implies $(K_{\tilde{E}} \cdot C_0) = -(C_0^2)$, and hence $K_{\tilde{E}} = \tau^*K_E - C_0$. Moreover, $(K_{\tilde{E}}^2) = 0$ [Har77, Corollary V.2.11]. Thus

$$d = (K_E^2) = ((K_{\tilde{E}} + C_0)^2) = (K_{\tilde{E}}^2) + 2(K_{\tilde{E}} \cdot C_0) + (C_0^2) = \deg \mathcal{L}.$$

Let v be the vertex of E. Since the affine cone $\mathrm{Spec} \bigoplus_{m\ge 0} H^0(C, \mathcal{L}^{\otimes m})$ is an open subset of E containing v [Kol13, 3.8], the local ring $\mathcal{O}_{E,v}$ is the localization of the graded ring $\bigoplus_{m\ge 0} H^0(C, \mathcal{L}^{\otimes m})$ at the ideal $\bigoplus_{m>0} H^0(C, \mathcal{L}^{\otimes m})$. Let $\mathfrak{m}_E$ be the maximal ideal of $\mathcal{O}_{E,v}$. The degree-one part $H^0(C, \mathcal{L})$ injects into $\mathfrak{m}_E/\mathfrak{m}_E^2$, and since C has genus one and $\deg \mathcal{L} > 0$, the Riemann–Roch theorem [Har77, Theorem IV.1.3] implies $\dim_{\mathbb{C}} H^0(C, \mathcal{L}) = d$. On the other hand, since E is a Cartier divisor on Y and Y is locally a hypersurface along E, $\dim_{\mathbb{C}} \mathfrak{m}_E/\mathfrak{m}_E^2 \le 4$. Thus $d \le 4 < 7$, a contradiction.

Therefore E is rational, so the minimal resolution $\tilde{E}$ of E satisfies $H^1(\tilde{E}, \mathcal{O}_{\tilde{E}}) = 0$. By [HW81, Theorem 3.4], E has at most rational double points, and $d \le 9$. Since $d > 7$ is an integer, $d \in \{8, 9\}$. By [HW81, Theorem 3.4] again, if $d = 9$, then $E \cong \mathbb{P}^2$, and if $d = 8$, then E is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, $\mathbb{F}_1$, or the cone over a smooth conic, that is, a quadric surface of rank 3 in $\mathbb{P}^3$.

Step 3. In this step, we show that E is not a quadric surface of rank 3, and that $X \ni x$ is analytically isomorphic to $C_E \ni o$.

Suppose that $E \subset \mathbb{P}^3$ is a quadric surface of rank 3, and let $\mathcal{O}_E(1)$ be the hyperplane bundle. By the adjunction formula [KM98, Proposition 5.73], $\omega_E \cong \mathcal{O}_E(-2)$, that is, $\mathcal{O}_E(-K_E) \cong \mathcal{O}_E(2)$. Since $f(E) = \{x\}$ and K_X is Cartier, $\mathcal{O}_Y(K_Y)|_E \cong \mathcal{O}_E$, so the adjunction formula [Kol13, (4.1.3)] implies $\mathcal{O}_E(E) \cong \omega_E$. Thus $\mathcal{O}_E(-E) \cong \mathcal{O}_E(2)$. Since E has only a rational double point, E is klt [KM98, Theorem 4.20]. Since Y is Gorenstein along E and E is Cartier, $(K_Y + E)|_E = K_E$,

so (Y, E) is plt near E by inversion of adjunction [KM98, Theorem 5.50]. Since $Y \setminus E \cong X \setminus \{x\}$ is canonical, (Y, E) is lc. Thus Proposition 8.5(2) implies $\widehat{\mathrm{vol}}(X \ni x) = \frac{27}{4} < 7$, a contradiction.

Suppose that $E \cong \mathbb{F}_1$ or $E \cong \mathbb{P}^1 \times \mathbb{P}^1$. By Proposition 8.9, $X \ni x$ is analytically isomorphic to $C_E \ni o$.

Suppose that $E \cong \mathbb{P}^2$. Then $\omega_E \cong \mathcal{O}_{\mathbb{P}^2}(-3)$. Let $k \geq 1$ be an integer. Twisting the dual of the Euler sequence [Har77, Theorem II.8.13] by $\mathcal{O}_{\mathbb{P}^2}(3k)$, we obtain the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^2}(3k) \to \mathcal{O}_{\mathbb{P}^2}(3k+1)^{\oplus 3} \to T_{\mathbb{P}^2}(3k) \to 0.$$

Since $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3k+1)) = 0$ and $H^2(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3k)) = 0$ [Har77, Theorem III.5.1], we have $H^1(E, T_E \otimes \omega_E^{\otimes(-k)}) = H^1(\mathbb{P}^2, T_{\mathbb{P}^2}(3k)) = 0$. Moreover, $H^1(E, \omega_E^{\otimes(1-k)}) = H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3k-3)) = 0$ [Har77, Theorem III.5.1]. Since E is a smooth Cartier divisor and $-K_E$ is ample, Lemma 8.8 implies that the complex analytic germs $X \ni x$ and $C_{\mathbb{P}^2} \ni o$ are isomorphic. By Lemma 2.5(2), $X \ni x$ is analytically isomorphic to $C_{\mathbb{P}^2} \ni o$.

Step 4. In this step, we conclude the proof.

By **Steps 2 and 3**, $X \ni x$ is analytically isomorphic to $C_S \ni o$ for some $S \in \{\mathbb{P}^2, \mathbb{P}^1 \times \mathbb{P}^1, \mathbb{F}_1\}$. By Lemma 4.3, $7 < q < 8$. By **Step 1**, the local volumes of $C_{\mathbb{P}^2} \ni o$, $C_{\mathbb{P}^1 \times \mathbb{P}^1} \ni o$, and $C_{\mathbb{F}_1} \ni o$ are 9, 8, and q, which are pairwise distinct. By Proposition 2.7(1), $C_{\mathbb{P}^2} \ni o$, $C_{\mathbb{P}^1 \times \mathbb{P}^1} \ni o$, and $C_{\mathbb{F}_1} \ni o$ are pairwise not analytically isomorphic, so $X \ni x$ is analytically isomorphic to exactly one of them. Together with **Step 1**, this proves the classification and the converse.

Let $X \ni x$ be a Gorenstein canonical threefold germ with $\mathrm{crex}(X \ni x) = 1$ which is not LCIQ. If $\widehat{\mathrm{vol}}(X \ni x) \leq 7$, then $\widehat{\mathrm{vol}}(X \ni x) < q$. If $\widehat{\mathrm{vol}}(X \ni x) > 7$, then $X \ni x$ is analytically isomorphic to exactly one of $C_{\mathbb{P}^2} \ni o$, $C_{\mathbb{P}^1 \times \mathbb{P}^1} \ni o$, and $C_{\mathbb{F}_1} \ni o$. Since LCIQ is defined up to analytic isomorphism and $C_{\mathbb{P}^2} \ni o$ and $C_{\mathbb{P}^1 \times \mathbb{P}^1} \ni o$ are LCIQ, $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$, and $\widehat{\mathrm{vol}}(X \ni x) = q$ by Proposition 2.7(1). Thus $\widehat{\mathrm{vol}}(X \ni x) \leq q$, and if equality holds, then $\widehat{\mathrm{vol}}(X \ni x) = q > 7$, so $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$. Conversely, by **Step 1**, $C_{\mathbb{F}_1} \ni o$ is a Gorenstein canonical threefold germ with $\mathrm{crex}(C_{\mathbb{F}_1} \ni o) = 1$ which is not LCIQ and has local volume q. □

9. Singularities with two crepant divisors

In this section, we prove Theorem 9.1, which bounds the local volume of a Gorenstein canonical threefold germ with exactly two crepant divisors (Definition 2.14). LCIQ germs are defined in Definition 2.13.

Theorem 9.1. *Let $X \ni x$ be a Gorenstein canonical threefold germ with $\mathrm{crex}(X \ni x) = 2$. Then*

$$\widehat{\mathrm{vol}}(X \ni x) \leq \frac{250}{33} < q.$$

In particular, if $X \ni x$ is a klt threefold germ which is not LCIQ and $\widehat{\mathrm{vol}}(X \ni x) > q$, then $X \ni x$ is Gorenstein canonical and $\mathrm{crex}(X \ni x) \geq 3$.

The proof of Theorem 9.1 is given in Subsection 9.5. Together with Theorem 8.1, the last sentence of Theorem 9.1 gives at least three crepant divisors, and it is used only once, in the proof of Proposition 10.14, to show that the quotient pair there is not exceptional. We follow the division of [Liu25, Section 4.3] according to the blow-up of the maximal ideal of x and bound the local volume in each case separately. When the degree of the exceptional divisor does not suffice, we mainly use a Kollár component whose different is half of a curve (Lemma 9.4).

9.1. Two exceptional components.

In this subsection, we treat the case when the exceptional divisor of the blow-up of the maximal ideal has two irreducible components. We start with three lemmas on a double curve of the exceptional divisor.

Lemma 9.2. *Notation and conditions as in Set-up 8.2. Let $C \subset E$ be a smooth rational curve, let E' and E'' be distinct prime divisors contained in E with $C \subset E' \cap E''$, and let $\Sigma \subset C$ be a set of at most one closed point. Assume that Y is locally a hypersurface along E, that $\operatorname{Sing} Y \cap E$ is finite, that E' and E'' are smooth along C, and that the following hold.*

(i) For any closed point $p \in C \backslash \Sigma$, there exists an isomorphism

$$\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u_1, u_2, u_3, u_4]]/(u_1u_2 + u_3\psi_p)$$

which sends the ideals of E, E', E'', and C to (u_3), (u_1, u_3), (u_2, u_3), and (u_1, u_2, u_3), respectively.

(ii) For $y \in \Sigma$, there exists an isomorphism

$$\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[s_1, s_2, s_3, s_4]]/(s_3(s_1s_2 - s_3) + s_4\psi_y)$$

which sends the ideals of E, E', E'', and C to (s_4), (s_3, s_4), $(s_3 - s_1s_2, s_4)$, and (s_2, s_3, s_4), respectively.

Then the following hold.

(1) For any closed point $p \in C \backslash \Sigma$, the integer $l_p := \operatorname{ord}_{u_4} \psi_p(0, 0, 0, u_4) \geq 0$ does not depend on the choice of the isomorphism in (i). For $y \in \Sigma$, the integer $l_y := \operatorname{ord}_{s_1} \psi_y(s_1, 0, 0, 0) \geq 0$ does not depend on the choice of the isomorphism in (ii). Moreover, for any closed point $p \in C$, we have $l_p > 0$ if and only if Y is singular at p.

(2) We have

$$\sum_{p \in C} l_p = 2 + |\Sigma| + (C^2)_{E'} + (C^2)_{E''},$$

where $(C^2)_{E'}$ and $(C^2)_{E''}$ are the self-intersection numbers of C on E' and on E''.

Proof. For a variety V containing C and an open subset $C_0 \subset C$ along which V is smooth, we denote by $T_V|_{C_0}$ the restriction to C_0 of the tangent bundle of the smooth locus of V, and we write $T_V|_C$ when $C_0 = C$. Since E' and E'' are smooth along C, the homomorphism

$$\iota : T_C \to T_{E'}|_C \oplus T_{E''}|_C, \qquad v \to (v, -v)$$

is injective, and $\mathcal{T} := \operatorname{coker} \iota$ is locally free of rank 3. Since $\deg T_{E'}|_C = 2 + (C^2)_{E'}$ and $\deg T_{E''}|_C = 2 + (C^2)_{E''}$ by the exact sequence $0 \to T_C \to T_{E'}|_C \to N_{C/E'} \to 0$ and the analogous sequence for E'', we have

$$\deg \mathcal{T} = 2 + (C^2)_{E'} + (C^2)_{E''}.$$

Since Y is locally a hypersurface along E, the dualizing sheaf ω_Y is invertible along E. Since $\operatorname{Sing} Y \cap E$ is finite, the set C° of points of C at which Y is smooth is a dense open subset of C. Over C°, the homomorphism $T_{E'}|_{C^\circ} \oplus T_{E''}|_{C^\circ} \to T_Y|_{C^\circ}$, $(v', v'') \to v' + v''$, vanishes on $\iota(T_C)$, and hence induces a homomorphism $\mathcal{T}|_{C^\circ} \to T_Y|_{C^\circ}$. Taking determinants and using $\det T_Y = \omega_Y^{-1}$ over C°, we obtain a section θ over C° of the line bundle

$$\mathcal{L} := (\det \mathcal{T})^{-1} \otimes \omega_Y^{-1}|_C.$$

Since C° is dense in C, θ is a rational section of $\mathcal{L}$. Since $K_Y = f^*K_X$ and $f(C) = \{x\}$, we have $\deg \omega_Y|_C = (K_Y \cdot C) = 0$. Thus

$$\deg \mathcal{L} = -2 - (C^2)_{E'} - (C^2)_{E''}. \tag{9.1}$$

The order $\operatorname{ord}_p \theta$ of θ at a closed point $p \in C$ can be computed after completion at p.

Let $p \in C \backslash \Sigma$, fix an isomorphism as in (i), and let $F_p := u_1u_2 + u_3\psi_p$. The linear part of F_p is $\psi_p(0)u_3$. Since a Noetherian local ring R is regular if and only if $\widehat{R}$ is regular [Mat86, Theorem 23.7], Y is smooth at p if and only if $\psi_p(0) \neq 0$, that is, if and only if $\operatorname{ord}_{u_4} \psi_p(0, 0, 0, u_4) = 0$. Since E', E'', and C correspond to $\mathbb{C}[[u_2, u_4]]$, $\mathbb{C}[[u_1, u_4]]$, and $\mathbb{C}[[u_4]]$, the completion of $\mathcal{T}$ at p is generated by the images of $(\partial_{u_2}, 0)$, $(0, \partial_{u_1})$, and $(\partial_{u_4}, 0)$, which are sent to ∂_{u_2}, ∂_{u_1}, and ∂_{u_4}. By the adjunction formula [Kol13, (4.1.3)], the completion of ω_Y at p is generated by the

residue ω_p of $du_1 \wedge du_2 \wedge du_3 \wedge du_4/F_p$, and $\omega_p = du_1 \wedge du_2 \wedge du_4/\partial_{u_3} F_p$ where $\partial_{u_3} F_p \neq 0$. Since $\partial_{u_3} F_p = \psi_p(0,0,0,u_4)$ along C, we obtain

$$\langle \omega_p, \partial_{u_2} \wedge \partial_{u_1} \wedge \partial_{u_4} \rangle = -\frac{1}{\psi_p(0,0,0,u_4)}$$

along C.

Let $y \in \Sigma$, fix an isomorphism as in (ii), and let $F_y := s_3(s_1 s_2 - s_3) + s_4 \psi_y$. The linear part of F_y is $\psi_y(0)s_4$, so Y is smooth at y if and only if $\operatorname{ord}_{s_1} \psi_y(s_1,0,0,0) = 0$. The surface E'' is the image of $(s_1, s_2) \to (s_1, s_2, s_1 s_2, 0)$, and $C = (s_2 = s_3 = s_4 = 0)$. Hence the completions of $T_{E'}|_C$, $T_{E''}|_C$, and T_C at y are generated by $\{\partial_{s_1}, \partial_{s_2}\}$, $\{\partial_{s_1}, \partial_{s_2} + s_1 \partial_{s_3}\}$, and ∂_{s_1}, respectively, and the completion of $\mathcal{T}$ at y is generated by the images of $(\partial_{s_2}, 0)$, $(0, \partial_{s_2} + s_1 \partial_{s_3})$, and $(\partial_{s_1}, 0)$. The completion of ω_Y at y is generated by the residue $\omega_y = ds_1 \wedge ds_2 \wedge ds_3/\partial_{s_4} F_y$ of $ds_1 \wedge ds_2 \wedge ds_3 \wedge ds_4/F_y$, and $\partial_{s_4} F_y = \psi_y(s_1,0,0,0)$ along C. Since

$$\langle ds_1 \wedge ds_2 \wedge ds_3, \partial_{s_2} \wedge (\partial_{s_2} + s_1 \partial_{s_3}) \wedge \partial_{s_1} \rangle = s_1,$$

we obtain

$$\langle \omega_y, \partial_{s_2} \wedge (\partial_{s_2} + s_1 \partial_{s_3}) \wedge \partial_{s_1} \rangle = \frac{s_1}{\psi_y(s_1,0,0,0)}$$

along C.

Since Y is smooth at a general point of C, the local computations above show that $\theta \neq 0$, that $\operatorname{ord}_p \theta = -l_p$ for $p \in C \backslash \Sigma$, and that $\operatorname{ord}_y \theta = 1 - l_y$ for $y \in \Sigma$. In particular, l_p and l_y are finite. Since θ is defined without the isomorphisms in (i) and (ii), the integers l_p and l_y do not depend on them. Together with the characterizations of the smooth points above, this proves (1). Since $\deg \operatorname{div} \theta = \deg \mathcal{L}$, (9.1) implies

$$|\Sigma| - \sum_{p \in C} l_p = -2 - (C^2)_{E'} - (C^2)_{E''},$$

which is (2). □

Lemma 9.3. *Let Y be a normal threefold, let E be a reduced Cartier divisor on Y, let E_1 and E_2 be prime divisors on Y contained in E, let $p \in E_1 \cap E_2$ be a closed point, and let $C := E_1 \cap E_2$. Let $\rho : W \to Y$ be the blow-up of the ideal of E_1, and let S_1 and S_2 be the strict transforms of E_1 and E_2 on W. Assume that there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u_1,u_2,u_3,u_4]]/(u_1 u_2 + u_3 \psi)$ which sends the ideals of E, E_1, and E_2 to (u_3), (u_1, u_3), and (u_2, u_3), respectively, where $\psi \in \mathbb{C}[[u_1,u_2,u_3,u_4]]$ and $\psi(0,0,0,u_4) = u_4^l \epsilon$ for an integer $l \geq 1$ and a unit $\epsilon \in \mathbb{C}[[u_4]]$. Then, after shrinking Y around p, the following hold.*

1. *There exists such an isomorphism for which, in addition, $\psi \in \mathbb{C}[[u_3,u_4]]$.*
2. *W is normal, ρ is an isomorphism over $Y \backslash \{p\}$, $\rho^{-1}(p) \cong \mathbb{P}^1$, and $K_W = \rho^* K_Y$. Moreover, S_1 and S_2 are Cartier, $\rho^* E = S_1 + S_2$, W is smooth along S_2, and $S_2 \to E_2$ is an isomorphism.*
3. *Let $\pi : S_1 \to E_1$ be the induced morphism. Then π is the blow-up of the ideal sheaf $\mathcal{I} \subset \mathcal{O}_{E_1}$ such that $\mathcal{O}_{E_1}/\mathcal{I}$ is supported at p and $\mathcal{I}\widehat{\mathcal{O}}_{E_1,p} = (u_2, u_4^l)$. The exceptional curve T_p of π satisfies $(T_p^2) = -\frac{1}{l}$ and $K_{S_1} = \pi^* K_{E_1} + l T_p$, S_1 has at most one singular point on T_p, which is a Du Val singularity of type A_{l-1}, and the strict transform C' of C satisfies $\pi^* C = C' + l T_p$ and intersects T_p at one smooth point of S_1. Moreover, for any irreducible curve $\Gamma \subset E_1$ with $\Gamma \neq C$, we have $\pi^* \Gamma = \Gamma' + r T_p$ for an integer $0 \leq r \leq (\Gamma \cdot C)_p$, where Γ' is the strict transform of Γ and $(\Gamma \cdot C)_p$ is the local intersection number at p.*

Proof. We first prove (1). Let $R := \mathbb{C}[[u_3,u_4]]$, $\Lambda := \mathbb{C}[[u_1,u_2,u_3,u_4]] = R[[u_1,u_2]]$, and $F := u_1 u_2 + u_3 \psi \in \Lambda$. Modulo u_3, the matrix of second derivatives of F with respect to (u_1, u_2) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. By the formal implicit function theorem, the system $\partial_{u_1} F = \partial_{u_2} F = 0$ has a unique solution $(u_1, u_2) = (u_1^0, u_2^0)$ with $u_1^0, u_2^0 \in u_3 R$. Let $v_i := u_i - u_i^0$ for $i = 1, 2$, so that

$\Lambda = R[[v_1, v_2]]$, and let $c := F(u_1^0, u_2^0, u_3, u_4) \in R$. Since $u_1^0 u_2^0 \in u_3^2 R$, we have $c = u_3\phi$ for some $\phi \in R$ with $\phi(0, u_4) = \psi(0, 0, 0, u_4)$. Since (u_1^0, u_2^0) is a critical point, $F - c \in (v_1, v_2)^2$. Since $F \equiv v_1 v_2 \mod u_3\Lambda$ and u_3 is a nonzerodivisor modulo $(v_1, v_2)^2$, we may write

$$F - c = \beta v_1^2 + \alpha v_1 v_2 + \gamma v_2^2$$

with $\alpha - 1, \beta, \gamma \in u_3\Lambda$. By Hensel's lemma, the equation $\beta\mu^2 - \alpha\mu + \gamma = 0$ has a solution $\mu \in u_3\Lambda$, since modulo u_3 it reads $-\mu = 0$, whose root 0 is simple. Since $\alpha - \beta\mu \equiv 1 \mod u_3\Lambda$ is a unit and $\mu(\alpha - \beta\mu) = \gamma$, we have $\mu = \gamma(\alpha - \beta\mu)^{-1} \in \gamma\Lambda$. Let $u_1' := v_1 + \mu v_2$ and $u_2' := \beta v_1 + (\alpha - \beta\mu)v_2$. Then

$$F = u_1' u_2' + u_3\phi(u_3, u_4),$$

and $u_i' \equiv u_i \mod u_3\Lambda$ for $i = 1, 2$. Thus (u_1', u_2', u_3, u_4) is a coordinate system of Λ, and the ideals (u_1', u_3) and (u_2', u_3) are equal to (u_1, u_3) and (u_2, u_3). This proves (1). For the case without the parameters u_3, u_4, compare the splitting lemma [GLS07, Theorem I.2.47].

By (1), we may assume that $\psi = \psi(u_3, u_4) \in \mathbb{C}[[u_3, u_4]]$ and $\psi(0, u_4) = u_4^l\epsilon$. Since $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[u_1, u_2, u_4]]/(u_1 u_2)$, after shrinking Y around p we have $E = E_1 + E_2$. The partial derivatives u_2, u_1, $\psi + u_3\partial_{u_3}\psi$, and $u_3\partial_{u_4}\psi$ of F vanish at a point of $(u_3 = u_1 u_2 = 0)$ only at the origin. By the Jacobian criterion [Mat86, Theorem 30.8 and the remark following it], the singular locus of $\operatorname{Spec}\widehat{\mathcal{O}}_{Y,p}$ intersects $(u_3 = 0)$ only at the closed point. Since $\widehat{\mathcal{O}}_{Y,p}$ is faithfully flat over $\mathcal{O}_{Y,p}$ and regularity descends along flat local homomorphisms [Mat86, Theorem 23.7], after shrinking Y around p, Y is smooth along $E\backslash\{p\}$. Thus the ideal of E_1 is invertible on $Y\backslash\{p\}$: it is the ideal of E near $E_1\backslash C$, and Y is smooth along $C\backslash\{p\}$. Hence ρ is an isomorphism over $Y\backslash\{p\}$.

Since blowing up commutes with flat base change, $W' := W \times_Y \operatorname{Spec}\widehat{\mathcal{O}}_{Y,p}$ is the blow-up of $\operatorname{Spec}\widehat{\mathcal{O}}_{Y,p}$ along (u_1, u_3). It is covered by the charts

$$u_3 = u_1\eta, \quad u_2 = -\eta\psi(u_1\eta, u_4) \qquad \text{and} \qquad u_1 = u_3\eta', \quad \eta' u_2 + \psi(u_3, u_4) = 0,$$

with coordinates (u_1, η, u_4) and (η', u_2, u_3, u_4), respectively. We call them the *first chart* and the *second chart*. For any point $w \in \rho^{-1}(p)$, the homomorphism from $\mathcal{O}_{W,w}$ to the local ring of W' at the unique point over w is flat with closed fiber a field. Thus W is regular at w if and only if W' is regular at the point over w [Mat86, Theorem 23.7], and W is normal at w if W' is normal there [Mat86, Corollary to Theorem 23.9].

The first chart is regular. The second chart is a hypersurface whose singular locus is contained in $(\eta' = u_2 = \psi = \partial_{u_3}\psi = \partial_{u_4}\psi = 0)$, and $\dim(\eta' = u_2 = \psi = \partial_{u_3}\psi = \partial_{u_4}\psi = 0) \le 1$ since $\psi \ne 0$. Hence the second chart is normal by Serre's criterion. Thus W is normal. The fiber $\rho^{-1}(p)$ is the union of the line $(u_1 = u_4 = 0)$ of the first chart and the line $(u_2 = u_3 = u_4 = 0)$ of the second chart, so $\rho^{-1}(p) \cong \mathbb{P}^1$. In particular, ρ is small. Since K_Y is Cartier near p and ρ is small, $K_W = \rho^* K_Y$.

Since ρ is the blow-up of the ideal of E_1, the inverse image of the ideal of E_1 is invertible [Har77, Proposition II.7.13(a)], and it is generated by u_1 on the first chart and by u_3 on the second chart. Since ρ is small, it defines the Cartier divisor S_1. In the first chart, $S_2 = (\eta = 0)$ with coordinates (u_1, u_4), which is mapped isomorphically onto $E_2 = (u_2 = u_3 = 0)$, and $S_2 \cap \rho^{-1}(p)$ is the point $\eta = u_1 = u_4 = 0$. Hence $S_2 \to E_2$ is an isomorphism, and W is smooth along S_2. Since a regular local ring is a UFD [Mat86, Theorem 20.3], S_2 is Cartier. Since $u_3 = u_1\eta$ on the first chart and u_3 defines S_1 on the second chart, $\rho^* E = S_1 + S_2$. This proves (2).

We prove (3). Since $\widehat{\mathcal{O}}_{E_1,p} = \mathbb{C}[[u_2, u_4]]$, the surface S_1 is covered by

$$u_2 = -\eta u_4^l\epsilon(u_4) \qquad \text{and} \qquad \eta' u_2 + u_4^l\epsilon(u_4) = 0,$$

with coordinates (η, u_4) and (η', u_2, u_4), and π is given by (u_2, u_4). They are the charts of the blow-up of (u_2, u_4^l), with $\eta = -u_2/(u_4^l\epsilon)$ and $\eta' = -u_4^l\epsilon/u_2$. Since (u_2, u_4^l) is primary to the maximal ideal, there exists an ideal sheaf $\mathcal{I} \subset \mathcal{O}_{E_1}$ such that $\mathcal{O}_{E_1}/\mathcal{I}$ is supported at p and $\mathcal{I}\widehat{\mathcal{O}}_{E_1,p} = (u_2, u_4^l)$. Then $\mathcal{I}\mathcal{O}_{S_1}$ is invertible, since the extension of $\mathcal{I}\mathcal{O}_{S_1}$ to each chart is principal, and an ideal of a Noetherian local ring is principal if the extension of the ideal along a flat

local homomorphism with closed fiber a field is principal, by Nakayama's lemma. Hence π factors through the blow-up of $\mathcal{I}$, and the induced morphism from S_1 to the blow-up of $\mathcal{I}$ is an isomorphism, since it is an isomorphism over $E_1\backslash\{p\}$ and after base change to $\operatorname{Spec}\widehat{\mathcal{O}}_{E_1,p}$.

The exceptional curve T_p is $(u_4 = 0)$ in the first chart and $(u_2 = u_4 = 0)$ in the second chart. The first chart is smooth. Replacing η' with $\eta'\epsilon(u_4)$, the second chart becomes $\eta' u_2 + u_4^l = 0$, which is smooth if $l = 1$ and has a Du Val singularity of type A_{l-1} at the origin if $l \geq 2$. On the first chart, $\pi^* u_2 = -\eta u_4^l \epsilon$, and on the second chart, $u_2 = 0$ implies $u_4 = 0$. Thus $\pi^* C = C' + lT_p$, where $C' = (\eta = 0)$ in the first chart, and C' intersects T_p transversally at the smooth point $\eta = u_4 = 0$ and at no other point. Hence $(C' \cdot T_p) = 1$, and $(\pi^* C \cdot T_p) = 0$ implies $(T_p^2) = -\frac{1}{l}$. On the first chart,

$$\pi^*(du_2 \wedge du_4) = -u_4^l \epsilon d\eta \wedge du_4,$$

so $K_{S_1} = \pi^* K_{E_1} + lT_p$.

Let $\Gamma \subset E_1$ be an irreducible curve with $\Gamma \neq C$, and let $\chi = \sum_{i,j} c_{ij} u_2^i u_4^j \in \mathbb{C}[[u_2, u_4]]$ be a local equation of Γ at p. On the first chart,

$$\pi^* \chi = \sum_{i,j} c_{ij} (-\eta\epsilon)^i u_4^{li+j},$$

and the coefficient of the lowest power of u_4 is a non-zero polynomial in η. Thus $\pi^*\Gamma = \Gamma' + rT_p$ with $r = \min\{li + j \mid c_{ij} \neq 0\} \geq 0$. Since $\Gamma \neq C = (u_2 = 0)$, we have $\chi(0, u_4) \neq 0$, and $j_0 := \operatorname{ord}_{u_4} \chi(0, u_4) = (\Gamma \cdot C)_p$. Since $c_{0j_0} \neq 0$, we have $r \leq j_0$. This proves (3). □

Lemma 9.4. *Let $X \ni x$ be a Gorenstein canonical threefold germ, and let $f : W \to X$ be a projective birational morphism from a normal variety W with $K_W = f^* K_X$ which is an isomorphism over $X\backslash\{x\}$. Let S_1 and S_2 be Cartier prime divisors on W such that $f^{-1}(x)_{\mathrm{red}} = S_1 \cup S_2$ and $S_2 \cong \mathbb{F}_a$ for an integer $a \geq 0$, and let $M := -2S_1 - S_2$ and $C' := S_2|_{S_1}$. Assume that $M|_{S_2}$ is numerically equivalent to a positive multiple of a fiber of the ruling of S_2, that $M|_{S_1}$ is nef and big, that $(S_1, \frac{1}{2}C')$ is klt, and that W is smooth near $S_2\backslash S_1$. Then there exist a normal variety Z and projective birational morphisms $g : W \to Z$ and $h : Z \to X$ with $h \circ g = f$ such that the following hold.*

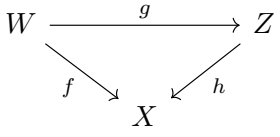
(1) The curves contracted by g are the fibers of the ruling of S_2 and the curves $\Gamma \subset S_1$ with $(M \cdot \Gamma) = 0$.

(2) h is a plt blow-up with $K_Z = h^* K_X$ whose Kollár component is $P := g(S_1)$, and $g^* P = S_1 + \frac{1}{2} S_2$.

(3) $(g|_{S_1})^*(K_P + \operatorname{Diff}_P(0)) = K_{S_1} + \frac{1}{2}C'$, and $\widehat{\mathrm{vol}}(\mathrm{ord}_P) = ((K_{S_1} + \frac{1}{2}C')^2)$.

Proof. Since $K_W = f^* K_X$ and X is canonical, W is canonical [KM98, Lemma 2.30], hence klt and Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. After shrinking X, $K_W = f^* K_X \sim 0$. Let A be a fiber of the ruling of S_2. By the adjunction formula [Kol13, (4.1.3)], $S_2|_{S_2} \sim K_{S_2}$, so $(S_2 \cdot A) = (K_{S_2} \cdot A) = -2$. Since $M|_{S_2} \equiv cA$ for some $c > 0$, we have $(M \cdot A) = 0$, and hence $(S_1 \cdot A) = 1$.

Since f is an isomorphism over $X\backslash\{x\}$, any curve contracted by f is contained in $S_1 \cup S_2$. For any irreducible curve $\Gamma \subset S_2$, we have $(M \cdot \Gamma) = c(A \cdot \Gamma)_{S_2} \geq 0$, with equality if and only if Γ is a fiber of the ruling. For any irreducible curve $\Gamma \subset S_1$, we have $(M \cdot \Gamma) \geq 0$ since $M|_{S_1}$ is nef. Thus the Cartier divisor M is f-nef, and $M - K_W \equiv_f M$ is f-nef and f-big since f is birational. By the relative base-point-free theorem [KM98, Theorem 3.24], bM is f-free for any sufficiently large integer b. Fix such b, let $g : W \to Z$ be the Stein factorization of the morphism $W \to \mathbb{P}_X(f_* \mathcal{O}_W(bM))$ over X, and let $h : Z \to X$ be the induced morphism. Then Z is normal, $h \circ g = f$, and there exists an h-ample Cartier divisor H on Z with $g^* H \sim bM$:



A curve $\Gamma \subset W$ is contracted by g if and only if Γ is contracted by f and $(M \cdot \Gamma) = 0$. This proves (1).

Since $(S_1 \cdot A) = 1$, any fiber of the ruling of S_2 intersects S_1, so $g(S_2) \subset g(S_1) = P$. Since $M|_{S_1}$ is big, $g|_{S_1}$ is generically finite, so g contracts only finitely many curves on S_1. Thus P is a prime divisor, $g|_{S_1} : S_1 \to P$ is birational, S_2 is the only prime divisor contracted by g, and $h^{-1}(x) = P$. Moreover, h is an isomorphism over $X\backslash\{x\}$, since $h \circ g = f$ is and g is surjective onto the normal variety Z. Pushing forward $g^*H \sim bM$ by g, we obtain $H \sim -2bP$. Thus P is $\mathbb{Q}$-Cartier and $-P$ is h-ample. Write $g^*P = S_1 + eS_2$. Since A is contracted by g, we have $0 = (g^*P \cdot A) = 1 - 2e$, so $e = \frac{1}{2}$. Since $K_W = g^*h^*K_X$, pushing forward by g implies $K_Z = h^*K_X$, and hence

$$K_W + S_1 + \tfrac{1}{2}S_2 = g^*(K_Z + P). \tag{9.2}$$

By the adjunction formula, $(K_W + S_1 + \frac{1}{2}S_2)|_{S_1} = K_{S_1} + \frac{1}{2}C'$. Since $(S_1, \frac{1}{2}C')$ is klt, $(W, S_1 + \frac{1}{2}S_2)$ is plt near S_1 by inversion of adjunction [KM98, Theorem 5.50]. Near $S_2\backslash S_1$, both W and S_2 are smooth, so $(W, \frac{1}{2}S_2)$ is klt there. Away from $S_1 \cup S_2$, W is canonical. Thus $(W, S_1 + \frac{1}{2}S_2)$ is plt, and so is (Z, P) by (9.2) and [KM98, Lemma 2.30]. Therefore h is a plt blow-up with $K_Z = h^*K_X$ and Kollár component P. This proves (2).

Restricting (9.2) to S_1, we obtain $(g|_{S_1})^*(K_P + \mathrm{Diff}_P(0)) = K_{S_1} + \frac{1}{2}C'$. Since $g|_{S_1}$ is birational, Lemma 2.19(1) and the projection formula [Laz04, Remark 1.1.13] imply

$$\widehat{\mathrm{vol}}(\mathrm{ord}_P) = ((K_P + \mathrm{Diff}_P(0))^2) = ((K_{S_1} + \tfrac{1}{2}C')^2).$$

This proves (3). □

Lemma 9.5 treats two Hirzebruch surfaces glued along their negative sections. In this case, [Liu25, Propositions 4.6–4.9] show that the local volume is less than 9, and the degree of E only gives bounds of at least 8. Keeping half of the conductor as the different of a Kollár component lowers the bound to $\frac{27}{4}$. In Lemmas 9.5, 9.6, and 9.8, the volume of such a Kollár component is computed by Lemma 9.4(3).

Lemma 9.5. *Notation and conditions as in Set-up 8.2. Assume that Y is locally a hypersurface along E, that $E = E_1 + E_2$ has normal crossings with two smooth components $E_i \cong \mathbb{F}_{a_i}$ for integers $a_i \geq 0$, that $C := E_1 \cap E_2$ is a smooth rational curve which is the negative section B_i of $\mathbb{F}_{a_i}$ for $i = 1, 2$ (a section with $(B_i^2) = 0$ if $a_i = 0$), and that Y is smooth at the generic point of C. Let $L := 2 - a_1 - a_2$, and choose $i \in \{1, 2\}$ with $a_i = \min\{a_1, a_2\}$. Then the following hold.*

1. *$a_1 + a_2 \leq 2$, that is, $L \geq 0$.*
2. *There exist a small crepant morphism $W \to Y$ and a projective birational morphism $g : W \to Z$ over X, such that the induced morphism $h : Z \to X$ is a plt blow-up with $K_Z = h^*K_X$ whose Kollár component P is the image of the strict transform S_i of E_i on W.*
3. *The morphism $S_i \to E_i$ is the composition of the blow-ups of the ideals $(t_p, w_p^{l_p})$ at distinct points $p \in C$, where t_p, w_p are local coordinates of E_i at p with $C = (t_p = 0)$ near p, and the l_p are positive integers with $\sum_p l_p = L$. In particular, it is an isomorphism if $L = 0$.*
4. *The strict transform C' of C on S_i is contained in the smooth locus of S_i, $P \cong S_i$, $\mathrm{Diff}_P(0) = \frac{1}{2}C'$, and*

$$\widehat{\mathrm{vol}}(\mathrm{ord}_{E_i}) = 6 + \frac{3}{4}a_i - \frac{1}{4}L \leq \frac{27}{4}.$$

In particular, $\widehat{\mathrm{vol}}(X \ni x) \leq \frac{27}{4}$.

Proof. Let $j \in \{1, 2\}$ be the index with $j \neq i$. Let $p \in C$ be a closed point, and let u_3 be a local equation of E at p. Since E has normal crossings at p, $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[u_1, u_2, u_4]]/(u_1u_2)$, where E_i and E_j correspond to (u_1) and (u_2). Lifting u_1, u_2, u_4 to $\widehat{\mathcal{O}}_{Y,p}$, we obtain a surjection $\mathbb{C}[[u_1, u_2, u_3, u_4]] \to \widehat{\mathcal{O}}_{Y,p}$, whose kernel is generated by one element F since Y is a hypersurface at p. Since $F \equiv \lambda u_1u_2 \mod u_3$ for a unit λ, replacing F with $\lambda^{-1}F$, we may assume that

$F = u_1u_2 + u_3\psi_p$. Thus condition (i) of Lemma 9.2 holds for $(E', E'') := (E_i, E_j)$ at any closed point of C. Since E is Cartier and E is smooth away from C, Y is smooth along $E\backslash C$. Hence $\operatorname{Sing} Y \cap E \subset C$, and $\operatorname{Sing} Y \cap E$ is finite since Y is smooth at the generic point of C. Since $(C^2)_{E_1} = -a_1$ and $(C^2)_{E_2} = -a_2$, Lemma 9.2 with $\Sigma = \emptyset$ implies

$$\sum_{p\in C} l_p = 2 - a_1 - a_2 = L. \tag{9.3}$$

Thus $L \geq 0$, which proves (1). In particular, $a_i \leq 1$.

Let $p_1, \dots, p_k$ be the points $p \in C$ with $l_p > 0$, which are the singular points of Y on E by Lemma 9.2(1). Let $\rho : W \to Y$ be the blow-up of the ideal of E_i, and let S_i and S_j be the strict transforms of E_i and E_j. Since E_i is Cartier on the smooth locus of Y and is equal to E near $E_i\backslash C$, ρ is an isomorphism over $Y\backslash\{p_1, \dots, p_k\}$. By Lemma 9.3, applied at each p_m to E_i and E_j, W is normal, ρ is small, $K_W = \rho^*K_Y$, both S_i and S_j are Cartier, $\rho^*E = S_i + S_j$, W is smooth along S_j, and $S_j \to E_j$ is an isomorphism. Moreover, the morphism $\pi : S_i \to E_i$ is the composition of the blow-ups of the ideals $(u_2, u_4^{l_{p_m}})$ at p_m for $m = 1, \dots, k$, in the coordinates of Lemma 9.3, where $C = (u_2 = 0)$ on E_i. Together with (9.3), this proves (3).

For $p = p_1, \dots, p_k$, let T_p be the exceptional curve of π over p. By Lemma 9.3(3), the curves T_p are pairwise disjoint, $(T_p^2) = -\frac{1}{l_p}$, $(C' \cdot T_p) = 1$, the singular points of S_i are Du Val and are contained in $\bigcup_p T_p\backslash C'$, and

$$\pi^*C = C' + \sum_p l_pT_p, \qquad K_{S_i} = \pi^*K_{E_i} + \sum_p l_pT_p.$$

In particular, C' is contained in the smooth locus of S_i. Since $(C^2)_{E_i} = -a_i$ and $(K_{E_i} \cdot C) = a_i - 2$, we obtain

$$((C')^2) = -a_i - L, \qquad (K_{S_i}^2) = 8 - L, \qquad (K_{S_i} \cdot C') = a_i - 2 + L. \tag{9.4}$$

Since $S_j \cong E_j$ contains no curve contracted by ρ, we have $S_i \cap S_j = C'$, and S_i and S_j intersect transversally at a general point of C'. Thus $S_j|_{S_i} = C'$ and $S_i|_{S_j} = B_j$. Since $K_W = (f \circ \rho)^*K_X$ is trivial near $(f \circ \rho)^{-1}(x)$ after shrinking X, the adjunction formula [Kol13, (4.1.3)] implies $S_i|_{S_i} \sim K_{S_i}$ and $S_j|_{S_j} \sim K_{S_j}$. Let A_i and A_j be fibers of the rulings of E_i and E_j, and let $M := -2S_i - S_j$. Then

$$M|_{S_j} \sim -2B_j - K_{\mathbb{F}_{a_j}} = (a_j + 2)A_j, \qquad M|_{S_i} \sim -2K_{S_i} - C' = \pi^*(3B_i + 2(a_i + 2)A_i) - \sum_p l_pT_p.$$

We have $(M|_{S_i} \cdot T_p) = 1$ and $(M|_{S_i} \cdot C') = 4 - a_i - L = 2 + a_j > 0$. Let $\Gamma' \subset S_i$ be an irreducible curve other than C' and the T_p. Then Γ' is the strict transform of an irreducible curve $\Gamma \subset E_i$ with $\Gamma \neq C$, and $\Gamma \equiv uB_i + vA_i$ with $u \geq 0$, $v \geq a_iu$, and $(u, v) \neq (0, 0)$ [Har77, Proposition V.2.20]. By Lemma 9.3(3), $\pi^*\Gamma = \Gamma' + \sum_p r_pT_p$ with $0 \leq r_p \leq (\Gamma \cdot C)_p$, so $\sum_p r_p \leq (\Gamma \cdot C) = v - a_iu$. Thus

$$(M|_{S_i} \cdot \Gamma') = 3v + (4 - a_i)u - \sum_p r_p \geq 2v + 4u > 0.$$

Moreover, $((M|_{S_i})^2) = 3a_i + 24 - L > 0$. Hence $M|_{S_i}$ is nef and big, and $(M|_{S_i} \cdot \Gamma') > 0$ for any curve $\Gamma' \subset S_i$. Since the singularities of S_i are Du Val and C' is a smooth curve contained in the smooth locus of S_i, $(S_i, \frac{1}{2}C')$ is klt.

Since $\rho^{-1}(p_m) = T_{p_m} \subset S_i$, we have $(f \circ \rho)^{-1}(x)_{\mathrm{red}} = S_i \cup S_j$, and W is smooth near S_j. By Lemma 9.4 for $f \circ \rho$ and $(S_1, S_2) := (S_i, S_j)$, we obtain a normal variety Z and projective birational morphisms $g : W \to Z$ and $h : Z \to X$ with

$$\begin{array}{ccc} W & \xrightarrow{\rho} & Y \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{h} & X \end{array}$$

commutative, such that h is a plt blow-up with $K_Z = h^*K_X$ whose Kollár component is $P = g(S_i)$. This proves (2). By Lemma 9.4(1), g contracts no curve on S_i, so $g|_{S_i} : S_i \to P$ is finite and birational, hence an isomorphism since P is normal. By Lemma 9.4(3), $\mathrm{Diff}_P(0) = \frac{1}{2}C'$ under the isomorphism $g|_{S_i}$. Since S_i is the strict transform of E_i and $P = g(S_i)$, $\mathrm{ord}_{E_i} = \mathrm{ord}_P$ as valuations of $K(X)$. By Lemma 9.4(3) and (9.4),

$$\widehat{\mathrm{vol}}(\mathrm{ord}_{E_i}) = ((K_{S_i} + \tfrac{1}{2}C')^2) = (8 - L) + (a_i - 2 + L) - \tfrac{1}{4}(a_i + L) = 6 + \tfrac{3}{4}a_i - \tfrac{1}{4}L.$$

Since $a_i \le 1$ and $L \ge 0$, $\widehat{\mathrm{vol}}(\mathrm{ord}_{E_i}) \le \frac{27}{4}$, and $\widehat{\mathrm{vol}}(X \ni x) \le \widehat{\mathrm{vol}}(\mathrm{ord}_{E_i})$ by Definition 2.6. This proves (4). □

Lemma 9.6 treats two Hirzebruch surfaces glued crosswise, where slc surfaces are in the sense of [Kol13, Definition–Lemma 5.10]. In the cases of small degree we use the degree of E, and in the other cases we use Lemma 9.4 again.

Lemma 9.6. *Notation and conditions as in Set-up 8.2. Assume that Y is locally a hypersurface along E, that $\mathrm{Sing}\, Y \cap E$ is finite, and that $E = E_1 \cup E_2$ is the reduced Gorenstein slc surface glued from $E_i \cong \mathbb{F}_{a_i}$ with integers $a_i \ge 0$ by identifying a fiber A_1 of E_1 with the negative section B_2 of E_2, the negative section B_1 of E_1 with a fiber A_2 of E_2, and the point $A_1 \cap B_1$ with the point $B_2 \cap A_2$ (for $a_i = 0$, B_i is the section with $(B_i^2) = 0$ used in the gluing). Let $y \in E$ be the image of $A_1 \cap B_1$. Assume moreover that E has double crossings away from y, and that $\widehat{\mathcal{O}}_{E,y} \cong \mathbb{C}[[s_1, s_2, s_3]]/(s_3(s_1s_2 - s_3))$ with components $\{s_3 = 0\}$ and $\{s_3 = s_1s_2\}$. Then $a_1, a_2 \le 3$ and $\widehat{\mathrm{vol}}(X \ni x) \le 7$. More precisely, the following hold.*

(1) If $a_1 + a_2 \le 3$, then $\widehat{\mathrm{vol}}(X \ni x) \le a_1 + a_2 + 4$.

(2) Assume that $a_1 \le a_2$ and $(a_1, a_2) \in \{(1,3), (2,2), (2,3)\}$. Then there exist a small crepant morphism $W \to Y$ and a projective birational morphism $g : W \to Z$ over X such that the induced morphism $h : Z \to X$ is a plt blow-up with $K_Z = h^*K_X$ whose Kollár component P is the image of the strict transform S_1 of E_1, and $g^*P = S_1 + \frac{1}{2}S_2$, where S_2 is the strict transform of E_2. Moreover, $\widehat{\mathrm{vol}}(\mathrm{ord}_{E_1}) = 3 + a_1 + \frac{a_2}{4}$, which is $\frac{19}{4}$, $\frac{11}{2}$, and $\frac{23}{4}$, respectively.

(3) If $(a_1, a_2) = (3, 3)$, then $\widehat{\mathrm{vol}}(X \ni x) \le \frac{27}{4}$.

Proof. The proof is long, so we divide it into several steps. Let $C_1 := B_1 = A_2$ and $C_2 := A_1 = B_2$ in E. Then $E_1 \cap E_2 = C_1 \cup C_2$ and $C_1 \cap C_2 = \{y\}$.

Step 1. In this step, we prove that $a_1, a_2 \le 3$, and we compute the singular points of Y on E.

Let s_4 be a local equation of E at y. Lifting s_1, s_2, s_3 to $\widehat{\mathcal{O}}_{Y,y}$ and using that Y is a hypersurface at y, we obtain $\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[s_1, s_2, s_3, s_4]]/(F)$ with $F \equiv \lambda s_3(s_1s_2 - s_3) \mod s_4$ for a unit λ. Replacing F with $\lambda^{-1}F$, we may assume that

$$F = s_3(s_1s_2 - s_3) + s_4\psi$$

for some $\psi \in \mathbb{C}[[s_1, s_2, s_3, s_4]]$, where E_1 and E_2 correspond to (s_3, s_4) and $(s_3 - s_1s_2, s_4)$ (Lemma 9.7 below, after replacing s_3 by $s_1s_2 - s_3$ if necessary). After exchanging s_1 and s_2 if necessary, C_1 and C_2 correspond to (s_2, s_3, s_4) and (s_1, s_3, s_4). Thus condition (ii) of Lemma 9.2 holds at y for $C = C_1$, and, after exchanging s_1 and s_2, for $C = C_2$, where $(E', E'') := (E_1, E_2)$. At a closed point $p \in C_k \backslash \{y\}$, E has double crossings, and lifting the coordinates of $\widehat{\mathcal{O}}_{E,p}$ together with a local equation of E yields an isomorphism as in condition (i) of Lemma 9.2 for $C = C_k$ and $(E', E'') := (E_1, E_2)$. For $k = 1, 2$, let $l_{y,k}$ be the integer l_y of Lemma 9.2(1) for $C = C_k$, so that $l_{y,1} = \mathrm{ord}_{s_1} \psi(s_1, 0, 0, 0)$ and $l_{y,2} = \mathrm{ord}_{s_2} \psi(0, s_2, 0, 0)$. Since $(C_1^2)_{E_1} = (B_1^2) = -a_1$, $(C_1^2)_{E_2} = (A_2^2) = 0$, $(C_2^2)_{E_1} = (A_1^2) = 0$, and $(C_2^2)_{E_2} = (B_2^2) = -a_2$, Lemma 9.2(2) with $\Sigma = \{y\}$ implies

$$l_{y,1} + \sum_{p \in B_1 \backslash \{y\}} l_p = 3 - a_1, \qquad l_{y,2} + \sum_{p \in A_1 \backslash \{y\}} l_p = 3 - a_2. \tag{9.5}$$

Hence $a_1, a_2 \leq 3$. By Lemma 9.2(1), Y is singular at y if and only if $l_{y,1} > 0$, if and only if $l_{y,2} > 0$, and a point $p \in (A_1 \cup B_1)\backslash\{y\}$ is a singular point of Y if and only if $l_p > 0$. Since E is Cartier and E is smooth away from $C_1 \cup C_2$, Y is smooth along $E\backslash(C_1 \cup C_2)$.

Step 2. In this step, we prove (1).

Since $K_Y = f^*K_X$ is trivial near E after shrinking X, the adjunction formula [Kol13, (4.1.3)] implies $\omega_E \cong \mathcal{O}_E(E)$, and $\mathcal{O}_E(-E)$ is ample since $-E$ is f-ample. Thus E is a Gorenstein del Pezzo surface in the sense of [Rei94, §0.1]. Let $\nu : E_1 \sqcup E_2 \to E$ be the normalization. The conductor of ν is $(A_1 + B_1) + (A_2 + B_2)$, so [Rei94, Proposition 2.3] implies

$$\nu^*K_E \sim (K_{E_1} + A_1 + B_1) + (K_{E_2} + A_2 + B_2).$$

Since $((K_{\mathbb{F}_a} + A + B)^2) = a + 2$ for a fiber A and the negative section B of $\mathbb{F}_a$, the projection formula [Laz04, Remark 1.1.13] implies $(K_E^2) = a_1 + a_2 + 4$. Thus Lemma 2.19(2) implies $\widehat{\mathrm{vol}}(X \ni x) \leq a_1 + a_2 + 4$. This proves (1).

Step 3. In this step, we construct W in the remaining cases.

By (1), we may assume that $a_1 \leq a_2$ and $(a_1, a_2) \in \{(1,3), (2,2), (2,3), (3,3)\}$. If $a_2 = 3$, then (9.5) implies $l_{y,2} = 0$, so Y is smooth at y. Thus either Y is smooth at y, or $(a_1, a_2) = (2,2)$ and Y is singular at y. If Y is singular at y, then (9.5) implies $l_{y,1} = l_{y,2} = 1$, and Y is smooth along $E\backslash\{y\}$.

Let $\rho : W \to Y$ be the blow-up of the ideal of E_1, let S_1 and S_2 be the strict transforms of E_1 and E_2, and let $\pi : S_1 \to E_1$ be the induced morphism. Since E_1 is Cartier on the smooth locus of Y and is equal to E near $E_1\backslash(C_1 \cup C_2)$, ρ is an isomorphism over the complement of the singular points of Y on E. Let A_1' and B_1' be the strict transforms of A_1 and B_1 on S_1, and let $C' := A_1' + B_1'$.

First assume that Y is smooth at y. Then the singular points of Y on E are the points $p \in (A_1 \cup B_1)\backslash\{y\}$ with $l_p > 0$. If $(a_1, a_2) = (3,3)$, then (9.5) implies that Y is smooth along E, so E_1 is Cartier, $W = Y$, and $S_k = E_k$ for $k = 1, 2$. In general, at each p with $l_p > 0$, after shrinking Y around p we have $E_1 \cap E_2 = C_k$ for some k, and Lemma 9.3 for E_1, E_2, and p implies the following: W is normal, ρ is small, $K_W = \rho^*K_Y$, both S_1 and S_2 are Cartier, $\rho^*E = S_1 + S_2$, W is smooth along S_2, and $S_2 \to E_2$ is an isomorphism. Moreover, for the exceptional curve T_p of π over p, we have $(T_p^2) = -\frac{1}{l_p}$, $(C' \cdot T_p) = 1$, the singular points of S_1 are Du Val and are contained in $\bigcup_p T_p\backslash C'$, and

$$K_{S_1} = \pi^*K_{E_1} + \sum_p l_p T_p, \qquad \pi^*(A_1 + B_1) = C' + \sum_p l_p T_p,$$

where the sums run over the singular points p of Y on E.

Next assume that Y is singular at y. Then ρ is an isomorphism over $Y\backslash\{y\}$. Since $l_{y,1} = l_{y,2} = 1$, we may write

$$\psi_0 := \psi(s_1, s_2, 0, 0) = s_1\tilde{\alpha} + s_2\tilde{\beta}$$

with units $\tilde{\alpha}, \tilde{\beta} \in \mathbb{C}[[s_1, s_2]]$. After completion at y, the blow-up of (s_3, s_4) is covered by the charts

$$s_4 = s_3\eta, \quad s_1s_2 - s_3 + \eta\psi(s_1, s_2, s_3, s_3\eta) = 0$$

and

$$s_3 = s_4\eta', \quad \eta'(s_1s_2 - s_4\eta') + \psi(s_1, s_2, s_4\eta', s_4) = 0,$$

with coordinates (s_1, s_2, s_3, η) and (s_1, s_2, s_4, η'), respectively. The fiber $T := \rho^{-1}(y)$ is the union of the η-line and the η'-line, so $T \cong \mathbb{P}^1$. For any point $w \in T$, the homomorphism from $\mathcal{O}_{W,w}$ to the local ring of the blow-up of $\operatorname{Spec}\widehat{\mathcal{O}}_{Y,y}$ at the unique point over w is flat with closed fiber a field. Thus regularity and normality of W and S_1 along T can be checked on the blow-up of $\operatorname{Spec}\widehat{\mathcal{O}}_{Y,y}$ [Mat86, Theorem 23.7], [Mat86, Corollary to Theorem 23.9]. At the point $(0,0,0,\eta)$ of the first chart, the partial derivatives of the equation with respect to s_1 and s_3 are $\tilde{\alpha}(0)\eta$ and $-1 + \eta\big(\partial_{s_3}\psi(0) + \eta\partial_{s_4}\psi(0)\big)$, and they do not vanish simultaneously. At the point $(0,0,0,\eta')$

of the second chart, the partial derivative with respect to s_1 is $\tilde{\alpha}(0) \neq 0$. Since a Noetherian local ring R is regular if and only if $\widehat{R}$ is regular [Mat86, Theorem 23.7], W is smooth along T. In particular, W is normal and ρ is small, so $K_W = \rho^* K_Y$. The divisor S_1 is $(s_3 = 0)$ in the first chart and $(s_4 = 0)$ in the second chart, and $S_2 = (\eta = 0)$ in the first chart, which is mapped isomorphically onto E_2 with coordinates (s_1, s_2). Since $s_4 = s_3\eta$, we have $\rho^* E = S_1 + S_2$. Since W is smooth along T, ρ is an isomorphism over $Y\backslash\{y\}$, and Y is smooth along $E\backslash\{y\}$, W is smooth along $S_1 \cup S_2$, so S_1 and S_2 are Cartier, as a regular local ring is a UFD [Mat86, Theorem 20.3].

The surface S_1 is covered by $s_1 s_2 + \eta\psi_0 = 0$ and $\eta' s_1 s_2 + \psi_0 = 0$, so near y, π is the blow-up of $(s_1 s_2, \psi_0)$. Since $s_1 s_2 \equiv -s_1^2 \tilde{\alpha}/\tilde{\beta} \mod \psi_0$, we have $(s_1 s_2, \psi_0) = (\psi_0, s_1^2)$, and (ψ_0, s_1) is a coordinate system of E_1 at y. In the chart $\psi_0 = s_1^2\zeta$ of the blow-up of (ψ_0, s_1^2), with coordinates (s_1, ζ), we have $T = (s_1 = 0)$, $\pi^*(d\psi_0 \wedge ds_1) = s_1^2 d\zeta \wedge ds_1$, and $s_2 = s_1(s_1\zeta - \tilde{\alpha})/\tilde{\beta}$. Moreover, the strict transform of the smooth curve $(\psi_0 = 0)$ is $(\zeta = 0)$, which intersects T transversally at one point and is disjoint from T in the other chart $s_1^2 = \psi_0\zeta'$. Thus $\pi^*(\psi_0 = 0) = (\zeta = 0) + 2T$ with $((\zeta = 0) \cdot T) = 1$, and $(\pi^*(\psi_0 = 0) \cdot T) = 0$ gives $(T^2) = -\frac{1}{2}$. Since $A_1 = (s_1 = 0)$ and $B_1 = (s_2 = 0)$ on E_1, we obtain

$$K_{S_1} = \pi^* K_{E_1} + 2T, \qquad \pi^* A_1 = A_1' + T, \qquad \pi^* B_1 = B_1' + T, \qquad (T^2) = -\tfrac{1}{2}. \tag{9.6}$$

On S_1, the partial derivatives of $s_1 s_2 + \eta\psi_0$ with respect to s_1 and s_2 are $\tilde{\alpha}(0)\eta$ and $\tilde{\beta}(0)\eta$ along T, and the partial derivative of $\eta' s_1 s_2 + \psi_0$ with respect to s_2 is $\tilde{\beta}(0) \neq 0$ along T. Thus S_1 is smooth along T except at the point $y' := (s_1 = s_2 = \eta = 0)$ of the first chart. Near y', let $U := s_1 + \eta\tilde{\beta}$ and $V := s_2 + \eta\tilde{\alpha}$. Then (U, V, η) is a coordinate system of $\mathbb{C}[[s_1, s_2, \eta]]$, and

$$s_1 s_2 + \eta\psi_0 = UV - \eta^2 \tilde{\alpha}\tilde{\beta}.$$

Let κ be a square root of the unit $\tilde{\alpha}\tilde{\beta}$, and let $\tau := \kappa\eta$. Then near y', $S_1 \cong (UV = \tau^2)$, which is a Du Val singularity of type A_1, and $S_2|_{S_1} = (\tau = 0)$.

In both cases, since $S_2 \cong E_2$ contains no curve contracted by ρ, we have $S_1 \cap S_2 = C'$, and S_1 and S_2 intersect transversally at a general point of C'. Thus $S_2|_{S_1} = C'$ and $S_1|_{S_2} = A_2 + B_2$. Since $\rho^{-1}(p) \subset S_1$ for any point p over which ρ is not an isomorphism, we have $(f \circ \rho)^{-1}(x)_{\mathrm{red}} = S_1 \cup S_2$. Since Y is smooth along $E_2\backslash(A_2 \cup B_2)$, W is smooth near $S_2\backslash S_1$.

Step 4. In this step, we prove that $(S_1, \frac{1}{2}C')$ is klt and that $M|_{S_1}$ is nef and big, where $M := -2S_1 - S_2$.

If Y is smooth at y, then A_1' and B_1' are smooth curves intersecting transversally at y, and the singularities of S_1 are Du Val and are not contained in C'. Thus $(S_1, \frac{1}{2}C')$ is klt. If Y is singular at y, then Y is smooth along $E\backslash\{y\}$, so S_1 is smooth away from y', and A_1' and B_1' are disjoint smooth curves away from y'. Thus $(S_1, \frac{1}{2}C')$ is klt away from y'. Near y', consider the quotient map $\mathbb{C}^2 \to (UV = \tau^2)$, $(z_1, z_2) \to (z_1^2, z_2^2, z_1 z_2)$, which is étale in codimension one. The pullback of $C' = (\tau = 0)$ is $(z_1 z_2 = 0)$, and $(\mathbb{C}^2, \frac{1}{2}(z_1 z_2 = 0))$ is klt. Thus $(S_1, \frac{1}{2}C')$ is klt near y' by [KM98, Proposition 5.20(4)] (the klt property near y' is determined by the completion of $\mathcal{O}_{S_1, y'}$ together with the ideal of C', since the blow-up of y' is a log resolution of (S_1, C') near y' and blowing up commutes with completion).

Since $K_W = (f \circ \rho)^* K_X$ is trivial near $(f \circ \rho)^{-1}(x)$ after shrinking X, the adjunction formula [Kol13, (4.1.3)] implies $S_k|_{S_k} \sim K_{S_k}$ for $k = 1, 2$. Let A and B denote a fiber and the negative section of $E_1 \cong \mathbb{F}_{a_1}$, so that $A \sim A_1$ and $B = B_1$. Thus

$$M|_{S_2} \sim -2(A_2 + B_2) - K_{\mathbb{F}_{a_2}} = a_2 A_2, \qquad M|_{S_1} \sim -2K_{S_1} - C'.$$

In particular, $M|_{S_2}$ is a positive multiple of a fiber of the ruling of S_2. Let $\Gamma' \subset S_1$ be an irreducible curve which is the strict transform of an irreducible curve $\Gamma \subset E_1$ with $\Gamma \neq A_1, B_1$, and write $\Gamma \equiv uB + vA$ with $u \geq 0$, $v \geq a_1 u$, and $(u, v) \neq (0, 0)$ [Har77, Proposition V.2.20]. Then $(\Gamma \cdot (A_1 + B_1)) = v + (1 - a_1)u$.

First assume that Y is smooth at y. Then

$$M|_{S_1} \sim \pi^*(3B + (2a_1+3)A) - \sum_p l_p T_p.$$

Since $l_{y,1} = l_{y,2} = 0$, (9.5) implies

$$(M|_{S_1} \cdot T_p) = 1, \qquad (M|_{S_1} \cdot B_1') = 3 - a_1 - \sum_{p \in B_1} l_p = 0, \qquad (M|_{S_1} \cdot A_1') = 3 - \sum_{p \in A_1} l_p = a_2 > 0.$$

By Lemma 9.3(3), $\pi^*\Gamma = \Gamma' + \sum_p r_p T_p$ with $0 \le r_p \le (\Gamma \cdot (A_1 + B_1))_p$, so $\sum_p r_p \le v + (1-a_1)u$. Thus

$$(M|_{S_1} \cdot \Gamma') = 3v + (3-a_1)u - \sum_p r_p \ge 2v + 2u > 0.$$

Moreover, since $\sum_p l_p = 6 - a_1 - a_2$ by (9.5),

$$((M|_{S_1})^2) = 3a_1 + 18 - \sum_p l_p = 4a_1 + a_2 + 12 > 0.$$

Next assume that Y is singular at y. Then $a_1 = a_2 = 2$, and (9.6) implies

$$M|_{S_1} \sim \pi^*(3B + 7A) - 2T, \qquad (T \cdot A_1') = (T \cdot B_1') = \tfrac{1}{2}.$$

Thus $(M|_{S_1} \cdot T) = 1$, $(M|_{S_1} \cdot B_1') = 0$, and $(M|_{S_1} \cdot A_1') = 2$. Let $\chi = \sum_{i,j} c_{ij} \psi_0^i s_1^j$ be a local equation of Γ at y. In the chart $\psi_0 = s_1^2 \zeta$, we have $\pi^*\chi = \sum_{i,j} c_{ij} \zeta^i s_1^{2i+j}$, so $\pi^*\Gamma = \Gamma' + rT$ with

$$r = \min\{2i + j \mid c_{ij} \ne 0\} \le 2 \operatorname{mult}_y \Gamma \le (\Gamma \cdot A_1)_y + (\Gamma \cdot B_1)_y,$$

where the last inequality holds since A_1 and B_1 are smooth at y. Since $(2T \cdot \Gamma') = r$, we obtain

$$(M|_{S_1} \cdot \Gamma') = 3v + u - r \ge 3v + u - (v - u) = 2v + 2u > 0,$$

and $((M|_{S_1})^2) = 24 - 2 = 22 > 0$.

In both cases, $M|_{S_1}$ is nef and big, and B_1' is the only curve $\Gamma' \subset S_1$ with $(M|_{S_1} \cdot \Gamma') = 0$.

Step 5. In this step, we prove (2) and (3), and conclude the proof.

By **Steps 3 and 4**, the conditions of Lemma 9.4 hold for $f \circ \rho$ and (S_1, S_2). Thus Lemma 9.4 implies that there exist a normal variety Z and projective birational morphisms $g : W \to Z$ and $h : Z \to X$ with $h \circ g = f \circ \rho$, such that h is a plt blow-up with $K_Z = h^* K_X$ whose Kollár component is $P = g(S_1)$, and $g^*P = S_1 + \frac{1}{2} S_2$. Since S_1 is the strict transform of E_1 and $P = g(S_1)$, $\operatorname{ord}_{E_1} = \operatorname{ord}_P$ as valuations of $K(X)$. Since $((K_{\mathbb{F}_{a_1}} + \frac{1}{2}(A+B))^2) = \frac{9}{2} + \frac{3}{4} a_1$, Lemma 9.4(3) implies the following.

If Y is smooth at y, then $K_{S_1} + \frac{1}{2} C' = \pi^*(K_{E_1} + \frac{1}{2}(A_1 + B_1)) + \frac{1}{2} \sum_p l_p T_p$, so

$$\widehat{\operatorname{vol}}(\operatorname{ord}_{E_1}) = \frac{9}{2} + \frac{3}{4} a_1 - \frac{1}{4} \sum_p l_p = \frac{9}{2} + \frac{3}{4} a_1 - \frac{1}{4}(6 - a_1 - a_2) = 3 + a_1 + \frac{a_2}{4}.$$

If Y is singular at y, then $K_{S_1} + \frac{1}{2} C' = \pi^*(K_{E_1} + \frac{1}{2}(A_1 + B_1)) + T$ by (9.6), so

$$\widehat{\operatorname{vol}}(\operatorname{ord}_{E_1}) = \frac{9}{2} + \frac{3}{2} - \frac{1}{2} = \frac{11}{2} = 3 + a_1 + \frac{a_2}{4}.$$

For $(a_1, a_2) = (1,3)$, $(2,2)$, and $(2,3)$, the value $3 + a_1 + \frac{a_2}{4}$ is $\frac{19}{4}$, $\frac{11}{2}$, and $\frac{23}{4}$, respectively. This proves (2). For $(a_1, a_2) = (3,3)$, we obtain $\widehat{\operatorname{vol}}(X \ni x) \le \widehat{\operatorname{vol}}(\operatorname{ord}_{E_1}) = \frac{27}{4}$, which proves (3).

Finally, after exchanging E_1 and E_2 if necessary, any pair (a_1, a_2) with $a_1, a_2 \le 3$ satisfies $a_1 + a_2 \le 3$ or is one of the pairs in (2) and (3). Since $a_1 + a_2 + 4 \le 7$ in (1), and $\frac{19}{4}, \frac{11}{2}, \frac{23}{4}, \frac{27}{4} \le 7$, we obtain $\widehat{\operatorname{vol}}(X \ni x) \le 7$. □

The following lemma lifts a presentation of the completed local ring of E at a closed point to a hypersurface presentation of the completed local ring of Y whose equation is contained in the ideal of E, compatibly with the ideals of closed subschemes of E. The proof of Lemma 9.5 lifts the presentations directly, and the proof of Lemma 9.6 uses Lemma 9.7 at the point y of Lemma 9.6. Below we use it in the proofs of Lemmas 9.8 and 9.9 and of Proposition 9.10.

Lemma 9.7. *Let Y be a normal threefold, let E be a Cartier divisor on Y, and let $p \in E$ be a closed point. Let $R := \mathbb{C}[[x_1, x_2, x_3, x_4]]$, and let $\varphi : R \to \widehat{\mathcal{O}}_{E,p}$ be a surjective homomorphism with kernel I. Assume that Y is locally a hypersurface at p, and that either $I \subset (x_1, x_2, x_3, x_4)^2$ or $I = (g, x_4)$ for some $g \in \mathbb{C}[[x_1, x_2, x_3]]$. Then there exist a surjective homomorphism $\Phi : R \to \widehat{\mathcal{O}}_{Y,p}$ and an element $F \in I$ such that $\operatorname{Ker}\Phi = (F)$ and φ is the composite of Φ and the quotient map $\widehat{\mathcal{O}}_{Y,p} \to \widehat{\mathcal{O}}_{E,p}$. In particular, for any closed subscheme $V \subset E$, the preimage under Φ of the ideal of V in $\widehat{\mathcal{O}}_{Y,p}$ is equal to the preimage under φ of the ideal of V in $\widehat{\mathcal{O}}_{E,p}$. Moreover, if $I = (g, x_4)$, then we may choose Φ and F such that $F = g + x_4\psi$ for some $\psi \in R$ and $\Phi(x_4)$ generates the ideal of E in $\widehat{\mathcal{O}}_{Y,p}$.*

Proof. Let $\mathfrak{m}_Y$ and $\mathfrak{m}_E$ be the maximal ideals of $\widehat{\mathcal{O}}_{Y,p}$ and $\widehat{\mathcal{O}}_{E,p}$, and let $e \in \widehat{\mathcal{O}}_{Y,p}$ be a local equation of E. Since Y is normal, $\widehat{\mathcal{O}}_{Y,p}$ is a normal domain of dimension three [Mat86, §32, Remark 1 after Theorem 32.6].

First assume that $I \subset (x_1, x_2, x_3, x_4)^2$. Then $\dim_{\mathbb{C}} \mathfrak{m}_E/\mathfrak{m}_E^2 = 4$. Since Y is locally a hypersurface at p, $\dim_{\mathbb{C}} \mathfrak{m}_Y/\mathfrak{m}_Y^2 \le 4$, so the surjection $\mathfrak{m}_Y/\mathfrak{m}_Y^2 \to \mathfrak{m}_E/\mathfrak{m}_E^2$ is an isomorphism. We choose $y_i \in \widehat{\mathcal{O}}_{Y,p}$ whose image in $\widehat{\mathcal{O}}_{E,p}$ is $\varphi(x_i)$ for $1 \le i \le 4$. Next assume that $I = (g, x_4)$. We choose $y_i \in \widehat{\mathcal{O}}_{Y,p}$ whose image in $\widehat{\mathcal{O}}_{E,p}$ is $\varphi(x_i)$ for $1 \le i \le 3$, and let $y_4 := e$. In both cases, y_1, y_2, y_3, y_4 generate $\mathfrak{m}_Y$ by Nakayama's lemma, so $\Phi(x_i) := y_i$ defines a surjective homomorphism $\Phi : R \to \widehat{\mathcal{O}}_{Y,p}$, and φ is the composite of Φ and the quotient map. Since $\operatorname{Ker}\Phi$ is a prime ideal of height one in the factorial ring R, we have $\operatorname{Ker}\Phi = (F)$ for some $F \in R$, and $F \in \operatorname{Ker}\varphi = I$. The statement on V holds since the ideal of V in $\widehat{\mathcal{O}}_{Y,p}$ is the preimage of the ideal of V in $\widehat{\mathcal{O}}_{E,p}$.

Assume that $I = (g, x_4)$. Then

$$(F, x_4) = \Phi^{-1}(e\widehat{\mathcal{O}}_{Y,p}) = \operatorname{Ker}\varphi = (g, x_4).$$

Thus $(F(x_1, x_2, x_3, 0)) = (g)$ as ideals of the domain $\mathbb{C}[[x_1, x_2, x_3]]$. Since $\dim \widehat{\mathcal{O}}_{E,p} = 2$, we have $g \neq 0$, so $F(x_1, x_2, x_3, 0) = \epsilon g$ for a unit $\epsilon \in \mathbb{C}[[x_1, x_2, x_3]]$. Replacing F with $\epsilon^{-1}F$, we have $F = g + x_4\psi$ for some $\psi \in R$. □

When exactly one component of E is of type (e) in [Rei94, Theorem 1.1], [Liu25, Proposition 4.9] shows that the local volume is less than 9. In Lemma 9.8, we contract the component of type (e) by Lemma 9.4 and obtain a Kollár component with half of the double curve as its different. For each of the five possibilities for the other component, the volume of the Kollár component is at most 7, the bound of Proposition 9.10.

Lemma 9.8. *Let $X \ni x$ be a Gorenstein canonical threefold germ which is not a hypersurface germ, with $\operatorname{crex}(X \ni x) = 2$. Let $f : Y \to X$ be the blow-up of the maximal ideal $\mathfrak{m}_x$, and let $E := f^{-1}(x)_{\mathrm{red}} = E_1 + E_2$. Assume that Y has cDV singularities near E, that E is Cartier, and that $-E$ is f-ample. Assume moreover that E is a gluing of type (B) in* [Rei94, Section 1.3] *along a smooth curve C with exactly one component of type (e), namely $E_2 \cong \mathbb{F}_a$ for an integer $a \ge 0$, C is the negative section B_2 of E_2, and $\mathcal{O}_E(-E)|_{E_2} \cong \mathcal{O}_{\mathbb{F}_a}(B_2 + (a+2)A_2)$, where A_2 is a fiber (for $a = 0$, B_2 is the section with $(B_2^2) = 0$ used in the gluing). Assume that the pair (E_1, C) is one of the following: $\mathbb{P}^2$ with a smooth conic, $\mathbb{P}^2$ with a line, the quadric cone $\mathbb{P}(1,1,2) \subset \mathbb{P}^3$ with a smooth hyperplane section, $\mathbb{F}_0$ with a smooth curve in $|A_1 + B_1|$, and $\mathbb{F}_1$ with a smooth curve in $|A_1 + B_1|$, where on $E_1 \cong \mathbb{F}_b$ ($b \in \{0, 1\}$) A_1 is a fiber and B_1 is a section with $(B_1^2) = -b$. Then $\widehat{\mathrm{vol}}(X \ni x)$ is at most 4, $\frac{25}{4}$, $\frac{9}{2}$, $\frac{9}{2}$, and $\frac{21}{4}$, respectively. In particular, $\widehat{\mathrm{vol}}(X \ni x) \le \frac{25}{4}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we compute $\mathcal{O}_E(-E)|_{E_1}$ and the integers l_p of Lemma 9.2 along C.

By Theorem 2.15(2), $K_Y = f^*K_X$, and Y is normal and Gorenstein canonical. Thus $X \ni x$, f, and E satisfy the conditions of Set-up 8.2. Possibly shrinking X around x, we may assume that $K_X \sim 0$, so $\omega_Y \cong \mathcal{O}_Y$.

We show that $\operatorname{Sing} Y \cap E$ is finite. Suppose that Y is singular along a curve $\Gamma \subset E$. By Lemma 2.16, there exists a prime divisor F over Y with $\operatorname{center}_Y F = \Gamma$ and $a(F, Y) = 1$. Since $K_Y = f^*K_X$, we have $a(F, X) = 1$ [KM98, Lemma 2.30], and the center of F on X is $f(\Gamma) = \{x\}$. Hence F, E_1, and E_2 are three distinct crepant divisors over $X \ni x$, which contradicts $\operatorname{crex}(X \ni x) = 2$. Moreover, a general hyperplane section of Y through a point of E is a Du Val singularity, which has embedding dimension three [KM98, Definition 5.32 and Theorem 4.22]. Thus Y has embedding dimension at most four at any point of E, that is, Y is locally a hypersurface along E.

The curve $C \cong B_2$ is a smooth rational curve, and E_1 and E_2 are smooth along C, since a smooth hyperplane section of $\mathbb{P}(1, 1, 2) \subset \mathbb{P}^3$ does not contain the vertex. Let $p \in C$ be a closed point. By [Rei94, Section 1.3], in type (B), E is the pushout $E_1 \sqcup_C E_2$ of the inclusions $C \subset E_1$ and $C = B_2 \subset E_2$. Thus

$$\widehat{\mathcal{O}}_{E,p} \cong \widehat{\mathcal{O}}_{E_1,p} \times_{\widehat{\mathcal{O}}_{C,p}} \widehat{\mathcal{O}}_{E_2,p} \cong \mathbb{C}[[x_1, x_2, x_3]]/(x_1x_2),$$

where the ideals of E_1, E_2, and C correspond to (x_1), (x_2), and (x_1, x_2). By Lemma 9.7 for $I = (x_1x_2, x_4)$ and after renaming (x_1, x_2, x_3, x_4) as (u_1, u_2, u_4, u_3), there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u_1, u_2, u_3, u_4]]/(u_1u_2 + u_3\psi_p)$ which sends the ideals of E, E_1, E_2, and C to (u_3), (u_1, u_3), (u_2, u_3), and (u_1, u_2, u_3), respectively. Thus Lemma 9.2 for $E' := E_1$, $E'' := E_2$, and $\Sigma = \emptyset$ implies that the integers $l_p := \operatorname{ord}_{u_4} \psi_p(0, 0, 0, u_4) \geq 0$ satisfy $l_p > 0$ if and only if Y is singular at p, and, since $(C^2)_{E_2} = (B_2^2) = -a$,

$$\sum_{p \in C} l_p = 2 + (C^2)_{E_1} - a. \tag{9.7}$$

Let $N := -(K_{E_1} + C)$, which is a Cartier divisor on E_1. Then N is H, $2H$, H, $A_1 + B_1$, and $B_1 + 2A_1$ in the five cases, where H is the hyperplane class of $\mathbb{P}^2$ or of $\mathbb{P}(1, 1, 2) \subset \mathbb{P}^3$, so N is ample. We show that $\mathcal{O}_E(-E)|_{E_1} \cong \mathcal{O}_{E_1}(N)$. By adjunction [Kol13, (4.1.3)] and $\omega_Y \cong \mathcal{O}_Y$, we have $\omega_E \cong \mathcal{O}_E(E)$. Both E_1 and E_2 are Cartier on the smooth locus U of Y, and $E_2|_{E_1} = C$ by the local description above, so adjunction implies $\omega_E|_{E_1 \cap U} \cong \omega_{E_1}(C)|_{E_1 \cap U}$. Since $E_1 \backslash U$ is finite, E_1 is normal, and $\mathcal{O}_E(E)|_{E_1}$ and $\mathcal{O}_{E_1}(K_{E_1} + C)$ are invertible, we have $\mathcal{O}_E(E)|_{E_1} \cong \mathcal{O}_{E_1}(K_{E_1} + C)$.

Step 2. In this step, we construct a small modification $\rho : W \to Y$ and compute $M := -2S_1 - S_2$ on the strict transforms S_1 and S_2 of E_1 and E_2.

Let $\rho : W \to Y$ be the blow-up of the ideal sheaf of E_1, and let $\Sigma_Y := \operatorname{Sing} Y \cap C$, which is finite by **Step 1**. Since E_1 is Cartier on the smooth locus of Y and coincides with E near any point of $E_1 \backslash C$, E_1 is Cartier on $Y \backslash \Sigma_Y$, so ρ is an isomorphism over $Y \backslash \Sigma_Y$. For any $p \in \Sigma_Y$, we have $l_p \geq 1$, and the isomorphism of **Step 1** at p satisfies the conditions of Lemma 9.3 with $l = l_p$. Thus Lemma 9.3 implies the following.

- W is normal, $K_W = \rho^*K_Y$, both S_1 and S_2 are Cartier, $\rho^*E = S_1 + S_2$, W is smooth along S_2 over Σ_Y, and $S_2 \to E_2$ is an isomorphism.
- The induced morphism $\pi : S_1 \to E_1$ is an isomorphism over $E_1 \backslash \Sigma_Y$, and for any $p \in \Sigma_Y$, $T_p := \rho^{-1}(p) \subset S_1$ is the exceptional curve of π over p with $(T_p^2) = -\frac{1}{l_p}$.
- We have

$$K_{S_1} = \pi^*K_{E_1} + \sum_{p \in \Sigma_Y} l_pT_p \quad \text{and} \quad \pi^*C = C' + \sum_{p \in \Sigma_Y} l_pT_p,$$

  where C' is the strict transform of C, and C' intersects any T_p at one smooth point of S_1.

- For any $p \in \Sigma_Y$, S_1 has at most one singular point on T_p, which is a Du Val singularity of type A_{l_p-1}.

Since $l_p = 0$ for $p \in C \backslash \Sigma_Y$, we may write the sums above over $p \in C$.

Since $S_2 \to E_2$ is an isomorphism, S_2 contains no T_p. Thus $S_1 \cap S_2 = C'$ and $S_2|_{S_1} = C'$. Under the isomorphism $S_2 \cong E_2$, we have $S_1|_{S_2} = B_2$ and $\rho^*E|_{S_2} = E|_{E_2} \sim -B_2 - (a+2)A_2$. Hence

$$M|_{S_2} = -2S_1|_{S_2} - (\rho^*E - S_1)|_{S_2} \sim -2B_2 + B_2 + (a+2)A_2 + B_2 = (a+2)A_2.$$

Since $\rho^*E|_{S_1} = \pi^*(E|_{E_1}) \sim -\pi^*N$ by **Step 1**, we have

$$M|_{S_1} = -2(\rho^*E - S_2)|_{S_1} - S_2|_{S_1} \sim 2\pi^*N + C'.$$

Step 3. In this step, we contract S_2 by Lemma 9.4 and conclude.

Recall that C' is the strict transform of C on S_1 and $N = -(K_{E_1} + C)$. We have

$$((\pi^*C - l_pT_p) \cdot T_p) = 1,$$

so $(C' \cdot T_p) = 1$. By (9.7),

$$(C'^2) = (C^2)_{E_1} - \sum_{p \in C} l_p = a - 2.$$

Since C is a smooth rational curve in the smooth locus of E_1, $((K_{E_1} + C) \cdot C) = -2$, so $(N \cdot C) = 2$. Thus, for any $p \in \Sigma_Y$ and any irreducible curve $\Gamma' \subset S_1$ which is neither C' nor some T_p, with image $\Gamma := \pi(\Gamma')$, we have

$$(M \cdot T_p) = 1, \quad (M \cdot C') = 2(N \cdot C) + (C'^2) = a + 2, \quad (M \cdot \Gamma') = 2(N \cdot \Gamma) + (C' \cdot \Gamma') \geq 2(N \cdot \Gamma) > 0,$$

and

$$(M|_{S_1}^2) = 2(M \cdot \pi^*N) + (M \cdot C') = 4(N^2) + 2(N \cdot C) + a + 2 > 0.$$

Since M is Cartier, $M|_{S_1}$ is ample by the Nakai–Moishezon criterion [KM98, Theorem 1.37].

The singular points of S_1 are the Du Val points on the curves T_p and, if $E_1 \cong \mathbb{P}(1,1,2)$, the vertex. None of them is on C'. Since $(C' \cdot T_p) = 1$ at a smooth point of S_1 and $C' \backslash \bigcup_p T_p \cong C \backslash \Sigma_Y$, C' is smooth. Thus $(S_1, \frac{1}{2}C')$ is klt. Moreover, W is smooth near $S_2 \backslash S_1$: over Σ_Y, W is smooth along S_2 by **Step 2**, Y is smooth along $C \backslash \Sigma_Y$, and near any point of $E_2 \backslash C$, $E = E_2$ is a smooth Cartier divisor, so Y is smooth there.

Hence $f \circ \rho : W \to X$, S_1, and $S_2 \cong \mathbb{F}_a$ satisfy the conditions of Lemma 9.4, where $K_W = (f \circ \rho)^*K_X$, $f \circ \rho$ is an isomorphism over $X \backslash \{x\}$, and $(f \circ \rho)^{-1}(x)_{\mathrm{red}} = S_1 \cup S_2$. By Lemma 9.4, there exists a plt blow-up $h : Z \to X$ with $K_Z = h^*K_X$ whose Kollár component P satisfies $\widehat{\mathrm{vol}}(\mathrm{ord}_P) = ((K_{S_1} + \frac{1}{2}C')^2)$. By **Step 2**,

$$K_{S_1} + \frac{1}{2}C' = \pi^*\Big(K_{E_1} + \frac{1}{2}C\Big) + \frac{1}{2}\sum_{p \in C} l_pT_p.$$

Since $(T_p^2) = -\frac{1}{l_p}$, $(T_p \cdot T_{p'}) = 0$ for $p \neq p'$, and $(\pi^*D \cdot T_p) = 0$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on E_1, we have

$$\widehat{\mathrm{vol}}(X \ni x) \leq \widehat{\mathrm{vol}}(\mathrm{ord}_P) = \Big(\Big(K_{E_1} + \frac{1}{2}C\Big)^2\Big) - \frac{1}{4}\sum_{p \in C} l_p \leq \Big(\Big(K_{E_1} + \frac{1}{2}C\Big)^2\Big).$$

It remains to compute $((K_{E_1} + \frac{1}{2}C)^2)$. If $E_1 \cong \mathbb{P}^2$, then $K_{E_1} \sim -3H$ and $C \sim 2H$ or $C \sim H$, so $((K_{E_1} + \frac{1}{2}C)^2)$ is 4 or $\frac{25}{4}$. If $E_1 \cong \mathbb{P}(1,1,2)$, then $(H^2) = 2$, $K_{E_1} \sim -2H$, and $C \sim H$, so $((K_{E_1} + \frac{1}{2}C)^2) = \frac{9}{4}(H^2) = \frac{9}{2}$. If $E_1 \cong \mathbb{F}_b$ for $b \in \{0,1\}$, then $K_{E_1} \sim -2B_1 - (b+2)A_1$ and $C \sim A_1 + B_1$, so

$$\Big(\Big(K_{E_1} + \frac{1}{2}C\Big)^2\Big) = \Big(\Big(\frac{3}{2}B_1 + \Big(b + \frac{3}{2}\Big)A_1\Big)^2\Big) = -\frac{9}{4}b + 3\Big(b + \frac{3}{2}\Big) = \frac{3}{4}b + \frac{9}{2},$$

which is $\frac{9}{2}$ for $b = 0$ and $\frac{21}{4}$ for $b = 1$. Table 2 lists the five values. □

(E_1, C)	$-(K_{E_1}+C)$	$((K_{E_1}+\frac{1}{2}C)^2)$
$\mathbb{P}^2$, smooth conic	H	4
$\mathbb{P}^2$, line	$2H$	$\frac{25}{4}$
$\mathbb{P}(1,1,2)$, smooth hyperplane section	H	$\frac{9}{2}$
$\mathbb{F}_0$, smooth curve in $\lvert A_1+B_1\rvert$	A_1+B_1	$\frac{9}{2}$
$\mathbb{F}_1$, smooth curve in $\lvert A_1+B_1\rvert$	B_1+2A_1	$\frac{21}{4}$

TABLE 2. The five pairs (E_1, C) of Lemma 9.8 and the upper bounds for $\widehat{\mathrm{vol}}(X \ni x)$, where H is the hyperplane class of $\mathbb{P}^2$ or of $\mathbb{P}(1,1,2) \subset \mathbb{P}^3$.

Next, we treat the gluing of a cone over a rational normal curve with a Hirzebruch surface along two lines.

Lemma 9.9. *Notation and conditions as in Set-up 8.2. Assume that Y is locally a hypersurface along E, that $\operatorname{Sing} Y \cap E$ is finite, and that $E = E_1 + E_2$ is a Gorenstein slc del Pezzo surface, where E_1 is the projective cone over a rational normal curve of degree $a_1 \in \{2,3\}$ and $E_2 \cong \mathbb{F}_{a_2}$ for an integer $a_2 \geq 0$. Assume moreover that two distinct lines $\ell, \ell' \subset E_1$ through the vertex are glued to the negative section B and to a fiber A of E_2, respectively, with the intersection points identified, and that $\ell + \ell'$ and $A + B$ are the conductor divisors on E_1 and on E_2, respectively. Let C_1 and C_2 be the images of ℓ and ℓ' in E, and let $y \in E$ be their common point. Assume that E has double crossings along $C_1 \cup C_2$ away from y, and that there exists an isomorphism $\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[x_1,x_2,x_3,x_4]]/(f_0)$ for some f_0 which sends the ideals of C_1, C_2, E_2, and E_1 to (x_2,x_3,x_4), (x_1,x_2,x_4), (x_2,x_4), and*

$$\begin{cases} (x_2^2 - x_1x_3, x_4) & \text{if } a_1 = 2, \\ (x_2^2 - x_1x_4, x_4^2 - x_2x_3, x_1x_3 - x_2x_4) & \text{if } a_1 = 3, \end{cases}$$

respectively. Then $a_1 + a_2 \leq 5$ and

$$\widehat{\mathrm{vol}}(X \ni x) \leq (K_E^2) = a_1 + a_2 + 2 \leq 7.$$

Proof. Possibly shrinking X around x, we may assume that $K_X \sim 0$, so $\omega_Y \cong \mathcal{O}_Y$. Let $\sigma : \tilde{E}_1 \to E_1$ be the minimal resolution, where $\tilde{E}_1 \cong \mathbb{F}_{a_1}$, and let $\tilde{\ell}$ be the strict transform of ℓ, which is a fiber of the ruling of $\tilde{E}_1$, so $(\tilde{\ell}^2) = 0$. Lemma 9.2 does not apply here, since E_1 is singular at the vertex y. We repeat its degree computation with the minimal resolution $\tilde{E}_1$ in place of E_1. We identify $\tilde{\ell}$, ℓ, C_1, and B via σ and the gluing. Let $\Theta := \mathcal{H}om(\Omega_Y|_{C_1}, \mathcal{O}_{C_1})$. The differentials of σ and of the inclusion $E_2 \subset Y$ induce homomorphisms

$$\alpha : T_{\tilde{E}_1}|_{\tilde{\ell}} \to \Theta \quad \text{and} \quad \beta : T_{E_2}|_B \to \Theta,$$

and we let $T := \alpha(T_{\tilde{E}_1}|_{\tilde{\ell}}) + \beta(T_{E_2}|_B) \subset \Theta$.

We describe T near y. Fix an isomorphism $\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[x_1,x_2,x_3,x_4]]/(f_0)$ as in the statement, and let $\partial_i := \partial/\partial x_i$. Then $\Theta \otimes \widehat{\mathcal{O}}_{C_1,y}$ is the module of vectors $\sum_i c_i\partial_i$ with $c_i \in \mathbb{C}[[x_1]]$ and $\sum_i c_i \frac{\partial f_0}{\partial x_i}|_{C_1} = 0$. Near the point of $\tilde{\ell}$ over y, there exist local coordinates (u,v) of $\tilde{E}_1$ with $\tilde{\ell} = (v = 0)$ and $u|_{\tilde{\ell}} = x_1$, in which

$$\sigma(u,v) = \begin{cases} (u, uv, uv^2, 0) & \text{if } a_1 = 2, \\ (u, uv, uv^3, uv^2) & \text{if } a_1 = 3. \end{cases}$$

Thus the image of α is generated by ∂_1 and $x_1\partial_2$ near y, the image of β is generated by ∂_1 and ∂_3 since $E_2 = (x_2 = x_4 = 0)$, and T is generated by ∂_1, $x_1\partial_2$, and ∂_3. In particular, $\frac{\partial f_0}{\partial x_i}|_{C_1} = 0$

for $i = 1, 2, 3$. Since Y is smooth at the generic point of C_1, the restriction of $f_{0,x_4} := \frac{\partial f_0}{\partial x_4}$ to C_1 is non-zero. Let $l := \operatorname{ord}_y(f_{0,x_4}|_{C_1}) \geq 0$.

We describe T near a closed point $p \in C_1 \backslash \{y\}$. Since E has double crossings at p, $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[x_1, x_2, x_3]]/(x_1x_2)$, where the ideals of E_1, E_2, and C_1 correspond to (x_1), (x_2), and (x_1, x_2). By Lemma 9.7 for $I = (x_1x_2, x_4)$ and after renaming (x_1, x_2, x_3, x_4) as (u_1, u_2, u_4, u_3), there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u_1, u_2, u_3, u_4]]/(u_1u_2 + u_3\psi_p)$ which sends the ideals of E, E_1, E_2, and C_1 to (u_3), (u_1, u_3), (u_2, u_3), and (u_1, u_2, u_3), respectively. On C_1, all partial derivatives of $u_1u_2 + u_3\psi_p$ vanish except $\psi_p(0, 0, 0, u_4)$. Since Y is smooth at the generic point of C_1, $l_p := \operatorname{ord}_{u_4} \psi_p(0, 0, 0, u_4)$ is a non-negative integer, and $l_p = 0$ if and only if Y is smooth at p. Near p, the image of α is generated by ∂_{u_2} and ∂_{u_4}, the image of β is generated by ∂_{u_1} and ∂_{u_4}, and T is generated by ∂_{u_1}, ∂_{u_2}, and ∂_{u_4}.

By the local descriptions near y and near the points of $C_1\backslash\{y\}$, α and β are injective, T is a locally free sheaf of rank three, and the intersection of the images of α and β is the image of T_{C_1}. Thus the sequence

$$0 \to T_{C_1} \to T_{\tilde{E}_1}|_{\tilde{\ell}} \oplus T_{E_2}|_B \to T \to 0$$

is exact. Since $\deg T_S|_\Gamma = 2 + (\Gamma^2)_S$ for a smooth rational curve Γ on a smooth surface S [Har77, Proposition V.1.5], we have

$$\deg T = (2 + (\tilde{\ell}^2)) + (2 + (B^2)) - 2 = 2 - a_2.$$

We compare $\det T$ with $\omega_Y|_{C_1}$. Let $C_1^\circ := C_1 \setminus \operatorname{Sing} Y$. Over C_1°, Θ is the restriction of the tangent bundle of Y, so the inclusion $T \subset \Theta$ induces a homomorphism $\det T \to \omega_Y^{-1}|_{C_1^\circ}$, that is, a section s of $(\det T)^{-1} \otimes \omega_Y^{-1}|_{C_1}$ over C_1°. Near y, by adjunction [Kol13, (4.1.3)], ω_Y is generated by the residue of $dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4/f_0$, which is equal to $dx_1 \wedge dx_2 \wedge dx_3/f_{0,x_4}$ up to sign where $f_{0,x_4} \neq 0$. Since $\det T$ is generated by $x_1\partial_1 \wedge \partial_2 \wedge \partial_3$ near y, the order of s at y is

$$\operatorname{ord}_y \left(\frac{x_1}{f_{0,x_4}|_{C_1}} \right) = 1 - l.$$

Similarly, near $p \in C_1\backslash\{y\}$, ω_Y is generated by $du_1 \wedge du_2 \wedge du_4/(\psi_p + u_3\partial_{u_3}\psi_p)$ up to sign, and $\det T$ is generated by $\partial_{u_1} \wedge \partial_{u_2} \wedge \partial_{u_4}$, so the order of s at p is $-l_p$. Thus s extends to a rational section of $(\det T)^{-1} \otimes \omega_Y^{-1}|_{C_1}$ whose divisor is $(1-l)y - \sum_{p \neq y} l_p p$. Since $\omega_Y \cong \mathcal{O}_Y$, taking degrees, we have $1 - l - \sum_{p \neq y} l_p = -\deg T = a_2 - 2$, that is,

$$a_2 = 3 - l - \sum_{p \in C_1 \backslash \{y\}} l_p. \tag{9.8}$$

If $a_1 = 2$, then (9.8) implies $a_2 \leq 3$. Assume that $a_1 = 3$. The ideal of E_1 is generated by three quadratic polynomials, so E_1 has embedding dimension four at y. Since E_1 is a closed subscheme of Y and a smooth threefold has embedding dimension three, Y is singular at y. Thus all first partial derivatives of f_0 vanish at y, so $l \geq 1$, and (9.8) implies $a_2 \leq 2$. In both cases, $a_1 + a_2 \leq 5$.

Finally, we compute (K_E^2). By adjunction [Kol13, (4.1.3)] and $\omega_Y \cong \mathcal{O}_Y$, $\omega_E \cong \mathcal{O}_E(E)$. Let $D_1 := \ell + \ell'$ and $D_2 := A + B$. Both E_1 and E_2 are Cartier on Y away from the finite set $\operatorname{Sing} Y \cup \{y\}$, and $E_j|_{E_i} = D_i$ for $\{i, j\} = \{1, 2\}$ by the double crossings, so adjunction implies $\omega_E|_{E_i} \cong \omega_{E_i}(D_i)$ there. Since E_i is normal and both sheaves are reflexive of rank one on E_i, we have $\mathcal{O}_E(-E)|_{E_i} \cong \mathcal{O}_{E_i}(-(K_{E_i} + D_i))$. Since $E_1 \cong \mathbb{P}(1, 1, a_1)$, $\ell, \ell' \in |\mathcal{O}(1)|$, $K_{E_1} \sim -(a_1 + 2)\ell$, and the hyperplane class H_1 of E_1 satisfies $H_1 \sim a_1\ell$, we have $-(K_{E_1} + D_1) \sim H_1$ and $(H_1^2) = a_1$. On $E_2 \cong \mathbb{F}_{a_2}$, $-(K_{E_2} + D_2) \sim B + (a_2 + 1)A$, whose self-intersection number is $-a_2 + 2(a_2 + 1) = a_2 + 2$. By the projection formula for the normalization of E [Laz04, Remark 1.1.13],

$$(K_E^2) = ((K_{E_1} + D_1)^2) + ((K_{E_2} + D_2)^2) = a_1 + a_2 + 2 \leq 7.$$

By Lemma 2.19(2), $\widehat{\operatorname{vol}}(X \ni x) \leq (K_E^2)$. □

We now bound the local volume in the branch where the blow-up of the maximal ideal has two exceptional prime divisors.

Proposition 9.10. *Let $X \ni x$ be a Gorenstein canonical threefold germ which is not a hypersurface germ, with $\operatorname{crex}(X \ni x) = 2$. Assume that the blow-up of the maximal ideal $\mathfrak{m}_x$ has two distinct exceptional prime divisors. Then $\widehat{\operatorname{vol}}(X \ni x) \le 7$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that E is a reduced Cartier divisor with $\mathcal{O}_E(-E) \cong \omega_E^{-1}$ ample, and we reduce to three gluings of [Rei94, Section 1.3].

Let $f : Y \to X$ be the blow-up of $\mathfrak{m}_x$, let E_1 and E_2 be the exceptional prime divisors of f, and let $E := f^{-1}(x)_{\mathrm{red}} = E_1 + E_2$. By Theorem 2.15(2), $K_Y = f^*K_X$, and Y is normal and Gorenstein canonical. Thus $a(E_1, X) = a(E_2, X) = 1$, and E_1 and E_2 are the only crepant divisors over $X \ni x$. Possibly shrinking X around x, we may assume that $K_X \sim 0$, so $\omega_Y \cong \mathcal{O}_Y$.

Let $y \in E$ be a closed point. If $Y \ni y$ is not a cDV hypersurface germ, then $\operatorname{crex}(Y \ni y) > 0$ by Theorem 2.15(1). Since $K_Y = f^*K_X$, a crepant divisor over $Y \ni y$ is a crepant divisor over $X \ni x$ [KM98, Lemma 2.30], and it is neither E_1 nor E_2, which contradicts $\operatorname{crex}(X \ni x) = 2$. Thus Y is cDV at any point of E. Since Y has only finitely many non-cDV points [KM98, Corollary 5.40], possibly shrinking X around x, we may assume that Y has cDV singularities. Since a Du Val singularity has embedding dimension three [KM98, Definition 5.32 and Theorem 4.22] and a hyperplane section through a point lowers the embedding dimension by at most one, Y is locally a hypersurface along E. We show that $\operatorname{Sing} Y \cap E$ is finite. If Y is singular along a curve $\Gamma \subset E$, then by Lemma 2.16, there exists a prime divisor F over Y with $\operatorname{center}_Y F = \Gamma$ and $a(F, Y) = 1$. Then $a(F, X) = 1$ [KM98, Lemma 2.30] and the center of F on X is x, so F is a third crepant divisor over $X \ni x$, a contradiction.

Let $h \in \mathfrak{m}_x$ be a general element, and let $H := (h = 0) \ni x$. By Theorem 2.15(2), H is a Gorenstein elliptic surface singularity. Write $\mathfrak{m}_x\mathcal{O}_Y = \mathcal{O}_Y(-F)$, where F is a Cartier divisor with $\operatorname{Supp} F = E$. The strict transform H_Y of H on Y is $\operatorname{Bl}_x H$ [Har77, Corollary II.7.15], and H_Y is a general member of the base-point-free linear system on Y defined by $\mathfrak{m}_x\mathcal{O}_Y$. Thus H_Y avoids the finite set $\operatorname{Sing} Y \cap E$, and H_Y is smooth near E by Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2]. Moreover, $f^{-1}(x)$ is a closed subscheme of $\mathbb{P}(\mathfrak{m}_x/\mathfrak{m}_x^2)$, and $H_Y \cap E_i$ is a general hyperplane section of the irreducible surface $E_i \subset \mathbb{P}(\mathfrak{m}_x/\mathfrak{m}_x^2)$ for $i = 1, 2$. Thus $\Gamma_i := H_Y \cap E_i$ is an irreducible curve by Bertini's theorem [Laz04, Theorem 3.3.1], and $H_Y \to H$ is a resolution with exactly two exceptional curves Γ_1 and Γ_2. Since $X \ni x$ is not a hypersurface germ, X has embedding dimension at least five at x, so H has embedding dimension at least four at x, and $H \ni x$ is not a hypersurface germ. By [Liu25, Theorem 2.22], which follows from [Lau77, Theorems 3.4, 3.13, and 3.15], $H_Y \to H$ is the canonical modification. Since H_Y is smooth near $\Gamma_1 \cup \Gamma_2$, $H_Y \to H$ is the minimal resolution. By [Liu25, Proposition 2.23], the fundamental cycle, that is, the smallest cycle $Z > 0$ supported on the exceptional curves with $(Z \cdot \Gamma) \le 0$ for any exceptional curve Γ [Art66, pp. 131–132], is $Z = \Gamma_1 + \Gamma_2$, and [Lau77, Theorem 3.13] shows that $(Z^2) \le -4$ and $\mathfrak{m}_H\mathcal{O}_{H_Y} = \mathcal{O}_{H_Y}(-Z)$. Since $\mathfrak{m}_H\mathcal{O}_{H_Y} = \mathcal{O}_{H_Y}(-F|_{H_Y})$ and $F|_{H_Y} = (\operatorname{mult}_{E_1} F)\Gamma_1 + (\operatorname{mult}_{E_2} F)\Gamma_2$, we have $\operatorname{mult}_{E_1} F = \operatorname{mult}_{E_2} F = 1$, that is, $F = E$. Thus E is Cartier, $f^{-1}(x) = E$ as schemes, and $\mathcal{O}_Y(-E) = \mathfrak{m}_x\mathcal{O}_Y$ is f-ample. By adjunction [Kol13, (4.1.3)] and $\omega_Y \cong \mathcal{O}_Y$, we have $\omega_E \cong \mathcal{O}_E(E)$. Since E is connected [Har77, Corollary III.11.4], projective, and Gorenstein with ω_E^{-1} ample, E is a Gorenstein del Pezzo surface [Rei94, §0.1]. Since E is connected and has two irreducible components, E is not normal.

Let $d := (K_E^2)$. Since $X \ni x$, f, and E satisfy the conditions of Set-up 8.2, Lemma 2.19(2) implies

$$\widehat{\operatorname{vol}}(X \ni x) \le d. \tag{9.9}$$

Thus we may assume that $d \ge 8$. By [Rei94, Theorem 1.5], whose tameness assumption holds in characteristic zero, E is one of the surfaces of [Rei94, Sections 1.3–1.4]. The surfaces of [Rei94,

Section 1.4] have degree at most two, and the surfaces of [Rei94, Section 1.3] with exactly two irreducible components are the gluings of types (B), (C_2), and (D_2). Their components E_i are the normal surfaces of [Rei94, Theorem 1.1], polarized by $\mathcal{O}_E(-E)|_{E_i} = \omega_E^{-1}|_{E_i}$, with conductor divisors D_i. Writing $\mathbb{F}_{a;k}$ for $\mathbb{F}_a$ embedded by $\mathcal{O}((a+k)A+B)$, a scroll of degree $a+2k$ [Rei94, Theorem 1.1], the pairs (E_i, D_i) are as follows.

- (a): $\mathbb{P}^2$ with $\mathcal{O}(1)$, of degree 1, where D_i is a smooth conic (a1), two distinct lines (a2), or a double line (a3).
- (b): $\mathbb{P}^2$ with $\mathcal{O}(2)$, of degree 4, where D_i is a line.
- (c): the cone $\mathbb{F}_{a;0}$ over a rational normal curve of degree $a \geq 2$, of degree a, where D_i is a smooth hyperplane section and $a = 2$ (c0), two distinct lines through the vertex (c1), or a double line through the vertex (c2).
- (d): $\mathbb{F}_{a;1}$, of degree $a+2$, where D_i is a smooth member of $|A+B|$ and $a \leq 1$ (d0), or $D_i = A + B$ (d1).
- (e): $\mathbb{F}_{a;2}$, of degree $a+4$, where $D_i = B$.

By the projection formula for the normalization of E [Laz04, Remark 1.1.13], d is the sum of the degrees of E_1 and E_2. If E_i is of type (c), then E_i has embedding dimension $a+1$ at the vertex. Since E_i is a closed subscheme of Y, which has embedding dimension at most four, we have $a \leq 3$.

Step 2. In this step, we assume that E is of type (B) or (C_2), show that E is slc and that (Y, E) is lc near E, and describe $\widehat{\mathcal{O}}_{E,p}$ for closed points $p \in E_1 \cap E_2$.

Since Y is canonical, Y is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. Thus the Cartier divisor E is S_2, and E is Gorenstein. Let η be the generic point of an irreducible component of $E_1 \cap E_2$. By **Step 1**, Y is smooth at η. Thus E is a curve germ in the regular surface $\operatorname{Spec} \mathcal{O}_{Y,\eta}$ with two smooth branches. Since the conductor divisors D_1 and D_2 are reduced in types (B) and (C_2), the branches are transverse, and E has a node at η. Hence E is demi-normal [Kol13, Definition 5.1]. Any pair (E_i, D_i) is lc: it is a smooth surface with a smooth curve or with two transverse smooth curves in types (a1), (a2), (b), (d0), (d1), and (e), $\mathbb{P}(1,1,2)$ with a smooth hyperplane section avoiding the vertex in type (c0), and a cone over a rational normal curve with two lines through the vertex, which is a toric surface with toric boundary, in type (c1). Since K_E is Cartier, E is slc [Kol13, Definition–Lemma 5.10].

Since $\omega_Y \cong \mathcal{O}_Y$ and E is Cartier, the different of 0 on the normalization $E_1 \sqcup E_2$ of E is a $\mathbb{Z}$-divisor [Kol13, Definition 4.2]. Let V be a prime divisor on E_i. If $V \not\subset D_i$, then Y and E are regular at the generic point of the image of V, so the coefficient of V in the different is 0 [Kol13, Proposition 4.5(1)]. If $V \subset D_i$, then E has a node at the generic point of the image of V, so E is not regular there and (Y, E) is lc there, and the coefficient of V in the different is 1 [Kol13, Proposition 4.5(1) and (2)]. Thus the different of 0 on $E_1 \sqcup E_2$ is the conductor $D_1 + D_2$. Thus (Y, E) is lc near E by [Kaw07, Theorem].

Let $p \in E_1 \cap E_2$ be a closed point. By [Kol13, Proposition 5.3] applied to $\operatorname{Spec} \widehat{\mathcal{O}}_{E,p}$ ($\operatorname{Spec} \widehat{\mathcal{O}}_{E,p}$ is again demi-normal and its normalization and conductor are the completions of those of E, since E is excellent [Gro65, Scholie 7.8.3(v)(vii)]), the ring $\widehat{\mathcal{O}}_{E,p}$ is determined by the germs of the normalization of E over p, the conductor divisors on them, and the gluing involution. We use [Kol13, Proposition 5.3] in the following three situations.

(1) Assume that p is a smooth point of D_1 and of D_2, which holds for any p in type (B) and for any $p \neq y$ in type (C_2), where y is the image of the singular points of D_1 and of D_2. Then the normalization of E over p consists of two smooth surface germs, each with one smooth conductor curve, glued by an isomorphism. The normalization of the demi-normal hypersurface $(x_1 x_2 = 0) \subset \operatorname{Spec} \mathbb{C}[[x_1, x_2, x_3]]$ also consists of two smooth surface germs $(x_1 = 0)$ and $(x_2 = 0)$, each with the smooth conductor curve $(x_1 = x_2 = 0)$, glued by the identity. Thus $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[x_1, x_2, x_3]]/(x_1 x_2)$, where the ideals of E_1 and E_2 correspond to (x_1) and (x_2). In particular, E has double crossings along $E_1 \cap E_2$ away from y.

(2) Assume that E is of type (C_2) and that E_1 and E_2 are smooth at y. Then the normalization of E over y consists of two smooth surface germs, each with two transverse smooth conductor curves, and the gluing sends each conductor curve on one germ to a conductor curve on the other one. The hypersurface $(s_3(s_1s_2 - s_3) = 0) \subset \operatorname{Spec} \mathbb{C}[[s_1, s_2, s_3]]$ is demi-normal, since it has a node at the generic points of the s_1-axis and the s_2-axis. The normalization of $(s_3(s_1s_2 - s_3) = 0)$ consists of the germs $(s_3 = 0)$ and $(s_3 = s_1s_2)$ with coordinates (s_1, s_2), each with conductor $(s_1s_2 = 0)$, glued by the identity on (s_1, s_2). After a change of coordinates on the germ of E_2, the gluing of E is also the identity. Thus $\widehat{\mathcal{O}}_{E,y} \cong \mathbb{C}[[s_1, s_2, s_3]]/(s_3(s_1s_2 - s_3))$, where the ideals of E_1 and E_2 correspond to (s_3) and $(s_3 - s_1s_2)$, and we may choose which conductor curve corresponds to (s_2, s_3).

(3) Assume that E is of type (C_2), E_1 is of type (c1) with $a_1 := a \in \{2, 3\}$, and $E_2 \cong \mathbb{F}_{a_2}$ is of type (d1), with the lines ℓ, ℓ' of D_1 glued to B and A, respectively. Let C_1 and C_2 be the images of ℓ and ℓ' in E. Let $V_1, V_2 \subset \operatorname{Spec} \mathbb{C}[[x_1, x_2, x_3, x_4]]$ be the subschemes defined by the equations of E_1 and of E_2 in Lemma 9.9, and let $V := V_1 \cup V_2$. Then $V_1 \cap V_2$ is the reduced pair of lines $(x_2 = x_4 = x_1x_3 = 0)$, and V_1 and V_2 are Cohen–Macaulay, so V is S_2 by the exact sequence $0 \to \mathcal{O}_V \to \mathcal{O}_{V_1} \oplus \mathcal{O}_{V_2} \to \mathcal{O}_{V_1 \cap V_2} \to 0$. The surfaces V_1 and V_2 are smooth and transverse at the generic points of $V_1 \cap V_2$, so V is demi-normal. The normalization of V is $V_1 \sqcup V_2$, a cone germ with the lines $V_1 \cap V_2$ and a smooth germ with the transverse lines $V_1 \cap V_2$, glued by the identity. Since any pair of distinct lines through the vertex of E_1 is mapped to any other such pair by an automorphism of E_1, after a change of coordinates on the germ of E_2, the normalization data of E at y and of V at the origin are isomorphic. Thus $\widehat{\mathcal{O}}_{E,y} \cong \widehat{\mathcal{O}}_{V,0}$, where the ideals of E_1, E_2, C_1, and C_2 correspond to those of V_1, V_2, $(x_2 = x_3 = x_4 = 0)$, and $(x_1 = x_2 = x_4 = 0)$.

Step 3. In this step, we treat type (B).

In type (B), E_1 and E_2 are of types (a1), (b), (c0), (d0), and (e), not both of type (a1), and E is obtained by identifying the smooth rational curves D_1 and D_2 [Rei94, Section 1.3]. Let $C \subset E$ be the image of D_1 and D_2.

Assume that neither E_1 nor E_2 is of type (e). Since $d \geq 8$ and the degrees of types (a1), (b), (c0), and (d0) are 1, 4, 2, and at most 3, both E_1 and E_2 are of type (b). Thus E_1 and E_2 are planes, and E is obtained by identifying lines $D_1 \subset E_1$ and $D_2 \subset E_2$. Let $\nu : E_1 \sqcup E_2 \to E$ be the normalization. The sequence

$$1 \to \mathcal{O}_E^* \to \nu_* \mathcal{O}_{E_1 \sqcup E_2}^* \to \mathcal{O}_C^* \to 1,$$

where the last homomorphism sends (g_1, g_2) to $g_1|_C \cdot g_2|_C^{-1}$, is exact, since $\mathcal{O}_E = \mathcal{O}_{E_1} \times_{\mathcal{O}_C} \mathcal{O}_{E_2}$ by **Step 2**(1). Since the homomorphism

$$H^0(E_1 \sqcup E_2, \mathcal{O}^*) = \mathbb{C}^* \times \mathbb{C}^* \to H^0(C, \mathcal{O}_C^*) = \mathbb{C}^*$$

is surjective, the homomorphism $\operatorname{Pic} E \to \operatorname{Pic} E_1 \times \operatorname{Pic} E_2$ is injective. Let L be the line bundle on E obtained by gluing $\mathcal{O}_{E_1}(1)$ and $\mathcal{O}_{E_2}(1)$ along C. Since $\mathcal{O}_E(-E)|_{E_i} \cong \mathcal{O}_{E_i}(2)$ in type (b), we have $L^{\otimes 2} \cong \mathcal{O}_E(-E)$. The space $H^0(E, L)$ consists of the pairs $(t_1, t_2) \in H^0(E_1, \mathcal{O}(1)) \times H^0(E_2, \mathcal{O}(1))$ with $t_1|_C = t_2|_C$, so $\dim H^0(E, L) = 3 + 3 - 2 = 4$. The morphism $E \to \mathbb{P}^3$ defined by $H^0(E, L)$ is a linear embedding on E_1 and on E_2, and the images of E_1 and E_2 are the distinct planes defined by the sections $(0, t_2)$ and $(t_1, 0)$ with $t_2|_C = 0$ and $t_1|_C = 0$. Thus the image Q of E is the union of two distinct planes, which has double crossings and is seminormal, and $E \to Q$ is a finite birational bijective morphism. Hence $E \to Q$ is an isomorphism, E is a quadric surface of rank two, $\mathcal{O}_E(1) \cong L$, and $\mathcal{O}_E(-E) \cong \mathcal{O}_E(2)$. Since (Y, E) is lc by **Step 2**, Proposition 8.5(3) implies $\widehat{\operatorname{vol}}(X \ni x) \leq 4$.

Assume that exactly one of E_1 and E_2, say E_2, is of type (e). Then $E_2 \cong \mathbb{F}_a$, C is the negative section B_2 of E_2, and $\mathcal{O}_E(-E)|_{E_2} \cong \mathcal{O}_{\mathbb{F}_a}(B_2 + (a+2)A_2)$, where A_2 is a fiber. The pair (E_1, C) is of type (a1), (b), (c0), or (d0), that is, one of the five pairs of Lemma 9.8. By **Step 1**, Y has cDV singularities near E, E is Cartier, and $-E$ is f-ample. Thus Lemma 9.8 implies $\widehat{\operatorname{vol}}(X \ni x) \leq \frac{25}{4}$.

Assume that both E_1 and E_2 are of type (e). Then $E_i \cong \mathbb{F}_{a_i}$, and C is the negative section B_i of E_i for $i = 1, 2$. By **Step 1**, Y is locally a hypersurface along E and smooth at the generic point of C. By **Step 2**(1), which holds at any $p \in C$ in type (B), E has double crossings along C. Thus E has normal crossings with smooth components E_1 and E_2, and Lemma 9.5 implies $\widehat{\mathrm{vol}}(X \ni x) \leq \frac{27}{4}$.

Step 4. In this step, we treat type (D_2).

In type (D_2), E_1 and E_2 are of types (a3) and (c2), not both of type (a3). By **Step 1**, a component of type (c2) has degree at most 3, and a component of type (a3) has degree 1. Thus $d \leq 6$, which contradicts $d \geq 8$.

Step 5. In this step, we treat type (C_2).

In type (C_2), E_1 and E_2 are of types (a2), (c1), and (d1), not both of type (a2), and each line of D_1 is glued to a line of D_2 with the intersection points identified [Rei94, Section 1.3]. Let $y \in E$ be the image of the intersection points. By **Step 1**, a component of type (a2) or (c1) has degree at most 3. If neither E_1 nor E_2 is of type (d1), then $d \leq 6$, which contradicts $d \geq 8$.

Assume that E_1 is of type (a2) and $E_2 \cong \mathbb{F}_a$ is of type (d1). After renaming the lines of D_1, a line $\ell \subset E_1$ is glued to $B \subset E_2$. Let $C \subset E$ be the image of ℓ and B. Then C is a smooth rational curve, E_1 and E_2 are smooth along C, $(C^2)_{E_1} = 1$, and $(C^2)_{E_2} = -a$. For $p \in C \backslash \{y\}$, **Step 2**(1) and Lemma 9.7 for $I = (x_1x_2, x_4)$, after renaming (x_1, x_2, x_3, x_4) as (u_1, u_2, u_4, u_3), imply condition (i) of Lemma 9.2 for $E' := E_1$ and $E'' := E_2$. By **Step 2**(2) with C corresponding to (s_2, s_3) and Lemma 9.7 for $I = (s_3(s_1s_2 - s_3), s_4)$, there exists an isomorphism $\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[s_1, s_2, s_3, s_4]]/(s_3(s_1s_2 - s_3) + s_4\psi_y)$ which sends the ideals of E, E_1, E_2, and C to (s_4), (s_3, s_4), $(s_3 - s_1s_2, s_4)$, and (s_2, s_3, s_4), respectively, which is condition (ii) of Lemma 9.2 for $\Sigma = \{y\}$. By **Step 1**, Y is locally a hypersurface along E and $\operatorname{Sing} Y \cap E$ is finite. Thus Lemma 9.2 implies

$$0 \leq \sum_{p \in C} l_p = 2 + 1 + 1 - a = 4 - a.$$

Hence $a \leq 4$, and $d = 1 + (a + 2) \leq 7$, which contradicts $d \geq 8$.

Assume that both $E_1 \cong \mathbb{F}_{a_1}$ and $E_2 \cong \mathbb{F}_{a_2}$ are of type (d1), and let A_i and B_i be a fiber and the negative section of E_i with $D_i = A_i + B_i$. Then either A_1 is glued to A_2 and B_1 is glued to B_2, or A_1 is glued to B_2 and B_1 is glued to A_2.

In the first case, let $C \subset E$ be the image of B_1 and B_2. Then C is a smooth rational curve, E_1 and E_2 are smooth along C, and $(C^2)_{E_i} = -a_i$. As in the previous paragraph, **Step 2**(1), **Step 2**(2) with C corresponding to (s_2, s_3), and Lemma 9.7 imply conditions (i) and (ii) of Lemma 9.2 for $E' := E_1$, $E'' := E_2$, and $\Sigma = \{y\}$. Thus Lemma 9.2 implies

$$0 \leq \sum_{p \in C} l_p = 2 + 1 - a_1 - a_2 = 3 - a_1 - a_2.$$

Hence $d = (a_1 + 2) + (a_2 + 2) \leq 7$, which contradicts $d \geq 8$.

In the second case, y is the image of $A_1 \cap B_1$. By **Step 1**, Y is locally a hypersurface along E and $\operatorname{Sing} Y \cap E$ is finite. By **Step 2**, E is a reduced Gorenstein slc surface with double crossings away from y, and $\widehat{\mathcal{O}}_{E,y} \cong \mathbb{C}[[s_1, s_2, s_3]]/(s_3(s_1s_2 - s_3))$ with components $\{s_3 = 0\}$ and $\{s_3 = s_1s_2\}$. Thus Lemma 9.6 implies $\widehat{\mathrm{vol}}(X \ni x) \leq 7$.

Assume that E_1 is of type (c1) and $E_2 \cong \mathbb{F}_{a_2}$ is of type (d1). Then E_1 is the cone over a rational normal curve of degree $a_1 \in \{2, 3\}$ by **Step 1**. After renaming the lines of D_1, the lines $\ell, \ell' \subset E_1$ are glued to B and A, respectively. As in **Step 2**(3), C_1 and C_2 denote the images of ℓ and ℓ'. By **Steps 1 and 2**, Y is locally a hypersurface along E, $\operatorname{Sing} Y \cap E$ is finite, E is a Gorenstein slc del Pezzo surface, and E has double crossings along $C_1 \cup C_2$ away from y. By **Step 2**(3), $\widehat{\mathcal{O}}_{E,y} \cong \widehat{\mathcal{O}}_{V,0}$. If $a_1 = 2$, then the ideal of V is $(x_2(x_2^2 - x_1x_3), x_4)$, and if $a_1 = 3$, then the ideal of V is contained in $(x_1, x_2, x_3, x_4)^2$. In both cases, Lemma 9.7 implies

$\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[x_1,x_2,x_3,x_4]]/(f_0)$ for some f_0, where the ideals of C_1, C_2, E_2, and E_1 are the ideals in Lemma 9.9. Thus Lemma 9.9 implies $\widehat{\mathrm{vol}}(X \ni x) \le 7$.

Hence $\widehat{\mathrm{vol}}(X \ni x) \le 7$ in all cases. □

9.2. **Non-isolated hypersurface singularities.** In this subsection, we classify the klt hypersurface threefold germs whose singular locus is not isolated and whose local volume is greater than $\frac{250}{33}$: by Proposition 9.12, such a germ falls into one of four families of normal forms, with local volume $\frac{27}{2}$, $6\sqrt{3}$, 9, or 8. We also study the blow-up of a singular curve of such a germ. The classification and the blow-up description are used in Proposition 9.17 at the closed points of the non-normal locus of the exceptional surface. Lemma 9.15 excludes a curve of transversal A_2 singularities, since the exceptional locus of the blow-up of such a curve consists of two crepant divisors.

We first describe the singular locus of a germ whose completed local ring is $\mathbb{C}[[u,v,z,t]]/(uv+f)$ with $f \in \mathbb{C}[[z,t]]$.

Lemma 9.11. *Let $X \ni x$ be a threefold germ, and let $0 \ne f \in (z,t) \subset \mathbb{C}[[z,t]]$. Assume that $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+f)$. Then the following hold.*

(1) $\mathrm{Sing}(\operatorname{Spec}\widehat{\mathcal{O}}_{X,x})$ is the preimage of $\mathrm{Sing}(\operatorname{Spec}\mathcal{O}_{X,x})$.
(2) The one-dimensional irreducible components of $\mathrm{Sing}(\operatorname{Spec}\widehat{\mathcal{O}}_{X,x})$ are the closed subsets $V(u,v,g)$, where g runs over the prime elements of $\mathbb{C}[[z,t]]$ with $g^2 \mid f$.
(3) The singular locus of X is not isolated at x if and only if f is not reduced.

Proof. Let $R := \mathcal{O}_{X,x}$ and $\widehat{R} := \widehat{\mathcal{O}}_{X,x}$.

(1) Since $\mathbb{C}$ is excellent and a localization of a $\mathbb{C}$-algebra of finite type is excellent, R is excellent [Gro65, Scholie 7.8.3(ii)(iii)]. Hence $\operatorname{Spec}\widehat{R} \to \operatorname{Spec} R$ is a regular morphism and $\mathrm{Reg}(\operatorname{Spec}\widehat{R})$ is the preimage of $\mathrm{Reg}(\operatorname{Spec} R)$ [Gro65, Scholie 7.8.3(v)]. Taking complements, we obtain (1).

(2) By the Jacobian criterion in $\mathbb{C}[[u,v,z,t]]$ [Mat86, Theorem 30.8 and the remark following it],

$$\mathrm{Sing}(\operatorname{Spec}\widehat{R}) = V(u,v,f,\partial_z f,\partial_t f).$$

Let g be a prime element of $\mathbb{C}[[z,t]]$. We show that $f,\partial_z f,\partial_t f \in (g)$ if and only if $g^2 \mid f$. If $g^2 \mid f$, then g divides $\partial_z f$ and $\partial_t f$ by the Leibniz rule. Conversely, assume that $f,\partial_z f,\partial_t f \in (g)$, and suppose that $f = gf_1$ with $g \nmid f_1$. Since $\partial_z f = f_1\partial_z g + g\partial_z f_1$, we have $g \mid \partial_z g$, and similarly $g \mid \partial_t g$. Let $d := \operatorname{ord} g \ge 1$, and let g_d be the homogeneous part of degree d of g. By the Euler formula $z\partial_z g_d + t\partial_t g_d = dg_d \ne 0$, one of $\partial_z g$ and $\partial_t g$ is non-zero of order $d-1$, hence not divisible by g, a contradiction. Thus $g^2 \mid f$. Since $f \ne 0$, the prime ideals of $\mathbb{C}[[z,t]]$ containing $(f,\partial_z f,\partial_t f)$, other than the maximal ideal, are exactly the ideals (g) with g prime and $g^2 \mid f$. Hence (2) holds.

(3) The singular locus of X is not isolated at x if and only if $\mathrm{Sing}(\operatorname{Spec} R)$ contains a non-maximal prime ideal. Since $R \to \widehat{R}$ is faithfully flat [Mat86, Theorem 8.14], any prime ideal of R is the contraction of a prime ideal of $\widehat{R}$, and the maximal ideal of $\widehat{R}$ is the only prime ideal over $\mathfrak{m}_x$. By (1), $\mathrm{Sing}(\operatorname{Spec} R)$ contains a non-maximal prime ideal if and only if $\mathrm{Sing}(\operatorname{Spec}\widehat{R})$ does, that is, by (2), if and only if $g^2 \mid f$ for some prime element g. Since $\mathbb{C}[[z,t]]$ is a unique factorization domain, the last condition means that f is not reduced. □

Proposition 9.12. *Let $X \ni x$ be a klt hypersurface threefold germ whose singular locus is not isolated at x, and let $V := \widehat{\mathrm{vol}}(X \ni x)$. Assume that $V > \frac{250}{33}$. Then $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+z^2h)$ for some $h \in \mathbb{C}[[z,t]]$ such that exactly one of the following holds.*

(1) $h = 1$, and $V = \frac{27}{2}$.
(2) $h = t$, and $V = 6\sqrt{3}$.
(3) $h = z + t^k$ for an integer $k \ge 2$, or $h = z$, and $V = 9$.
(4) $\operatorname{ord} h = 2$ and the quadratic part of h has non-zero coefficient of t^2, and $V = 8$.

Conversely, let $X \ni x$ be a klt germ such that $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[u,v,z,t]]/(uv+z^2h)$ for some h as in one of (1)–(4). Then $X \ni x$ is a hypersurface germ whose singular locus is not isolated at x, and $\widehat{\mathrm{vol}}(X \ni x)$ is the value given in that item.

Proof. The proof is long, so we divide it into several steps. Let $P := \mathbb{C}[[u,v,z,t]]$. For $w \in \mathbb{R}^4_{>0}$, let v_w be the monomial valuation on P with weights w for (u,v,z,t). For a unit c of $\mathbb{C}[[z,t]]$, we have $(uv+f) = (c^{-1}uv + c^{-1}f)$ and $(c^{-1}u, v, z, t)$ is a coordinate system of P. Thus, for $f \in \mathbb{C}[[z,t]]$, replacing u with $c^{-1}u$, we may replace f with $c^{-1}f$ in a presentation $P/(uv+f)$. We call the replacement of (u,f) by $(c^{-1}u, c^{-1}f)$ *absorbing c into u*.

Step 1. In this step, we show that $\widehat{\mathcal{O}}_{X,x} \cong P/(uv+f)$ for some non-reduced $f \in \mathbb{C}[[z,t]]$ with $\operatorname{ord} f \geq 2$.

Since the singular locus of X is not isolated at x, x is a singular point of X. Thus $\widehat{\mathcal{O}}_{X,x} \cong P/(F)$ for some $F \in (u,v,z,t)^2$. Let $m := \operatorname{ord} F$. If $m \geq 3$, then $v_w(F) = m$ for $w = (1,1,1,1)$, and Proposition 2.20 implies that

$$V \leq (4-m)^3 m \leq 3 < \frac{250}{33},$$

a contradiction. Thus $m = 2$. Assume that the quadratic part of F has rank one. By the splitting lemma [GP26, Theorem 2.1(2)], after a coordinate change, $F = u^2 + G(v,z,t)$ with $\operatorname{ord} G \geq 3$. For $w = (3,2,2,2)$, $v_w(F) = 6$, and Proposition 2.20 implies that

$$V \leq (9-6)^3 \cdot \frac{6}{3 \cdot 2 \cdot 2 \cdot 2} = \frac{27}{4} < \frac{250}{33},$$

a contradiction. Thus the quadratic part of F has rank at least two. By the splitting lemma [GP26, Theorem 2.1(2)] and $y_1^2 + y_2^2 = (y_1 + \sqrt{-1}y_2)(y_1 - \sqrt{-1}y_2)$, after a coordinate change, $F = uv + f(z,t)$ with $\operatorname{ord} f \geq 2$. Since X is normal and $\mathcal{O}_{X,x}$ is excellent, $\widehat{\mathcal{O}}_{X,x}$ is a normal domain [Mat86, Corollary to Theorem 23.9]. Since $P/(uv)$ is not a domain, $f \neq 0$. By Lemma 9.11(3), f is not reduced.

From now on until the end of **Step 3**, let $e := \operatorname{ord} f$, and let g be a prime element of $\mathbb{C}[[z,t]]$ with $g^2 \mid f$.

Step 2. In this step, we show that $e \leq 4$, and that $\widehat{\mathcal{O}}_{X,x}$ has one of the presentations in (1)–(3) if $e \leq 3$.

Assume that $e \geq 5$. For $w = (e,e,2,2)$, any monomial of f has weight at least $2e$, so $v_w(uv+f) = 2e$, and Proposition 2.20 implies that

$$V \leq 4^3 \cdot \frac{2e}{e \cdot e \cdot 2 \cdot 2} = \frac{32}{e} \leq \frac{32}{5} < \frac{250}{33},$$

a contradiction. Thus $e \leq 4$.

Assume that $e = 2$. Since $2 \operatorname{ord} g \leq e$, we have $\operatorname{ord} g = 1$ and $f = cg^2$ for a unit c. After a coordinate change of $\mathbb{C}[[z,t]]$, $g = z$. Absorbing c into u, we obtain $f = z^2$, that is, $h = 1$.

Assume that $e = 3$. Then $\operatorname{ord} g = 1$, and after a coordinate change of $\mathbb{C}[[z,t]]$, we have $g = z$ and $f = z^2 h$ with $\operatorname{ord} h = 1$. If $\partial_t h(0) \neq 0$, then (z,h) is a coordinate system of $\mathbb{C}[[z,t]]$, and replacing t with h, we obtain $h = t$. Assume that $\partial_t h(0) = 0$. Then $\partial_z h(0) \neq 0$. By the Weierstrass preparation theorem [GP08, Corollary 6.2.8], $h = c \cdot (z + \varphi(t))$ for a unit c and some $\varphi \in t\mathbb{C}[[t]]$. Since the coefficient of t in h is zero, $\varphi \in t^2\mathbb{C}[[t]]$. Absorbing c into u, we may assume that $c = 1$. If $\varphi = 0$, then $h = z$. Otherwise $\varphi = t^k \epsilon(t)$ for an integer $k \geq 2$ and a unit ϵ, and replacing t with $\epsilon^{1/k} t$, we obtain $h = z + t^k$.

Step 3. In this step, we show that, if $e = 4$, then $\widehat{\mathcal{O}}_{X,x}$ has a presentation as in (4).

Assume first that $\operatorname{ord} g = 1$. After a coordinate change of $\mathbb{C}[[z,t]]$, $g = z$ and $f = z^2 h$ with $\operatorname{ord} h = 2$. Let $h_2 = \alpha t^2 + \beta zt + \gamma z^2$ be the quadratic part of h. If $\alpha \neq 0$, then we are in case (4). Suppose that $\alpha = 0$ and $\beta \neq 0$. Replacing t with $\beta t + \gamma z$, we have $h_2 = zt$. For $w = (11,11,6,4)$,

$v_w(uv) = v_w(z^3t) = 22$, and any monomial of $z^2(h - h_2)$ has weight at least $12 + 3 \cdot 4 = 24$. Thus $v_w(uv + f) = 22$, and Proposition 2.20 implies that

$$V \leq (32 - 22)^3 \cdot \frac{22}{11 \cdot 11 \cdot 6 \cdot 4} = \frac{250}{33},$$

a contradiction. Suppose that $\alpha = \beta = 0$. Then $\gamma \neq 0$ and $h_2 = \gamma z^2$. For $w = (6, 6, 3, 2)$, $v_w(uv) = v_w(z^4) = 12$, and any monomial of $z^2(h - h_2)$ has weight at least $6 + 3 \cdot 2 = 12$. Thus $v_w(uv + f) = 12$, and Proposition 2.20 implies that

$$V \leq (17 - 12)^3 \cdot \frac{12}{6 \cdot 6 \cdot 3 \cdot 2} = \frac{125}{18} < \frac{250}{33}, \tag{9.10}$$

a contradiction.

Assume next that f has no prime factor g with $g^2 \mid f$ and $\operatorname{ord} g = 1$. Then $\operatorname{ord} g = 2$ since $2 \operatorname{ord} g \leq 4$, and $f = cg^2$ for a unit c. Absorbing c into u, we may assume that $f = g^2$. If the quadratic part g_2 of g had rank two, then g would be right equivalent to zt by the splitting lemma [GP26, Theorem 2.1(2)], contradicting the irreducibility of g. Thus $g_2 = \lambda L^2$ for a non-zero linear form L and some $\lambda \in \mathbb{C}^*$. After a linear coordinate change of $\mathbb{C}[[z, t]]$, $L = z$. For $w = (6, 6, 3, 2)$, any monomial of g has weight at least 6, so $v_w(uv + g^2) = 12$, and Proposition 2.20 implies (9.10), a contradiction.

Step 4. In this step, we assume only that $X \ni x$ is a klt germ with $\widehat{\mathcal{O}}_{X,x} \cong P/(F)$, where $F := uv + z^2h$ for some h as in one of (1)–(4), and we show that V is $\frac{27}{2}$, $6\sqrt{3}$, or 9 in cases (1), (2), and (3), respectively.

In case (1), $V = \frac{27}{2}$ by Proposition 3.1(2). In case (2), $V = 6\sqrt{3}$ by Proposition 3.1(4). Assume that we are in case (3), and set $k := +\infty$ and $t^k := 0$ if $h = z$.

We show that $X \ni x$ is of type cA_2. Let H be a general hyperplane section of X through x. The image of a local equation of H in $\mathfrak{m}_x/\mathfrak{m}_x^2$ has non-zero coefficient of t, so it can replace t in the coordinate system. Thus $\widehat{\mathcal{O}}_{H,x} \cong \mathbb{C}[[u, v, z]]/(F_H)$, where $F_H := uv + z^3 + z^2\ell^k$ for some $\ell \in (u, v, z)$. The quadratic part of F_H is uv, so by the splitting lemma [GP26, Theorem 2.1(2)], F_H is right equivalent to $uv + g(z)$ for some $g \in z^3\mathbb{C}[[z]]$. For $G \in \mathbb{C}[[u, v, z]]$, let

$$\mu(G) := \dim_{\mathbb{C}} \mathbb{C}[[u, v, z]]/(\partial_u G, \partial_v G, \partial_z G).$$

By the chain rule, μ does not change under right equivalence, and $\mu(uv + g) = \operatorname{ord} g - 1$. On the other hand, $\partial_u F_H = v + kz^2\ell^{k-1}\partial_u\ell$ and $\partial_v F_H = u + kz^2\ell^{k-1}\partial_v\ell$, so by the formal implicit function theorem, $\mathbb{C}[[u, v, z]]/(\partial_u F_H, \partial_v F_H) \cong \mathbb{C}[[z]]$, and the images of u and v belong to $z^2\mathbb{C}[[z]]$. Thus the image of ℓ belongs to $z\mathbb{C}[[z]]$, and since $k \geq 2$, the image of

$$\partial_z F_H = 3z^2 + 2z\ell^k + kz^2\ell^{k-1}\partial_z\ell$$

is $3z^2$ modulo z^3. Hence $\mu(F_H) = 2$, $\operatorname{ord} g = 3$, and $H \ni x$ is a Du Val singularity of type A_2. Therefore $X \ni x$ is of type cA_2 (Definition 2.14), and $V \geq 9$ by Theorem 2.17(4).

For $w = (3, 3, 2, 1)$, $v_w(uv) = v_w(z^3) = 6$ and $v_w(z^2t^k) = 4 + k \geq 6$, so $v_w(F) = 6$, and Proposition 2.20 implies that

$$V \leq (9 - 6)^3 \cdot \frac{6}{3 \cdot 3 \cdot 2 \cdot 1} = 9.$$

Thus $V = 9$.

Step 5. In this step, under the assumption of **Step 4**, we show that $V = 8$ in case (4).

Let $\theta \neq 0$ be the coefficient of t^2 in h. For $w = (2, 2, 1, 1)$, $v_w(uv) = 4$ and any monomial of z^2h has weight at least 4, so $v_w(F) = 4$, and Proposition 2.20 implies that

$$V \leq (6 - 4)^3 \cdot \frac{4}{2 \cdot 2 \cdot 1 \cdot 1} = 8.$$

For the reverse inequality, we degenerate $X \ni x$ to a cone of local volume 8. In the rest of this step, let $w := (3, 3, 2, 1)$. For $0 \neq G \in P$, let G_w be the sum of the terms of G of weight $v_w(G)$.

The monomials uv and z^2t^2 have weight 6, and the other monomials of F have weight at least 7, so

$$F_w = uv + \theta z^2 t^2.$$

Let $\widehat{R} := \widehat{\mathcal{O}}_{X,x} = P/(F)$, and for $j \in \mathbb{Z}$, let $\widehat{\mathfrak{a}}_j \subset \widehat{R}$ be the image of $\{G \in P \mid v_w(G) \geq j\}$. Since $(GF)_w = G_w F_w$ for any $0 \neq G \in P$, the associated graded ring of $\{\widehat{\mathfrak{a}}_j\}_j$ is $\mathbb{C}[u,v,z,t]/(uv+\theta z^2t^2)$. The ring $\mathbb{C}[u,v,z,t]/(uv+\theta z^2t^2)$ is a domain, since $uv+\theta z^2t^2$ is linear in v with coprime coefficients u and θz^2t^2. Moreover, $\bigcap_j \widehat{\mathfrak{a}}_j \subset \bigcap_j \widehat{\mathfrak{m}}^j = 0$, where $\widehat{\mathfrak{m}}$ is the maximal ideal of $\widehat{R}$. Hence the function $v_w(r) := \max\{v_w(G) \mid G \in P \text{ maps to } r\} = \max\{j \mid r \in \widehat{\mathfrak{a}}_j\}$ on $\widehat{R}\backslash\{0\}$ is a valuation of $\widehat{R}$ induced by the monomial valuation v_w of P, and we denote by v_w also its restriction to $R := \mathcal{O}_{X,x}$. Since $\widehat{\mathfrak{a}}_j \supset \widehat{\mathfrak{m}}^j$ and $R/\mathfrak{m}_x^j \cong \widehat{R}/\widehat{\mathfrak{m}}^j$ for any $j \geq 1$, we obtain

$$\operatorname{gr}_{v_w} \mathcal{O}_{X,x} \cong \operatorname{gr}_{v_w} \widehat{R} \cong \mathbb{C}[u,v,z,t]/(uv+\theta z^2t^2). \tag{9.11}$$

Let $g_1, g_2, g_3, g_4 \in R$ be elements whose classes in $\operatorname{gr}_{v_w} \mathcal{O}_{X,x}$ are u, v, z, t, so that $v_w(g_i) = w_i$. Let $(y_1, y_2, y_3, y_4) := (u, v, z, t)$. Then $g_i - y_i$ is the image of a sum of monomials of weight greater than w_i, and such a monomial either has degree at least two or is y_l for some $l < i$. Thus, in $\mathfrak{m}_x/\mathfrak{m}_x^2$, the class of g_i is the class of y_i plus a $\mathbb{C}$-linear combination of the classes of y_l with $l < i$, so $g_1, \dots, g_4$ generate $\mathfrak{m}_x$. Since $X \ni x$ is a hypersurface germ, possibly shrinking X around x, we may assume that $X = \operatorname{Spec} R_0$ is an affine hypersurface in a smooth affine fourfold, hence Gorenstein, that $g_1, \dots, g_4 \in R_0$, and that x is the only common zero of $g_1, \dots, g_4$. For $j \geq 1$, let

$$\mathfrak{a}_j := \{r \in R_0 \mid v_w(r) \geq j\}, \quad J_j := \left(g_1^{b_1} g_2^{b_2} g_3^{b_3} g_4^{b_4} \;\middle|\; \sum_{i=1}^4 b_i w_i \geq j\right) \subset R_0,$$

and let $\mathfrak{a}_j := R_0$ for $j \leq 0$.

We show that $\mathfrak{a}_j = J_j$ for $j \geq 1$. Since $v_w(g_i) = w_i$, $J_j \subset \mathfrak{a}_j$. Let $r \in \widehat{\mathfrak{a}}_j$ with $d := v_w(r) < +\infty$. By (9.11), the class of r in $\widehat{\mathfrak{a}}_d/\widehat{\mathfrak{a}}_{d+1}$ is a $\mathbb{C}$-linear combination of the classes of the monomials $g_1^{b_1} \cdots g_4^{b_4}$ with $\sum_i b_i w_i = d \geq j$. Subtracting the corresponding linear combination of $g_1^{b_1} \cdots g_4^{b_4}$, we obtain an element of $r + J_j\widehat{R}$ of larger value. Repeating this, $r \in J_j\widehat{R} + \widehat{\mathfrak{a}}_N$ for any N. Since $\widehat{\mathfrak{a}}_N \subset \widehat{\mathfrak{m}}^{\lceil N/3 \rceil}$, the Krull intersection theorem implies that $r \in J_j\widehat{R}$. Thus $\widehat{\mathfrak{a}}_j = J_j\widehat{R}$, and by [Mat86, Theorems 7.5(ii) and 8.14],

$$\mathfrak{a}_j R = \widehat{\mathfrak{a}}_j \cap R = J_j\widehat{R} \cap R = J_j R.$$

Since $g_i^j \in J_j$, the ideals $J_j \subset \mathfrak{a}_j$ of R_0 are primary to the maximal ideal of x, so they are determined by their localizations at x. Thus $\mathfrak{a}_j = J_j$.

Let s be a variable, and let $\mathcal{A} := \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j s^{-j} \subset R_0[s, s^{-1}]$ be the extended Rees algebra. Since $\mathfrak{a}_j = J_j$, $\mathcal{A}$ is generated over $R_0[s]$ by $g_1 s^{-3}, g_2 s^{-3}, g_3 s^{-2}, g_4 s^{-1}$. Let $\mathcal{X} := \operatorname{Spec} \mathcal{A}$ and $\pi : \mathcal{X} \to \mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[s]$. Then $\mathcal{X}$ is a variety, and π is flat and surjective, since $\mathcal{A}$ is a domain of finite type over $\mathbb{C}$ and is torsion free over $\mathbb{C}[s]$. Since $\mathcal{A}[s^{-1}] = R_0[s, s^{-1}]$, we have $\pi^{-1}(\mathbb{A}^1\backslash\{0\}) \cong X \times \mathbb{G}_m$. Since $\mathfrak{a}_j/\mathfrak{a}_{j+1}$ is annihilated by the maximal ideal of x for $j \geq 0$, (9.11) implies that the fiber $\mathcal{X}_0 := \pi^{-1}(0)$ satisfies

$$\mathcal{X}_0 = \operatorname{Spec} \bigoplus_{j \geq 0} \mathfrak{a}_j/\mathfrak{a}_{j+1} \cong \operatorname{Spec} \mathbb{C}[u,v,z,t]/(uv+\theta z^2t^2).$$

The ring homomorphism $\mathcal{A} \to \mathbb{C}[s]$, $\sum_j r_j s^{-j} \to \sum_{j \leq 0} r_j(x) s^{-j}$, defines a section σ of π with $\sigma(a) = (x, a)$ for $a \neq 0$, and $\sigma(0)$ is the vertex o of $\mathcal{X}_0$.

We show that $\mathcal{X}_0$ is normal and klt, and that $\widehat{\operatorname{vol}}(\mathcal{X}_0 \ni o) = 8$. Let $\{u'v' = zt\} \subset \mathbb{A}^4$ be the cone of the ordinary double point, and let ι be the involution $(u', v', z, t) \to (-u', -v', z, t)$ of $\{u'v' = zt\}$. A direct calculation shows that the ring of ι-invariant functions on $\{u'v' = zt\}$ is generated by u'^2, v'^2, z, t, and that $u \to -\theta u'^2$, $v \to v'^2$, $z \to z$, $t \to t$ induces an isomorphism from $\mathbb{C}[u,v,z,t]/(uv+\theta z^2t^2)$ to the ring of ι-invariant functions. Thus $\mathcal{X}_0$ is the quotient of $\{u'v' = zt\}$ by ι, and in particular $\mathcal{X}_0$ is normal. The fixed locus of ι is $\{u' = v' = zt = 0\}$,

so $\dim \mathrm{Fix}(\iota) = 1$, and the quotient morphism $\{u'v' = zt\} \to \mathcal{X}_0$ is a finite Galois morphism of degree 2 which is étale in codimension one. Since the ordinary double point is klt, $\mathcal{X}_0$ is klt [KM98, Proposition 5.20(4)]. By Theorem 2.17(1), the ordinary double point has local volume 16, so Proposition 2.7(2) implies that $\widehat{\mathrm{vol}}(\mathcal{X}_0 \ni o) = \frac{16}{2} = 8$.

We check that π and σ satisfy the conditions of [BL21, Definition 7]. The base $\mathbb{A}^1$ is normal. The fibers of π are X and $\mathcal{X}_0$, which are connected, normal, and klt, and the generic fiber $\operatorname{Spec} R_0 \otimes_{\mathbb{C}} \mathbb{C}(s)$ is connected, normal, and klt, since it is a localization of the normal domain $R_0[s]$ [Mat86, Corollary to Theorem 23.9], and a log resolution of $\operatorname{Spec} R_0$ base-changes to a log resolution of it with equal discrepancies [KM98, Corollary 2.32]. Since $\mathcal{X}_0$ is a hypersurface and s is a nonzerodivisor of $\mathcal{A}$, the local rings of $\mathcal{X}$ at the points of $\mathcal{X}_0$ are Gorenstein [BH98, Proposition 3.1.19(b)]. Since $X \times \mathbb{G}_m$ is Gorenstein, $\mathcal{X}$ is Gorenstein. Moreover, $\mathcal{X}$ is normal [Mat86, Corollary to Theorem 23.9], since $\mathbb{A}^1$, the fibers of π, and the generic fiber $\operatorname{Spec} R_0 \otimes_{\mathbb{C}} \mathbb{C}(s)$ are normal. Thus $K_{\mathcal{X}/\mathbb{A}^1}$ is Cartier. Hence, by [BL21, Theorem 1], the function $a \to \widehat{\mathrm{vol}}(\mathcal{X}_a \ni \sigma(a))$ on $\mathbb{A}^1(\mathbb{C})$ is lower semicontinuous. Since $\widehat{\mathrm{vol}}(\mathcal{X}_a \ni \sigma(a)) = V$ for $a \neq 0$,

$$8 = \widehat{\mathrm{vol}}(\mathcal{X}_0 \ni o) \leq V.$$

Thus $V = 8$.

Step 6. In this step, we conclude the proof.

By **Steps 1–3**, $\widehat{\mathcal{O}}_{X,x} \cong P/(uv + z^2 h)$ for some h as in one of (1)–(4), and by **Steps 4 and 5**, V is the value given in that item. Since $\frac{27}{2}$, $6\sqrt{3}$, 9, and 8 are distinct, exactly one of (1)–(4) holds.

Conversely, let $X \ni x$ be a klt germ such that $\widehat{\mathcal{O}}_{X,x} \cong P/(uv + z^2 h)$ for some h as in one of (1)–(4). Then $\dim \mathfrak{m}_x/\mathfrak{m}_x^2 \leq 4$, so $X \ni x$ is a hypersurface germ. Since $h \neq 0$ and $z^2 \mid z^2 h$, the singular locus of X is not isolated at x by Lemma 9.11(3). By **Steps 4 and 5**, $\widehat{\mathrm{vol}}(X \ni x)$ is the value given in that item. □

We next describe the blow-up of a smooth singular curve of a hypersurface threefold germ whose completed local ring is $\mathbb{C}[[u, v, z, t]]/(uv + z^2 h)$. Lemma 9.13 provides a normal form along the curve, Lemma 9.14 is local over a point p of the curve, and Lemma 9.15 is the global consequence along a rational curve.

Lemma 9.13. *Let $Y \ni p$ be a klt hypersurface threefold germ with $\widehat{\mathrm{vol}}(Y \ni p) > \frac{250}{33}$, and let $C \subset \operatorname{Sing} Y$ be a smooth curve germ through p with ideal I_C. Then there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u, v, z, t]]/(uv + z^2 h)$ which sends $I_C \widehat{\mathcal{O}}_{Y,p}$ to (u, v, z), where $h \in \mathbb{C}[[z, t]]$ is a unit, $h = t$, $h = z + t^k$ for an integer $k \geq 2$, $h = z$, or $\operatorname{ord} h = 2$ and the quadratic part of h has non-zero coefficient of t^2.*

Proof. Let $P := \mathbb{C}[[u, v, z, t]]$. Since $C \subset \operatorname{Sing} Y$, the singular locus of Y is not isolated at p. By Proposition 9.12, there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong P/(uv + z^2 h_0)$ for some h_0 as in one of Proposition 9.12(1)–(4). Since C is smooth, $\widehat{\mathcal{O}}_{Y,p}/I_C \widehat{\mathcal{O}}_{Y,p} \cong \widehat{\mathcal{O}}_{C,p} \cong \mathbb{C}[[t]]$, so $I_C \widehat{\mathcal{O}}_{Y,p}$ is a prime ideal and $V(I_C \widehat{\mathcal{O}}_{Y,p})$ is irreducible with $\dim V(I_C \widehat{\mathcal{O}}_{Y,p}) = 1$. By Lemma 9.11(1), $V(I_C \widehat{\mathcal{O}}_{Y,p}) \subset \operatorname{Sing}(\operatorname{Spec} \widehat{\mathcal{O}}_{Y,p})$, and $\dim \operatorname{Sing}(\operatorname{Spec} \widehat{\mathcal{O}}_{Y,p}) \leq 1$ since Y is normal. Thus $V(I_C \widehat{\mathcal{O}}_{Y,p})$ is a one-dimensional irreducible component of $\operatorname{Sing}(\operatorname{Spec} \widehat{\mathcal{O}}_{Y,p})$, and by Lemma 9.11(2), $I_C \widehat{\mathcal{O}}_{Y,p} = (u, v, g)$ for a prime element g of $\mathbb{C}[[z, t]]$ with $g^2 \mid z^2 h_0$.

If g is a unit multiple of z, then the isomorphism sends $I_C \widehat{\mathcal{O}}_{Y,p}$ to (u, v, z), and $h := h_0$ is a unit, t, $z + t^k$, z, or satisfies $\operatorname{ord} h = 2$ with non-zero coefficient of t^2. Otherwise $g^2 \mid h_0$. In cases (1)–(3) of Proposition 9.12, h_0 is 1, t, $z + t^k$, or z, and no square of a prime element divides h_0. Thus we are in case (4), $\operatorname{ord} h_0 = 2$, and $h_0 = cg^2$ for a unit c and some g with $\operatorname{ord} g = 1$. Since the coefficient of t^2 in h_0 is non-zero, the coefficient of t in g is non-zero, so (g, z) is a coordinate system of $\mathbb{C}[[z, t]]$. Replacing (z, t) with (g, z) and absorbing c into u, we obtain an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong P/(uv + z^2 t^2)$ which sends $I_C \widehat{\mathcal{O}}_{Y,p}$ to (u, v, z), and $h := t^2$ is as in the statement. □

Lemma 9.14. *Let $Y \ni p$ be a normal hypersurface threefold germ, let $C \subset \operatorname{Sing} Y$ be a smooth curve germ through p with ideal I_C, let $\phi : W := \operatorname{Bl}_C Y \to Y$ be the blow-up of I_C, and let S be the reduced exceptional divisor. Assume that there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u,v,z,t]]/(uv+z^2h)$ which sends $I_C\widehat{\mathcal{O}}_{Y,p}$ to (u,v,z), where $0 \neq h \in \mathbb{C}[[z,t]]$. Then the following hold.*

(1) W is normal, $K_W = \phi^* K_Y$, and S is Cartier. The fiber $\phi^{-1}(p)$ is the reduced conic $\{U_1U_2+h(0,0)U_3^2=0\}$ in $\mathbb{P}^2$ with homogeneous coordinates $[U_1 : U_2 : U_3]$ corresponding to u,v,z, and W is smooth at any point of $\phi^{-1}(p)$ with $U_3 = 0$. For any point $w = [a_1 : a_2 : 1]$ of $\phi^{-1}(p)$, we have

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[z,t,U_1-a_1,U_2-a_2]]/(U_1U_2+h(z,t)),$$

where $u = zU_1$ and $v = zU_2$, and S corresponds to (z). If $h(0,t) \neq 0$, then exactly one prime component of S dominates C.

(2) If h is a unit, then W and S are smooth near $\phi^{-1}(p)$, and $\phi^{-1}(p)$ is a smooth conic.

(3) If $h = t$, then W and S are smooth near $\phi^{-1}(p)$, and $\phi^{-1}(p)$ is the union of two lines.

(4) If $h = z + t^k$ for an integer $k \geq 2$, then W is smooth near $\phi^{-1}(p)$, and S has exactly one singular point over p, which is a Du Val singularity of type A_{k-1}.

(5) If $\operatorname{ord} h = 2$ and the quadratic part of h has non-zero coefficient of t^2, then W has exactly one singular point w over p, namely $w = [0 : 0 : 1]$. Moreover, S has an A_1 singularity at w, $W \ni w$ is of type A_j for some $j \geq 1$ or a transversal A_1 singularity, and w is not contained in any smooth Cartier divisor on W.

(6) If $h = z$, then W is smooth near $\phi^{-1}(p)$, and $S \times_Y \operatorname{Spec} \widehat{\mathcal{O}}_{Y,p}$ has exactly two irreducible components, each of which intersects $\phi^{-1}(p)$ in a line. Moreover, S has two prime components if and only if the generic fiber of $S \to C$ is the union of two lines defined over $K(C)$.

Proof. The proof is long, so we divide it into several steps. Let $P := \mathbb{C}[[u,v,z,t]]$ and $F := uv + z^2h \in P$.

Step 1. In this step, we compute the completed local rings of W at the points of $\phi^{-1}(p)$.

Since $Y \ni p$ is a hypersurface germ, the embedding dimension of Y at p is at most four, so possibly shrinking Y around p, we may assume that Y is a hypersurface in a smooth fourfold M, and that C is a smooth closed curve in M. Lifting u,v,z,t to $\widehat{\mathcal{O}}_{M,p}$, we obtain an isomorphism $\widehat{\mathcal{O}}_{M,p} \cong P$ under which the ideals of Y and C correspond to (F) and (u,v,z). Let $\pi : \widetilde{M} := \operatorname{Bl}_C M \to M$, and let E_M be the exceptional divisor of π. Since Y is integral, W is the strict transform of Y in $\widetilde{M}$ [Har77, Corollary II.7.15]. Since $\widehat{\mathcal{O}}_{M,p}$ is flat over $\mathcal{O}_{M,p}$, we have $I^n \otimes_{\mathcal{O}_{M,p}} \widehat{\mathcal{O}}_{M,p} = (I\widehat{\mathcal{O}}_{M,p})^n$ for any ideal I of $\mathcal{O}_{M,p}$ and any n. Thus $\widehat{M} := \widetilde{M} \times_M \operatorname{Spec} P$ is the blow-up of (u,v,z) in $\operatorname{Spec} P$, and $\widehat{W} := W \times_M \operatorname{Spec} P$ is the strict transform of $\operatorname{Spec} P/(F)$ in $\widehat{M}$. Let y be a closed point of $\phi^{-1}(p)$. Since $P/(u,v,z,t) = \mathbb{C}$, there exists exactly one point $\widehat{y}$ of $\widehat{W}$ over y, the homomorphism $\mathcal{O}_{W,y} \to \mathcal{O}_{\widehat{W},\widehat{y}}$ is flat, and $\mathfrak{m}_y\mathcal{O}_{\widehat{W},\widehat{y}}$ is the maximal ideal of $\mathcal{O}_{\widehat{W},\widehat{y}}$. Hence $\widehat{\mathcal{O}}_{W,y} \cong \widehat{\mathcal{O}}_{\widehat{W},\widehat{y}}$. Replacing W and $\widehat{W}$ with $\widetilde{M}$ and $\widehat{M}$, the argument of the previous two sentences shows that $\widehat{\mathcal{O}}_{\widetilde{M},y} \cong \widehat{\mathcal{O}}_{\widehat{M},\widehat{y}}$. Moreover, $\mathcal{O}_{W,y}$ is regular if and only if $\widehat{\mathcal{O}}_{W,y}$ is regular, and $\mathcal{O}_{W,y}$ is normal if $\widehat{\mathcal{O}}_{W,y}$ is normal [Mat86, Theorem 23.7 and Corollary to Theorem 23.9]. For any closed subscheme Z of W containing y, applying the argument above to Z and $Z \times_M \operatorname{Spec} P$ in place of W and $\widehat{W}$, we obtain $\widehat{\mathcal{O}}_{Z,y} \cong \widehat{\mathcal{O}}_{Z\times_M \operatorname{Spec} P,\widehat{y}}$. Moreover, $\mathcal{O}_{Z,y}$ is regular if and only if $\widehat{\mathcal{O}}_{Z,y}$ is regular, and $\mathcal{O}_{Z,y}$ is normal if $\widehat{\mathcal{O}}_{Z,y}$ is normal.

We use the homogeneous coordinates $[U_1 : U_2 : U_3]$ of the fiber of $\widehat{M}$ over the closed point of $\operatorname{Spec} P$, corresponding to u,v,z. Let $w = [a_1 : a_2 : 1]$. On the chart $U_3 \neq 0$ of $\widehat{M}$, we have $u = zU_1$ and $v = zU_2$, and the completed local ring of $\widehat{M}$ at w is $\mathbb{C}[[z,t,U_1-a_1,U_2-a_2]]$. There

$F = z^2 G$ with

$$G := U_1 U_2 + h(z,t).$$

Since $G \equiv U_1 U_2 + h(0,t) \not\equiv 0$ modulo z, $z \nmid G$. The local ring of $\widehat{M}$ at w is a unique factorization domain in which z is prime, so the ideal of the strict transform at w is (G). Thus

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[z,t,U_1-a_1,U_2-a_2]]/(U_1U_2+h(z,t)).$$

On the chart $U_1 \neq 0$ of $\widehat{M}$, we have $v = uV$ and $z = uZ$, and $F = u^2(V + Z^2 h(uZ,t))$. Since $V + Z^2 h(uZ,t) \equiv V + Z^2 h(0,t) \not\equiv 0$ modulo u, and u is prime in the local ring of $\widehat{M}$ at any point of the chart $U_1 \neq 0$, the strict transform is defined by $V + Z^2 h(uZ,t)$, whose derivative with respect to V is 1. Thus W is smooth at the points of $\phi^{-1}(p)$ on the chart $U_1 \neq 0$, and similarly on the chart $U_2 \neq 0$. Since $\mathfrak{m}_p \mathcal{O}_W$ is generated by z and t on the chart $U_3 \neq 0$ and by u and t on the chart $U_1 \neq 0$, the fiber $\phi^{-1}(p)$ is the conic

$$\{U_1U_2 + h(0,0)U_3^2 = 0\} \subset \mathbb{P}^2,$$

which is reduced since the quadratic form $U_1U_2 + h(0,0)U_3^2$ has rank at least two. A point of $\phi^{-1}(p)$ with $U_3 = 0$ is $[1:0:0]$ or $[0:1:0]$, so W is smooth at the points of $\phi^{-1}(p)$ with $U_3 = 0$.

Finally, the associated graded ring of P with respect to (u,v,z) is $\mathbb{C}[[t]][U_1,U_2,U_3]$, where U_1, U_2, U_3 are the classes of u, v, z, and the leading form of F is $U_1U_2 + h(0,t)U_3^2$, which is a nonzerodivisor of $\mathbb{C}[[t]][U_1,U_2,U_3]$. Thus

$$\bigoplus_{n\geq 0} \frac{I_C^n}{I_C^{n+1}} \otimes_{\mathcal{O}_{Y,p}} \widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[t]][U_1,U_2,U_3]/(U_1U_2+h(0,t)U_3^2). \tag{9.12}$$

Step 2. In this step, we prove (1).

By **Step 1**, the completed local rings of W at the points of $\phi^{-1}(p)$ with $U_3 = 0$ are regular. Let $w = [a_1 : a_2 : 1] \in \phi^{-1}(p)$. The ring $\widehat{\mathcal{O}}_{W,w}$ is a hypersurface ring, hence Cohen–Macaulay. By the Jacobian criterion [Mat86, Theorem 30.8 and the remark following it], the singular locus of $\operatorname{Spec} \widehat{\mathcal{O}}_{W,w}$ is contained in $V(U_1,U_2,h,\partial_z h,\partial_t h)$, and $\dim V(h,\partial_z h,\partial_t h) \leq 1$ in $\operatorname{Spec} \mathbb{C}[[z,t]]$ as $h \neq 0$. Thus $\widehat{\mathcal{O}}_{W,w}$ satisfies R_1 and S_2, so it is normal [Har77, Theorem II.8.22A], and by **Step 1**, W is normal at w. Since the normal locus of W is open and ϕ is proper, possibly shrinking Y around p, W is normal.

By [KM98, Lemma 2.29], $K_{\widetilde{M}} = \pi^* K_M + 2E_M$. By **Step 1**, the pullback of a local equation of Y is $z^2 G$ with $z \nmid G$ on the chart $U_3 \neq 0$, and $u^2(V + Z^2 h(uZ,t))$ with $u \nmid V + Z^2 h(uZ,t)$ on the chart $U_1 \neq 0$. Thus $\pi^* Y = W + 2E_M$ near $\pi^{-1}(p)$. By adjunction,

$$K_W = (K_{\widetilde{M}} + W)|_W = \pi^*(K_M + Y)|_W = \phi^* K_Y$$

near $\phi^{-1}(p)$, and possibly shrinking Y around p, $K_W = \phi^* K_Y$.

The ideal $I_C \mathcal{O}_W$ is invertible, and it is generated by z on the chart $U_3 \neq 0$. By **Step 1**, the completed local ring of $\phi^{-1}(C)$ at w is

$$\mathbb{C}[[t,U_1-a_1,U_2-a_2]]/(U_1U_2+h(0,t)). \tag{9.13}$$

The element $U_1U_2 + h(0,t)$ is square-free, since a square factor of it divides both ∂_{U_1} and ∂_{U_2} of it, that is, U_2 and U_1. On the chart $U_1 \neq 0$, the completed local ring of $\phi^{-1}(C)$ at a point over p is $\mathbb{C}[[u,V,Z,t]]/(u, V + Z^2 h(uZ,t)) \cong \mathbb{C}[[Z,t]]$. Thus $\phi^{-1}(C)$ is reduced near $\phi^{-1}(p)$, and possibly shrinking Y around p, $S = \phi^{-1}(C)$ is Cartier and corresponds to (z) in $\widehat{\mathcal{O}}_{W,w}$. Moreover, S is smooth at the points of $\phi^{-1}(p)$ with $U_3 = 0$.

Since S has finitely many prime components and $\dim \phi^{-1}(p) = 1$, possibly shrinking Y around p, any prime component of S dominates C. Thus the prime components of S correspond to the irreducible components of the generic fiber S_η of $S \to C$, which is a conic over $K(C)$. By (9.12),

$$S \times_Y \operatorname{Spec} \widehat{\mathcal{O}}_{Y,p} \cong \operatorname{Proj} \mathbb{C}[[t]][U_1,U_2,U_3]/(U_1U_2+h(0,t)U_3^2),$$

so $S_\eta \otimes_{K(C)} \mathbb{C}((t))$ is the conic $\{U_1U_2 + h(0,t)U_3^2 = 0\}$ over $\mathbb{C}((t))$. If $h(0,t) \neq 0$, then $\{U_1U_2 + h(0,t)U_3^2 = 0\}$ is a smooth conic over $\mathbb{C}((t))$, hence irreducible, so S_η is irreducible, and exactly one prime component of S dominates C. This proves (1).

Step 3. In this step, we prove (2)–(6).

By **Steps 1 and 2**, it suffices to consider the points $w = [a_1 : a_2 : 1] \in \phi^{-1}(p)$, where $\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[z, t, U_1 - a_1, U_2 - a_2]]/(G)$ and the completed local ring of S at w is the ring in (9.13).

(2) If h is a unit, then $a_1a_2 = -h(0,0) \neq 0$, so $\partial_{U_2}G = U_1$ and $\partial_{U_2}(U_1U_2 + h(0,t)) = U_1$ are units at w. Thus W and S are smooth at w. The conic $\phi^{-1}(p)$ is smooth since $U_1U_2 + h(0,0)U_3^2$ has rank three.

(3) If $h = t$, then $\partial_t G = 1$ and $\partial_t(U_1U_2 + t) = 1$, so W and S are smooth at w. Since $h(0,0) = 0$, $\phi^{-1}(p) = \{U_1U_2 = 0\}$ is the union of two lines.

(4) If $h = z + t^k$, then $\partial_z G = 1$, so W is smooth at w. The surface $\{U_1U_2 + t^k = 0\}$ is smooth at w unless $a_1 = a_2 = 0$, and at $[0 : 0 : 1]$ it is a Du Val singularity of type A_{k-1}.

(5) Assume that $\operatorname{ord} h = 2$ and that the coefficient of t^2 in h is non-zero. If $(a_1, a_2) \neq (0, 0)$, then $\partial_{U_1}G = U_2$ or $\partial_{U_2}G = U_1$ is a unit at w, so W is smooth at w. Let $w := [0 : 0 : 1]$. Since $h(0,t)$ has order 2 in t, the Weierstrass preparation theorem [GP08, Corollary 6.2.8] implies that $h = c \cdot (t^2 + b_1(z)t + b_2(z))$ for a unit c and some $b_1, b_2 \in z\mathbb{C}[[z]]$. Let $T := t + \frac{1}{2}b_1(z)$ and $a(z) := b_2(z) - \frac{1}{4}b_1(z)^2$, so that $h = c \cdot (T^2 + a(z))$. Since h has no linear term, $b_2 \in z^2\mathbb{C}[[z]]$, so $a \in z^2\mathbb{C}[[z]]$. Replacing U_1 with $c^{-1}U_1$, we obtain

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[z, T, U_1, U_2]]/(U_1U_2 + T^2 + a(z)),$$

and S still corresponds to (z). If $a \neq 0$, then $a = z^m\epsilon(z)$ for an integer $m \geq 2$ and a unit ϵ, and replacing z with $\epsilon^{1/m}z$, we obtain $U_1U_2 + T^2 + z^m$, so $W \ni w$ is of type A_{m-1} [Liu25, Definition 2.12]. If $a = 0$, then $W \ni w$ is a transversal A_1 singularity [Liu25, Definition 2.15]. In this case, the singular locus of $\operatorname{Spec} \widehat{\mathcal{O}}_{W,w}$ is the curve $V(U_1, U_2, T)$, and w is the only point of $\phi^{-1}(p)$ on it. In both cases W is singular at w, and the completed local ring of S at w is $\mathbb{C}[[T, U_1, U_2]]/(U_1U_2 + T^2)$, so S has an A_1 singularity at w.

Let D be a smooth Cartier divisor on W through a point w', and let s be a local equation of D at w'. Then $\mathcal{O}_{W,w'}/(s)$ is regular of dimension two, so s and lifts of two generators of the maximal ideal of $\mathcal{O}_{W,w'}/(s)$ generate the maximal ideal of $\mathcal{O}_{W,w'}$. Since $\dim \mathcal{O}_{W,w'} = 3$, $\mathcal{O}_{W,w'}$ is regular. Thus w is not contained in any smooth Cartier divisor on W.

(6) If $h = z$, then $\partial_z G = 1$, so W is smooth at w. By (9.12),

$$S \times_Y \operatorname{Spec} \widehat{\mathcal{O}}_{Y,p} \cong \operatorname{Proj} \mathbb{C}[[t]][U_1, U_2, U_3]/(U_1U_2),$$

which has exactly two irreducible components $\{U_1 = 0\}$ and $\{U_2 = 0\}$, and they intersect $\phi^{-1}(p) = \{t = 0\}$ in the lines $\{U_1 = 0\}$ and $\{U_2 = 0\}$ of $\mathbb{P}^2$. By **Step 2**, $S_\eta \otimes_{K(C)} \mathbb{C}((t)) = \{U_1U_2 = 0\}$, so the quadratic form defining S_η has rank two, and over an algebraic closure $\overline{K(C)}$ of $K(C)$, S_η is the union of two distinct lines L_1 and L_2. The Galois group of $\overline{K(C)}/K(C)$ either exchanges L_1 and L_2, in which case S_η is irreducible, or fixes both, in which case L_1 and L_2 are defined over $K(C)$ and $S_\eta = L_1 \cup L_2$ has two irreducible components. By **Step 2**, the prime components of S correspond to the irreducible components of S_η. This proves (6). □

Lemma 9.15. *Let Y be a normal klt threefold, and let $C \subset \operatorname{Sing} Y$ be an irreducible component with $C \cong \mathbb{P}^1$. Assume that $Y \ni p$ is a hypersurface germ with $\widehat{\operatorname{vol}}(Y \ni p) > \frac{250}{33}$ for any closed point $p \in C$, and that $Y \ni q$ is a transversal A_2 singularity for a general closed point $q \in C$. Then the following hold.*

(1) $\operatorname{Bl}_C Y$ is smooth near the preimage of C, and the reduced exceptional divisor of $\operatorname{Bl}_C Y \to Y$ is $S_1 \cup S_2$, where S_1 and S_2 are smooth prime divisors, each a $\mathbb{P}^1$-bundle over C, intersecting transversally along a section over C. Moreover, $a(S_1, Y) = a(S_2, Y) = 1$.

(2) *If $f : Y \to X$ is a projective birational morphism with $K_Y = f^*K_X$ such that $f(C)$ is a point x and f has an exceptional prime divisor $E \supset C$ with $f(E) = \{x\}$, then $X \ni x$ is klt, E, S_1, and S_2 are three distinct crepant divisors over $X \ni x$, and* $\operatorname{crex}(X \ni x) \neq 2$.

Proof. The proof is long, so we divide it into several steps. Let $P := \mathbb{C}[[u, v, z, t]]$, let $\phi : W := \operatorname{Bl}_C Y \to Y$, let S be the reduced exceptional divisor of ϕ, and let $\mathcal{F} := I_C/I_C^2$.

Step 1. In this step, we show that, for any closed point $p \in C$, any isomorphism provided by Lemma 9.13 has $h = z$.

Let $p \in C$ be a closed point. By Lemma 9.13, there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong P/(uv+z^2h_p)$ which sends $I_C\widehat{\mathcal{O}}_{Y,p}$ to (u, v, z), where h_p is a unit, t, $z + t^k$ for an integer $k \geq 2$, z, or satisfies $\operatorname{ord} h_p = 2$ with non-zero coefficient of t^2 in the quadratic part of h_p. Since I_C^n/I_C^{n+1} is a finite $\mathcal{O}_{C,p}$-module and $\mathcal{O}_{Y,p} \to \mathcal{O}_{C,p}$ is surjective, the completions $(I_C^n/I_C^{n+1}) \otimes_{\mathcal{O}_{Y,p}} \widehat{\mathcal{O}}_{Y,p}$ and $(I_C^n/I_C^{n+1}) \otimes_{\mathcal{O}_{C,p}} \widehat{\mathcal{O}}_{C,p}$ coincide for any n. Thus, by (9.12) with $h = h_p$, the ring $\bigoplus_{n\geq 0}(I_C^n/I_C^{n+1}) \otimes_{\mathcal{O}_{C,p}} \widehat{\mathcal{O}}_{C,p}$ is $\mathbb{C}[[t]][U_1, U_2, U_3]/(U_1U_2 + h_p(0, t)U_3^2)$, where U_1, U_2, U_3 are the classes of u, v, z. Since p is arbitrary, $\mathcal{F}$ is locally free of rank three, and the kernel of the surjection $\operatorname{Sym}^2 \mathcal{F} \to I_C^2/I_C^3$ is a line subbundle $\mathcal{K} \subset \operatorname{Sym}^2 \mathcal{F}$, generated by $U_1U_2 + h_p(0, t)U_3^2$ after completion at p. Let $\mathcal{L} := \mathcal{K}^\vee$, and let $Q \in H^0(C, \operatorname{Sym}^2 \mathcal{F} \otimes \mathcal{L})$ correspond to the inclusion $\mathcal{K} \subset \operatorname{Sym}^2 \mathcal{F}$. Thus Q is a quadratic form on $\mathcal{F}^\vee$ with values in $\mathcal{L}$. Then $S \subset \mathbb{P}_C(\mathcal{F}) := \operatorname{Proj}_C \operatorname{Sym} \mathcal{F}$ is the zero scheme of Q, whose fiber over a closed point $q' \in C$ is the conic of isotropic lines of $Q(q')$ in $\mathcal{F}^\vee(q')$. The determinant of Q is a section $\det Q$ of a line bundle on C, and after completion at p, $\det Q$ is $-\frac{1}{4}h_p(0, t)$ up to a unit of $\widehat{\mathcal{O}}_{C,p}$.

Let $q \in C$ be a general closed point. Since $Y \ni q$ is a transversal A_2 singularity [Liu25, Definition 2.15], $\widehat{\mathcal{O}}_{Y,q} \cong P/(uv + z^3)$. By Lemma 9.11(1) and (2), the ideal $I_C\widehat{\mathcal{O}}_{Y,q}$ of the one-dimensional irreducible component $V(I_C\widehat{\mathcal{O}}_{Y,q})$ of $\operatorname{Sing}(\operatorname{Spec} \widehat{\mathcal{O}}_{Y,q})$ is (u, v, z), since z is the only prime element whose square divides z^3. Thus we may take $h_q = z$, and $\det Q$ vanishes in $\widehat{\mathcal{O}}_{C,q}$. Since C is integral, $\det Q = 0$. If $h_p(0, t) \neq 0$ for some closed point $p \in C$, then $\det Q \neq 0$ in $\widehat{\mathcal{O}}_{C,p}$, a contradiction. Among the possibilities for h_p, only $h_p = z$ satisfies $h_p(0, t) = 0$. Thus $h_p = z$ for any closed point $p \in C$.

Step 2. In this step, we prove (1).

By **Step 1** and Lemma 9.14(6), W is smooth near $\phi^{-1}(p)$ for any closed point $p \in C$, so W is smooth near $\phi^{-1}(C)$. By **Step 1**, after completion at p, Q is U_1U_2, so $Q(p)$ has rank two for any closed point $p \in C$. Thus the radical $\operatorname{rad} Q \subset \mathcal{F}^\vee$ of Q is a line subbundle, and Q induces a non-degenerate quadratic form $\overline{Q}$ on the rank-two bundle $\mathcal{F}^\vee/\operatorname{rad} Q$ with values in $\mathcal{L}$, whose discriminant is locally a unit. Let $H \to C$ be the scheme of isotropic lines of $\overline{Q}$. Locally on C, H is obtained by adjoining a square root of the discriminant of $\overline{Q}$, so $H \to C$ is finite étale of degree two. A connected finite étale double cover $D \to C \cong \mathbb{P}^1$ would satisfy $2g(D) - 2 = 2 \cdot (-2)$ by the Hurwitz formula [Har77, Corollary IV.2.4], which is impossible. Thus H is the disjoint union of two sections, which correspond to line subbundles $\mathcal{M}_1, \mathcal{M}_2 \subset \mathcal{F}^\vee/\operatorname{rad} Q$.

For $i = 1, 2$, let $\mathcal{F}_i \subset \mathcal{F}^\vee$ be the preimage of $\mathcal{M}_i$, which is a subbundle of rank two, and let $S_i \subset \mathbb{P}_C(\mathcal{F})$ be the $\mathbb{P}^1$-subbundle whose fiber over a closed point $q' \in C$ consists of the lines in $\mathcal{F}_i(q')$. Since $Q(\xi)$, for a local section ξ of $\mathcal{F}^\vee$, depends only on the class of ξ modulo $\operatorname{rad} Q$, and $\mathcal{M}_i$ is isotropic, Q vanishes on $\mathcal{F}_i$, so $S_i \subset S$. For any closed point $q' \in C$, the isotropic lines of $Q(q')$ are the lines in $\mathcal{F}_1(q') \cup \mathcal{F}_2(q')$. Since S is reduced by Lemma 9.14(1), $S = S_1 \cup S_2$. The divisors S_1 and S_2 are smooth and prime, as $\mathbb{P}^1$-bundles over C, and $S_1 \cap S_2 = \mathbb{P}_C((\operatorname{rad} Q)^\vee)$ is a section over C. Let $w \in S_1 \cap S_2$ be a point over a closed point $p \in C$. Then $w = [0 : 0 : 1]$ in the coordinates of Lemma 9.14(1), since the radical of U_1U_2 is spanned by the U_3-direction. By Lemma 9.14(1),

$$\widehat{\mathcal{O}}_{W,w} \cong \mathbb{C}[[z, t, U_1, U_2]]/(U_1U_2 + z) \cong \mathbb{C}[[t, U_1, U_2]],$$

and S corresponds to $(z) = (U_1U_2)$. Thus S_1 and S_2 intersect transversally at w. Finally, $K_W = \phi^*K_Y$ by Lemma 9.14(1), so $a(S_1, Y) = a(S_2, Y) = 1$.

Step 3. In this step, we prove (2).

By [KM98, Lemma 2.30], $a(D,X) = a(D,Y)$ for any prime divisor D over X. Since Y is klt, $X \ni x$ is klt, and the crepant divisors over $X \ni x$ are defined (Definition 2.14). We have $a(E,X) = a(E,Y) = 1$ since E is a prime divisor on Y, and $a(S_i,X) = a(S_i,Y) = 1$ for $i = 1,2$ by (1). Moreover, $\operatorname{center}_X E = f(E) = \{x\}$ and $\operatorname{center}_X S_i = f(\phi(S_i)) = f(C) = \{x\}$. The center of E on Y is the divisor E, while the center of S_i on Y is the curve C, and $S_1 \neq S_2$ on W. Thus E, S_1, and S_2 are three distinct crepant divisors over $X \ni x$, and $\operatorname{crex}(X \ni x) \geq 3$. In particular, $\operatorname{crex}(X \ni x) \neq 2$. □

9.3. **A non-normal Cartier exceptional surface.** In this subsection, we bound the local volume when $\operatorname{crex}(X \ni x) = 2$ and the surface E of Set-up 8.2 is integral and not normal. The exceptional surface of the blow-up used in the proof is a conic bundle in the following sense.

Definition 9.16. Let C be a smooth curve, let $\mathcal{F}$ and $\mathcal{L}$ be locally free sheaves on C of rank three and one, respectively, and let $Q \in H^0(C, \operatorname{Sym}^2 \mathcal{F} \otimes \mathcal{L})$. We denote by $\det Q \in H^0(C, (\det \mathcal{F})^{\otimes 2} \otimes \mathcal{L}^{\otimes 3})$ the determinant of the symmetric homomorphism $\mathcal{F}^\vee \to \mathcal{F} \otimes \mathcal{L}$ induced by Q. Assume that Q does not vanish at any closed point of C and that $\det Q \neq 0$. Then the zero scheme $S \subset \mathbb{P}_C(\mathcal{F})$ of Q, with the projection $S \to C$, is called a *conic bundle* over C, $\det Q$ is called the *discriminant* of the conic bundle, and $\operatorname{ord}_p(\det Q)$ is called the *vanishing order of the discriminant* at a closed point $p \in C$.

When the degree of E is at least 8, the degree alone does not imply the bound. If moreover the local volume of Y at any closed point of the non-normal curve of E is greater than $\frac{250}{33}$, we blow up the non-normal curve of E and contract the normalization of E. The image of the exceptional divisor of the blow-up is then a Kollár component over $X \ni x$ whose valuation has normalized volume at most 3.

Proposition 9.17. *Notation and conditions as in Set-up 8.2. Assume that* $\operatorname{crex}(X \ni x) = 2$, *that E is integral and not normal, and that Y has cDV singularities near E. Then*

$$\widehat{\operatorname{vol}}(X \ni x) \leq \frac{250}{33}.$$

Moreover, let $m := (K_E^2)$, and let C be the reduced non-normal locus of E. Assume that $m \geq 8$ and that $\widehat{\operatorname{vol}}(Y \ni p) > \frac{250}{33}$ for any closed point $p \in C$. Then the following hold.

1. *$C \cong \mathbb{P}^1$, and the blow-up $W := \operatorname{Bl}_C Y$ is normal with K_W equal to the pullback of K_Y.*
2. *The reduced exceptional divisor S of $W \to Y$ is a normal surface with only Du Val singularities, and S is a conic bundle over C.*
3. *The strict transform T of E on W is the normalization of E, and T is a smooth Hirzebruch surface.*
4. *$m = 10 - \delta$, where δ is the sum of the vanishing orders of the discriminant of the conic bundle $S \to C$.*
5. *The Cartier divisor $-2S - T$ defines a projective morphism $g : W \to Z$ over X which contracts T along the ruling, the induced morphism $h : Z \to X$ is a plt blow-up with $K_Z = h^* K_X$ and Kollár component $P = g(S)$, and $g^* P = S + \frac{1}{2}T$.*
6. *$\widehat{\operatorname{vol}}(\operatorname{ord}_P) = \frac{m+2}{4} \leq 3$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we determine the normalization of E and the singularities of E along C. In **Steps 1–6**, we assume that $m \geq 8$ and that $\widehat{\operatorname{vol}}(Y \ni p) > \frac{250}{33}$ for any closed point $p \in C$.

Since K_X is Cartier, possibly shrinking X around x, we may assume that $K_X \sim 0$. Then $K_Y = f^* K_X \sim 0$, and Y is canonical by [KM98, Lemma 2.30] and [KM98, Definition 2.34]. In particular, Y is klt. Since E is Cartier, adjunction [Kol13, (4.1.3)] implies that $\omega_E \cong \mathcal{O}_E(E)$. Since $-E$ is f-ample, E is an integral Gorenstein del Pezzo surface [Rei94, §0.1] of degree $(K_E^2) = m$.

We use the classification of non-normal del Pezzo surfaces in [Rei94, Theorem 1.5 and §1.3]. The tame assumption of [Rei94, Theorem 1.5] holds for any surface in characteristic zero, as Reid remarks after the statement of [Rei94, Theorem 1.5].

The surfaces of type (B), (C_r), and (D_r) with $r \geq 2$ in [Rei94, §1.3] have at least two irreducible components, so E is not of type (B), (C_r), or (D_r) with $r \geq 2$, since E is integral. Since $m \geq 8$, E is not one of the surfaces of degree 1 and 2 in [Rei94, §1.4], and not the surface of type (A) whose normalization is the Veronese surface, which has degree 4. Thus E is one of the following:

- the projection of $\mathbb{F}_{a;2}$ of type (A), where $\mathbb{F}_{a;2}$ denotes $\mathbb{F}_a$ embedded by $\mathcal{O}_{\mathbb{F}_a}((a+2)A + B)$ [Rei94, Theorem 1.1], with A and B a fiber and the negative section of $\mathbb{F}_a$, with normalization as in item (e) of the table of [Rei94, Theorem 1.1] and $a = m - 4$;
- the cone over a nodal rational curve of degree m, obtained in (C_1) from (c1);
- the projected scroll obtained in (C_1) from (d1), with $a = m - 2$;
- the cone over a cuspidal rational curve of degree m, obtained in (D_1) from (c2).

Since Y is cDV near E and E is Cartier, E has embedding dimension at most 4 at any point. Suppose that E is the cone over a rational curve Γ of degree m with a node or a cusp. Then Γ is a hyperplane section of $E \subset \mathbb{P}^m$ not through the vertex, where E is embedded by the complete linear system $|\omega_E^{-1}|$ [Rei94, Corollary 4.10], so Γ is embedded by $\omega_E^{-1}|_\Gamma = \mathcal{O}_\Gamma(1)$, a line bundle of degree m, as a non-degenerate curve in $\mathbb{P}^{m-1}$ (since E spans $\mathbb{P}^m$). Hence the homogeneous ideal of Γ contains no linear form, and E has embedding dimension $m \geq 8$ at the vertex. Thus E is not a cone. We call the first item *case (e)* and the third item *case (d1)*.

Let $\nu : T \to E$ be the normalization, let $D_T \subset T$ be the conductor divisor, and let A_T and B_T be a fiber and the negative section of $T \cong \mathbb{F}_a$. By [Rei94, Theorem 1.1 and §1.3], we have the following.

- In case (d1), $T \cong \mathbb{F}_a$ with $a = m - 2$, and $D_T = A_T + B_T$. The morphism ν maps both A_T and B_T isomorphically onto C, and maps $\tilde{p}_0 := A_T \cap B_T$ to a point $p_0 \in C$.
- In case (e), $T \cong \mathbb{F}_a$ with $a = m - 4$, and $D_T = B_T$. The morphism $\nu|_{B_T} : B_T \to C$ is finite of degree 2.

In case (e), the projection from a point in the plane of the conic B_T not on B_T maps B_T two-to-one onto a line, and in case (d1), the projection from a point in the plane spanned by the lines A_T and B_T not on $A_T \cup B_T$ maps $A_T \cup B_T$ onto a line [Rei94, §1.3]. Hence in both cases, C is a line in the anti-canonical embedding of E. Thus $C \cong \mathbb{P}^1$, and since $K_E \sim E|_E$,

$$(E \cdot C) = (K_E \cdot C) = -1. \tag{9.14}$$

On T, we have

$$(A_T^2) = 0, \qquad (A_T \cdot B_T) = 1, \qquad (B_T^2) = -a, \qquad K_T \sim -2B_T - (a+2)A_T$$

by [Har77, Propositions V.2.3 and V.2.9 and Lemma V.2.10].

The conductor ideal of ν is an ideal of both $\mathcal{O}_E$ and $\nu_*\mathcal{O}_T$, and it defines D_T in T [Rei94, §2.1]. By [Rei94, §2.1], the structure sheaf of the subscheme of E defined by the conductor ideal is a subring of $\nu_*\mathcal{O}_{D_T}$. Since D_T is reduced, the conductor ideal defines C in E. Thus $\mathcal{O}_E$ consists of the sections of $\nu_*\mathcal{O}_T$ whose restrictions to D_T are pulled back from $\mathcal{O}_C$. Let $p \in C$ be a closed point. We show that there exist $t, y_1, y_2 \in \widehat{\mathcal{O}}_{E,p}$, where t restricts to a local parameter of C at p and (y_1, y_2) is the ideal of C, such that one of the following holds.

(a) p has two preimages in D_T at which $\nu|_{D_T}$ is étale, and $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[t, y_1, y_2]]/(y_1 y_2)$.
(b) We are in case (e), p is a branch point of $\nu|_{B_T}$, and $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[t, y_1, y_2]]/(y_2^2 - y_1^2 t)$.
(c) We are in case (d1), $p = p_0$, and $\widehat{\mathcal{O}}_{E,p} \cong \mathbb{C}[[t, y_1, y_2]]/(y_2^2 + y_1^3 - t y_1 y_2)$.

In case (a), let $\tilde{p}_1, \tilde{p}_2 \in D_T$ be the preimages of p, and choose formal coordinates (w_i, t) of T at $\tilde{p}_i$ such that $D_T = \{w_i = 0\}$ and t is the pullback of a local parameter of C at p. Then $\widehat{\mathcal{O}}_{E,p}$ consists of the pairs $(f_1, f_2) \in \mathbb{C}[[w_1, t]] \times \mathbb{C}[[w_2, t]]$ with $f_1(0, t) = f_2(0, t)$ (completion is exact and the completion of the semilocal ring $(\nu_*\mathcal{O}_T)_p$ is the product of the completions at the points

over p, [Mat86, Theorems 8.7, 8.8, 8.15]), that is,

$$\widehat{\mathcal{O}}_{E,p} = \mathbb{C}[[t]] \oplus w_1\mathbb{C}[[w_1,t]] \oplus w_2\mathbb{C}[[w_2,t]] \cong \mathbb{C}[[t,y_1,y_2]]/(y_1y_2)$$

with $y_1 = (w_1, 0)$ and $y_2 = (0, w_2)$.

In case (b), let $\tilde{p} \in B_T$ be the preimage of p, and choose formal coordinates (w, s) of T at $\tilde{p}$ such that $B_T = \{w = 0\}$ and s^2 restricts to the pullback of a local parameter of C at p. Then $\widehat{\mathcal{O}}_{E,p}$ consists of the series $f \in \mathbb{C}[[w,s]]$ with $f(0,s) \in \mathbb{C}[[s^2]]$ (completion is exact and commutes with ν_* as in case (a), [Mat86, Theorems 8.7, 8.8, 8.15]). Let $t := s^2$, $y_1 := w$, and $y_2 := ws$. Since $\mathbb{C}[[w,s]] = \mathbb{C}[[w,s^2]] \oplus s\mathbb{C}[[w,s^2]]$, we obtain $\widehat{\mathcal{O}}_{E,p} = \mathbb{C}[[t,y_1]] \oplus y_2\mathbb{C}[[t,y_1]]$, and $y_2^2 = y_1^2 t$. The ring $\mathbb{C}[[t,y_1,y_2]]/(y_2^2 - y_1^2 t)$ is a free $\mathbb{C}[[t,y_1]]$-module with basis $1, y_2$, so the natural surjection $\mathbb{C}[[t,y_1,y_2]]/(y_2^2 - y_1^2 t) \to \widehat{\mathcal{O}}_{E,p}$ is an isomorphism.

In case (c), there exist formal coordinates (U, t) of T at $\tilde{p}_0$ such that $A_T = \{U = 0\}$, $B_T = \{U = t\}$, and t restricted to A_T and to B_T is the pullback of a local parameter of C at p_0, because A_T and B_T are smooth curves intersecting transversally at $\tilde{p}_0$ and ν maps both of them isomorphically onto C. Then

$$\widehat{\mathcal{O}}_{E,p_0} = \{f \in \mathbb{C}[[U,t]] \mid f(0,t) = f(t,t)\} = \mathbb{C}[[t]] + U(t-U)\mathbb{C}[[U,t]]$$

(completion is exact and commutes with ν_* as in case (a), [Mat86, Theorems 8.7, 8.8, 8.15]). Let $y_1 := U(t-U)$ and $y_2 := y_1 U$. Since $U^2 - tU + y_1 = 0$, the Weierstrass preparation theorem implies that $\mathbb{C}[[U,t]]$ is a free $\mathbb{C}[[t,y_1]]$-module with basis $1, U$. Hence $\widehat{\mathcal{O}}_{E,p_0} = \mathbb{C}[[t,y_1]] \oplus y_2\mathbb{C}[[t,y_1]]$. A direct calculation shows that

$$y_2^2 + y_1^3 - ty_1y_2 = y_1^2(U^2 - tU + y_1) = 0.$$

The ring $\mathbb{C}[[t,y_1,y_2]]/(y_2^2 + y_1^3 - ty_1y_2)$ is also a free $\mathbb{C}[[t,y_1]]$-module with basis $1, y_2$, so the natural surjection $\mathbb{C}[[t,y_1,y_2]]/(y_2^2 + y_1^3 - ty_1y_2) \to \widehat{\mathcal{O}}_{E,p_0}$ is an isomorphism.

In case (e), $\nu|_{B_T} : B_T \to C$ is finite of degree 2, so a point p which is not a branch point has two preimages in $D_T = B_T$, at which $\nu|_{D_T}$ is étale. In case (d1), a point $p \neq p_0$ has one preimage in A_T and one in B_T, which are distinct since $A_T \cap B_T = \{\tilde{p}_0\}$, and $\nu|_{D_T}$ is étale at both since ν maps A_T and B_T isomorphically onto C. Hence any closed point $p \in C$ is as in (a), (b), or (c). In each case, y_1 and y_2 vanish on D_T, and the quotient by (y_1, y_2) is $\mathbb{C}[[t]]$, so (y_1, y_2) is the ideal of C.

Step 2. In this step, we prove (1) and (2).

We first show that $C \subset \operatorname{Sing} Y$. Since $a(E,X) = 1$ and $\operatorname{center}_X E = \{x\}$, E is a crepant divisor over $X \ni x$. Let E' be the other crepant divisor over $X \ni x$. Then $\operatorname{center}_Y E'$ is a proper closed subset of $f^{-1}(x) = E$, and $a(E', Y) = a(E', X) = 1$ by [KM98, Lemma 2.30]. If $\operatorname{center}_Y E'$ is a point p, then $Y \ni p$ is a cDV germ with K_Y Cartier, so $\operatorname{crex}(Y \ni p) = 0$ by Theorem 2.15(1), a contradiction. Thus $\operatorname{center}_Y E'$ is a curve $\Gamma \subset E$. If Γ intersects the smooth locus of Y, then $a(E', Y) \geq 2$ by [KM98, Corollary 2.31(2)] applied on the smooth locus of Y, a contradiction. On the other hand, Y is smooth at any point of $E \backslash C$: indeed, E is Cartier and $E \backslash C \cong T \backslash D_T$ is smooth, and if R is a Noetherian local domain and $0 \neq \epsilon \in \mathfrak{m}_R$ is such that $R/(\epsilon)$ is regular, then R is regular, since $\dim R/(\epsilon) = \dim R - 1$ and the embedding dimension of R is at most the embedding dimension of $R/(\epsilon)$ plus one. Hence $\Gamma = C$, and $C \subset \operatorname{Sing} Y$. Since $\dim \operatorname{Sing} Y \leq 1$, C is an irreducible component of $\operatorname{Sing} Y$.

Let $p \in C$ be a closed point. Since Y is cDV near E, $Y \ni p$ is a klt hypersurface germ. Since $C \cong \mathbb{P}^1$ is smooth by **Step 1**, $C \subset \operatorname{Sing} Y$, and $\widehat{\operatorname{vol}}(Y \ni p) > \frac{250}{33}$, Lemma 9.13 implies that there exists an isomorphism $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[u,v,z,t]]/(uv + z^2 h_p)$ with $h_p \in \mathbb{C}[[z,t]]$ which sends $I_C\widehat{\mathcal{O}}_{Y,p}$ to (u,v,z) such that we are in one of the cases (2)–(6) of Lemma 9.14.

Since Y is locally a hypersurface in a smooth fourfold whose equation is contained in the square of the ideal of C, $\mathcal{F} := I_C/I_C^2$ is a locally free sheaf of rank 3 on C. At p, the classes U_1, U_2, U_3 of u, v, z form a basis of $\mathcal{F} \otimes \widehat{\mathcal{O}}_{C,p}$, and t is a local parameter of C at p. The multiplication

homomorphism $\operatorname{Sym}^2 \mathcal{F} \to I_C^2/I_C^3$ is surjective, and the kernel $\mathcal{K}$ of $\operatorname{Sym}^2 \mathcal{F} \to I_C^2/I_C^3$ is locally generated by the class of the local equation of Y, which is

$$U_1U_2 + h_p(0,t)U_3^2$$

at p, a non-zero element and hence a non-zero-divisor of the domain $\mathbb{C}[[t]][U_1,U_2,U_3]$. Let $\mathcal{L} := \mathcal{K}^\vee$, and let $Q \in H^0(C, \operatorname{Sym}^2 \mathcal{F} \otimes \mathcal{L})$ correspond to the inclusion $\mathcal{K} \subset \operatorname{Sym}^2 \mathcal{F}$. Then Q does not vanish at any closed point of C, and the germ of $\det Q$ at p is $h_p(0,t)$ times a unit of $\widehat{\mathcal{O}}_{C,p}$, since the symmetric matrix of $U_1U_2 + h_p(0,t)U_3^2$ in the basis U_1, U_2, U_3 has determinant $-\frac{1}{4}h_p(0,t)$.

Assume that some closed point $p \in C$ is in case (6) of Lemma 9.14, that is, $h_p = z$. Then $\det Q$ vanishes in $\widehat{\mathcal{O}}_{C,p}$, hence $\det Q = 0$ since C is integral. Thus $h_q(0,t) = 0$ for any closed point $q \in C$, and among the cases (2)–(6) of Lemma 9.14 this holds only in case (6). Hence $\widehat{\mathcal{O}}_{Y,q} \cong \mathbb{C}[[u,v,z,t]]/(uv+z^3)$ for any closed point $q \in C$, and by Lemma 2.5(2), Y has a transversal A_2 singularity [Liu25, Definition 2.15] at any closed point of C. Then Lemma 9.15(2) implies that $\operatorname{crex}(X \ni x) \neq 2$, a contradiction. Thus any closed point $p \in C$ is in one of the cases (2)–(5). Let $k_p := 0$, 1, k, and 2 in cases (2), (3), (4), and (5), respectively. Then $h_p(0,t)$ is t^{k_p} times a unit of $\mathbb{C}[[t]]$, so

$$\operatorname{ord}_p(\det Q) = k_p$$

for any closed point $p \in C$. In particular, $\det Q \neq 0$.

By Lemma 9.14(1) at any closed point of C, W is normal, $K_W = \phi^* K_Y$, where $\phi : W \to Y$ is the blow-up, and W has only hypersurface singularities near $\phi^{-1}(C)$. Together with $C \cong \mathbb{P}^1$ from **Step 1**, this proves (1). In particular, W is Gorenstein near $\phi^{-1}(C)$ and $K_W \sim 0$.

Since the associated graded ring of $\mathbb{C}[[u,v,z,t]]$ with respect to (u,v,z) is the polynomial ring $\mathbb{C}[[t]][U_1,U_2,U_3]$, which is a domain, we have $\bigoplus_{j\geq 0} I_C^j/I_C^{j+1} = \operatorname{Sym}\mathcal{F}/(Q)$. Thus the exceptional divisor $\phi^{-1}(C) = \operatorname{Proj}_C \bigoplus_{j\geq 0} I_C^j/I_C^{j+1}$ is the zero scheme of Q in the smooth threefold $\mathbb{P}_C(\mathcal{F})$.

Near the fiber over p, $\phi^{-1}(C)$ is defined by $U_1U_2 + t^{k_p}\cdot(\text{unit})\cdot U_3^2 = 0$. The surface $\phi^{-1}(C)$ is smooth on the charts $U_1 \neq 0$ and $U_2 \neq 0$. On the chart $U_3 \neq 0$, it is smooth if $k_p \leq 1$, and it has a Du Val singularity of type A_{k_p-1} at $[0:0:1]$ if $k_p \geq 2$. Thus $\phi^{-1}(C)$ has finitely many singular points, all of which are Du Val. Since $\phi^{-1}(C)$ is a hypersurface in $\mathbb{P}_C(\mathcal{F})$, it is normal by Serre's criterion. Any irreducible component of $\phi^{-1}(C)$ is two-dimensional, and it dominates C because the fibers over closed points are conics. The generic fiber is a smooth conic over the function field of C because $\det Q \neq 0$, so exactly one irreducible component dominates C. Hence $\phi^{-1}(C)$ is a normal integral surface. Thus $S = \phi^{-1}(C)$ as schemes,

$$I_C\mathcal{O}_W = \mathcal{O}_W(-S),$$

and S is a conic bundle over C in the sense of Definition 9.16, with discriminant $\det Q$ and

$$\delta = \sum_{p\in C} k_p.$$

Let $\pi := \phi|_S : S \to C$. Then π is flat [Har77, Proposition III.9.7], and $\pi^{-1}(p)$ is a smooth conic if $k_p = 0$ and the union of two distinct lines if $k_p \geq 1$. This proves (2).

Step 3. In this step, we prove (3) and compute $S|_T$ and $T|_S$.

Let $p \in C$ be a general closed point. Then $k_p = 0$, so p is in case (2) of Lemma 9.14, and p is as in (a) of **Step 1**. Let ϵ be a local equation of E at p. Since $C \subset E$, we may write $\epsilon = \alpha u + \beta v + \gamma z$ with $\alpha, \beta, \gamma \in \widehat{\mathcal{O}}_{Y,p}$. The ring $\widehat{\mathcal{O}}_{Y,p}$ has embedding dimension 4, and $\widehat{\mathcal{O}}_{E,p}$ has embedding dimension 3 by (a) of **Step 1**. Hence $\epsilon \notin \mathfrak{m}_p^2$, and $(\alpha(0),\beta(0),\gamma(0)) \neq (0,0,0)$. On the chart $U_3 \neq 0$ of Lemma 9.14(1), we have $u = zU_1$, $v = zU_2$, and $S = (z)$, so

$$\phi^*\epsilon = z(\alpha U_1 + \beta U_2 + \gamma).$$

The function $\alpha U_1 + \beta U_2 + \gamma$ restricts to the non-zero function $\alpha(0)U_1 + \beta(0)U_2 + \gamma(0)$ on the smooth affine conic $\phi^{-1}(p) \cap \{U_3 \neq 0\}$, which is not contained in a line. Thus $\alpha U_1 + \beta U_2 + \gamma$

does not vanish identically on S, and $\operatorname{ord}_S(\phi^*E) = 1$. Hence $\phi^*E - S \geq 0$ is a Cartier divisor on W which does not contain S. Since W is Cohen–Macaulay near S and ϕ is an isomorphism at the generic point of the strict transform of E, $\phi^*E - S$ is reduced, and it is the strict transform T of E. Thus

$$\phi^*E = S + T,$$

and T is Cartier.

By [Har77, Corollary II.7.15], T is the blow-up of E along $I_C\mathcal{O}_E$. We compute it with the local forms of **Step 1**, where $I_C\mathcal{O}_E$ corresponds to (y_1, y_2). For (a), (b), and (c), the equation of E, the charts of the blow-up of (y_1, y_2) containing points over p, and the inverse image of (y_1, y_2) are

(a)	$y_1y_2 = 0$,	$\{y_2 = 0\} \sqcup \{y_1 = 0\}$,	(y_1) and (y_2),
(b)	$y_2^2 = y_1^2t$,	$y_2 = y_1V,\ V^2 = t$,	(y_1),
(c)	$y_2^2 + y_1^3 = ty_1y_2$,	$y_2 = y_1U,\ y_1 = U(t - U)$,	$(U(t - U))$.

In (a), the blow-up separates the branches $\{y_2 = 0\}$ and $\{y_1 = 0\}$, which are smooth with coordinates (y_1, t) and (y_2, t). In (b) and (c), the other chart $y_1 = y_2U'$ has the equation $1 - U'^2t = 0$, resp. $1 + y_2U'^3 - tU' = 0$, which has no solution with $t = y_2 = 0$, and the chart in the table is smooth with coordinates (y_1, V), resp. (U, t). In each case, the blow-up is smooth with finitely many points over p, and the inverse image of (y_1, y_2) is the reduced preimage of C. Thus $T \to E$ is a finite birational morphism from a smooth surface, so T is the normalization of E, $T \cong \mathbb{F}_a$, and

$$I_C\mathcal{O}_T = \mathcal{O}_T(-D_T).$$

This proves (3). Since T is a smooth Cartier divisor, W is smooth at any closed point $q \in T$: the quotient of $\mathcal{O}_{W,q}$ by a local equation of T at q is regular of dimension two, so the maximal ideal of $\mathcal{O}_{W,q}$ is generated by three elements, and $\mathcal{O}_{W,q}$ is regular since $\dim \mathcal{O}_{W,q} = 3$. Since $\mathcal{O}_W(-S)|_T = I_C\mathcal{O}_T = \mathcal{O}_T(-D_T)$, we obtain

$$S|_T = D_T.$$

Let $D := T|_S$. Then $D \geq 0$ is a Cartier divisor on S, and $D = S \cap T = D_T$ as subschemes of W.

Since $\mathcal{O}_W(-S) = I_C\mathcal{O}_W$ is the tautological sheaf of the blow-up, $\mathcal{O}_W(-S)|_S \cong \mathcal{O}_{\mathbb{P}_C(\mathcal{F})}(1)|_S$. Let $\Gamma \subset S$ be a curve contained in a fiber of π. Then $(\phi^*E \cdot \Gamma) = 0$, so

$$(D \cdot \Gamma)_S = (T \cdot \Gamma) = -(S \cdot \Gamma) = \deg \mathcal{O}_{\mathbb{P}_C(\mathcal{F})}(1)|_\Gamma.$$

Let $F := \pi^*p$ for a closed point $p \in C$, a Cartier divisor on S whose linear equivalence class does not depend on p since $C \cong \mathbb{P}^1$. Since the fibers of π are conics in $\mathbb{P}^2$,

$$(D \cdot \ell)_S = 1, \qquad (D \cdot F)_S = 2, \qquad (F^2)_S = 0 \tag{9.15}$$

for any line ℓ contained in a fiber of π.

Step 4. In this step, we prove (4).

Since W is Gorenstein near $S \cup T$ and $K_W \sim 0$, adjunction [Kol13, (4.1.3)] implies that $K_S \sim S|_S$ and $K_T \sim T|_T$. Since $(S + T)|_S = (\phi^*E)|_S = \pi^*(E|_C)$, (9.14) implies

$$K_S + D \sim -F. \tag{9.16}$$

By **Step 1**, in case (d1) we have $(K_T \cdot (A_T + B_T))_T = -2 + (a - 2) = m - 6$, and in case (e) we have $(K_T \cdot B_T)_T = a - 2 = m - 6$. Hence

$$(D^2)_S = (S \cdot T \cdot T) = (D_T \cdot K_T)_T = m - 6.$$

By (9.15) and (9.16),

$$(K_S^2) = ((D + F)^2)_S = (m - 6) + 4 + 0 = m - 2. \tag{9.17}$$

Let $\rho : \tilde{S} \to S$ be the minimal resolution, and let $\tilde{\pi} := \pi \circ \rho : \tilde{S} \to C$. Since S has only Du Val singularities, $K_{\tilde{S}} = \rho^*K_S$. Let $p \in C$ be a closed point. If $k_p = 0$, then $\tilde{\pi}^{-1}(p) = \pi^{-1}(p)$ is a smooth conic. If $k_p = 1$, then $\tilde{\pi}^{-1}(p) = \pi^{-1}(p)$ is the union of two lines intersecting at one

point. If $k_p \geq 2$, then the minimal resolution of the A_{k_p-1} singularity $U_1U_2 + t^{k_p} \cdot (\text{unit}) = 0$ is a chain of $k_p - 1$ smooth rational (-2)-curves [KM98, Theorem 4.22]. The minimal resolution of the A_{k_p-1} singularity is obtained by blowing up the singular point repeatedly, and at each step, on the chart $U_1 = tU_1'$, $U_2 = tU_2'$ of the blow-up, which intersects any new exceptional curve, an equation $U_1U_2 + t^j \cdot (\text{unit}) = 0$ with $j \geq 2$ becomes $U_1'U_2' + t^{j-2} \cdot (\text{unit}) = 0$, and the new exceptional curves are the components of the reduced curve $\{t = 0\}$. Hence t vanishes to order one along any of the $k_p - 1$ curves. Thus, for $k_p \geq 1$, the fiber $\tilde{\pi}^* p$ is a reduced chain of $k_p + 1$ smooth rational curves.

A reduced chain of smooth rational curves has arithmetic genus 0, so $H^1(\tilde{\pi}^{-1}(p), \mathcal{O}) = 0$ for any closed point $p \in C$. Since $\tilde{\pi}$ is flat [Har77, Proposition III.9.7] and the fibers are connected and reduced, $\tilde{\pi}_* \mathcal{O}_{\tilde{S}} = \mathcal{O}_C$ and $R^1\tilde{\pi}_* \mathcal{O}_{\tilde{S}} = 0$ by [Har77, Corollary III.12.9], hence $\chi(\mathcal{O}_{\tilde{S}}) = \chi(\mathcal{O}_{\mathbb{P}^1}) = 1$ by [Har77, Exercise III.8.1]. Over the complement of the finitely many points p with $k_p \geq 1$, $\tilde{\pi}$ is a $\mathbb{P}^1$-bundle. The topological Euler number is additive for the decomposition of $\tilde{S}$ into the $\mathbb{P}^1$-bundle and the singular fibers, and multiplicative for the $\mathbb{P}^1$-bundle, and a chain of $k_p + 1$ smooth rational curves has topological Euler number $k_p + 2$. Thus

$$e_{\text{top}}(\tilde{S}) = 2 \cdot 2 + \sum_{p \in C} k_p = 4 + \delta.$$

By Noether's formula [Har77, Remark V.1.6.1], where $c_2 = e_{\text{top}}$ for a smooth projective surface,

$$(K_S^2) = (K_{\tilde{S}}^2) = 12\chi(\mathcal{O}_{\tilde{S}}) - e_{\text{top}}(\tilde{S}) = 8 - \delta.$$

By (9.17), $m = 10 - \delta$. This proves (4).

Step 5. In this step, we construct g and h, and we compute g^*P.

Let $M := -2S - T$, which is a Cartier divisor on W. By **Steps 1 and 3**, in both cases (d1) and (e),

$$M|_T \sim -2D_T - K_T \sim (m-2)A_T,$$

and by (9.16),

$$M|_S \sim -2K_S - D \sim D + 2F.$$

We show that M is nef over X and that $(M \cdot \Gamma) = 0$ for a curve Γ contracted by $f \circ \phi$ if and only if Γ is a fiber of the ruling of T. Such a curve Γ is contained in $(f \circ \phi)^{-1}(x) = S \cup T$. If $\Gamma \subset T$, then $(M \cdot \Gamma) = (m-2)(A_T \cdot \Gamma)_T \geq 0$, with equality if and only if Γ is a fiber of the ruling. Assume that $\Gamma \subset S$ and $\Gamma \not\subset T$. If $\pi(\Gamma)$ is a point, then $(M \cdot \Gamma) = (D \cdot \Gamma)_S > 0$ by (9.15). If $\pi(\Gamma) = C$, then $(D \cdot \Gamma)_S \geq 0$ since $\Gamma \not\subset D$, and $(F \cdot \Gamma)_S > 0$, so $(M \cdot \Gamma) > 0$. This proves the claim on M, since $m - 2 > 0$.

By [KM98, Lemma 2.30], W is canonical, hence klt. The Cartier divisor M is nef over X, and $M - K_W \sim M$ is nef and big over X since $f \circ \phi$ is birational. By [Fuj12, Theorem 2.1] with zero boundary and $r = 1$, M is semi-ample over X. Thus there exist a positive integer n, a projective morphism $g : W \to Z$ over X with $g_* \mathcal{O}_W = \mathcal{O}_Z$, and an h-ample Cartier divisor H on Z, where $h : Z \to X$ is the induced morphism, such that $g^*H \sim nM$. Since W is normal, Z is normal. We have the commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g} & Z \\ {\scriptstyle\phi}\downarrow & & \downarrow{\scriptstyle h} \\ Y & \xrightarrow[f]{} & X \end{array}$$

and h is a projective birational morphism which is an isomorphism over $X \backslash \{x\}$.

A curve on W is contracted by g if and only if it is contracted by $f \circ \phi$ and has zero intersection with M, that is, if and only if it is a fiber of the ruling of T. Thus g contracts T along the ruling. The restriction $g|_S$ contracts no curve in case (e), and contracts only A_T in case (d1). Let $P := g(S)$, which is a prime divisor on Z. Since any fiber of the ruling of T intersects $B_T \subset S$,

we have $g(T) \subset P$. Hence $h^{-1}(x) = g(S \cup T) = P$ as sets. Since $g_*M = -2P$ and $g_*g^*H = H$, we obtain $H \sim -2nP$. Thus P is $\mathbb{Q}$-Cartier and $-P$ is h-ample.

Write $g^*P = S + cT$ for some $c \in \mathbb{Q}$. Let ℓ_T be a fiber of the ruling of T. Then

$$(S \cdot \ell_T) = (D_T \cdot A_T)_T = 1, \qquad (T \cdot \ell_T) = (K_T \cdot A_T)_T = -2.$$

Since g contracts ℓ_T, $0 = (g^*P \cdot \ell_T) = 1 - 2c$. Thus

$$g^*P = S + \frac{1}{2}T.$$

Since $K_W = (f \circ \phi)^*K_X$, we obtain $K_Z = g_*K_W = h^*K_X$, and $K_W = g^*K_Z$.

Step 6. In this step, we prove that h is a plt blow-up, which finishes the proof of (5), and we prove (6).

Since S is a normal Cartier divisor on W with only Du Val singularities, (W, S) is plt near S by inversion of adjunction [KM98, Theorem 5.50]. Since W is smooth along T and S is Cartier, $(K_W + T + S)|_T = K_T + D_T$. The pair (T, D_T) is lc, since D_T is the smooth curve B_T in case (e) and the simple normal crossing curve $A_T + B_T$ in case (d1). Hence $(W, S + T)$ is lc near T by inversion of adjunction [Kaw07, Theorem]. Since $a(G, W, \cdot)$ is affine in the boundary for any prime divisor G over W,

$$a\Big(G, W, S + \frac{1}{2}T\Big) = \frac{1}{2}a(G, W, S) + \frac{1}{2}a(G, W, S + T).$$

Near $S \cap T$, the right-hand side is positive for any prime divisor G exceptional over W, because (W, S) is plt and $(W, S + T)$ is lc there. Near $T \backslash S$, $(W, \frac{1}{2}T)$ is klt because W is smooth and T is a smooth divisor with coefficient $\frac{1}{2} < 1$. Near $S \backslash T$, $(W, S + \frac{1}{2}T) = (W, S)$ is plt. Thus $(W, S + \frac{1}{2}T)$ is plt near $S \cup T$.

By **Step 5**, $K_W + S + \frac{1}{2}T = g^*(K_Z + P)$ and $g_*(S + \frac{1}{2}T) = P$. Hence $a(G, Z, P) = a(G, W, S + \frac{1}{2}T)$ for any prime divisor G over Z [KM98, Lemma 2.30]. In particular, $a(T, Z, P) = \frac{1}{2}$, and (Z, P) is plt near $P = h^{-1}(x)$. Since h is proper, possibly shrinking X around x, we may assume that (Z, P) is plt. Thus h is a plt blow-up of $X \ni x$ with Kollár component P (Definition 2.18), and $K_Z = h^*K_X$. This proves (5).

Let $\tau := g|_S : S \to P$, which is a birational morphism between projective surfaces. By adjunction,

$$\tau^*(K_P + \mathrm{Diff}_P(0)) \sim_{\mathbb{Q}} (g^*(K_Z + P))|_S = \Big(K_W + S + \frac{1}{2}T\Big)\Big|_S = K_S + \frac{1}{2}D.$$

By Lemma 2.19(1), the projection formula for the birational morphism τ, (9.16), (9.15), and **Step 4**,

$$\widehat{\mathrm{vol}}(\mathrm{ord}_P) = ((K_P + \mathrm{Diff}_P(0))^2) = \Big(\Big(K_S + \frac{1}{2}D\Big)^2\Big)_S = \Big(\Big(F + \frac{1}{2}D\Big)^2\Big)_S = 2 + \frac{m-6}{4} = \frac{m+2}{4}.$$

Since $m = 10 - \delta \le 10$, $\widehat{\mathrm{vol}}(\mathrm{ord}_P) \le 3$. This proves (6).

Step 7. In this step, we prove that $\widehat{\mathrm{vol}}(X \ni x) \le \frac{250}{33}$, without the assumptions of **Steps 1–6**.

By Lemma 2.19(2), $\widehat{\mathrm{vol}}(X \ni x) \le (K_E^2) = m$. Since $7 < \frac{250}{33}$, we may assume that $m \ge 8$. The variety Y is klt by [KM98, Lemma 2.30], and any closed point $p \in C$ is contained in $E = \mathrm{Exc}(f)$. If $\widehat{\mathrm{vol}}(Y \ni p) \le \frac{250}{33}$ for some closed point $p \in C$, then $\widehat{\mathrm{vol}}(X \ni x) < \widehat{\mathrm{vol}}(Y \ni p) \le \frac{250}{33}$ by Proposition 2.7(3). Otherwise, h and P as in (5) exist, and by (6) and Definition 2.6,

$$\widehat{\mathrm{vol}}(X \ni x) \le \widehat{\mathrm{vol}}(\mathrm{ord}_P) \le 3 < \frac{250}{33}.$$

□

9.4. **Normal exceptional surfaces and non-cDV points.** In this subsection, we prove Propositions 9.20 and 9.21. In both proofs, the exceptional surface is a cone over a curve of arithmetic genus one, and at the vertex of the cone, Y is analytically isomorphic to the anti-canonical cone over $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or $\mathbb{F}_1$. Lemma 9.18 describes the blow-up of such a point, and Lemma 9.19, the common last step of both proofs, constructs from it a Kollár component over $X \ni x$ of normalized volume at most $\frac{9}{4}$.

Lemma 9.18. *Let Y be a normal threefold, and let $y \in Y$ be a closed point such that $Y \ni y$ is analytically isomorphic to $C_F \ni o$, where F is $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or $\mathbb{F}_1$. Let $d := (K_F^2)$, and let $\rho : W \to Y$ be the blow-up of y. Then the following hold.*

(1) W is smooth near $\rho^{-1}(y)$, $K_W = \rho^* K_Y$, Y has embedding dimension $d+1$ at y, and the exceptional divisor of ρ is isomorphic to F. Identifying it with F, we have $\mathcal{O}_W(F)|_F \cong \mathcal{O}_F(K_F)$.

(2) For any integral Cartier divisor $E \subset Y$ which is the projective cone, with vertex y, over an integral curve D of arithmetic genus one embedded in a projective space by a complete linear system of degree at least 3, we have $\rho^* E = F + T$, where $T := \rho_*^{-1} E$. Moreover, D has degree d, $C := T \cap F$ is an integral member of $|-K_F|$ which is isomorphic to D, and $T \cong \mathbb{P}_C(\mathcal{O}_C \oplus \mathcal{O}_C(K_F|_C))$ with C as the negative section.

Proof. Let $A_F := \bigoplus_{m \geq 0} H^0(F, \mathcal{O}_F(-mK_F))$, so that $C_F = \operatorname{Spec} A_F$. Since F is a smooth projective toric surface, A_F is generated by $(A_F)_1 = H^0(F, \mathcal{O}_F(-K_F))$ [CLS11, Proposition 4.3.3 and Corollary 2.2.13]. By [Kol13, Section 3.1 (3.8)], the morphism $\operatorname{Spec}_F \bigoplus_{m \geq 0} \mathcal{O}_F(-mK_F) \to C_F$ is the weighted blow-up of the vertex o, and its exceptional divisor is isomorphic to F. Since A_F is generated in degree one, the j-th power of the ideal $\bigoplus_{i \geq 1} (A_F)_i$ of o is $\bigoplus_{i \geq j} (A_F)_i$ for any $j \geq 1$, so $\operatorname{Spec}_F \bigoplus_{m \geq 0} \mathcal{O}_F(-mK_F) \to C_F$ is the blow-up $\rho_F : W_F \to C_F$ of o. Moreover, W_F is the total space of the line bundle $\mathcal{O}_F(K_F)$, and the exceptional divisor of ρ_F is the zero section, which has normal bundle $\mathcal{O}_F(K_F)$. In particular, W_F is smooth, and ω_{W_F} is trivial, being the pullback of $\omega_F \otimes \mathcal{O}_F(-K_F) \cong \mathcal{O}_F$. Since ρ_F is an isomorphism over $C_F \backslash \{o\}$ and C_F is normal, ω_{C_F} is trivial as well. Moreover, A_F is a domain, $\dim(A_F)_1 = h^0(F, \mathcal{O}_F(-K_F)) = d+1$ by the Riemann–Roch theorem and the Kodaira vanishing theorem, and $\operatorname{gr}_{\mathfrak{m}_o} \mathcal{O}_{C_F,o} \cong A_F$ as graded rings, where $\mathfrak{m}_o$ is the maximal ideal of $\mathcal{O}_{C_F,o}$.

We prove (1). Let $\mathfrak{m}_y$ be the maximal ideal of $\mathcal{O}_{Y,y}$. By Definition 2.4, $\widehat{\mathcal{O}}_{Y,y} \cong \widehat{\mathcal{O}}_{C_F,o}$. Thus $\operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{Y,y} \cong A_F$. Since $\dim(A_F)_1 = d+1$, Y has embedding dimension $d+1$ at y. Blowing up commutes with flat base change [Sta, Tag 0805]. Hence the base change W' of ρ along $\operatorname{Spec} \widehat{\mathcal{O}}_{Y,y} \to Y$ is also the base change of ρ_F along $\operatorname{Spec} \widehat{\mathcal{O}}_{C_F,o} \to C_F$, both being the blow-up of the closed point of $\operatorname{Spec} \widehat{\mathcal{O}}_{Y,y} \cong \operatorname{Spec} \widehat{\mathcal{O}}_{C_F,o}$. The local rings $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{C_F,o}$ are excellent, so the completion morphisms are faithfully flat with geometrically regular fibers.

- Since $\mathcal{O}_{Y,y}/\mathfrak{m}_y^j \cong \widehat{\mathcal{O}}_{Y,y}/\widehat{\mathfrak{m}}_y^j$ for any $j \geq 1$, where $\widehat{\mathfrak{m}}_y := \mathfrak{m}_y \widehat{\mathcal{O}}_{Y,y}$, the morphisms $W' \to W$ and $W' \to W_F$ induce an isomorphism between $\rho^{-1}(y)$ and $\rho_F^{-1}(o)$, under which $\mathcal{O}_W(-jF)|_{\rho^{-1}(y)}$ corresponds to $\mathcal{O}_{W_F}(-jF)|_F$ for any $j \geq 1$. Thus the exceptional divisor of ρ is isomorphic to F, and $\mathcal{O}_W(F)|_F \cong \mathcal{O}_F(K_F)$.
- Since W_F is smooth and $\operatorname{Spec} \widehat{\mathcal{O}}_{C_F,o} \to C_F$ is flat with geometrically regular fibers, W' is regular at any point over F [Mat86, Theorem 23.7(ii)]. Since $W' \to W$ is flat, W is regular at any point of F [Mat86, Theorem 23.7(i)]. As the regular locus of W is open, W is smooth near $\rho^{-1}(y)$.
- Since ρ_F is a resolution with ω_{W_F} and ω_{C_F} trivial (the discrepancy of the zero section is 0, by the adjunction argument below applied to ρ_F), C_F is canonical, hence Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. As ω_{C_F} is trivial, $\widehat{\mathcal{O}}_{C_F,o} \cong \widehat{\mathcal{O}}_{Y,y}$ is Gorenstein. The morphism $\mathcal{O}_{Y,y} \to \widehat{\mathcal{O}}_{Y,y}$ is flat and its closed fiber is the residue field, so $\mathcal{O}_{Y,y}$ is Gorenstein [Mat86, Theorem 23.4]. Thus K_Y is Cartier near y.

Write $K_W = \rho^* K_Y + aF$ near F for an integer a. Since $(\rho^* K_Y)|_F \sim 0$, adjunction implies

$$K_F \sim (K_W + F)|_F \sim (a+1)F|_F \sim (a+1)K_F.$$

As $(K_F^2) = d > 0$, we have $a = 0$. Since ρ is an isomorphism over $Y \backslash \{y\}$, $K_W = \rho^* K_Y$. This proves (1).

We prove (2). Let s be a local equation of E at y, let $k \geq 1$ be the integer such that $s \in \mathfrak{m}_y^k \backslash \mathfrak{m}_y^{k+1}$, and let $s_k \in (A_F)_k$ be the class of s in $\mathfrak{m}_y^k / \mathfrak{m}_y^{k+1}$. For any non-zero $u \in \mathcal{O}_{Y,y}$ whose class in $\operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{Y,y}$ is $u_j \in (A_F)_j$, the class of us is $u_j s_k \neq 0$, since A_F is a domain. Hence

$$\operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{E,y} \cong A_F/(s_k). \tag{9.18}$$

Since $\dim (A_F)_j = h^0(F, \mathcal{O}_F(-jK_F)) = 1 + \frac{j(j+1)}{2} d$ for any $j \geq 0$, (9.18) implies that $\operatorname{mult}_y E = kd$. Moreover, writing $\operatorname{edim}(Z, z) := \dim_{\mathbb{C}} \mathfrak{m}_{Z,z}/\mathfrak{m}_{Z,z}^2$ for the embedding dimension of a scheme Z at a closed point z, where $\mathfrak{m}_{Z,z}$ is the maximal ideal of $\mathcal{O}_{Z,z}$, we have $\operatorname{edim}(E, y) = \dim (A_F/(s_k))_1$, which is d if $k = 1$ and $d+1$ if $k \geq 2$.

On the other hand, let $D \subset \mathbb{P}^N$ be the embedding in the statement, given by the complete linear system $|\mathcal{O}_D(1)|$ with $\deg D \geq 3$, so that E is the projective cone over D with vertex y. Let ω_D° be the dualizing sheaf of D [Har77, Proposition III.7.5]. For any closed point $q \in D$, the definition of the dualizing sheaf implies $\operatorname{Hom}(k(q), \omega_D^\circ) \cong H^1(D, k(q))^\vee = 0$, so ω_D° has no non-zero subsheaf supported at points. Hence any non-zero homomorphism $\mathcal{O}_D(1) \to \omega_D^\circ$ is injective, and it would imply

$$h^0(D, \mathcal{O}_D(1)) \leq h^0(D, \omega_D^\circ) = h^1(D, \mathcal{O}_D) = p_a(D) = 1,$$

which contradicts $h^0(D, \mathcal{O}_D(1)) \geq 2$. Thus $H^1(D, \mathcal{O}_D(1)) \cong \operatorname{Hom}(\mathcal{O}_D(1), \omega_D^\circ)^\vee = 0$, and by [Har77, Exercise IV.1.9(a)(c)],

$$h^0(D, \mathcal{O}_D(1)) = \chi(D, \mathcal{O}_D(1)) = \deg D + 1 - p_a(D) = \deg D.$$

Thus $N = \deg D - 1$. Since E is the affine cone over $D \subset \mathbb{P}^{\deg D - 1}$ near y, the ring $\operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{E,y}$ is the homogeneous coordinate ring of $D \subset \mathbb{P}^{\deg D - 1}$, which has no linear relations since $|\mathcal{O}_D(1)|$ is complete. Hence

$$\operatorname{mult}_y E = \operatorname{edim}(E, y) = \deg D.$$

If $k \geq 2$, then $kd = d + 1$, which is impossible since $d \in \{8, 9\}$. Thus $k = 1$, and $\deg D = d$.

Near F, we have $\mathfrak{m}_y^j \mathcal{O}_W = \mathcal{O}_W(-jF)$ for any $j \geq 0$, and by (1), restriction to F induces isomorphisms

$$\mathfrak{m}_y^j / \mathfrak{m}_y^{j+1} \cong H^0(F, \mathcal{O}_W(-jF)|_F) = (A_F)_j.$$

Since $s \in \mathfrak{m}_y \backslash \mathfrak{m}_y^2$, the pullback $\rho^* s$ vanishes along F to order exactly one, and the restriction of $\rho^* s$, viewed as a section of $\mathcal{O}_W(-F)$, to F is s_1. Hence $\rho^* E = F + T$, and $C = T \cap F$ is the divisor $V_F(s_1) \in |-K_F|$.

By [Har77, Corollary II.7.15] and the definition following it, T is the blow-up of E at y, and $C = T \cap \rho^{-1}(y)$ is the exceptional curve of $T \to E$. Thus $C \cong \operatorname{Proj} \operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{E,y}$. By (9.18), $\operatorname{Proj}(A_F/(s_1))$ is the curve $V_F(s_1) = C \subset F$, and by the previous paragraph, $\operatorname{Proj} \operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{E,y}$ is D. Therefore there exists an isomorphism $C \cong D$ under which $\mathcal{O}_C(-K_F|_C)$ corresponds to $\mathcal{O}_D(1)$, since under (9.18) both are the sheaf $\mathcal{O}(1)$ of the graded ring $\operatorname{gr}_{\mathfrak{m}_y} \mathcal{O}_{E,y}$. In particular, C is integral of arithmetic genus one. By [Har77, Example V.2.11.4], whose proof is valid for any closed subvariety of a projective space, the blow-up of the vertex of the projective cone over $D \subset \mathbb{P}^{d-1}$ is $\mathbb{P}_D(\mathcal{O}_D \oplus \mathcal{O}_D(-1))$, and the inverse image of the vertex is the negative section. Hence $T \cong \mathbb{P}_C(\mathcal{O}_C \oplus \mathcal{O}_C(K_F|_C))$ with C as the negative section. □

Lemma 9.19. *Let $X \ni x$ be a Gorenstein canonical threefold germ, and let $f : Y \to X$ be a projective birational morphism from a normal variety Y with $K_Y = f^* K_X$ which is an isomorphism over $X \backslash \{x\}$, such that $E := f^{-1}(x)_{\mathrm{red}}$ is an integral Cartier divisor and $-E$ is f-ample. Let $y \in E$ be a closed point such that $Y \ni y$ is analytically isomorphic to $C_F \ni o$, where F is $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or $\mathbb{F}_1$, and let $d := (K_F^2)$. Let $\rho : W \to Y$ be the blow-up of y, identify the*

exceptional divisor of ρ with F, and let $T := \rho_*^{-1}E$. Assume that $\rho^*E = F+T$, that $C := T\cap F$ is an integral curve of arithmetic genus one in $|-K_F|$, and that $T \cong \mathbb{P}_C(\mathcal{O}_C \oplus \mathcal{O}_C(K_F|_C))$ with C as the negative section. Then there exist a normal variety Z and projective morphisms $g : W \to Z$ and $h : Z \to X$ with $h\circ g = f\circ\rho$, such that g contracts T along the ruling and contracts no curve on F, h is a plt blow-up with $K_Z = h^*K_X$ whose Kollár component is $P := g(F)$, $g|_F : F \to P$ is an isomorphism, $g^*P = F + \frac{1}{2}T$, and $\mathrm{Diff}_P(0) = \frac{1}{2}g(C)$. In particular,

$$\widehat{\mathrm{vol}}(\mathrm{ord}_P) = \frac{d}{4} \le \frac{9}{4}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct g and h.

Since K_X is Cartier and the statement is local around x, possibly shrinking X around x, we may assume that $K_X \sim 0$. Then $K_Y = f^*K_X \sim 0$. By Lemma 9.18(1), W is smooth near F, $K_W = \rho^*K_Y \sim 0$, and $\mathcal{O}_W(F)|_F \cong \mathcal{O}_F(K_F)$. Since $\rho^*E = F+T$ and E is Cartier, T is a Cartier divisor on W. Let $p : T \to C$ be the $\mathbb{P}^1$-bundle, and for any $c \in C$, let $\ell_c := p^{-1}(c)$. Let $L := -K_F|_C$, a Cartier divisor on C with

$$\deg L = (-K_F \cdot C)_F = (K_F^2) = d.$$

As $C \sim -K_F$, adjunction implies $\omega_C \cong \mathcal{O}_C$ [KM98, Proposition 5.73]. Since p is smooth and projective of relative dimension one, we have $\omega_T \cong \Omega_{T/C} \otimes p^*\omega_C$ [Sta, Tag 0AU3, (1) and (9)]. Let $\mathcal{E} := \mathcal{O}_C \oplus \mathcal{O}_C(-L)$, so that $T = \mathbb{P}_C(\mathcal{E})$. Then $\Omega_{T/C} \cong p^*(\wedge^2\mathcal{E}) \otimes \mathcal{O}_T(-2)$ [Har77, Exercise III.8.4(b)], $\wedge^2\mathcal{E} = \mathcal{O}_C(-L)$, and $\mathcal{O}_T(1) \cong \mathcal{O}_T(C)$, since C is the zero locus of the section of $\mathcal{O}_T(1)$ given by the direct summand $\mathcal{O}_C \subset \mathcal{E}$. Hence

$$\omega_T \cong \Omega_{T/C} \otimes p^*\omega_C \cong \mathcal{O}_T(-2C) \otimes p^*\mathcal{O}_C(-L).$$

On the other hand, Y is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10], hence so is W, as W is smooth near F and isomorphic to $Y\backslash\{y\}$ away from F, and adjunction implies $\omega_T \cong \mathcal{O}_W(K_W+T)|_T \cong \mathcal{O}_W(T)|_T$ [Kol13, (4.1.3)]. Thus

$$T|_T \sim -2C - p^*L, \tag{9.19}$$

and moreover $F|_T = C$ and $T|_F = C$ as Cartier divisors.

Let $M := -2F - T$. By (9.19),

$$M|_F \sim -2K_F - C \sim -K_F, \qquad M|_T \sim -2C + 2C + p^*L = p^*L.$$

The support of $(f\circ\rho)^{-1}(x)$ is $F\cup T$. For any curve $\Gamma \subset F$ and any curve $\Gamma' \subset T$ which is not contained in F, we have

$$(M\cdot\Gamma) = (-K_F\cdot\Gamma)_F > 0, \qquad (M\cdot\Gamma') = (p^*L\cdot\Gamma')_T \ge 0,$$

where degrees on curves in T are computed on the normalizations of the curves, and since $\deg L > 0$, $(M\cdot\Gamma') = 0$ if and only if $\Gamma' = \ell_c$ for some $c \in C$. Thus M is $(f\circ\rho)$-nef. Since $K_W = (f\circ\rho)^*K_X$ and X is canonical, W is canonical [KM98, Lemma 2.30], hence klt. The divisor $M - K_W$ is $(f\circ\rho)$-nef, since K_W is the pullback of a divisor on X, and $(f\circ\rho)$-big, since $f\circ\rho$ is birational. By the relative base-point-free theorem [KM98, Theorem 3.24], bM is $(f\circ\rho)$-free for any $b \gg 0$. Let $g : W \to Z$ be the Stein factorization of the morphism over X defined by bM for some $b \gg 0$, and let $h : Z \to X$ be the induced morphism. Then Z is normal, $g_*\mathcal{O}_W = \mathcal{O}_Z$, h is projective, and there exists a positive integer n such that nM is linearly equivalent to the pullback of an h-ample Cartier divisor on Z. We obtain the commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g} & Z \\ {\scriptstyle\rho}\downarrow & & \downarrow{\scriptstyle h} \\ Y & \xrightarrow[f]{} & X. \end{array}$$

A curve in W is contracted by g if and only if it is contained in $(f\circ\rho)^{-1}(x)$ and has M-degree zero, that is, if and only if it is ℓ_c for some $c\in C$. So g contracts T along the ruling and contracts no curve on F.

Since g is birational with finite fibers over $Z\backslash g(T)$ and Z is normal, g is an isomorphism over $Z\backslash g(T)$ by Zariski's Main Theorem. In particular, h is an isomorphism over $X\backslash\{x\}$. For any $c\in C$, the fiber $g^{-1}(g(c))$ is connected, consists of curves $\ell_{c'}$ and finitely many points of F, and any point of F in $g^{-1}(g(c))$ is contained in some $\ell_{c'}$ by connectedness. Since the curves $\ell_{c'}$ are pairwise disjoint, $g^{-1}(g(c))=\ell_c$, and $\ell_c\cap F=\{c\}$. Thus $g|_F: F\to P$ is finite, injective, and birational. Moreover, $g(T)=g(C)\subset P$, so $h^{-1}(x)=g(F\cup T)=P$.

Step 2. In this step, we show that $-P$ is $\mathbb{Q}$-Cartier and h-ample, and compute g^*P and K_Z.

Since $g_*F=P$ and $g_*T=0$, we have $g_*(nM)=-2nP$. Thus the h-ample Cartier divisor whose pullback is linearly equivalent to nM is linearly equivalent to $-2nP$. Hence $2nP$ is Cartier, and $-P$ is h-ample. The only g-exceptional prime divisor is T, so $g^*P=F+aT$ for some $a\in\mathbb{Q}$. By (9.19), $(T\cdot\ell_c)=((-2C-p^*L)\cdot\ell_c)_T=-2$, and $(F\cdot\ell_c)=(C\cdot\ell_c)_T=1$. Since ℓ_c is contracted by g,

$$0=(g^*P\cdot\ell_c)=(F\cdot\ell_c)+a(T\cdot\ell_c)=1-2a.$$

Thus $g^*P=F+\frac{1}{2}T$. Since $K_W=(f\circ\rho)^*K_X=g^*h^*K_X$, we have $K_Z=g_*K_W=h^*K_X$ and $K_W=g^*K_Z$. Therefore

$$K_W+F+\tfrac{1}{2}T=g^*(K_Z+P). \tag{9.20}$$

Step 3. In this step, we show that (Z,P) is plt.

We first show that $(F,\frac{1}{2}C)$ is klt. Let $c\in C$, let $m:=\operatorname{mult}_c C$, let $\sigma: F'\to F$ be the blow-up of c with exceptional curve G_0, and let C' be the strict transform of C. Then $C'\sim\sigma^*C-mG_0$ and $K_{F'}\sim\sigma^*K_F+G_0$. By the adjunction formula [Har77, Exercise V.1.3(a)],

$$\begin{aligned}2p_a(C')-2&=(C'\cdot(C'+K_{F'}))\\&=(C\cdot(C+K_F))-m(m-1)=2p_a(C)-2-m(m-1)=-m(m-1).\end{aligned}$$

Since C' is integral, $p_a(C')\geq 0$, so $m\leq 2$. Thus $\operatorname{mult}_c(\frac{1}{2}C)\leq 1$ for any $c\in C$, and $(F,\frac{1}{2}C)$ is canonical [KM98, Theorem 4.5(2)]. As the coefficient of C is $\frac{1}{2}<1$, $(F,\frac{1}{2}C)$ is klt.

Since W is smooth near F, F is a smooth Cartier divisor, and $T|_F=C$, inversion of adjunction [KM98, Theorem 5.50(1)] implies that $(W,F+\frac{1}{2}T)$ is plt near F. In particular, there exists an open neighborhood U of F such that $\operatorname{Nklt}(W,F+\frac{1}{2}T)\cap U=F$, where $\operatorname{Nklt}(W,F+\frac{1}{2}T)$ denotes the non-klt locus of $(W,F+\frac{1}{2}T)$, cf. [KM98, Corollary 5.49]. Away from $(f\circ\rho)^{-1}(x)$, the pair $(W,F+\frac{1}{2}T)$ is isomorphic to $(X\backslash\{x\},0)$, which is canonical.

It remains to show that $(W,F+\frac{1}{2}T)$ is klt at any point of $T\backslash C$. By (9.20), $-(K_W+F+\frac{1}{2}T)$ is g-trivial, hence g-nef, and $g_*(F+\frac{1}{2}T)=P\geq 0$. By the connectedness lemma [KM98, Corollary 5.49], $\operatorname{Nklt}(W,F+\frac{1}{2}T)$ is connected in a neighborhood of any fiber of g. Since g is proper, $\operatorname{Nklt}(W,F+\frac{1}{2}T)\cap g^{-1}(z)$ is connected for any $z\in Z$. Otherwise, it is the union of two disjoint non-empty closed subsets, and for disjoint open neighborhoods U_1 and U_2 of them, the properness of g implies that $\operatorname{Nklt}(W,F+\frac{1}{2}T)\cap g^{-1}(V)$ is contained in $U_1\cup U_2$ and intersects both U_1 and U_2 for any sufficiently small neighborhood V of z, a contradiction. Let $c\in C$. By **Step 1**, $g^{-1}(g(c))=\ell_c\cong\mathbb{P}^1$, so $\operatorname{Nklt}(W,F+\frac{1}{2}T)\cap\ell_c$ is a connected closed subset of ℓ_c containing c. It is not ℓ_c, since $\operatorname{Nklt}(W,F+\frac{1}{2}T)\cap\ell_c\cap U=\{c\}$. Thus $\operatorname{Nklt}(W,F+\frac{1}{2}T)\cap\ell_c=\{c\}$. Since $T=\bigcup_{c\in C}\ell_c$, the pair $(W,F+\frac{1}{2}T)$ is klt at any point of $T\backslash C$. Therefore $(W,F+\frac{1}{2}T)$ is plt.

By (9.20) and $g_*(F+\frac{1}{2}T)=P$, we have $a(G,W,F+\frac{1}{2}T)=a(G,Z,P)$ for any prime divisor G over Z [KM98, Lemma 2.30]. The divisor T is exceptional over Z with $a(T,Z,P)=\frac{1}{2}$, and any other prime divisor exceptional over Z is exceptional over W. Hence (Z,P) is plt. Together with **Steps 1** and 2, h is a plt blow-up of $X\ni x$ with Kollár component P, and $K_Z=h^*K_X$. By Definition 2.18, P is normal. Since $g|_F: F\to P$ is finite and birational, $g|_F$ is an isomorphism by Zariski's Main Theorem.

Step 4. In this step, we compute $\mathrm{Diff}_P(0)$ and $\widehat{\mathrm{vol}}(\mathrm{ord}_P)$.

Restricting (9.20) to F, and using $(K_W+F)|_F \sim K_F$ and $T|_F = C$, we obtain

$$K_F + \tfrac{1}{2}C = (g|_F)^*\big(K_P + \mathrm{Diff}_P(0)\big) = K_F + (g|_F)^*\,\mathrm{Diff}_P(0).$$

Thus $\mathrm{Diff}_P(0) = \frac{1}{2}g(C)$ (Z near $P\backslash g(C)$ is isomorphic to W near $F\backslash C$, which is smooth with P Cartier, so $\mathrm{Supp}\,\mathrm{Diff}_P(0) \subset g(C)$, and compare $(C^2)_F = d$). Since $C \sim -K_F$ and $K_Z = h^*K_X$, Lemma 2.19(1) implies

$$\widehat{\mathrm{vol}}(\mathrm{ord}_P) = \big((K_P + \mathrm{Diff}_P(0))^2\big) = \big((K_F + \tfrac{1}{2}C)^2\big) = \tfrac{1}{4}(K_F^2) = \frac{d}{4}.$$

As $d \in \{8, 9\}$, $\widehat{\mathrm{vol}}(\mathrm{ord}_P) \le \frac{9}{4}$. □

Proposition 9.20. *Notation and conditions as in Set-up 8.2. Assume that* $\mathrm{crex}(X \ni x) = 2$ *and that E is a normal prime divisor. Then* $\widehat{\mathrm{vol}}(X \ni x) \le 7$.

Proof. Suppose that $\widehat{\mathrm{vol}}(X \ni x) > 7$. Since K_X is Cartier and the statement is local around x, possibly shrinking X around x, we may assume that $K_X \sim 0$. Then $K_Y = f^*K_X \sim 0$, and Y is canonical [KM98, Lemma 2.30]. By Lemma 2.19(2), E is Gorenstein, and by adjunction, $\omega_E \cong \mathcal{O}_Y(K_Y+E)|_E \cong \mathcal{O}_E(E)$ [Kol13, (4.1.3)]. Since $-E$ is f-ample, $-K_E$ is ample. Thus E is a normal projective Gorenstein surface with $-K_E$ ample. By [Bre80, Theorem 1], either E has only Du Val singularities, or E is the contraction of a section with negative self-intersection on a $\mathbb{P}^1$-bundle over a smooth elliptic curve.

Suppose that E has only Du Val singularities. Then E is klt, and $K_Y + E$ is Cartier, so (Y, E) is plt near E [KM98, Theorem 5.50(1)]. Let G be a crepant divisor over $X \ni x$ other than E. Then G is exceptional over Y, $\mathrm{center}_Y G \subset f^{-1}(x) = E$, and $a(G, Y) = a(G, X) = 1$ [KM98, Lemma 2.30]. Let $\mathrm{mult}_G E$ be the multiplicity along G of the pullback of E. Since E is Cartier and contains $\mathrm{center}_Y G$, $\mathrm{mult}_G E \ge 1$, so

$$a(G, Y) = a(G, Y, E) + \mathrm{mult}_G E > 1,$$

a contradiction. Thus $\mathrm{crex}(X \ni x) = 1$, which contradicts $\mathrm{crex}(X \ni x) = 2$.

Hence E is the contraction of the negative section of $\mathbb{P}_D(\mathcal{O}_D \oplus \mathcal{O}_D(-L))$, where D is a smooth elliptic curve and L is a divisor on D with $\deg L > 0$, and E has exactly one singular point y [Bre80, Theorem 1 and pp. 117–120]. Let $S_L := \bigoplus_{m\ge 0} H^0(D, \mathcal{O}_D(mL))$, and let w be a variable of degree one. By [Kol13, Section 3.1 (3.8)], there exists a morphism

$$\mathbb{P}_D(\mathcal{O}_D \oplus \mathcal{O}_D(-L)) \cong \mathrm{Proj}_D\,\mathrm{Sym}(\mathcal{O}_D(L) \oplus \mathcal{O}_D) \to \mathrm{Proj}\,S_L[w]$$

which contracts the negative section to the vertex and is an isomorphism elsewhere. Since E and $\mathrm{Proj}\,S_L[w]$ are normal surfaces obtained by contracting the negative section of $\mathbb{P}_D(\mathcal{O}_D \oplus \mathcal{O}_D(-L))$, they are isomorphic by Zariski's Main Theorem. Thus E is the projective cone $\mathrm{Proj}\,S_L[w]$ with vertex y. At a smooth point of the Cartier divisor E, the threefold Y is smooth, so $\mathrm{Sing}\,Y \cap E \subset \{y\}$. Let G be a crepant divisor over $X \ni x$ other than E. By [KM98, Lemma 2.30], G is exceptional over Y with $a(G, Y) = a(G, X) = 1$ and $\mathrm{center}_Y G \subset f^{-1}(x) = E$. Since $a(G', Y) \ge 2$ for any prime divisor G' exceptional over Y whose center is not contained in $\mathrm{Sing}\,Y$ [KM98, Corollary 2.31(2)], we have $\mathrm{center}_Y G = \{y\}$. Conversely, a crepant divisor over $Y \ni y$ is a crepant divisor over $X \ni x$ other than E [KM98, Lemma 2.30]. Hence

$$\mathrm{crex}(Y \ni y) = \mathrm{crex}(X \ni x) - 1 = 1.$$

Since $Y \ni y$ is a Gorenstein canonical threefold germ and $y \in E = \mathrm{Exc}(f)$, Proposition 2.7(3) implies $\widehat{\mathrm{vol}}(Y \ni y) > \widehat{\mathrm{vol}}(X \ni x) > 7$. By Theorem 8.1, $Y \ni y$ is analytically isomorphic to $C_F \ni o$, where F is $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or $\mathbb{F}_1$. Let $d := (K_F^2) \in \{8, 9\}$.

By Lemma 9.18(1), $\mathrm{edim}(Y, y) = d + 1$. As E is a Cartier divisor on Y,

$$\mathrm{edim}(E, y) \ge \mathrm{edim}(Y, y) - 1 = d \ge 8.$$

If $\deg L \leq 2$, then a direct calculation using the Riemann–Roch theorem shows that S_L is generated by three homogeneous elements, of degrees 1, 2, 3 if $\deg L = 1$ and of degrees 1, 1, 2 if $\deg L = 2$, so $\operatorname{edim}(E, y) = 3$, a contradiction. Thus $\deg L \geq 3$. Then S_L is generated by $H^0(D, \mathcal{O}_D(L))$, since $\deg L \geq 3 = 2p_a(D) + 1$ [Laz04, Definition 1.8.50 and Theorem 1.8.53], and $h^0(D, \mathcal{O}_D(L)) = \deg L$ [Har77, Theorem IV.1.3]. Hence E is the projective cone, with vertex y, over D embedded in $\mathbb{P}^{\deg L - 1}$ by the complete linear system $|L|$ of degree $\deg L \geq 3$.

Let $\rho : W \to Y$ be the blow-up of y, identify the exceptional divisor of ρ with F, and let $T := \rho_*^{-1}E$. By Lemma 9.18(2), $\rho^*E = F + T$, $C := T \cap F$ is an integral member of $|-K_F|$ which is isomorphic to D, hence of arithmetic genus one, and $T \cong \mathbb{P}_C(\mathcal{O}_C \oplus \mathcal{O}_C(K_F|_C))$ with C as the negative section. Lemma 9.19 implies that there exists a Kollár component P over $X \ni x$ with $\widehat{\operatorname{vol}}(\operatorname{ord}_P) \leq \frac{9}{4}$. By Definition 2.6,

$$\widehat{\operatorname{vol}}(X \ni x) \leq \widehat{\operatorname{vol}}(\operatorname{ord}_P) \leq \frac{9}{4} < 7,$$

a contradiction. Therefore $\widehat{\operatorname{vol}}(X \ni x) \leq 7$. □

Proposition 9.21. *Let $X \ni x$ be a Gorenstein canonical threefold germ which is not a hypersurface germ, with $\operatorname{crex}(X \ni x) = 2$, and let $f : Y \to X$ be the blow-up of the maximal ideal $\mathfrak{m}_x$. Assume that Y is not cDV at some point of $f^{-1}(x)$. Then $f^{-1}(x)_{\mathrm{red}}$ is a prime divisor E, $\operatorname{Sing} Y \cap E$ is finite, and $\mathfrak{m}_x\mathcal{O}_Y = \mathcal{O}_Y(-E)$. In particular, E is Cartier. Moreover,*

$$\widehat{\operatorname{vol}}(X \ni x) \leq 7.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that $f^{-1}(x)_{\mathrm{red}}$ is a prime divisor E and that $\operatorname{Sing} Y \cap E$ is finite.

By Theorem 2.15(2), f is a crepant projective birational morphism, and Y is normal and Gorenstein canonical. The ideal $\mathfrak{m}_x\mathcal{O}_Y$ is invertible, and the closed subscheme of Y defined by $\mathfrak{m}_x\mathcal{O}_Y$ is a Cartier divisor with support $f^{-1}(x)$. Since $K_Y = f^*K_X$, any irreducible component of $f^{-1}(x)$ is a crepant divisor over $X \ni x$. Let $y \in f^{-1}(x)$ be a closed point at which Y is not cDV. By Theorem 2.15(1), $\operatorname{crex}(Y \ni y) > 0$. A crepant divisor over $Y \ni y$ has center $\{y\}$ on Y, and it is a crepant divisor over $X \ni x$ [KM98, Lemma 2.30], different from the irreducible components of $f^{-1}(x)$. Since $\operatorname{crex}(X \ni x) = 2$, $f^{-1}(x)_{\mathrm{red}}$ is a prime divisor E, and $\operatorname{crex}(Y \ni y) = 1$. In particular, there exists an integer $r \geq 1$ such that

$$\mathfrak{m}_x\mathcal{O}_Y = \mathcal{O}_Y(-rE).$$

Let $h_1, \dots, h_N$ be generators of $\mathfrak{m}_x$. For a general element h of the vector space spanned by $h_1, \dots, h_N$, let $H := \operatorname{div}(h) \subset X$, and let H_Y be the strict transform of H on Y, so that $f^*H = H_Y + rE$. The divisors H_Y form a base-point-free linear system V on Y with $\dim V < +\infty$, since $f^*h_1, \dots, f^*h_N$ generate $\mathcal{O}_Y(-rE)$. Since $h_1, \dots, h_N$ generate the Rees algebra $\bigoplus_{j \geq 0} \mathfrak{m}_x^j$ in degree one, they embed $Y = \operatorname{Proj}_X \bigoplus_{j \geq 0} \mathfrak{m}_x^j$ in $\mathbb{P}_X^{N-1}$, so E is a closed subvariety of $\mathbb{P}^{N-1}$, and the restrictions of the members of V to E are hyperplane sections of E. By Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], H_Y is smooth outside $\operatorname{Sing} Y$.

Suppose that $\operatorname{Sing} Y \cap E$ is infinite. Then there exists a curve $\Gamma \subset \operatorname{Sing} Y \cap E$. By Lemma 2.16, there exists a prime divisor G over Y with $\operatorname{center}_Y G = \Gamma$ and $a(G, Y) = 1$. By [KM98, Lemma 2.30], G is a crepant divisor over $X \ni x$, different from E and from the crepant divisor over $Y \ni y$, whose center on Y is $\{y\}$. Hence $\operatorname{crex}(X \ni x) \geq 3$, a contradiction. Therefore $\operatorname{Sing} Y \cap E$ is finite.

Step 2. In this step, we show that $r = 1$.

We keep the notation H and H_Y of **Step 1**. By Theorem 2.15(2), $H \ni x$ is a Gorenstein elliptic surface singularity. Since $X \ni x$ is not a hypersurface germ, $\operatorname{edim}(X, x) \geq 5$, so $\operatorname{edim}(H, x) = \operatorname{edim}(X, x) - 1 \geq 4$, and $H \ni x$ is not a hypersurface singularity. By [Har77, Corollary II.7.15] and the definition following it, $H_Y = \operatorname{Bl}_x H$. Let Z be the fundamental cycle of $H \ni x$ [Art66, pp. 131–132], and let $e := -(Z^2)$. By [Liu25, Theorem 2.22], if $e \leq 3$, then $H \ni x$

is a hypersurface singularity, and if $e \geq 3$, then $\mathrm{Bl}_x H \to H$ is the canonical modification of $H \ni x$. As $H \ni x$ is not a hypersurface singularity, $e \geq 4$, so $H_Y \to H$ is the canonical modification of $H \ni x$. Since V is base-point-free and $\operatorname{Sing} Y \cap E$ is finite by **Step 1**, the general member H_Y does not intersect $\operatorname{Sing} Y \cap E$. Thus H_Y is smooth near $f^{-1}(x)$, and possibly shrinking X around x, $f_H := f|_{H_Y} : H_Y \to H$ is a resolution. Since K_{H_Y} is f_H-ample, $(K_{H_Y} \cdot C_0) > 0$ for any f_H-exceptional curve C_0, whereas $(K_{H_Y} \cdot C_0) = -1$ if C_0 is a smooth rational curve with $(C_0^2) = -1$. Thus f_H is the minimal resolution.

The exceptional set of f_H is $H_Y \cap E$, and $H_Y \cap E$ is a general hyperplane section of the integral surface $E \subset \mathbb{P}^{N-1}$. Hence the scheme $C_H := H_Y \cap E$ is an integral curve [Jou83, Théorème 6.3(3)(4)]. Since Y is smooth along $H_Y \cap E$, E is Cartier there, and $E|_{H_Y} = C_H$. Thus f_H has exactly one exceptional curve C_H. Since the exceptional set of f_H is irreducible, the fundamental cycle Z is C_H. As $H \ni x$ is minimally elliptic [Lau77, Theorem 3.10] (see also [Liu25, Section 2.4]), we have $p_a(C_H) = p_a(Z) = 1$ [Lau77, Definitions 3.1, 3.2 and Theorem 3.4]. Possibly shrinking X around x, we may assume that $K_X \sim 0$, since K_X is Cartier. By adjunction [Kol13, (4.1.3)] and $f^*H = H_Y + rE$,

$$K_{H_Y} = (K_Y + H_Y)|_{H_Y} = \big(f^*(K_X + H) - rE\big)|_{H_Y} = f_H^* K_H - rC_H.$$

Hence

$$(K_{H_Y} \cdot C_H) + r(C_H^2) = (f_H^* K_H \cdot C_H) = 0.$$

By adjunction, $\omega_{C_H} \cong \omega_{H_Y}(C_H)|_{C_H}$ [Kol13, (4.1.3)], and $\deg \omega_{C_H} = 2p_a(C_H) - 2$ [Har77, Exercise IV.1.9], so

$$(K_{H_Y} \cdot C_H) + (C_H^2) = 2p_a(C_H) - 2 = 0.$$

Subtracting, we obtain $(r-1)(C_H^2) = 0$. Since C_H is the exceptional curve of a resolution of a normal surface singularity, $(C_H^2) < 0$. Thus $r = 1$, that is, $\mathfrak{m}_x \mathcal{O}_Y = \mathcal{O}_Y(-E)$. In particular, E is Cartier, and $-E$ is f-ample.

Step 3. In this step, we show that $\widehat{\mathrm{vol}}(X \ni x) \leq 7$.

Suppose that $\widehat{\mathrm{vol}}(X \ni x) > 7$. By **Step 2**, $X \ni x$, f, and E satisfy the conditions of Set-up 8.2. If E is normal, then $\widehat{\mathrm{vol}}(X \ni x) \leq 7$ by Proposition 9.20, a contradiction. Thus E is not normal. Possibly shrinking X around x, we may assume that $K_X \sim 0$. By Lemma 2.19(2), E is Gorenstein, and $\omega_E \cong \mathcal{O}_E(E)$ [Kol13, (4.1.3)], so $-K_E$ is ample. Hence E is an integral non-normal Gorenstein del Pezzo surface in the sense of [Rei94, Section 0.1].

By **Step 1**, $\mathrm{crex}(Y \ni y) = 1$. Since $y \in \mathrm{Exc}(f)$, Proposition 2.7(3) implies $\widehat{\mathrm{vol}}(Y \ni y) > \widehat{\mathrm{vol}}(X \ni x) > 7$. By Theorem 8.1, $Y \ni y$ is analytically isomorphic to $C_F \ni o$, where F is $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or $\mathbb{F}_1$. Let $d := (K_F^2) \in \{8, 9\}$. By Lemma 9.18(1), $\mathrm{edim}(Y, y) = d + 1$. As E is a Cartier divisor on Y,

$$\mathrm{edim}(E, y) \geq d \geq 8. \tag{9.21}$$

By [Rei94, Theorem 1.5], E is obtained by one of the constructions of [Rei94, Sections 1.3 and 1.4]. Since E is irreducible, [Rei94, Theorem 1.1 and Section 1.3] shows that E is one of the following: of type (A) applied to the scroll of type (e); of type (A) applied to the Veronese surface of type (b); of type (C_1) applied to (d1); one of the irreducible surfaces of [Rei94, Section 1.4]; or of type (C_1) applied to (c1), or of type (D_1). We bound $\mathrm{edim}(E, z)$ for the closed points $z \in E$ in each case.

- Type (A), from the scroll of type (e): a direct calculation from the definition of type (A) in [Rei94, Section 1.3] shows that the completed local ring of E at any singular point is defined by $uv = 0$ or $u^2 - v^2 w = 0$ in three variables. Thus $\mathrm{edim}(E, z) \leq 3$.
- Type (A), from the Veronese surface of type (b): then E is the Bordello surface in $\mathbb{P}^4$ [Rei94, Section 1.3, (A)]. Thus $\mathrm{edim}(E, z) \leq 4$.
- Type (C_1), from (d1): a direct calculation from the definition of type (C_1) in [Rei94, Section 1.3] shows that the completed local ring of E at any singular point is defined by $uv = 0$ or by $u_3^2 + u_2^3 - u_1 u_2 u_3 = 0$ in three variables. Thus $\mathrm{edim}(E, z) \leq 3$.

- The surfaces of [Rei94, Section 1.4]: E is a hypersurface in $\mathbb{P}(1,1,2,3)$ or $\mathbb{P}(1,1,1,2)$ which, by its equation, does not pass through the singular points of the weighted projective space, so $\operatorname{edim}(E,z)\le 3$.
- Type (C_1) from (c1), and type (D_1): then E is the projective cone over an integral rational curve D with one node, respectively one cusp, embedded in a projective space by a complete linear system [Rei94, Theorem 1.1 and Section 1.3] (E is Gorenstein, hence S_2, at the vertex, and an affine cone over a curve is S_2 if and only if all the restriction maps on $H^0(\mathcal{O}(j))$ are surjective [Kol13, Section 3.1 (3.8)]). In this case $p_a(D)=1$, and at any closed point z other than the vertex, E is locally the product of D with a smooth curve, so $\operatorname{edim}(E,z)\le 3$.

By (9.21), E is the projective cone of the last case, and y is the vertex. Since (C_1) is built from (c1) of degree at least 3, and (D_1) is the projective cone over a cuspidal rational curve of degree at least 3, we have $\deg D\ge 3$ [Rei94, Section 1.3].

Let $\rho: W\to Y$ be the blow-up of y, identify the exceptional divisor of ρ with F, and let $T:=\rho_*^{-1}E$. By Lemma 9.18(2), $\rho^*E=F+T$, $C:=T\cap F$ is an integral member of $|-K_F|$ which is isomorphic to D, hence of arithmetic genus one, and $T\cong\mathbb{P}_C(\mathcal{O}_C\oplus\mathcal{O}_C(K_F|_C))$ with C as the negative section. Lemma 9.19 implies that there exists a Kollár component P over $X\ni x$ with $\widehat{\operatorname{vol}}(\operatorname{ord}_P)\le\frac{9}{4}$. By Definition 2.6,

$$\widehat{\operatorname{vol}}(X\ni x)\le\widehat{\operatorname{vol}}(\operatorname{ord}_P)\le\frac{9}{4}<7,$$

a contradiction. Therefore $\widehat{\operatorname{vol}}(X\ni x)\le 7$. □

9.5. **The bound for two crepant divisors.** In this subsection, we prove Proposition 9.22, and then we prove Theorem 9.1. In the proof of Proposition 9.22, Theorem 1.3 determines the local volumes of the local index-one covers at the points where the exceptional surface is not Cartier.

Proposition 9.22. *Let $X\ni x$ be a Gorenstein canonical threefold germ which is not a hypersurface germ, with $\operatorname{crex}(X\ni x)=2$. Assume that the blow-up $f: Y\to X$ of the maximal ideal $\mathfrak{m}_x$ is normal with $K_Y=f^*K_X$, that $f^{-1}(x)_{\mathrm{red}}$ is a prime divisor E, and that Y has cDV singularities along E. If E is not Cartier, then*

$$\widehat{\operatorname{vol}}(X\ni x)\le\frac{125}{18}.$$

Proof. The proof is long, so we divide it into several steps. Let $V:=\widehat{\operatorname{vol}}(X\ni x)$, and suppose that $V>\frac{125}{18}$.

Step 1. In this step, we describe Y near the points at which E is not Cartier.

Since $\mathfrak{m}_x\mathcal{O}_Y$ is invertible and $f^{-1}(x)_{\mathrm{red}}=E$, there exists a positive integer r such that $\mathfrak{m}_x\mathcal{O}_Y=\mathcal{O}_Y(-rE)$. Thus rE is Cartier and $-E$ is f-ample. Since E is not Cartier, $r\ge 2$. Let $N_E\subset E$ be the set of points at which E is not Cartier. Then N_E is a non-empty closed subset of E, and $N_E\subset\operatorname{Sing}Y$. Since $K_Y=f^*K_X$, Y is canonical [KM98, Lemma 2.30]. Possibly shrinking X around x, we may assume that $K_X\sim 0$, and hence $K_Y\sim 0$.

Let $p\in N_E$, let $d\ge 2$ be the Cartier index of E at p, and let $\pi:(\tilde{Y}\ni\tilde{p})\to(Y\ni p)$ be the index-one cover of E at p [KM98, Definition 2.52]. Then π is a cyclic quasi-étale cover of degree d with Galois group μ_d. Moreover, $\pi^{-1}(p)=\{\tilde{p}\}$ [Kol13, 2.48(1)], since $\mathcal{O}_Y(-jE)$ is not invertible at p for $0<j<d$. The decomposition $\pi_*\mathcal{O}_{\tilde{Y}}=\bigoplus_{i=0}^{d-1}\mathcal{O}_Y(-iE)$ is the decomposition into μ_d-eigensheaves, the divisor π^*E is Cartier, and a local equation s of π^*E at $\tilde{p}$ can be taken in the eigensheaf $\mathcal{O}_Y(-E)$ [Kol13, 2.44]. Thus $\gamma^*s=\chi(\gamma)s$ for any $\gamma\in\mu_d$, where χ is the character of degree one by which μ_d acts on $\mathcal{O}_Y(-E)$. The group μ_d acts on the summand $\mathcal{O}_Y(-iE)$ through χ^i [Kol13, 2.44]. If $\chi(\gamma)=1$, then γ acts trivially on $\pi_*\mathcal{O}_{\tilde{Y}}$, and hence $\gamma=1$, since μ_d is the Galois group of π. Thus χ is faithful. Since π is quasi-étale, $K_{\tilde{Y}}=\pi^*K_Y$ is Cartier, so $\tilde{Y}$ is

canonical [KM98, Proposition 5.20(3)], hence Cohen–Macaulay [KM98, Theorems 5.22 and 5.10], and Gorenstein. Since $p \in \operatorname{Exc}(f)$, Proposition 2.7(2) and (3) imply that

$$\widehat{\operatorname{vol}}(\tilde{Y} \ni \tilde{p}) = d \cdot \widehat{\operatorname{vol}}(Y \ni p) > dV > \frac{125d}{18}.$$

By Theorem 2.17(1), $\widehat{\operatorname{vol}}(\tilde{Y} \ni \tilde{p}) \leq 27 < \frac{125 \cdot 4}{18}$. Thus $d \in \{2, 3\}$. If $d = 3$, then $\widehat{\operatorname{vol}}(\tilde{Y} \ni \tilde{p}) > \frac{125}{6} > 16$, and $\tilde{Y} \ni \tilde{p}$ is smooth by Theorem 2.17(1). If $d = 2$, then $\widehat{\operatorname{vol}}(\tilde{Y} \ni \tilde{p}) > \frac{125}{9}$. Hence $\widehat{\operatorname{vol}}(\tilde{Y} \ni \tilde{p}) \in \{16, 27\}$ by Theorem 1.3, and $\tilde{Y} \ni \tilde{p}$ is an ordinary double point or smooth by Theorem 2.17(1).

Let ω be a generator of $\omega_{\tilde{Y}}$ at $\tilde{p}$. Since K_Y is Cartier and $K_{\tilde{Y}} = \pi^* K_Y$, the pullback of a generator of ω_Y at p is $u\omega$ for a unit u at $\tilde{p}$, and $u\omega$ is μ_d-invariant. Thus

$$\frac{\gamma^*\omega}{\omega}(\tilde{p}) = \frac{u}{\gamma^* u}(\tilde{p}) = 1 \quad \text{for any } \gamma \in \mu_d. \tag{9.22}$$

Let ζ be a generator of μ_d.

Suppose that $\tilde{Y} \ni \tilde{p}$ is smooth. By [Car57, Section 4, pp. 97–98], there exist local analytic coordinates z_1, z_2, z_3 of $\tilde{Y}$ at $\tilde{p}$ and integers $0 \leq a_i < d$ such that ζ acts by $z_i \to \zeta^{a_i} z_i$. Taking $\omega = dz_1 \wedge dz_2 \wedge dz_3$, (9.22) implies that $a_1 + a_2 + a_3 \equiv 0 \pmod{d}$. Since μ_d acts faithfully, up to a permutation of the coordinates and the choice of ζ, we have $(a_1, a_2, a_3) = (1, 1, 0)$ if $d = 2$, and $(a_1, a_2, a_3) \in \{(1, 2, 0), (1, 1, 1)\}$ if $d = 3$. If $(a_1, a_2, a_3) = (1, 1, 1)$, then $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[z_1, z_2, z_3]]^{\mu_3}$, whose maximal ideal is minimally generated by the ten cubic monomials. Thus $Y \ni p$ has embedding dimension 10, which contradicts the assumption that Y is cDV at p, since a cDV germ has embedding dimension at most 4 [KM98, Definition 5.32 and Theorem 4.22], as a hyperplane section through the point lowers the embedding dimension by at most one. Hence $Y \ni p$ is analytically isomorphic to the product of a smooth curve germ and an A_{d-1} surface singularity, and the fixed locus $\{z_1 = z_2 = 0\}$ of any non-trivial element of μ_d maps onto $\operatorname{Sing} Y$ near p.

Suppose that $d = 2$ and $\tilde{Y} \ni \tilde{p}$ is an ordinary double point, and let ι be the non-trivial element of μ_2. Lifting an eigenbasis of the cotangent space of $\tilde{Y}$ at $\tilde{p}$ by the projections $\frac{1}{2}(1 \pm \iota^*)$, we obtain an embedding of $\tilde{Y} \ni \tilde{p}$ into $(\mathbb{C}^4, 0)$ with coordinates $z_1, \dots, z_4$ such that ι acts by $z_i \to \epsilon_i z_i$ with $\epsilon_i \in \{1, -1\}$ and $\epsilon_1 \leq \epsilon_2 \leq \epsilon_3 \leq \epsilon_4$. Since the ideal of $\tilde{Y}$ is ι-stable and generated by one element, by Nakayama's lemma it is generated by an element h with $\iota^* h = \lambda h$ for some $\lambda \in \{1, -1\}$. The quadratic part h_2 of h is a non-degenerate quadratic form.

Let ω be the residue of $dz_1 \wedge \cdots \wedge dz_4/h$. Then $\iota^*\omega = \lambda^{-1}\epsilon_1\epsilon_2\epsilon_3\epsilon_4\omega$, and (9.22) implies that $\epsilon_1\epsilon_2\epsilon_3\epsilon_4 = \lambda$. If $\lambda = -1$, then $\iota^* h_2 = -h_2$, so $\epsilon_i \neq \epsilon_j$ for any monomial $z_i z_j$ of h_2. Since h_2 is non-degenerate, exactly two of $\epsilon_1, \dots, \epsilon_4$ are equal to -1, which implies that $\epsilon_1\epsilon_2\epsilon_3\epsilon_4 = 1$, a contradiction. Thus $\lambda = 1$ and $\epsilon_1\epsilon_2\epsilon_3\epsilon_4 = 1$. Since ι acts non-trivially, $\epsilon_1 = \epsilon_2 = -1$.

- If $\epsilon_3 = \epsilon_4 = -1$, then h is invariant and $\widehat{\mathcal{O}}_{Y,p} \cong \mathbb{C}[[z_1, \dots, z_4]]^\iota/(h)$. The ten monomials $z_i z_j$ form a basis of the cotangent space of $\mathbb{C}[[z_1, \dots, z_4]]^\iota$, and the image of h in it is $h_2 \neq 0$. Thus $Y \ni p$ has embedding dimension 9, which contradicts the assumption that Y is cDV at p.
- If $\epsilon_3 = \epsilon_4 = 1$, then $\iota^* h_2 = h_2$ implies that $h_2 = h_2^-(z_1, z_2) + h_2^+(z_3, z_4)$ for non-degenerate quadratic forms h_2^- and h_2^+. The fixed locus of ι on $\tilde{Y}$ is the curve $\{h(0, 0, z_3, z_4) = 0\}$ in the (z_3, z_4)-plane. Since h_2^+ is non-degenerate, it is the union of two smooth curve germs Θ_1 and Θ_2 intersecting transversally at $\tilde{p}$ [GLS07, Theorem I.2.47]. Since $\tilde{Y}$ is smooth outside $\tilde{p}$, for any point $y' \in (\Theta_1 \cup \Theta_2) \backslash \{\tilde{p}\}$, the action of ι is linearizable at y' [Car57, Section 4, pp. 97–98], with fixed locus of dimension one. Thus ι acts on the tangent space at y' with eigenvalues $(-1, -1, 1)$, and Y is analytically isomorphic to the product of a smooth curve germ and an A_1 surface singularity at $\pi(y')$.

Finally, let $\gamma \in \mu_d$ be non-trivial and let y' be a fixed point of γ. Then $s(y') = \chi(\gamma) s(y')$, so $s(y') = 0$. If E were Cartier at $\pi(y')$ with a local equation e, then $\pi^* e = \theta s$ near y' for a unit θ, and $\gamma^*\theta = \chi(\gamma)^{-1}\theta$ would imply $\theta(y') = 0$, a contradiction. Thus π maps the fixed locus of

γ into N_E. Since μ_d acts freely outside the fixed loci of the non-trivial elements of μ_d and $\tilde{Y}$ is smooth outside $\tilde{p}$, near p we have $N_E = \operatorname{Sing} Y$, and N_E is the image of the fixed loci of the non-trivial elements of μ_d. In conclusion, for any $p \in N_E$, one of the following holds near p.

(1) $d = 2$, $Y \ni p$ is analytically isomorphic to the product of a smooth curve germ and an A_1 surface singularity, and N_E is a smooth curve.
(2) $d = 3$, $Y \ni p$ is analytically isomorphic to the product of a smooth curve germ and an A_2 surface singularity, and N_E is a smooth curve.
(3) $d = 2$, $N_E = \pi(\Theta_1) \cup \pi(\Theta_2)$ is a curve with two smooth branches at p, since one of the ι-invariant coordinates z_3, z_4 restricts to a local parameter of Θ_i, and Y is analytically isomorphic to the product of a smooth curve germ and an A_1 surface singularity at any point of $N_E \backslash \{p\}$.

Step 2. In this step, we show that N_E is an irreducible curve C, and that E has Cartier index 2 at any point of C.

We first show that any irreducible curve $C' \subset \operatorname{Sing} Y \cap E$ is the center on Y of a crepant divisor over $X \ni x$ which is different from E. By Lemma 2.16, there exists a prime divisor F over Y with $\operatorname{center}_Y F = C'$ and $a(F, Y) = 1$. Since $K_Y = f^* K_X$, we have $a(F, X) = a(F, Y) = 1$ [KM98, Lemma 2.30], and $\operatorname{center}_X F = f(C') = \{x\}$. Thus F is a crepant divisor over $X \ni x$, and $F \neq E$.

Since $\operatorname{crex}(X \ni x) = 2$ and E is a crepant divisor over $X \ni x$, it follows that $\operatorname{Sing} Y \cap E$ contains at most one irreducible curve, and that for such a curve C', there exists exactly one exceptional prime divisor F over Y with $\operatorname{center}_Y F = C'$ and $a(F, Y) = 1$. By **Step 1**, N_E is a non-empty union of curves contained in $\operatorname{Sing} Y \cap E$. Thus N_E is an irreducible curve, which we denote by C, and $(\operatorname{Sing} Y \cap E) \backslash C$ is finite.

By **Step 1**, E has Cartier index 2 or 3 at any point of C. The subsets of C at which $2E$, resp. $3E$, is Cartier are open in C. They cover C, and they are disjoint since E is not Cartier at any point of C. Since C is irreducible, either E has Cartier index 2 at any point of C, or E has Cartier index 3 at any point of C.

Suppose that E has Cartier index 3 at any point of C. By **Step 1**(2), C is smooth, and $\operatorname{Sing} Y = C$ near any point of C. Thus $Y^\circ := Y \backslash \overline{\operatorname{Sing} Y \backslash C}$ is an open subset of Y containing C. Let $g : Y' \to Y^\circ$ be the blow-up of C. Let $p \in C$, and let z_1, z_2, z_3, ζ, and s be as in **Step 1**, so that ζ acts by $(z_1, z_2, z_3) \to (\zeta z_1, \zeta^2 z_2, z_3)$, and $\chi(\zeta) = \zeta^j$ for some $j \in \{1, 2\}$. Since blowing up commutes with flat base change, near $g^{-1}(p)$, the morphism g is analytically isomorphic to the product of a smooth curve germ and the blow-up of the A_2 surface singularity at its maximal ideal (for $\{y_1 y_2 = y_3^3\} \subset \mathbb{C}^3$, the three affine charts $\{y_2' = y_1 y_3'^3\}$, $\{y_1' = y_2 y_3'^3\}$, and $\{y_1' y_2' = y_3\}$ of the blow-up are smooth, and the exceptional set consists of two (-2)-curves), which is already the toric minimal resolution of the A_2 surface singularity. Thus $K_{Y'} = g^* K_{Y^\circ}$, and $g^{-1}(p)$ is the union of two smooth rational curves Γ_1 and Γ_2. Moreover, let v_1 and v_2 be the monomial valuations on $\mathbb{C}\{z_1, z_2, z_3\}$ with weights $(\frac{1}{3}, \frac{2}{3}, 0)$ and $(\frac{2}{3}, \frac{1}{3}, 0)$ respectively. Then for any prime divisor $F \subset g^{-1}(C)$ containing Γ_i and any $\varphi \in \mathcal{O}_{Y,p}$, we have $\operatorname{ord}_F(g^* \varphi) = v_i(\pi^* \varphi)$. The A_2 surface singularity $\mathbb{C}^2/\mu_3$, where μ_3 acts with weights $(1, 2)$, is an affine toric surface [CLS11, Proposition 10.1.2]. The minimal resolution of $\mathbb{C}^2/\mu_3$ is a toric resolution, and the rays added in it are generated by $(\frac{1}{3}, \frac{2}{3})$ and $(\frac{2}{3}, \frac{1}{3})$ in the coordinates dual to z_1, z_2 [CLS11, Theorem 10.2.3 and Corollary 10.4.9]. For the torus-invariant prime divisor D_ρ corresponding to a ray ρ with generator u_ρ, the order of a monomial $z_1^{m_1} z_2^{m_2}$ along D_ρ is $\langle (m_1, m_2), u_\rho \rangle$ [CLS11, Proposition 4.1.2]. Since $\operatorname{ord}_{D_\rho}$ is invariant under the torus action, $\operatorname{ord}_{D_\rho}$ is the monomial valuation with weights u_ρ. Taking the product with the smooth curve germ, on which the coordinate z_3 has weight 0, we obtain v_1 and v_2, and hence $\operatorname{ord}_F(g^* \varphi) = v_i(\pi^* \varphi)$. Since any prime divisor $F \subset g^{-1}(C)$ satisfies $a(F, Y) = 1$ and $\operatorname{center}_Y F = C$, the uniqueness above implies that $g^{-1}(C)$ is a prime divisor F containing Γ_1 and Γ_2. Let e be a local equation of $3E$ at p. Then $\pi^* e = \theta s^3$ for a unit θ, so

$$3 v_1(s) = \operatorname{ord}_F(g^* e) = 3 v_2(s).$$

Any monomial $z_1^{m_1}z_2^{m_2}z_3^{m_3}$ of s satisfies $m_1+2m_2\equiv j \pmod 3$, so $\frac{m_1+2m_2}{3}\in\frac{j}{3}+\mathbb{Z}$ and $\frac{2m_1+m_2}{3}\in\frac{2j}{3}+\mathbb{Z}$. Thus $v_1(s)\in\frac{j}{3}+\mathbb{Z}$ and $v_2(s)\in\frac{2j}{3}+\mathbb{Z}$. Since $3\nmid j$, we have $v_1(s)\neq v_2(s)$, a contradiction.

Therefore E has Cartier index 2 at any point of C. By **Step 1**, for any $p\in C$, either **Step 1**(1) holds and C is smooth at p, or **Step 1**(3) holds and C is singular at p. Let

$$\Sigma := \operatorname{Sing} C \cup \big((\operatorname{Sing} Y\cap E)\backslash C\big),$$

which is a finite set. Then Y is smooth at any point of $E\backslash(C\cup\Sigma)$, and Y is analytically isomorphic to the product of a smooth curve germ and an A_1 surface singularity at any point $p\in C\backslash\Sigma$, where C corresponds to the curve factor.

Step 3. In this step, we show that $r=2$ and that E is a surface of minimal degree in the linear span $\langle E\rangle$ of E in $\mathbb{P}(\mathfrak{m}_x/\mathfrak{m}_x^2):=\operatorname{Proj}\operatorname{Sym}(\mathfrak{m}_x/\mathfrak{m}_x^2)$.

Recall that $f^{-1}(x)=\operatorname{Proj}\bigoplus_{k\geq 0}\mathfrak{m}_x^k/\mathfrak{m}_x^{k+1}$ is a closed subscheme of $\mathbb{P}(\mathfrak{m}_x/\mathfrak{m}_x^2)$, and that $\mathcal{O}(1)$ restricts to $\mathfrak{m}_x\mathcal{O}_Y|_{f^{-1}(x)}$. Fix generators $g_1,\dots,g_n$ of $\mathfrak{m}_x$, let $H\subset X$ be the surface defined by a general $\mathbb{C}$-linear combination ℓ of $g_1,\dots,g_n$, and let H_Y be the strict transform of H on Y. Then $f^*H=H_Y+rE$, and H_Y is a general member of the linear system spanned by the sections $f^*g_1,\dots,f^*g_n$ of $\mathfrak{m}_x\mathcal{O}_Y=\mathcal{O}_Y(-rE)$, which is base-point-free. Let $\Lambda\subset\mathbb{P}(\mathfrak{m}_x/\mathfrak{m}_x^2)$ be the hyperplane defined by the image of ℓ in $\mathfrak{m}_x/\mathfrak{m}_x^2$, and let $C_H:=E\cap\Lambda$ as schemes, so that $C_H=E\cap H_Y$. Since ℓ is general and the linear system above is base-point-free, we may assume the following, where (1) and (2) follow from [Har77, Corollary III.10.9 and Remark III.10.9.2] applied to $Y\setminus\operatorname{Sing}Y$ and to $C\backslash\Sigma$.

(1) H_Y is smooth outside $\operatorname{Sing}Y$.
(2) $H_Y\cap\Sigma=\emptyset$, and H_Y intersects the smooth curve $C\backslash\Sigma$ transversally.
(3) $H\ni x$ is a Gorenstein elliptic surface singularity (Theorem 2.15(2)).
(4) C_H is an integral curve [Jou83, Théorème 6.3(3)(4)], and $\langle E\rangle\not\subset\Lambda$.

Let t be the number of points of $C\cap H_Y$. Since $\mathfrak{m}_x\mathcal{O}_Y|_C$ is ample and $C\not\subset H_Y$, we have $t\geq 1$. Let $p\in C\cap H_Y$. By **Step 2**, Y is analytically isomorphic near p to the product of a smooth curve germ and an A_1 surface singularity, with C corresponding to the curve factor, and by (2), a local equation of H_Y restricts to a local coordinate of C at p. Thus H_Y has an A_1 singularity at p. By (1), (2), and **Step 2**, H_Y is smooth at any point of $C_H\backslash C$. In particular, H_Y is normal, since it is a Cartier divisor on the Cohen–Macaulay variety Y with isolated singularities near C_H. Since f is an isomorphism over $X\backslash\{x\}$, the morphism $H_Y\to H$ is an isomorphism over $H\backslash\{x\}$, and $\operatorname{Exc}(H_Y\to H)=C_H$.

Let $\rho_H:\tilde{H}\to H$ be the composition of the minimal resolution $\sigma:\tilde{H}\to H_Y$ of the t singular points of H_Y near C_H with $H_Y\to H$. The exceptional curves of ρ_H are the strict transform C_0 of C_H and pairwise disjoint (-2)-curves $D_1,\dots,D_t$. Since $K_Y\sim 0$ and $f^*H\sim 0$, adjunction implies that $\mathcal{O}_{H_Y}(K_{H_Y})\cong\mathcal{O}_Y(H_Y)|_{H_Y}\cong\mathfrak{m}_x\mathcal{O}_Y|_{H_Y}$. Since σ is crepant and $\mathfrak{m}_x\mathcal{O}_Y$ is f-ample,

$$(K_{\tilde{H}}\cdot C_0)=(K_{H_Y}\cdot C_H)=\deg\big(\mathfrak{m}_x\mathcal{O}_Y|_{C_H}\big)\geq 1.$$

Thus no exceptional curve of ρ_H is a (-1)-curve, and ρ_H is the minimal resolution of H. By (3), $H\ni x$ is minimally elliptic [Lau77, Theorem 3.10]. Let Z be the fundamental cycle of ρ_H [Art66, pp. 131–132]. By [Lau77, Definitions 3.1–3.2 and Theorem 3.4], $\chi(\mathcal{O}_Z)=0$, $\chi(\mathcal{O}_D)>0$ for any cycle $0<D<Z$, and $(Z\cdot\Gamma)=-(K_{\tilde{H}}\cdot\Gamma)$ for any exceptional curve Γ of ρ_H.

Since $X\ni x$ is not a hypersurface germ, the embedding dimension of $H\ni x$ is equal to the embedding dimension of $X\ni x$ minus one, which is at least 4. Thus $H\ni x$ is not a double point, since a double point of a normal surface has embedding dimension at most 3. By [Lau77, Theorem 3.13], $(Z^2)\leq -3$ and $\mathfrak{m}_H\mathcal{O}_{\tilde{H}}=\mathcal{O}_{\tilde{H}}(-Z)$. Since sections of $\mathcal{O}_{\tilde{H}}(-Z)$ vanish along $\operatorname{Exc}(\rho_H)$, we have

$$\mathfrak{m}_H\subset\rho_{H*}(\mathfrak{m}_H\mathcal{O}_{\tilde{H}})=\rho_{H*}\mathcal{O}_{\tilde{H}}(-Z)\subset\mathfrak{m}_H,$$

so $\rho_{H*}\mathcal{O}_{\tilde{H}}(-Z) = \mathfrak{m}_H$. At a general point of C_H, the divisor E is Cartier, $C_H = E \cap H_Y$ is reduced by (4), and $\mathfrak{m}_H\mathcal{O}_{H_Y} = \mathfrak{m}_x\mathcal{O}_Y|_{H_Y} = \mathcal{O}_Y(-rE)|_{H_Y}$. Thus the coefficient of C_0 in Z is $r \geq 2$.

Let $Z_{\mathrm{red}} := C_0 + \sum_{j=1}^t D_j$. Since $r \geq 2$, we have $0 < Z_{\mathrm{red}} < Z$, so $\chi(\mathcal{O}_{Z_{\mathrm{red}}}) > 0$. Since Z_{red} is reduced and connected [Har77, Corollary III.11.4], $h^0(\mathcal{O}_{Z_{\mathrm{red}}}) = 1$, and hence $p_a(Z_{\mathrm{red}}) = 1 - \chi(\mathcal{O}_{Z_{\mathrm{red}}}) \leq 0$. Since $D_1, \dots, D_t$ are pairwise disjoint and Z_{red} is connected, $(C_0 \cdot D_j) \geq 1$ for any j. By the formula $p_a(D + D') = p_a(D) + p_a(D') + (D \cdot D') - 1$ and $p_a(D_j) = 0$,

$$0 \geq p_a(Z_{\mathrm{red}}) = p_a(C_0) + \sum_{j=1}^{t} \big((C_0 \cdot D_j) - 1\big) \geq p_a(C_0) \geq 0.$$

Thus $p_a(C_0) = 0$, so $C_0 \cong \mathbb{P}^1$, and $(C_0 \cdot D_j) = 1$ for any j. Let $b := -(C_0^2)$. By adjunction [Har77, Exercise V.1.3(a)], $(K_{\tilde{H}} \cdot C_0) = b - 2$. Since $(K_{\tilde{H}} \cdot C_0) \geq 1$, we have $b \geq 3$.

Let $Z' := 2C_0 + \sum_{j=1}^t D_j$. Since Z has coefficient $r \geq 2$ along C_0 and positive coefficients along $D_1, \dots, D_t$, we have $Z' \leq Z$, so $\chi(\mathcal{O}_{Z'}) \geq 0$. By the Riemann–Roch formula $\chi(\mathcal{O}_D) = -\frac{1}{2}\big((D^2) + (K_{\tilde{H}} \cdot D)\big)$ [Rei97, 4.11], together with $(Z'^2) = -4b + 2t$ and $(K_{\tilde{H}} \cdot Z') = 2b - 4$, we obtain

$$0 \leq \chi(\mathcal{O}_{Z'}) = 2 + b - t.$$

Thus $t \leq b + 2$. Moreover, $(Z' \cdot D_j) = 2 - 2 = 0$ for any j, and

$$(Z' \cdot C_0) = -2b + t \leq 2 - b < 0.$$

Hence $(Z' \cdot \Gamma) \leq 0$ for any exceptional curve Γ of ρ_H, and the minimality of Z implies that $Z \leq Z'$. Therefore $Z = Z'$, $r = 2$, and $\chi(\mathcal{O}_Z) = 0$ implies that $t = b + 2$. In particular,

$$\deg \mathcal{O}_{C_0}(-Z) = -(Z \cdot C_0) = b - 2 \geq 1, \qquad \deg \mathcal{O}_{D_j}(-Z) = -(Z \cdot D_j) = 0,$$

and

$$\deg \mathcal{O}_{C_0}(-Z_{\mathrm{red}} - Z) = -(Z_{\mathrm{red}} \cdot C_0) - (Z \cdot C_0) = (b - t) + (b - 2) = b - 4 \geq -1.$$

Since $Z = C_0 + Z_{\mathrm{red}}$, there exists an exact sequence

$$0 \to \mathcal{O}_{C_0}(-Z_{\mathrm{red}} - Z) \to \mathcal{O}_Z(-Z) \to \mathcal{O}_{Z_{\mathrm{red}}}(-Z) \to 0.$$

Since

$$H^1(C_0, \mathcal{O}_{C_0}(-Z_{\mathrm{red}} - Z)) = H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(b - 4)) = 0,$$

the restriction $H^0(\mathcal{O}_Z(-Z)) \to H^0(\mathcal{O}_{Z_{\mathrm{red}}}(-Z))$ is surjective. Since $\mathcal{O}_{\tilde{H}}(-Z)|_{D_j}$ is trivial and D_j intersects C_0 transversally at one point for any j, the restriction $H^0(\mathcal{O}_{Z_{\mathrm{red}}}(-Z)) \to H^0(\mathcal{O}_{C_0}(-Z))$ is an isomorphism. The invertible sheaf $\mathcal{O}_{\tilde{H}}(-Z)$ is ρ_H-nef, and $(Z \cdot \Gamma) = -(K_{\tilde{H}} \cdot \Gamma)$ for any exceptional curve Γ of ρ_H, so $Z = -K_{\tilde{H}}$ by the negative definiteness of the intersection matrix [KM98, Lemma 3.40], that is, Z is the cycle of [KM98, Notation 4.44], and by [KM98, Proposition 4.45(1)], the restriction

$$\mathfrak{m}_H = H^0(\tilde{H}, \mathcal{O}_{\tilde{H}}(-Z)) \to H^0(Z, \mathcal{O}_Z(-Z))$$

is surjective. Since $\mathfrak{m}_H^2\mathcal{O}_{\tilde{H}} = \mathcal{O}_{\tilde{H}}(-2Z)$ maps to zero in $\mathcal{O}_Z(-Z)$, we obtain a surjection

$$\mathfrak{m}_H/\mathfrak{m}_H^2 \to H^0(C_0, \mathcal{O}_{C_0}(-Z)) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(b - 2)).$$

Since H_Y is the blow-up of $\mathfrak{m}_H$ [Har77, Corollary II.7.15] and $\mathbb{P}(\mathfrak{m}_H/\mathfrak{m}_H^2) = \Lambda$, the composition $C_0 \to C_H \subset \Lambda$ is defined by the image of $\mathfrak{m}_H/\mathfrak{m}_H^2$ in $H^0(C_0, \sigma^*(\mathfrak{m}_H\mathcal{O}_{H_Y})|_{C_0}) = H^0(C_0, \mathcal{O}_{C_0}(-Z))$. Thus $C_0 \to C_H$ is an isomorphism, and C_H is a rational normal curve of degree $b - 2$ which spans a linear subspace of Λ of dimension $b - 2$.

We show that C_H spans $\langle E \rangle \cap \Lambda$. We regard the image of ℓ in $\mathfrak{m}_x/\mathfrak{m}_x^2$ as a linear form defining Λ, and let ℓ' be a linear form vanishing on C_H. Since $C_H = E \cap \Lambda$ is reduced by (4), the restriction $\ell'|_E$ vanishes on the Cartier divisor $C_H = \{\ell|_E = 0\}$ of E. Thus $\ell'|_E = \alpha\ell|_E$ for some

$\alpha \in H^0(E, \mathcal{O}_E) = \mathbb{C}$, so $\ell' - \alpha\ell$ vanishes on $\langle E\rangle$. Hence C_H spans $\langle E\rangle \cap \Lambda$, and $\dim\langle E\rangle = b - 1$ by (4). Let

$$L := \mathfrak{m}_x \mathcal{O}_Y|_E = \mathcal{O}_Y(-2E)|_E,$$

which is the restriction of $\mathcal{O}(1)$ to $E \subset \langle E\rangle$. Then $(L^2) = \deg C_H = b - 2 = \dim\langle E\rangle - 1$. Thus E is a non-degenerate integral surface of minimal degree in $\langle E\rangle$.

Step 4. In this step, we show that $V \le 2$, which contradicts $V > \frac{125}{18}$.

By [EH87, Theorem 1 and §1] and **Step 3**, one of the following holds: $E \cong \mathbb{P}^2$ with $L = \mathcal{O}(1)$ or $L = \mathcal{O}(2)$; $E \cong \mathbb{F}_a$ for some $a \ge 0$ and $L \sim B + (a + c)A$ for some integer $c \ge 1$, where A is a fiber of the ruling of $\mathbb{F}_a$ and B is the negative section, with $(B^2) = -a$; or E is the cone over a rational normal curve of degree δ with $(L^2) = \delta$. In particular, E is normal. In the last case, the vertex of E has embedding dimension $\delta + 1$. Since E is a closed subscheme of Y, Y is cDV along E, and a hyperplane section through a point lowers the embedding dimension by at most one, we have $\delta + 1 \le 4$ [KM98, Definition 5.32 and Theorem 4.22], that is, $\delta \le 3$.

We compute $\mathrm{Diff}_E(0)$. Let η be the generic point of C. The coefficient of $\mathrm{Diff}_E(0)$ along C depends only on the two-dimensional pair $(\operatorname{Spec}\mathcal{O}_{Y,\eta}, E)$ [Kol13, Definition 4.2]. Recall from **Step 3** the general surface $H \subset X$, its strict transform H_Y, the minimal resolution $\sigma : \tilde{H} \to H_Y$ near $C_H = E \cap H_Y$, the strict transform C_0 of C_H, and the (-2)-curves $D_1, \dots, D_t$ over the points of $C \cap H_Y$. Let $p \in C \cap H_Y$, and let D_j be the (-2)-curve with $\sigma(D_j) = p$. By **Step 3**(2), $p \notin \Sigma$, so by **Step 2**, there exists an open neighborhood $U \subset Y\backslash\Sigma$ of p with $\operatorname{Sing} U = C \cap U$.

Let $\beta : Y'' \to U$ be the blow-up of $C \cap U$. By **Step 2**, at any point of $C \cap U$, U is analytically isomorphic to the product of a smooth curve germ and an A_1 surface singularity, where C corresponds to the curve factor. Since blowing up commutes with flat base change, near any point of $C \cap U$, the morphism β is analytically isomorphic to the product of a smooth curve germ and the blow-up of the A_1 surface singularity at its singular point, which is its minimal resolution. Thus Y'' is smooth, $K_{Y''} = \beta^* K_U$, and the exceptional divisor F of β is irreducible. Since a local equation of H_Y restricts to a local coordinate of C at p (**Step 3**), near $\beta^{-1}(p)$ the strict transform of H_Y on Y'' is the blow-up of H_Y at p, that is, $\tilde{H}$ near D_j. Thus the fiber $\beta^{-1}(p)$ of F over p is D_j, and $F|_{\tilde{H}} = D_j$ near D_j. Let E' be the strict transform of E on Y''. Then $E'|_{\tilde{H}} = C_0$ near D_j (in general $E'|_{\tilde{H}} = C_0 + kD_j$ with $k \ge 0$, and $(\beta^* E \cdot D_j) = 0$ gives $1 - 2k = 2a$, where $a > 0$ is the coefficient of F in $\beta^* E$, so $k = 0$). By **Step 3**, C_0 intersects D_j transversally at one point, so

$$(E' \cdot D_j) = (C_0 \cdot D_j) = 1, \qquad (F \cdot D_j) = (D_j^2)_{\tilde{H}} = -2.$$

Write $\beta^* E = E' + aF$. Since $(\beta^* E \cdot D_j) = 0$, we have $1 - 2a = 0$. Thus $\beta^* E = E' + \frac{1}{2}F$, and

$$K_{Y''} + E' + \frac{1}{2}F = \beta^*(K_U + E|_U).$$

Localizing at η, the curve F_η is the unique exceptional curve of the minimal resolution of the two-dimensional pair $(\operatorname{Spec}\mathcal{O}_{Y,\eta}, E)$, and its discrepancy is $-\frac{1}{2}$. Thus the coefficient of $\mathrm{Diff}_E(0)$ along C is $\frac{1}{2}$ [Kol13, Proposition 2.35(5)]. At any other point of codimension one of E, both Y and E are regular by **Step 2**, so the coefficient of $\mathrm{Diff}_E(0)$ is 0 [Kol13, Proposition 2.35(3)]. Since $K_Y \sim 0$ and $2E|_E \sim -L$, by [Kol13, Definition 4.2],

$$K_E + \frac{1}{2}C \sim_{\mathbb{Q}} (K_Y + E)|_E \sim_{\mathbb{Q}} -\frac{1}{2}L. \tag{9.23}$$

We compute $\widehat{\mathrm{vol}}(\mathrm{ord}_E)$. Since f is an isomorphism over $X\backslash\{x\}$ and E is the only prime divisor on Y over x, the valuation ideals of ord_E are $\mathfrak{a}_k := f_*\mathcal{O}_Y(-kE)$ for $k \ge 0$. For any $j \ge 0$, $\mathcal{O}_Y(-2jE) = \mathfrak{m}_x^j \mathcal{O}_Y$ is invertible, and $\mathcal{O}_Y(-2E)$ is f-ample. The ideal of the scheme $f^{-1}(x)$ is $\mathcal{O}_Y(-2E)$, and $[f^{-1}(x)] = 2[E]$ as cycles. Thus, by relative Serre vanishing and the asymptotic Riemann–Roch theorem on the projective scheme $f^{-1}(x)$, for $j \gg 0$,

$$\dim_{\mathbb{C}} \mathfrak{a}_{2j}/\mathfrak{a}_{2j+2} = h^0\big(f^{-1}(x), \mathcal{O}_Y(-2jE)|_{f^{-1}(x)}\big) = \frac{2(L^2)}{2} j^2 + O(j) = (L^2) j^2 + O(j).$$

Summing over j, we obtain

$$\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{a}_{2j} = \frac{(L^2)}{4} \cdot \frac{(2j)^3}{6} + O(j^2),$$

so $\operatorname{vol}(\operatorname{ord}_E) = \frac{(L^2)}{4}$. Since $a(\operatorname{ord}_E, X) = 1$,

$$V \le \widehat{\operatorname{vol}}(\operatorname{ord}_E) = \frac{(L^2)}{4}.$$

The computation above only uses that $-E$ is f-ample (compare [LX20, Lemma 2.11]).

If $E \cong \mathbb{P}^2$, then $(L^2) \le 4$, so $V \le 1$. If E is a cone, then $(L^2) = \delta \le 3$, so $V \le \frac{3}{4}$. Suppose that $E \cong \mathbb{F}_a$. Since $K_{\mathbb{F}_a} \sim -2B - (a+2)A$ and $\operatorname{Pic}(\mathbb{F}_a)$ is torsion free, (9.23) implies that

$$C \sim -2K_E - L \sim 3B + (a + 4 - c)A.$$

Since $(C \cdot A) = 3 \ne 1 = (B \cdot A)$, we have $C \ne B$, and hence

$$0 \le (C \cdot B) = -3a + (a + 4 - c) = 4 - 2a - c.$$

Thus $(L^2) = a + 2c \le a + 2(4 - 2a) = 8 - 3a \le 8$, so $V \le 2$.

In any case, $V \le 2 < \frac{125}{18}$, which contradicts the assumption $V > \frac{125}{18}$. Therefore $V \le \frac{125}{18}$. □

Proof of Theorem 9.1. First suppose that $X \ni x$ is a hypersurface germ. Since $\operatorname{crex}(X \ni x) = 2$, $X \ni x$ is not cDV by Theorem 2.15(1), and in particular $X \ni x$ is singular. Write $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[z_1, \dots, z_4]]/(F)$, and let $m := \operatorname{mult}_0 F \ge 2$. If $m \ge 3$, then Proposition 2.20 with weights $(1,1,1,1)$ implies that $\widehat{\operatorname{vol}}(X \ni x) \le (4-m)^3 m$. Since $\widehat{\operatorname{vol}}(X \ni x) > 0$, we have $m = 3$ and $\widehat{\operatorname{vol}}(X \ni x) \le 3$.

Suppose that $m = 2$ and the quadratic part of F has rank at least 2. Let h be a general element of $\mathfrak{m}_x$, and let $H := (h = 0) \subset X$. Since X is normal, $\dim \operatorname{Sing} X \le 1$ near x, and h does not vanish identically on any irreducible component of $\operatorname{Sing} X$ through x. Since the generators of $\mathfrak{m}_x$ span a base-point-free linear system on $X \backslash \{x\}$, H is smooth outside $\operatorname{Sing} X$ [Har77, Corollary III.10.9 and Remark III.10.9.2]. Thus $H \ni x$ is an isolated singularity. The linear part of h is a general linear form, and the restriction of the quadratic part of F to a general hyperplane has rank at least 2. Thus, by [GLS07, Theorem I.2.47], after a coordinate change, the equation of $H \ni x$ is $uv + \psi(w)$. Here $\psi \ne 0$, since otherwise H is singular along $\{u = v = 0\}$, which contradicts that $H \ni x$ is an isolated singularity. Thus $H \ni x$ is a Du Val singularity of type A, and $X \ni x$ is cDV [KM98, Definition 5.32], a contradiction.

Suppose that $m = 2$ and the quadratic part of F has rank 1. Then by [GLS07, Theorem I.2.47], after a coordinate change, $F = z_1^2 + \varphi(z_2, z_3, z_4)$ with $\operatorname{ord} \varphi \ge 3$. Proposition 2.20 with weights $(3,2,2,2)$, for which F has weight 6, implies that

$$\widehat{\operatorname{vol}}(X \ni x) \le (9-6)^3 \cdot \frac{6}{3 \cdot 2 \cdot 2 \cdot 2} = \frac{27}{4}.$$

Now suppose that $X \ni x$ is not a hypersurface germ, and let $f : Y \to X$ be the blow-up of $\mathfrak{m}_x$. By Theorem 2.15(2), f is a crepant projective birational morphism, and Y is normal and Gorenstein canonical. Moreover, f is an isomorphism over $X \backslash \{x\}$, and $\mathfrak{m}_x \mathcal{O}_Y = \mathcal{O}_Y(-D)$ for a Cartier divisor $D > 0$ with $\operatorname{Supp} D = f^{-1}(x)$. Any prime component E_i of D satisfies $a(E_i, X) = 1$ and $\operatorname{center}_X E_i = \{x\}$, so D has at most $\operatorname{crex}(X \ni x) = 2$ prime components. If D has two prime components, then $\widehat{\operatorname{vol}}(X \ni x) \le 7$ by Proposition 9.10.

Suppose that $E := f^{-1}(x)_{\mathrm{red}}$ is a prime divisor. If Y is not cDV at some point of $f^{-1}(x)$, then $\widehat{\operatorname{vol}}(X \ni x) \le 7$ by Proposition 9.21. Suppose that Y is cDV at any point of E. By [KM98, Corollary 5.40], Y has only finitely many non-cDV points. Since none of them is contained in E and f is proper, their images form a finite subset $\Xi \subset X \backslash \{x\}$. Possibly replacing X with $X \backslash \Xi$, we may assume that Y has cDV singularities.

- If E is not Cartier, then the assumptions of Proposition 9.22 hold: $X \ni x$ is not a hypersurface germ, f is the blow-up of $\mathfrak{m}_x$ with Y normal and $K_Y = f^*K_X$, $f^{-1}(x)_{\mathrm{red}} = E$ is a prime divisor, and Y is cDV along E. Thus $\widehat{\mathrm{vol}}(X \ni x) \le \frac{125}{18}$.
- If E is Cartier, then $D = rE$ as Weil divisors for some positive integer r, and since the map from Cartier divisors to Weil divisors on the normal variety Y is injective [Har77, Remark II.6.11.2], $D = rE$ as Cartier divisors. Since $-D$ is f-ample, $-E$ is f-ample. Thus the conditions of Set-up 8.2 hold for $X \ni x$, f, and E. If E is normal, then $\widehat{\mathrm{vol}}(X \ni x) \le 7$ by Proposition 9.20. If E is not normal, then E is integral, Y is cDV near E, and $\widehat{\mathrm{vol}}(X \ni x) \le \frac{250}{33}$ by Proposition 9.17.

Since 3, $\frac{27}{4}$, 7, and $\frac{125}{18}$ are less than $\frac{250}{33}$, we obtain $\widehat{\mathrm{vol}}(X \ni x) \le \frac{250}{33}$ in any case. By Lemma 4.3, $q > \frac{250}{33}$.

Finally, let $X \ni x$ be a klt threefold germ which is not LCIQ, with $\widehat{\mathrm{vol}}(X \ni x) > q$. Since $q > \frac{9}{2}$, Lemma 5.1(4) implies that $X \ni x$ is Gorenstein canonical and is not a hypersurface germ. By Theorem 2.15(1), $\mathrm{crex}(X \ni x) \ne 0$. By Theorem 8.1, $\mathrm{crex}(X \ni x) \ne 1$, since otherwise $\widehat{\mathrm{vol}}(X \ni x) \le q$. By the first part of the theorem, $\mathrm{crex}(X \ni x) \ne 2$. Thus $\mathrm{crex}(X \ni x) \ge 3$. □

10. Quasi-regular Fano cones

In this section, we prove the upper bound of the local volume for quasi-regular Fano cones, that is, in the case when the minimizing valuation has rational rank one (Theorem 10.3 and Proposition 10.6). The local analysis at the points of the quotient surface S (Set-up 10.1) with non-trivial stabilizers (Subsections 10.1, 10.3, and 10.4) gives upper bounds of $\widehat{\mathrm{vol}}(X \ni o)$, but all of them exceed q when the canonical degree α is large. Thus we prove Theorem 10.3 globally on S, using complements of bounded index (Subsection 10.5, where Theorems 8.1 and 9.1 are used), the Kawamata–Viehweg vanishing, and Proposition 5.6 (Subsections 10.6 and 10.7).

Set-up 10.1. Let R, X, o, S, η, and α be as follows.

(1) $R = \bigoplus_{j \ge 0} R_j$ is a finitely generated normal $\mathbb{Z}_{\ge 0}$-graded $\mathbb{C}$-domain of dimension three with $R_0 = \mathbb{C}$ and $\gcd\{j \mid R_j \ne 0\} = 1$, $X := \operatorname{Spec} R$, $o \in X$ is the vertex defined by $\bigoplus_{j>0} R_j$, and $S := \operatorname{Proj} R$.
(2) X is Gorenstein and klt, and the canonical module of R, with the natural grading, is $\omega_R = R\eta$ for a homogeneous element η of degree $\alpha \in \mathbb{N}^+$.

We call α the *canonical degree* of R, and we denote by v_R the grading valuation of R, that is, $v_R(g) := \min\{j \mid g_j \ne 0\}$ for $g = \sum_j g_j$ with $g_j \in R_j$.

Lemma 10.2. *Let (X, ξ) be a K-semistable Fano cone with respect to a torus $T = \operatorname{Spec} \mathbb{C}[M]$, and let x be the vertex. Assume that X is a Gorenstein threefold, that ξ is quasi-regular, and that the canonical module of $\mathbb{C}[X]$ is generated by a T-homogeneous element η. Let ξ_0 be the primitive element of $\mathrm{Hom}(M, \mathbb{Z})$ in $\mathbb{R}_{>0}\xi$, and let $R := \mathbb{C}[X]$ with the grading*

$$R_j := \bigoplus_{\langle u, \xi_0 \rangle = j} \mathbb{C}[X]_u \quad \text{for } j \in \mathbb{Z}.$$

Then $R = \bigoplus_{j \ge 0} R_j$, and R, X, $o := x$, $S := \operatorname{Proj} R$, η, and $\alpha := a(\mathrm{wt}_{\xi_0}, X)$ satisfy the conditions of Set-up 10.1. Moreover, $v_R = \mathrm{wt}_{\xi_0}$, and v_R computes $\widehat{\mathrm{vol}}(X \ni x)$.

Proof. Write $\xi_0 = c\xi$ with $c > 0$. Since ξ_0 is primitive in $\mathrm{Hom}(M, \mathbb{Z})$, we have $\langle M, \xi_0 \rangle = \mathbb{Z}$. Since the action of T is faithful, the weights u with $\mathbb{C}[X]_u \ne 0$ generate M (if the weights generated a proper sublattice M', then $\operatorname{Spec} \mathbb{C}[M/M'] \subset T$ would act trivially). Thus the integers j with $R_j \ne 0$ generate $\mathbb{Z}$, that is, $\gcd\{j \mid R_j \ne 0\} = 1$. Since the action is good and ξ is a Reeb vector, we have $\langle u, \xi_0 \rangle = c\langle u, \xi \rangle > 0$ for any $u \ne 0$ with $\mathbb{C}[X]_u \ne 0$. Thus $R_j = 0$ for $j < 0$, $R_0 = \mathbb{C}[X]_0 = \mathbb{C}$, and $\bigoplus_{j>0} R_j = \bigoplus_{u \ne 0} \mathbb{C}[X]_u$ is a T-invariant maximal ideal, which defines

the vertex x. Since X is a normal affine threefold, R is a finitely generated normal domain of dimension three. This proves Set-up 10.1(1) with $o = x$.

Let $u_\eta \in M$ be the weight of η. Then η is homogeneous of degree $\langle u_\eta, \xi_0\rangle \in \mathbb{Z}$ for the natural grading of ω_R induced by the grading of R. Since ξ_0 is a Reeb vector, Lemma 2.9 implies that

$$\langle u_\eta, \xi_0\rangle = a(\mathrm{wt}_{\xi_0}, X) = \alpha,$$

and $\alpha > 0$ since X is klt (Definition 2.6). Since X is Gorenstein and klt, Set-up 10.1(2) holds.

For $g = \sum_u g_u \in R$, let $g_j := \sum_{\langle u,\xi_0\rangle = j} g_u$. Then

$$\mathrm{wt}_{\xi_0}(g) = \min\{\langle u, \xi_0\rangle \mid g_u \neq 0\} = \min\{j \mid g_j \neq 0\} = v_R(g),$$

so $v_R = \mathrm{wt}_{\xi_0}$.

By Definition 2.8, $\mathrm{wt}_{\xi_0} = c\,\mathrm{wt}_\xi$, so $\mathfrak{a}_\lambda(\mathrm{wt}_{\xi_0}) = \mathfrak{a}_{\lambda/c}(\mathrm{wt}_\xi)$ for any $\lambda \geq 0$, and Definition 2.6 implies that $\mathrm{vol}(\mathrm{wt}_{\xi_0}) = c^{-3}\,\mathrm{vol}(\mathrm{wt}_\xi)$. By Lemma 2.9 for ξ and for ξ_0, we have

$$a(\mathrm{wt}_{\xi_0}, X) = \langle u_\eta, \xi_0\rangle = c\langle u_\eta, \xi\rangle = ca(\mathrm{wt}_\xi, X).$$

Thus

$$\widehat{\mathrm{vol}}(\mathrm{wt}_{\xi_0}) = c^3 a(\mathrm{wt}_\xi, X)^3 \cdot c^{-3}\,\mathrm{vol}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(X \ni x),$$

where the last equality holds since (X, ξ) is K-semistable (Definition 2.8). □

The main result of this section is the following theorem. We prove it in Subsection 10.7.

Theorem 10.3. *Notation and conditions as in Set-up 10.1. Assume that $\alpha \geq 2$, that v_R computes $\widehat{\mathrm{vol}}(X \ni o)$, and that $X \ni o$ is not LCIQ. Then*

$$\widehat{\mathrm{vol}}(X \ni o) \leq q.$$

10.1. **Seifert bundles.** In this subsection, we describe the cone X in Set-up 10.1 by data on the surface S, following [Kol04] and [Wat81], and we bound the stabilizers of the $\mathbb{G}_m$-action on $X\backslash\{o\}$. The arguments in the rest of this section take place on S.

Construction-Proposition 10.4. Notation and conditions as in Set-up 10.1. Let $\mathbb{G}_m$ act on X by $\lambda \cdot g := \lambda^j g$ for $g \in R_j$. The reduced fibers of the quotient morphism $X\backslash\{o\} \to S$ over the closed points of S are the $\mathbb{G}_m$-orbits in $X\backslash\{o\}$ [Kol04, Definitions 1 and 3, Theorem 7], and we call the reduced fiber over a closed point $p \in S$ the *orbit over* p. There exist a $\mathbb{Q}$-divisor D on S such that kD is an ample Cartier divisor for some positive integer k, and a homogeneous element u of degree one in the fraction field of R, such that $R_j = H^0(S, \mathcal{O}_S(\lfloor jD\rfloor))u^j$ for any $j \geq 0$. Let

$$f : Y := \mathrm{Spec}_S \bigoplus_{j\geq 0} \mathcal{O}_S(\lfloor jD\rfloor) \to X$$

be the natural morphism, and let $E \subset Y$ be the zero section. For an open subset $U \subset S$, let $Y_U \subset Y$ be the preimage of U under the natural projection $Y \to S$. For a closed point $p \in S$, let $m(p)$ be the order of the stabilizer in $\mathbb{G}_m$ of the orbit over p in $X\backslash\{o\}$. Let $\Delta := \sum_P (1 - \frac{1}{m_P})P$, where P runs over the prime divisors on S and m_P is the order of the stabilizer of the orbits over the generic point of P. Then the following hold.

(1) f is a projective birational morphism, f is an isomorphism over $X\backslash\{o\}$, $f^{-1}(o)_{\mathrm{red}} = E$, and $E \cong S$.
(2) For any closed point $p \in S$, there exist an open neighborhood U of p in S and a normal klt surface T with a faithful action of $\mu_{m(p)}$ such that $U = T/\mu_{m(p)}$, the quotient morphism $T \to U$ is a cyclic cover of degree $m(p)$ with one point over p, and the following hold.
   (i) There exists an isomorphism $Y_U \cong (T \times \mathbb{A}^1)/\mu_{m(p)}$ over U under which $E \cap Y_U$ is the image of $T \times \{0\}$, where $\mu_{m(p)}$ acts on $\mathbb{A}^1$ with weight -1 and the morphism $(T \times \mathbb{A}^1)/\mu_{m(p)} \to U$ is induced by the first projection.
   (ii) $m(p)E$ is Cartier on Y_U.
   (iii) K_T is the pullback of $K_U + \Delta|_U$.

(iv) The pullback of u under the composite $T \times \mathbb{A}^1 \to (T \times \mathbb{A}^1)/\mu_{m(p)} \cong Y_U$ of (i) is ψt for a rational function ψ on T whose divisor is the pullback of $D|_U$ under $T \to U$, where t is the coordinate of $\mathbb{A}^1$.

(3) $\mathrm{ord}_E = v_R$, (Y, E) is plt, $a(E, X) = a(v_R, X) = \alpha$, $K_Y = f^*K_X + (\alpha - 1)E$, and $a(F, X) = a(F, Y, E) + \alpha \,\mathrm{mult}_F E$ for any prime divisor F over Y.

(4) Let ω be the rational 2-form on S with $\eta = u^{\alpha-1}du \wedge \omega$, and let $K_S := \mathrm{div}(\omega)$. Then $K_S + \Delta + \alpha D = 0$.

(5) If v_R computes $\widehat{\mathrm{vol}}(X \ni o)$, then (S, Δ) is a K-semistable klt log Fano pair, and

$$\widehat{\mathrm{vol}}(X \ni o) = \alpha((K_S + \Delta)^2).$$

(6) The following hold.

(i) For any prime divisor P on S, the coefficient of D along P, written as a fraction in lowest terms, has denominator m_P.

(ii) For any integer $j \geq 0$, $0 \leq jD - \lfloor jD \rfloor \leq \Delta$.

(iii) For any integer $j \geq 0$ and any prime divisor P on S with $m_P \mid j$, the coefficient of $jD - \lfloor jD \rfloor$ along P is zero.

(iv) For any closed point $p \in S$, $m(p)D$ is a Cartier divisor near p.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct D and u, and we prove (6).

Since R is normal and $\gcd\{j \mid R_j \neq 0\} = 1$, there exist a homogeneous element u of degree one in the fraction field of R and a $\mathbb{Q}$-divisor D on S such that kD is an ample Cartier divisor for some positive integer k and $R_j = H^0(S, \mathcal{O}_S(\lfloor jD \rfloor))u^j$ for any $j \geq 0$, by Demazure's theorem [Wat81, Introduction and (1.1)]. Moreover,

$$X \backslash \{o\} \cong \mathrm{Spec}_S \bigoplus_{j \in \mathbb{Z}} \mathcal{O}_S(\lfloor jD \rfloor)$$

over S [Wat81, Section 1], so $\mathcal{O}_S(\lfloor jD \rfloor)$ is the degree-j eigensheaf of the Seifert $\mathbb{G}_m$-bundle $X \backslash \{o\} \to S$. By [Kol04, Theorem 7(3)], for any point $p \in S$, $m(p)$ is the smallest positive integer N such that N times the coefficient of D along any prime divisor through p is an integer and $\mathcal{O}_S(ND)$ is invertible at p.

Let P be a prime divisor on S, and write the coefficient of D along P as $\frac{p_0}{q_0}$ in lowest terms with $q_0 > 0$. At the generic point of P, S is smooth, so any rank one reflexive sheaf is invertible there, and by [Kol04, Theorem 7(3)], m_P is the smallest positive integer N such that $N\frac{p_0}{q_0} \in \mathbb{Z}$. Thus $m_P = q_0$. For any integer $j \geq 0$, the coefficient of $jD - \lfloor jD \rfloor$ along P belongs to $\{0, \frac{1}{q_0}, \ldots, \frac{q_0-1}{q_0}\}$, and it is zero if q_0 divides j. Since $1 - \frac{1}{m_P} = \frac{q_0-1}{q_0}$, we have $0 \leq jD - \lfloor jD \rfloor \leq \Delta$. Finally, for a closed point $p \in S$, [Kol04, Theorem 7(3)] implies that $m(p)D$ has integral coefficients along the prime divisors through p and that $\mathcal{O}_S(m(p)D)$ is invertible at p, that is, $m(p)D$ is Cartier near p. This proves (6).

Step 2. In this step, we prove (1).

By [Wat81, Section 1], f is proper, f maps $Y \backslash E$ isomorphically onto $X \backslash \{o\}$, and $f(E) = \{o\}$. Thus f is birational, f is an isomorphism over $X \backslash \{o\}$, and $f^{-1}(o)_{\mathrm{red}} = E$. Since the ideal of E is $\bigoplus_{j>0} \mathcal{O}_S(\lfloor jD \rfloor)$ and $\mathcal{O}_S(\lfloor 0 \cdot D \rfloor) = \mathcal{O}_S$, we have $E \cong S$. Since Y is affine over the projective surface S, Y is quasi-projective, so the proper morphism f is projective. This proves (1).

Step 3. In this step, we prove (2).

Fix a closed point $p \in S$ and let $m := m(p)$. By (6), mD is Cartier near p, so there exists an affine open neighborhood U of p such that $mD|_U$ is Cartier and $\mathcal{O}_U(mD) \cong \mathcal{O}_U$. We fix such an isomorphism. For $j \in \mathbb{Z}$, let $L_j := \mathcal{O}_U(\lfloor jD \rfloor)$. Then $L_{j+m} = L_j \otimes \mathcal{O}_U(mD) \cong L_j$ for any j.

Following [Kol04, Paragraph 18], let

$$\pi : T := \operatorname{Spec}_U \bigoplus_{i=0}^{m-1} L_i \to U,$$

where the product of L_i and $L_{i'}$ is followed by the isomorphism $L_{i+i'} \cong L_{i+i'-m}$ when $i+i' \geq m$, and let $\lambda \in \mu_m$ act on L_i by multiplication by λ^i. Then π is a cyclic cover of degree m with group μ_m, $\pi^{-1}(p)$ is one point, and the natural morphism $T \times \mathbb{G}_m \to Y_U \backslash E$ is the quotient by μ_m, where μ_m acts on $\mathbb{G}_m$ with weight -1, and it is étale [Kol04, Paragraphs 18–21 and Proposition 19]. The μ_m-invariant part of $\bigoplus_{i=0}^{m-1} L_i$ is $L_0 = \mathcal{O}_U$, so $U = T/\mu_m$ and π is the quotient morphism. The action of μ_m on T is faithful, since for $m \geq 2$ any $\lambda \in \mu_m$ acts on the non-zero summand L_1 by multiplication by λ. Since $Y_U \backslash E$ is isomorphic to an open subset of X by (1), $T \times \mathbb{G}_m$ is normal and klt. Since the projection $T \times \mathbb{G}_m \to T$ is smooth and surjective, T is normal [Mat86, Corollary to Theorem 23.9]. Since T is normal, its connected components are irreducible. Each irreducible component of T surjects onto U, since $\bigoplus_{i=0}^{m-1} L_i$ is torsion free over $\mathcal{O}_U$ and π is finite, so it contains the only point of T over p. Thus T is connected, hence irreducible. Since $K_{T \times \mathbb{G}_m}$ is the pullback of K_T and $T \cong T \times \{1\}$, K_T is $\mathbb{Q}$-Cartier. For a log resolution $T' \to T$, the morphism $T' \times \mathbb{G}_m \to T \times \mathbb{G}_m$ is a log resolution, and $a(F \times \mathbb{G}_m, T \times \mathbb{G}_m) = a(F, T)$ for any prime divisor F on T', so T is klt [KM98, Corollary 2.32].

Let μ_m act on $T \times \mathbb{A}^1$ by the action on T and with weight -1 on the coordinate t of $\mathbb{A}^1$. For $0 \leq i \leq m-1$ and $j \geq 0$, an element ht^j with $h \in L_i$ is μ_m-invariant if and only if $i \equiv j \pmod m$. Thus the degree-j part of the invariant ring is $L_{\bar{j}} t^j \cong L_j$, where $0 \leq \bar{j} \leq m-1$ and $\bar{j} \equiv j \pmod m$, and the multiplications match since both use the fixed isomorphism $\mathcal{O}_U(mD) \cong \mathcal{O}_U$. Therefore

$$Y_U = \operatorname{Spec}_U \bigoplus_{j \geq 0} L_j \cong (T \times \mathbb{A}^1)/\mu_m,$$

see also [Kol04, Paragraph 14]. The isomorphism is induced by an isomorphism of $\mathcal{O}_U$-algebras, so it is an isomorphism over U, where $(T \times \mathbb{A}^1)/\mu_m \to U$ is induced by the first projection. The ideal $\bigoplus_{j>0} L_j$ of $E \cap Y_U$ corresponds to the ideal generated by the invariant elements of positive degree in t. The invariant elements of positive degree in t vanish on $T \times \{0\}$, and t^m is one of them, so the preimage of $E \cap Y_U$ in $T \times \mathbb{A}^1$ is $T \times \{0\}$, that is, $E \cap Y_U$ is the image of $T \times \{0\}$. Under the isomorphism $Y_U \cong (T \times \mathbb{A}^1)/\mu_m$, a homogeneous element of degree j in the fraction field of R pulls back to ht^j for some rational function h on T. Let e be the rational function on U such that the fixed isomorphism $\mathcal{O}_U(mD) \cong \mathcal{O}_U$ is the multiplication by e, so that $\operatorname{div}(e) = mD|_U$. If $m \geq 2$, let ψ be the rational function on T given by the rational section 1 of the summand L_1 of $\bigoplus_{i=0}^{m-1} L_i$. Then the pullback of u is ψt, and $\psi^m = \pi^* e$, since products in $\bigoplus_{i=0}^{m-1} L_i$ of degree at least m are computed through the fixed isomorphism $\mathcal{O}_U(mD) \cong \mathcal{O}_U$. If $m = 1$, then $T = U$, π is the identity, and L_1 corresponds to $L_0 t$ through the multiplication by e, so the pullback of u is ψt with $\psi := e = \pi^* e$, the image of the rational section 1 of L_1 under the fixed isomorphism $L_1 \cong L_0 = \mathcal{O}_U$. In both cases, $\psi^m = \pi^* e$, and hence $\operatorname{div}_T(\psi) = \pi^*(D|_U)$.

Let $\tilde{\pi} : T \times \mathbb{A}^1 \to Y_U$ be the quotient morphism. Since the action of μ_m on T is faithful, it is generically free. Thus $\tilde{\pi}$ is unramified at the generic point of $T \times \{0\}$, and $\tilde{\pi}$ is étale over $Y_U \backslash E$. Since t^m is μ_m-invariant, t^m is a regular function on Y_U. Since $\tilde{\pi}^{-1}(E) = T \times \{0\}$ and $\tilde{\pi}^* \operatorname{div}(t^m) = m(T \times \{0\})$, we have $\operatorname{div}(t^m) = mE$ on Y_U. Thus mE is Cartier on Y_U.

Let P be a prime divisor on U, and let P' be a prime divisor on T over P. The morphism $T \times \mathbb{G}_m \to Y_U \backslash E$ is a $\mathbb{G}_m$-equivariant quotient by μ_m, and μ_m acts freely on the factor $\mathbb{G}_m$. Thus, for a general closed point $x \in P'$, the stabilizer of x in μ_m is isomorphic to the stabilizer in $\mathbb{G}_m$ of the orbit over $\pi(x)$, whose order is m_P [Kol04, Theorem 7(3)]. The stabilizer in μ_m of a general point of P' is the inertia group of P' (an element of μ_m outside the inertia group fixes only finitely many points of P', and in characteristic zero the ramification index along P' is the order of the inertia group), so the ramification index of π along P' is m_P. In particular, if

$P \not\subset \operatorname{Supp}\Delta$, then π is étale at the general points of P'. Thus the Hurwitz formula implies that $K_T = \pi^*(K_U + \Delta|_U)$. This proves (2).

Step 4. In this step, we prove (3).

Since $Y_U \cong (T \times \mathbb{A}^1)/\mu_{m(p)}$ for an open cover of S by such U (**Step 3**), Y is normal. Over the generic point of S, the sheaf $\mathcal{O}_S(\lfloor jD \rfloor)$ is $\mathbb{C}(S)$, so the local ring of Y at the generic point of E is $\mathbb{C}(S)[u]_{(u)}$. Thus $\operatorname{ord}_E(u) = 1$, and any non-zero element of $R_j = H^0(S, \mathcal{O}_S(\lfloor jD \rfloor))u^j$ has order j along E. Hence $\operatorname{ord}_E(g) = \min\{j \mid g_j \neq 0\}$ for $g = \sum_j g_j \in R$, that is, $\operatorname{ord}_E = v_R$.

The fraction field of R is $\mathbb{C}(S)(u)$, du has degree one, and the rational 2-forms on S have degree zero. Since η is homogeneous of degree α, there exists a unique rational 2-form ω on S such that $\eta = u^{\alpha-1}du \wedge \omega$. Let $z_1, z_2 \in \mathbb{C}(S)$ be a transcendence basis of $\mathbb{C}(S)$ over $\mathbb{C}$, and write $\omega = c dz_1 \wedge dz_2$ with $c \in \mathbb{C}(S)^*$. Then c is a unit at the generic point of E, and $du \wedge dz_1 \wedge dz_2$ generates the canonical module of $\mathbb{C}(S)[u]_{(u)}$. Thus η vanishes to order $\alpha - 1$ along E. Since η generates ω_R, we may take $K_X := \operatorname{div}_X(\eta) = 0$ and $K_Y := \operatorname{div}_Y(\eta)$. Since f is an isomorphism over $X \backslash \{o\}$, K_Y is supported on E, so

$$K_Y = f^*K_X + (\alpha - 1)E \quad \text{and} \quad a(E, X) = \alpha.$$

Since $\operatorname{ord}_E = v_R$, we also have $a(v_R, X) = a(E, X) = \alpha$.

By (2), E is $\mathbb{Q}$-Cartier, so $K_Y + E$ is $\mathbb{Q}$-Cartier. Fix a closed point $p \in S$, and let U, T, and $\tilde{\pi}$ be as in **Step 3**. Since $\tilde{\pi}$ is étale over $Y_U \backslash E$, $\tilde{\pi}$ is unramified at the generic point of $T \times \{0\}$, and $\tilde{\pi}^*E = T \times \{0\}$, we have

$$K_{T \times \mathbb{A}^1} + T \times \{0\} = \tilde{\pi}^*(K_{Y_U} + E|_{Y_U}).$$

Since $T \times \{0\} \cong T$ is klt and $T \times \{0\}$ is a Cartier divisor, $(T \times \mathbb{A}^1, T \times \{0\})$ is plt near $T \times \{0\}$ [KM98, Theorem 5.50]. Since $T \times (\mathbb{A}^1 \backslash \{0\})$ is klt, $(T \times \mathbb{A}^1, T \times \{0\})$ is plt. Since $\tilde{\pi}$ is a finite morphism of degree m between normal varieties, [KM98, Proposition 5.20(2)] gives $\operatorname{discrep}(Y_U, E|_{Y_U}) + 1 \geq \frac{1}{m}(\operatorname{discrep}(T \times \mathbb{A}^1, T \times \{0\}) + 1) > 0$, so $(Y_U, E|_{Y_U})$ is plt [KM98, Definition 2.34]. Since p is arbitrary, (Y, E) is plt.

Let F be a prime divisor over Y. Since $K_Y - (\alpha - 1)E = f^*K_X$, [KM98, Lemma 2.30] implies that

$$a(F, X) = a(F, Y, -(\alpha - 1)E) = a(F, Y, E) + \alpha \operatorname{mult}_F E.$$

This proves (3).

Step 5. In this step, we prove (4).

Let ω be as in **Step 4**. Let P be a prime divisor on S, write the coefficient of D along P as $\frac{p_0}{q_0}$ in lowest terms with $q_0 > 0$, and let $k_P := \operatorname{ord}_P(\omega)$. By (6), $q_0 = m_P$. It suffices to show that

$$k_P + \left(1 - \frac{1}{q_0}\right) + \alpha\frac{p_0}{q_0} = 0.$$

Fix a closed point $p \in P$, let U, T, and π be as in **Step 3** for p, and let P' be a prime divisor on T over $P \cap U$. By **Step 3**, the ramification index of π along P' is q_0. Let $z \in \mathbb{C}(S)$ be a uniformizer of the local ring of S at the generic point of P. Then $\operatorname{ord}_{P'}(\pi^*z) = q_0$. By (2), the pullback of u to $T \times \mathbb{G}_m$ is ψt for a rational function ψ on T with $\operatorname{div}_T(\psi) = \pi^*(D|_U)$. Since the ramification index of π along P' is q_0 and the coefficient of D along P is $\frac{p_0}{q_0}$, we have $\operatorname{ord}_{P'}(\psi) = q_0 \cdot \frac{p_0}{q_0} = p_0$.

By the Hurwitz formula, $\operatorname{ord}_{P'}(\pi^*\omega) = q_0k_P + q_0 - 1$. Since $d\psi \wedge \pi^*\omega$ is a rational 3-form on the surface T, it is zero. Thus the pullback of η to $T \times \mathbb{G}_m$ is

$$(\psi t)^{\alpha-1}(\psi dt + t d\psi) \wedge \pi^*\omega = \psi^\alpha t^{\alpha-1} dt \wedge \pi^*\omega,$$

which vanishes to order $\alpha p_0 + q_0k_P + q_0 - 1$ along $P' \times \mathbb{G}_m$. Since η generates the canonical module of X and $T \times \mathbb{G}_m \to Y_U \backslash E$ is étale, the pullback of η generates the canonical module of $T \times \mathbb{G}_m$ at the generic point of $P' \times \mathbb{G}_m$. Thus

$$\alpha p_0 + q_0k_P + q_0 - 1 = 0.$$

Dividing by q_0, we obtain the desired equality. Since P is arbitrary, $K_S + \Delta + \alpha D = 0$. This proves (4).

Step 6. In this step, we prove (5).

Assume that v_R computes $\widehat{\mathrm{vol}}(X \ni o)$. By (2), $K_T = \pi^*(K_U + \Delta|_U)$ for a finite morphism $\pi : T \to U$ from a klt surface T, so $(U, \Delta|_U)$ is klt [KM98, Proposition 5.20(4)]. Thus (S, Δ) is klt. By (4), $-(K_S + \Delta) = \alpha D$, which is ample since kD is ample and $\alpha > 0$. Thus (S, Δ) is a klt log Fano pair.

We show that E is a Kollár component over $X \ni o$ (Definition 2.18). By (1), f is an isomorphism over $X \backslash \{o\}$ and $f^{-1}(o)_{\mathrm{red}} = E$. By (3), (Y, E) is plt, and by (2), $-E$ is $\mathbb{Q}$-Cartier. Fix a positive integer k such that kD is an ample Cartier divisor. By **Step 4**, a local section $\sum_j g_j u^j$ of $\mathcal{O}_Y$ has order $\min\{j \mid g_j \neq 0\}$ along E, so

$$\mathcal{O}_Y(-kE) = \bigoplus_{j \geq k} \mathcal{O}_S(\lfloor jD \rfloor) u^j = \mathcal{O}_S(kD) u^k \cdot \mathcal{O}_Y,$$

where the second equality holds since kD is Cartier. Thus $\mathcal{O}_Y(-kE)$ is the pullback of $\mathcal{O}_S(kD)$ under the projection $Y \to S$, and the restriction of $\mathcal{O}_Y(-kE)$ to $E \cong S$ is $\mathcal{O}_S(kD)$, which is ample. Since E is the only positive-dimensional fiber of f, $-E$ is f-ample. Thus E is a Kollár component over $X \ni o$.

We show that $\mathrm{Diff}_E(0) = \Delta$ under the isomorphism $E \cong S$. The different is determined by the local structure of Y at the codimension one points of E [Kol13, Proposition 4.5]. Let P be a prime divisor on S, and let p be a general closed point of P. Then S is smooth at p, and P is the only prime divisor through p along which D has non-integral coefficient, so $m(p) = m_P$ by [Kol04, Theorem 7(3)] and **Step 1**. Let U, T, and π be as in **Step 3** for p. Since π has degree m_P and the ramification index of π along any prime divisor over $P \cap U$ is m_P, the group μ_{m_P} acts faithfully on a transverse coordinate of T along the preimage of P. Thus, at the point of Y given by the generic point of $P \subset E$, Y is a two-dimensional cyclic quotient singularity by μ_{m_P}, where μ_{m_P} acts faithfully on the transverse coordinate of T and with weight -1 on t, and E is the image of $\{t = 0\}$. Thus the coefficient of $\mathrm{Diff}_E(0)$ along P is $1 - \frac{1}{m_P}$ [Kol13, Theorem 3.32 and (3.36.1)], see also [Kol13, Example 4.3]. Hence $\mathrm{Diff}_E(0) = \Delta$.

Since $v_R = \mathrm{ord}_E$ (**Step 4**) computes $\widehat{\mathrm{vol}}(X \ni o)$, [LX20, Theorem 1.2] implies that (S, Δ) is K-semistable in the sense of test configurations, hence K-semistable in the sense of Definition 2.11 [ADL24, Theorem 2.9(1)]. By Lemma 2.19(1), the equality $\mathrm{Diff}_E(0) = \Delta$, and (3),

$$\widehat{\mathrm{vol}}(X \ni o) = \widehat{\mathrm{vol}}(\mathrm{ord}_E) = a(E, X) \cdot ((K_S + \Delta)^2) = \alpha((K_S + \Delta)^2).$$

This proves (5). □

The following lemma provides the numerical bound used in the local arguments of this section.

Lemma 10.5. *Notation and conditions as in Set-up 10.1, and let $m(p)$ be as in Construction-Proposition 10.4. Assume that v_R computes $V := \widehat{\mathrm{vol}}(X \ni o)$. Let $p \in S$ be a closed point, let U, T, and $T \to U$ be as in Construction-Proposition 10.4(2) for p, and let $\tilde{p} \in T$ be the point over p. Then the germ $T \ni \tilde{p}$ is analytically isomorphic to $\mathbb{C}^2/H_p$ at the origin for a finite subgroup $H_p \subset SL_2(\mathbb{C})$, and*

$$m(p)|H_p| \leq \frac{9\alpha}{V}.$$

Proof. Let $\pi : T \to U$ be the cyclic cover. By Construction-Proposition 10.4(2) and [Kol04, Proposition 19], the morphism $T \times \mathbb{G}_m \to Y_U \backslash E$ is étale, where $\mu_{m(p)}$ acts freely on the factor $\mathbb{G}_m$ with weight -1. Since f is an isomorphism over $X \backslash \{o\}$ and X is Gorenstein, $T \times \mathbb{G}_m$ is Gorenstein. Since the projection $T \times \mathbb{G}_m \to T$ is flat with smooth fibers, T is Gorenstein [Mat86, Theorem 23.4]. Since T is klt and K_T is Cartier, the discrepancies of T are integers greater than -1, so T is canonical [KM98, Definition 2.34]. Thus $T \ni \tilde{p}$ is analytically isomorphic to $\mathbb{C}^2/H_p$

at the origin for a finite subgroup $H_p \subset SL_2(\mathbb{C})$ [KM98, Theorem 4.20 and Remark 4.21(1)]. By [LL19, Proposition 4.10] and Proposition 2.7(1), $\widehat{\mathrm{vol}}(T \ni \tilde{p}) = \frac{4}{|H_p|}$.

By Construction-Proposition 10.4(2), π is a finite Galois morphism of degree $m(p)$ with $\pi^{-1}(p) = \{\tilde{p}\}$ and $K_T = \pi^*(K_U + \Delta|_U)$, and $(U, \Delta|_U)$ is klt by Construction-Proposition 10.4(5). Thus Proposition 2.7(2) implies that

$$\widehat{\mathrm{vol}}(S \ni p, \Delta) = \frac{\widehat{\mathrm{vol}}(T \ni \tilde{p})}{m(p)} = \frac{4}{m(p)|H_p|}.$$

By Construction-Proposition 10.4(5), (S, Δ) is a K-semistable log Fano pair and

$$V = \alpha((K_S + \Delta)^2).$$

By [ADL24, Theorem 2.9(1)], (S, Δ) is K-semistable in the sense of test configurations, hence log-Ding-semistable [Fuj19, Theorem 6.5] (with $\delta = 0$), see also [LL19, Remark 2.4]. By [LL19, Proposition 4.6], $\widehat{\mathrm{vol}}(v) \geq \frac{4}{9}((K_S + \Delta)^2)$ for any $v \in \mathrm{Val}_{S,p}$, where the normalized volume is taken with respect to (S, Δ). Thus

$$\frac{4V}{9\alpha} = \frac{4}{9}((K_S + \Delta)^2) \leq \widehat{\mathrm{vol}}(S \ni p, \Delta) = \frac{4}{m(p)|H_p|},$$

that is, $m(p)|H_p| \leq \frac{9\alpha}{V}$. □

10.2. **Canonical degree one.** In this subsection, we treat the case $\alpha = 1$ of Set-up 10.1. The anti-canonical cone $C_{\mathbb{F}_1}$, graded by $H^0(\mathbb{F}_1, \mathcal{O}_{\mathbb{F}_1}(-jK_{\mathbb{F}_1}))$ in degree j, satisfies the conditions of Set-up 10.1 with $\alpha = 1$, but the grading valuation of $C_{\mathbb{F}_1}$ does not compute the local volume (Construction-Proposition 4.2). In the following proposition, the assumption that v_R computes the local volume excludes $C_{\mathbb{F}_1}$.

Proposition 10.6. *Notation and conditions as in Set-up 10.1. Assume that $\alpha = 1$ and that v_R computes $V := \widehat{\mathrm{vol}}(X \ni o)$. If $V > \frac{9}{2}$, then $m(p) = 1$ for any closed point $p \in S$, S is a smooth del Pezzo surface, and $R \cong \bigoplus_{j \geq 0} H^0(S, \mathcal{O}_S(-jK_S))$ as graded $\mathbb{C}$-algebras. In particular, $V = (K_S^2)$ is an integer. If $V > 7$, then $S \cong \mathbb{P}^2$ or $S \cong \mathbb{P}^1 \times \mathbb{P}^1$, hence $V = 9$ or $V = 8$, and $X \ni o$ is the quotient of $\mathbb{A}^3$ by μ_3 acting by scalars or the quotient of the threefold ordinary double point by μ_2 acting by -1, respectively. In particular, if $X \ni o$ is not LCIQ, then $V \leq 7$.*

Proof. Let D, u, Δ, and ω be as in Construction-Proposition 10.4, and let $K_S := \mathrm{div}(\omega)$. Assume that $V > \frac{9}{2}$. Let $p \in S$ be a closed point, let U, T, and $T \to U$ be as in Construction-Proposition 10.4(2) for p, and let $\tilde{p} \in T$ be the point over p. By Lemma 10.5, the germ $T \ni \tilde{p}$ is analytically isomorphic to $\mathbb{C}^2/H_p$ at the origin for a finite subgroup $H_p \subset SL_2(\mathbb{C})$, and

$$m(p)|H_p| \leq \frac{9}{V} < 2,$$

so $m(p) = 1$ and $|H_p| = 1$. Thus the cyclic cover $T \to U$ of Construction-Proposition 10.4(2) has degree one, so it is an isomorphism, and T is smooth at $\tilde{p}$. Hence S is smooth at p. Since p is arbitrary, S is smooth, and D is Cartier by Construction-Proposition 10.4(6). In particular, D has integral coefficients, so $m_P = 1$ for any prime divisor P on S (Construction-Proposition 10.4(6)), and $\Delta = 0$. By Construction-Proposition 10.4(4), $K_S + D = 0$, that is, $D = -K_S$ as divisors. Thus

$$R_j = H^0(S, \mathcal{O}_S(-jK_S))u^j$$

for any $j \geq 0$, and $-K_S$ is ample since kD is ample for some positive integer k. Hence S is a smooth del Pezzo surface, and the map $\varphi u^j \to \varphi$ induces an isomorphism of graded $\mathbb{C}$-algebras $R \cong \bigoplus_{j \geq 0} H^0(S, \mathcal{O}_S(-jK_S))$. By Construction-Proposition 10.4(5), $V = (K_S^2)$, which is an integer.

Assume that $V > 7$. Then $(K_S^2) \geq 8$. Since S is smooth and projective and $-K_S$ is ample, $H^1(S, \mathcal{O}_S) = H^1(S, \mathcal{O}_S(-K_S) \otimes \omega_S) = 0$ by the Kodaira vanishing theorem [Har77,

Remark III.7.15(a)]. Since S is smooth, [HW81, Theorem 3.4(i)(iii)(iv)] implies that $S \cong \mathbb{P}^2$, $S \cong \mathbb{P}^1 \times \mathbb{P}^1$, or $S \cong \mathbb{F}_1$. Suppose that $S \cong \mathbb{F}_1$. Then $X \ni o$ is $C_{\mathbb{F}_1} \ni o$ (Definition 2.10), and Construction-Proposition 10.4(5) implies that

$$\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = \widehat{\mathrm{vol}}(v_R) = (K^2_{\mathbb{F}_1}) = 8.$$

However, $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q$ by Construction-Proposition 4.2, and $q < 8$ by Lemma 4.3. This is a contradiction. Thus $S \cong \mathbb{P}^2$ or $S \cong \mathbb{P}^1 \times \mathbb{P}^1$.

If $S \cong \mathbb{P}^2$, then $V = (K^2_{\mathbb{P}^2}) = 9$, and

$$R \cong \bigoplus_{j \geq 0} H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3j)) = \bigoplus_{j \geq 0} \mathbb{C}[z_0, z_1, z_2]_{3j} = \mathbb{C}[z_0, z_1, z_2]^{\mu_3},$$

where $\mathbb{C}[z_0, z_1, z_2]_{3j}$ is the space of homogeneous polynomials of degree $3j$ and μ_3 acts on $\mathbb{A}^3$ by scalars. Thus X is the quotient of $\mathbb{A}^3$ by μ_3 acting by scalars.

If $S \cong \mathbb{P}^1 \times \mathbb{P}^1$, then $V = (K^2_{\mathbb{P}^1 \times \mathbb{P}^1}) = 8$. Let

$$B := \mathbb{C}[y_{00}, y_{01}, y_{10}, y_{11}]/(y_{00}y_{11} - y_{01}y_{10}),$$

graded by $\deg y_{ij} = 1$. Then $\operatorname{Spec} B$ is the threefold ordinary double point, and the Segre embedding implies that $B_j \cong H^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(j, j))$ for any $j \geq 0$, compatibly with the multiplications. Since $-K_{\mathbb{P}^1 \times \mathbb{P}^1} = \mathcal{O}(2, 2)$, we have

$$R \cong \bigoplus_{j \geq 0} H^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(2j, 2j)) \cong \bigoplus_{j \geq 0} B_{2j} = B^{\mu_2},$$

where μ_2 acts on $\operatorname{Spec} B$ by -1. Thus X is the quotient of the threefold ordinary double point by μ_2 acting by -1.

In both cases, the group acts freely outside the origin, the origin is mapped to o, and $\mathbb{A}^3$ and the threefold ordinary double point are normal local complete intersections. Thus $X \ni o$ is LCIQ (Definition 2.13). Therefore, if $X \ni o$ is not LCIQ, then $V \leq 7$. □

10.3. **Crepant divisors and stabilizers.** In this subsection, we show that a crepant divisor over the vertex of a graded cone forces a large stabilizer, and we describe the crepant divisors over the vertex locally when the local volume is larger than $\frac{54}{7}$.

Proposition 10.7. *Notation and conditions as in Set-up 10.1, and let $f : Y \to X$, $E \cong S$, and $m(p)$ be as in Construction-Proposition 10.4. Then the following hold.*

1. *If $m(p) \leq \alpha$ for any closed point $p \in S$, then $\operatorname{crex}(X \ni o) = 1$ when $\alpha = 1$, and $\operatorname{crex}(X \ni o) = 0$ when $\alpha \geq 2$. If moreover $\alpha \geq 2$, then $X \ni o$ is a cDV hypersurface germ, and in particular $X \ni o$ is LCIQ.*
2. *Assume that v_R computes $V := \widehat{\mathrm{vol}}(X \ni o)$. If $\alpha \geq 2$ and $X \ni o$ is not cDV, then $V \leq \frac{9\alpha}{\alpha+1}$. Moreover, if $X \ni o$ is not LCIQ and $V > q$, then $\alpha \geq 7$.*
3. *For any crepant divisor F over $X \ni o$ with $F \neq E$ and any closed point $p \in S$ contained in the center of F on Y, we have $m(p) > \alpha$.*

Proof. We identify E with S by Construction-Proposition 10.4(1). We first prove (3). Let F be a crepant divisor over $X \ni o$ with $F \neq E$, and let $p \in S$ be a closed point contained in $\operatorname{center}_Y F$. Since f is projective, $\operatorname{center}_Y F$ is non-empty, and since $f(\operatorname{center}_Y F) = \{o\}$, we have $\operatorname{center}_Y F \subset f^{-1}(o)_{\mathrm{red}} = E$ by Construction-Proposition 10.4(1). Since Y is normal, $\mathcal{O}_{Y,E}$ is a discrete valuation ring, so $\operatorname{center}_Y F \neq E$ as $F \neq E$. Thus F is exceptional over Y, and since (Y, E) is plt by Construction-Proposition 10.4(3), we have $a(F, Y, E) > 0$.

Let U be the open neighborhood of p in Construction-Proposition 10.4(2), so that $m(p)E$ is Cartier on Y_U. Since Y_U is open in Y and contains the point p of $\operatorname{center}_Y F$, it contains the generic point of $\operatorname{center}_Y F$. Thus there exists a regular function g on a neighborhood of the

generic point of $\mathrm{center}_Y F$ such that $\mathrm{div}(g) = m(p)E$ there. Since $\mathrm{center}_Y F \subset E$, the function g vanishes along $\mathrm{center}_Y F$, so $\mathrm{ord}_F(g)$ is a positive integer. Hence

$$\mathrm{mult}_F E = \frac{\mathrm{ord}_F(g)}{m(p)} \geq \frac{1}{m(p)}.$$

By Construction-Proposition 10.4(3), we have

$$1 = a(F, X) = a(F, Y, E) + \alpha \,\mathrm{mult}_F E > \frac{\alpha}{m(p)},$$

so $m(p) > \alpha$. This proves (3).

Next we prove (1). Assume that $m(p) \leq \alpha$ for any closed point $p \in S$. By (3), any crepant divisor over $X \ni o$ is equal to E. By Construction-Proposition 10.4(1)(3), E is a prime divisor over X with $\mathrm{center}_X E = \{o\}$ and $a(E, X) = \alpha$. Thus $\mathrm{crex}(X \ni o) = 1$ if $\alpha = 1$, and $\mathrm{crex}(X \ni o) = 0$ if $\alpha \geq 2$. Assume that $\alpha \geq 2$. Since K_X is Cartier, $a(P, X)$ is an integer for any prime divisor P over X, and since X is klt, $a(P, X) \geq 1$. Thus $X \ni o$ is a Gorenstein canonical threefold germ with $\mathrm{crex}(X \ni o) = 0$, and $X \ni o$ is a cDV hypersurface germ by Theorem 2.15(1). In particular, $X \ni o$ is a local complete intersection klt germ, hence LCIQ (Definition 2.13). This proves (1).

Finally we prove (2). Assume that v_R computes V, $\alpha \geq 2$, and $X \ni o$ is not cDV. By (1), there exists a closed point $p \in S$ such that $m(p) > \alpha$, so $m(p) \geq \alpha + 1$. By Lemma 10.5,

$$\alpha + 1 \leq m(p) \leq m(p)|H_p| \leq \frac{9\alpha}{V},$$

hence $V \leq \frac{9\alpha}{\alpha+1}$.

Now assume that $X \ni o$ is not LCIQ and $V > q$. By Lemma 4.3, $q > \frac{54}{7} > 7$. If $\alpha = 1$, then $V \leq 7 < q$ by Proposition 10.6, which is not possible. Suppose that $2 \leq \alpha \leq 6$. Then

$$\frac{9\alpha}{\alpha+1} = 9 - \frac{9}{\alpha+1} \leq 9 - \frac{9}{7} = \frac{54}{7} < q < V.$$

By Lemma 10.5, for any closed point $p \in S$, we have

$$m(p) \leq \frac{9\alpha}{V} < \alpha + 1,$$

so $m(p) \leq \alpha$. By (1), $X \ni o$ is LCIQ, which is not possible. Therefore $\alpha \geq 7$. □

Definition 10.8. Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, and let N be a positive integer. A *monotonic N-complement* of (X, B) is a $\mathbb{Q}$-divisor $B^+ \geq B$ on X such that (X, B^+) is lc and $N(K_X + B^+) \sim 0$. Assume moreover that (X, B) is klt and $-(K_X + B)$ is ample. We say that (X, B) is *exceptional* if (X, B') is klt for any $\mathbb{Q}$-divisor $B' \geq B$ such that (X, B') is lc and $K_X + B' \sim_{\mathbb{Q}} 0$.

For a monotonic N-complement Δ^+ of (S, Δ), the relation $K_S + \Delta + \alpha D = 0$ turns $N(K_S + \Delta^+) \sim 0$ into a homogeneous element of $R_{N\alpha}$ (Proposition 10.14).

Remark 10.9. A monotonic N-complement in the sense of Definition 10.8 is a monotonic N-complement in the sense of [BL23, Definition 2.10]. Indeed, $N(K_X + B^+) \sim 0$ implies that NB^+ has integral coefficients. Let P be a prime divisor on X, and let b and b^+ be the coefficients of P in B and B^+ respectively. If $b = 1$, then $Nb^+ \geq N$. If $b < 1$, then the integer Nb^+ satisfies $Nb^+ \geq Nb > (N+1)b - 1$, so $Nb^+ \geq \lfloor (N+1)b \rfloor$. Thus $NB^+ \geq \lfloor (N+1)\{B\} \rfloor + N\lfloor B \rfloor$, where $\{B\} := B - \lfloor B \rfloor$.

Lemma 10.10 describes the crepant divisors over $X \ni o$ by their restrictions to S, and Lemma 10.10(4) follows from the bound on the number of prime divisors of small log discrepancy over an exceptional log del Pezzo surface with standard coefficients in [Pro01, Corollary 10.1.2].

Lemma 10.10. *Notation and conditions as in Set-up 10.1, and let $f : Y \to X$, $E \cong S$, Δ, $m(p)$, and the surfaces T be as in Construction-Proposition 10.4. Assume that $\alpha \geq 2$ and that v_R*

computes $V := \widehat{\mathrm{vol}}(X \ni o) > \frac{54}{7}$. *For a crepant divisor* F *over* $X \ni o$, *let* $Z_F \subset S$ *be the center of* F *on* Y. *Then the following hold.*

(1) *The subvarieties* Z_F *are non-empty and proper, and* $Z_F \cap Z_{F'} = \emptyset$ *for distinct crepant divisors* F *and* F'.
(2) *For any crepant divisor* F *over* $X \ni o$, *the restriction of* ord_F *to* $K(S) \subset K(X)$ *is a positive multiple of* ord_W *for a prime divisor* W *over* S *with* $0 < a(W, S, \Delta) < \frac{1}{7}$, *and the prime divisors* W *for distinct crepant divisors over* $X \ni o$ *are distinct.*
(3) *Let* $p \in Z_F$ *be a closed point and* $m := m(p)$. *Then* $\alpha < m < \frac{7\alpha}{6}$, T *is smooth at the point over* p, *and, after an analytic localization at* p, Y *is isomorphic to* $(\mathbb{A}^2 \times \mathbb{A}^1)/\mu_m$, *where* μ_m *acts on the coordinates* (s_1, s_2, t) *with weights* $(-b, -c, -1)$ *for integers* $0 \le b, c < m$ *with* $b + c = m - \alpha$, *and the point of* E *over* p *is the image of the origin. Under the local isomorphism* $Y \cong (\mathbb{A}^2 \times \mathbb{A}^1)/\mu_m$, F *has exactly one local branch at the point of* E *over* p, *which is the toric divisor of the ray spanned by* $\frac{1}{m}(b, c, 1)$. *In particular, any point of* S *is contained in at most one* Z_F.
(4) *If* (S, Δ) *is exceptional, then* $\mathrm{crex}(X \ni o) \le 2$.

Proof. The proof is long, so we divide it into several steps. We identify E with S by Construction-Proposition 10.4(1).

Step 1. In this step, we show that Z_F is a non-empty proper subvariety of S, and that $\alpha < m(p) < \frac{7\alpha}{6}$ and T is smooth over p for any closed point $p \in Z_F$.

Let F be a crepant divisor over $X \ni o$. Since $a(E, X) = \alpha \ge 2$ by Construction-Proposition 10.4(3), we have $F \ne E$. Since f is projective and $f^{-1}(o)_{\mathrm{red}} = E$, the center Z_F is non-empty and contained in E. Since $\mathcal{O}_{Y,E}$ is a discrete valuation ring and $F \ne E$, we have $Z_F \ne E$.

Let $p \in Z_F$ be a closed point and $m := m(p)$. Let U, T, and $T \to U$ be as in Construction-Proposition 10.4(2) for p, and let $\tilde{p} \in T$ be the point over p. By Proposition 10.7(3), $m > \alpha$. By Lemma 10.5, the germ $T \ni \tilde{p}$ is analytically isomorphic to $\mathbb{C}^2/H_p$ at the origin for a finite subgroup $H_p \subset SL_2(\mathbb{C})$, and since $V > \frac{54}{7}$,

$$m|H_p| \le \frac{9\alpha}{V} < \frac{7 \cdot 9\alpha}{54} = \frac{7\alpha}{6}.$$

If $|H_p| \ge 2$, then $m < \frac{7\alpha}{12} < \alpha$, which is not possible. Thus $|H_p| = 1$, T is smooth at $\tilde{p}$, and $\alpha < m < \frac{7\alpha}{6}$.

Step 2. In this step, we describe the germ of Y at the point y_p of E over p as an affine toric variety, and we compute the log discrepancies of the toric divisors over X.

Since $\tilde{p}$ is the only point of T over p, the group μ_m fixes $\tilde{p}$. Since $T \to U$ is a cyclic cover of degree m, μ_m acts faithfully on T. Since T is smooth at $\tilde{p}$, after an analytic localization at $\tilde{p}$ there exist local coordinates of T at $\tilde{p}$ on which μ_m acts linearly [Car57, Section 4, pp. 97–98]. Since μ_m is cyclic, we may choose the linear coordinates s_1, s_2 to be eigenvectors of the action. Thus μ_m acts on the coordinates (s_1, s_2, t) of $\mathbb{A}^2 \times \mathbb{A}^1$ with weights $(-b, -c, -1)$ for integers $0 \le b, c < m$, where t is the coordinate of $\mathbb{A}^1$ in Construction-Proposition 10.4(2). By Construction-Proposition 10.4(2), the germ of Y at y_p is analytically isomorphic to the germ of $\mathbb{A}^3/\mu_m$ at the image 0 of the origin, and E is the image of $\{t = 0\}$.

Let $\lambda \in \mu_m$ be a non-trivial element. Then λ acts non-trivially on t, and since μ_m acts faithfully on T, λ acts non-trivially on (s_1, s_2). Thus the fixed locus of λ on $\mathbb{A}^3$ has codimension at least two, so μ_m acts freely in codimension one on $\mathbb{A}^3$. Hence $\mathbb{A}^3/\mu_m$ is the affine toric variety of the positive octant $\sigma \subset \mathbb{R}^3$ with respect to the lattice

$$N := \mathbb{Z}^3 + \mathbb{Z} \cdot \frac{1}{m}(b, c, 1),$$

by [CLS11, Proposition 1.3.18] applied to the sublattice $\mathbb{Z}^3 \subset N$, whose quotient $N/\mathbb{Z}^3$ is cyclic of order m and generated by the class of $\frac{1}{m}(b, c, 1)$, and the standard basis vectors e_1, e_2, e_3 of $\mathbb{R}^3$ are

primitive in N (for e_3, since $\gcd(b,c,m)=1$ by the faithfulness of the action on (s_1,s_2)). For a ray $\rho\subset\sigma$ with $\rho\cap N\neq\{0\}$, let $w_\rho=(w_{\rho,1},w_{\rho,2},w_{\rho,3})$ be the primitive generator of ρ in N and P_ρ the toric prime divisor over $\mathbb{A}^3/\mu_m$ corresponding to ρ. Then $\mathrm{ord}_{P_\rho}(s_1^is_2^jt^k)=iw_{\rho,1}+jw_{\rho,2}+kw_{\rho,3}$ for any invariant monomial $s_1^is_2^jt^k$, and ord_{P_ρ} of an invariant power series is the minimum of the values of the monomials appearing in it. Since e_1, e_2, and e_3 are primitive in N, we have

$$a(P_\rho,Y)=w_{\rho,1}+w_{\rho,2}+w_{\rho,3}$$

by [CLS11, Lemma 11.4.10] (see also [Rei87, Section 4]). The monomial t^m is invariant, and $\mathrm{div}(t^m)=mP_{e_3}$ on $\mathbb{A}^3/\mu_m$. Since E is the image of $\{t=0\}$, we have $P_{e_3}=E$ on the germ, so $\mathrm{div}(t^m)=mE$ and $\mathrm{mult}_{P_\rho}E=w_{\rho,3}$. By Construction-Proposition 10.4(3),

$$a(P_\rho,X)=a(P_\rho,Y)-\mathrm{mult}_{P_\rho}E+\alpha\,\mathrm{mult}_{P_\rho}E=w_{\rho,1}+w_{\rho,2}+\alpha w_{\rho,3}. \tag{10.1}$$

Step 3. In this step, we show that the local branch of F at y_p is the toric divisor of the ray spanned by $\frac{1}{m}(b,c,1)$ and that $b+c=m-\alpha$, which proves (3) and the rest of (1).

Since K_X is Cartier and X is klt, $a(P,X)$ is a positive integer for any prime divisor P over the germ of X at o. Let $h:Y'\to\mathbb{A}^3/\mu_m$ be the toric resolution given by a regular subdivision of σ with respect to N, and let ρ run over the rays of the fan of Y'. By (10.1) and [KM98, Lemma 2.30],

$$K_{Y'}=(f\circ h)^*K_X+\sum_\rho(w_{\rho,1}+w_{\rho,2}+\alpha w_{\rho,3}-1)P_\rho,$$

and $w_{\rho,1}+w_{\rho,2}+\alpha w_{\rho,3}-1=a(P_\rho,X)-1\geq 0$ for any ρ. Let G be a prime divisor exceptional over Y'. Since Y' is smooth, $a(G,Y')\geq 2$ by [KM98, Corollary 2.31(2)], so

$$a(G,X)\geq a(G,Y')\geq 2.$$

Thus any prime divisor P over the germ of $\mathbb{A}^3/\mu_m$ at 0 with $a(P,X)=1$ is a prime divisor on Y'. Assume moreover that the center of P on $\mathbb{A}^3/\mu_m$ is contained in E and is not equal to E. Then P is exceptional over $\mathbb{A}^3/\mu_m$. Since h is an isomorphism over the open torus orbit, $P=P_\rho$ for a ray ρ of the fan of Y'. Write $w_\rho=(w_1,w_2,w_3)$. Since the center of P_ρ is contained in E, we have $mw_3=\mathrm{ord}_{P_\rho}(t^m)>0$. By (10.1),

$$w_1+w_2+\alpha w_3=1,\quad w_1,w_2\geq 0,\quad w_3>0.$$

The third coordinates of the vectors of N are in $\frac{1}{m}\mathbb{Z}$, so $w_3=\frac{j}{m}$ for a positive integer j. Since $j\leq\frac{m}{\alpha}<\frac{7}{6}$ by **Step 1**, we have $j=1$. Any vector of N with third coordinate $\frac{1}{m}$ is congruent to $\frac{1}{m}(b,c,1)$ modulo $\mathbb{Z}^2\times\{0\}$, so $w_1\equiv\frac{b}{m}$ and $w_2\equiv\frac{c}{m}$ modulo $\mathbb{Z}$. Since $w_1,w_2\geq 0$ and $w_1+w_2=1-\frac{\alpha}{m}<1$, we have $0\leq w_1,w_2<1$. Since $0\leq\frac{b}{m},\frac{c}{m}<1$, we have

$$w_\rho=\frac{1}{m}(b,c,1)\quad\text{and}\quad b+c=m-\alpha.$$

Conversely, $\frac{1}{m}(b,c,1)$ is primitive in N, since the third coordinates of the vectors of N are in $\frac{1}{m}\mathbb{Z}$. Hence there exists at most one prime divisor P over the germ of $\mathbb{A}^3/\mu_m$ at 0 such that $a(P,X)=1$ and the center of P is a proper subset of E, and if such P exists, then P is the toric divisor of the ray spanned by $\frac{1}{m}(b,c,1)$ and $b+c=m-\alpha$.

We pass from the algebraic divisor F to the germ. Let $g:Y''\to Y$ be a log resolution such that F is a prime divisor on Y''. The relative canonical divisor $K_{Y''}-(f\circ g)^*K_X$ and the exceptional divisors of g are preserved under the analytic localization at y_p, so any local branch of F at y_p is a prime divisor P over the germ of $\mathbb{A}^3/\mu_m$ at 0 with $a(P,X)=1$, and the center of P is contained in Z_F, which is a proper subset of E by **Step 1**. Since $p\in Z_F$, F has a local branch at y_p. Thus F has exactly one local branch at y_p, the local branch is the toric divisor of the ray spanned by $\frac{1}{m}(b,c,1)$, and $b+c=m-\alpha$. This proves (3) except for the last sentence.

Let F and F' be crepant divisors over $X\ni o$ and assume that there exists a closed point $p\in Z_F\cap Z_{F'}$. Then by the previous paragraph, the local branch of F at y_p and the local branch of F' at y_p are both the toric divisor of the ray spanned by $\frac{1}{m}(b,c,1)$. Since ord_F and $\mathrm{ord}_{F'}$ are

the restrictions to $K(X)$ of the valuation of the toric divisor of the ray spanned by $\frac{1}{m}(b,c,1)$, we have $\mathrm{ord}_F = \mathrm{ord}_{F'}$, so $F = F'$. This proves the last sentence of (3), and together with **Step 1**, it proves (1).

Step 4. In this step, we prove (2).

Let $\pi : Y \to S$ be the structure morphism of $Y = \mathrm{Spec}_S \bigoplus_{j\geq 0} \mathcal{O}_S(\lfloor jD \rfloor)$. Then $\pi|_E : E \to S$ is the isomorphism of Construction-Proposition 10.4(1). We regard $K(S)$ as the subfield of $K(X) = \mathrm{Frac}\, R$ of homogeneous fractions of degree zero, which is $\pi^* K(S)$. Since $\pi^* \mathcal{O}_{S,\pi(\zeta)} \subset \mathcal{O}_{Y,\zeta}$ for the generic point ζ of Z_F, the center of the restriction of ord_F to $K(S)$ is $\pi(Z_F) = Z_F$. In particular, the restriction is a non-zero discrete valuation of $K(S)$. The residue field of the restriction is contained in the residue field of ord_F, which has transcendence degree 2 over $\mathbb{C}$, and $\mathrm{trdeg}_{K(S)} K(X) = 1$, so the residue field of the restriction has transcendence degree at least 1 over $\mathbb{C}$ by the Abhyankar inequality, and at most $1 = \dim S - 1$ since the restriction is a non-trivial valuation of $K(S)$, hence equal to 1. Hence the restriction of ord_F to $K(S)$ is $d\,\mathrm{ord}_W$ for a prime divisor W over S and a positive integer d [KM98, Lemma 2.45].

We compute W and d at a closed point $p \in Z_F$. Notation as in **Step 2**. On the germ at y_p, the morphism π is the projection $(s_1, s_2, t) \to (s_1, s_2)$ to $\mathbb{A}^2/\mu_m$, which is the germ of U at p. The quotient $\mathbb{A}^2/\mu_m$ is the affine toric variety of the positive quadrant with respect to the lattice

$$N_S := \mathbb{Z}^2 + \mathbb{Z} \cdot \frac{1}{m}(b,c),$$

whose dual lattice is the lattice of exponents of the invariant monomials in s_1, s_2. By **Step 3**, for any invariant power series $\phi(s_1, s_2)$, $\mathrm{ord}_F(\phi)$ is the minimum of $\frac{ib+jc}{m}$ over the monomials $s_1^i s_2^j$ appearing in ϕ. Thus the restriction of ord_F to $K(S)$ is the toric valuation of the vector $w_S := \frac{1}{m}(b,c) \in N_S$. Since $b + c = m - \alpha > 0$, we have $w_S \neq 0$, so $w_S = d' w_W$ for a primitive vector $w_W \in N_S$ and a positive integer d'. Thus $d = d'$, and ord_W is the toric valuation of w_W. If $bc = 0$, then w_W spans a coordinate ray and W is a prime divisor on S through p.

By Construction-Proposition 10.4(2), K_T is the pullback of $K_U + \Delta|_U$, so the quotient morphism $\mathbb{A}^2 \to \mathbb{A}^2/\mu_m$ is log crepant for $(\mathbb{A}^2, 0)$ and the germ of $(U, \Delta|_U)$ at p. The monomial valuation of $\mathbb{A}^2$ with weights w_W on (s_1, s_2) restricts to ord_W on the invariant functions, and the log discrepancy of the monomial valuation with respect to $\mathbb{A}^2$ is the sum of the coordinates of w_W [CLS11, Lemma 11.4.10]. Since the ramification is tame in characteristic zero, log discrepancies of valuations are preserved under a finite log crepant morphism [Kol13, 2.41 and (2.42.4)]. Since $m < \frac{7\alpha}{6}$ by **Step 1**, we have

$$a(W, S, \Delta) = \frac{b+c}{md} = \frac{m-\alpha}{md} \leq 1 - \frac{\alpha}{m} < 1 - \frac{6}{7} = \frac{1}{7}.$$

Since $m > \alpha$, we have $a(W, S, \Delta) > 0$. If $bc = 0$, then $a(W, S, \Delta) < 1$, so W is a component of Δ.

The center of W on S is Z_F. By (1), $Z_F \cap Z_{F'} = \emptyset$ for distinct crepant divisors F and F', and $Z_F \neq \emptyset$, so the prime divisors W for distinct crepant divisors are distinct. This proves (2).

Step 5. In this step, we prove (4).

Assume that (S, Δ) is exceptional. For any crepant divisor F over $X \ni o$, we denote by W_F the prime divisor W over S in (2), so that $0 < a(W_F, S, \Delta) < \frac{1}{7}$ and $F \to W_F$ is injective. If $\mathrm{crex}(X \ni o) = 0$, then there is nothing to prove, so we may assume that there exists a crepant divisor F over $X \ni o$.

We check the hypotheses of [Pro01, Corollary 10.1.2] for (S, Δ). By Construction-Proposition 10.4(5), (S, Δ) is klt and $-(K_S + \Delta)$ is ample, so (S, Δ) is a log del Pezzo surface in the sense of [Pro01, Definition 5.1.1]. The coefficients of $\Delta = \sum_P (1 - \frac{1}{m_P})P$ are standard. A $\mathbb{Q}$-complement of $K_S + \Delta$ in the sense of [Pro01, Definition 4.1.2] is $K_S + B'$ for a $\mathbb{Q}$-divisor $B' \geq \Delta$ such that (S, B') is lc and $K_S + B' \sim_{\mathbb{Q}} 0$, and such a B' exists by Bertini's theorem, as $-(K_S + \Delta)$ is ample. Thus (S, Δ) is exceptional in the sense of [Pro01, Definition 4.5.1].

Condition (v) of [Pro01, Section 9.1] requires a boundary $B^\nabla \le \Delta$ such that (S, B^∇) is klt and $-(K_S + B^\nabla)$ is nef and big, and $B^\nabla = \Delta$ satisfies it by Construction-Proposition 10.4(5).

It remains to show that $K_S + \Delta$ has no regular complements, that is, no n-complements with $n \in \{1, 2, 3, 4, 6\}$. Suppose that Δ^+ is an n-complement of $K_S + \Delta$ in the sense of [Pro01, Definition 4.1.3] for some $n \in \{1, 2, 3, 4, 6\}$. Since the coefficients of Δ are standard, $\Delta^+ \ge \Delta$ by [Pro01, Remark 4.2.8]. Since $n(K_S + \Delta^+) \sim 0$, we have $na(W_F, S, \Delta^+) \in \mathbb{Z}$, and since (S, Δ^+) is lc, $a(W_F, S, \Delta^+) \ge 0$. Since $K_S + \Delta$ and $K_S + \Delta^+$ are $\mathbb{Q}$-Cartier, $\Delta^+ - \Delta \ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, so

$$a(W_F, S, \Delta^+) \le a(W_F, S, \Delta) < \frac{1}{7} < \frac{1}{n}.$$

Thus $a(W_F, S, \Delta^+) = 0$, so (S, Δ^+) is not klt. Since $\Delta^+ \ge \Delta$, (S, Δ^+) is lc, and $K_S + \Delta^+ \sim_{\mathbb{Q}} 0$, we get a contradiction to the assumption that (S, Δ) is exceptional. Thus $K_S + \Delta$ has no regular complements.

Let $\delta(S, \Delta)$ be the number of prime divisors P over S, with the prime divisors on S counted, such that $a(P, S, \Delta) \le \frac{1}{7}$ [Pro01, Section 9.3]. By [Pro01, Corollary 10.1.2], $\delta(S, \Delta) \le 2$. Since $F \to W_F$ is injective and $a(W_F, S, \Delta) < \frac{1}{7}$ for any crepant divisor F over $X \ni o$, we have

$$\operatorname{crex}(X \ni o) \le \delta(S, \Delta) \le 2.$$

This proves (4). □

10.4. **Centers of crepant divisors.** In this subsection, we study the centers on the zero section E of the crepant divisors over the cones of Set-up 10.1 whose grading valuation computes a local volume larger than q. The main result is Proposition 10.13. Parts (1) and (2) of Proposition 10.13 are the inputs of Proposition 10.17 and Theorem 10.3.

Definition 10.11. Notation and conditions as in Set-up 10.1, and let $f : Y \to X$ and $E \cong S$ be as in Construction-Proposition 10.4. For a crepant divisor F over $X \ni o$, we regard $\operatorname{center}_Y F \subset E$ as a closed subvariety of S, and we call it a *crepant point center* if it is a closed point, and a *crepant curve center* if it is a curve.

The following lemma describes Y near a closed point $p \in S$ with $m(p) > \alpha$, together with the crepant divisors whose centers on Y contain p.

Lemma 10.12. *Notation and conditions as in Set-up 10.1, let $f : Y \to X$, $E \cong S$, and $m(p)$ be as in Construction-Proposition 10.4, let $p \in S$ be a closed point, let $m := m(p)$, let U, T, and $T \to U$ be as in Construction-Proposition 10.4(2) for p, and let $\tilde{p} \in T$ be the point over p. Assume that $\alpha \ge 2$, that v_R computes $V := \widehat{\operatorname{vol}}(X \ni o) > q$, and that $m > \alpha$. Then the following hold.*

(1) T is smooth at $\tilde{p}$, and there exist local coordinates s_1, s_2 of T at $\tilde{p}$ and integers $0 \le b, c < m$ with $\gcd(m, b, c) = 1$ and $b + c + \alpha \equiv 0 \pmod m$, such that μ_m acts on (s_1, s_2, t) with weights $(-b, -c, -1)$, where t is the coordinate of $\mathbb{A}^1$. Let Y' be the affine toric variety of the non-negative octant in the lattice $\mathbb{Z}^3 + \mathbb{Z}\frac{1}{m}(b, c, 1)$, let $0 \in Y'$ be the torus-fixed point, and let E' be the toric divisor of the ray spanned by $(0, 0, 1)$. Then there exist an open neighborhood $Y^\circ \subset Y_U$ of p and an étale morphism $g : Y^\circ \to Y'$ such that $g^{-1}(0) = \{p\}$, $g^*E' = E|_{Y^\circ}$, and $g \circ \pi = \pi' \circ (s_1, s_2, t)$ on $\pi^{-1}(Y^\circ)$, where $\pi : T \times \mathbb{A}^1 \to Y_U$ is the composite $T \times \mathbb{A}^1 \to (T \times \mathbb{A}^1)/\mu_m \cong Y_U$ of Construction-Proposition 10.4(2)(i) and $\pi' : \mathbb{A}^3 \to Y'$ is the quotient morphism by μ_m acting on (s_1, s_2, t) with weights $(-b, -c, -1)$. In particular, the germ of Y at $p \in E \cong S$ is analytically isomorphic to $Y' \ni 0$. For a primitive vector $w = (w_1, w_2, w_3)$ of the lattice in the octant, let F_w be the toric prime divisor over Y' of the ray spanned by w, let Y'_w be the toric variety of the star subdivision of the octant at w, and let $F'_w \subset Y'_w$ be the toric divisor of w, so that F'_w is F_w realized on Y'_w. We call an irreducible component of $F'_w \times_{Y'} Y^\circ \subset Y'_w \times_{Y'} Y^\circ$ whose center on Y contains p a toric divisor over Y of the ray spanned by w. *There exists exactly one toric divisor G over Y of the ray spanned by w, and $a(G, X) = w_1 + w_2 + \alpha w_3$. Moreover, the value of ord_G on*

any non-zero element of $\mathcal{O}_{Y,p}$ is the minimum of $j_1w_1+j_2w_2+j_3w_3$ over the monomials $s_1^{j_1}s_2^{j_2}t^{j_3}$ with non-zero coefficient in its pullback to $\widehat{\mathcal{O}}_{T\times\mathbb{A}^1,(\tilde{p},0)}=\mathbb{C}[[s_1,s_2,t]]$ under π.

(2) *There exists a crepant divisor over $X\ni o$ whose center on Y contains p if and only if $b+c=m-\alpha$. In this case, there exists exactly one such crepant divisor F, namely the toric divisor over Y of the ray spanned by $\frac{1}{m}(b,c,1)$, and the center of F on $E\cong S$ is $\{p\}$ if $b,c>0$, the irreducible curve in S through p which coincides near p with the image of the curve $\{s_2=0\}$ of T if $b=0$, and the irreducible curve in S through p which coincides near p with the image of the curve $\{s_1=0\}$ of T if $c=0$.*

Proof. Let π be as in (1). Since $\tilde{p}$ is the only point of T over p, $\tilde{p}$ is fixed by μ_m. Through $E\cong S$ we regard p as a point of Y, so that $p=\pi(\tilde{p},0)$ and $\pi^{-1}(p)=\{(\tilde{p},0)\}$ by Construction-Proposition 10.4(2).

We prove (1). By Lemma 10.5, $T\ni\tilde{p}$ is analytically isomorphic to $\mathbb{C}^2/H_p$ at the origin for a finite subgroup $H_p\subset SL_2(\mathbb{C})$ with $m|H_p|\le\frac{9\alpha}{V}<\frac{9\alpha}{q}$. By Lemma 4.3, $q>7$. Since $m>\alpha$, we obtain $|H_p|<\frac{9}{q}<2$. Thus H_p is trivial, T is smooth at $\tilde{p}$, and

$$m<\frac{9\alpha}{q}<\frac{3\alpha}{2}. \tag{10.2}$$

Since μ_m is a finite abelian group fixing $\tilde{p}$, we may lift a basis $\bar{s}_1,\bar{s}_2$ of the cotangent space $\mathfrak{m}_{\tilde{p}}/\mathfrak{m}_{\tilde{p}}^2$ of T at $\tilde{p}$ consisting of eigenvectors of μ_m to elements $s_1,s_2\in\mathcal{O}_{T,\tilde{p}}$ such that μ_m acts on s_i by the character by which it acts on $\bar{s}_i$ for $i=1,2$. Then s_1,s_2 are local coordinates of T at $\tilde{p}$, and there exist integers $0\le b,c<m$ such that μ_m acts on (s_1,s_2,t) with weights $(-b,-c,-1)$. If $d:=\gcd(m,b,c)>1$, then $\mu_d\subset\mu_m$ acts trivially on $\widehat{\mathcal{O}}_{T,\tilde{p}}=\mathbb{C}[[s_1,s_2]]$, hence on $\mathcal{O}_{T,\tilde{p}}$ and on the fraction field of $\mathcal{O}_{T,\tilde{p}}$, which contradicts the fact that $T\to U$ is a cyclic cover of degree m with one point over p. Thus $\gcd(m,b,c)=1$.

Let Y', 0, and E' be as in (1). Then Y' is the quotient $\mathbb{A}^3/\mu_m$, where μ_m acts on the coordinates (s_1,s_2,t) of $\mathbb{A}^3$ with weights $(-b,-c,-1)$, 0 is the image of the origin, and E' is the image of $\{t=0\}$. Let D_1' and D_2' be the images of $\{s_1=0\}$ and $\{s_2=0\}$. Then D_1', D_2', and E' are the toric divisors of the rays spanned by $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$, where $(0,0,1)$ is primitive since $\gcd(m,b,c)=1$. For a primitive vector $w=(w_1,w_2,w_3)$ of $\mathbb{Z}^3+\mathbb{Z}\frac{1}{m}(b,c,1)$ in the octant, let F_w be as in (1). Then

$$\operatorname{ord}_{F_w}(s_1^{u_1}s_2^{u_2}t^{u_3})=u_1w_1+u_2w_2+u_3w_3$$

for any μ_m-invariant monomial $s_1^{u_1}s_2^{u_2}t^{u_3}$.

Let T° be a μ_m-stable affine open neighborhood of $(\tilde{p},0)$ in $T\times\mathbb{A}^1$ on which s_1, s_2, and t are regular. Then (s_1,s_2,t) defines a μ_m-equivariant morphism $T^\circ\to\mathbb{A}^3$ which is étale at $(\tilde{p},0)$, and hence induces a morphism $g:Y^\circ:=\pi(T^\circ)\to Y'$ with $g(p)=0$ and $g\circ\pi=\pi'\circ(s_1,s_2,t)$ on $\pi^{-1}(Y^\circ)=T^\circ$, where $\pi':\mathbb{A}^3\to Y'$ is the quotient morphism. Since $\pi^{-1}(p)=\{(\tilde{p},0)\}$, g induces an isomorphism

$$\widehat{\mathcal{O}}_{Y',0}=\mathbb{C}[[s_1,s_2,t]]^{\mu_m}\to\widehat{\mathcal{O}}_{Y,p}.$$

Thus the germ of Y at p is analytically isomorphic to $Y'\ni 0$ in the sense of Definition 2.4, and g is étale at p. Since g is quasi-finite near p, possibly shrinking Y°, we may assume that g is étale and $g^{-1}(0)=\{p\}$. Since μ_m acts faithfully on T, acts on t with weight -1, and $\gcd(m,b,c)=1$, both π and the quotient morphism $\mathbb{A}^3\to Y'$ are étale in codimension one. Hence

$$\operatorname{div}(s_1^m)=mD_1',\qquad \operatorname{div}(s_2^m)=mD_2',\qquad \operatorname{div}(t^m)=mE'$$

on Y', $\operatorname{div}(t^m)=mE$ on Y_U, and $g^*E'=E|_{Y^\circ}$.

The μ_m-invariant form $\frac{ds_1}{s_1}\wedge\frac{ds_2}{s_2}\wedge\frac{dt}{t}$ descends to a torus-invariant 3-form on Y', which has a pole of order one along any toric prime divisor over Y' [CLS11, Theorem 8.2.3 and the discussion after Corollary 8.2.8]. Hence $a(F_w,Y',D_1'+D_2'+E')=0$. Since $\operatorname{ord}_{F_w}(s_1^m)=mw_1$, $\operatorname{ord}_{F_w}(s_2^m)=mw_2$, and $\operatorname{ord}_{F_w}(t^m)=mw_3$, we have $\operatorname{mult}_{F_w}D_1'=w_1$, $\operatorname{mult}_{F_w}D_2'=w_2$, and $\operatorname{mult}_{F_w}E'=w_3$. Thus

$$a(F_w,Y',(1-\alpha)E')=w_1+w_2+\alpha w_3. \tag{10.3}$$

Let Y'_w and F'_w be as in (1). Since the support of the star subdivision is the octant, $Y'_w \to Y'$ is a proper birational toric morphism. Hence $Y'_w \times_{Y'} Y^\circ \to Y^\circ$ is proper and birational: $Y'_w \times_{Y'} Y^\circ$ is normal, being étale over Y'_w, any connected component of $Y'_w \times_{Y'} Y^\circ$ has open image in Y'_w and so intersects the preimage of the torus of Y', and over the torus $Y'_w \times_{Y'} Y^\circ$ is isomorphic to an open subset of Y°, so $Y'_w \times_{Y'} Y^\circ$ is irreducible. Thus the irreducible components of $F'_w \times_{Y'} Y^\circ$ are prime divisors over Y. The center of F'_w on Y' is the closure of a torus orbit, so it contains 0. Since $Y'_w \times_{Y'} Y^\circ \to Y^\circ$ is proper, the image of $F'_w \times_{Y'} Y^\circ$ in Y° is the closed subset $g^{-1}(\mathrm{center}_{Y'} F'_w)$, which contains p. Hence some irreducible component of $F'_w \times_{Y'} Y^\circ \subset Y'_w \times_{Y'} Y^\circ$ has center on Y containing p, that is, a toric divisor over Y of the ray spanned by w exists. Since $F'_w \times_{Y'} Y^\circ$ is normal, being étale over the normal toric surface F'_w [CLS11, Proposition 3.2.7], its irreducible components are pairwise disjoint. They are proper over Y°, and the fiber of $F'_w \times_{Y'} Y^\circ$ over p is isomorphic to the fiber of F'_w over 0, which is F'_w, the closure of a one-dimensional torus orbit, or a point [CLS11, Theorem 3.2.6 and Lemma 3.3.21], hence connected. Thus exactly one of them has center containing p, that is, the toric divisor over Y of the ray spanned by w is unique. Let G be the toric divisor over Y of the ray spanned by w. Since g is étale and induced by (s_1, s_2, t), and G is the only irreducible component of $F'_w \times_{Y'} Y^\circ$ whose center on Y contains p, flat base change [Har77, Proposition III.9.3] shows that near p the ideals $\{\mathrm{ord}_G \geq k\}$, $k \geq 0$, are generated by the pullbacks of the ideals $\{\mathrm{ord}_{F_w} \geq k\}$ of $\mathcal{O}_{Y'}$, which are spanned by monomials. Since $\widehat{\mathcal{O}}_{Y,p}$ is faithfully flat over $\mathcal{O}_{Y,p}$ [Mat86, Theorem 8.14], the description of ord_G in (1) follows. Since g is étale and $g^*E' = E|_{Y^\circ}$, we have $a(G, Y, (1-\alpha)E) = a(F_w, Y', (1-\alpha)E')$. By Construction-Proposition 10.4(3) and (10.3),

$$a(G, X) = a(G, Y, E) + \alpha \,\mathrm{mult}_G E = a(G, Y, (1-\alpha)E) = w_1 + w_2 + \alpha w_3.$$

Since X is Gorenstein, $a(G', X)$ is an integer for any prime divisor G' over X. The vector $\frac{1}{m}(b, c, 1)$ is contained in the lattice and in the octant, so $\frac{1}{m}(b, c, 1) = \lambda w$ for a primitive vector w and a positive integer λ. Hence

$$\frac{b + c + \alpha}{m} = \lambda(w_1 + w_2 + \alpha w_3) \in \mathbb{Z},$$

that is, $b + c + \alpha \equiv 0 \pmod{m}$. This proves (1).

We prove (2). By (1), the linear function $w_1 + w_2 + \alpha w_3$ takes integer values on $\mathbb{Z}^3 + \mathbb{Z}\frac{1}{m}(b, c, 1)$, and it is positive on the non-zero vectors of the octant. Hence

$$w_1 + w_2 + \alpha w_3 \geq 1 \tag{10.4}$$

for any non-zero vector w of $\mathbb{Z}^3 + \mathbb{Z}\frac{1}{m}(b, c, 1)$ in the octant. Let w be such a vector with $w_3 > 0$ and $w_1 + w_2 + \alpha w_3 = 1$. Then $w_3 \in \frac{1}{m}\mathbb{N}^+$ and $\alpha w_3 \leq 1$. Since $m < 2\alpha$ by (10.2), we have $w_3 = \frac{1}{m}$. Hence $w_1 = \frac{b}{m} + n_1$ and $w_2 = \frac{c}{m} + n_2$ for non-negative integers n_1, n_2, and

$$1 = \frac{b + c + \alpha}{m} + n_1 + n_2.$$

Since $\frac{b+c+\alpha}{m}$ is a positive integer by (1), we have $n_1 = n_2 = 0$ and $b + c = m - \alpha$. Thus a vector w of $\mathbb{Z}^3 + \mathbb{Z}\frac{1}{m}(b, c, 1)$ in the octant with $w_3 > 0$ and $w_1 + w_2 + \alpha w_3 = 1$ exists if and only if $b + c = m - \alpha$, and in this case $w = \frac{1}{m}(b, c, 1)$, which is primitive since $w_1 + w_2 + \alpha w_3 = 1$.

Let Σ_0 be the fan of faces of the octant if $b + c \neq m - \alpha$, and the star subdivision of the octant at $\frac{1}{m}(b, c, 1)$ if $b + c = m - \alpha$. By [CLS11, Theorem 11.1.9], Σ_0 has a smooth refinement, which contains every ray of Σ_0. Let $h : W \to Y'$ be the corresponding toric resolution. Write

$$K_W + B_W = h^*(K_{Y'} + (1-\alpha)E').$$

By (10.3), $B_W = \sum_\rho (1 - w_{\rho,1} - w_{\rho,2} - \alpha w_{\rho,3}) F_{w_\rho}$, where ρ runs over the rays of the fan and $w_\rho = (w_{\rho,1}, w_{\rho,2}, w_{\rho,3})$ is the primitive vector of ρ. Hence $B_W \leq 0$ by (10.4). Let $W^\circ := W \times_{Y'} Y^\circ$, and let $h^\circ : W^\circ \to Y^\circ$ and $g_W : W^\circ \to W$ be the projections. Then W° is smooth, g_W is étale,

W° is irreducible (it is smooth, and each connected component maps onto an open set meeting the torus, over which W° is isomorphic to an open subset of Y°), and

$$K_{W^\circ} + g_W^* B_W = (h^\circ)^*(K_{Y^\circ} + (1-\alpha)E|_{Y^\circ}).$$

Let G be a crepant divisor over $X \ni o$ whose center on Y contains p. By Construction-Proposition 10.4(3) and [KM98, Lemma 2.30],

$$a(G, W^\circ, g_W^* B_W) = a(G, Y, (1-\alpha)E) = a(G, X) = 1.$$

If G is not a divisor on W°, then $a(G, W^\circ, g_W^* B_W) \geq a(G, W^\circ) \geq 2$ by $g_W^* B_W \leq 0$ and [KM98, Corollary 2.31(2)], a contradiction. Thus G is a prime divisor on W°. Since the center of G on Y is contained in E, G is exceptional over Y°. Hence G is an irreducible component of $g_W^{-1}(F_{w_\rho})$ for a ray ρ, and

$$1 = a(G, W^\circ, g_W^* B_W) = w_{\rho,1} + w_{\rho,2} + \alpha w_{\rho,3}.$$

Since t^m vanishes along the center of G on Y, we have $m w_{\rho,3} = \mathrm{ord}_{F_{w_\rho}}(t^m) = \mathrm{ord}_G(t^m) > 0$. By the previous paragraph, $b + c = m - \alpha$ and $w_\rho = \frac{1}{m}(b, c, 1)$.

Conversely, assume that $b + c = m - \alpha$, and let $w := \frac{1}{m}(b, c, 1)$. Then F_w is a divisor on W, and any irreducible component G of $g_W^{-1}(F_w)$ satisfies $a(G, X) = w_1 + w_2 + \alpha w_3 = 1$ and $\mathrm{ord}_G(t^m) = m w_3 > 0$, so G is a crepant divisor over $X \ni o$. The center of G on Y contains p if and only if G intersects $(h^\circ)^{-1}(p)$. Since g is étale and $g^{-1}(0) = \{p\}$, g_W induces an isomorphism from $g_W^{-1}(F_w) \cap (h^\circ)^{-1}(p)$ onto $F_w \cap h^{-1}(0)$. We show that $F_w \cap h^{-1}(0)$ is non-empty and connected. Let Y'_w and F'_w be as in (1) for $w = \frac{1}{m}(b, c, 1)$. Since $W \to Y'_w$ is birational and $F_w \to F'_w$ is a proper birational morphism onto a normal surface, the fibers of $F_w \to F'_w$ are connected by Zariski's main theorem, so it suffices to show that the fiber of F'_w over 0 is non-empty and connected. If $b, c > 0$, then w is contained in the interior of the octant, and the fiber of F'_w over 0 is F'_w. By [CLS11, Theorem 3.2.6 and Lemma 3.3.21], the fiber of $Y'_w \to Y'$ over 0 is the union of the orbits of the cones of the star subdivision whose relative interiors are contained in the interior of the octant. If $b = 0$ (resp. $c = 0$), then the fiber of F'_w over 0 is the closure of the torus orbit of the cone spanned by w and $(1, 0, 0)$ (resp. $(0, 1, 0)$), which is an irreducible curve. Since $g_W^{-1}(F_w)$ is smooth, the irreducible components of $g_W^{-1}(F_w)$ are pairwise disjoint, so exactly one of them, denoted by F, intersects $(h^\circ)^{-1}(p)$. Since $W \to Y'_w$ is birational, the irreducible components of $g_W^{-1}(F_w)$ are the strict transforms on W° of the irreducible components of $F'_w \times_{Y'} Y^\circ$, so F is the only toric divisor over Y of the ray spanned by w. Together with the previous paragraph, F is the only crepant divisor over $X \ni o$ whose center on Y contains p.

Finally, the center of F on Y° is the irreducible component containing p of g^{-1} of the center of F_w on Y'. The center of F_w on Y' is $\{0\}$ if $b, c > 0$, $D'_2 \cap E'$ if $b = 0$, and $D'_1 \cap E'$ if $c = 0$, and $b = c = 0$ is impossible since $b + c = m - \alpha > 0$. Since $g^{-1}(0) = \{p\}$ and $g^{-1}(D'_2 \cap E')$ is the image of $\{s_2 = t = 0\}$, which is the image in $E \cong S$ of the curve $\{s_2 = 0\}$ of T, the center of F on Y° is $\{p\}$ if $b, c > 0$, and the irreducible component containing p of the image of the curve $\{s_2 = 0\}$ of T if $b = 0$ (resp. of the curve $\{s_1 = 0\}$ if $c = 0$). The center of F on Y is the closure in Y of the center of F on Y°, so the center of F on $E \cong S$ is as in (2). □

The next proposition divides the centers of crepant divisors into points and curves. Part (2) controls the stabilizer orders along a crepant curve center by the β-invariant of the curve, and part (3) bounds α from below when $X \ni o$ is not LCIQ.

Proposition 10.13. *Notation and conditions as in Set-up 10.1, let $f : Y \to X$, $E \cong S$, Δ, D, and $m(p)$ be as in Construction-Proposition 10.4, and let $A := -(K_S + \Delta)$. Assume that $\alpha \geq 2$ and that v_R computes $V := \widehat{\mathrm{vol}}(X \ni o) > q$. Then the following hold.*

(1) The center on E of any crepant divisor over $X \ni o$ is either a point or one of finitely many pairwise disjoint smooth projective curves C contained in the smooth locus of S. Any such curve C is Cartier, C is disjoint from the other components of Δ, and there

exists an integer $m > \alpha$ such that the coefficient of Δ along C is $1 - \frac{1}{m}$ and $m(p) = m$ for any $p \in C$. Distinct crepant divisors have disjoint centers. Moreover, $m(p) > \alpha$ for any closed point $p \in S$ contained in the center of a crepant divisor, and $m(p) \geq \alpha + 2$ if $\{p\}$ is a crepant point center.

(2) *For any crepant curve center C, let $m := m(p)$ for $p \in C$, let $k := m - \alpha$, and let g be the genus of C. Then there exists an integer l such that*

$$(C^2) = k(2g-2) + \alpha l, \qquad (A \cdot C) = \frac{\alpha}{m}(l + 2 - 2g), \qquad 1 \leq k < \frac{\alpha}{2}.$$

If $(C^2) \leq 0$, then $g = 0$, $l = 0$, $(C^2) = -2k$,

$$V \leq \frac{24\alpha^2 + 16\alpha k}{3(\alpha + k)^2}, \quad \text{and} \quad 0 < k < \frac{\alpha}{39}.$$

In particular, if there exist two crepant curve centers, then $(C^2) \leq 0$ for at least one of them, and hence $\alpha \geq 40$ and $\alpha < m < \frac{40\alpha}{39}$ for that curve.

(3) *If $X \ni o$ is not LCIQ, then there exist at least two crepant divisors over $X \ni o$, and either some crepant divisor has a crepant point center, in which case $\alpha \geq 13$, or there exist at least two crepant curve centers, in which case $\alpha \geq 40$. In particular, if $2 \leq \alpha \leq 12$, then $X \ni o$ is LCIQ.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

Let F be a crepant divisor over $X \ni o$, and let Z be the center of F on Y. Since $a(E, X) = \alpha \geq 2$ by Construction-Proposition 10.4(3), we have $F \neq E$. Since $f^{-1}(o)_{\mathrm{red}} = E$ by Construction-Proposition 10.4(1), Z is an irreducible closed subset of E different from E, so Z is a closed point or a curve. By Proposition 10.7(3), $m(p) > \alpha$ for any closed point $p \in Z$. Hence Lemma 10.12 holds at any closed point $p \in Z$, and Lemma 10.12(2) shows that F is the only crepant divisor over $X \ni o$ whose center on Y contains p. Thus distinct crepant divisors have disjoint centers. If $Z = \{p\}$ is a crepant point center, then by Lemma 10.12, $m(p) - \alpha = b + c$ for integers $b, c > 0$, so $m(p) \geq \alpha + 2$.

Suppose that $Z = C$ is a curve. Let $p \in C$ be a closed point, let $m := m(p)$, let U, T, and $T \to U$ be as in Construction-Proposition 10.4(2) for p, and let $\tilde{p} \in T$ be the point over p. By Lemma 10.12(1), there exist local coordinates s_1, s_2 of T at $\tilde{p}$ and integers $0 \leq b, c < m$ with $\gcd(m, b, c) = 1$ such that μ_m acts on (s_1, s_2) with weights $(-b, -c)$. Since $\{p\} \neq C$, Lemma 10.12(2) shows that $b = 0$ or $c = 0$. Since the roles of s_1 and s_2 are symmetric, possibly exchanging them, we may assume that $b = 0$. Then $c = m - \alpha$, $\gcd(m, c) = 1$, and C coincides near p with the image of $\{s_2 = 0\}$. Since $T \to U$ is the quotient by μ_m (Construction-Proposition 10.4(2)), μ_m acts trivially on s_1, and μ_m acts faithfully on s_2, we have

$$\widehat{\mathcal{O}}_{S,p} = \mathbb{C}[[s_1, s_2]]^{\mu_m} = \mathbb{C}[[s_1, s_2^m]].$$

Thus S is smooth at p, and C is defined near p by s_2^m, so C is smooth at p and Cartier near p. Moreover, near $\tilde{p}$, $T \to U$ is ramified along $\{s_2 = 0\}$ with ramification index m and is étale outside $\{s_2 = 0\}$ (μ_m acts trivially on s_1 and faithfully on s_2, and (s_1, s_2) is biholomorphic on a μ_m-stable analytic neighborhood, so stabilizers are trivial off $\{s_2 = 0\}$ and equal to μ_m on it). By the Hurwitz formula and since K_T is the pullback of $K_U + \Delta|_U$ (Construction-Proposition 10.4(2)), we have $\Delta = (1 - \frac{1}{m})C$ near p. Since $p \in C$ is arbitrary, C is a smooth projective curve contained in the smooth locus of S, C is Cartier, C is disjoint from the other components of Δ, and the coefficient of Δ along C is $1 - \frac{1}{m(p)}$ for any $p \in C$. Hence $m(p)$ is constant along C, and $m(p) > \alpha$ for $p \in C$ by the first paragraph of this step.

Finally, since X is Gorenstein and klt, X is canonical, so $\operatorname{crex}(X \ni o)$ is finite by [KM98, Proposition 2.36(2)]. Hence there exist only finitely many crepant curve centers, and the crepant curve centers are pairwise disjoint. This proves (1).

Step 2. In this step, we prove the formulas for (C^2) and $(A \cdot C)$ in (2), together with $1 \le k < \frac{\alpha}{2}$.

For any closed point $p \in S$, by Lemma 10.5,

$$m(p) \le m(p)|H_p| \le \frac{9\alpha}{V}, \tag{10.5}$$

where H_p is the finite subgroup of $SL_2(\mathbb{C})$ in Lemma 10.5. Let C be a crepant curve center, and let m, k, and g be as in (2). Since $V > q > 6$ by Lemma 4.3, (10.5) implies that $m < \frac{3\alpha}{2}$, so $k < \frac{\alpha}{2}$. By (1), $m > \alpha$, so $k \ge 1$.

By (1), the coefficient of Δ along C is $1 - \frac{1}{m}$. Hence, by Construction-Proposition 10.4(6), the coefficient of D along C is $\frac{e}{m}$ for an integer e with $\gcd(e, m) = 1$. Let $D_C := D - \frac{e}{m}C$. By Construction-Proposition 10.4(6), the coefficient of D along a prime divisor P is an integer if P is not a component of Δ. Since C is disjoint from the other components of Δ, D_C has integer coefficients near C, so D_C is Cartier near C. Since $K_S = \mathrm{div}(\omega)$ has integer coefficients and $K_S + \Delta + \alpha D = 0$ by Construction-Proposition 10.4(4), comparing the coefficients along C shows that

$$n := \frac{\alpha e - 1}{m}$$

is an integer. Near C, we have

$$K_S + \alpha D_C + (n+1)C = K_S + \Big(1 - \frac{1}{m}\Big)C + \alpha D_C + \frac{\alpha e}{m}C = K_S + \Delta + \alpha D = 0.$$

By the adjunction formula on the minimal resolution of S, which is an isomorphism near C, $(K_S \cdot C) + (C^2) = 2g - 2$ [Har77, Exercise V.1.3(a)]. Intersecting with C (globally $K_S + \alpha D_C + (n+1)C$ equals $-(\Delta - (1 - \frac{1}{m})C)$, whose support is disjoint from C, so its intersection number with C is 0), we obtain

$$2g - 2 + \alpha(D_C \cdot C) + n(C^2) = 0, \tag{10.6}$$

where $(D_C \cdot C)$ is an integer. Since $nm = \alpha e - 1$ and $m = \alpha + k$, we have $nk \equiv -1 \pmod{\alpha}$. Multiplying (10.6) by k, we obtain $k(2g-2) - (C^2) \equiv 0 \pmod{\alpha}$. Thus there exists an integer l such that

$$(C^2) = k(2g-2) + \alpha l.$$

Since $A = -K_S - (1 - \frac{1}{m})C$ near C,

$$\begin{aligned}(A \cdot C) &= -(K_S \cdot C) - (C^2) + \frac{(C^2)}{m} = 2 - 2g + \frac{k(2g-2) + \alpha l}{m} \\ &= \frac{(2-2g)(m-k) + \alpha l}{m} = \frac{\alpha}{m}(l + 2 - 2g).\end{aligned}$$

Step 3. In this step, we show that if $(C^2) \le 0$, then $g = 0$ and $l \in \{-1, 0\}$.

Since (S, Δ) is a log Fano pair by Construction-Proposition 10.4(5), A is ample, so $(A \cdot C) > 0$, that is, $l \ge 2g - 1$. If $g \ge 1$, then

$$(C^2) \ge k(2g-2) + \alpha(2g-1) = (2g-2)m + \alpha \ge \alpha > 0.$$

Thus $(C^2) \le 0$ implies $g = 0$. Then $(C^2) = -2k + \alpha l$ and $l \ge -1$. If $l \ge 1$, then $(C^2) \ge \alpha - 2k > 0$ since $k < \frac{\alpha}{2}$. Thus $l \in \{-1, 0\}$.

Step 4. In this step, we show that if $(C^2) \le 0$, then

$$(A^2) \le \frac{4(A \cdot C)}{m} - \frac{8(C^2)}{3m^2}. \tag{10.7}$$

By **Step 3**, $C \cong \mathbb{P}^1$, and C is Cartier by (1). Let N be a positive integer such that NA is Cartier. For any integer $i \ge 0$, the restriction of $NA - iC$ to C has degree $N(A \cdot C) - i(C^2) \ge 0$, so

$$h^0(C, (NA - iC)|_C) = N(A \cdot C) - i(C^2) + 1.$$

By the exact sequences

$$0 \to \mathcal{O}_S(NA-(i+1)C) \to \mathcal{O}_S(NA-iC) \to \mathcal{O}_C((NA-iC)|_C) \to 0,$$

for any integer $j \geq 0$,

$$\begin{aligned} h^0(S, NA-jC) &\geq h^0(S, NA) - \sum_{i=0}^{j-1} \big(N(A\cdot C) - i(C^2) + 1\big) \\ &= h^0(S, NA) - j\big(N(A\cdot C)+1\big) + \frac{j(j-1)}{2}(C^2). \end{aligned}$$

Let $t \geq 0$ be a rational number. Taking N divisible enough, $j = Nt$, dividing by $\frac{N^2}{2}$, and letting $N \to +\infty$, we obtain

$$\mathrm{vol}(A-tC) \geq (A^2) - 2t(A\cdot C) + t^2(C^2), \tag{10.8}$$

where $h^0(S, NA) = \frac{(A^2)}{2}N^2 + O(N)$ by [Laz04, Corollary 1.1.25]. Since $C \geq 0$, $\mathrm{vol}(A-tC)$ is non-increasing in t. Hence for a real number $t \geq 0$ and rational numbers $t' > t$, we have $\mathrm{vol}(A-tC) \geq \mathrm{vol}(A-t'C) \geq (A^2) - 2t'(A\cdot C) + t'^2(C^2)$, and letting $t' \to t$ shows that (10.8) holds for any real number $t \geq 0$.

By Construction-Proposition 10.4(5), (S, Δ) is K-semistable, so $\beta_{(S,\Delta)}(C) \geq 0$ by Definition 2.11. Since the coefficient of Δ along C is $1-\frac{1}{m}$, we have $a(C, S, \Delta) = \frac{1}{m}$, and $\mathrm{vol}(A) = (A^2)$ since A is ample. Hence

$$\int_0^{+\infty} \mathrm{vol}(A-tC)dt \leq \frac{(A^2)}{m}.$$

Since $\mathrm{vol}(A-tC) \geq 0$ for any $t \geq 0$, truncating the integral at $t = \frac{2}{m}$ and applying (10.8) on $[0, \frac{2}{m}]$, we obtain

$$\int_0^{+\infty} \mathrm{vol}(A-tC)dt \geq \int_0^{\frac{2}{m}} \big((A^2) - 2t(A\cdot C) + t^2(C^2)\big)dt = \frac{2(A^2)}{m} - \frac{4(A\cdot C)}{m^2} + \frac{8(C^2)}{3m^3}.$$

Truncating at $t = \tau > \frac{1}{m}$ instead gives $(A^2) \leq \frac{\tau^2}{\tau-\frac{1}{m}}(A\cdot C) - \frac{\tau^3}{3(\tau-\frac{1}{m})}(C^2)$, and $\tau = \frac{2}{m}$ minimizes the coefficient $\frac{\tau^2}{\tau-\frac{1}{m}}$ of $(A\cdot C)$. Comparing the upper and lower bounds of $\int_0^{+\infty}\mathrm{vol}(A-tC)dt$, we obtain (10.7).

Step 5. In this step, we finish the proof of (2).

Assume that $(C^2) \leq 0$. By **Step 3**, $g = 0$ and $l \in \{-1, 0\}$. By Construction-Proposition 10.4(5), $V = \alpha(A^2)$.

Suppose that $l = -1$. Then $(C^2) = -\alpha - 2k$ and $(A\cdot C) = \frac{\alpha}{m}$ by **Step 2**, and (10.7) implies that $(A^2) \leq \frac{20\alpha+16k}{3m^2}$. Since $m = \alpha + k$ and $k > 0$,

$$V \leq \frac{20\alpha^2+16\alpha k}{3(\alpha+k)^2} < \frac{20\alpha^2+40\alpha k+20k^2}{3(\alpha+k)^2} = \frac{20}{3} < 7 < q,$$

a contradiction, since $q > 7$ by Lemma 4.3. Thus $l = 0$, $(C^2) = -2k$, and $(A\cdot C) = \frac{2\alpha}{m}$. Then (10.7) implies that $(A^2) \leq \frac{24\alpha+16k}{3m^2}$, that is,

$$V \leq \frac{24\alpha^2+16\alpha k}{3(\alpha+k)^2} = \psi\Big(\frac{k}{\alpha}\Big), \quad \text{where} \quad \psi(r) := \frac{8(1+\frac{2r}{3})}{(1+r)^2}.$$

For $r > 0$, we have

$$\frac{(1+r)^3}{8}\psi'(r) = \frac{2}{3}(1+r) - 2\big(1+\frac{2r}{3}\big) = -\frac{4}{3} - \frac{2r}{3} < 0,$$

so ψ is decreasing on $(0, +\infty)$. Moreover, $\psi(\frac{1}{39}) = \frac{1547}{200} < q$, since $q > \frac{1547}{200}$ is equivalent to $\sqrt{13} > \frac{2341}{650}$, which holds since $13\cdot 650^2 = 5492500 > 5480281 = 2341^2$. Hence $\frac{k}{\alpha} \geq \frac{1}{39}$ implies

$V \le \psi(\frac{1}{39}) < q$, a contradiction. Thus $0 < k < \frac{\alpha}{39}$. Since $k \ge 1$, we have $\alpha > 39$, that is, $\alpha \ge 40$, and $\alpha < m = \alpha + k < \frac{40\alpha}{39}$.

Let C and C' be two crepant curve centers, and suppose that $(C^2) > 0$ and $(C'^2) > 0$. By (1), C and C' are disjoint and contained in the smooth locus of S. Let $\tilde{S} \to S$ be the minimal resolution, which is an isomorphism near $C \cup C'$, and let $\tilde{C}$ and $\tilde{C}'$ be the strict transforms of C and C'. Then $(\tilde{C}^2) > 0$, $(\tilde{C}'^2) > 0$, and $(\tilde{C} \cdot \tilde{C}') = 0$, so the intersection form of $\tilde{S}$ is positive definite on the span of $\tilde{C}$ and $\tilde{C}'$, which contradicts the Hodge index theorem [Har77, Theorem V.1.9 and Remark V.1.9.1]. Thus $(C^2) \le 0$ or $(C'^2) \le 0$, which proves (2).

Step 6. In this step, we prove (3).

Assume that $X \ni o$ is not LCIQ. Since X is Gorenstein and klt, $X \ni o$ is a Gorenstein canonical threefold germ. If $\operatorname{crex}(X \ni o) = 0$, then $X \ni o$ is a cDV hypersurface germ by Theorem 2.15(1), so $X \ni o$ is LCIQ by Definition 2.13, a contradiction. If $\operatorname{crex}(X \ni o) = 1$, then $V \le q$ by Theorem 8.1, a contradiction. Thus there exist at least two crepant divisors over $X \ni o$.

Suppose that some crepant divisor has a crepant point center $\{p\}$. By (1), $m(p) \ge \alpha + 2$, so (10.5) implies that

$$V \le \frac{9\alpha}{\alpha + 2}.$$

If $\alpha \le 12$, then $V \le \frac{108}{14} = \frac{54}{7} < q$ by Lemma 4.3, a contradiction. Thus $\alpha \ge 13$. Otherwise, any crepant divisor has a crepant curve center, and distinct crepant divisors have disjoint centers by (1), so there exist at least two crepant curve centers. Then $\alpha \ge 40$ by (2). In particular, if $2 \le \alpha \le 12$, then $X \ni o$ is LCIQ. □

10.5. **Complements.** In this subsection, we construct a monotonic complement of bounded index of the quotient pair (S, Δ) when $X \ni o$ is not LCIQ and has large local volume, and we lift the complement to a homogeneous element of R.

We use the monotonic form [BL23, Proposition 5.3] of Shokurov's theorem [Pro01, Corollary 8.4.2] on complements of non-exceptional log del Pezzo surface pairs.

Proposition 10.14. *Notation and conditions as in Set-up 10.1, and let $f : Y \to X$, $E \cong S$, Δ, D, and u be as in Construction-Proposition 10.4. Assume that $\alpha \ge 2$, that v_R computes $\widehat{\operatorname{vol}}(X \ni o) > q$, and that $X \ni o$ is not LCIQ. Then (S, Δ) is not exceptional, and there exist $N \in \{1, 2, 3, 4, 6\}$, a monotonic N-complement Δ^+ of (S, Δ), and a non-zero element $h \in R_{N\alpha}$, such that the following hold.*

(1) For any prime divisor W over S with $a(W, S, \Delta) < \frac{1}{7}$, we have $a(W, S, \Delta^+) = 0$ and $\operatorname{mult}_W(\Delta^+ - \Delta) = a(W, S, \Delta)$. There exist at least three crepant divisors over $X \ni o$, and for any crepant divisor F over $X \ni o$, the restriction of ord_F to $K(S)$ is a positive multiple of ord_W for a prime divisor W over S with $a(W, S, \Delta) < \frac{1}{7}$. Any crepant curve center has coefficient one in Δ^+.

(2) The pair $(X, \frac{1}{N}\operatorname{div}(h))$ is lc, the log canonical threshold of $\operatorname{div}(h)$ at o is $\frac{1}{N}$, $\operatorname{ord}_E(h) = N\alpha$, and $\operatorname{ord}_F(h) = N$ for any crepant divisor F over $X \ni o$.

(3) For any integer $k \ge 1$, $R_{k\alpha} = H^0(S, \mathcal{O}_S(-kK_S - \lceil k\Delta \rceil))u^{k\alpha}$, where K_S is as in Construction-Proposition 10.4(4). Moreover, if $h = \varphi u^{N\alpha}$ with $\varphi \in K(S)$, then $\operatorname{div}_S(\varphi) + N\alpha D = N(\Delta^+ - \Delta)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that (S, Δ) is not exceptional, construct N and Δ^+, and prove (1).

By Lemma 4.3, $\widehat{\operatorname{vol}}(X \ni o) > q > \frac{54}{7}$, so the hypotheses of Lemma 10.10 hold. Since $X \ni o$ is a klt germ which is not LCIQ and $\widehat{\operatorname{vol}}(X \ni o) > q$, Theorem 9.1 implies that $\operatorname{crex}(X \ni o) \ge 3$. By Construction-Proposition 10.4(5), (S, Δ) is a klt log Fano pair, and since $\operatorname{crex}(X \ni o) \ge 3$, Lemma 10.10(4) implies that (S, Δ) is not exceptional in the sense of Definition 10.8. Thus there

exists a $\mathbb{Q}$-divisor $\Delta' \geq \Delta$ such that (S, Δ') is lc but not klt and $K_S + \Delta' \sim_{\mathbb{Q}} 0$. Let $G' := \Delta' - \Delta$. Then $G' \in |-(K_S + \Delta)|_{\mathbb{R}}$ and $(S, \Delta + G')$ is not klt. Hence (S, Δ) is $\mathbb{R}$-complementary [BL23, Definition 2.10], and (S, Δ) is not exceptional in the sense of [BL23, Definition 2.11].

By [BL23, Proposition 5.3], if a projective lc surface pair with boundary $B + B'$ (over a point) is $\mathbb{R}$-complementary and not exceptional, the coefficients of B belong to the set $\{1 - \frac{1}{r} \mid r \in \mathbb{N}^+\}$ of standard coefficients, and the non-zero coefficients of B' belong to $[\frac{6}{7}, 1)$, then it has a monotonic N-complement which is not klt for some $N \in \{1, 2, 3, 4, 6\}$. For (S, Δ) with $B = \Delta$ and $B' = 0$, the hypotheses of [BL23, Proposition 5.3] hold: (S, Δ) is klt, $\mathbb{R}$-complementary, and not exceptional by the previous paragraph, the coefficients of Δ are standard, and $B' = 0$ satisfies the condition on B', which concerns only the non-zero coefficients [BL23, Section 2.1]. Thus there exist $N \in \{1, 2, 3, 4, 6\}$ and a monotonic N-complement Δ^+ of (S, Δ) in the sense of Definition 10.8, that is, $\Delta^+ \geq \Delta$, (S, Δ^+) is lc, and $N(K_S + \Delta^+) \sim 0$. Let $G := \Delta^+ - \Delta \geq 0$. Since S is a klt surface, S is $\mathbb{Q}$-factorial [KM98, Proposition 4.11]. Thus G is $\mathbb{Q}$-Cartier, and for any prime divisor W over S,

$$\operatorname{mult}_W G = a(W, S, \Delta) - a(W, S, \Delta^+) \geq 0.$$

We show that $a(W, S, \Delta^+) \in \frac{1}{N}\mathbb{Z}$ for any prime divisor W over S. Let $g : S' \to S$ be a projective birational morphism from a normal surface S' such that W is a divisor on S', and write $K_{S'} + \Delta^+_{S'} := g^*(K_S + \Delta^+)$. Since $N(K_S + \Delta^+)$ is a principal divisor, $Ng^*(K_S + \Delta^+)$ is a principal divisor on S'. Thus $N\Delta^+_{S'}$ has integer coefficients, and $a(W, S, \Delta^+) = 1 - \operatorname{mult}_W \Delta^+_{S'} \in \frac{1}{N}\mathbb{Z}$.

Now let W be a prime divisor over S with $a(W, S, \Delta) < \frac{1}{7}$. Since (S, Δ^+) is lc,

$$0 \leq a(W, S, \Delta^+) \leq a(W, S, \Delta) < \frac{1}{7}.$$

Since $N \leq 6$, any positive element of $\frac{1}{N}\mathbb{Z}$ is at least $\frac{1}{N} \geq \frac{1}{6} > \frac{1}{7}$. Thus $a(W, S, \Delta^+) = 0$ and $\operatorname{mult}_W G = a(W, S, \Delta)$. This proves the first sentence of (1). The second sentence of (1) follows from $\operatorname{crex}(X \ni o) \geq 3$ and Lemma 10.10(2).

Let C be a crepant curve center, and let F be a crepant divisor over $X \ni o$ with $\operatorname{center}_Y F = C$ (Definition 10.11). By Lemma 10.10(2), the restriction of ord_F to $K(S)$ is a positive multiple of ord_W for a prime divisor W over S with $a(W, S, \Delta) < \frac{1}{7}$. The structure morphism $Y \to S$ restricts to the isomorphism $E \cong S$ and induces the inclusion $K(S) \subset K(Y) = K(X)$. Hence the center of ord_W on S is the image of $\operatorname{center}_Y F = C$, that is, C. Thus $W = C$ is a prime divisor on S with $a(C, S, \Delta) < \frac{1}{7}$, and $a(C, S, \Delta^+) = 0$. In particular, C has coefficient one in Δ^+. This proves (1).

Step 2. In this step, we prove (3) and construct h.

By Construction-Proposition 10.4(4), $K_S + \Delta + \alpha D = 0$. Since $K_S = \operatorname{div}(\omega)$ has integer coefficients, for any integer $k \geq 1$ we have

$$\lfloor k\alpha D \rfloor = \lfloor -kK_S - k\Delta \rfloor = -kK_S - \lceil k\Delta \rceil.$$

Since $R_j = H^0(S, \mathcal{O}_S(\lfloor jD \rfloor))u^j$ for any $j \geq 0$ (Construction-Proposition 10.4), we obtain $R_{k\alpha} = H^0(S, \mathcal{O}_S(-kK_S - \lceil k\Delta \rceil))u^{k\alpha}$.

Since $N(K_S + \Delta^+) \sim 0$, there exists $\varphi \in K(S)\backslash\{0\}$ such that $\operatorname{div}_S(\varphi) = N(K_S + \Delta^+)$. Then

$$\operatorname{div}_S(\varphi) + N\alpha D = N(K_S + \Delta^+) - N(K_S + \Delta) = NG \geq 0.$$

Since $\operatorname{div}_S(\varphi)$ has integer coefficients, $\operatorname{div}_S(\varphi) + \lfloor N\alpha D \rfloor \geq 0$, that is, $\varphi \in H^0(S, \mathcal{O}_S(\lfloor N\alpha D \rfloor))$. Let $h := \varphi u^{N\alpha}$. Then h is a non-zero element of $R_{N\alpha}$, and φ is the unique element of $K(S)$ with $h = \varphi u^{N\alpha}$. This proves (3).

Step 3. In this step, we show that $\operatorname{lct}(X \ni o; \operatorname{div}(h)) = \frac{1}{N}$, that $\operatorname{ord}_E(h) = N\alpha$, and that $(X, \frac{1}{N}\operatorname{div}(h))$ is lc.

Let $p \in S$ be a closed point, let U, T, and the cyclic cover $\rho : T \to U$ be as in Construction-Proposition 10.4(2) for p, let $\tilde{\rho} : T \times \mathbb{A}^1 \to Y_U$ be the composite $T \times \mathbb{A}^1 \to (T \times \mathbb{A}^1)/\mu_{m(p)} \cong Y_U$

of Construction-Proposition 10.4(2)(i), and let t be the coordinate of $\mathbb{A}^1$. By Construction-Proposition 10.4(2)(i) and (iv), $\tilde{\rho}^{-1}(E) = T \times \{0\}$ (as $T \times \{0\}$ is $\mu_{m(p)}$-stable), and $\tilde{\rho}^* u = t\psi$ for a rational function ψ on T with $\operatorname{div}_T(\psi) = \rho^* D$. In particular, $\tilde{\rho}^* h = t^{N\alpha} h_T$, where $h_T := (\rho^*\varphi)\psi^{N\alpha}$ satisfies

$$\operatorname{div}_T(h_T) = \rho^*(\operatorname{div}_S(\varphi) + N\alpha D) = N\rho^* G \geq 0.$$

Since T is normal, h_T is a non-zero regular function on T.

By Construction-Proposition 10.4(2), $\mu_{m(p)}$ acts faithfully on T, and $\mu_{m(p)}$ acts on t with weight -1. Thus for any $1 \neq \zeta \in \mu_{m(p)}$, the fixed locus of ζ on $T \times \mathbb{A}^1$ is contained in $Z \times \{0\}$ for a proper closed subset $Z \subsetneq T$, and $\tilde{\rho}$ is étale in codimension one. In particular, ord_E is the restriction of $\operatorname{ord}_{T\times\{0\}}$ to $K(Y)$, and $K_{T\times\mathbb{A}^1} = \tilde{\rho}^* K_{Y_U}$. Since h_T is a non-zero function on T, we obtain

$$\operatorname{ord}_E(h) = \operatorname{ord}_{T\times\{0\}}(t^{N\alpha} h_T) = N\alpha.$$

Let H_Y be the strict transform of $\operatorname{div}_X(h)$ on Y. By Construction-Proposition 10.4(1), f is an isomorphism over $X \backslash \{o\}$ and $f^{-1}(o)_{\mathrm{red}} = E$, so

$$\operatorname{div}_Y(f^* h) = N\alpha E + H_Y.$$

Let $\Gamma := \rho^*(G|_U) = \frac{1}{N}\operatorname{div}_T(h_T) \geq 0$. Since $\tilde{\rho}^* \operatorname{div}_Y(f^* h) = \operatorname{div}(t^{N\alpha} h_T)$ and $\tilde{\rho}^* E = T \times \{0\}$, we have

$$\tilde{\rho}^*\Big(E + \frac{1}{N} H_Y\Big) = T \times \{0\} + \Gamma \times \mathbb{A}^1.$$

By Construction-Proposition 10.4(2), $K_T = \rho^*(K_U + \Delta|_U)$, so $K_T + \Gamma = \rho^*(K_U + \Delta^+|_U)$. Since (S, Δ^+) is lc, (T, Γ) is lc [KM98, Proposition 5.20]. Let $T' \to T$ be a log resolution of (T, Γ), and write $K_{T'} + \Gamma'$ for the pullback of $K_T + \Gamma$. Then $T' \times \mathbb{A}^1 \to T \times \mathbb{A}^1$ is a log resolution of $(T \times \mathbb{A}^1, \Gamma \times \mathbb{A}^1 + T \times \{0\})$, the pullback of $K_{T\times\mathbb{A}^1} + \Gamma \times \mathbb{A}^1 + T \times \{0\}$ is $K_{T'\times\mathbb{A}^1} + \Gamma' \times \mathbb{A}^1 + T' \times \{0\}$, and $\Gamma' \times \mathbb{A}^1 + T' \times \{0\}$ is a simple normal crossing divisor with coefficients at most one, as (T, Γ) is lc. Hence $(T \times \mathbb{A}^1, \Gamma \times \mathbb{A}^1 + T \times \{0\})$ is lc [KM98, Lemma 2.30 and Corollary 2.31(3)]. Since

$$K_{T\times\mathbb{A}^1} + \Gamma \times \mathbb{A}^1 + T \times \{0\} = \tilde{\rho}^*\Big(K_Y + E + \frac{1}{N} H_Y\Big)\Big|_{Y_U},$$

the pair $(Y_U, (E + \frac{1}{N} H_Y)|_{Y_U})$ is lc [KM98, Proposition 5.20]. The open subsets Y_U cover Y as p varies, so $(Y, E + \frac{1}{N} H_Y)$ is lc.

By Construction-Proposition 10.4(3), $K_Y = f^* K_X + (\alpha - 1)E$. Thus

$$K_Y + E + \frac{1}{N} H_Y = f^* K_X + \frac{1}{N}(N\alpha E + H_Y) = f^*\Big(K_X + \frac{1}{N}\operatorname{div}_X(h)\Big).$$

Since f is a projective birational morphism and $(Y, E + \frac{1}{N} H_Y)$ is lc, $(X, \frac{1}{N}\operatorname{div}(h))$ is lc. Hence for any prime divisor F over X,

$$a(F, X) - \frac{1}{N}\operatorname{mult}_F \operatorname{div}(h) = a\Big(F, X, \frac{1}{N}\operatorname{div}(h)\Big) \geq 0.$$

Moreover, $\operatorname{center}_X E = \{o\}$, and by Construction-Proposition 10.4(3),

$$\frac{a(E, X)}{\operatorname{ord}_E(h)} = \frac{\alpha}{N\alpha} = \frac{1}{N}.$$

By Definition 2.3, $\operatorname{lct}(X \ni o; \operatorname{div}(h)) = \frac{1}{N}$.

Step 4. In this step, we show that $\operatorname{ord}_F(h) = N$ for any crepant divisor F over $X \ni o$.

Let F be a crepant divisor over $X \ni o$. Since $a(E, X) = \alpha \geq 2$ and $a(F, X) = 1$, $F \neq E$. Since $f(\operatorname{center}_Y F) = \{o\}$, $\operatorname{center}_Y F \subset E$. Let $p \in \operatorname{center}_Y F$ be a closed point, let $m := m(p)$, and let U, T, ρ, $\tilde{\rho}$, t, and h_T be as in **Step 3** for p. By Lemma 10.10(3), $\alpha < m < \frac{7\alpha}{6}$. In particular, the hypotheses of Lemma 10.12 hold. By Lemma 10.12(1)(2), there exist local coordinates s_1, s_2 of T at the point $\tilde{p}$ over p and integers $0 \leq b, c < m$ with $b + c = m - \alpha$, such that μ_m acts on (s_1, s_2, t) with weights $(-b, -c, -1)$, and F is the toric divisor over Y of the ray spanned by

$\frac{1}{m}(b,c,1)$. Since $\frac{1}{m}(b,c,1)$ is primitive in $\mathbb{Z}^3+\mathbb{Z}\frac{1}{m}(b,c,1)$, after an analytic localization at p, ord_F is the restriction to $K(Y)$ of the monomial valuation of $\mathbb{C}[[s_1,s_2,t]]$ (Lemma 10.12(1)) with

$$\mathrm{ord}_F(s_1)=\frac{b}{m},\quad \mathrm{ord}_F(s_2)=\frac{c}{m},\quad \mathrm{ord}_F(t)=\frac{1}{m}.$$

Since h_T is regular at $\tilde{p}$, $h_T\in\mathbb{C}[[s_1,s_2]]$, and $\mathrm{ord}_F(h_T)\geq 0$. Since $\tilde{\rho}^*h=t^{N\alpha}h_T$ and $N\leq 6$,

$$\mathrm{ord}_F(h)=\frac{N\alpha}{m}+\mathrm{ord}_F(h_T)\geq\frac{N\alpha}{m}>\frac{6N}{7}>N-1.$$

Since $h\in R$ is a non-zero regular function on X and F is a prime divisor over X, $\mathrm{ord}_F(h)$ is an integer. Thus $\mathrm{ord}_F(h)\geq N$. On the other hand, by **Step 3**, $(X,\frac{1}{N}\,\mathrm{div}(h))$ is lc, and $a(F,X)=1$, so

$$0\leq a\Big(F,X,\frac{1}{N}\,\mathrm{div}(h)\Big)=1-\frac{1}{N}\,\mathrm{ord}_F(h).$$

Thus $\mathrm{ord}_F(h)\leq N$, and hence $\mathrm{ord}_F(h)=N$. Together with **Step 3**, this proves (2). □

10.6. **Curve centers.** In this subsection, we prove Proposition 10.17, which shows that there exists at most one crepant curve center. We first compute the orders of homogeneous elements of R along the prime divisor on X over a crepant curve center.

Lemma 10.15. *Notation and conditions as in Set-up 10.1, and let $f:Y\to X$, $E\cong S$, D, u, m_P, and $m(p)$ be as in Construction-Proposition 10.4. Let $C\subset S$ be a prime divisor, let $m:=m_C$, let $k:=m-\alpha$, and let $P_C\subset X$ be the closure of the preimage of C under the quotient morphism $X\backslash\{o\}\to S$. Assume that $\alpha\geq 2$, that v_R computes $\widehat{\mathrm{vol}}(X\ni o)>q$, and that C is a crepant curve center. Then P_C is a prime divisor on X containing o, and $m(p)=m$ for any closed point $p\in C$. Moreover, for any integer $j\geq 0$ and any non-zero $g\in R_j$, written as $g=\varphi u^j$ with $\varphi\in K(S)$, the following hold.*

(1) $\mathrm{ord}_{P_C}(g)=m\,\mathrm{ord}_C(\varphi)+jm\,\mathrm{mult}_C D$, and $k\,\mathrm{ord}_{P_C}(g)+j\equiv 0 \pmod m$.

(2) If m divides j, then m divides $\mathrm{ord}_{P_C}(g)$, and the image of φ in $H^0(C,\mathcal{O}_S(\lfloor jD\rfloor)|_C)$ is zero if and only if $\mathrm{ord}_{P_C}(g)\geq m$.

Proof. Let Δ be as in Construction-Proposition 10.4, and let K_S be as in Construction-Proposition 10.4(4). By Proposition 10.13(1), C is a smooth projective curve contained in the smooth locus of S, C is disjoint from the other components of Δ, and there exists an integer $m'>\alpha$ such that the coefficient of Δ along C is $1-\frac{1}{m'}$ and $m(p)=m'$ for any closed point $p\in C$. Since the coefficient of Δ along C is $1-\frac{1}{m_C}$, we have $m'=m_C=m$. Thus $m(p)=m$ for any closed point $p\in C$.

Let $p\in C$ be a closed point, let U and the cyclic cover $\rho:T\to U$ be as in Construction-Proposition 10.4(2) for p, and let $\tilde{p}\in T$ be the point over p. By Lemma 10.12(1), T is smooth at $\tilde{p}$, and there exist local coordinates s_1,s_2 of T at $\tilde{p}$ and integers $0\leq b,c<m$ with $\gcd(m,b,c)=1$, such that μ_m acts on (s_1,s_2,t) with weights $(-b,-c,-1)$, where t is the coordinate of $\mathbb{A}^1$. The center on $E\cong S$ of the crepant divisor over $X\ni o$ with center C contains p. Thus $b+c=m-\alpha$ by Lemma 10.12(2), and $b=0$ or $c=0$, because C is a curve. Possibly exchanging s_1 and s_2, we may assume that $b=0$. Then $c=m-\alpha=k$, $\gcd(m,k)=1$, and near p the curve C is the image of the curve $\tilde{C}:=\{s_2=0\}$ of T.

The group μ_m acts trivially on $\tilde{C}$, and it acts freely on $T\backslash\tilde{C}$ near $\tilde{p}$ because $\gcd(m,k)=1$. Thus ρ is étale on $T\backslash\tilde{C}$ near $\tilde{p}$, and $\rho^*C=m\tilde{C}$ near $\tilde{p}$.

We show that P_C is a prime divisor containing o. Let $\mathfrak{p}\subset R$ be the homogeneous prime ideal not containing $\bigoplus_{j>0}R_j$ that is the generic point of C in $S=\mathrm{Proj}\,R$, so that $C=V_+(\mathfrak{p})$. Since $\mathfrak{p}$ is generated by homogeneous elements, a prime ideal $\mathfrak{q}$ of R contains $\mathfrak{p}$ if and only if the ideal generated by the homogeneous elements of $\mathfrak{q}$ contains $\mathfrak{p}$. Hence the preimage of C under the quotient morphism $X\backslash\{o\}\to S$ is $V(\mathfrak{p})\backslash\{o\}$, and $P_C=V(\mathfrak{p})$ is irreducible. The preimage of C is the union of the orbits over the closed points of C, and the orbit over any closed point of S

is an irreducible curve, so $\dim P_C = 2$. Since $R_0 = \mathbb{C}$, $\mathfrak{p} \subset \bigoplus_{j>0} R_j$, so $o \in P_C$. Hence P_C is a prime divisor on X containing o.

Next we compute ord_{P_C} on the chart at p. Since μ_m acts on t with weight -1, it acts freely on $T \times (\mathbb{A}^1\backslash\{0\})$. By Construction-Proposition 10.4(1)(2), the quotient morphism

$$T \times (\mathbb{A}^1\backslash\{0\}) \to Y_U\backslash E$$

is étale, f maps $Y_U\backslash E$ isomorphically onto the preimage of U in $X\backslash\{o\}$, and the induced morphism $T \times (\mathbb{A}^1\backslash\{0\}) \to X\backslash\{o\} \to S$ is the composite of the first projection and ρ. Thus, near $\tilde{p}$, the preimage of P_C in $T \times (\mathbb{A}^1\backslash\{0\})$ is the reduced divisor $\tilde{C} \times (\mathbb{A}^1\backslash\{0\})$, and for any non-zero rational function g on X, $\mathrm{ord}_{P_C}(g)$ is the order of the pullback of g along $\tilde{C} \times (\mathbb{A}^1\backslash\{0\})$. By Construction-Proposition 10.4(2)(iv), the pullback of u to $T \times \mathbb{A}^1$ is $t\psi$ for a rational function ψ on T with $\mathrm{div}_T(\psi) = \rho^* D$.

(1) The pullback of $g = \varphi u^j$ is $(\rho^*\varphi)\psi^j t^j$, and t is a unit on $T \times (\mathbb{A}^1\backslash\{0\})$. Since $\rho^* C = m\tilde{C}$ near $\tilde{p}$, we obtain

$$\mathrm{ord}_{P_C}(g) = \mathrm{ord}_{\tilde{C}}(\rho^*\varphi) + j\,\mathrm{ord}_{\tilde{C}}(\psi) = m\,\mathrm{ord}_C(\varphi) + jm\,\mathrm{mult}_C D. \tag{10.9}$$

By Construction-Proposition 10.4(6), $\mathrm{mult}_C D = \frac{e}{m}$ for an integer e with $\gcd(e, m) = 1$. Comparing the coefficients along C in $K_S + \Delta + \alpha D = 0$ (Construction-Proposition 10.4(4)), where $\mathrm{mult}_C K_S \in \mathbb{Z}$ and $\mathrm{mult}_C \Delta = 1 - \frac{1}{m}$, we obtain

$$m\,\mathrm{mult}_C K_S + m - 1 + \alpha e = 0, \quad \text{hence} \quad \alpha e \equiv 1 \pmod{m}.$$

Since $k \equiv -\alpha \pmod{m}$, (10.9) implies that

$$k\,\mathrm{ord}_{P_C}(g) + j = km\,\mathrm{ord}_C(\varphi) + kje + j \equiv j(1 - \alpha e) \equiv 0 \pmod{m}.$$

(2) Assume that m divides j. Since $m(p) = m$ for any closed point $p \in C$ and m divides j, Construction-Proposition 10.4(6) implies that jD is Cartier near C. Thus $\lfloor jD \rfloor = jD$ near C, $\mathcal{O}_S(\lfloor jD \rfloor)$ is invertible near C, and $\mathrm{ord}_C(\varphi) + \mathrm{mult}_C(jD)$ is a non-negative integer because $\varphi \in H^0(S, \mathcal{O}_S(\lfloor jD \rfloor))$. By (10.9),

$$\mathrm{ord}_{P_C}(g) = m\big(\mathrm{ord}_C(\varphi) + \mathrm{mult}_C(jD)\big),$$

so m divides $\mathrm{ord}_{P_C}(g)$. Since $\mathcal{O}_S(\lfloor jD \rfloor)$ is invertible near C, the image of φ in $H^0(C, \mathcal{O}_S(\lfloor jD \rfloor)|_C)$ is zero if and only if $\mathrm{ord}_C(\varphi) + \mathrm{mult}_C(jD) \geq 1$, that is, if and only if $\mathrm{ord}_{P_C}(g) \geq m$. □

To prove Proposition 10.17, we perturb the complement of Proposition 10.14 along the crepant curve centers to a klt boundary, and the Kawamata–Viehweg vanishing theorem then lifts sections from the crepant curve centers simultaneously (Lemma 10.16). Together with Proposition 5.6(3) and Lemma 10.15, the lifted sections exclude a second crepant curve center, so at least two crepant divisors over $X \ni o$ have crepant point centers, which is an input of Theorem 10.3.

Lemma 10.16. *Notation and conditions as in Set-up 10.1, and let $f : Y \to X$, $E \cong S$, D, u, and $m(p)$ be as in Construction-Proposition 10.4. For any integer $d \geq 0$, let $F_d := \lfloor dD \rfloor$. For any integer $j \geq 0$, through the isomorphism $\varphi \to \varphi u^j$ from $H^0(S, \mathcal{O}_S(F_j))$ onto R_j, we regard the restriction maps below as maps on R_j. Assume that $\alpha \geq 2$, that v_R computes $\widehat{\mathrm{vol}}(X \ni o) > q$, and that $X \ni o$ is not LCIQ. Then the following hold.*

(1) Let $C_1, \dots, C_s$ be distinct crepant curve centers, let m_i be the constant value of $m(p)$ on C_i, and let d be a positive integer divisible by $m_1, \dots, m_s$. Then $\mathcal{O}_S(F_d)$ is invertible near $C_1 \cup \cdots \cup C_s$, and the restriction map

$$H^0(S, \mathcal{O}_S(F_d)) \to \bigoplus_{i=1}^{s} H^0(C_i, \mathcal{O}_S(F_d)|_{C_i})$$

is surjective.

(2) *If C is a crepant curve center with $(C^2) \leq 0$ and m is the value of $m(p)$ on C, then $C \cong \mathbb{P}^1$, $\mathcal{O}_S(F_m)|_C \cong \mathcal{O}_{\mathbb{P}^1}(2)$, and the restriction map induces an isomorphism $R_m \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2))$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we collect the properties of S, of the crepant curve centers, and of a complement of (S, Δ) that we use below.

Let Δ be as in Construction-Proposition 10.4, and let K_S be as in Construction-Proposition 10.4(4), so that $K_S+\Delta+\alpha D = 0$. By Construction-Proposition 10.4(5), (S, Δ) is a klt log Fano pair. Thus S is a projective klt surface, hence S is $\mathbb{Q}$-factorial [KM98, Proposition 4.11], and $\alpha D = -(K_S + \Delta)$ is ample. For any integer $j \geq 0$, let $\{jD\} := jD - \lfloor jD \rfloor$.

Let C be a crepant curve center, and let m be the value of $m(p)$ on C. By Proposition 10.13(1), C is a smooth projective curve contained in the smooth locus of S, C is Cartier, C is disjoint from the other components of Δ, $m > \alpha$, and $\operatorname{mult}_C \Delta = 1 - \frac{1}{m}$. By Construction-Proposition 10.4(6), $0 \leq \{jD\} \leq \Delta$ for any integer $j \geq 0$, and jD is Cartier near C if m divides j, since $m(p) = m$ for any closed point $p \in C$. Thus, if m divides j, then $\lfloor jD \rfloor = jD$ near C, and $\lfloor jD \rfloor$ is Cartier near C. Taking $C = C_i$ for $1 \leq i \leq s$ and $j = d$, we see that $\mathcal{O}_S(F_d)$ is invertible and $\{dD\} = 0$ near $C_1 \cup \cdots \cup C_s$.

By Proposition 10.14, there exist $N \in \{1, 2, 3, 4, 6\}$ and a monotonic N-complement Δ^+ of (S, Δ) such that $\operatorname{mult}_C \Delta^+ = 1$ for any crepant curve center C. By Definition 10.8, $\Delta^+ \geq \Delta$, (S, Δ^+) is lc, and $N(K_S + \Delta^+) \sim 0$, so $K_S + \Delta^+ \sim_{\mathbb{Q}} 0$.

Finally, $a(v_R, X) = \alpha$ by Construction-Proposition 10.4(3), and v_R is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,o}$. Thus Proposition 5.6(3), applied with $n = 3$, implies that

$$\alpha \operatorname{ord}_P(g) \leq 3j \tag{10.10}$$

for any integer $j > 0$, any non-zero $g \in R_j$, and any prime divisor P on X through o.

Step 2. In this step, we prove (1).

Let $C := C_1 + \cdots + C_s$, let $J := \sum_{i=1}^s \frac{1}{m_i} C_i$, and let $B_0 := \Delta + J$. By **Step 1**, the curves $C_1, \ldots, C_s$ are pairwise disjoint Cartier divisors, so $C \geq 0$ is a Cartier divisor, and

$$\operatorname{mult}_{C_i} \Delta^+ = \operatorname{mult}_{C_i} B_0 = \operatorname{mult}_{C_i} C = 1, \qquad \operatorname{mult}_{C_i}\{dD\} = 0$$

for $1 \leq i \leq s$. For a rational number $0 < \epsilon < 1$, let

$$B_\epsilon := (1-\epsilon)\Delta^+ + \epsilon B_0 - C - \{dD\}.$$

We show that (S, B_ϵ) is klt. Along C_i, the coefficient of B_ϵ is $(1-\epsilon)+\epsilon-1-0 = 0$. Along any other prime divisor on S, the coefficient of $(1-\epsilon)\Delta^+ + \epsilon B_0$ is at least that of Δ, because $\Delta^+ \geq \Delta$ and $B_0 \geq \Delta$, and the coefficient of C is zero. Since $\{dD\} \leq \Delta$, $B_\epsilon \geq 0$. Moreover, $\Delta^+ - C \geq 0$ and $(S, \Delta^+ - C)$ is lc, because $C \geq 0$ and (S, Δ^+) is lc. Also $B_0 - C = \Delta - \sum_{i=1}^s (1 - \frac{1}{m_i}) C_i$ satisfies $0 \leq B_0 - C \leq \Delta$, so $(S, B_0 - C)$ is klt. Since log discrepancies are affine in the boundary, for any prime divisor W over S,

$$a\big(W, S, (1-\epsilon)(\Delta^+ - C) + \epsilon(B_0 - C)\big) = (1-\epsilon)a(W, S, \Delta^+ - C) + \epsilon a(W, S, B_0 - C) > 0.$$

Since $B_\epsilon = (1-\epsilon)(\Delta^+ - C) + \epsilon(B_0 - C) - \{dD\}$ and $\{dD\} \geq 0$, the pair (S, B_ϵ) is klt.

Since $F_d = dD - \{dD\}$, $K_S + \Delta^+ \sim_{\mathbb{Q}} 0$, and $K_S + B_0 = -\alpha D + J$, we obtain

$$(F_d - C) - (K_S + B_\epsilon) = dD - (1-\epsilon)(K_S + \Delta^+) - \epsilon(K_S + B_0) \sim_{\mathbb{Q}} dD + \epsilon(\alpha D - J). \tag{10.11}$$

Since dD is ample and the ample cone of S is open [Laz04, Theorem 1.4.23], the right-hand side of (10.11) is ample for $0 < \epsilon \ll 1$. We fix such an ϵ.

Let $L := F_d - C - K_S$. Since S is $\mathbb{Q}$-factorial, L is a $\mathbb{Q}$-Cartier Weil divisor, and $L - B_\epsilon \sim_{\mathbb{Q}} dD + \epsilon(\alpha D - J)$ is ample by (10.11). By the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.70] for the klt pair (S, B_ϵ), $H^1(S, \mathcal{O}_S(-L)) = 0$. By Serre duality for the projective

dlt pair (S, B_ϵ) [KM98, Corollary 5.27], $H^1(S, \mathcal{O}_S(F_d - C))$ is dual to $H^1(S, \mathcal{O}_S(K_S - F_d + C)) = H^1(S, \mathcal{O}_S(-L))$. Thus

$$H^1(S, \mathcal{O}_S(F_d - C)) = 0. \tag{10.12}$$

Since $C \geq 0$ is a Cartier divisor and $\mathcal{O}_S(F_d)$ is invertible near C (**Step 1**), the sequence

$$0 \to \mathcal{O}_S(F_d - C) \to \mathcal{O}_S(F_d) \to \mathcal{O}_C(F_d|_C) \to 0$$

is exact, because near C it is the restriction sequence of a line bundle to C and away from C the first map is an isomorphism. Since $C_1, \dots, C_s$ are pairwise disjoint,

$$H^0(C, \mathcal{O}_C(F_d|_C)) = \bigoplus_{i=1}^{s} H^0(C_i, \mathcal{O}_S(F_d)|_{C_i}).$$

Thus the long exact sequence of cohomology and (10.12) imply (1).

Step 3. In this step, we prove (2).

Let $k := m - \alpha$. By Proposition 10.13(2), $C \cong \mathbb{P}^1$, $(C^2) = -2k$,

$$\big(-(K_S + \Delta) \cdot C\big) = \frac{2\alpha}{m}, \qquad 0 < k < \frac{\alpha}{39}.$$

In particular, $\alpha > 39k \geq 39$, so $\alpha \geq 40$. By **Step 1**, $F_m = mD$ near C, so

$$(F_m \cdot C) = m(D \cdot C) = \frac{m}{\alpha}\big(-(K_S + \Delta) \cdot C\big) = 2.$$

Thus $\mathcal{O}_S(F_m)|_C \cong \mathcal{O}_{\mathbb{P}^1}(2)$. By (1) applied to C and $d = m$, the restriction map $R_m \to H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2))$ is surjective.

We show that the restriction map is injective. Suppose that a non-zero $g \in R_m$ has zero image in $H^0(C, \mathcal{O}_S(F_m)|_C)$. Let P_C be the closure of the preimage of C under the quotient morphism $X \backslash \{o\} \to S$, which is a prime divisor on X through o by Lemma 10.15. By Lemma 10.15(2) with $j = m$, $\operatorname{ord}_{P_C}(g) \geq m$. By (10.10), $\alpha m \leq \alpha \operatorname{ord}_{P_C}(g) \leq 3m$, so $\alpha \leq 3$, which contradicts $\alpha \geq 40$. Hence the restriction map is an isomorphism, which proves (2). □

Proposition 10.17. *Notation and conditions as in Set-up 10.1, and let $f : Y \to X$ and $E \cong S$ be as in Construction-Proposition 10.4. Assume that $\alpha \geq 2$, that v_R computes $\widehat{\operatorname{vol}}(X \ni o) > q$, and that $X \ni o$ is not LCIQ. Then there exists at most one crepant curve center. In particular, at least two crepant divisors over $X \ni o$ have crepant point centers.*

Proof. Since v_R computes $\widehat{\operatorname{vol}}(X \ni o)$, v_R is a minimizer of $\widehat{\operatorname{vol}}$ on $\operatorname{Val}_{X,o}$, and $a(v_R, X) = \alpha$ by Construction-Proposition 10.4(3). Thus Proposition 5.6(3), applied with $n = 3$, implies that

$$\alpha \operatorname{ord}_P(g) \leq 3j \tag{10.13}$$

for any integer $j > 0$, any non-zero $g \in R_j$, and any prime divisor P on X through o. Let D and $m(p)$ be as in Construction-Proposition 10.4, and for any integer $d \geq 0$, let $F_d := \lfloor dD \rfloor$.

Step 1. In this step, we assume that there exist two crepant curve centers, and we show that the values of $m(p)$ on them are equal.

By Proposition 10.13(2), there exist distinct crepant curve centers C_1 and C_2 with $(C_1^2) \leq 0$. Let m and m_2 be the values of $m(p)$ on C_1 and C_2, respectively, and let $k := m - \alpha$ and $k_2 := m_2 - \alpha$. By Proposition 10.13(1)(2),

$$\alpha \geq 40, \qquad 0 < k < \frac{\alpha}{39}, \qquad \alpha < m < \frac{40\alpha}{39}, \qquad m_2 > \alpha, \qquad 1 \leq k_2 < \frac{\alpha}{2}.$$

By Lemma 10.16(2), $R_m \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2)) \neq 0$. Let $g \in R_m$ be a non-zero element, let P_{C_2} be the closure of the preimage of C_2 under the quotient morphism $X \backslash \{o\} \to S$, which is a prime divisor on X through o by Lemma 10.15, and let $r := \operatorname{ord}_{P_{C_2}}(g)$. By (10.13),

$$\alpha r \leq 3m < \frac{40\alpha}{13},$$

so $r \in \{0,1,2,3\}$. By Lemma 10.15(1), applied to C_2 and $j = m$,

$$m + rk_2 \equiv 0 \pmod{m_2}. \tag{10.14}$$

Suppose that $r \in \{1,2,3\}$. Then

$$m + rk_2 = m_2 + \big(k + (r-1)k_2\big), \qquad 0 < k + (r-1)k_2 \le k + 2k_2 < m_2,$$

where the last inequality holds because $k + k_2 < \frac{\alpha}{39} + \frac{\alpha}{2} < \alpha$. Thus $m_2 < m + rk_2 < 2m_2$, which contradicts (10.14). Hence $r = 0$, and (10.14) implies that m_2 divides m. Since $\alpha < m_2$ and $m < \frac{40\alpha}{39} < 2m_2$, we obtain $m = m_2$.

Step 2. In this step, we finish the proof.

Suppose that there exist two crepant curve centers, and let C_1, C_2, and m be as in **Step 1**. By **Step 1**, $m(p) = m$ on C_1 and on C_2. By Lemma 10.16(2), $\mathcal{O}_S(F_m)|_{C_1} \cong \mathcal{O}_{\mathbb{P}^1}(2)$. By Lemma 10.16(1), applied to C_1, C_2, and $d = m$, there exists $g' \in R_m$ whose image in $H^0(C_1, \mathcal{O}_S(F_m)|_{C_1}) \oplus H^0(C_2, \mathcal{O}_S(F_m)|_{C_2})$ is the pair of a non-zero section of $\mathcal{O}_S(F_m)|_{C_1}$ and the zero section of $\mathcal{O}_S(F_m)|_{C_2}$. In particular, $g' \neq 0$. Let P_{C_2} be as in **Step 1**. By Lemma 10.15(2) with $j = m$, $\mathrm{ord}_{P_{C_2}}(g') \ge m$. By (10.13), $\alpha m \le 3m$, so $\alpha \le 3$, which contradicts $\alpha \ge 40$. Hence there exists at most one crepant curve center.

By Proposition 10.14(1), there exist at least three crepant divisors over $X \ni o$. By Proposition 10.13(1), the center on $E \cong S$ of any crepant divisor over $X \ni o$ is a point or a crepant curve center, and distinct crepant divisors have distinct centers. Since there exists at most one crepant curve center, at least two crepant divisors over $X \ni o$ have crepant point centers. □

10.7. **Canonical degree at least two.** In this subsection, we prove Theorem 10.3. Assuming $\widehat{\mathrm{vol}}(X \ni o) > q$, we obtain a contradiction from two incompatible properties of homogeneous elements of degree m, where $p_1 \neq p_2$ are the centers of two crepant divisors and $m := m(p_1) = m(p_2)$ by **Step 4**. On a higher model of S extracting divisors W_1, W_2 centered at p_1, p_2, the Kawamata–Viehweg vanishing for a klt boundary obtained by perturbing the complement of Proposition 10.14 yields such an element that vanishes at p_2 and does not vanish at p_1 (**Step 2**), whereas Proposition 5.6 forces any non-zero element of degree m to be non-zero at both p_1 and p_2 (**Steps 3 and 5**).

Proof of Theorem 10.3. The proof is long, so we divide it into several steps. Let $V := \widehat{\mathrm{vol}}(X \ni o)$, and suppose that $V > q$.

Step 1. In this step, we fix two crepant point centers and the local data at them.

By Lemma 4.3, $V > q > \frac{54}{7}$, so the hypothesis $V > \frac{54}{7}$ of Lemma 10.10 holds. Let $f : Y \to X$, $E \cong S$, D, Δ, u, and $m(p)$ be as in Construction-Proposition 10.4, and let $F_j := \lfloor jD \rfloor$ for any integer $j \ge 0$. Thus $R_j = H^0(S, \mathcal{O}_S(F_j))u^j$ for any $j \ge 0$.

By Proposition 10.17, there exist crepant divisors G_1 and G_2 over $X \ni o$ whose centers on Y are crepant point centers $\{p_1\}$ and $\{p_2\}$ (Definition 10.11), and $p_1 \neq p_2$ by Proposition 10.13(1). Through $E \cong S$, we regard p_1 and p_2 as points of Y. For $i = 1, 2$, let $m_i := m(p_i)$. By Lemma 10.10(3),

$$\alpha < m_i < \frac{7\alpha}{6} \qquad \text{for } i = 1, 2. \tag{10.15}$$

For $i = 1, 2$, let U_i, T_i, and $T_i \to U_i$ be as in Construction-Proposition 10.4(2) for p_i, and let $\tilde{p}_i \in T_i$ be the point over p_i. Since $\alpha \ge 2$, $V > q$, and $m_i > \alpha$, Lemma 10.12(1) implies that T_i is smooth at $\tilde{p}_i$, and that there exist local coordinates s_1, s_2 of T_i at $\tilde{p}_i$ and integers $0 \le b_i, c_i < m_i$ such that μ_{m_i} acts on (s_1, s_2, t) with weights $(-b_i, -c_i, -1)$, where t is the coordinate of $\mathbb{A}^1$. For both $i = 1$ and $i = 2$, we write s_1, s_2 for the coordinates of T_i at $\tilde{p}_i$. For $i = 1, 2$, since

$\operatorname{center}_Y G_i = \{p_i\}$, Lemma 10.12(2) implies that $b_i + c_i = m_i - \alpha$, that $b_i, c_i > 0$, and that G_i is the toric divisor over Y of the ray spanned by $\frac{1}{m_i}(b_i, c_i, 1)$. Therefore

$$b_i > 0, \qquad c_i > 0, \qquad b_i + c_i = m_i - \alpha \qquad \text{for } i = 1, 2. \tag{10.16}$$

By Lemma 10.10(2), for $i = 1, 2$, the restriction of ord_{G_i} to $K(S) \subset K(X)$ is a positive multiple of ord_{W_i}, where $W_1 \neq W_2$ are prime divisors over S such that

$$0 < \epsilon_i := a(W_i, S, \Delta) < \frac{1}{7} \qquad \text{for } i = 1, 2.$$

Here $K(S) \subset K(Y) = K(X)$ is given by the projection $Y \to S$. For $i = 1, 2$, since $\operatorname{center}_Y G_i = \{p_i\}$, ord_{G_i} is non-negative on $\mathcal{O}_{S,p_i} \subset \mathcal{O}_{Y,p_i}$ and positive on $\mathfrak{m}_{S,p_i} \subset \mathfrak{m}_{Y,p_i}$. Thus $\operatorname{center}_S W_i = \{p_i\}$ for $i = 1, 2$.

By Proposition 10.14 and Definition 10.8, there exist a positive integer N and a $\mathbb{Q}$-divisor $\Delta^+ \geq \Delta$ on S such that (S, Δ^+) is lc, $N(K_S + \Delta^+) \sim 0$, and $a(W_i, S, \Delta^+) = 0$ for $i = 1, 2$. Moreover, (S, Δ) is a klt log Fano pair and $K_S + \Delta + \alpha D = 0$ by Construction-Proposition 10.4(4)(5).

Step 2. In this step, we show the following. Let $I \subset \{1, 2\}$ be a non-empty subset, and let d be a positive integer such that $m_i \mid d$ for any $i \in I$. Then $\mathcal{O}_S(F_d)$ is invertible near p_i for any $i \in I$, and the map

$$H^0(S, \mathcal{O}_S(F_d)) \to \bigoplus_{i \in I} \mathcal{O}_S(F_d) \otimes k(p_i), \qquad \sigma \to (\sigma(p_i))_{i \in I} \tag{10.17}$$

is surjective, where $\sigma(p_i)$ denotes the image of σ in $\mathcal{O}_S(F_d) \otimes k(p_i)$.

By Construction-Proposition 10.4(6), $m_i D$ is Cartier near p_i. Since $m_i \mid d$, dD is Cartier near p_i, so $F_d = dD$ near p_i and $\mathcal{O}_S(F_d)$ is invertible near p_i. Moreover, $0 \leq dD - F_d \leq \Delta$ by Construction-Proposition 10.4(6).

The prime divisors W_i, $i \in I$, are exceptional over S and $a(W_i, S, \Delta) = \epsilon_i < 1$. By [Kol13, Corollary 1.38] (see also [BCHM10, Corollary 1.4.3]), applied to the klt, hence dlt, pair (S, Δ) and the prime divisors W_i, $i \in I$, whose discrepancies satisfy $-1 < \epsilon_i - 1 < 0$, there exists a proper birational morphism $h : S' \to S$ (the model in [Kol13, Corollary 1.38] is a minimal model obtained in [Kol13, 1.35], which is proper over the base by [Kol13, Definition 1.19(1)]) such that S' is $\mathbb{Q}$-factorial and the exceptional curves of h are the curves E'_i, $i \in I$, with $\operatorname{ord}_{E'_i} = \operatorname{ord}_{W_i}$. Since $\operatorname{center}_S W_i = \{p_i\}$, we have $h(E'_i) = p_i$. Since S is normal, Zariski's main theorem implies that h is an isomorphism over $S \backslash \{p_i \mid i \in I\}$ and that $h^{-1}(p_i) = E'_i$ for any $i \in I$. In particular, $E'_1 \cap E'_2 = \emptyset$ if $I = \{1, 2\}$.

Let $D' := h^*D$, and let Δ' and Δ'^+ be the crepant pullbacks of Δ and Δ^+, that is, $K_{S'} + \Delta' = h^*(K_S + \Delta)$, $K_{S'} + \Delta'^+ = h^*(K_S + \Delta^+)$, $h_*\Delta' = \Delta$, and $h_*\Delta'^+ = \Delta^+$. Then

$$\Delta' = h_*^{-1}\Delta + \sum_{i \in I} (1 - \epsilon_i) E'_i, \qquad \Delta'^+ = h_*^{-1}\Delta^+ + \sum_{i \in I} E'_i.$$

By [KM98, Lemma 2.30], (S', Δ') is klt and (S', Δ'^+) is lc. Moreover, $K_{S'} + \Delta'^+ \sim_{\mathbb{Q}} 0$ and $K_{S'} + \Delta' = -\alpha D'$. Let

$$J := \sum_{i \in I} \epsilon_i E'_i, \qquad \{dD'\} := dD' - \lfloor dD' \rfloor.$$

Since dD is Cartier near p_i, $dD' = h^*(dD)$ is Cartier near $E'_i = h^{-1}(p_i)$, so $\lfloor dD' \rfloor = h^*F_d$ near E'_i and $\{dD'\} = 0$ near E'_i. Since h is an isomorphism over $S \backslash \{p_i \mid i \in I\}$, $\{dD'\}$ is the strict transform of $dD - F_d$. Hence

$$0 \leq \{dD'\} \leq h_*^{-1}\Delta. \tag{10.18}$$

We show that $(E'^2_i) < 0$ for any $i \in I$. Let $h' : S'' \to S'$ be a resolution. Since S' is $\mathbb{Q}$-factorial, $h'^*E'_i \geq 0$ is a non-zero $\mathbb{Q}$-divisor whose support consists of curves contracted by $h \circ h'$ to p_i. By [KM98, Lemma 3.40] applied to the proper birational morphism $h \circ h'$ from the smooth surface

S'', the intersection matrix of the exceptional curves of $h \circ h'$ over p_i is negative definite. Thus $(E_i'^2) = ((h'^* E_i')^2) < 0$.

We show that $-J + \gamma D'$ is ample for some rational number $\gamma > \alpha + d$. Fix a positive integer n such that nD is a very ample Cartier divisor. Then there exists a Cartier divisor $L \geq 0$ on S with $L \sim nD$ whose support contains p_1 and p_2. For $i \in I$, let $\mu_i := \operatorname{mult}_{E_i'} h^* L$. Then $\mu_i > 0$ because $h(E_i') = p_i \in \operatorname{Supp} L$. Let $C_1, \dots, C_l$ be the irreducible components of $h_*^{-1} L$. For any curve C on S' that is not one of the E_i', $h_* C$ is a curve and $(D' \cdot C) = (D \cdot h_* C) > 0$. Fix a rational number $\gamma > \alpha + d$ such that

$$\gamma > \sum_{i \in I} \frac{n \epsilon_i}{\mu_i}, \qquad \gamma (D' \cdot C_j) > (J \cdot C_j) \text{ for } 1 \leq j \leq l, \qquad \gamma^2 (D^2) + (J^2) > 0.$$

We check the conditions of the Nakai–Moishezon criterion for $-J + \gamma D'$. Since the curves E_i' are pairwise disjoint and $(D' \cdot E_i') = 0$, we have

$$((-J + \gamma D') \cdot E_i') = -\epsilon_i (E_i'^2) > 0.$$

By the choice of γ, $((-J + \gamma D') \cdot C_j) > 0$ for $1 \leq j \leq l$. Let C be any other curve on S'. Then $C \not\subset \operatorname{Supp} h^* L$, so for any $i \in I$,

$$n (D' \cdot C) = (h^* L \cdot C) \geq \mu_i (E_i' \cdot C).$$

Thus

$$(J \cdot C) = \sum_{i \in I} \epsilon_i (E_i' \cdot C) \leq \Big(\sum_{i \in I} \frac{n \epsilon_i}{\mu_i} \Big) (D' \cdot C) < \gamma (D' \cdot C),$$

that is, $((-J + \gamma D') \cdot C) > 0$. Finally, $(D' \cdot J) = 0$ and $(D'^2) = (D^2)$, so

$$((-J + \gamma D')^2) = \gamma^2 (D^2) + (J^2) > 0.$$

Since S' is $\mathbb{Q}$-factorial and proper, [KM98, Theorem 1.37] applied to a Cartier multiple of $-J + \gamma D'$ implies that $-J + \gamma D'$ is ample. In particular, S' is projective.

Let $\lambda := \frac{d}{\gamma - \alpha}$. Then $0 < \lambda < 1$ and $d + \lambda \alpha = \lambda \gamma$. Let

$$B_\lambda := (1 - \lambda) h_*^{-1} \Delta^+ + \lambda h_*^{-1} \Delta - \{ d D' \}.$$

Since $\Delta^+ \geq \Delta$, (10.18) implies that $B_\lambda \geq h_*^{-1} \Delta - \{ d D' \} \geq 0$. The pair $(S', h_*^{-1} \Delta^+)$ is lc because $h_*^{-1} \Delta^+ = \Delta'^+ - \sum_{i \in I} E_i'$, and the pair $(S', h_*^{-1} \Delta)$ is klt because $h_*^{-1} \Delta \leq \Delta'$. Since log discrepancies are affine in the boundary, for any prime divisor P over S',

$$a(P, S', (1 - \lambda) h_*^{-1} \Delta^+ + \lambda h_*^{-1} \Delta) = (1 - \lambda) a(P, S', h_*^{-1} \Delta^+) + \lambda a(P, S', h_*^{-1} \Delta) > 0.$$

Since $B_\lambda \leq (1 - \lambda) h_*^{-1} \Delta^+ + \lambda h_*^{-1} \Delta$, (S', B_λ) is a projective klt pair. Since $\Delta'^+ = h_*^{-1} \Delta^+ + \sum_{i \in I} E_i'$ and $\Delta' + J = h_*^{-1} \Delta + \sum_{i \in I} E_i'$, we have

$$K_{S'} + B_\lambda + \sum_{i \in I} E_i' + \{ d D' \} = (1 - \lambda)(K_{S'} + \Delta'^+) + \lambda (K_{S'} + \Delta' + J).$$

Since $K_{S'} + \Delta'^+ \sim_{\mathbb{Q}} 0$ and $K_{S'} + \Delta' = -\alpha D'$, we obtain

$$\lfloor d D' \rfloor - \sum_{i \in I} E_i' - (K_{S'} + B_\lambda) \sim_{\mathbb{Q}} (d + \lambda \alpha) D' - \lambda J = \lambda (-J + \gamma D'),$$

which is ample. By [KM98, Theorem 2.70] applied to (S', B_λ) and the $\mathbb{Q}$-Cartier divisor $\lfloor d D' \rfloor - \sum_{i \in I} E_i' - K_{S'}$, together with Serre duality [KM98, Corollary 5.27],

$$H^1 \Big(S', \mathcal{O}_{S'} \Big(\lfloor d D' \rfloor - \sum_{i \in I} E_i' \Big) \Big) = 0. \tag{10.19}$$

The sheaf $\mathcal{O}_{S'}(-\sum_{i \in I} E_i')$ is the ideal sheaf of the reduced curve $\sum_{i \in I} E_i'$, and $\lfloor d D' \rfloor$ is Cartier near $\sum_{i \in I} E_i'$. Since the curves E_i' are pairwise disjoint, we obtain an exact sequence

$$0 \to \mathcal{O}_{S'} \Big(\lfloor d D' \rfloor - \sum_{i \in I} E_i' \Big) \to \mathcal{O}_{S'}(\lfloor d D' \rfloor) \to \bigoplus_{i \in I} \mathcal{O}_{S'}(\lfloor d D' \rfloor)|_{E_i'} \to 0.$$

By (10.19), the map

$$H^0(S', \mathcal{O}_{S'}(\lfloor dD' \rfloor)) \to \bigoplus_{i \in I} H^0(E'_i, \mathcal{O}_{S'}(\lfloor dD' \rfloor)|_{E'_i})$$

is surjective. For $i \in I$, let e_i be a generator of $\mathcal{O}_S(F_d)$ on a neighborhood U'_i of p_i. Then $\mathcal{O}_{S'}(\lfloor dD' \rfloor) = \mathcal{O}_{S'} \cdot h^* e_i$ on $h^{-1}(U'_i)$, so $\mathcal{O}_{S'}(\lfloor dD' \rfloor)|_{E'_i} \cong \mathcal{O}_{E'_i}$. Since E'_i is an integral proper curve, $H^0(E'_i, \mathcal{O}_{E'_i}) = \mathbb{C}$.

We show that $h_* \mathcal{O}_{S'}(\lfloor dD' \rfloor) = \mathcal{O}_S(F_d)$. Over $S \backslash \{p_i \mid i \in I\}$, h is an isomorphism and $\lfloor dD' \rfloor = h_*^{-1} F_d$. Over U'_i, $\lfloor dD' \rfloor = h^* F_d$ with F_d Cartier. Since h is proper and birational and S is normal, $h_* \mathcal{O}_{S'}$ is a coherent $\mathcal{O}_S$-algebra contained in $K(S)$, so $h_* \mathcal{O}_{S'} = \mathcal{O}_S$. Thus the projection formula [Har77, Exercise II.5.1(d)] implies that $h_* \mathcal{O}_{S'}(h^* F_d) = \mathcal{O}_S(F_d)$ over U'_i.

Therefore $H^0(S', \mathcal{O}_{S'}(\lfloor dD' \rfloor)) = H^0(S, \mathcal{O}_S(F_d))$. If $\sigma \in H^0(S, \mathcal{O}_S(F_d))$ and $\sigma = a e_i$ on U'_i with $a \in \mathcal{O}_S(U'_i)$, then the restriction of $h^* \sigma$ to E'_i is $a(p_i) \cdot h^* e_i|_{E'_i}$. Under the isomorphisms

$$H^0(E'_i, \mathcal{O}_{S'}(\lfloor dD' \rfloor)|_{E'_i}) \cong \mathbb{C} \cong \mathcal{O}_S(F_d) \otimes k(p_i)$$

given by $h^* e_i$ and e_i, the surjective map above is (10.17). This proves the surjectivity of (10.17).

In particular, applying **Step 2** to $I = \{i\}$ and $d = m_i$, we obtain $H^0(S, \mathcal{O}_S(F_{m_i})) \neq 0$. Thus

$$R_{m_1} \neq 0 \qquad \text{and} \qquad R_{m_2} \neq 0. \tag{10.20}$$

Step 3. In this step, we fix $i \in \{1, 2\}$, describe homogeneous elements of R near p_i, and bound their values along a valuation transverse to E at p_i.

Shrinking U_i, we may assume that U_i is affine, so T_i is affine. By [Kol04, Paragraph 18 and Proposition 19], the isomorphism $Y_{U_i} \cong (T_i \times \mathbb{A}^1)/\mu_{m_i}$ of Construction-Proposition 10.4(2) is compatible with the projections to U_i, and it is $\mathbb{G}_m$-equivariant, where $\mathbb{G}_m$ acts on Y_{U_i} through the grading of $\bigoplus_{j \geq 0} \mathcal{O}_{U_i}(F_j)$ and acts on $T_i \times \mathbb{A}^1$ by scaling t. Since $Y_{U_i} = \operatorname{Spec}_{U_i} \bigoplus_{j \geq 0} \mathcal{O}_{U_i}(F_j)$, the regular functions on Y_{U_i} on which $\mathbb{G}_m$ acts with weight $j \geq 0$ form $H^0(U_i, \mathcal{O}_{U_i}(F_j)) u^j$, and they correspond to the μ_{m_i}-invariant elements of $\mathcal{O}(T_i) t^j$. We show the following.

(1) Let d be a positive integer, and let $0 \neq g \in R_d$. Then the pullback of g to $T_i \times \mathbb{A}^1$ is $t^d \varphi$ for some $0 \neq \varphi \in \mathcal{O}(T_i)$, and any monomial $s_1^{j_1} s_2^{j_2}$ with non-zero coefficient in the expansion of φ in $\widehat{\mathcal{O}}_{T_i, \tilde{p}_i} = \mathbb{C}[[s_1, s_2]]$ satisfies

$$b_i j_1 + c_i j_2 \equiv -d \pmod{m_i}.$$

(2) Assume moreover that $m_i \mid d$, and write $g = \sigma u^d$ with $\sigma \in H^0(S, \mathcal{O}_S(F_d))$. Then $\sigma(p_i) \neq 0$ if and only if $\varphi(\tilde{p}_i) \neq 0$.

For (1), g restricts to a non-zero element of $H^0(U_i, \mathcal{O}_{U_i}(F_d)) u^d$, so the pullback of g is $t^d \varphi$ for some $0 \neq \varphi \in \mathcal{O}(T_i)$. Since s_1, s_2 are eigenfunctions, μ_{m_i} acts on $\mathbb{C}[[s_1, s_2]]$ monomial by monomial. Since $t^d \varphi$ is μ_{m_i}-invariant and μ_{m_i} acts on (s_1, s_2, t) with weights $(-b_i, -c_i, -1)$, any monomial $s_1^{j_1} s_2^{j_2}$ with non-zero coefficient in φ satisfies $-b_i j_1 - c_i j_2 - d \equiv 0 \pmod{m_i}$.

For (2), shrinking U_i, we may assume that $\mathcal{O}_{U_i}(F_d) = \mathcal{O}_{U_i} \cdot e$ for some $e \in K(S)$, since $\mathcal{O}_S(F_d)$ is invertible near p_i by **Step 2**. Let $t^d \varphi_e$ be the pullback of $e u^d$, where $\varphi_e \in \mathcal{O}(T_i)$. Since $m_i \mid d$, t^d is μ_{m_i}-invariant, so t^d is the pullback of $a_0 e u^d$ for some $a_0 \in \mathcal{O}(U_i)$. Since the isomorphism is compatible with the projections to U_i, we obtain $t^d = a_0 t^d \varphi_e$ on $T_i \times \mathbb{A}^1$, where a_0 is regarded as a function on T_i. Thus $a_0 \varphi_e = 1$, and φ_e is a unit on T_i. Write $\sigma = ae$ on U_i with $a \in \mathcal{O}(U_i)$. Then $\varphi = a \varphi_e$, so $\varphi(\tilde{p}_i) = a(p_i) \varphi_e(\tilde{p}_i)$. Hence $\varphi(\tilde{p}_i) \neq 0$ if and only if $a(p_i) \neq 0$, that is, $\sigma(p_i) \neq 0$.

Now we construct the valuation. By Lemma 10.12(1), the germ of Y at p_i is analytically isomorphic, through the coordinates (s_1, s_2, t), to the germ at the origin of the affine toric variety of the non-negative octant in the lattice $\mathbb{Z}^3 + \mathbb{Z} \frac{1}{m_i}(b_i, c_i, 1)$. The vector $(b_i, c_i, 0)$ is a positive integer multiple k_i of a primitive vector of the lattice. Let Q_i be the toric divisor over Y of the ray spanned by $\frac{1}{k_i}(b_i, c_i, 0)$ (Lemma 10.12(1)), and let $\nu_i := k_i \operatorname{ord}_{Q_i}$. By the formula for log

discrepancies in Lemma 10.12(1),

$$a(\nu_i, X) = k_i a(Q_i, X) = b_i + c_i = m_i - \alpha.$$

By Lemma 10.12(1), the value of ν_i on any non-zero element of $\mathcal{O}_{Y,p_i}$ is the minimum of $b_i j_1 + c_i j_2$ over the monomials $s_1^{j_1} s_2^{j_2} t^{j_3}$ with non-zero coefficient in its pullback to $\mathbb{C}[[s_1, s_2, t]]$ (Q_i is an irreducible component of the base change of the toric divisor of the ray along an étale morphism induced by (s_1, s_2, t), and for a primitive lattice vector $w = (w_1, w_2, w_3)$ in the octant, the valuation of the toric divisor of the ray spanned by w is the torus-invariant valuation with value $j_1 w_1 + j_2 w_2 + j_3 w_3$ on the monomial $s_1^{j_1} s_2^{j_2} t^{j_3}$, that is, the monomial valuation with weights w, cf. [CLS11, Proposition 4.1.1]). In particular, for d, g, and φ as in (1),

$$\nu_i(g) = \min\{b_i j_1 + c_i j_2 \mid s_1^{j_1} s_2^{j_2} \text{ has non-zero coefficient in } \varphi\}. \tag{10.21}$$

Since $b_i, c_i > 0$, $\nu_i(g) = 0$ if and only if $\varphi(\tilde{p}_i) \neq 0$. Moreover, ν_i is non-negative on $\mathcal{O}_{Y,p_i}$, so the center of ν_i on Y contains p_i. Since $f(p_i) = o$ by Construction-Proposition 10.4(1), we have $o \in \operatorname{center}_X Q_i$.

Let d be a positive integer and $0 \neq g \in R_d$. Since v_R computes V, v_R is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,o}$, and $a(v_R, X) = \alpha$ by Construction-Proposition 10.4(3). Let $\mathfrak{b} := g\mathcal{O}_X$. Then $v_R(\mathfrak{b}) = d$ and $\mathrm{mult}_{Q_i} \mathfrak{b} = \mathrm{ord}_{Q_i}(g)$. By Proposition 5.6(1) applied to $(X \ni o, 0)$, v_R, $\mathfrak{b}$, and $w = \mathrm{ord}_{Q_i}$, we have $3da(Q_i, X) \geq \alpha \,\mathrm{ord}_{Q_i}(g)$. Multiplying by k_i, we obtain

$$\alpha \nu_i(g) \leq 3d(m_i - \alpha). \tag{10.22}$$

Step 4. In this step, we show that $m_1 = m_2$.

Suppose that $m_1 > m_2$. By (10.20), there exists $0 \neq g \in R_{m_1}$. By (10.15), $m_2 < m_1 < \frac{7\alpha}{6} < 2\alpha < 2m_2$. Thus the least non-negative integer congruent to $-m_1$ modulo m_2 is $2m_2 - m_1$. By **Step 3**(1) for $i = 2$ and $d = m_1$, together with (10.21), $\nu_2(g)$ is a non-negative integer congruent to $-m_1$ modulo m_2. Hence, by (10.15),

$$\nu_2(g) \geq 2m_2 - m_1 > 2\alpha - \frac{7\alpha}{6} = \frac{5\alpha}{6}.$$

On the other hand, $m_2 - \alpha < \frac{\alpha}{6}$ by (10.15), so (10.22) for $i = 2$ implies that

$$\nu_2(g) \leq \frac{3m_1(m_2 - \alpha)}{\alpha} < \frac{3}{\alpha} \cdot \frac{7\alpha}{6} \cdot \frac{\alpha}{6} = \frac{7\alpha}{12},$$

which is impossible since $\frac{7\alpha}{12} < \frac{5\alpha}{6}$. Thus $m_1 \leq m_2$. Interchanging p_1 and p_2, we obtain $m_2 \leq m_1$. Therefore $m_1 = m_2$. Let $m := m_1 = m_2$.

Step 5. In this step, we show that the surjectivity of **Step 2** in degree m contradicts the inequality (10.22), which forces any non-zero element of R_m to be non-zero at both p_1 and p_2.

Let $0 \neq g \in R_m$, write $g = \sigma u^m$ with $\sigma \in H^0(S, \mathcal{O}_S(F_m))$, and let $i \in \{1, 2\}$. By **Step 3**(1) for $d = m$ and (10.21), $\nu_i(g)$ is a non-negative integer divisible by m. By (10.15), $m - \alpha < \frac{\alpha}{6}$, so (10.22) implies that

$$\nu_i(g) \leq \frac{3m(m - \alpha)}{\alpha} < \frac{m}{2} < m.$$

Thus $\nu_i(g) = 0$, so $\varphi(\tilde{p}_i) \neq 0$ by **Step 3**, and $\sigma(p_i) \neq 0$ by **Step 3**(2). Therefore any non-zero $\sigma \in H^0(S, \mathcal{O}_S(F_m))$ satisfies $\sigma(p_1) \neq 0$ and $\sigma(p_2) \neq 0$.

On the other hand, by **Step 2** applied to $I = \{1, 2\}$ and $d = m$, there exists $\sigma \in H^0(S, \mathcal{O}_S(F_m))$ such that $\sigma(p_1) \neq 0$ and $\sigma(p_2) = 0$. Then $\sigma \neq 0$ and σu^m is a non-zero element of R_m, which contradicts the previous paragraph. Therefore $V \leq q$. □

11. The equality case and the proof of Theorem 1.4

In this section, we prove Theorem 1.4. Theorem 1.3 was proved in Section 3.

11.1. **The minimizing cone at equality.** In this subsection, we show that a minimizer on a klt threefold germ of local volume q has rational rank two, and we identify the minimizing cone when it is Gorenstein and not LCIQ.

We first relate the degrees of the associated graded ring of a minimizer to the value group of the minimizer, and we find a one-parameter subgroup which contracts the minimizing cone to the vertex.

Lemma 11.1. *Let $X \ni x$ be a klt germ with $\dim X = n$, let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, let $X_0 := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$ with vertex x_0, and let $M \subset \mathbb{R}$ be the subgroup generated by the degrees λ with $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$. Then the following hold.*

(1) $\dim X_0 = n$, $M = v(K(X)^*)$, and M is a free abelian group whose rank is the rational rank of v.

(2) There exists a one-parameter subgroup of the torus $\operatorname{Spec} \mathbb{C}[M]$ which contracts X_0 to x_0.

Proof. By [XZ25, Theorem 1.2(1)], $\dim X_0 = n$. Let $g \in \mathcal{O}_{X,x}$ be a non-zero element. Then the image of g in $\mathfrak{a}_{v(g)}(v)/\mathfrak{a}_{>v(g)}(v)$ is non-zero, so $v(g) \in M$. Since $K(X)$ is the fraction field of $\mathcal{O}_{X,x}$, we have $v(K(X)^*) \subset M$. Conversely, if $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$, then $\lambda = v(g)$ for any $g \in \mathfrak{a}_\lambda(v) \backslash \mathfrak{a}_{>\lambda}(v)$. Thus $M = v(K(X)^*)$.

By Theorem 2.12(2), the $\mathbb{C}$-algebra $\operatorname{gr}_v \mathcal{O}_{X,x}$ is generated by finitely many homogeneous elements of positive degrees $u_1, \dots, u_s$. Any degree λ with $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$ is a sum of elements of $\{u_1, \dots, u_s\}$, so M is generated by $u_1, \dots, u_s$. Hence M is a finitely generated torsion-free abelian group. Thus M is free of rank $\dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} M)$, which is the rational rank of v. This proves (1).

By Theorem 2.12(2), the inclusion $M \subset \mathbb{R}$ defines a Reeb vector $\xi \in \operatorname{Hom}(M, \mathbb{R})$, so that $\langle u, \xi \rangle = u$ for any $u \in M$. The set

$$\{\varphi \in \operatorname{Hom}(M, \mathbb{R}) \mid \langle u_i, \varphi \rangle > 0 \text{ for any } 1 \leq i \leq s\}$$

is open and contains ξ. Hence it contains an element $\varphi_1 \in \operatorname{Hom}(M, \mathbb{Q})$, and a positive integer multiple of φ_1 is a homomorphism $\varphi_0 \in \operatorname{Hom}(M, \mathbb{Z})$ with $\langle u_i, \varphi_0 \rangle > 0$ for any i. Let $\gamma : \mathbb{G}_m \to \operatorname{Spec} \mathbb{C}[M]$ be the one-parameter subgroup corresponding to φ_0. Then $\gamma(t)$ acts on the homogeneous piece of degree u by multiplication by $t^{\langle u, \varphi_0 \rangle}$. Any degree $u > 0$ with a non-zero homogeneous piece is a non-zero sum of elements of $\{u_1, \dots, u_s\}$, so $\langle u, \varphi_0 \rangle > 0$. Since the homogeneous piece of degree 0 is $\mathbb{C}$, for any closed point $y \in X_0$ the orbit map $t \to \gamma(t) \cdot y$ extends to $\mathbb{A}^1$ and sends 0 to the point defined by the ideal of elements of positive degree, which is x_0. Thus γ contracts X_0 to x_0. □

The following lemma excludes minimizers of rational rank one in the equality case, since $q \notin \mathbb{Q}$.

Lemma 11.2. *Let $X \ni x$ be a klt germ, and let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$. Assume that v has rational rank one. Then $\widehat{\mathrm{vol}}(X \ni x) \in \mathbb{Q}$.*

Proof. Let $n := \dim X$. By Lemma 11.1(1), the value group $v(K(X)^*)$ is free of rank one, so $v(K(X)^*) = \lambda\mathbb{Z}$ for a real number $\lambda > 0$. Let $v' := v/\lambda$. Then $v'(K(X)^*) = \mathbb{Z}$.

By Theorem 2.12(1), v is quasi-monomial, and hence so is v'. Thus there exist a log smooth pair (Y, D) with a proper birational morphism $\pi : Y \to X$ which is an isomorphism outside $\operatorname{Supp} D$, and a point $\eta \in Y$, such that $v' \in \mathrm{QM}_\eta(Y, D)$ [JM12, Section 3.1]. Let $E_1, \dots, E_r$ be the components of D through η, and let $z_1, \dots, z_r$ be local equations of $E_1, \dots, E_r$ at η. Then v' is the monomial valuation in $z_1, \dots, z_r$ with weights $a_i := v'(z_i) \geq 0$. Since $z_i \in K(X)^*$, we have $a_i \in \mathbb{Z}$. By [Li18, Section 2.1],

$$a(v', X) = \sum_{i=1}^{r} a_i (1 + \operatorname{mult}_{E_i} K_{Y/X}) = \sum_{i=1}^{r} a_i a(E_i, X).$$

Since K_X is $\mathbb{Q}$-Cartier, $a(E_i, X) \in \mathbb{Q}$ for any i. Thus $a(v', X) \in \mathbb{Q}$.

Since the values of v' are integers, $\mathfrak{a}_{>j}(v') = \mathfrak{a}_{j+1}(v')$ for any integer $j \geq 0$. Let

$$G := \operatorname{gr}_{v'} \mathcal{O}_{X,x} = \bigoplus_{j\geq 0} G_j, \quad G_j := \mathfrak{a}_j(v')/\mathfrak{a}_{j+1}(v').$$

The ring G is $\operatorname{gr}_v \mathcal{O}_{X,x}$ with the degrees divided by λ. By Theorem 2.12(2) and Lemma 11.1(1), G is a finitely generated $\mathbb{C}$-algebra with $G_0 = \mathbb{C}$ and $\dim G = n$. By [BH98, Proposition 4.4.1], there exist positive integers $d_1, \dots, d_n$ and a Laurent polynomial $P(z) = \sum_l p_l z^l \in \mathbb{Z}[z, z^{-1}]$ such that

$$\sum_{j\geq 0} \dim_{\mathbb{C}} G_j z^j = \frac{P(z)}{\prod_{i=1}^n (1 - z^{d_i})}.$$

For a positive integer m, the chain $\mathcal{O}_{X,x} = \mathfrak{a}_0(v') \supset \cdots \supset \mathfrak{a}_m(v')$ has the factors $G_0, \dots, G_{m-1}$. Since the coefficient of z^j in the power series $\prod_{i=1}^n (1 - z^{d_i})^{-1}$ is $\#\{b \in \mathbb{Z}^n_{\geq 0} \mid \sum_i d_i b_i = j\}$, we obtain

$$\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{a}_m(v') = \sum_{j=0}^{m-1} \dim_{\mathbb{C}} G_j = \sum_l p_l \cdot \#\Big\{b \in \mathbb{Z}^n_{\geq 0} \;\Big|\; \sum_{i=1}^n d_i b_i < m - l\Big\}.$$

For a real number $k > 0$, let $\Pi_k := \{y \in \mathbb{R}^n_{\geq 0} \mid \sum_{i=1}^n d_i y_i < k\}$, so that $\operatorname{vol}(\Pi_k) = \frac{k^n}{n! \prod_i d_i}$. For $m > l$, the cubes $b + [0,1)^n$ with $b \in \mathbb{Z}^n_{\geq 0} \cap \Pi_{m-l}$ are pairwise disjoint, cover Π_{m-l}, and are contained in $\Pi_{m-l+d_1+\cdots+d_n}$. Hence

$$\frac{(m-l)^n}{n! \prod_i d_i} \leq \#(\mathbb{Z}^n_{\geq 0} \cap \Pi_{m-l}) \leq \frac{(m - l + d_1 + \cdots + d_n)^n}{n! \prod_i d_i}.$$

Thus, for any l,

$$\#(\mathbb{Z}^n_{\geq 0} \cap \Pi_{m-l}) = \frac{m^n}{n! \prod_i d_i} + O(m^{n-1}),$$

and

$$\operatorname{vol}(v') = \lim_{m\to+\infty} \frac{\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{a}_m(v')}{m^n/n!} = \frac{P(1)}{\prod_{i=1}^n d_i} \in \mathbb{Q}.$$

Since $a(v, X) = \lambda a(v', X)$ and $\operatorname{vol}(v) = \lambda^{-n} \operatorname{vol}(v')$, we have

$$\widehat{\operatorname{vol}}(X \ni x) = \widehat{\operatorname{vol}}(v) = a(v', X)^n \operatorname{vol}(v') \in \mathbb{Q}. \quad \square$$

Proposition 11.3. *Let $X \ni x$ be a klt threefold germ with $\widehat{\operatorname{vol}}(X \ni x) = q$, let v be a minimizer of $\widehat{\operatorname{vol}}$ on $\operatorname{Val}_{X,x}$, and let $X_0 := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$ with vertex x_0. Then the following hold.*

(1) The valuation v has rational rank two.

(2) If $X_0 \ni x_0$ is Gorenstein and not LCIQ, then there exists an isomorphism $X_0 \cong C_{\mathbb{F}_1}$ of affine varieties which sends x_0 to o.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

By Proposition 6.4, the rational rank of v is at most two. Since v is centered at x, v is positive on the maximal ideal of $\mathcal{O}_{X,x}$, so the rational rank of v is at least one. If the rational rank of v is one, then $q = \widehat{\operatorname{vol}}(X \ni x) \in \mathbb{Q}$ by Lemma 11.2. Since $\sqrt{13} \notin \mathbb{Q}$, we have $q \notin \mathbb{Q}$. Thus v has rational rank two.

Step 2. In this step, we show that X_0 is as in Set-up 7.1 and Gorenstein.

Assume that $X_0 \ni x_0$ is Gorenstein and not LCIQ. Let $M \subset \mathbb{R}$ be the subgroup generated by the degrees of $\operatorname{gr}_v \mathcal{O}_{X,x}$, let $T := \operatorname{Spec} \mathbb{C}[M]$, and let ξ be the Reeb vector of X_0 defined by the inclusion $M \subset \mathbb{R}$ (Theorem 2.12(2)). By **Step 1** and Lemma 11.1(1), $M \cong \mathbb{Z}^2$, so $T \cong (\mathbb{C}^*)^2$ is a torus with character lattice M. By Theorem 2.12(2), X_0 is a normal affine klt variety, and T acts faithfully on X_0. By Lemma 11.1, $\dim X_0 = 3$, and a one-parameter subgroup γ of T

contracts X_0 to x_0. Thus X_0, T, M, and x_0 satisfy the conditions of Set-up 7.1 with X_0 and x_0 in place of X and o. By Theorem 2.12(3), wt_ξ computes $\widehat{\mathrm{vol}}(X_0 \ni x_0) = q$.

We show that X_0 is Gorenstein. Let ω be the canonical module of $\mathbb{C}[X_0]$ with the M-grading, and let $\mathfrak{m}_{x_0}$ be the ideal of x_0, which is generated by the elements of positive degree. Since $\mathcal{O}_{X_0,x_0}$ is Gorenstein, $\dim_{\mathbb{C}} \omega/\mathfrak{m}_{x_0}\omega = 1$, so there exists a homogeneous element $\eta \in \omega$ whose image spans $\omega/\mathfrak{m}_{x_0}\omega$. By Nakayama's lemma, the cokernel Q of $\mathbb{C}[X_0]\eta \to \omega$ vanishes near x_0. The support of Q is closed and T-invariant, and the closure of the γ-orbit of any point of X_0 contains x_0. Hence $Q = 0$. Since ω is torsion-free of rank one and $\eta \neq 0$, we have $\omega = \mathbb{C}[X_0]\eta \cong \mathbb{C}[X_0]$. Since X_0 is klt, X_0 is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. Thus X_0 is Gorenstein [BH98, Theorem 3.3.7(a)]. Let κ be the canonical character of X_0, and write $\kappa = \zeta\theta$ with $\theta \in M$ primitive and ζ a positive integer (Definition-Lemma 7.2).

Step 3. In this step, we show that X_0 is toric.

Suppose that X_0 is not toric. If X_0 is $\mathbb{Q}$-factorial at x_0, then $X_0 \ni x_0$ is LCIQ by Proposition 7.5, which is not possible. Thus X_0 is not $\mathbb{Q}$-factorial at x_0. If $\zeta = 1$, then κ is primitive in M, and Proposition 7.12, applied with X_0 and x_0 in place of X and o since X_0 is Gorenstein and not toric and X_0 is not $\mathbb{Q}$-factorial at x_0, implies that $\widehat{\mathrm{vol}}(X_0 \ni x_0) \leq \frac{20}{3}$. If $\zeta \geq 2$, then κ is not primitive in M, and the last statement of Theorem 7.3, applied with X_0 and x_0 in place of X and o since X_0 is Gorenstein and not toric, $X_0 \ni x_0$ is not LCIQ, and wt_ξ computes $\widehat{\mathrm{vol}}(X_0 \ni x_0)$ by **Step 2**, implies that $\widehat{\mathrm{vol}}(X_0 \ni x_0) \leq \frac{27}{4}$. By Lemma 4.3, $q > 7 > \frac{27}{4} > \frac{20}{3}$. Both bounds contradict $\widehat{\mathrm{vol}}(X_0 \ni x_0) = q$. Thus X_0 is toric.

Step 4. In this step, we show that $X_0 \cong C_{\mathbb{F}_1}$ with x_0 sent to o.

By **Step 3**, there exist a lattice N of rank three and a strongly convex rational polyhedral cone $\sigma_0 \subset N_{\mathbb{R}}$ such that X_0 is isomorphic to the affine toric variety X_{σ_0}, since X_0 is normal [CLS11, Theorem 1.3.5]. Let τ be a face of σ_0 with $\dim \tau \leq 2$, and let $y \in X_{\sigma_0}$ be a point of the torus orbit corresponding to τ. Since τ has at most two rays, τ is simplicial, and the ray generators of τ are primitive in N. Let $k := \dim \tau$, let $u_1, \dots, u_k$ be the ray generators of τ, and let $N_\tau := N \cap \mathrm{span}(\tau)$. Let U_τ be the affine toric variety of τ with respect to N, and U_{τ,N_τ} the affine toric variety of τ with respect to N_τ. Then $U_\tau \subset X_{\sigma_0}$ is a torus-invariant open subset containing the orbit corresponding to τ [CLS11, Theorem 3.2.6(c)]. Since N_τ is saturated in N, we have $U_\tau \cong U_{\tau,N_\tau} \times (\mathbb{C}^*)^{3-k}$ [CLS11, Proposition 3.3.11(a)], and $U_{\tau,N_\tau} \cong \mathbb{C}^k/G$ with $G := N_\tau/(\mathbb{Z}u_1 + \cdots + \mathbb{Z}u_k)$ [CLS11, Example 1.3.20]. By Lemma 6.2, applied to the sublattice $\mathbb{Z}u_1 + \cdots + \mathbb{Z}u_k \subset N_\tau$ and the cone τ, the toric morphism $\mathbb{C}^k \to U_{\tau,N_\tau}$ is the quotient morphism by G, and G acts freely on $\mathbb{C}^k$ outside a closed subset of codimension at least two. Hence y has a torus-invariant neighborhood which is the product of a torus and the quotient of an affine space by a finite abelian group acting freely in codimension one, with y the image of a fixed point. In particular, $X_{\sigma_0} \ni y$ is LCIQ.

Since $X_0 \ni x_0$ is not LCIQ, the image of x_0 belongs to the orbit of a face of dimension three. Thus $\dim \sigma_0 = 3$, and x_0 is sent to the torus-fixed point of X_{σ_0}. The variety X_{σ_0} is Gorenstein and klt by **Step 2**, the germ at the torus-fixed point is not LCIQ, and the local volume there is q. By Theorem 6.1, applied with σ_0 in place of σ, (σ_0, N) is isomorphic to $(\sigma_{\mathbb{F}_1}, \mathbb{Z}^3)$. By Construction-Proposition 4.2, there exists an isomorphism $X_0 \to C_{\mathbb{F}_1}$ which sends x_0 to o. This proves (2).

□

11.2. **Proof of Theorem 1.4.** In this subsection, we prove Theorem 1.4.

Proof of Theorem 1.4. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that $\widehat{\mathrm{vol}}(X \ni x) \leq q$.

Let $V := \widehat{\mathrm{vol}}(X \ni x)$, and suppose that $V > q$. Let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$ (Theorem 2.12(1)), let $X_0 := \operatorname{Spec} \mathrm{gr}_v \mathcal{O}_{X,x}$ with vertex x_0, let $M \subset \mathbb{R}$ be as in Theorem 2.12(2), and let $T := \operatorname{Spec} \mathbb{C}[M]$. By Proposition 5.8(2)(3), X_0 is a normal Gorenstein canonical affine

threefold, the action of T on X_0 is faithful and good (Definition 2.8), the inclusion $M \subset \mathbb{R}$ defines a Reeb vector ξ such that (X_0, ξ) is a K-semistable Fano cone with $\widehat{\mathrm{vol}}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(X_0 \ni x_0) = V$, the canonical module of $\mathbb{C}[X_0]$ is generated by a homogeneous element η, and $X_0 \ni x_0$ is not LCIQ. By Lemma 11.1(1) and Proposition 6.4, M is free of rank at most two. Since $\dim X_0 = 3$, we have $M \neq 0$.

Suppose that $\operatorname{rank} M = 2$. Then $T \cong (\mathbb{C}^*)^2$, and by Lemma 11.1(2), X_0, T, M, and x_0 satisfy the conditions of Set-up 7.1 with X_0 and x_0 in place of X and o. Since X_0 is Gorenstein, $X_0 \ni x_0$ is not LCIQ, and wt_ξ computes $\widehat{\mathrm{vol}}(X_0 \ni x_0)$, Theorem 7.3 implies that $V \leq q$, a contradiction.

Suppose that $\operatorname{rank} M = 1$. Then $M = \lambda\mathbb{Z}$ for a real number $\lambda > 0$, and the homomorphism $\xi_0 \in \operatorname{Hom}(M, \mathbb{Z})$ with $\langle \lambda, \xi_0 \rangle = 1$ satisfies $\xi = \lambda\xi_0$. Thus ξ is quasi-regular, and ξ_0 is the primitive element of $\operatorname{Hom}(M, \mathbb{Z})$ in $\mathbb{R}_{>0}\xi$. Let $R := \mathbb{C}[X_0]$ with the grading $R_j := \bigoplus_{\langle u, \xi_0 \rangle = j} \mathbb{C}[X_0]_u$, and let $\alpha := a(\mathrm{wt}_{\xi_0}, X_0)$. By Lemma 10.2, applied with X_0 and x_0 in place of X and x, the data R, X_0, x_0, $\operatorname{Proj} R$, η, and α satisfy the conditions of Set-up 10.1 with X_0 and x_0 in place of X and o, and v_R computes $\widehat{\mathrm{vol}}(X_0 \ni x_0) = V$. In particular, α is a positive integer. If $\alpha = 1$, then Proposition 10.6 implies that $V \leq 7$, since $X_0 \ni x_0$ is not LCIQ. By Lemma 4.3, $q > 7$, a contradiction. If $\alpha \geq 2$, then Theorem 10.3 implies that $V \leq q$, a contradiction. Thus $V \leq q$.

Step 2. In this step, we prove that $C_{\mathbb{F}_1} \ni o$ is not LCIQ, and that equality holds if $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$.

By Construction-Proposition 4.2, $C_{\mathbb{F}_1} \ni o$ is not LCIQ and $\widehat{\mathrm{vol}}(C_{\mathbb{F}_1} \ni o) = q$. If $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$, then $\widehat{\mathrm{vol}}(X \ni x) = q$ by Proposition 2.7(1).

Step 3. In this step, we prove that $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$ if $\widehat{\mathrm{vol}}(X \ni x) = q$. Recognizing the minimizing cone does not yet recognize the germ. We use the formal rigidity of $C_{\mathbb{F}_1}$ over reduced complete bases on the completion of the Rees family at the section, and we return to the germ by Proposition 5.9.

Assume that $\widehat{\mathrm{vol}}(X \ni x) = q$. Let v be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, and let $X_0 := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$ with vertex x_0. By Proposition 5.8(2)(3), X_0 is Gorenstein and $X_0 \ni x_0$ is not LCIQ. By Proposition 11.3(2), there exists an isomorphism $X_0 \cong C_{\mathbb{F}_1}$ which sends x_0 to o. By Proposition 5.8(4), there exist an affine open neighborhood U of x in X and a prime divisor F over X with $\operatorname{center}_X F = \{x\}$ such that, for $w := \mathrm{ord}_F$, the morphism $f : \mathcal{X} \to \mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[t]$ defined by the extended Rees algebra of w is flat, affine, and of finite type, and f has a section σ such that $\mathcal{X}_0 \ni \sigma(0)$ is isomorphic to $X_0 \ni x_0$, and $\mathcal{X}_t \ni \sigma(t)$ is isomorphic to $U \ni x$ for any $t \neq 0$. In particular, $\mathcal{X}_0 \ni \sigma(0)$ is isomorphic to $C_{\mathbb{F}_1} \ni o$.

We check the conditions of Proposition 4.4(3) with $\Lambda = \mathbb{C}[[t]]$ and $\mathcal{A} = \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$, where $\widehat{\mathcal{O}}_{C_{\mathbb{F}_1},o}$ is the ring $\widehat{R}$ of Proposition 4.4. The ring $\mathbb{C}[[t]]$ is a reduced complete Noetherian local $\mathbb{C}$-algebra with residue field $\mathbb{C}$. Since $f(\sigma(0)) = 0$, the map $\mathbb{C}[t] \to \mathcal{O}_{\mathcal{X},\sigma(0)}$ extends to a local homomorphism $\mathbb{C}[[t]] \to \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$. Since $\mathcal{X}$ is affine and of finite type over $\mathbb{A}^1$, there exists a closed embedding of $\mathcal{X}$ into $\mathbb{A}^n \times \mathbb{A}^1$ over $\mathbb{A}^1$ which sends $\sigma(0)$ to the origin. Hence $\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} = \mathbb{C}[[t]][[z_1, \dots, z_n]]/I$ for an ideal I.

Since $\mathcal{O}_{\mathcal{X},\sigma(0)}$ is Noetherian and local, the completion $\mathcal{O}_{\mathcal{X},\sigma(0)} \to \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$ is flat, and the completion of a finitely generated $\mathcal{O}_{\mathcal{X},\sigma(0)}$-module is the tensor product with $\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$ [Mat86, Theorems 8.7 and 8.8]. Since f is flat, t is a nonzerodivisor in $\mathcal{O}_{\mathcal{X},\sigma(0)}$, and hence t is a nonzerodivisor in $\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$. Moreover,

$$\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}/t\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} \cong \widehat{\mathcal{O}}_{\mathcal{X}_0,\sigma(0)} \cong \widehat{\mathcal{O}}_{C_{\mathbb{F}_1},o}.$$

Let $k \geq 0$ be an integer. If $g \in \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$ and $tg \in t^{k+1}\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$, then $g \in t^k\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$, since t is a nonzerodivisor. Thus, for $\mathcal{A}_k := \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}/t^{k+1}\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$, the kernel of the multiplication by t on $\mathcal{A}_k$ is $t^k\mathcal{A}_k$, so $\mathcal{A}_k$ is flat over $\mathbb{C}[t]/(t^{k+1})$ [Mat86, Theorem 22.3].

Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$. We show that the natural map

$$\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} \to \varprojlim_k \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}/t^{k+1}\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$$

is an isomorphism. Since $t^{k+1}\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} \subset \mathfrak{m}^{k+1}$, the map is injective by the Krull intersection theorem [Mat86, Theorem 8.10(i)]. Let $(\bar{g}_k)_k$ be a compatible sequence in the inverse limit. Then there exist lifts $g_k \in \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$ of $\bar{g}_k$ with $g_{k+1} - g_k \in t^{k+1}\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} \subset \mathfrak{m}^{k+1}$. Hence $(g_k)_k$ converges $\mathfrak{m}$-adically to an element $g \in \widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$. Since any ideal of $\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$ is closed in the $\mathfrak{m}$-adic topology [Mat86, Theorem 8.10(i)], we have $g - g_k \in t^{k+1}\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)}$ for any k. Thus g maps to $(\bar{g}_k)_k$, and the map is surjective.

By Proposition 4.4(3), there exists an isomorphism of $\mathbb{C}[[t]]$-algebras

$$\widehat{\mathcal{O}}_{\mathcal{X},\sigma(0)} \cong \mathbb{C}[[t]]\widehat{\otimes}_{\mathbb{C}}\widehat{\mathcal{O}}_{C_{\mathbb{F}_1},o} = \widehat{\mathcal{O}}_{C_{\mathbb{F}_1}\times\mathbb{A}^1,(o,0)},$$

where $\mathbb{C}[[t]] = \widehat{\mathcal{O}}_{\mathbb{A}^1,0}$. By Construction-Proposition 4.2, $C_{\mathbb{F}_1} \ni o$ is an isolated singularity. Hence the hypotheses of Proposition 5.9 hold for the smooth pointed curve $(\mathbb{A}^1, 0)$, the morphism f, the section σ, and $C_{\mathbb{F}_1} \ni o$ in place of $X' \ni x'$. Since a smooth germ is LCIQ (Definition 2.13) and $X \ni x$ is not LCIQ, the germ $X \ni x$ is singular. For any $t \neq 0$, the germ $\mathcal{X}_t \ni \sigma(t)$ is isomorphic to $U \ni x$, which is the germ $X \ni x$. Thus Proposition 5.9 implies that $X \ni x$ is analytically isomorphic to $C_{\mathbb{F}_1} \ni o$.

Steps 1, 2, and 3 prove Theorem 1.4. □

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